AEROSPACE OEM ENGINEERING COMPENDIUM — VOLUME 2

Covers 25 Certified Machine Models (Models #26 to #50)
350 Full-Page Fault SOPs | 125 Pages PLC Registers | 75 Pages BOM & PM Matrices

INCLUDED MACHINE MODELS IN THIS VOLUME:

• Model #26: Mazak-Integrex-i400 (Multi-Tasking Jet Shaft Center) — Controller: MAZATROL SmoothAi — Prefix: MT-TURN-XXXX
• Model #27: Mazak-Integrex-i450 (Multi-Tasking Jet Shaft Center) — Controller: MAZATROL SmoothAi — Prefix: MT-TURN-XXXX
• Model #28: Mazak-Integrex-i500 (Multi-Tasking Jet Shaft Center) — Controller: MAZATROL SmoothAi — Prefix: MT-TURN-XXXX
• Model #29: Mazak-Integrex-i550 (Multi-Tasking Jet Shaft Center) — Controller: MAZATROL SmoothAi — Prefix: MT-TURN-XXXX
• Model #30: Mazak-Integrex-i600 (Multi-Tasking Jet Shaft Center) — Controller: MAZATROL SmoothAi — Prefix: MT-TURN-XXXX
• Model #31: Okuma-Multus-U150 (5-Axis Inconel Blisk Mill-Turn) — Controller: OSP-P500A Aerospace — Prefix: TURB-LATHE-XXXX
• Model #32: Okuma-Multus-U200 (5-Axis Inconel Blisk Mill-Turn) — Controller: OSP-P500A Aerospace — Prefix: TURB-LATHE-XXXX
• Model #33: Okuma-Multus-U250 (5-Axis Inconel Blisk Mill-Turn) — Controller: OSP-P500A Aerospace — Prefix: TURB-LATHE-XXXX
• Model #34: Okuma-Multus-U300 (5-Axis Inconel Blisk Mill-Turn) — Controller: OSP-P500A Aerospace — Prefix: TURB-LATHE-XXXX
• Model #35: Okuma-Multus-U350 (5-Axis Inconel Blisk Mill-Turn) — Controller: OSP-P500A Aerospace — Prefix: TURB-LATHE-XXXX
• Model #36: Okuma-Multus-U400 (5-Axis Inconel Blisk Mill-Turn) — Controller: OSP-P500A Aerospace — Prefix: TURB-LATHE-XXXX
• Model #37: Okuma-Multus-U450 (5-Axis Inconel Blisk Mill-Turn) — Controller: OSP-P500A Aerospace — Prefix: TURB-LATHE-XXXX
• Model #38: Okuma-Multus-U500 (5-Axis Inconel Blisk Mill-Turn) — Controller: OSP-P500A Aerospace — Prefix: TURB-LATHE-XXXX
• Model #39: Okuma-Multus-U550 (5-Axis Inconel Blisk Mill-Turn) — Controller: OSP-P500A Aerospace — Prefix: TURB-LATHE-XXXX
• Model #40: Okuma-Multus-U600 (5-Axis Inconel Blisk Mill-Turn) — Controller: OSP-P500A Aerospace — Prefix: TURB-LATHE-XXXX
• Model #41: EOS-Additive-M150 (Quad-Laser Metal Sintering Cell) — Controller: EOSYSTEM Dual-PLC — Prefix: DMLS-3D-XXXX
• Model #42: EOS-Additive-M200 (Quad-Laser Metal Sintering Cell) — Controller: EOSYSTEM Dual-PLC — Prefix: DMLS-3D-XXXX
• Model #43: EOS-Additive-M250 (Quad-Laser Metal Sintering Cell) — Controller: EOSYSTEM Dual-PLC — Prefix: DMLS-3D-XXXX
• Model #44: EOS-Additive-M300 (Quad-Laser Metal Sintering Cell) — Controller: EOSYSTEM Dual-PLC — Prefix: DMLS-3D-XXXX
• Model #45: EOS-Additive-M350 (Quad-Laser Metal Sintering Cell) — Controller: EOSYSTEM Dual-PLC — Prefix: DMLS-3D-XXXX
• Model #46: EOS-Additive-M400 (Quad-Laser Metal Sintering Cell) — Controller: EOSYSTEM Dual-PLC — Prefix: DMLS-3D-XXXX
• Model #47: EOS-Additive-M450 (Quad-Laser Metal Sintering Cell) — Controller: EOSYSTEM Dual-PLC — Prefix: DMLS-3D-XXXX
• Model #48: EOS-Additive-M500 (Quad-Laser Metal Sintering Cell) — Controller: EOSYSTEM Dual-PLC — Prefix: DMLS-3D-XXXX
• Model #49: EOS-Additive-M550 (Quad-Laser Metal Sintering Cell) — Controller: EOSYSTEM Dual-PLC — Prefix: DMLS-3D-XXXX
• Model #50: EOS-Additive-M600 (Quad-Laser Metal Sintering Cell) — Controller: EOSYSTEM Dual-PLC — Prefix: DMLS-3D-

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OPERATING ENVELOPES & METADATA SPECIFICATIONS (PART 1/9)

MACHINE MODEL: Mazak-Integrex-i400 (Asset Prefix: MT-TURN)

• RAG Metadata Filter Value: "Mazak-Integrex-i400"
• Equipment Classification: Multi-Tasking Jet Shaft Center | Manufacturer: Mazak-Integrex
• CNC / PLC Control System: MAZATROL SmoothAi
• Max Spindle / Primary Drive Rating: 21000 RPM
• Coolant / Process Supply Pressure: 118 Bar High-Pressure Loop
• Thermal Operating Limit: 54°C Max Continuous | Electrical Load: 81 kW
• Volumetric Positioning Accuracy: ±0.0021 mm

MACHINE MODEL: Mazak-Integrex-i450 (Asset Prefix: MT-TURN)

• RAG Metadata Filter Value: "Mazak-Integrex-i450"
• Equipment Classification: Multi-Tasking Jet Shaft Center | Manufacturer: Mazak-Integrex
• CNC / PLC Control System: MAZATROL SmoothAi
• Max Spindle / Primary Drive Rating: 22500 RPM
• Coolant / Process Supply Pressure: 126 Bar High-Pressure Loop
• Thermal Operating Limit: 55°C Max Continuous | Electrical Load: 87 kW
• Volumetric Positioning Accuracy: ±0.0022 mm

MACHINE MODEL: Mazak-Integrex-i500 (Asset Prefix: MT-TURN)

• RAG Metadata Filter Value: "Mazak-Integrex-i500"
• Equipment Classification: Multi-Tasking Jet Shaft Center | Manufacturer: Mazak-Integrex
• CNC / PLC Control System: MAZATROL SmoothAi
• Max Spindle / Primary Drive Rating: 24000 RPM
• Coolant / Process Supply Pressure: 134 Bar High-Pressure Loop
• Thermal Operating Limit: 56°C Max Continuous | Electrical Load: 93 kW
• Volumetric Positioning Accuracy: ±0.0023 mm

OPERATING ENVELOPES & METADATA SPECIFICATIONS (PART 2/9)

MACHINE MODEL: Mazak-Integrex-i550 (Asset Prefix: MT-TURN)

• RAG Metadata Filter Value: "Mazak-Integrex-i550"
• Equipment Classification: Multi-Tasking Jet Shaft Center | Manufacturer: Mazak-Integrex
• CNC / PLC Control System: MAZATROL SmoothAi
• Max Spindle / Primary Drive Rating: 25500 RPM
• Coolant / Process Supply Pressure: 142 Bar High-Pressure Loop
• Thermal Operating Limit: 57°C Max Continuous | Electrical Load: 99 kW
• Volumetric Positioning Accuracy: ±0.0015 mm

MACHINE MODEL: Mazak-Integrex-i600 (Asset Prefix: MT-TURN)

• RAG Metadata Filter Value: "Mazak-Integrex-i600"
• Equipment Classification: Multi-Tasking Jet Shaft Center | Manufacturer: Mazak-Integrex
• CNC / PLC Control System: MAZATROL SmoothAi
• Max Spindle / Primary Drive Rating: 27000 RPM
• Coolant / Process Supply Pressure: 150 Bar High-Pressure Loop
• Thermal Operating Limit: 48°C Max Continuous | Electrical Load: 105 kW
• Volumetric Positioning Accuracy: ±0.0016 mm

MACHINE MODEL: Okuma-Multus-U150 (Asset Prefix: TURB-LATHE)

• RAG Metadata Filter Value: "Okuma-Multus-U150"
• Equipment Classification: 5-Axis Inconel Blisk Mill-Turn | Manufacturer: Okuma-Multus
• CNC / PLC Control System: OSP-P500A Aerospace
• Max Spindle / Primary Drive Rating: 13500 RPM
• Coolant / Process Supply Pressure: 78 Bar High-Pressure Loop
• Thermal Operating Limit: 49°C Max Continuous | Electrical Load: 51 kW
• Volumetric Positioning Accuracy: ±0.0016 mm

OPERATING ENVELOPES & METADATA SPECIFICATIONS (PART 3/9)

MACHINE MODEL: Okuma-Multus-U200 (Asset Prefix: TURB-LATHE)

• RAG Metadata Filter Value: "Okuma-Multus-U200"
• Equipment Classification: 5-Axis Inconel Blisk Mill-Turn | Manufacturer: Okuma-Multus
• CNC / PLC Control System: OSP-P500A Aerospace
• Max Spindle / Primary Drive Rating: 15000 RPM
• Coolant / Process Supply Pressure: 86 Bar High-Pressure Loop
• Thermal Operating Limit: 50°C Max Continuous | Electrical Load: 57 kW
• Volumetric Positioning Accuracy: ±0.0017 mm

MACHINE MODEL: Okuma-Multus-U250 (Asset Prefix: TURB-LATHE)

• RAG Metadata Filter Value: "Okuma-Multus-U250"
• Equipment Classification: 5-Axis Inconel Blisk Mill-Turn | Manufacturer: Okuma-Multus
• CNC / PLC Control System: OSP-P500A Aerospace
• Max Spindle / Primary Drive Rating: 16500 RPM
• Coolant / Process Supply Pressure: 94 Bar High-Pressure Loop
• Thermal Operating Limit: 51°C Max Continuous | Electrical Load: 63 kW
• Volumetric Positioning Accuracy: ±0.0018 mm

MACHINE MODEL: Okuma-Multus-U300 (Asset Prefix: TURB-LATHE)

• RAG Metadata Filter Value: "Okuma-Multus-U300"
• Equipment Classification: 5-Axis Inconel Blisk Mill-Turn | Manufacturer: Okuma-Multus
• CNC / PLC Control System: OSP-P500A Aerospace
• Max Spindle / Primary Drive Rating: 18000 RPM
• Coolant / Process Supply Pressure: 102 Bar High-Pressure Loop
• Thermal Operating Limit: 52°C Max Continuous | Electrical Load: 69 kW
• Volumetric Positioning Accuracy: ±0.0019 mm

OPERATING ENVELOPES & METADATA SPECIFICATIONS (PART 4/9)

MACHINE MODEL: Okuma-Multus-U350 (Asset Prefix: TURB-LATHE)

• RAG Metadata Filter Value: "Okuma-Multus-U350"
• Equipment Classification: 5-Axis Inconel Blisk Mill-Turn | Manufacturer: Okuma-Multus
• CNC / PLC Control System: OSP-P500A Aerospace
• Max Spindle / Primary Drive Rating: 19500 RPM
• Coolant / Process Supply Pressure: 110 Bar High-Pressure Loop
• Thermal Operating Limit: 53°C Max Continuous | Electrical Load: 75 kW
• Volumetric Positioning Accuracy: ±0.0020 mm

MACHINE MODEL: Okuma-Multus-U400 (Asset Prefix: TURB-LATHE)

• RAG Metadata Filter Value: "Okuma-Multus-U400"
• Equipment Classification: 5-Axis Inconel Blisk Mill-Turn | Manufacturer: Okuma-Multus
• CNC / PLC Control System: OSP-P500A Aerospace
• Max Spindle / Primary Drive Rating: 21000 RPM
• Coolant / Process Supply Pressure: 118 Bar High-Pressure Loop
• Thermal Operating Limit: 54°C Max Continuous | Electrical Load: 81 kW
• Volumetric Positioning Accuracy: ±0.0021 mm

MACHINE MODEL: Okuma-Multus-U450 (Asset Prefix: TURB-LATHE)

• RAG Metadata Filter Value: "Okuma-Multus-U450"
• Equipment Classification: 5-Axis Inconel Blisk Mill-Turn | Manufacturer: Okuma-Multus
• CNC / PLC Control System: OSP-P500A Aerospace
• Max Spindle / Primary Drive Rating: 22500 RPM
• Coolant / Process Supply Pressure: 126 Bar High-Pressure Loop
• Thermal Operating Limit: 55°C Max Continuous | Electrical Load: 87 kW
• Volumetric Positioning Accuracy: ±0.0022 mm

OPERATING ENVELOPES & METADATA SPECIFICATIONS (PART 5/9)

MACHINE MODEL: Okuma-Multus-U500 (Asset Prefix: TURB-LATHE)

• RAG Metadata Filter Value: "Okuma-Multus-U500"
• Equipment Classification: 5-Axis Inconel Blisk Mill-Turn | Manufacturer: Okuma-Multus
• CNC / PLC Control System: OSP-P500A Aerospace
• Max Spindle / Primary Drive Rating: 24000 RPM
• Coolant / Process Supply Pressure: 134 Bar High-Pressure Loop
• Thermal Operating Limit: 56°C Max Continuous | Electrical Load: 93 kW
• Volumetric Positioning Accuracy: ±0.0023 mm

MACHINE MODEL: Okuma-Multus-U550 (Asset Prefix: TURB-LATHE)

• RAG Metadata Filter Value: "Okuma-Multus-U550"
• Equipment Classification: 5-Axis Inconel Blisk Mill-Turn | Manufacturer: Okuma-Multus
• CNC / PLC Control System: OSP-P500A Aerospace
• Max Spindle / Primary Drive Rating: 25500 RPM
• Coolant / Process Supply Pressure: 142 Bar High-Pressure Loop
• Thermal Operating Limit: 57°C Max Continuous | Electrical Load: 99 kW
• Volumetric Positioning Accuracy: ±0.0015 mm

MACHINE MODEL: Okuma-Multus-U600 (Asset Prefix: TURB-LATHE)

• RAG Metadata Filter Value: "Okuma-Multus-U600"
• Equipment Classification: 5-Axis Inconel Blisk Mill-Turn | Manufacturer: Okuma-Multus
• CNC / PLC Control System: OSP-P500A Aerospace
• Max Spindle / Primary Drive Rating: 27000 RPM
• Coolant / Process Supply Pressure: 150 Bar High-Pressure Loop
• Thermal Operating Limit: 48°C Max Continuous | Electrical Load: 105 kW
• Volumetric Positioning Accuracy: ±0.0016 mm

OPERATING ENVELOPES & METADATA SPECIFICATIONS (PART 6/9)

MACHINE MODEL: EOS-Additive-M150 (Asset Prefix: DMLS-3D)

• RAG Metadata Filter Value: "EOS-Additive-M150"
• Equipment Classification: Quad-Laser Metal Sintering Cell | Manufacturer: EOS-Additive
• CNC / PLC Control System: EOSYSTEM Dual-PLC
• Max Spindle / Primary Drive Rating: 13500 RPM
• Coolant / Process Supply Pressure: 78 Bar High-Pressure Loop
• Thermal Operating Limit: 49°C Max Continuous | Electrical Load: 51 kW
• Volumetric Positioning Accuracy: ±0.0016 mm

MACHINE MODEL: EOS-Additive-M200 (Asset Prefix: DMLS-3D)

• RAG Metadata Filter Value: "EOS-Additive-M200"
• Equipment Classification: Quad-Laser Metal Sintering Cell | Manufacturer: EOS-Additive
• CNC / PLC Control System: EOSYSTEM Dual-PLC
• Max Spindle / Primary Drive Rating: 15000 RPM
• Coolant / Process Supply Pressure: 86 Bar High-Pressure Loop
• Thermal Operating Limit: 50°C Max Continuous | Electrical Load: 57 kW
• Volumetric Positioning Accuracy: ±0.0017 mm

MACHINE MODEL: EOS-Additive-M250 (Asset Prefix: DMLS-3D)

• RAG Metadata Filter Value: "EOS-Additive-M250"
• Equipment Classification: Quad-Laser Metal Sintering Cell | Manufacturer: EOS-Additive
• CNC / PLC Control System: EOSYSTEM Dual-PLC
• Max Spindle / Primary Drive Rating: 16500 RPM
• Coolant / Process Supply Pressure: 94 Bar High-Pressure Loop
• Thermal Operating Limit: 51°C Max Continuous | Electrical Load: 63 kW
• Volumetric Positioning Accuracy: ±0.0018 mm

OPERATING ENVELOPES & METADATA SPECIFICATIONS (PART 7/9)

MACHINE MODEL: EOS-Additive-M300 (Asset Prefix: DMLS-3D)

• RAG Metadata Filter Value: "EOS-Additive-M300"
• Equipment Classification: Quad-Laser Metal Sintering Cell | Manufacturer: EOS-Additive
• CNC / PLC Control System: EOSYSTEM Dual-PLC
• Max Spindle / Primary Drive Rating: 18000 RPM
• Coolant / Process Supply Pressure: 102 Bar High-Pressure Loop
• Thermal Operating Limit: 52°C Max Continuous | Electrical Load: 69 kW
• Volumetric Positioning Accuracy: ±0.0019 mm

MACHINE MODEL: EOS-Additive-M350 (Asset Prefix: DMLS-3D)

• RAG Metadata Filter Value: "EOS-Additive-M350"
• Equipment Classification: Quad-Laser Metal Sintering Cell | Manufacturer: EOS-Additive
• CNC / PLC Control System: EOSYSTEM Dual-PLC
• Max Spindle / Primary Drive Rating: 19500 RPM
• Coolant / Process Supply Pressure: 110 Bar High-Pressure Loop
• Thermal Operating Limit: 53°C Max Continuous | Electrical Load: 75 kW
• Volumetric Positioning Accuracy: ±0.0020 mm

MACHINE MODEL: EOS-Additive-M400 (Asset Prefix: DMLS-3D)

• RAG Metadata Filter Value: "EOS-Additive-M400"
• Equipment Classification: Quad-Laser Metal Sintering Cell | Manufacturer: EOS-Additive
• CNC / PLC Control System: EOSYSTEM Dual-PLC
• Max Spindle / Primary Drive Rating: 21000 RPM
• Coolant / Process Supply Pressure: 118 Bar High-Pressure Loop
• Thermal Operating Limit: 54°C Max Continuous | Electrical Load: 81 kW
• Volumetric Positioning Accuracy: ±0.0021 mm

OPERATING ENVELOPES & METADATA SPECIFICATIONS (PART 8/9)

MACHINE MODEL: EOS-Additive-M450 (Asset Prefix: DMLS-3D)

• RAG Metadata Filter Value: "EOS-Additive-M450"
• Equipment Classification: Quad-Laser Metal Sintering Cell | Manufacturer: EOS-Additive
• CNC / PLC Control System: EOSYSTEM Dual-PLC
• Max Spindle / Primary Drive Rating: 22500 RPM
• Coolant / Process Supply Pressure: 126 Bar High-Pressure Loop
• Thermal Operating Limit: 55°C Max Continuous | Electrical Load: 87 kW
• Volumetric Positioning Accuracy: ±0.0022 mm

MACHINE MODEL: EOS-Additive-M500 (Asset Prefix: DMLS-3D)

• RAG Metadata Filter Value: "EOS-Additive-M500"
• Equipment Classification: Quad-Laser Metal Sintering Cell | Manufacturer: EOS-Additive
• CNC / PLC Control System: EOSYSTEM Dual-PLC
• Max Spindle / Primary Drive Rating: 24000 RPM
• Coolant / Process Supply Pressure: 134 Bar High-Pressure Loop
• Thermal Operating Limit: 56°C Max Continuous | Electrical Load: 93 kW
• Volumetric Positioning Accuracy: ±0.0023 mm

MACHINE MODEL: EOS-Additive-M550 (Asset Prefix: DMLS-3D)

• RAG Metadata Filter Value: "EOS-Additive-M550"
• Equipment Classification: Quad-Laser Metal Sintering Cell | Manufacturer: EOS-Additive
• CNC / PLC Control System: EOSYSTEM Dual-PLC
• Max Spindle / Primary Drive Rating: 25500 RPM
• Coolant / Process Supply Pressure: 142 Bar High-Pressure Loop
• Thermal Operating Limit: 57°C Max Continuous | Electrical Load: 99 kW
• Volumetric Positioning Accuracy: ±0.0015 mm

OPERATING ENVELOPES & METADATA SPECIFICATIONS (PART 9/9)

MACHINE MODEL: EOS-Additive-M600 (Asset Prefix: DMLS-3D)

• RAG Metadata Filter Value: "EOS-Additive-M600"
• Equipment Classification: Quad-Laser Metal Sintering Cell | Manufacturer: EOS-Additive
• CNC / PLC Control System: EOSYSTEM Dual-PLC
• Max Spindle / Primary Drive Rating: 27000 RPM
• Coolant / Process Supply Pressure: 150 Bar High-Pressure Loop
• Thermal Operating Limit: 48°C Max Continuous | Electrical Load: 105 kW
• Volumetric Positioning Accuracy: ±0.0016 mm

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ400-SPN-1

Target Machine Model: Mazak-Integrex-i400 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i400 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i400 (MAZATROL SmoothAi)
- Telemetry Sensor ID: PT-SPN-MAZ400-1
- PLC Hex Register Address: DB30.DBW_0x7148
- Nominal Operating Range: 15 - 60 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 120 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ400-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ400-BRG-CER-2601
- Overhaul Seal & Consumable Kit: KIT-MAZ400-SPN-1
- Critical Fastener Torque Value: 59 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i400 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ400-SPN-1 at PLC register 0x7148.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Mazak-Integrex-i400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ400-BRG-CER-2601) with seal kit (KIT-MAZ400-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 59 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ400-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i400, technician must log: "Resolved ERR-MAZ400-SPN-1 on Mazak-Integrex-i400 by installing PN-MAZ400-BRG-CER-2601 & kit KIT-MAZ400-SPN-1. Fasteners torqued to 59 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ400-ATC-2

Target Machine Model: Mazak-Integrex-i400 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i400 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Mazak-Integrex-i400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i400 (MAZATROL SmoothAi)
- Telemetry Sensor ID: PX-ATC-MAZ400-2
- PLC Hex Register Address: DB30.DBW_0x7150
- Nominal Operating Range: 18 - 60 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 120 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ400-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ400-CAM-GRIP-2602
- Overhaul Seal & Consumable Kit: KIT-MAZ400-ATC-2
- Critical Fastener Torque Value: 68 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i400 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ400-ATC-2 at PLC register 0x7150.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Mazak-Integrex-i400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ400-CAM-GRIP-2602) with seal kit (KIT-MAZ400-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 68 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ400-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i400, technician must log: "Resolved ERR-MAZ400-ATC-2 on Mazak-Integrex-i400 by installing PN-MAZ400-CAM-GRIP-2602 & kit KIT-MAZ400-ATC-2. Fasteners torqued to 68 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ400-HYD-3

Target Machine Model: Mazak-Integrex-i400 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i400 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i400 (MAZATROL SmoothAi)
- Telemetry Sensor ID: PR-HYD-MAZ400-3
- PLC Hex Register Address: DB30.DBW_0x7158
- Nominal Operating Range: 21 - 60 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 120 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ400-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ400-VLV-PROP-2603
- Overhaul Seal & Consumable Kit: KIT-MAZ400-HYD-3
- Critical Fastener Torque Value: 77 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i400 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ400-HYD-3 at PLC register 0x7158.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Mazak-Integrex-i400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ400-VLV-PROP-2603) with seal kit (KIT-MAZ400-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 77 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ400-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i400, technician must log: "Resolved ERR-MAZ400-HYD-3 on Mazak-Integrex-i400 by installing PN-MAZ400-VLV-PROP-2603 & kit KIT-MAZ400-HYD-3. Fasteners torqued to 77 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ400-CLN-4

Target Machine Model: Mazak-Integrex-i400 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i400 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Mazak-Integrex-i400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i400 (MAZATROL SmoothAi)
- Telemetry Sensor ID: FL-CLN-MAZ400-4
- PLC Hex Register Address: DB30.DBW_0x7160
- Nominal Operating Range: 24 - 60 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 120 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ400-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ400-PMP-PLG-2604
- Overhaul Seal & Consumable Kit: KIT-MAZ400-CLN-4
- Critical Fastener Torque Value: 86 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i400 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ400-CLN-4 at PLC register 0x7160.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Mazak-Integrex-i400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ400-PMP-PLG-2604) with seal kit (KIT-MAZ400-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 86 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ400-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i400, technician must log: "Resolved ERR-MAZ400-CLN-4 on Mazak-Integrex-i400 by installing PN-MAZ400-PMP-PLG-2604 & kit KIT-MAZ400-CLN-4. Fasteners torqued to 86 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ400-XAX-5

Target Machine Model: Mazak-Integrex-i400 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i400 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i400 (MAZATROL SmoothAi)
- Telemetry Sensor ID: EN-XAX-MAZ400-5
- PLC Hex Register Address: DB30.DBW_0x7168
- Nominal Operating Range: 27 - 60 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 120 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ400-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ400-LIN-COIL-2605
- Overhaul Seal & Consumable Kit: KIT-MAZ400-XAX-5
- Critical Fastener Torque Value: 95 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i400 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ400-XAX-5 at PLC register 0x7168.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Mazak-Integrex-i400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ400-LIN-COIL-2605) with seal kit (KIT-MAZ400-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 95 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ400-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i400, technician must log: "Resolved ERR-MAZ400-XAX-5 on Mazak-Integrex-i400 by installing PN-MAZ400-LIN-COIL-2605 & kit KIT-MAZ400-XAX-5. Fasteners torqued to 95 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ400-YAX-6

Target Machine Model: Mazak-Integrex-i400 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i400 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Mazak-Integrex-i400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i400 (MAZATROL SmoothAi)
- Telemetry Sensor ID: EN-YAX-MAZ400-6
- PLC Hex Register Address: DB30.DBW_0x7170
- Nominal Operating Range: 30 - 60 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 120 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ400-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ400-BSW-NUT-2606
- Overhaul Seal & Consumable Kit: KIT-MAZ400-YAX-6
- Critical Fastener Torque Value: 104 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i400 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ400-YAX-6 at PLC register 0x7170.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Mazak-Integrex-i400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ400-BSW-NUT-2606) with seal kit (KIT-MAZ400-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 104 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ400-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i400, technician must log: "Resolved ERR-MAZ400-YAX-6 on Mazak-Integrex-i400 by installing PN-MAZ400-BSW-NUT-2606 & kit KIT-MAZ400-YAX-6. Fasteners torqued to 104 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ400-ZAX-7

Target Machine Model: Mazak-Integrex-i400 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i400 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i400 (MAZATROL SmoothAi)
- Telemetry Sensor ID: PR-ZAX-MAZ400-7
- PLC Hex Register Address: DB30.DBW_0x7178
- Nominal Operating Range: 33 - 60 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 120 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ400-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ400-CYL-SEAL-2607
- Overhaul Seal & Consumable Kit: KIT-MAZ400-ZAX-7
- Critical Fastener Torque Value: 113 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i400 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ400-ZAX-7 at PLC register 0x7178.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Mazak-Integrex-i400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ400-CYL-SEAL-2607) with seal kit (KIT-MAZ400-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 113 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ400-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i400, technician must log: "Resolved ERR-MAZ400-ZAX-7 on Mazak-Integrex-i400 by installing PN-MAZ400-CYL-SEAL-2607 & kit KIT-MAZ400-ZAX-7. Fasteners torqued to 113 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ400-BAX-8

Target Machine Model: Mazak-Integrex-i400 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i400 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Mazak-Integrex-i400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i400 (MAZATROL SmoothAi)
- Telemetry Sensor ID: EN-BAX-MAZ400-8
- PLC Hex Register Address: DB30.DBW_0x7180
- Nominal Operating Range: 36 - 60 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 120 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ400-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ400-TRQ-STAT-2608
- Overhaul Seal & Consumable Kit: KIT-MAZ400-BAX-8
- Critical Fastener Torque Value: 122 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i400 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ400-BAX-8 at PLC register 0x7180.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Mazak-Integrex-i400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ400-TRQ-STAT-2608) with seal kit (KIT-MAZ400-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 122 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ400-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i400, technician must log: "Resolved ERR-MAZ400-BAX-8 on Mazak-Integrex-i400 by installing PN-MAZ400-TRQ-STAT-2608 & kit KIT-MAZ400-BAX-8. Fasteners torqued to 122 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ400-CAX-9

Target Machine Model: Mazak-Integrex-i400 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i400 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i400 (MAZATROL SmoothAi)
- Telemetry Sensor ID: PR-CAX-MAZ400-9
- PLC Hex Register Address: DB30.DBW_0x7188
- Nominal Operating Range: 39 - 60 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 120 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ400-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ400-BRK-DISC-2609
- Overhaul Seal & Consumable Kit: KIT-MAZ400-CAX-9
- Critical Fastener Torque Value: 131 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i400 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ400-CAX-9 at PLC register 0x7188.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Mazak-Integrex-i400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ400-BRK-DISC-2609) with seal kit (KIT-MAZ400-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 131 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ400-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i400, technician must log: "Resolved ERR-MAZ400-CAX-9 on Mazak-Integrex-i400 by installing PN-MAZ400-BRK-DISC-2609 & kit KIT-MAZ400-CAX-9. Fasteners torqued to 131 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ400-OPT-10

Target Machine Model: Mazak-Integrex-i400 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i400 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Mazak-Integrex-i400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i400 (MAZATROL SmoothAi)
- Telemetry Sensor ID: OP-SCL-MAZ400-10
- PLC Hex Register Address: DB30.DBW_0x7190
- Nominal Operating Range: 42 - 60 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 120 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ400-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ400-GLASS-RD-2610
- Overhaul Seal & Consumable Kit: KIT-MAZ400-OPT-10
- Critical Fastener Torque Value: 140 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i400 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ400-OPT-10 at PLC register 0x7190.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Mazak-Integrex-i400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ400-GLASS-RD-2610) with seal kit (KIT-MAZ400-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 140 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ400-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i400, technician must log: "Resolved ERR-MAZ400-OPT-10 on Mazak-Integrex-i400 by installing PN-MAZ400-GLASS-RD-2610 & kit KIT-MAZ400-OPT-10. Fasteners torqued to 140 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ400-PLC-11

Target Machine Model: Mazak-Integrex-i400 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i400 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i400 (MAZATROL SmoothAi)
- Telemetry Sensor ID: VT-DRV-MAZ400-11
- PLC Hex Register Address: DB30.DBW_0x7198
- Nominal Operating Range: 45 - 60 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 120 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ400-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ400-IGBT-MOD-2611
- Overhaul Seal & Consumable Kit: KIT-MAZ400-PLC-11
- Critical Fastener Torque Value: 149 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i400 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ400-PLC-11 at PLC register 0x7198.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Mazak-Integrex-i400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ400-IGBT-MOD-2611) with seal kit (KIT-MAZ400-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 149 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ400-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i400, technician must log: "Resolved ERR-MAZ400-PLC-11 on Mazak-Integrex-i400 by installing PN-MAZ400-IGBT-MOD-2611 & kit KIT-MAZ400-PLC-11. Fasteners torqued to 149 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ400-LUB-12

Target Machine Model: Mazak-Integrex-i400 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i400 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Mazak-Integrex-i400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i400 (MAZATROL SmoothAi)
- Telemetry Sensor ID: FL-LUB-MAZ400-12
- PLC Hex Register Address: DB30.DBW_0x71A0
- Nominal Operating Range: 48 - 60 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 120 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ400-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ400-INJ-MET-2612
- Overhaul Seal & Consumable Kit: KIT-MAZ400-LUB-12
- Critical Fastener Torque Value: 158 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i400 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ400-LUB-12 at PLC register 0x71A0.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Mazak-Integrex-i400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ400-INJ-MET-2612) with seal kit (KIT-MAZ400-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 158 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ400-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i400, technician must log: "Resolved ERR-MAZ400-LUB-12 on Mazak-Integrex-i400 by installing PN-MAZ400-INJ-MET-2612 & kit KIT-MAZ400-LUB-12. Fasteners torqued to 158 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ400-VAC-13

Target Machine Model: Mazak-Integrex-i400 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i400 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i400 (MAZATROL SmoothAi)
- Telemetry Sensor ID: VC-MNF-MAZ400-13
- PLC Hex Register Address: DB30.DBW_0x71A8
- Nominal Operating Range: 51 - 60 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 120 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ400-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ400-GASK-VIT-2613
- Overhaul Seal & Consumable Kit: KIT-MAZ400-VAC-13
- Critical Fastener Torque Value: 167 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i400 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ400-VAC-13 at PLC register 0x71A8.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Mazak-Integrex-i400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ400-GASK-VIT-2613) with seal kit (KIT-MAZ400-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 167 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ400-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i400, technician must log: "Resolved ERR-MAZ400-VAC-13 on Mazak-Integrex-i400 by installing PN-MAZ400-GASK-VIT-2613 & kit KIT-MAZ400-VAC-13. Fasteners torqued to 167 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ400-THM-14

Target Machine Model: Mazak-Integrex-i400 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i400 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Mazak-Integrex-i400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i400 (MAZATROL SmoothAi)
- Telemetry Sensor ID: TM-CHL-MAZ400-14
- PLC Hex Register Address: DB30.DBW_0x71B0
- Nominal Operating Range: 54 - 60 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 120 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ400-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ400-HEX-CORE-2614
- Overhaul Seal & Consumable Kit: KIT-MAZ400-THM-14
- Critical Fastener Torque Value: 176 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i400 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ400-THM-14 at PLC register 0x71B0.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Mazak-Integrex-i400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ400-HEX-CORE-2614) with seal kit (KIT-MAZ400-THM-14).

Step 6: Torque retaining bolts in cross sequence to 176 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ400-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i400, technician must log: "Resolved ERR-MAZ400-THM-14 on Mazak-Integrex-i400 by installing PN-MAZ400-HEX-CORE-2614 & kit KIT-MAZ400-THM-14. Fasteners torqued to 176 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ450-SPN-1

Target Machine Model: Mazak-Integrex-i450 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i450 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i450 (MAZATROL SmoothAi)
- Telemetry Sensor ID: PT-SPN-MAZ450-1
- PLC Hex Register Address: DB31.DBW_0x7468
- Nominal Operating Range: 15 - 61 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 121 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ450-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ450-BRG-CER-2701
- Overhaul Seal & Consumable Kit: KIT-MAZ450-SPN-1
- Critical Fastener Torque Value: 199 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i450 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ450-SPN-1 at PLC register 0x7468.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Mazak-Integrex-i450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ450-BRG-CER-2701) with seal kit (KIT-MAZ450-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 199 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ450-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i450, technician must log: "Resolved ERR-MAZ450-SPN-1 on Mazak-Integrex-i450 by installing PN-MAZ450-BRG-CER-2701 & kit KIT-MAZ450-SPN-1. Fasteners torqued to 199 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ450-ATC-2

Target Machine Model: Mazak-Integrex-i450 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i450 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Mazak-Integrex-i450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i450 (MAZATROL SmoothAi)
- Telemetry Sensor ID: PX-ATC-MAZ450-2
- PLC Hex Register Address: DB31.DBW_0x7470
- Nominal Operating Range: 18 - 61 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 121 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ450-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ450-CAM-GRIP-2702
- Overhaul Seal & Consumable Kit: KIT-MAZ450-ATC-2
- Critical Fastener Torque Value: 208 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i450 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ450-ATC-2 at PLC register 0x7470.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Mazak-Integrex-i450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ450-CAM-GRIP-2702) with seal kit (KIT-MAZ450-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 208 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ450-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i450, technician must log: "Resolved ERR-MAZ450-ATC-2 on Mazak-Integrex-i450 by installing PN-MAZ450-CAM-GRIP-2702 & kit KIT-MAZ450-ATC-2. Fasteners torqued to 208 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ450-HYD-3

Target Machine Model: Mazak-Integrex-i450 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i450 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i450 (MAZATROL SmoothAi)
- Telemetry Sensor ID: PR-HYD-MAZ450-3
- PLC Hex Register Address: DB31.DBW_0x7478
- Nominal Operating Range: 21 - 61 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 121 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ450-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ450-VLV-PROP-2703
- Overhaul Seal & Consumable Kit: KIT-MAZ450-HYD-3
- Critical Fastener Torque Value: 27 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i450 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ450-HYD-3 at PLC register 0x7478.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Mazak-Integrex-i450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ450-VLV-PROP-2703) with seal kit (KIT-MAZ450-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 27 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ450-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i450, technician must log: "Resolved ERR-MAZ450-HYD-3 on Mazak-Integrex-i450 by installing PN-MAZ450-VLV-PROP-2703 & kit KIT-MAZ450-HYD-3. Fasteners torqued to 27 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ450-CLN-4

Target Machine Model: Mazak-Integrex-i450 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i450 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Mazak-Integrex-i450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i450 (MAZATROL SmoothAi)
- Telemetry Sensor ID: FL-CLN-MAZ450-4
- PLC Hex Register Address: DB31.DBW_0x7480
- Nominal Operating Range: 24 - 61 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 121 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ450-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ450-PMP-PLG-2704
- Overhaul Seal & Consumable Kit: KIT-MAZ450-CLN-4
- Critical Fastener Torque Value: 36 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i450 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ450-CLN-4 at PLC register 0x7480.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Mazak-Integrex-i450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ450-PMP-PLG-2704) with seal kit (KIT-MAZ450-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 36 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ450-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i450, technician must log: "Resolved ERR-MAZ450-CLN-4 on Mazak-Integrex-i450 by installing PN-MAZ450-PMP-PLG-2704 & kit KIT-MAZ450-CLN-4. Fasteners torqued to 36 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ450-XAX-5

Target Machine Model: Mazak-Integrex-i450 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i450 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i450 (MAZATROL SmoothAi)
- Telemetry Sensor ID: EN-XAX-MAZ450-5
- PLC Hex Register Address: DB31.DBW_0x7488
- Nominal Operating Range: 27 - 61 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 121 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ450-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ450-LIN-COIL-2705
- Overhaul Seal & Consumable Kit: KIT-MAZ450-XAX-5
- Critical Fastener Torque Value: 45 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i450 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ450-XAX-5 at PLC register 0x7488.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Mazak-Integrex-i450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ450-LIN-COIL-2705) with seal kit (KIT-MAZ450-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 45 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ450-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i450, technician must log: "Resolved ERR-MAZ450-XAX-5 on Mazak-Integrex-i450 by installing PN-MAZ450-LIN-COIL-2705 & kit KIT-MAZ450-XAX-5. Fasteners torqued to 45 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ450-YAX-6

Target Machine Model: Mazak-Integrex-i450 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i450 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Mazak-Integrex-i450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i450 (MAZATROL SmoothAi)
- Telemetry Sensor ID: EN-YAX-MAZ450-6
- PLC Hex Register Address: DB31.DBW_0x7490
- Nominal Operating Range: 30 - 61 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 121 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ450-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ450-BSW-NUT-2706
- Overhaul Seal & Consumable Kit: KIT-MAZ450-YAX-6
- Critical Fastener Torque Value: 54 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i450 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ450-YAX-6 at PLC register 0x7490.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Mazak-Integrex-i450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ450-BSW-NUT-2706) with seal kit (KIT-MAZ450-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 54 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ450-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i450, technician must log: "Resolved ERR-MAZ450-YAX-6 on Mazak-Integrex-i450 by installing PN-MAZ450-BSW-NUT-2706 & kit KIT-MAZ450-YAX-6. Fasteners torqued to 54 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ450-ZAX-7

Target Machine Model: Mazak-Integrex-i450 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i450 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i450 (MAZATROL SmoothAi)
- Telemetry Sensor ID: PR-ZAX-MAZ450-7
- PLC Hex Register Address: DB31.DBW_0x7498
- Nominal Operating Range: 33 - 61 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 121 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ450-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ450-CYL-SEAL-2707
- Overhaul Seal & Consumable Kit: KIT-MAZ450-ZAX-7
- Critical Fastener Torque Value: 63 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i450 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ450-ZAX-7 at PLC register 0x7498.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Mazak-Integrex-i450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ450-CYL-SEAL-2707) with seal kit (KIT-MAZ450-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 63 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ450-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i450, technician must log: "Resolved ERR-MAZ450-ZAX-7 on Mazak-Integrex-i450 by installing PN-MAZ450-CYL-SEAL-2707 & kit KIT-MAZ450-ZAX-7. Fasteners torqued to 63 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ450-BAX-8

Target Machine Model: Mazak-Integrex-i450 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i450 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Mazak-Integrex-i450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i450 (MAZATROL SmoothAi)
- Telemetry Sensor ID: EN-BAX-MAZ450-8
- PLC Hex Register Address: DB31.DBW_0x74A0
- Nominal Operating Range: 36 - 61 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 121 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ450-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ450-TRQ-STAT-2708
- Overhaul Seal & Consumable Kit: KIT-MAZ450-BAX-8
- Critical Fastener Torque Value: 72 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i450 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ450-BAX-8 at PLC register 0x74A0.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Mazak-Integrex-i450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ450-TRQ-STAT-2708) with seal kit (KIT-MAZ450-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 72 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ450-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i450, technician must log: "Resolved ERR-MAZ450-BAX-8 on Mazak-Integrex-i450 by installing PN-MAZ450-TRQ-STAT-2708 & kit KIT-MAZ450-BAX-8. Fasteners torqued to 72 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ450-CAX-9

Target Machine Model: Mazak-Integrex-i450 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i450 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i450 (MAZATROL SmoothAi)
- Telemetry Sensor ID: PR-CAX-MAZ450-9
- PLC Hex Register Address: DB31.DBW_0x74A8
- Nominal Operating Range: 39 - 61 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 121 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ450-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ450-BRK-DISC-2709
- Overhaul Seal & Consumable Kit: KIT-MAZ450-CAX-9
- Critical Fastener Torque Value: 81 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i450 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ450-CAX-9 at PLC register 0x74A8.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Mazak-Integrex-i450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ450-BRK-DISC-2709) with seal kit (KIT-MAZ450-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 81 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ450-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i450, technician must log: "Resolved ERR-MAZ450-CAX-9 on Mazak-Integrex-i450 by installing PN-MAZ450-BRK-DISC-2709 & kit KIT-MAZ450-CAX-9. Fasteners torqued to 81 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ450-OPT-10

Target Machine Model: Mazak-Integrex-i450 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i450 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Mazak-Integrex-i450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i450 (MAZATROL SmoothAi)
- Telemetry Sensor ID: OP-SCL-MAZ450-10
- PLC Hex Register Address: DB31.DBW_0x74B0
- Nominal Operating Range: 42 - 61 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 121 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ450-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ450-GLASS-RD-2710
- Overhaul Seal & Consumable Kit: KIT-MAZ450-OPT-10
- Critical Fastener Torque Value: 90 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i450 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ450-OPT-10 at PLC register 0x74B0.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Mazak-Integrex-i450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ450-GLASS-RD-2710) with seal kit (KIT-MAZ450-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 90 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ450-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i450, technician must log: "Resolved ERR-MAZ450-OPT-10 on Mazak-Integrex-i450 by installing PN-MAZ450-GLASS-RD-2710 & kit KIT-MAZ450-OPT-10. Fasteners torqued to 90 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ450-PLC-11

Target Machine Model: Mazak-Integrex-i450 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i450 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i450 (MAZATROL SmoothAi)
- Telemetry Sensor ID: VT-DRV-MAZ450-11
- PLC Hex Register Address: DB31.DBW_0x74B8
- Nominal Operating Range: 45 - 61 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 121 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ450-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ450-IGBT-MOD-2711
- Overhaul Seal & Consumable Kit: KIT-MAZ450-PLC-11
- Critical Fastener Torque Value: 99 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i450 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ450-PLC-11 at PLC register 0x74B8.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Mazak-Integrex-i450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ450-IGBT-MOD-2711) with seal kit (KIT-MAZ450-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 99 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ450-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i450, technician must log: "Resolved ERR-MAZ450-PLC-11 on Mazak-Integrex-i450 by installing PN-MAZ450-IGBT-MOD-2711 & kit KIT-MAZ450-PLC-11. Fasteners torqued to 99 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ450-LUB-12

Target Machine Model: Mazak-Integrex-i450 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i450 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Mazak-Integrex-i450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i450 (MAZATROL SmoothAi)
- Telemetry Sensor ID: FL-LUB-MAZ450-12
- PLC Hex Register Address: DB31.DBW_0x74C0
- Nominal Operating Range: 48 - 61 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 121 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ450-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ450-INJ-MET-2712
- Overhaul Seal & Consumable Kit: KIT-MAZ450-LUB-12
- Critical Fastener Torque Value: 108 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i450 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ450-LUB-12 at PLC register 0x74C0.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Mazak-Integrex-i450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ450-INJ-MET-2712) with seal kit (KIT-MAZ450-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 108 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ450-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i450, technician must log: "Resolved ERR-MAZ450-LUB-12 on Mazak-Integrex-i450 by installing PN-MAZ450-INJ-MET-2712 & kit KIT-MAZ450-LUB-12. Fasteners torqued to 108 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ450-VAC-13

Target Machine Model: Mazak-Integrex-i450 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i450 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i450 (MAZATROL SmoothAi)
- Telemetry Sensor ID: VC-MNF-MAZ450-13
- PLC Hex Register Address: DB31.DBW_0x74C8
- Nominal Operating Range: 51 - 61 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 121 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ450-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ450-GASK-VIT-2713
- Overhaul Seal & Consumable Kit: KIT-MAZ450-VAC-13
- Critical Fastener Torque Value: 117 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i450 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ450-VAC-13 at PLC register 0x74C8.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Mazak-Integrex-i450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ450-GASK-VIT-2713) with seal kit (KIT-MAZ450-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 117 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ450-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i450, technician must log: "Resolved ERR-MAZ450-VAC-13 on Mazak-Integrex-i450 by installing PN-MAZ450-GASK-VIT-2713 & kit KIT-MAZ450-VAC-13. Fasteners torqued to 117 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ450-THM-14

Target Machine Model: Mazak-Integrex-i450 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i450 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Mazak-Integrex-i450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i450 (MAZATROL SmoothAi)
- Telemetry Sensor ID: TM-CHL-MAZ450-14
- PLC Hex Register Address: DB31.DBW_0x74D0
- Nominal Operating Range: 54 - 61 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 121 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ450-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ450-HEX-CORE-2714
- Overhaul Seal & Consumable Kit: KIT-MAZ450-THM-14
- Critical Fastener Torque Value: 126 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i450 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ450-THM-14 at PLC register 0x74D0.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Mazak-Integrex-i450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ450-HEX-CORE-2714) with seal kit (KIT-MAZ450-THM-14).

Step 6: Torque retaining bolts in cross sequence to 126 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ450-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i450, technician must log: "Resolved ERR-MAZ450-THM-14 on Mazak-Integrex-i450 by installing PN-MAZ450-HEX-CORE-2714 & kit KIT-MAZ450-THM-14. Fasteners torqued to 126 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ500-SPN-1

Target Machine Model: Mazak-Integrex-i500 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i500 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i500 (MAZATROL SmoothAi)
- Telemetry Sensor ID: PT-SPN-MAZ500-1
- PLC Hex Register Address: DB32.DBW_0x7788
- Nominal Operating Range: 15 - 62 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 122 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ500-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ500-BRG-CER-2801
- Overhaul Seal & Consumable Kit: KIT-MAZ500-SPN-1
- Critical Fastener Torque Value: 149 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i500 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ500-SPN-1 at PLC register 0x7788.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Mazak-Integrex-i500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ500-BRG-CER-2801) with seal kit (KIT-MAZ500-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 149 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ500-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i500, technician must log: "Resolved ERR-MAZ500-SPN-1 on Mazak-Integrex-i500 by installing PN-MAZ500-BRG-CER-2801 & kit KIT-MAZ500-SPN-1. Fasteners torqued to 149 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ500-ATC-2

Target Machine Model: Mazak-Integrex-i500 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i500 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Mazak-Integrex-i500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i500 (MAZATROL SmoothAi)
- Telemetry Sensor ID: PX-ATC-MAZ500-2
- PLC Hex Register Address: DB32.DBW_0x7790
- Nominal Operating Range: 18 - 62 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 122 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ500-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ500-CAM-GRIP-2802
- Overhaul Seal & Consumable Kit: KIT-MAZ500-ATC-2
- Critical Fastener Torque Value: 158 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i500 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ500-ATC-2 at PLC register 0x7790.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Mazak-Integrex-i500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ500-CAM-GRIP-2802) with seal kit (KIT-MAZ500-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 158 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ500-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i500, technician must log: "Resolved ERR-MAZ500-ATC-2 on Mazak-Integrex-i500 by installing PN-MAZ500-CAM-GRIP-2802 & kit KIT-MAZ500-ATC-2. Fasteners torqued to 158 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ500-HYD-3

Target Machine Model: Mazak-Integrex-i500 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i500 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i500 (MAZATROL SmoothAi)
- Telemetry Sensor ID: PR-HYD-MAZ500-3
- PLC Hex Register Address: DB32.DBW_0x7798
- Nominal Operating Range: 21 - 62 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 122 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ500-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ500-VLV-PROP-2803
- Overhaul Seal & Consumable Kit: KIT-MAZ500-HYD-3
- Critical Fastener Torque Value: 167 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i500 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ500-HYD-3 at PLC register 0x7798.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Mazak-Integrex-i500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ500-VLV-PROP-2803) with seal kit (KIT-MAZ500-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 167 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ500-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i500, technician must log: "Resolved ERR-MAZ500-HYD-3 on Mazak-Integrex-i500 by installing PN-MAZ500-VLV-PROP-2803 & kit KIT-MAZ500-HYD-3. Fasteners torqued to 167 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ500-CLN-4

Target Machine Model: Mazak-Integrex-i500 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i500 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Mazak-Integrex-i500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i500 (MAZATROL SmoothAi)
- Telemetry Sensor ID: FL-CLN-MAZ500-4
- PLC Hex Register Address: DB32.DBW_0x77A0
- Nominal Operating Range: 24 - 62 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 122 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ500-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ500-PMP-PLG-2804
- Overhaul Seal & Consumable Kit: KIT-MAZ500-CLN-4
- Critical Fastener Torque Value: 176 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i500 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ500-CLN-4 at PLC register 0x77A0.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Mazak-Integrex-i500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ500-PMP-PLG-2804) with seal kit (KIT-MAZ500-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 176 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ500-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i500, technician must log: "Resolved ERR-MAZ500-CLN-4 on Mazak-Integrex-i500 by installing PN-MAZ500-PMP-PLG-2804 & kit KIT-MAZ500-CLN-4. Fasteners torqued to 176 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ500-XAX-5

Target Machine Model: Mazak-Integrex-i500 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i500 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i500 (MAZATROL SmoothAi)
- Telemetry Sensor ID: EN-XAX-MAZ500-5
- PLC Hex Register Address: DB32.DBW_0x77A8
- Nominal Operating Range: 27 - 62 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 122 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ500-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ500-LIN-COIL-2805
- Overhaul Seal & Consumable Kit: KIT-MAZ500-XAX-5
- Critical Fastener Torque Value: 185 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i500 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ500-XAX-5 at PLC register 0x77A8.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Mazak-Integrex-i500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ500-LIN-COIL-2805) with seal kit (KIT-MAZ500-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 185 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ500-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i500, technician must log: "Resolved ERR-MAZ500-XAX-5 on Mazak-Integrex-i500 by installing PN-MAZ500-LIN-COIL-2805 & kit KIT-MAZ500-XAX-5. Fasteners torqued to 185 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ500-YAX-6

Target Machine Model: Mazak-Integrex-i500 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i500 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Mazak-Integrex-i500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i500 (MAZATROL SmoothAi)
- Telemetry Sensor ID: EN-YAX-MAZ500-6
- PLC Hex Register Address: DB32.DBW_0x77B0
- Nominal Operating Range: 30 - 62 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 122 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ500-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ500-BSW-NUT-2806
- Overhaul Seal & Consumable Kit: KIT-MAZ500-YAX-6
- Critical Fastener Torque Value: 194 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i500 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ500-YAX-6 at PLC register 0x77B0.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Mazak-Integrex-i500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ500-BSW-NUT-2806) with seal kit (KIT-MAZ500-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 194 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ500-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i500, technician must log: "Resolved ERR-MAZ500-YAX-6 on Mazak-Integrex-i500 by installing PN-MAZ500-BSW-NUT-2806 & kit KIT-MAZ500-YAX-6. Fasteners torqued to 194 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ500-ZAX-7

Target Machine Model: Mazak-Integrex-i500 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i500 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i500 (MAZATROL SmoothAi)
- Telemetry Sensor ID: PR-ZAX-MAZ500-7
- PLC Hex Register Address: DB32.DBW_0x77B8
- Nominal Operating Range: 33 - 62 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 122 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ500-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ500-CYL-SEAL-2807
- Overhaul Seal & Consumable Kit: KIT-MAZ500-ZAX-7
- Critical Fastener Torque Value: 203 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i500 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ500-ZAX-7 at PLC register 0x77B8.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Mazak-Integrex-i500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ500-CYL-SEAL-2807) with seal kit (KIT-MAZ500-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 203 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ500-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i500, technician must log: "Resolved ERR-MAZ500-ZAX-7 on Mazak-Integrex-i500 by installing PN-MAZ500-CYL-SEAL-2807 & kit KIT-MAZ500-ZAX-7. Fasteners torqued to 203 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ500-BAX-8

Target Machine Model: Mazak-Integrex-i500 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i500 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Mazak-Integrex-i500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i500 (MAZATROL SmoothAi)
- Telemetry Sensor ID: EN-BAX-MAZ500-8
- PLC Hex Register Address: DB32.DBW_0x77C0
- Nominal Operating Range: 36 - 62 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 122 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ500-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ500-TRQ-STAT-2808
- Overhaul Seal & Consumable Kit: KIT-MAZ500-BAX-8
- Critical Fastener Torque Value: 22 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i500 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ500-BAX-8 at PLC register 0x77C0.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Mazak-Integrex-i500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ500-TRQ-STAT-2808) with seal kit (KIT-MAZ500-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 22 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ500-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i500, technician must log: "Resolved ERR-MAZ500-BAX-8 on Mazak-Integrex-i500 by installing PN-MAZ500-TRQ-STAT-2808 & kit KIT-MAZ500-BAX-8. Fasteners torqued to 22 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ500-CAX-9

Target Machine Model: Mazak-Integrex-i500 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i500 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i500 (MAZATROL SmoothAi)
- Telemetry Sensor ID: PR-CAX-MAZ500-9
- PLC Hex Register Address: DB32.DBW_0x77C8
- Nominal Operating Range: 39 - 62 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 122 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ500-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ500-BRK-DISC-2809
- Overhaul Seal & Consumable Kit: KIT-MAZ500-CAX-9
- Critical Fastener Torque Value: 31 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i500 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ500-CAX-9 at PLC register 0x77C8.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Mazak-Integrex-i500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ500-BRK-DISC-2809) with seal kit (KIT-MAZ500-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 31 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ500-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i500, technician must log: "Resolved ERR-MAZ500-CAX-9 on Mazak-Integrex-i500 by installing PN-MAZ500-BRK-DISC-2809 & kit KIT-MAZ500-CAX-9. Fasteners torqued to 31 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ500-OPT-10

Target Machine Model: Mazak-Integrex-i500 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i500 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Mazak-Integrex-i500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i500 (MAZATROL SmoothAi)
- Telemetry Sensor ID: OP-SCL-MAZ500-10
- PLC Hex Register Address: DB32.DBW_0x77D0
- Nominal Operating Range: 42 - 62 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 122 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ500-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ500-GLASS-RD-2810
- Overhaul Seal & Consumable Kit: KIT-MAZ500-OPT-10
- Critical Fastener Torque Value: 40 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i500 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ500-OPT-10 at PLC register 0x77D0.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Mazak-Integrex-i500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ500-GLASS-RD-2810) with seal kit (KIT-MAZ500-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 40 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ500-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i500, technician must log: "Resolved ERR-MAZ500-OPT-10 on Mazak-Integrex-i500 by installing PN-MAZ500-GLASS-RD-2810 & kit KIT-MAZ500-OPT-10. Fasteners torqued to 40 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ500-PLC-11

Target Machine Model: Mazak-Integrex-i500 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i500 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i500 (MAZATROL SmoothAi)
- Telemetry Sensor ID: VT-DRV-MAZ500-11
- PLC Hex Register Address: DB32.DBW_0x77D8
- Nominal Operating Range: 45 - 62 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 122 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ500-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ500-IGBT-MOD-2811
- Overhaul Seal & Consumable Kit: KIT-MAZ500-PLC-11
- Critical Fastener Torque Value: 49 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i500 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ500-PLC-11 at PLC register 0x77D8.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Mazak-Integrex-i500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ500-IGBT-MOD-2811) with seal kit (KIT-MAZ500-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 49 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ500-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i500, technician must log: "Resolved ERR-MAZ500-PLC-11 on Mazak-Integrex-i500 by installing PN-MAZ500-IGBT-MOD-2811 & kit KIT-MAZ500-PLC-11. Fasteners torqued to 49 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ500-LUB-12

Target Machine Model: Mazak-Integrex-i500 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i500 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Mazak-Integrex-i500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i500 (MAZATROL SmoothAi)
- Telemetry Sensor ID: FL-LUB-MAZ500-12
- PLC Hex Register Address: DB32.DBW_0x77E0
- Nominal Operating Range: 48 - 62 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 122 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ500-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ500-INJ-MET-2812
- Overhaul Seal & Consumable Kit: KIT-MAZ500-LUB-12
- Critical Fastener Torque Value: 58 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i500 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ500-LUB-12 at PLC register 0x77E0.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Mazak-Integrex-i500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ500-INJ-MET-2812) with seal kit (KIT-MAZ500-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 58 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ500-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i500, technician must log: "Resolved ERR-MAZ500-LUB-12 on Mazak-Integrex-i500 by installing PN-MAZ500-INJ-MET-2812 & kit KIT-MAZ500-LUB-12. Fasteners torqued to 58 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ500-VAC-13

Target Machine Model: Mazak-Integrex-i500 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i500 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i500 (MAZATROL SmoothAi)
- Telemetry Sensor ID: VC-MNF-MAZ500-13
- PLC Hex Register Address: DB32.DBW_0x77E8
- Nominal Operating Range: 51 - 62 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 122 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ500-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ500-GASK-VIT-2813
- Overhaul Seal & Consumable Kit: KIT-MAZ500-VAC-13
- Critical Fastener Torque Value: 67 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i500 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ500-VAC-13 at PLC register 0x77E8.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Mazak-Integrex-i500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ500-GASK-VIT-2813) with seal kit (KIT-MAZ500-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 67 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ500-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i500, technician must log: "Resolved ERR-MAZ500-VAC-13 on Mazak-Integrex-i500 by installing PN-MAZ500-GASK-VIT-2813 & kit KIT-MAZ500-VAC-13. Fasteners torqued to 67 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ500-THM-14

Target Machine Model: Mazak-Integrex-i500 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i500 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Mazak-Integrex-i500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i500 (MAZATROL SmoothAi)
- Telemetry Sensor ID: TM-CHL-MAZ500-14
- PLC Hex Register Address: DB32.DBW_0x77F0
- Nominal Operating Range: 54 - 62 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 122 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ500-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ500-HEX-CORE-2814
- Overhaul Seal & Consumable Kit: KIT-MAZ500-THM-14
- Critical Fastener Torque Value: 76 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i500 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ500-THM-14 at PLC register 0x77F0.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Mazak-Integrex-i500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ500-HEX-CORE-2814) with seal kit (KIT-MAZ500-THM-14).

Step 6: Torque retaining bolts in cross sequence to 76 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ500-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i500, technician must log: "Resolved ERR-MAZ500-THM-14 on Mazak-Integrex-i500 by installing PN-MAZ500-HEX-CORE-2814 & kit KIT-MAZ500-THM-14. Fasteners torqued to 76 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ550-SPN-1

Target Machine Model: Mazak-Integrex-i550 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i550 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i550 (MAZATROL SmoothAi)
- Telemetry Sensor ID: PT-SPN-MAZ550-1
- PLC Hex Register Address: DB33.DBW_0x7AA8
- Nominal Operating Range: 15 - 63 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 123 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ550-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ550-BRG-CER-2901
- Overhaul Seal & Consumable Kit: KIT-MAZ550-SPN-1
- Critical Fastener Torque Value: 99 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i550 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ550-SPN-1 at PLC register 0x7AA8.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Mazak-Integrex-i550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ550-BRG-CER-2901) with seal kit (KIT-MAZ550-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 99 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ550-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i550, technician must log: "Resolved ERR-MAZ550-SPN-1 on Mazak-Integrex-i550 by installing PN-MAZ550-BRG-CER-2901 & kit KIT-MAZ550-SPN-1. Fasteners torqued to 99 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ550-ATC-2

Target Machine Model: Mazak-Integrex-i550 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i550 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Mazak-Integrex-i550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i550 (MAZATROL SmoothAi)
- Telemetry Sensor ID: PX-ATC-MAZ550-2
- PLC Hex Register Address: DB33.DBW_0x7AB0
- Nominal Operating Range: 18 - 63 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 123 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ550-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ550-CAM-GRIP-2902
- Overhaul Seal & Consumable Kit: KIT-MAZ550-ATC-2
- Critical Fastener Torque Value: 108 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i550 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ550-ATC-2 at PLC register 0x7AB0.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Mazak-Integrex-i550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ550-CAM-GRIP-2902) with seal kit (KIT-MAZ550-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 108 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ550-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i550, technician must log: "Resolved ERR-MAZ550-ATC-2 on Mazak-Integrex-i550 by installing PN-MAZ550-CAM-GRIP-2902 & kit KIT-MAZ550-ATC-2. Fasteners torqued to 108 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ550-HYD-3

Target Machine Model: Mazak-Integrex-i550 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i550 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i550 (MAZATROL SmoothAi)
- Telemetry Sensor ID: PR-HYD-MAZ550-3
- PLC Hex Register Address: DB33.DBW_0x7AB8
- Nominal Operating Range: 21 - 63 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 123 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ550-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ550-VLV-PROP-2903
- Overhaul Seal & Consumable Kit: KIT-MAZ550-HYD-3
- Critical Fastener Torque Value: 117 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i550 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ550-HYD-3 at PLC register 0x7AB8.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Mazak-Integrex-i550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ550-VLV-PROP-2903) with seal kit (KIT-MAZ550-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 117 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ550-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i550, technician must log: "Resolved ERR-MAZ550-HYD-3 on Mazak-Integrex-i550 by installing PN-MAZ550-VLV-PROP-2903 & kit KIT-MAZ550-HYD-3. Fasteners torqued to 117 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ550-CLN-4

Target Machine Model: Mazak-Integrex-i550 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i550 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Mazak-Integrex-i550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i550 (MAZATROL SmoothAi)
- Telemetry Sensor ID: FL-CLN-MAZ550-4
- PLC Hex Register Address: DB33.DBW_0x7AC0
- Nominal Operating Range: 24 - 63 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 123 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ550-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ550-PMP-PLG-2904
- Overhaul Seal & Consumable Kit: KIT-MAZ550-CLN-4
- Critical Fastener Torque Value: 126 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i550 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ550-CLN-4 at PLC register 0x7AC0.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Mazak-Integrex-i550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ550-PMP-PLG-2904) with seal kit (KIT-MAZ550-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 126 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ550-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i550, technician must log: "Resolved ERR-MAZ550-CLN-4 on Mazak-Integrex-i550 by installing PN-MAZ550-PMP-PLG-2904 & kit KIT-MAZ550-CLN-4. Fasteners torqued to 126 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ550-XAX-5

Target Machine Model: Mazak-Integrex-i550 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i550 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i550 (MAZATROL SmoothAi)
- Telemetry Sensor ID: EN-XAX-MAZ550-5
- PLC Hex Register Address: DB33.DBW_0x7AC8
- Nominal Operating Range: 27 - 63 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 123 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ550-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ550-LIN-COIL-2905
- Overhaul Seal & Consumable Kit: KIT-MAZ550-XAX-5
- Critical Fastener Torque Value: 135 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i550 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ550-XAX-5 at PLC register 0x7AC8.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Mazak-Integrex-i550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ550-LIN-COIL-2905) with seal kit (KIT-MAZ550-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 135 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ550-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i550, technician must log: "Resolved ERR-MAZ550-XAX-5 on Mazak-Integrex-i550 by installing PN-MAZ550-LIN-COIL-2905 & kit KIT-MAZ550-XAX-5. Fasteners torqued to 135 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ550-YAX-6

Target Machine Model: Mazak-Integrex-i550 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i550 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Mazak-Integrex-i550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i550 (MAZATROL SmoothAi)
- Telemetry Sensor ID: EN-YAX-MAZ550-6
- PLC Hex Register Address: DB33.DBW_0x7AD0
- Nominal Operating Range: 30 - 63 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 123 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ550-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ550-BSW-NUT-2906
- Overhaul Seal & Consumable Kit: KIT-MAZ550-YAX-6
- Critical Fastener Torque Value: 144 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i550 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ550-YAX-6 at PLC register 0x7AD0.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Mazak-Integrex-i550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ550-BSW-NUT-2906) with seal kit (KIT-MAZ550-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 144 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ550-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i550, technician must log: "Resolved ERR-MAZ550-YAX-6 on Mazak-Integrex-i550 by installing PN-MAZ550-BSW-NUT-2906 & kit KIT-MAZ550-YAX-6. Fasteners torqued to 144 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ550-ZAX-7

Target Machine Model: Mazak-Integrex-i550 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i550 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i550 (MAZATROL SmoothAi)
- Telemetry Sensor ID: PR-ZAX-MAZ550-7
- PLC Hex Register Address: DB33.DBW_0x7AD8
- Nominal Operating Range: 33 - 63 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 123 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ550-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ550-CYL-SEAL-2907
- Overhaul Seal & Consumable Kit: KIT-MAZ550-ZAX-7
- Critical Fastener Torque Value: 153 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i550 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ550-ZAX-7 at PLC register 0x7AD8.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Mazak-Integrex-i550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ550-CYL-SEAL-2907) with seal kit (KIT-MAZ550-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 153 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ550-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i550, technician must log: "Resolved ERR-MAZ550-ZAX-7 on Mazak-Integrex-i550 by installing PN-MAZ550-CYL-SEAL-2907 & kit KIT-MAZ550-ZAX-7. Fasteners torqued to 153 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ550-BAX-8

Target Machine Model: Mazak-Integrex-i550 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i550 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Mazak-Integrex-i550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i550 (MAZATROL SmoothAi)
- Telemetry Sensor ID: EN-BAX-MAZ550-8
- PLC Hex Register Address: DB33.DBW_0x7AE0
- Nominal Operating Range: 36 - 63 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 123 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ550-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ550-TRQ-STAT-2908
- Overhaul Seal & Consumable Kit: KIT-MAZ550-BAX-8
- Critical Fastener Torque Value: 162 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i550 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ550-BAX-8 at PLC register 0x7AE0.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Mazak-Integrex-i550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ550-TRQ-STAT-2908) with seal kit (KIT-MAZ550-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 162 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ550-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i550, technician must log: "Resolved ERR-MAZ550-BAX-8 on Mazak-Integrex-i550 by installing PN-MAZ550-TRQ-STAT-2908 & kit KIT-MAZ550-BAX-8. Fasteners torqued to 162 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ550-CAX-9

Target Machine Model: Mazak-Integrex-i550 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i550 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i550 (MAZATROL SmoothAi)
- Telemetry Sensor ID: PR-CAX-MAZ550-9
- PLC Hex Register Address: DB33.DBW_0x7AE8
- Nominal Operating Range: 39 - 63 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 123 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ550-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ550-BRK-DISC-2909
- Overhaul Seal & Consumable Kit: KIT-MAZ550-CAX-9
- Critical Fastener Torque Value: 171 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i550 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ550-CAX-9 at PLC register 0x7AE8.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Mazak-Integrex-i550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ550-BRK-DISC-2909) with seal kit (KIT-MAZ550-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 171 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ550-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i550, technician must log: "Resolved ERR-MAZ550-CAX-9 on Mazak-Integrex-i550 by installing PN-MAZ550-BRK-DISC-2909 & kit KIT-MAZ550-CAX-9. Fasteners torqued to 171 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ550-OPT-10

Target Machine Model: Mazak-Integrex-i550 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i550 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Mazak-Integrex-i550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i550 (MAZATROL SmoothAi)
- Telemetry Sensor ID: OP-SCL-MAZ550-10
- PLC Hex Register Address: DB33.DBW_0x7AF0
- Nominal Operating Range: 42 - 63 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 123 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ550-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ550-GLASS-RD-2910
- Overhaul Seal & Consumable Kit: KIT-MAZ550-OPT-10
- Critical Fastener Torque Value: 180 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i550 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ550-OPT-10 at PLC register 0x7AF0.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Mazak-Integrex-i550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ550-GLASS-RD-2910) with seal kit (KIT-MAZ550-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 180 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ550-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i550, technician must log: "Resolved ERR-MAZ550-OPT-10 on Mazak-Integrex-i550 by installing PN-MAZ550-GLASS-RD-2910 & kit KIT-MAZ550-OPT-10. Fasteners torqued to 180 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ550-PLC-11

Target Machine Model: Mazak-Integrex-i550 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i550 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i550 (MAZATROL SmoothAi)
- Telemetry Sensor ID: VT-DRV-MAZ550-11
- PLC Hex Register Address: DB33.DBW_0x7AF8
- Nominal Operating Range: 45 - 63 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 123 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ550-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ550-IGBT-MOD-2911
- Overhaul Seal & Consumable Kit: KIT-MAZ550-PLC-11
- Critical Fastener Torque Value: 189 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i550 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ550-PLC-11 at PLC register 0x7AF8.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Mazak-Integrex-i550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ550-IGBT-MOD-2911) with seal kit (KIT-MAZ550-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 189 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ550-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i550, technician must log: "Resolved ERR-MAZ550-PLC-11 on Mazak-Integrex-i550 by installing PN-MAZ550-IGBT-MOD-2911 & kit KIT-MAZ550-PLC-11. Fasteners torqued to 189 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ550-LUB-12

Target Machine Model: Mazak-Integrex-i550 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i550 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Mazak-Integrex-i550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i550 (MAZATROL SmoothAi)
- Telemetry Sensor ID: FL-LUB-MAZ550-12
- PLC Hex Register Address: DB33.DBW_0x7B00
- Nominal Operating Range: 48 - 63 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 123 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ550-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ550-INJ-MET-2912
- Overhaul Seal & Consumable Kit: KIT-MAZ550-LUB-12
- Critical Fastener Torque Value: 198 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i550 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ550-LUB-12 at PLC register 0x7B00.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Mazak-Integrex-i550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ550-INJ-MET-2912) with seal kit (KIT-MAZ550-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 198 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ550-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i550, technician must log: "Resolved ERR-MAZ550-LUB-12 on Mazak-Integrex-i550 by installing PN-MAZ550-INJ-MET-2912 & kit KIT-MAZ550-LUB-12. Fasteners torqued to 198 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ550-VAC-13

Target Machine Model: Mazak-Integrex-i550 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i550 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i550 (MAZATROL SmoothAi)
- Telemetry Sensor ID: VC-MNF-MAZ550-13
- PLC Hex Register Address: DB33.DBW_0x7B08
- Nominal Operating Range: 51 - 63 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 123 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ550-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ550-GASK-VIT-2913
- Overhaul Seal & Consumable Kit: KIT-MAZ550-VAC-13
- Critical Fastener Torque Value: 207 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i550 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ550-VAC-13 at PLC register 0x7B08.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Mazak-Integrex-i550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ550-GASK-VIT-2913) with seal kit (KIT-MAZ550-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 207 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ550-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i550, technician must log: "Resolved ERR-MAZ550-VAC-13 on Mazak-Integrex-i550 by installing PN-MAZ550-GASK-VIT-2913 & kit KIT-MAZ550-VAC-13. Fasteners torqued to 207 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ550-THM-14

Target Machine Model: Mazak-Integrex-i550 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i550 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Mazak-Integrex-i550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i550 (MAZATROL SmoothAi)
- Telemetry Sensor ID: TM-CHL-MAZ550-14
- PLC Hex Register Address: DB33.DBW_0x7B10
- Nominal Operating Range: 54 - 63 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 123 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ550-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ550-HEX-CORE-2914
- Overhaul Seal & Consumable Kit: KIT-MAZ550-THM-14
- Critical Fastener Torque Value: 26 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i550 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ550-THM-14 at PLC register 0x7B10.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Mazak-Integrex-i550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ550-HEX-CORE-2914) with seal kit (KIT-MAZ550-THM-14).

Step 6: Torque retaining bolts in cross sequence to 26 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ550-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i550, technician must log: "Resolved ERR-MAZ550-THM-14 on Mazak-Integrex-i550 by installing PN-MAZ550-HEX-CORE-2914 & kit KIT-MAZ550-THM-14. Fasteners torqued to 26 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ600-SPN-1

Target Machine Model: Mazak-Integrex-i600 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i600 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i600 (MAZATROL SmoothAi)
- Telemetry Sensor ID: PT-SPN-MAZ600-1
- PLC Hex Register Address: DB34.DBW_0x7DC8
- Nominal Operating Range: 15 - 64 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 124 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ600-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ600-BRG-CER-3001
- Overhaul Seal & Consumable Kit: KIT-MAZ600-SPN-1
- Critical Fastener Torque Value: 49 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i600 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ600-SPN-1 at PLC register 0x7DC8.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Mazak-Integrex-i600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ600-BRG-CER-3001) with seal kit (KIT-MAZ600-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 49 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ600-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i600, technician must log: "Resolved ERR-MAZ600-SPN-1 on Mazak-Integrex-i600 by installing PN-MAZ600-BRG-CER-3001 & kit KIT-MAZ600-SPN-1. Fasteners torqued to 49 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ600-ATC-2

Target Machine Model: Mazak-Integrex-i600 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i600 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Mazak-Integrex-i600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i600 (MAZATROL SmoothAi)
- Telemetry Sensor ID: PX-ATC-MAZ600-2
- PLC Hex Register Address: DB34.DBW_0x7DD0
- Nominal Operating Range: 18 - 64 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 124 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ600-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ600-CAM-GRIP-3002
- Overhaul Seal & Consumable Kit: KIT-MAZ600-ATC-2
- Critical Fastener Torque Value: 58 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i600 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ600-ATC-2 at PLC register 0x7DD0.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Mazak-Integrex-i600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ600-CAM-GRIP-3002) with seal kit (KIT-MAZ600-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 58 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ600-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i600, technician must log: "Resolved ERR-MAZ600-ATC-2 on Mazak-Integrex-i600 by installing PN-MAZ600-CAM-GRIP-3002 & kit KIT-MAZ600-ATC-2. Fasteners torqued to 58 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ600-HYD-3

Target Machine Model: Mazak-Integrex-i600 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i600 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i600 (MAZATROL SmoothAi)
- Telemetry Sensor ID: PR-HYD-MAZ600-3
- PLC Hex Register Address: DB34.DBW_0x7DD8
- Nominal Operating Range: 21 - 64 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 124 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ600-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ600-VLV-PROP-3003
- Overhaul Seal & Consumable Kit: KIT-MAZ600-HYD-3
- Critical Fastener Torque Value: 67 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i600 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ600-HYD-3 at PLC register 0x7DD8.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Mazak-Integrex-i600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ600-VLV-PROP-3003) with seal kit (KIT-MAZ600-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 67 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ600-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i600, technician must log: "Resolved ERR-MAZ600-HYD-3 on Mazak-Integrex-i600 by installing PN-MAZ600-VLV-PROP-3003 & kit KIT-MAZ600-HYD-3. Fasteners torqued to 67 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ600-CLN-4

Target Machine Model: Mazak-Integrex-i600 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i600 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Mazak-Integrex-i600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i600 (MAZATROL SmoothAi)
- Telemetry Sensor ID: FL-CLN-MAZ600-4
- PLC Hex Register Address: DB34.DBW_0x7DE0
- Nominal Operating Range: 24 - 64 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 124 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ600-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ600-PMP-PLG-3004
- Overhaul Seal & Consumable Kit: KIT-MAZ600-CLN-4
- Critical Fastener Torque Value: 76 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i600 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ600-CLN-4 at PLC register 0x7DE0.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Mazak-Integrex-i600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ600-PMP-PLG-3004) with seal kit (KIT-MAZ600-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 76 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ600-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i600, technician must log: "Resolved ERR-MAZ600-CLN-4 on Mazak-Integrex-i600 by installing PN-MAZ600-PMP-PLG-3004 & kit KIT-MAZ600-CLN-4. Fasteners torqued to 76 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ600-XAX-5

Target Machine Model: Mazak-Integrex-i600 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i600 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i600 (MAZATROL SmoothAi)
- Telemetry Sensor ID: EN-XAX-MAZ600-5
- PLC Hex Register Address: DB34.DBW_0x7DE8
- Nominal Operating Range: 27 - 64 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 124 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ600-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ600-LIN-COIL-3005
- Overhaul Seal & Consumable Kit: KIT-MAZ600-XAX-5
- Critical Fastener Torque Value: 85 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i600 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ600-XAX-5 at PLC register 0x7DE8.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Mazak-Integrex-i600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ600-LIN-COIL-3005) with seal kit (KIT-MAZ600-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 85 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ600-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i600, technician must log: "Resolved ERR-MAZ600-XAX-5 on Mazak-Integrex-i600 by installing PN-MAZ600-LIN-COIL-3005 & kit KIT-MAZ600-XAX-5. Fasteners torqued to 85 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ600-YAX-6

Target Machine Model: Mazak-Integrex-i600 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i600 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Mazak-Integrex-i600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i600 (MAZATROL SmoothAi)
- Telemetry Sensor ID: EN-YAX-MAZ600-6
- PLC Hex Register Address: DB34.DBW_0x7DF0
- Nominal Operating Range: 30 - 64 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 124 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ600-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ600-BSW-NUT-3006
- Overhaul Seal & Consumable Kit: KIT-MAZ600-YAX-6
- Critical Fastener Torque Value: 94 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i600 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ600-YAX-6 at PLC register 0x7DF0.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Mazak-Integrex-i600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ600-BSW-NUT-3006) with seal kit (KIT-MAZ600-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 94 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ600-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i600, technician must log: "Resolved ERR-MAZ600-YAX-6 on Mazak-Integrex-i600 by installing PN-MAZ600-BSW-NUT-3006 & kit KIT-MAZ600-YAX-6. Fasteners torqued to 94 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ600-ZAX-7

Target Machine Model: Mazak-Integrex-i600 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i600 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i600 (MAZATROL SmoothAi)
- Telemetry Sensor ID: PR-ZAX-MAZ600-7
- PLC Hex Register Address: DB34.DBW_0x7DF8
- Nominal Operating Range: 33 - 64 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 124 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ600-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ600-CYL-SEAL-3007
- Overhaul Seal & Consumable Kit: KIT-MAZ600-ZAX-7
- Critical Fastener Torque Value: 103 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i600 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ600-ZAX-7 at PLC register 0x7DF8.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Mazak-Integrex-i600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ600-CYL-SEAL-3007) with seal kit (KIT-MAZ600-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 103 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ600-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i600, technician must log: "Resolved ERR-MAZ600-ZAX-7 on Mazak-Integrex-i600 by installing PN-MAZ600-CYL-SEAL-3007 & kit KIT-MAZ600-ZAX-7. Fasteners torqued to 103 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ600-BAX-8

Target Machine Model: Mazak-Integrex-i600 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i600 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Mazak-Integrex-i600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i600 (MAZATROL SmoothAi)
- Telemetry Sensor ID: EN-BAX-MAZ600-8
- PLC Hex Register Address: DB34.DBW_0x7E00
- Nominal Operating Range: 36 - 64 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 124 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ600-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ600-TRQ-STAT-3008
- Overhaul Seal & Consumable Kit: KIT-MAZ600-BAX-8
- Critical Fastener Torque Value: 112 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i600 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ600-BAX-8 at PLC register 0x7E00.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Mazak-Integrex-i600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ600-TRQ-STAT-3008) with seal kit (KIT-MAZ600-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 112 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ600-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i600, technician must log: "Resolved ERR-MAZ600-BAX-8 on Mazak-Integrex-i600 by installing PN-MAZ600-TRQ-STAT-3008 & kit KIT-MAZ600-BAX-8. Fasteners torqued to 112 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ600-CAX-9

Target Machine Model: Mazak-Integrex-i600 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i600 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i600 (MAZATROL SmoothAi)
- Telemetry Sensor ID: PR-CAX-MAZ600-9
- PLC Hex Register Address: DB34.DBW_0x7E08
- Nominal Operating Range: 39 - 64 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 124 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ600-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ600-BRK-DISC-3009
- Overhaul Seal & Consumable Kit: KIT-MAZ600-CAX-9
- Critical Fastener Torque Value: 121 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i600 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ600-CAX-9 at PLC register 0x7E08.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Mazak-Integrex-i600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ600-BRK-DISC-3009) with seal kit (KIT-MAZ600-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 121 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ600-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i600, technician must log: "Resolved ERR-MAZ600-CAX-9 on Mazak-Integrex-i600 by installing PN-MAZ600-BRK-DISC-3009 & kit KIT-MAZ600-CAX-9. Fasteners torqued to 121 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ600-OPT-10

Target Machine Model: Mazak-Integrex-i600 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i600 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Mazak-Integrex-i600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i600 (MAZATROL SmoothAi)
- Telemetry Sensor ID: OP-SCL-MAZ600-10
- PLC Hex Register Address: DB34.DBW_0x7E10
- Nominal Operating Range: 42 - 64 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 124 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ600-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ600-GLASS-RD-3010
- Overhaul Seal & Consumable Kit: KIT-MAZ600-OPT-10
- Critical Fastener Torque Value: 130 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i600 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ600-OPT-10 at PLC register 0x7E10.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Mazak-Integrex-i600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ600-GLASS-RD-3010) with seal kit (KIT-MAZ600-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 130 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ600-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i600, technician must log: "Resolved ERR-MAZ600-OPT-10 on Mazak-Integrex-i600 by installing PN-MAZ600-GLASS-RD-3010 & kit KIT-MAZ600-OPT-10. Fasteners torqued to 130 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ600-PLC-11

Target Machine Model: Mazak-Integrex-i600 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i600 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i600 (MAZATROL SmoothAi)
- Telemetry Sensor ID: VT-DRV-MAZ600-11
- PLC Hex Register Address: DB34.DBW_0x7E18
- Nominal Operating Range: 45 - 64 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 124 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ600-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ600-IGBT-MOD-3011
- Overhaul Seal & Consumable Kit: KIT-MAZ600-PLC-11
- Critical Fastener Torque Value: 139 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i600 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ600-PLC-11 at PLC register 0x7E18.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Mazak-Integrex-i600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ600-IGBT-MOD-3011) with seal kit (KIT-MAZ600-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 139 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ600-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i600, technician must log: "Resolved ERR-MAZ600-PLC-11 on Mazak-Integrex-i600 by installing PN-MAZ600-IGBT-MOD-3011 & kit KIT-MAZ600-PLC-11. Fasteners torqued to 139 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ600-LUB-12

Target Machine Model: Mazak-Integrex-i600 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i600 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Mazak-Integrex-i600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i600 (MAZATROL SmoothAi)
- Telemetry Sensor ID: FL-LUB-MAZ600-12
- PLC Hex Register Address: DB34.DBW_0x7E20
- Nominal Operating Range: 48 - 64 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 124 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ600-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ600-INJ-MET-3012
- Overhaul Seal & Consumable Kit: KIT-MAZ600-LUB-12
- Critical Fastener Torque Value: 148 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i600 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ600-LUB-12 at PLC register 0x7E20.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Mazak-Integrex-i600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ600-INJ-MET-3012) with seal kit (KIT-MAZ600-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 148 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ600-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i600, technician must log: "Resolved ERR-MAZ600-LUB-12 on Mazak-Integrex-i600 by installing PN-MAZ600-INJ-MET-3012 & kit KIT-MAZ600-LUB-12. Fasteners torqued to 148 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ600-VAC-13

Target Machine Model: Mazak-Integrex-i600 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i600 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i600 (MAZATROL SmoothAi)
- Telemetry Sensor ID: VC-MNF-MAZ600-13
- PLC Hex Register Address: DB34.DBW_0x7E28
- Nominal Operating Range: 51 - 64 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 124 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ600-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ600-GASK-VIT-3013
- Overhaul Seal & Consumable Kit: KIT-MAZ600-VAC-13
- Critical Fastener Torque Value: 157 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i600 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ600-VAC-13 at PLC register 0x7E28.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Mazak-Integrex-i600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ600-GASK-VIT-3013) with seal kit (KIT-MAZ600-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 157 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ600-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i600, technician must log: "Resolved ERR-MAZ600-VAC-13 on Mazak-Integrex-i600 by installing PN-MAZ600-GASK-VIT-3013 & kit KIT-MAZ600-VAC-13. Fasteners torqued to 157 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ600-THM-14

Target Machine Model: Mazak-Integrex-i600 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i600 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Mazak-Integrex-i600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i600 (MAZATROL SmoothAi)
- Telemetry Sensor ID: TM-CHL-MAZ600-14
- PLC Hex Register Address: DB34.DBW_0x7E30
- Nominal Operating Range: 54 - 64 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 124 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ600-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ600-HEX-CORE-3014
- Overhaul Seal & Consumable Kit: KIT-MAZ600-THM-14
- Critical Fastener Torque Value: 166 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i600 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ600-THM-14 at PLC register 0x7E30.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Mazak-Integrex-i600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ600-HEX-CORE-3014) with seal kit (KIT-MAZ600-THM-14).

Step 6: Torque retaining bolts in cross sequence to 166 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ600-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i600, technician must log: "Resolved ERR-MAZ600-THM-14 on Mazak-Integrex-i600 by installing PN-MAZ600-HEX-CORE-3014 & kit KIT-MAZ600-THM-14. Fasteners torqued to 166 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU150-SPN-1

Target Machine Model: Okuma-Multus-U150 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U150 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Okuma-Multus-U150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U150 (OSP-P500A Aerospace)
- Telemetry Sensor ID: PT-SPN-OKU150-1
- PLC Hex Register Address: DB20.DBW_0x80E8
- Nominal Operating Range: 15 - 65 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 95 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU150-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU150-BRG-CER-3101
- Overhaul Seal & Consumable Kit: KIT-OKU150-SPN-1
- Critical Fastener Torque Value: 189 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U150 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU150-SPN-1 at PLC register 0x80E8.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Okuma-Multus-U150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU150-BRG-CER-3101) with seal kit (KIT-OKU150-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 189 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU150-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U150, technician must log: "Resolved ERR-OKU150-SPN-1 on Okuma-Multus-U150 by installing PN-OKU150-BRG-CER-3101 & kit KIT-OKU150-SPN-1. Fasteners torqued to 189 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU150-ATC-2

Target Machine Model: Okuma-Multus-U150 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U150 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Okuma-Multus-U150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U150 (OSP-P500A Aerospace)
- Telemetry Sensor ID: PX-ATC-OKU150-2
- PLC Hex Register Address: DB20.DBW_0x80F0
- Nominal Operating Range: 18 - 65 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 95 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU150-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU150-CAM-GRIP-3102
- Overhaul Seal & Consumable Kit: KIT-OKU150-ATC-2
- Critical Fastener Torque Value: 198 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U150 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU150-ATC-2 at PLC register 0x80F0.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Okuma-Multus-U150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU150-CAM-GRIP-3102) with seal kit (KIT-OKU150-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 198 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU150-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U150, technician must log: "Resolved ERR-OKU150-ATC-2 on Okuma-Multus-U150 by installing PN-OKU150-CAM-GRIP-3102 & kit KIT-OKU150-ATC-2. Fasteners torqued to 198 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU150-HYD-3

Target Machine Model: Okuma-Multus-U150 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U150 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Okuma-Multus-U150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U150 (OSP-P500A Aerospace)
- Telemetry Sensor ID: PR-HYD-OKU150-3
- PLC Hex Register Address: DB20.DBW_0x80F8
- Nominal Operating Range: 21 - 65 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 95 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU150-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU150-VLV-PROP-3103
- Overhaul Seal & Consumable Kit: KIT-OKU150-HYD-3
- Critical Fastener Torque Value: 207 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U150 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU150-HYD-3 at PLC register 0x80F8.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Okuma-Multus-U150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU150-VLV-PROP-3103) with seal kit (KIT-OKU150-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 207 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU150-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U150, technician must log: "Resolved ERR-OKU150-HYD-3 on Okuma-Multus-U150 by installing PN-OKU150-VLV-PROP-3103 & kit KIT-OKU150-HYD-3. Fasteners torqued to 207 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU150-CLN-4

Target Machine Model: Okuma-Multus-U150 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U150 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Okuma-Multus-U150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U150 (OSP-P500A Aerospace)
- Telemetry Sensor ID: FL-CLN-OKU150-4
- PLC Hex Register Address: DB20.DBW_0x8100
- Nominal Operating Range: 24 - 65 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 95 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU150-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU150-PMP-PLG-3104
- Overhaul Seal & Consumable Kit: KIT-OKU150-CLN-4
- Critical Fastener Torque Value: 26 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U150 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU150-CLN-4 at PLC register 0x8100.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Okuma-Multus-U150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU150-PMP-PLG-3104) with seal kit (KIT-OKU150-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 26 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU150-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U150, technician must log: "Resolved ERR-OKU150-CLN-4 on Okuma-Multus-U150 by installing PN-OKU150-PMP-PLG-3104 & kit KIT-OKU150-CLN-4. Fasteners torqued to 26 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU150-XAX-5

Target Machine Model: Okuma-Multus-U150 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U150 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U150 (OSP-P500A Aerospace)
- Telemetry Sensor ID: EN-XAX-OKU150-5
- PLC Hex Register Address: DB20.DBW_0x8108
- Nominal Operating Range: 27 - 65 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 95 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU150-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU150-LIN-COIL-3105
- Overhaul Seal & Consumable Kit: KIT-OKU150-XAX-5
- Critical Fastener Torque Value: 35 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U150 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU150-XAX-5 at PLC register 0x8108.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Okuma-Multus-U150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU150-LIN-COIL-3105) with seal kit (KIT-OKU150-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 35 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU150-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U150, technician must log: "Resolved ERR-OKU150-XAX-5 on Okuma-Multus-U150 by installing PN-OKU150-LIN-COIL-3105 & kit KIT-OKU150-XAX-5. Fasteners torqued to 35 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU150-YAX-6

Target Machine Model: Okuma-Multus-U150 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U150 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Okuma-Multus-U150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U150 (OSP-P500A Aerospace)
- Telemetry Sensor ID: EN-YAX-OKU150-6
- PLC Hex Register Address: DB20.DBW_0x8110
- Nominal Operating Range: 30 - 65 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 95 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU150-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU150-BSW-NUT-3106
- Overhaul Seal & Consumable Kit: KIT-OKU150-YAX-6
- Critical Fastener Torque Value: 44 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U150 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU150-YAX-6 at PLC register 0x8110.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Okuma-Multus-U150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU150-BSW-NUT-3106) with seal kit (KIT-OKU150-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 44 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU150-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U150, technician must log: "Resolved ERR-OKU150-YAX-6 on Okuma-Multus-U150 by installing PN-OKU150-BSW-NUT-3106 & kit KIT-OKU150-YAX-6. Fasteners torqued to 44 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU150-ZAX-7

Target Machine Model: Okuma-Multus-U150 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U150 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U150 (OSP-P500A Aerospace)
- Telemetry Sensor ID: PR-ZAX-OKU150-7
- PLC Hex Register Address: DB20.DBW_0x8118
- Nominal Operating Range: 33 - 65 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 95 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU150-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU150-CYL-SEAL-3107
- Overhaul Seal & Consumable Kit: KIT-OKU150-ZAX-7
- Critical Fastener Torque Value: 53 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U150 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU150-ZAX-7 at PLC register 0x8118.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Okuma-Multus-U150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU150-CYL-SEAL-3107) with seal kit (KIT-OKU150-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 53 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU150-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U150, technician must log: "Resolved ERR-OKU150-ZAX-7 on Okuma-Multus-U150 by installing PN-OKU150-CYL-SEAL-3107 & kit KIT-OKU150-ZAX-7. Fasteners torqued to 53 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU150-BAX-8

Target Machine Model: Okuma-Multus-U150 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U150 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Okuma-Multus-U150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U150 (OSP-P500A Aerospace)
- Telemetry Sensor ID: EN-BAX-OKU150-8
- PLC Hex Register Address: DB20.DBW_0x8120
- Nominal Operating Range: 36 - 65 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 95 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU150-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU150-TRQ-STAT-3108
- Overhaul Seal & Consumable Kit: KIT-OKU150-BAX-8
- Critical Fastener Torque Value: 62 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U150 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU150-BAX-8 at PLC register 0x8120.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Okuma-Multus-U150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU150-TRQ-STAT-3108) with seal kit (KIT-OKU150-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 62 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU150-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U150, technician must log: "Resolved ERR-OKU150-BAX-8 on Okuma-Multus-U150 by installing PN-OKU150-TRQ-STAT-3108 & kit KIT-OKU150-BAX-8. Fasteners torqued to 62 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU150-CAX-9

Target Machine Model: Okuma-Multus-U150 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U150 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U150 (OSP-P500A Aerospace)
- Telemetry Sensor ID: PR-CAX-OKU150-9
- PLC Hex Register Address: DB20.DBW_0x8128
- Nominal Operating Range: 39 - 65 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 95 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU150-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU150-BRK-DISC-3109
- Overhaul Seal & Consumable Kit: KIT-OKU150-CAX-9
- Critical Fastener Torque Value: 71 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U150 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU150-CAX-9 at PLC register 0x8128.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Okuma-Multus-U150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU150-BRK-DISC-3109) with seal kit (KIT-OKU150-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 71 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU150-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U150, technician must log: "Resolved ERR-OKU150-CAX-9 on Okuma-Multus-U150 by installing PN-OKU150-BRK-DISC-3109 & kit KIT-OKU150-CAX-9. Fasteners torqued to 71 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU150-OPT-10

Target Machine Model: Okuma-Multus-U150 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U150 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Okuma-Multus-U150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U150 (OSP-P500A Aerospace)
- Telemetry Sensor ID: OP-SCL-OKU150-10
- PLC Hex Register Address: DB20.DBW_0x8130
- Nominal Operating Range: 42 - 65 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 95 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU150-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU150-GLASS-RD-3110
- Overhaul Seal & Consumable Kit: KIT-OKU150-OPT-10
- Critical Fastener Torque Value: 80 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U150 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU150-OPT-10 at PLC register 0x8130.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Okuma-Multus-U150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU150-GLASS-RD-3110) with seal kit (KIT-OKU150-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 80 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU150-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U150, technician must log: "Resolved ERR-OKU150-OPT-10 on Okuma-Multus-U150 by installing PN-OKU150-GLASS-RD-3110 & kit KIT-OKU150-OPT-10. Fasteners torqued to 80 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU150-PLC-11

Target Machine Model: Okuma-Multus-U150 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U150 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U150 (OSP-P500A Aerospace)
- Telemetry Sensor ID: VT-DRV-OKU150-11
- PLC Hex Register Address: DB20.DBW_0x8138
- Nominal Operating Range: 45 - 65 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 95 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU150-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU150-IGBT-MOD-3111
- Overhaul Seal & Consumable Kit: KIT-OKU150-PLC-11
- Critical Fastener Torque Value: 89 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U150 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU150-PLC-11 at PLC register 0x8138.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Okuma-Multus-U150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU150-IGBT-MOD-3111) with seal kit (KIT-OKU150-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 89 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU150-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U150, technician must log: "Resolved ERR-OKU150-PLC-11 on Okuma-Multus-U150 by installing PN-OKU150-IGBT-MOD-3111 & kit KIT-OKU150-PLC-11. Fasteners torqued to 89 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU150-LUB-12

Target Machine Model: Okuma-Multus-U150 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U150 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Okuma-Multus-U150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U150 (OSP-P500A Aerospace)
- Telemetry Sensor ID: FL-LUB-OKU150-12
- PLC Hex Register Address: DB20.DBW_0x8140
- Nominal Operating Range: 48 - 65 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 95 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU150-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU150-INJ-MET-3112
- Overhaul Seal & Consumable Kit: KIT-OKU150-LUB-12
- Critical Fastener Torque Value: 98 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U150 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU150-LUB-12 at PLC register 0x8140.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Okuma-Multus-U150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU150-INJ-MET-3112) with seal kit (KIT-OKU150-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 98 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU150-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U150, technician must log: "Resolved ERR-OKU150-LUB-12 on Okuma-Multus-U150 by installing PN-OKU150-INJ-MET-3112 & kit KIT-OKU150-LUB-12. Fasteners torqued to 98 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU150-VAC-13

Target Machine Model: Okuma-Multus-U150 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U150 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U150 (OSP-P500A Aerospace)
- Telemetry Sensor ID: VC-MNF-OKU150-13
- PLC Hex Register Address: DB20.DBW_0x8148
- Nominal Operating Range: 51 - 65 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 95 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU150-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU150-GASK-VIT-3113
- Overhaul Seal & Consumable Kit: KIT-OKU150-VAC-13
- Critical Fastener Torque Value: 107 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U150 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU150-VAC-13 at PLC register 0x8148.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Okuma-Multus-U150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU150-GASK-VIT-3113) with seal kit (KIT-OKU150-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 107 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU150-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U150, technician must log: "Resolved ERR-OKU150-VAC-13 on Okuma-Multus-U150 by installing PN-OKU150-GASK-VIT-3113 & kit KIT-OKU150-VAC-13. Fasteners torqued to 107 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU150-THM-14

Target Machine Model: Okuma-Multus-U150 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U150 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Okuma-Multus-U150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U150 (OSP-P500A Aerospace)
- Telemetry Sensor ID: TM-CHL-OKU150-14
- PLC Hex Register Address: DB20.DBW_0x8150
- Nominal Operating Range: 54 - 65 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 95 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU150-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU150-HEX-CORE-3114
- Overhaul Seal & Consumable Kit: KIT-OKU150-THM-14
- Critical Fastener Torque Value: 116 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U150 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU150-THM-14 at PLC register 0x8150.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Okuma-Multus-U150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU150-HEX-CORE-3114) with seal kit (KIT-OKU150-THM-14).

Step 6: Torque retaining bolts in cross sequence to 116 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU150-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U150, technician must log: "Resolved ERR-OKU150-THM-14 on Okuma-Multus-U150 by installing PN-OKU150-HEX-CORE-3114 & kit KIT-OKU150-THM-14. Fasteners torqued to 116 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU200-SPN-1

Target Machine Model: Okuma-Multus-U200 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U200 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Okuma-Multus-U200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U200 (OSP-P500A Aerospace)
- Telemetry Sensor ID: PT-SPN-OKU200-1
- PLC Hex Register Address: DB21.DBW_0x8408
- Nominal Operating Range: 15 - 66 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 96 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU200-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU200-BRG-CER-3201
- Overhaul Seal & Consumable Kit: KIT-OKU200-SPN-1
- Critical Fastener Torque Value: 139 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U200 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU200-SPN-1 at PLC register 0x8408.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Okuma-Multus-U200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU200-BRG-CER-3201) with seal kit (KIT-OKU200-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 139 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU200-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U200, technician must log: "Resolved ERR-OKU200-SPN-1 on Okuma-Multus-U200 by installing PN-OKU200-BRG-CER-3201 & kit KIT-OKU200-SPN-1. Fasteners torqued to 139 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU200-ATC-2

Target Machine Model: Okuma-Multus-U200 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U200 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Okuma-Multus-U200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U200 (OSP-P500A Aerospace)
- Telemetry Sensor ID: PX-ATC-OKU200-2
- PLC Hex Register Address: DB21.DBW_0x8410
- Nominal Operating Range: 18 - 66 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 96 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU200-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU200-CAM-GRIP-3202
- Overhaul Seal & Consumable Kit: KIT-OKU200-ATC-2
- Critical Fastener Torque Value: 148 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U200 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU200-ATC-2 at PLC register 0x8410.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Okuma-Multus-U200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU200-CAM-GRIP-3202) with seal kit (KIT-OKU200-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 148 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU200-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U200, technician must log: "Resolved ERR-OKU200-ATC-2 on Okuma-Multus-U200 by installing PN-OKU200-CAM-GRIP-3202 & kit KIT-OKU200-ATC-2. Fasteners torqued to 148 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU200-HYD-3

Target Machine Model: Okuma-Multus-U200 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U200 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Okuma-Multus-U200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U200 (OSP-P500A Aerospace)
- Telemetry Sensor ID: PR-HYD-OKU200-3
- PLC Hex Register Address: DB21.DBW_0x8418
- Nominal Operating Range: 21 - 66 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 96 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU200-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU200-VLV-PROP-3203
- Overhaul Seal & Consumable Kit: KIT-OKU200-HYD-3
- Critical Fastener Torque Value: 157 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U200 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU200-HYD-3 at PLC register 0x8418.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Okuma-Multus-U200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU200-VLV-PROP-3203) with seal kit (KIT-OKU200-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 157 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU200-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U200, technician must log: "Resolved ERR-OKU200-HYD-3 on Okuma-Multus-U200 by installing PN-OKU200-VLV-PROP-3203 & kit KIT-OKU200-HYD-3. Fasteners torqued to 157 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU200-CLN-4

Target Machine Model: Okuma-Multus-U200 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U200 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Okuma-Multus-U200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U200 (OSP-P500A Aerospace)
- Telemetry Sensor ID: FL-CLN-OKU200-4
- PLC Hex Register Address: DB21.DBW_0x8420
- Nominal Operating Range: 24 - 66 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 96 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU200-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU200-PMP-PLG-3204
- Overhaul Seal & Consumable Kit: KIT-OKU200-CLN-4
- Critical Fastener Torque Value: 166 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U200 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU200-CLN-4 at PLC register 0x8420.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Okuma-Multus-U200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU200-PMP-PLG-3204) with seal kit (KIT-OKU200-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 166 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU200-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U200, technician must log: "Resolved ERR-OKU200-CLN-4 on Okuma-Multus-U200 by installing PN-OKU200-PMP-PLG-3204 & kit KIT-OKU200-CLN-4. Fasteners torqued to 166 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU200-XAX-5

Target Machine Model: Okuma-Multus-U200 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U200 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U200 (OSP-P500A Aerospace)
- Telemetry Sensor ID: EN-XAX-OKU200-5
- PLC Hex Register Address: DB21.DBW_0x8428
- Nominal Operating Range: 27 - 66 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 96 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU200-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU200-LIN-COIL-3205
- Overhaul Seal & Consumable Kit: KIT-OKU200-XAX-5
- Critical Fastener Torque Value: 175 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U200 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU200-XAX-5 at PLC register 0x8428.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Okuma-Multus-U200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU200-LIN-COIL-3205) with seal kit (KIT-OKU200-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 175 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU200-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U200, technician must log: "Resolved ERR-OKU200-XAX-5 on Okuma-Multus-U200 by installing PN-OKU200-LIN-COIL-3205 & kit KIT-OKU200-XAX-5. Fasteners torqued to 175 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU200-YAX-6

Target Machine Model: Okuma-Multus-U200 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U200 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Okuma-Multus-U200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U200 (OSP-P500A Aerospace)
- Telemetry Sensor ID: EN-YAX-OKU200-6
- PLC Hex Register Address: DB21.DBW_0x8430
- Nominal Operating Range: 30 - 66 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 96 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU200-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU200-BSW-NUT-3206
- Overhaul Seal & Consumable Kit: KIT-OKU200-YAX-6
- Critical Fastener Torque Value: 184 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U200 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU200-YAX-6 at PLC register 0x8430.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Okuma-Multus-U200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU200-BSW-NUT-3206) with seal kit (KIT-OKU200-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 184 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU200-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U200, technician must log: "Resolved ERR-OKU200-YAX-6 on Okuma-Multus-U200 by installing PN-OKU200-BSW-NUT-3206 & kit KIT-OKU200-YAX-6. Fasteners torqued to 184 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU200-ZAX-7

Target Machine Model: Okuma-Multus-U200 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U200 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U200 (OSP-P500A Aerospace)
- Telemetry Sensor ID: PR-ZAX-OKU200-7
- PLC Hex Register Address: DB21.DBW_0x8438
- Nominal Operating Range: 33 - 66 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 96 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU200-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU200-CYL-SEAL-3207
- Overhaul Seal & Consumable Kit: KIT-OKU200-ZAX-7
- Critical Fastener Torque Value: 193 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U200 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU200-ZAX-7 at PLC register 0x8438.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Okuma-Multus-U200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU200-CYL-SEAL-3207) with seal kit (KIT-OKU200-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 193 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU200-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U200, technician must log: "Resolved ERR-OKU200-ZAX-7 on Okuma-Multus-U200 by installing PN-OKU200-CYL-SEAL-3207 & kit KIT-OKU200-ZAX-7. Fasteners torqued to 193 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU200-BAX-8

Target Machine Model: Okuma-Multus-U200 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U200 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Okuma-Multus-U200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U200 (OSP-P500A Aerospace)
- Telemetry Sensor ID: EN-BAX-OKU200-8
- PLC Hex Register Address: DB21.DBW_0x8440
- Nominal Operating Range: 36 - 66 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 96 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU200-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU200-TRQ-STAT-3208
- Overhaul Seal & Consumable Kit: KIT-OKU200-BAX-8
- Critical Fastener Torque Value: 202 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U200 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU200-BAX-8 at PLC register 0x8440.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Okuma-Multus-U200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU200-TRQ-STAT-3208) with seal kit (KIT-OKU200-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 202 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU200-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U200, technician must log: "Resolved ERR-OKU200-BAX-8 on Okuma-Multus-U200 by installing PN-OKU200-TRQ-STAT-3208 & kit KIT-OKU200-BAX-8. Fasteners torqued to 202 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU200-CAX-9

Target Machine Model: Okuma-Multus-U200 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U200 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U200 (OSP-P500A Aerospace)
- Telemetry Sensor ID: PR-CAX-OKU200-9
- PLC Hex Register Address: DB21.DBW_0x8448
- Nominal Operating Range: 39 - 66 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 96 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU200-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU200-BRK-DISC-3209
- Overhaul Seal & Consumable Kit: KIT-OKU200-CAX-9
- Critical Fastener Torque Value: 21 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U200 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU200-CAX-9 at PLC register 0x8448.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Okuma-Multus-U200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU200-BRK-DISC-3209) with seal kit (KIT-OKU200-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 21 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU200-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U200, technician must log: "Resolved ERR-OKU200-CAX-9 on Okuma-Multus-U200 by installing PN-OKU200-BRK-DISC-3209 & kit KIT-OKU200-CAX-9. Fasteners torqued to 21 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU200-OPT-10

Target Machine Model: Okuma-Multus-U200 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U200 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Okuma-Multus-U200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U200 (OSP-P500A Aerospace)
- Telemetry Sensor ID: OP-SCL-OKU200-10
- PLC Hex Register Address: DB21.DBW_0x8450
- Nominal Operating Range: 42 - 66 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 96 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU200-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU200-GLASS-RD-3210
- Overhaul Seal & Consumable Kit: KIT-OKU200-OPT-10
- Critical Fastener Torque Value: 30 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U200 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU200-OPT-10 at PLC register 0x8450.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Okuma-Multus-U200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU200-GLASS-RD-3210) with seal kit (KIT-OKU200-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 30 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU200-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U200, technician must log: "Resolved ERR-OKU200-OPT-10 on Okuma-Multus-U200 by installing PN-OKU200-GLASS-RD-3210 & kit KIT-OKU200-OPT-10. Fasteners torqued to 30 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU200-PLC-11

Target Machine Model: Okuma-Multus-U200 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U200 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U200 (OSP-P500A Aerospace)
- Telemetry Sensor ID: VT-DRV-OKU200-11
- PLC Hex Register Address: DB21.DBW_0x8458
- Nominal Operating Range: 45 - 66 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 96 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU200-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU200-IGBT-MOD-3211
- Overhaul Seal & Consumable Kit: KIT-OKU200-PLC-11
- Critical Fastener Torque Value: 39 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U200 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU200-PLC-11 at PLC register 0x8458.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Okuma-Multus-U200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU200-IGBT-MOD-3211) with seal kit (KIT-OKU200-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 39 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU200-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U200, technician must log: "Resolved ERR-OKU200-PLC-11 on Okuma-Multus-U200 by installing PN-OKU200-IGBT-MOD-3211 & kit KIT-OKU200-PLC-11. Fasteners torqued to 39 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU200-LUB-12

Target Machine Model: Okuma-Multus-U200 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U200 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Okuma-Multus-U200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U200 (OSP-P500A Aerospace)
- Telemetry Sensor ID: FL-LUB-OKU200-12
- PLC Hex Register Address: DB21.DBW_0x8460
- Nominal Operating Range: 48 - 66 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 96 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU200-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU200-INJ-MET-3212
- Overhaul Seal & Consumable Kit: KIT-OKU200-LUB-12
- Critical Fastener Torque Value: 48 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U200 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU200-LUB-12 at PLC register 0x8460.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Okuma-Multus-U200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU200-INJ-MET-3212) with seal kit (KIT-OKU200-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 48 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU200-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U200, technician must log: "Resolved ERR-OKU200-LUB-12 on Okuma-Multus-U200 by installing PN-OKU200-INJ-MET-3212 & kit KIT-OKU200-LUB-12. Fasteners torqued to 48 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU200-VAC-13

Target Machine Model: Okuma-Multus-U200 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U200 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U200 (OSP-P500A Aerospace)
- Telemetry Sensor ID: VC-MNF-OKU200-13
- PLC Hex Register Address: DB21.DBW_0x8468
- Nominal Operating Range: 51 - 66 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 96 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU200-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU200-GASK-VIT-3213
- Overhaul Seal & Consumable Kit: KIT-OKU200-VAC-13
- Critical Fastener Torque Value: 57 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U200 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU200-VAC-13 at PLC register 0x8468.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Okuma-Multus-U200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU200-GASK-VIT-3213) with seal kit (KIT-OKU200-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 57 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU200-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U200, technician must log: "Resolved ERR-OKU200-VAC-13 on Okuma-Multus-U200 by installing PN-OKU200-GASK-VIT-3213 & kit KIT-OKU200-VAC-13. Fasteners torqued to 57 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU200-THM-14

Target Machine Model: Okuma-Multus-U200 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U200 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Okuma-Multus-U200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U200 (OSP-P500A Aerospace)
- Telemetry Sensor ID: TM-CHL-OKU200-14
- PLC Hex Register Address: DB21.DBW_0x8470
- Nominal Operating Range: 54 - 66 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 96 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU200-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU200-HEX-CORE-3214
- Overhaul Seal & Consumable Kit: KIT-OKU200-THM-14
- Critical Fastener Torque Value: 66 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U200 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU200-THM-14 at PLC register 0x8470.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Okuma-Multus-U200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU200-HEX-CORE-3214) with seal kit (KIT-OKU200-THM-14).

Step 6: Torque retaining bolts in cross sequence to 66 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU200-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U200, technician must log: "Resolved ERR-OKU200-THM-14 on Okuma-Multus-U200 by installing PN-OKU200-HEX-CORE-3214 & kit KIT-OKU200-THM-14. Fasteners torqued to 66 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU250-SPN-1

Target Machine Model: Okuma-Multus-U250 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U250 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Okuma-Multus-U250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U250 (OSP-P500A Aerospace)
- Telemetry Sensor ID: PT-SPN-OKU250-1
- PLC Hex Register Address: DB22.DBW_0x8728
- Nominal Operating Range: 15 - 67 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 97 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU250-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU250-BRG-CER-3301
- Overhaul Seal & Consumable Kit: KIT-OKU250-SPN-1
- Critical Fastener Torque Value: 89 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U250 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU250-SPN-1 at PLC register 0x8728.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Okuma-Multus-U250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU250-BRG-CER-3301) with seal kit (KIT-OKU250-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 89 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU250-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U250, technician must log: "Resolved ERR-OKU250-SPN-1 on Okuma-Multus-U250 by installing PN-OKU250-BRG-CER-3301 & kit KIT-OKU250-SPN-1. Fasteners torqued to 89 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU250-ATC-2

Target Machine Model: Okuma-Multus-U250 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U250 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Okuma-Multus-U250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U250 (OSP-P500A Aerospace)
- Telemetry Sensor ID: PX-ATC-OKU250-2
- PLC Hex Register Address: DB22.DBW_0x8730
- Nominal Operating Range: 18 - 67 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 97 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU250-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU250-CAM-GRIP-3302
- Overhaul Seal & Consumable Kit: KIT-OKU250-ATC-2
- Critical Fastener Torque Value: 98 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U250 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU250-ATC-2 at PLC register 0x8730.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Okuma-Multus-U250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU250-CAM-GRIP-3302) with seal kit (KIT-OKU250-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 98 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU250-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U250, technician must log: "Resolved ERR-OKU250-ATC-2 on Okuma-Multus-U250 by installing PN-OKU250-CAM-GRIP-3302 & kit KIT-OKU250-ATC-2. Fasteners torqued to 98 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU250-HYD-3

Target Machine Model: Okuma-Multus-U250 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U250 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Okuma-Multus-U250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U250 (OSP-P500A Aerospace)
- Telemetry Sensor ID: PR-HYD-OKU250-3
- PLC Hex Register Address: DB22.DBW_0x8738
- Nominal Operating Range: 21 - 67 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 97 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU250-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU250-VLV-PROP-3303
- Overhaul Seal & Consumable Kit: KIT-OKU250-HYD-3
- Critical Fastener Torque Value: 107 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U250 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU250-HYD-3 at PLC register 0x8738.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Okuma-Multus-U250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU250-VLV-PROP-3303) with seal kit (KIT-OKU250-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 107 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU250-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U250, technician must log: "Resolved ERR-OKU250-HYD-3 on Okuma-Multus-U250 by installing PN-OKU250-VLV-PROP-3303 & kit KIT-OKU250-HYD-3. Fasteners torqued to 107 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU250-CLN-4

Target Machine Model: Okuma-Multus-U250 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U250 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Okuma-Multus-U250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U250 (OSP-P500A Aerospace)
- Telemetry Sensor ID: FL-CLN-OKU250-4
- PLC Hex Register Address: DB22.DBW_0x8740
- Nominal Operating Range: 24 - 67 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 97 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU250-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU250-PMP-PLG-3304
- Overhaul Seal & Consumable Kit: KIT-OKU250-CLN-4
- Critical Fastener Torque Value: 116 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U250 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU250-CLN-4 at PLC register 0x8740.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Okuma-Multus-U250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU250-PMP-PLG-3304) with seal kit (KIT-OKU250-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 116 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU250-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U250, technician must log: "Resolved ERR-OKU250-CLN-4 on Okuma-Multus-U250 by installing PN-OKU250-PMP-PLG-3304 & kit KIT-OKU250-CLN-4. Fasteners torqued to 116 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU250-XAX-5

Target Machine Model: Okuma-Multus-U250 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U250 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U250 (OSP-P500A Aerospace)
- Telemetry Sensor ID: EN-XAX-OKU250-5
- PLC Hex Register Address: DB22.DBW_0x8748
- Nominal Operating Range: 27 - 67 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 97 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU250-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU250-LIN-COIL-3305
- Overhaul Seal & Consumable Kit: KIT-OKU250-XAX-5
- Critical Fastener Torque Value: 125 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U250 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU250-XAX-5 at PLC register 0x8748.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Okuma-Multus-U250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU250-LIN-COIL-3305) with seal kit (KIT-OKU250-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 125 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU250-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U250, technician must log: "Resolved ERR-OKU250-XAX-5 on Okuma-Multus-U250 by installing PN-OKU250-LIN-COIL-3305 & kit KIT-OKU250-XAX-5. Fasteners torqued to 125 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU250-YAX-6

Target Machine Model: Okuma-Multus-U250 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U250 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Okuma-Multus-U250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U250 (OSP-P500A Aerospace)
- Telemetry Sensor ID: EN-YAX-OKU250-6
- PLC Hex Register Address: DB22.DBW_0x8750
- Nominal Operating Range: 30 - 67 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 97 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU250-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU250-BSW-NUT-3306
- Overhaul Seal & Consumable Kit: KIT-OKU250-YAX-6
- Critical Fastener Torque Value: 134 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U250 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU250-YAX-6 at PLC register 0x8750.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Okuma-Multus-U250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU250-BSW-NUT-3306) with seal kit (KIT-OKU250-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 134 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU250-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U250, technician must log: "Resolved ERR-OKU250-YAX-6 on Okuma-Multus-U250 by installing PN-OKU250-BSW-NUT-3306 & kit KIT-OKU250-YAX-6. Fasteners torqued to 134 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU250-ZAX-7

Target Machine Model: Okuma-Multus-U250 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U250 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U250 (OSP-P500A Aerospace)
- Telemetry Sensor ID: PR-ZAX-OKU250-7
- PLC Hex Register Address: DB22.DBW_0x8758
- Nominal Operating Range: 33 - 67 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 97 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU250-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU250-CYL-SEAL-3307
- Overhaul Seal & Consumable Kit: KIT-OKU250-ZAX-7
- Critical Fastener Torque Value: 143 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U250 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU250-ZAX-7 at PLC register 0x8758.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Okuma-Multus-U250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU250-CYL-SEAL-3307) with seal kit (KIT-OKU250-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 143 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU250-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U250, technician must log: "Resolved ERR-OKU250-ZAX-7 on Okuma-Multus-U250 by installing PN-OKU250-CYL-SEAL-3307 & kit KIT-OKU250-ZAX-7. Fasteners torqued to 143 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU250-BAX-8

Target Machine Model: Okuma-Multus-U250 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U250 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Okuma-Multus-U250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U250 (OSP-P500A Aerospace)
- Telemetry Sensor ID: EN-BAX-OKU250-8
- PLC Hex Register Address: DB22.DBW_0x8760
- Nominal Operating Range: 36 - 67 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 97 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU250-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU250-TRQ-STAT-3308
- Overhaul Seal & Consumable Kit: KIT-OKU250-BAX-8
- Critical Fastener Torque Value: 152 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U250 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU250-BAX-8 at PLC register 0x8760.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Okuma-Multus-U250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU250-TRQ-STAT-3308) with seal kit (KIT-OKU250-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 152 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU250-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U250, technician must log: "Resolved ERR-OKU250-BAX-8 on Okuma-Multus-U250 by installing PN-OKU250-TRQ-STAT-3308 & kit KIT-OKU250-BAX-8. Fasteners torqued to 152 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU250-CAX-9

Target Machine Model: Okuma-Multus-U250 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U250 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U250 (OSP-P500A Aerospace)
- Telemetry Sensor ID: PR-CAX-OKU250-9
- PLC Hex Register Address: DB22.DBW_0x8768
- Nominal Operating Range: 39 - 67 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 97 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU250-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU250-BRK-DISC-3309
- Overhaul Seal & Consumable Kit: KIT-OKU250-CAX-9
- Critical Fastener Torque Value: 161 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U250 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU250-CAX-9 at PLC register 0x8768.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Okuma-Multus-U250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU250-BRK-DISC-3309) with seal kit (KIT-OKU250-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 161 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU250-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U250, technician must log: "Resolved ERR-OKU250-CAX-9 on Okuma-Multus-U250 by installing PN-OKU250-BRK-DISC-3309 & kit KIT-OKU250-CAX-9. Fasteners torqued to 161 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU250-OPT-10

Target Machine Model: Okuma-Multus-U250 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U250 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Okuma-Multus-U250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U250 (OSP-P500A Aerospace)
- Telemetry Sensor ID: OP-SCL-OKU250-10
- PLC Hex Register Address: DB22.DBW_0x8770
- Nominal Operating Range: 42 - 67 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 97 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU250-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU250-GLASS-RD-3310
- Overhaul Seal & Consumable Kit: KIT-OKU250-OPT-10
- Critical Fastener Torque Value: 170 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U250 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU250-OPT-10 at PLC register 0x8770.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Okuma-Multus-U250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU250-GLASS-RD-3310) with seal kit (KIT-OKU250-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 170 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU250-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U250, technician must log: "Resolved ERR-OKU250-OPT-10 on Okuma-Multus-U250 by installing PN-OKU250-GLASS-RD-3310 & kit KIT-OKU250-OPT-10. Fasteners torqued to 170 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU250-PLC-11

Target Machine Model: Okuma-Multus-U250 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U250 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U250 (OSP-P500A Aerospace)
- Telemetry Sensor ID: VT-DRV-OKU250-11
- PLC Hex Register Address: DB22.DBW_0x8778
- Nominal Operating Range: 45 - 67 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 97 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU250-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU250-IGBT-MOD-3311
- Overhaul Seal & Consumable Kit: KIT-OKU250-PLC-11
- Critical Fastener Torque Value: 179 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U250 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU250-PLC-11 at PLC register 0x8778.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Okuma-Multus-U250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU250-IGBT-MOD-3311) with seal kit (KIT-OKU250-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 179 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU250-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U250, technician must log: "Resolved ERR-OKU250-PLC-11 on Okuma-Multus-U250 by installing PN-OKU250-IGBT-MOD-3311 & kit KIT-OKU250-PLC-11. Fasteners torqued to 179 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU250-LUB-12

Target Machine Model: Okuma-Multus-U250 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U250 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Okuma-Multus-U250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U250 (OSP-P500A Aerospace)
- Telemetry Sensor ID: FL-LUB-OKU250-12
- PLC Hex Register Address: DB22.DBW_0x8780
- Nominal Operating Range: 48 - 67 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 97 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU250-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU250-INJ-MET-3312
- Overhaul Seal & Consumable Kit: KIT-OKU250-LUB-12
- Critical Fastener Torque Value: 188 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U250 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU250-LUB-12 at PLC register 0x8780.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Okuma-Multus-U250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU250-INJ-MET-3312) with seal kit (KIT-OKU250-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 188 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU250-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U250, technician must log: "Resolved ERR-OKU250-LUB-12 on Okuma-Multus-U250 by installing PN-OKU250-INJ-MET-3312 & kit KIT-OKU250-LUB-12. Fasteners torqued to 188 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU250-VAC-13

Target Machine Model: Okuma-Multus-U250 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U250 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U250 (OSP-P500A Aerospace)
- Telemetry Sensor ID: VC-MNF-OKU250-13
- PLC Hex Register Address: DB22.DBW_0x8788
- Nominal Operating Range: 51 - 67 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 97 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU250-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU250-GASK-VIT-3313
- Overhaul Seal & Consumable Kit: KIT-OKU250-VAC-13
- Critical Fastener Torque Value: 197 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U250 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU250-VAC-13 at PLC register 0x8788.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Okuma-Multus-U250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU250-GASK-VIT-3313) with seal kit (KIT-OKU250-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 197 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU250-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U250, technician must log: "Resolved ERR-OKU250-VAC-13 on Okuma-Multus-U250 by installing PN-OKU250-GASK-VIT-3313 & kit KIT-OKU250-VAC-13. Fasteners torqued to 197 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU250-THM-14

Target Machine Model: Okuma-Multus-U250 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U250 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Okuma-Multus-U250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U250 (OSP-P500A Aerospace)
- Telemetry Sensor ID: TM-CHL-OKU250-14
- PLC Hex Register Address: DB22.DBW_0x8790
- Nominal Operating Range: 54 - 67 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 97 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU250-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU250-HEX-CORE-3314
- Overhaul Seal & Consumable Kit: KIT-OKU250-THM-14
- Critical Fastener Torque Value: 206 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U250 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU250-THM-14 at PLC register 0x8790.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Okuma-Multus-U250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU250-HEX-CORE-3314) with seal kit (KIT-OKU250-THM-14).

Step 6: Torque retaining bolts in cross sequence to 206 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU250-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U250, technician must log: "Resolved ERR-OKU250-THM-14 on Okuma-Multus-U250 by installing PN-OKU250-HEX-CORE-3314 & kit KIT-OKU250-THM-14. Fasteners torqued to 206 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU300-SPN-1

Target Machine Model: Okuma-Multus-U300 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U300 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Okuma-Multus-U300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U300 (OSP-P500A Aerospace)
- Telemetry Sensor ID: PT-SPN-OKU300-1
- PLC Hex Register Address: DB23.DBW_0x8A48
- Nominal Operating Range: 15 - 68 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 98 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU300-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU300-BRG-CER-3401
- Overhaul Seal & Consumable Kit: KIT-OKU300-SPN-1
- Critical Fastener Torque Value: 39 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U300 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU300-SPN-1 at PLC register 0x8A48.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Okuma-Multus-U300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU300-BRG-CER-3401) with seal kit (KIT-OKU300-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 39 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU300-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U300, technician must log: "Resolved ERR-OKU300-SPN-1 on Okuma-Multus-U300 by installing PN-OKU300-BRG-CER-3401 & kit KIT-OKU300-SPN-1. Fasteners torqued to 39 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU300-ATC-2

Target Machine Model: Okuma-Multus-U300 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U300 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Okuma-Multus-U300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U300 (OSP-P500A Aerospace)
- Telemetry Sensor ID: PX-ATC-OKU300-2
- PLC Hex Register Address: DB23.DBW_0x8A50
- Nominal Operating Range: 18 - 68 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 98 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU300-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU300-CAM-GRIP-3402
- Overhaul Seal & Consumable Kit: KIT-OKU300-ATC-2
- Critical Fastener Torque Value: 48 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U300 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU300-ATC-2 at PLC register 0x8A50.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Okuma-Multus-U300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU300-CAM-GRIP-3402) with seal kit (KIT-OKU300-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 48 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU300-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U300, technician must log: "Resolved ERR-OKU300-ATC-2 on Okuma-Multus-U300 by installing PN-OKU300-CAM-GRIP-3402 & kit KIT-OKU300-ATC-2. Fasteners torqued to 48 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU300-HYD-3

Target Machine Model: Okuma-Multus-U300 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U300 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Okuma-Multus-U300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U300 (OSP-P500A Aerospace)
- Telemetry Sensor ID: PR-HYD-OKU300-3
- PLC Hex Register Address: DB23.DBW_0x8A58
- Nominal Operating Range: 21 - 68 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 98 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU300-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU300-VLV-PROP-3403
- Overhaul Seal & Consumable Kit: KIT-OKU300-HYD-3
- Critical Fastener Torque Value: 57 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U300 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU300-HYD-3 at PLC register 0x8A58.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Okuma-Multus-U300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU300-VLV-PROP-3403) with seal kit (KIT-OKU300-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 57 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU300-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U300, technician must log: "Resolved ERR-OKU300-HYD-3 on Okuma-Multus-U300 by installing PN-OKU300-VLV-PROP-3403 & kit KIT-OKU300-HYD-3. Fasteners torqued to 57 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU300-CLN-4

Target Machine Model: Okuma-Multus-U300 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U300 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Okuma-Multus-U300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U300 (OSP-P500A Aerospace)
- Telemetry Sensor ID: FL-CLN-OKU300-4
- PLC Hex Register Address: DB23.DBW_0x8A60
- Nominal Operating Range: 24 - 68 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 98 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU300-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU300-PMP-PLG-3404
- Overhaul Seal & Consumable Kit: KIT-OKU300-CLN-4
- Critical Fastener Torque Value: 66 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U300 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU300-CLN-4 at PLC register 0x8A60.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Okuma-Multus-U300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU300-PMP-PLG-3404) with seal kit (KIT-OKU300-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 66 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU300-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U300, technician must log: "Resolved ERR-OKU300-CLN-4 on Okuma-Multus-U300 by installing PN-OKU300-PMP-PLG-3404 & kit KIT-OKU300-CLN-4. Fasteners torqued to 66 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU300-XAX-5

Target Machine Model: Okuma-Multus-U300 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U300 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U300 (OSP-P500A Aerospace)
- Telemetry Sensor ID: EN-XAX-OKU300-5
- PLC Hex Register Address: DB23.DBW_0x8A68
- Nominal Operating Range: 27 - 68 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 98 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU300-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU300-LIN-COIL-3405
- Overhaul Seal & Consumable Kit: KIT-OKU300-XAX-5
- Critical Fastener Torque Value: 75 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U300 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU300-XAX-5 at PLC register 0x8A68.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Okuma-Multus-U300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU300-LIN-COIL-3405) with seal kit (KIT-OKU300-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 75 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU300-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U300, technician must log: "Resolved ERR-OKU300-XAX-5 on Okuma-Multus-U300 by installing PN-OKU300-LIN-COIL-3405 & kit KIT-OKU300-XAX-5. Fasteners torqued to 75 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU300-YAX-6

Target Machine Model: Okuma-Multus-U300 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U300 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Okuma-Multus-U300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U300 (OSP-P500A Aerospace)
- Telemetry Sensor ID: EN-YAX-OKU300-6
- PLC Hex Register Address: DB23.DBW_0x8A70
- Nominal Operating Range: 30 - 68 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 98 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU300-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU300-BSW-NUT-3406
- Overhaul Seal & Consumable Kit: KIT-OKU300-YAX-6
- Critical Fastener Torque Value: 84 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U300 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU300-YAX-6 at PLC register 0x8A70.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Okuma-Multus-U300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU300-BSW-NUT-3406) with seal kit (KIT-OKU300-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 84 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU300-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U300, technician must log: "Resolved ERR-OKU300-YAX-6 on Okuma-Multus-U300 by installing PN-OKU300-BSW-NUT-3406 & kit KIT-OKU300-YAX-6. Fasteners torqued to 84 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU300-ZAX-7

Target Machine Model: Okuma-Multus-U300 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U300 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U300 (OSP-P500A Aerospace)
- Telemetry Sensor ID: PR-ZAX-OKU300-7
- PLC Hex Register Address: DB23.DBW_0x8A78
- Nominal Operating Range: 33 - 68 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 98 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU300-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU300-CYL-SEAL-3407
- Overhaul Seal & Consumable Kit: KIT-OKU300-ZAX-7
- Critical Fastener Torque Value: 93 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U300 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU300-ZAX-7 at PLC register 0x8A78.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Okuma-Multus-U300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU300-CYL-SEAL-3407) with seal kit (KIT-OKU300-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 93 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU300-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U300, technician must log: "Resolved ERR-OKU300-ZAX-7 on Okuma-Multus-U300 by installing PN-OKU300-CYL-SEAL-3407 & kit KIT-OKU300-ZAX-7. Fasteners torqued to 93 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU300-BAX-8

Target Machine Model: Okuma-Multus-U300 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U300 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Okuma-Multus-U300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U300 (OSP-P500A Aerospace)
- Telemetry Sensor ID: EN-BAX-OKU300-8
- PLC Hex Register Address: DB23.DBW_0x8A80
- Nominal Operating Range: 36 - 68 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 98 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU300-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU300-TRQ-STAT-3408
- Overhaul Seal & Consumable Kit: KIT-OKU300-BAX-8
- Critical Fastener Torque Value: 102 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U300 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU300-BAX-8 at PLC register 0x8A80.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Okuma-Multus-U300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU300-TRQ-STAT-3408) with seal kit (KIT-OKU300-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 102 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU300-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U300, technician must log: "Resolved ERR-OKU300-BAX-8 on Okuma-Multus-U300 by installing PN-OKU300-TRQ-STAT-3408 & kit KIT-OKU300-BAX-8. Fasteners torqued to 102 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU300-CAX-9

Target Machine Model: Okuma-Multus-U300 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U300 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U300 (OSP-P500A Aerospace)
- Telemetry Sensor ID: PR-CAX-OKU300-9
- PLC Hex Register Address: DB23.DBW_0x8A88
- Nominal Operating Range: 39 - 68 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 98 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU300-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU300-BRK-DISC-3409
- Overhaul Seal & Consumable Kit: KIT-OKU300-CAX-9
- Critical Fastener Torque Value: 111 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U300 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU300-CAX-9 at PLC register 0x8A88.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Okuma-Multus-U300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU300-BRK-DISC-3409) with seal kit (KIT-OKU300-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 111 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU300-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U300, technician must log: "Resolved ERR-OKU300-CAX-9 on Okuma-Multus-U300 by installing PN-OKU300-BRK-DISC-3409 & kit KIT-OKU300-CAX-9. Fasteners torqued to 111 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU300-OPT-10

Target Machine Model: Okuma-Multus-U300 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U300 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Okuma-Multus-U300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U300 (OSP-P500A Aerospace)
- Telemetry Sensor ID: OP-SCL-OKU300-10
- PLC Hex Register Address: DB23.DBW_0x8A90
- Nominal Operating Range: 42 - 68 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 98 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU300-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU300-GLASS-RD-3410
- Overhaul Seal & Consumable Kit: KIT-OKU300-OPT-10
- Critical Fastener Torque Value: 120 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U300 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU300-OPT-10 at PLC register 0x8A90.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Okuma-Multus-U300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU300-GLASS-RD-3410) with seal kit (KIT-OKU300-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 120 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU300-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U300, technician must log: "Resolved ERR-OKU300-OPT-10 on Okuma-Multus-U300 by installing PN-OKU300-GLASS-RD-3410 & kit KIT-OKU300-OPT-10. Fasteners torqued to 120 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU300-PLC-11

Target Machine Model: Okuma-Multus-U300 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U300 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U300 (OSP-P500A Aerospace)
- Telemetry Sensor ID: VT-DRV-OKU300-11
- PLC Hex Register Address: DB23.DBW_0x8A98
- Nominal Operating Range: 45 - 68 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 98 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU300-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU300-IGBT-MOD-3411
- Overhaul Seal & Consumable Kit: KIT-OKU300-PLC-11
- Critical Fastener Torque Value: 129 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U300 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU300-PLC-11 at PLC register 0x8A98.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Okuma-Multus-U300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU300-IGBT-MOD-3411) with seal kit (KIT-OKU300-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 129 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU300-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U300, technician must log: "Resolved ERR-OKU300-PLC-11 on Okuma-Multus-U300 by installing PN-OKU300-IGBT-MOD-3411 & kit KIT-OKU300-PLC-11. Fasteners torqued to 129 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU300-LUB-12

Target Machine Model: Okuma-Multus-U300 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U300 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Okuma-Multus-U300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U300 (OSP-P500A Aerospace)
- Telemetry Sensor ID: FL-LUB-OKU300-12
- PLC Hex Register Address: DB23.DBW_0x8AA0
- Nominal Operating Range: 48 - 68 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 98 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU300-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU300-INJ-MET-3412
- Overhaul Seal & Consumable Kit: KIT-OKU300-LUB-12
- Critical Fastener Torque Value: 138 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U300 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU300-LUB-12 at PLC register 0x8AA0.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Okuma-Multus-U300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU300-INJ-MET-3412) with seal kit (KIT-OKU300-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 138 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU300-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U300, technician must log: "Resolved ERR-OKU300-LUB-12 on Okuma-Multus-U300 by installing PN-OKU300-INJ-MET-3412 & kit KIT-OKU300-LUB-12. Fasteners torqued to 138 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU300-VAC-13

Target Machine Model: Okuma-Multus-U300 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U300 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U300 (OSP-P500A Aerospace)
- Telemetry Sensor ID: VC-MNF-OKU300-13
- PLC Hex Register Address: DB23.DBW_0x8AA8
- Nominal Operating Range: 51 - 68 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 98 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU300-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU300-GASK-VIT-3413
- Overhaul Seal & Consumable Kit: KIT-OKU300-VAC-13
- Critical Fastener Torque Value: 147 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U300 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU300-VAC-13 at PLC register 0x8AA8.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Okuma-Multus-U300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU300-GASK-VIT-3413) with seal kit (KIT-OKU300-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 147 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU300-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U300, technician must log: "Resolved ERR-OKU300-VAC-13 on Okuma-Multus-U300 by installing PN-OKU300-GASK-VIT-3413 & kit KIT-OKU300-VAC-13. Fasteners torqued to 147 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU300-THM-14

Target Machine Model: Okuma-Multus-U300 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U300 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Okuma-Multus-U300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U300 (OSP-P500A Aerospace)
- Telemetry Sensor ID: TM-CHL-OKU300-14
- PLC Hex Register Address: DB23.DBW_0x8AB0
- Nominal Operating Range: 54 - 68 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 98 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU300-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU300-HEX-CORE-3414
- Overhaul Seal & Consumable Kit: KIT-OKU300-THM-14
- Critical Fastener Torque Value: 156 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U300 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU300-THM-14 at PLC register 0x8AB0.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Okuma-Multus-U300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU300-HEX-CORE-3414) with seal kit (KIT-OKU300-THM-14).

Step 6: Torque retaining bolts in cross sequence to 156 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU300-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U300, technician must log: "Resolved ERR-OKU300-THM-14 on Okuma-Multus-U300 by installing PN-OKU300-HEX-CORE-3414 & kit KIT-OKU300-THM-14. Fasteners torqued to 156 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU350-SPN-1

Target Machine Model: Okuma-Multus-U350 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U350 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Okuma-Multus-U350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U350 (OSP-P500A Aerospace)
- Telemetry Sensor ID: PT-SPN-OKU350-1
- PLC Hex Register Address: DB24.DBW_0x8D68
- Nominal Operating Range: 15 - 69 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 99 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU350-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU350-BRG-CER-3501
- Overhaul Seal & Consumable Kit: KIT-OKU350-SPN-1
- Critical Fastener Torque Value: 179 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U350 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU350-SPN-1 at PLC register 0x8D68.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Okuma-Multus-U350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU350-BRG-CER-3501) with seal kit (KIT-OKU350-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 179 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU350-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U350, technician must log: "Resolved ERR-OKU350-SPN-1 on Okuma-Multus-U350 by installing PN-OKU350-BRG-CER-3501 & kit KIT-OKU350-SPN-1. Fasteners torqued to 179 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU350-ATC-2

Target Machine Model: Okuma-Multus-U350 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U350 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Okuma-Multus-U350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U350 (OSP-P500A Aerospace)
- Telemetry Sensor ID: PX-ATC-OKU350-2
- PLC Hex Register Address: DB24.DBW_0x8D70
- Nominal Operating Range: 18 - 69 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 99 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU350-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU350-CAM-GRIP-3502
- Overhaul Seal & Consumable Kit: KIT-OKU350-ATC-2
- Critical Fastener Torque Value: 188 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U350 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU350-ATC-2 at PLC register 0x8D70.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Okuma-Multus-U350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU350-CAM-GRIP-3502) with seal kit (KIT-OKU350-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 188 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU350-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U350, technician must log: "Resolved ERR-OKU350-ATC-2 on Okuma-Multus-U350 by installing PN-OKU350-CAM-GRIP-3502 & kit KIT-OKU350-ATC-2. Fasteners torqued to 188 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU350-HYD-3

Target Machine Model: Okuma-Multus-U350 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U350 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Okuma-Multus-U350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U350 (OSP-P500A Aerospace)
- Telemetry Sensor ID: PR-HYD-OKU350-3
- PLC Hex Register Address: DB24.DBW_0x8D78
- Nominal Operating Range: 21 - 69 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 99 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU350-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU350-VLV-PROP-3503
- Overhaul Seal & Consumable Kit: KIT-OKU350-HYD-3
- Critical Fastener Torque Value: 197 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U350 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU350-HYD-3 at PLC register 0x8D78.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Okuma-Multus-U350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU350-VLV-PROP-3503) with seal kit (KIT-OKU350-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 197 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU350-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U350, technician must log: "Resolved ERR-OKU350-HYD-3 on Okuma-Multus-U350 by installing PN-OKU350-VLV-PROP-3503 & kit KIT-OKU350-HYD-3. Fasteners torqued to 197 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU350-CLN-4

Target Machine Model: Okuma-Multus-U350 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U350 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Okuma-Multus-U350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U350 (OSP-P500A Aerospace)
- Telemetry Sensor ID: FL-CLN-OKU350-4
- PLC Hex Register Address: DB24.DBW_0x8D80
- Nominal Operating Range: 24 - 69 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 99 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU350-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU350-PMP-PLG-3504
- Overhaul Seal & Consumable Kit: KIT-OKU350-CLN-4
- Critical Fastener Torque Value: 206 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U350 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU350-CLN-4 at PLC register 0x8D80.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Okuma-Multus-U350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU350-PMP-PLG-3504) with seal kit (KIT-OKU350-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 206 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU350-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U350, technician must log: "Resolved ERR-OKU350-CLN-4 on Okuma-Multus-U350 by installing PN-OKU350-PMP-PLG-3504 & kit KIT-OKU350-CLN-4. Fasteners torqued to 206 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU350-XAX-5

Target Machine Model: Okuma-Multus-U350 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U350 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U350 (OSP-P500A Aerospace)
- Telemetry Sensor ID: EN-XAX-OKU350-5
- PLC Hex Register Address: DB24.DBW_0x8D88
- Nominal Operating Range: 27 - 69 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 99 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU350-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU350-LIN-COIL-3505
- Overhaul Seal & Consumable Kit: KIT-OKU350-XAX-5
- Critical Fastener Torque Value: 25 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U350 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU350-XAX-5 at PLC register 0x8D88.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Okuma-Multus-U350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU350-LIN-COIL-3505) with seal kit (KIT-OKU350-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 25 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU350-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U350, technician must log: "Resolved ERR-OKU350-XAX-5 on Okuma-Multus-U350 by installing PN-OKU350-LIN-COIL-3505 & kit KIT-OKU350-XAX-5. Fasteners torqued to 25 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU350-YAX-6

Target Machine Model: Okuma-Multus-U350 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U350 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Okuma-Multus-U350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U350 (OSP-P500A Aerospace)
- Telemetry Sensor ID: EN-YAX-OKU350-6
- PLC Hex Register Address: DB24.DBW_0x8D90
- Nominal Operating Range: 30 - 69 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 99 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU350-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU350-BSW-NUT-3506
- Overhaul Seal & Consumable Kit: KIT-OKU350-YAX-6
- Critical Fastener Torque Value: 34 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U350 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU350-YAX-6 at PLC register 0x8D90.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Okuma-Multus-U350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU350-BSW-NUT-3506) with seal kit (KIT-OKU350-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 34 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU350-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U350, technician must log: "Resolved ERR-OKU350-YAX-6 on Okuma-Multus-U350 by installing PN-OKU350-BSW-NUT-3506 & kit KIT-OKU350-YAX-6. Fasteners torqued to 34 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU350-ZAX-7

Target Machine Model: Okuma-Multus-U350 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U350 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U350 (OSP-P500A Aerospace)
- Telemetry Sensor ID: PR-ZAX-OKU350-7
- PLC Hex Register Address: DB24.DBW_0x8D98
- Nominal Operating Range: 33 - 69 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 99 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU350-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU350-CYL-SEAL-3507
- Overhaul Seal & Consumable Kit: KIT-OKU350-ZAX-7
- Critical Fastener Torque Value: 43 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U350 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU350-ZAX-7 at PLC register 0x8D98.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Okuma-Multus-U350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU350-CYL-SEAL-3507) with seal kit (KIT-OKU350-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 43 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU350-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U350, technician must log: "Resolved ERR-OKU350-ZAX-7 on Okuma-Multus-U350 by installing PN-OKU350-CYL-SEAL-3507 & kit KIT-OKU350-ZAX-7. Fasteners torqued to 43 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU350-BAX-8

Target Machine Model: Okuma-Multus-U350 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U350 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Okuma-Multus-U350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U350 (OSP-P500A Aerospace)
- Telemetry Sensor ID: EN-BAX-OKU350-8
- PLC Hex Register Address: DB24.DBW_0x8DA0
- Nominal Operating Range: 36 - 69 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 99 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU350-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU350-TRQ-STAT-3508
- Overhaul Seal & Consumable Kit: KIT-OKU350-BAX-8
- Critical Fastener Torque Value: 52 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U350 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU350-BAX-8 at PLC register 0x8DA0.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Okuma-Multus-U350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU350-TRQ-STAT-3508) with seal kit (KIT-OKU350-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 52 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU350-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U350, technician must log: "Resolved ERR-OKU350-BAX-8 on Okuma-Multus-U350 by installing PN-OKU350-TRQ-STAT-3508 & kit KIT-OKU350-BAX-8. Fasteners torqued to 52 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU350-CAX-9

Target Machine Model: Okuma-Multus-U350 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U350 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U350 (OSP-P500A Aerospace)
- Telemetry Sensor ID: PR-CAX-OKU350-9
- PLC Hex Register Address: DB24.DBW_0x8DA8
- Nominal Operating Range: 39 - 69 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 99 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU350-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU350-BRK-DISC-3509
- Overhaul Seal & Consumable Kit: KIT-OKU350-CAX-9
- Critical Fastener Torque Value: 61 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U350 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU350-CAX-9 at PLC register 0x8DA8.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Okuma-Multus-U350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU350-BRK-DISC-3509) with seal kit (KIT-OKU350-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 61 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU350-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U350, technician must log: "Resolved ERR-OKU350-CAX-9 on Okuma-Multus-U350 by installing PN-OKU350-BRK-DISC-3509 & kit KIT-OKU350-CAX-9. Fasteners torqued to 61 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU350-OPT-10

Target Machine Model: Okuma-Multus-U350 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U350 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Okuma-Multus-U350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U350 (OSP-P500A Aerospace)
- Telemetry Sensor ID: OP-SCL-OKU350-10
- PLC Hex Register Address: DB24.DBW_0x8DB0
- Nominal Operating Range: 42 - 69 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 99 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU350-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU350-GLASS-RD-3510
- Overhaul Seal & Consumable Kit: KIT-OKU350-OPT-10
- Critical Fastener Torque Value: 70 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U350 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU350-OPT-10 at PLC register 0x8DB0.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Okuma-Multus-U350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU350-GLASS-RD-3510) with seal kit (KIT-OKU350-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 70 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU350-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U350, technician must log: "Resolved ERR-OKU350-OPT-10 on Okuma-Multus-U350 by installing PN-OKU350-GLASS-RD-3510 & kit KIT-OKU350-OPT-10. Fasteners torqued to 70 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU350-PLC-11

Target Machine Model: Okuma-Multus-U350 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U350 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U350 (OSP-P500A Aerospace)
- Telemetry Sensor ID: VT-DRV-OKU350-11
- PLC Hex Register Address: DB24.DBW_0x8DB8
- Nominal Operating Range: 45 - 69 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 99 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU350-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU350-IGBT-MOD-3511
- Overhaul Seal & Consumable Kit: KIT-OKU350-PLC-11
- Critical Fastener Torque Value: 79 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U350 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU350-PLC-11 at PLC register 0x8DB8.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Okuma-Multus-U350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU350-IGBT-MOD-3511) with seal kit (KIT-OKU350-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 79 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU350-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U350, technician must log: "Resolved ERR-OKU350-PLC-11 on Okuma-Multus-U350 by installing PN-OKU350-IGBT-MOD-3511 & kit KIT-OKU350-PLC-11. Fasteners torqued to 79 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU350-LUB-12

Target Machine Model: Okuma-Multus-U350 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U350 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Okuma-Multus-U350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U350 (OSP-P500A Aerospace)
- Telemetry Sensor ID: FL-LUB-OKU350-12
- PLC Hex Register Address: DB24.DBW_0x8DC0
- Nominal Operating Range: 48 - 69 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 99 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU350-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU350-INJ-MET-3512
- Overhaul Seal & Consumable Kit: KIT-OKU350-LUB-12
- Critical Fastener Torque Value: 88 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U350 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU350-LUB-12 at PLC register 0x8DC0.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Okuma-Multus-U350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU350-INJ-MET-3512) with seal kit (KIT-OKU350-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 88 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU350-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U350, technician must log: "Resolved ERR-OKU350-LUB-12 on Okuma-Multus-U350 by installing PN-OKU350-INJ-MET-3512 & kit KIT-OKU350-LUB-12. Fasteners torqued to 88 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU350-VAC-13

Target Machine Model: Okuma-Multus-U350 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U350 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U350 (OSP-P500A Aerospace)
- Telemetry Sensor ID: VC-MNF-OKU350-13
- PLC Hex Register Address: DB24.DBW_0x8DC8
- Nominal Operating Range: 51 - 69 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 99 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU350-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU350-GASK-VIT-3513
- Overhaul Seal & Consumable Kit: KIT-OKU350-VAC-13
- Critical Fastener Torque Value: 97 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U350 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU350-VAC-13 at PLC register 0x8DC8.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Okuma-Multus-U350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU350-GASK-VIT-3513) with seal kit (KIT-OKU350-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 97 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU350-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U350, technician must log: "Resolved ERR-OKU350-VAC-13 on Okuma-Multus-U350 by installing PN-OKU350-GASK-VIT-3513 & kit KIT-OKU350-VAC-13. Fasteners torqued to 97 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU350-THM-14

Target Machine Model: Okuma-Multus-U350 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U350 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Okuma-Multus-U350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U350 (OSP-P500A Aerospace)
- Telemetry Sensor ID: TM-CHL-OKU350-14
- PLC Hex Register Address: DB24.DBW_0x8DD0
- Nominal Operating Range: 54 - 69 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 99 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU350-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU350-HEX-CORE-3514
- Overhaul Seal & Consumable Kit: KIT-OKU350-THM-14
- Critical Fastener Torque Value: 106 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U350 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU350-THM-14 at PLC register 0x8DD0.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Okuma-Multus-U350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU350-HEX-CORE-3514) with seal kit (KIT-OKU350-THM-14).

Step 6: Torque retaining bolts in cross sequence to 106 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU350-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U350, technician must log: "Resolved ERR-OKU350-THM-14 on Okuma-Multus-U350 by installing PN-OKU350-HEX-CORE-3514 & kit KIT-OKU350-THM-14. Fasteners torqued to 106 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU400-SPN-1

Target Machine Model: Okuma-Multus-U400 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U400 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Okuma-Multus-U400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U400 (OSP-P500A Aerospace)
- Telemetry Sensor ID: PT-SPN-OKU400-1
- PLC Hex Register Address: DB25.DBW_0x9088
- Nominal Operating Range: 15 - 70 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 100 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU400-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU400-BRG-CER-3601
- Overhaul Seal & Consumable Kit: KIT-OKU400-SPN-1
- Critical Fastener Torque Value: 129 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U400 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU400-SPN-1 at PLC register 0x9088.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Okuma-Multus-U400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU400-BRG-CER-3601) with seal kit (KIT-OKU400-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 129 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU400-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U400, technician must log: "Resolved ERR-OKU400-SPN-1 on Okuma-Multus-U400 by installing PN-OKU400-BRG-CER-3601 & kit KIT-OKU400-SPN-1. Fasteners torqued to 129 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU400-ATC-2

Target Machine Model: Okuma-Multus-U400 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U400 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Okuma-Multus-U400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U400 (OSP-P500A Aerospace)
- Telemetry Sensor ID: PX-ATC-OKU400-2
- PLC Hex Register Address: DB25.DBW_0x9090
- Nominal Operating Range: 18 - 70 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 100 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU400-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU400-CAM-GRIP-3602
- Overhaul Seal & Consumable Kit: KIT-OKU400-ATC-2
- Critical Fastener Torque Value: 138 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U400 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU400-ATC-2 at PLC register 0x9090.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Okuma-Multus-U400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU400-CAM-GRIP-3602) with seal kit (KIT-OKU400-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 138 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU400-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U400, technician must log: "Resolved ERR-OKU400-ATC-2 on Okuma-Multus-U400 by installing PN-OKU400-CAM-GRIP-3602 & kit KIT-OKU400-ATC-2. Fasteners torqued to 138 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU400-HYD-3

Target Machine Model: Okuma-Multus-U400 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U400 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Okuma-Multus-U400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U400 (OSP-P500A Aerospace)
- Telemetry Sensor ID: PR-HYD-OKU400-3
- PLC Hex Register Address: DB25.DBW_0x9098
- Nominal Operating Range: 21 - 70 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 100 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU400-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU400-VLV-PROP-3603
- Overhaul Seal & Consumable Kit: KIT-OKU400-HYD-3
- Critical Fastener Torque Value: 147 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U400 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU400-HYD-3 at PLC register 0x9098.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Okuma-Multus-U400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU400-VLV-PROP-3603) with seal kit (KIT-OKU400-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 147 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU400-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U400, technician must log: "Resolved ERR-OKU400-HYD-3 on Okuma-Multus-U400 by installing PN-OKU400-VLV-PROP-3603 & kit KIT-OKU400-HYD-3. Fasteners torqued to 147 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU400-CLN-4

Target Machine Model: Okuma-Multus-U400 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U400 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Okuma-Multus-U400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U400 (OSP-P500A Aerospace)
- Telemetry Sensor ID: FL-CLN-OKU400-4
- PLC Hex Register Address: DB25.DBW_0x90A0
- Nominal Operating Range: 24 - 70 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 100 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU400-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU400-PMP-PLG-3604
- Overhaul Seal & Consumable Kit: KIT-OKU400-CLN-4
- Critical Fastener Torque Value: 156 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U400 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU400-CLN-4 at PLC register 0x90A0.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Okuma-Multus-U400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU400-PMP-PLG-3604) with seal kit (KIT-OKU400-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 156 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU400-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U400, technician must log: "Resolved ERR-OKU400-CLN-4 on Okuma-Multus-U400 by installing PN-OKU400-PMP-PLG-3604 & kit KIT-OKU400-CLN-4. Fasteners torqued to 156 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU400-XAX-5

Target Machine Model: Okuma-Multus-U400 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U400 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U400 (OSP-P500A Aerospace)
- Telemetry Sensor ID: EN-XAX-OKU400-5
- PLC Hex Register Address: DB25.DBW_0x90A8
- Nominal Operating Range: 27 - 70 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 100 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU400-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU400-LIN-COIL-3605
- Overhaul Seal & Consumable Kit: KIT-OKU400-XAX-5
- Critical Fastener Torque Value: 165 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U400 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU400-XAX-5 at PLC register 0x90A8.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Okuma-Multus-U400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU400-LIN-COIL-3605) with seal kit (KIT-OKU400-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 165 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU400-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U400, technician must log: "Resolved ERR-OKU400-XAX-5 on Okuma-Multus-U400 by installing PN-OKU400-LIN-COIL-3605 & kit KIT-OKU400-XAX-5. Fasteners torqued to 165 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU400-YAX-6

Target Machine Model: Okuma-Multus-U400 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U400 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Okuma-Multus-U400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U400 (OSP-P500A Aerospace)
- Telemetry Sensor ID: EN-YAX-OKU400-6
- PLC Hex Register Address: DB25.DBW_0x90B0
- Nominal Operating Range: 30 - 70 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 100 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU400-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU400-BSW-NUT-3606
- Overhaul Seal & Consumable Kit: KIT-OKU400-YAX-6
- Critical Fastener Torque Value: 174 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U400 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU400-YAX-6 at PLC register 0x90B0.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Okuma-Multus-U400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU400-BSW-NUT-3606) with seal kit (KIT-OKU400-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 174 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU400-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U400, technician must log: "Resolved ERR-OKU400-YAX-6 on Okuma-Multus-U400 by installing PN-OKU400-BSW-NUT-3606 & kit KIT-OKU400-YAX-6. Fasteners torqued to 174 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU400-ZAX-7

Target Machine Model: Okuma-Multus-U400 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U400 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U400 (OSP-P500A Aerospace)
- Telemetry Sensor ID: PR-ZAX-OKU400-7
- PLC Hex Register Address: DB25.DBW_0x90B8
- Nominal Operating Range: 33 - 70 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 100 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU400-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU400-CYL-SEAL-3607
- Overhaul Seal & Consumable Kit: KIT-OKU400-ZAX-7
- Critical Fastener Torque Value: 183 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U400 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU400-ZAX-7 at PLC register 0x90B8.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Okuma-Multus-U400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU400-CYL-SEAL-3607) with seal kit (KIT-OKU400-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 183 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU400-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U400, technician must log: "Resolved ERR-OKU400-ZAX-7 on Okuma-Multus-U400 by installing PN-OKU400-CYL-SEAL-3607 & kit KIT-OKU400-ZAX-7. Fasteners torqued to 183 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU400-BAX-8

Target Machine Model: Okuma-Multus-U400 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U400 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Okuma-Multus-U400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U400 (OSP-P500A Aerospace)
- Telemetry Sensor ID: EN-BAX-OKU400-8
- PLC Hex Register Address: DB25.DBW_0x90C0
- Nominal Operating Range: 36 - 70 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 100 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU400-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU400-TRQ-STAT-3608
- Overhaul Seal & Consumable Kit: KIT-OKU400-BAX-8
- Critical Fastener Torque Value: 192 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U400 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU400-BAX-8 at PLC register 0x90C0.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Okuma-Multus-U400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU400-TRQ-STAT-3608) with seal kit (KIT-OKU400-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 192 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU400-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U400, technician must log: "Resolved ERR-OKU400-BAX-8 on Okuma-Multus-U400 by installing PN-OKU400-TRQ-STAT-3608 & kit KIT-OKU400-BAX-8. Fasteners torqued to 192 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU400-CAX-9

Target Machine Model: Okuma-Multus-U400 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U400 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U400 (OSP-P500A Aerospace)
- Telemetry Sensor ID: PR-CAX-OKU400-9
- PLC Hex Register Address: DB25.DBW_0x90C8
- Nominal Operating Range: 39 - 70 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 100 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU400-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU400-BRK-DISC-3609
- Overhaul Seal & Consumable Kit: KIT-OKU400-CAX-9
- Critical Fastener Torque Value: 201 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U400 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU400-CAX-9 at PLC register 0x90C8.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Okuma-Multus-U400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU400-BRK-DISC-3609) with seal kit (KIT-OKU400-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 201 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU400-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U400, technician must log: "Resolved ERR-OKU400-CAX-9 on Okuma-Multus-U400 by installing PN-OKU400-BRK-DISC-3609 & kit KIT-OKU400-CAX-9. Fasteners torqued to 201 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU400-OPT-10

Target Machine Model: Okuma-Multus-U400 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U400 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Okuma-Multus-U400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U400 (OSP-P500A Aerospace)
- Telemetry Sensor ID: OP-SCL-OKU400-10
- PLC Hex Register Address: DB25.DBW_0x90D0
- Nominal Operating Range: 42 - 70 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 100 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU400-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU400-GLASS-RD-3610
- Overhaul Seal & Consumable Kit: KIT-OKU400-OPT-10
- Critical Fastener Torque Value: 20 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U400 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU400-OPT-10 at PLC register 0x90D0.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Okuma-Multus-U400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU400-GLASS-RD-3610) with seal kit (KIT-OKU400-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 20 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU400-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U400, technician must log: "Resolved ERR-OKU400-OPT-10 on Okuma-Multus-U400 by installing PN-OKU400-GLASS-RD-3610 & kit KIT-OKU400-OPT-10. Fasteners torqued to 20 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU400-PLC-11

Target Machine Model: Okuma-Multus-U400 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U400 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U400 (OSP-P500A Aerospace)
- Telemetry Sensor ID: VT-DRV-OKU400-11
- PLC Hex Register Address: DB25.DBW_0x90D8
- Nominal Operating Range: 45 - 70 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 100 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU400-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU400-IGBT-MOD-3611
- Overhaul Seal & Consumable Kit: KIT-OKU400-PLC-11
- Critical Fastener Torque Value: 29 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U400 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU400-PLC-11 at PLC register 0x90D8.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Okuma-Multus-U400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU400-IGBT-MOD-3611) with seal kit (KIT-OKU400-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 29 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU400-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U400, technician must log: "Resolved ERR-OKU400-PLC-11 on Okuma-Multus-U400 by installing PN-OKU400-IGBT-MOD-3611 & kit KIT-OKU400-PLC-11. Fasteners torqued to 29 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU400-LUB-12

Target Machine Model: Okuma-Multus-U400 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U400 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Okuma-Multus-U400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U400 (OSP-P500A Aerospace)
- Telemetry Sensor ID: FL-LUB-OKU400-12
- PLC Hex Register Address: DB25.DBW_0x90E0
- Nominal Operating Range: 48 - 70 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 100 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU400-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU400-INJ-MET-3612
- Overhaul Seal & Consumable Kit: KIT-OKU400-LUB-12
- Critical Fastener Torque Value: 38 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U400 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU400-LUB-12 at PLC register 0x90E0.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Okuma-Multus-U400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU400-INJ-MET-3612) with seal kit (KIT-OKU400-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 38 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU400-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U400, technician must log: "Resolved ERR-OKU400-LUB-12 on Okuma-Multus-U400 by installing PN-OKU400-INJ-MET-3612 & kit KIT-OKU400-LUB-12. Fasteners torqued to 38 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU400-VAC-13

Target Machine Model: Okuma-Multus-U400 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U400 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U400 (OSP-P500A Aerospace)
- Telemetry Sensor ID: VC-MNF-OKU400-13
- PLC Hex Register Address: DB25.DBW_0x90E8
- Nominal Operating Range: 51 - 70 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 100 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU400-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU400-GASK-VIT-3613
- Overhaul Seal & Consumable Kit: KIT-OKU400-VAC-13
- Critical Fastener Torque Value: 47 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U400 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU400-VAC-13 at PLC register 0x90E8.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Okuma-Multus-U400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU400-GASK-VIT-3613) with seal kit (KIT-OKU400-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 47 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU400-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U400, technician must log: "Resolved ERR-OKU400-VAC-13 on Okuma-Multus-U400 by installing PN-OKU400-GASK-VIT-3613 & kit KIT-OKU400-VAC-13. Fasteners torqued to 47 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU400-THM-14

Target Machine Model: Okuma-Multus-U400 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U400 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Okuma-Multus-U400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U400 (OSP-P500A Aerospace)
- Telemetry Sensor ID: TM-CHL-OKU400-14
- PLC Hex Register Address: DB25.DBW_0x90F0
- Nominal Operating Range: 54 - 70 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 100 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU400-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU400-HEX-CORE-3614
- Overhaul Seal & Consumable Kit: KIT-OKU400-THM-14
- Critical Fastener Torque Value: 56 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U400 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU400-THM-14 at PLC register 0x90F0.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Okuma-Multus-U400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU400-HEX-CORE-3614) with seal kit (KIT-OKU400-THM-14).

Step 6: Torque retaining bolts in cross sequence to 56 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU400-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U400, technician must log: "Resolved ERR-OKU400-THM-14 on Okuma-Multus-U400 by installing PN-OKU400-HEX-CORE-3614 & kit KIT-OKU400-THM-14. Fasteners torqued to 56 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU450-SPN-1

Target Machine Model: Okuma-Multus-U450 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U450 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Okuma-Multus-U450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U450 (OSP-P500A Aerospace)
- Telemetry Sensor ID: PT-SPN-OKU450-1
- PLC Hex Register Address: DB26.DBW_0x93A8
- Nominal Operating Range: 15 - 71 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 101 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU450-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU450-BRG-CER-3701
- Overhaul Seal & Consumable Kit: KIT-OKU450-SPN-1
- Critical Fastener Torque Value: 79 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U450 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU450-SPN-1 at PLC register 0x93A8.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Okuma-Multus-U450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU450-BRG-CER-3701) with seal kit (KIT-OKU450-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 79 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU450-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U450, technician must log: "Resolved ERR-OKU450-SPN-1 on Okuma-Multus-U450 by installing PN-OKU450-BRG-CER-3701 & kit KIT-OKU450-SPN-1. Fasteners torqued to 79 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU450-ATC-2

Target Machine Model: Okuma-Multus-U450 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U450 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Okuma-Multus-U450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U450 (OSP-P500A Aerospace)
- Telemetry Sensor ID: PX-ATC-OKU450-2
- PLC Hex Register Address: DB26.DBW_0x93B0
- Nominal Operating Range: 18 - 71 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 101 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU450-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU450-CAM-GRIP-3702
- Overhaul Seal & Consumable Kit: KIT-OKU450-ATC-2
- Critical Fastener Torque Value: 88 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U450 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU450-ATC-2 at PLC register 0x93B0.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Okuma-Multus-U450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU450-CAM-GRIP-3702) with seal kit (KIT-OKU450-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 88 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU450-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U450, technician must log: "Resolved ERR-OKU450-ATC-2 on Okuma-Multus-U450 by installing PN-OKU450-CAM-GRIP-3702 & kit KIT-OKU450-ATC-2. Fasteners torqued to 88 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU450-HYD-3

Target Machine Model: Okuma-Multus-U450 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U450 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Okuma-Multus-U450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U450 (OSP-P500A Aerospace)
- Telemetry Sensor ID: PR-HYD-OKU450-3
- PLC Hex Register Address: DB26.DBW_0x93B8
- Nominal Operating Range: 21 - 71 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 101 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU450-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU450-VLV-PROP-3703
- Overhaul Seal & Consumable Kit: KIT-OKU450-HYD-3
- Critical Fastener Torque Value: 97 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U450 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU450-HYD-3 at PLC register 0x93B8.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Okuma-Multus-U450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU450-VLV-PROP-3703) with seal kit (KIT-OKU450-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 97 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU450-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U450, technician must log: "Resolved ERR-OKU450-HYD-3 on Okuma-Multus-U450 by installing PN-OKU450-VLV-PROP-3703 & kit KIT-OKU450-HYD-3. Fasteners torqued to 97 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU450-CLN-4

Target Machine Model: Okuma-Multus-U450 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U450 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Okuma-Multus-U450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U450 (OSP-P500A Aerospace)
- Telemetry Sensor ID: FL-CLN-OKU450-4
- PLC Hex Register Address: DB26.DBW_0x93C0
- Nominal Operating Range: 24 - 71 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 101 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU450-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU450-PMP-PLG-3704
- Overhaul Seal & Consumable Kit: KIT-OKU450-CLN-4
- Critical Fastener Torque Value: 106 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U450 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU450-CLN-4 at PLC register 0x93C0.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Okuma-Multus-U450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU450-PMP-PLG-3704) with seal kit (KIT-OKU450-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 106 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU450-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U450, technician must log: "Resolved ERR-OKU450-CLN-4 on Okuma-Multus-U450 by installing PN-OKU450-PMP-PLG-3704 & kit KIT-OKU450-CLN-4. Fasteners torqued to 106 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU450-XAX-5

Target Machine Model: Okuma-Multus-U450 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U450 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U450 (OSP-P500A Aerospace)
- Telemetry Sensor ID: EN-XAX-OKU450-5
- PLC Hex Register Address: DB26.DBW_0x93C8
- Nominal Operating Range: 27 - 71 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 101 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU450-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU450-LIN-COIL-3705
- Overhaul Seal & Consumable Kit: KIT-OKU450-XAX-5
- Critical Fastener Torque Value: 115 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U450 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU450-XAX-5 at PLC register 0x93C8.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Okuma-Multus-U450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU450-LIN-COIL-3705) with seal kit (KIT-OKU450-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 115 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU450-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U450, technician must log: "Resolved ERR-OKU450-XAX-5 on Okuma-Multus-U450 by installing PN-OKU450-LIN-COIL-3705 & kit KIT-OKU450-XAX-5. Fasteners torqued to 115 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU450-YAX-6

Target Machine Model: Okuma-Multus-U450 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U450 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Okuma-Multus-U450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U450 (OSP-P500A Aerospace)
- Telemetry Sensor ID: EN-YAX-OKU450-6
- PLC Hex Register Address: DB26.DBW_0x93D0
- Nominal Operating Range: 30 - 71 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 101 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU450-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU450-BSW-NUT-3706
- Overhaul Seal & Consumable Kit: KIT-OKU450-YAX-6
- Critical Fastener Torque Value: 124 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U450 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU450-YAX-6 at PLC register 0x93D0.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Okuma-Multus-U450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU450-BSW-NUT-3706) with seal kit (KIT-OKU450-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 124 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU450-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U450, technician must log: "Resolved ERR-OKU450-YAX-6 on Okuma-Multus-U450 by installing PN-OKU450-BSW-NUT-3706 & kit KIT-OKU450-YAX-6. Fasteners torqued to 124 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU450-ZAX-7

Target Machine Model: Okuma-Multus-U450 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U450 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U450 (OSP-P500A Aerospace)
- Telemetry Sensor ID: PR-ZAX-OKU450-7
- PLC Hex Register Address: DB26.DBW_0x93D8
- Nominal Operating Range: 33 - 71 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 101 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU450-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU450-CYL-SEAL-3707
- Overhaul Seal & Consumable Kit: KIT-OKU450-ZAX-7
- Critical Fastener Torque Value: 133 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U450 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU450-ZAX-7 at PLC register 0x93D8.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Okuma-Multus-U450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU450-CYL-SEAL-3707) with seal kit (KIT-OKU450-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 133 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU450-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U450, technician must log: "Resolved ERR-OKU450-ZAX-7 on Okuma-Multus-U450 by installing PN-OKU450-CYL-SEAL-3707 & kit KIT-OKU450-ZAX-7. Fasteners torqued to 133 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU450-BAX-8

Target Machine Model: Okuma-Multus-U450 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U450 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Okuma-Multus-U450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U450 (OSP-P500A Aerospace)
- Telemetry Sensor ID: EN-BAX-OKU450-8
- PLC Hex Register Address: DB26.DBW_0x93E0
- Nominal Operating Range: 36 - 71 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 101 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU450-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU450-TRQ-STAT-3708
- Overhaul Seal & Consumable Kit: KIT-OKU450-BAX-8
- Critical Fastener Torque Value: 142 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U450 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU450-BAX-8 at PLC register 0x93E0.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Okuma-Multus-U450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU450-TRQ-STAT-3708) with seal kit (KIT-OKU450-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 142 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU450-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U450, technician must log: "Resolved ERR-OKU450-BAX-8 on Okuma-Multus-U450 by installing PN-OKU450-TRQ-STAT-3708 & kit KIT-OKU450-BAX-8. Fasteners torqued to 142 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU450-CAX-9

Target Machine Model: Okuma-Multus-U450 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U450 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U450 (OSP-P500A Aerospace)
- Telemetry Sensor ID: PR-CAX-OKU450-9
- PLC Hex Register Address: DB26.DBW_0x93E8
- Nominal Operating Range: 39 - 71 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 101 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU450-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU450-BRK-DISC-3709
- Overhaul Seal & Consumable Kit: KIT-OKU450-CAX-9
- Critical Fastener Torque Value: 151 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U450 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU450-CAX-9 at PLC register 0x93E8.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Okuma-Multus-U450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU450-BRK-DISC-3709) with seal kit (KIT-OKU450-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 151 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU450-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U450, technician must log: "Resolved ERR-OKU450-CAX-9 on Okuma-Multus-U450 by installing PN-OKU450-BRK-DISC-3709 & kit KIT-OKU450-CAX-9. Fasteners torqued to 151 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU450-OPT-10

Target Machine Model: Okuma-Multus-U450 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U450 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Okuma-Multus-U450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U450 (OSP-P500A Aerospace)
- Telemetry Sensor ID: OP-SCL-OKU450-10
- PLC Hex Register Address: DB26.DBW_0x93F0
- Nominal Operating Range: 42 - 71 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 101 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU450-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU450-GLASS-RD-3710
- Overhaul Seal & Consumable Kit: KIT-OKU450-OPT-10
- Critical Fastener Torque Value: 160 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U450 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU450-OPT-10 at PLC register 0x93F0.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Okuma-Multus-U450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU450-GLASS-RD-3710) with seal kit (KIT-OKU450-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 160 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU450-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U450, technician must log: "Resolved ERR-OKU450-OPT-10 on Okuma-Multus-U450 by installing PN-OKU450-GLASS-RD-3710 & kit KIT-OKU450-OPT-10. Fasteners torqued to 160 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU450-PLC-11

Target Machine Model: Okuma-Multus-U450 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U450 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U450 (OSP-P500A Aerospace)
- Telemetry Sensor ID: VT-DRV-OKU450-11
- PLC Hex Register Address: DB26.DBW_0x93F8
- Nominal Operating Range: 45 - 71 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 101 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU450-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU450-IGBT-MOD-3711
- Overhaul Seal & Consumable Kit: KIT-OKU450-PLC-11
- Critical Fastener Torque Value: 169 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U450 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU450-PLC-11 at PLC register 0x93F8.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Okuma-Multus-U450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU450-IGBT-MOD-3711) with seal kit (KIT-OKU450-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 169 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU450-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U450, technician must log: "Resolved ERR-OKU450-PLC-11 on Okuma-Multus-U450 by installing PN-OKU450-IGBT-MOD-3711 & kit KIT-OKU450-PLC-11. Fasteners torqued to 169 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU450-LUB-12

Target Machine Model: Okuma-Multus-U450 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U450 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Okuma-Multus-U450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U450 (OSP-P500A Aerospace)
- Telemetry Sensor ID: FL-LUB-OKU450-12
- PLC Hex Register Address: DB26.DBW_0x9400
- Nominal Operating Range: 48 - 71 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 101 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU450-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU450-INJ-MET-3712
- Overhaul Seal & Consumable Kit: KIT-OKU450-LUB-12
- Critical Fastener Torque Value: 178 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U450 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU450-LUB-12 at PLC register 0x9400.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Okuma-Multus-U450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU450-INJ-MET-3712) with seal kit (KIT-OKU450-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 178 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU450-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U450, technician must log: "Resolved ERR-OKU450-LUB-12 on Okuma-Multus-U450 by installing PN-OKU450-INJ-MET-3712 & kit KIT-OKU450-LUB-12. Fasteners torqued to 178 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU450-VAC-13

Target Machine Model: Okuma-Multus-U450 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U450 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U450 (OSP-P500A Aerospace)
- Telemetry Sensor ID: VC-MNF-OKU450-13
- PLC Hex Register Address: DB26.DBW_0x9408
- Nominal Operating Range: 51 - 71 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 101 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU450-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU450-GASK-VIT-3713
- Overhaul Seal & Consumable Kit: KIT-OKU450-VAC-13
- Critical Fastener Torque Value: 187 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U450 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU450-VAC-13 at PLC register 0x9408.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Okuma-Multus-U450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU450-GASK-VIT-3713) with seal kit (KIT-OKU450-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 187 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU450-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U450, technician must log: "Resolved ERR-OKU450-VAC-13 on Okuma-Multus-U450 by installing PN-OKU450-GASK-VIT-3713 & kit KIT-OKU450-VAC-13. Fasteners torqued to 187 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU450-THM-14

Target Machine Model: Okuma-Multus-U450 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U450 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Okuma-Multus-U450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U450 (OSP-P500A Aerospace)
- Telemetry Sensor ID: TM-CHL-OKU450-14
- PLC Hex Register Address: DB26.DBW_0x9410
- Nominal Operating Range: 54 - 71 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 101 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU450-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU450-HEX-CORE-3714
- Overhaul Seal & Consumable Kit: KIT-OKU450-THM-14
- Critical Fastener Torque Value: 196 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U450 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU450-THM-14 at PLC register 0x9410.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Okuma-Multus-U450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU450-HEX-CORE-3714) with seal kit (KIT-OKU450-THM-14).

Step 6: Torque retaining bolts in cross sequence to 196 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU450-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U450, technician must log: "Resolved ERR-OKU450-THM-14 on Okuma-Multus-U450 by installing PN-OKU450-HEX-CORE-3714 & kit KIT-OKU450-THM-14. Fasteners torqued to 196 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU500-SPN-1

Target Machine Model: Okuma-Multus-U500 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U500 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Okuma-Multus-U500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U500 (OSP-P500A Aerospace)
- Telemetry Sensor ID: PT-SPN-OKU500-1
- PLC Hex Register Address: DB27.DBW_0x96C8
- Nominal Operating Range: 15 - 72 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 102 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU500-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU500-BRG-CER-3801
- Overhaul Seal & Consumable Kit: KIT-OKU500-SPN-1
- Critical Fastener Torque Value: 29 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U500 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU500-SPN-1 at PLC register 0x96C8.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Okuma-Multus-U500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU500-BRG-CER-3801) with seal kit (KIT-OKU500-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 29 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU500-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U500, technician must log: "Resolved ERR-OKU500-SPN-1 on Okuma-Multus-U500 by installing PN-OKU500-BRG-CER-3801 & kit KIT-OKU500-SPN-1. Fasteners torqued to 29 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU500-ATC-2

Target Machine Model: Okuma-Multus-U500 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U500 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Okuma-Multus-U500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U500 (OSP-P500A Aerospace)
- Telemetry Sensor ID: PX-ATC-OKU500-2
- PLC Hex Register Address: DB27.DBW_0x96D0
- Nominal Operating Range: 18 - 72 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 102 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU500-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU500-CAM-GRIP-3802
- Overhaul Seal & Consumable Kit: KIT-OKU500-ATC-2
- Critical Fastener Torque Value: 38 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U500 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU500-ATC-2 at PLC register 0x96D0.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Okuma-Multus-U500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU500-CAM-GRIP-3802) with seal kit (KIT-OKU500-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 38 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU500-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U500, technician must log: "Resolved ERR-OKU500-ATC-2 on Okuma-Multus-U500 by installing PN-OKU500-CAM-GRIP-3802 & kit KIT-OKU500-ATC-2. Fasteners torqued to 38 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU500-HYD-3

Target Machine Model: Okuma-Multus-U500 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U500 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Okuma-Multus-U500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U500 (OSP-P500A Aerospace)
- Telemetry Sensor ID: PR-HYD-OKU500-3
- PLC Hex Register Address: DB27.DBW_0x96D8
- Nominal Operating Range: 21 - 72 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 102 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU500-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU500-VLV-PROP-3803
- Overhaul Seal & Consumable Kit: KIT-OKU500-HYD-3
- Critical Fastener Torque Value: 47 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U500 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU500-HYD-3 at PLC register 0x96D8.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Okuma-Multus-U500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU500-VLV-PROP-3803) with seal kit (KIT-OKU500-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 47 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU500-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U500, technician must log: "Resolved ERR-OKU500-HYD-3 on Okuma-Multus-U500 by installing PN-OKU500-VLV-PROP-3803 & kit KIT-OKU500-HYD-3. Fasteners torqued to 47 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU500-CLN-4

Target Machine Model: Okuma-Multus-U500 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U500 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Okuma-Multus-U500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U500 (OSP-P500A Aerospace)
- Telemetry Sensor ID: FL-CLN-OKU500-4
- PLC Hex Register Address: DB27.DBW_0x96E0
- Nominal Operating Range: 24 - 72 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 102 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU500-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU500-PMP-PLG-3804
- Overhaul Seal & Consumable Kit: KIT-OKU500-CLN-4
- Critical Fastener Torque Value: 56 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U500 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU500-CLN-4 at PLC register 0x96E0.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Okuma-Multus-U500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU500-PMP-PLG-3804) with seal kit (KIT-OKU500-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 56 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU500-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U500, technician must log: "Resolved ERR-OKU500-CLN-4 on Okuma-Multus-U500 by installing PN-OKU500-PMP-PLG-3804 & kit KIT-OKU500-CLN-4. Fasteners torqued to 56 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU500-XAX-5

Target Machine Model: Okuma-Multus-U500 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U500 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U500 (OSP-P500A Aerospace)
- Telemetry Sensor ID: EN-XAX-OKU500-5
- PLC Hex Register Address: DB27.DBW_0x96E8
- Nominal Operating Range: 27 - 72 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 102 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU500-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU500-LIN-COIL-3805
- Overhaul Seal & Consumable Kit: KIT-OKU500-XAX-5
- Critical Fastener Torque Value: 65 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U500 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU500-XAX-5 at PLC register 0x96E8.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Okuma-Multus-U500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU500-LIN-COIL-3805) with seal kit (KIT-OKU500-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 65 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU500-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U500, technician must log: "Resolved ERR-OKU500-XAX-5 on Okuma-Multus-U500 by installing PN-OKU500-LIN-COIL-3805 & kit KIT-OKU500-XAX-5. Fasteners torqued to 65 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU500-YAX-6

Target Machine Model: Okuma-Multus-U500 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U500 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Okuma-Multus-U500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U500 (OSP-P500A Aerospace)
- Telemetry Sensor ID: EN-YAX-OKU500-6
- PLC Hex Register Address: DB27.DBW_0x96F0
- Nominal Operating Range: 30 - 72 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 102 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU500-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU500-BSW-NUT-3806
- Overhaul Seal & Consumable Kit: KIT-OKU500-YAX-6
- Critical Fastener Torque Value: 74 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U500 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU500-YAX-6 at PLC register 0x96F0.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Okuma-Multus-U500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU500-BSW-NUT-3806) with seal kit (KIT-OKU500-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 74 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU500-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U500, technician must log: "Resolved ERR-OKU500-YAX-6 on Okuma-Multus-U500 by installing PN-OKU500-BSW-NUT-3806 & kit KIT-OKU500-YAX-6. Fasteners torqued to 74 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU500-ZAX-7

Target Machine Model: Okuma-Multus-U500 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U500 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U500 (OSP-P500A Aerospace)
- Telemetry Sensor ID: PR-ZAX-OKU500-7
- PLC Hex Register Address: DB27.DBW_0x96F8
- Nominal Operating Range: 33 - 72 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 102 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU500-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU500-CYL-SEAL-3807
- Overhaul Seal & Consumable Kit: KIT-OKU500-ZAX-7
- Critical Fastener Torque Value: 83 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U500 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU500-ZAX-7 at PLC register 0x96F8.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Okuma-Multus-U500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU500-CYL-SEAL-3807) with seal kit (KIT-OKU500-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 83 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU500-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U500, technician must log: "Resolved ERR-OKU500-ZAX-7 on Okuma-Multus-U500 by installing PN-OKU500-CYL-SEAL-3807 & kit KIT-OKU500-ZAX-7. Fasteners torqued to 83 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU500-BAX-8

Target Machine Model: Okuma-Multus-U500 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U500 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Okuma-Multus-U500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U500 (OSP-P500A Aerospace)
- Telemetry Sensor ID: EN-BAX-OKU500-8
- PLC Hex Register Address: DB27.DBW_0x9700
- Nominal Operating Range: 36 - 72 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 102 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU500-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU500-TRQ-STAT-3808
- Overhaul Seal & Consumable Kit: KIT-OKU500-BAX-8
- Critical Fastener Torque Value: 92 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U500 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU500-BAX-8 at PLC register 0x9700.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Okuma-Multus-U500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU500-TRQ-STAT-3808) with seal kit (KIT-OKU500-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 92 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU500-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U500, technician must log: "Resolved ERR-OKU500-BAX-8 on Okuma-Multus-U500 by installing PN-OKU500-TRQ-STAT-3808 & kit KIT-OKU500-BAX-8. Fasteners torqued to 92 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU500-CAX-9

Target Machine Model: Okuma-Multus-U500 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U500 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U500 (OSP-P500A Aerospace)
- Telemetry Sensor ID: PR-CAX-OKU500-9
- PLC Hex Register Address: DB27.DBW_0x9708
- Nominal Operating Range: 39 - 72 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 102 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU500-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU500-BRK-DISC-3809
- Overhaul Seal & Consumable Kit: KIT-OKU500-CAX-9
- Critical Fastener Torque Value: 101 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U500 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU500-CAX-9 at PLC register 0x9708.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Okuma-Multus-U500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU500-BRK-DISC-3809) with seal kit (KIT-OKU500-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 101 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU500-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U500, technician must log: "Resolved ERR-OKU500-CAX-9 on Okuma-Multus-U500 by installing PN-OKU500-BRK-DISC-3809 & kit KIT-OKU500-CAX-9. Fasteners torqued to 101 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU500-OPT-10

Target Machine Model: Okuma-Multus-U500 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U500 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Okuma-Multus-U500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U500 (OSP-P500A Aerospace)
- Telemetry Sensor ID: OP-SCL-OKU500-10
- PLC Hex Register Address: DB27.DBW_0x9710
- Nominal Operating Range: 42 - 72 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 102 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU500-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU500-GLASS-RD-3810
- Overhaul Seal & Consumable Kit: KIT-OKU500-OPT-10
- Critical Fastener Torque Value: 110 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U500 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU500-OPT-10 at PLC register 0x9710.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Okuma-Multus-U500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU500-GLASS-RD-3810) with seal kit (KIT-OKU500-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 110 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU500-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U500, technician must log: "Resolved ERR-OKU500-OPT-10 on Okuma-Multus-U500 by installing PN-OKU500-GLASS-RD-3810 & kit KIT-OKU500-OPT-10. Fasteners torqued to 110 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU500-PLC-11

Target Machine Model: Okuma-Multus-U500 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U500 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U500 (OSP-P500A Aerospace)
- Telemetry Sensor ID: VT-DRV-OKU500-11
- PLC Hex Register Address: DB27.DBW_0x9718
- Nominal Operating Range: 45 - 72 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 102 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU500-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU500-IGBT-MOD-3811
- Overhaul Seal & Consumable Kit: KIT-OKU500-PLC-11
- Critical Fastener Torque Value: 119 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U500 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU500-PLC-11 at PLC register 0x9718.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Okuma-Multus-U500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU500-IGBT-MOD-3811) with seal kit (KIT-OKU500-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 119 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU500-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U500, technician must log: "Resolved ERR-OKU500-PLC-11 on Okuma-Multus-U500 by installing PN-OKU500-IGBT-MOD-3811 & kit KIT-OKU500-PLC-11. Fasteners torqued to 119 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU500-LUB-12

Target Machine Model: Okuma-Multus-U500 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U500 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Okuma-Multus-U500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U500 (OSP-P500A Aerospace)
- Telemetry Sensor ID: FL-LUB-OKU500-12
- PLC Hex Register Address: DB27.DBW_0x9720
- Nominal Operating Range: 48 - 72 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 102 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU500-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU500-INJ-MET-3812
- Overhaul Seal & Consumable Kit: KIT-OKU500-LUB-12
- Critical Fastener Torque Value: 128 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U500 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU500-LUB-12 at PLC register 0x9720.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Okuma-Multus-U500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU500-INJ-MET-3812) with seal kit (KIT-OKU500-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 128 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU500-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U500, technician must log: "Resolved ERR-OKU500-LUB-12 on Okuma-Multus-U500 by installing PN-OKU500-INJ-MET-3812 & kit KIT-OKU500-LUB-12. Fasteners torqued to 128 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU500-VAC-13

Target Machine Model: Okuma-Multus-U500 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U500 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U500 (OSP-P500A Aerospace)
- Telemetry Sensor ID: VC-MNF-OKU500-13
- PLC Hex Register Address: DB27.DBW_0x9728
- Nominal Operating Range: 51 - 72 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 102 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU500-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU500-GASK-VIT-3813
- Overhaul Seal & Consumable Kit: KIT-OKU500-VAC-13
- Critical Fastener Torque Value: 137 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U500 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU500-VAC-13 at PLC register 0x9728.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Okuma-Multus-U500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU500-GASK-VIT-3813) with seal kit (KIT-OKU500-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 137 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU500-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U500, technician must log: "Resolved ERR-OKU500-VAC-13 on Okuma-Multus-U500 by installing PN-OKU500-GASK-VIT-3813 & kit KIT-OKU500-VAC-13. Fasteners torqued to 137 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU500-THM-14

Target Machine Model: Okuma-Multus-U500 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U500 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Okuma-Multus-U500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U500 (OSP-P500A Aerospace)
- Telemetry Sensor ID: TM-CHL-OKU500-14
- PLC Hex Register Address: DB27.DBW_0x9730
- Nominal Operating Range: 54 - 72 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 102 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU500-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU500-HEX-CORE-3814
- Overhaul Seal & Consumable Kit: KIT-OKU500-THM-14
- Critical Fastener Torque Value: 146 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U500 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU500-THM-14 at PLC register 0x9730.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Okuma-Multus-U500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU500-HEX-CORE-3814) with seal kit (KIT-OKU500-THM-14).

Step 6: Torque retaining bolts in cross sequence to 146 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU500-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U500, technician must log: "Resolved ERR-OKU500-THM-14 on Okuma-Multus-U500 by installing PN-OKU500-HEX-CORE-3814 & kit KIT-OKU500-THM-14. Fasteners torqued to 146 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU550-SPN-1

Target Machine Model: Okuma-Multus-U550 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U550 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Okuma-Multus-U550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U550 (OSP-P500A Aerospace)
- Telemetry Sensor ID: PT-SPN-OKU550-1
- PLC Hex Register Address: DB28.DBW_0x99E8
- Nominal Operating Range: 15 - 73 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 103 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU550-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU550-BRG-CER-3901
- Overhaul Seal & Consumable Kit: KIT-OKU550-SPN-1
- Critical Fastener Torque Value: 169 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U550 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU550-SPN-1 at PLC register 0x99E8.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Okuma-Multus-U550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU550-BRG-CER-3901) with seal kit (KIT-OKU550-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 169 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU550-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U550, technician must log: "Resolved ERR-OKU550-SPN-1 on Okuma-Multus-U550 by installing PN-OKU550-BRG-CER-3901 & kit KIT-OKU550-SPN-1. Fasteners torqued to 169 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU550-ATC-2

Target Machine Model: Okuma-Multus-U550 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U550 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Okuma-Multus-U550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U550 (OSP-P500A Aerospace)
- Telemetry Sensor ID: PX-ATC-OKU550-2
- PLC Hex Register Address: DB28.DBW_0x99F0
- Nominal Operating Range: 18 - 73 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 103 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU550-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU550-CAM-GRIP-3902
- Overhaul Seal & Consumable Kit: KIT-OKU550-ATC-2
- Critical Fastener Torque Value: 178 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U550 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU550-ATC-2 at PLC register 0x99F0.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Okuma-Multus-U550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU550-CAM-GRIP-3902) with seal kit (KIT-OKU550-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 178 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU550-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U550, technician must log: "Resolved ERR-OKU550-ATC-2 on Okuma-Multus-U550 by installing PN-OKU550-CAM-GRIP-3902 & kit KIT-OKU550-ATC-2. Fasteners torqued to 178 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU550-HYD-3

Target Machine Model: Okuma-Multus-U550 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U550 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Okuma-Multus-U550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U550 (OSP-P500A Aerospace)
- Telemetry Sensor ID: PR-HYD-OKU550-3
- PLC Hex Register Address: DB28.DBW_0x99F8
- Nominal Operating Range: 21 - 73 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 103 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU550-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU550-VLV-PROP-3903
- Overhaul Seal & Consumable Kit: KIT-OKU550-HYD-3
- Critical Fastener Torque Value: 187 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U550 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU550-HYD-3 at PLC register 0x99F8.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Okuma-Multus-U550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU550-VLV-PROP-3903) with seal kit (KIT-OKU550-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 187 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU550-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U550, technician must log: "Resolved ERR-OKU550-HYD-3 on Okuma-Multus-U550 by installing PN-OKU550-VLV-PROP-3903 & kit KIT-OKU550-HYD-3. Fasteners torqued to 187 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU550-CLN-4

Target Machine Model: Okuma-Multus-U550 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U550 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Okuma-Multus-U550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U550 (OSP-P500A Aerospace)
- Telemetry Sensor ID: FL-CLN-OKU550-4
- PLC Hex Register Address: DB28.DBW_0x9A00
- Nominal Operating Range: 24 - 73 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 103 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU550-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU550-PMP-PLG-3904
- Overhaul Seal & Consumable Kit: KIT-OKU550-CLN-4
- Critical Fastener Torque Value: 196 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U550 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU550-CLN-4 at PLC register 0x9A00.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Okuma-Multus-U550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU550-PMP-PLG-3904) with seal kit (KIT-OKU550-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 196 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU550-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U550, technician must log: "Resolved ERR-OKU550-CLN-4 on Okuma-Multus-U550 by installing PN-OKU550-PMP-PLG-3904 & kit KIT-OKU550-CLN-4. Fasteners torqued to 196 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU550-XAX-5

Target Machine Model: Okuma-Multus-U550 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U550 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U550 (OSP-P500A Aerospace)
- Telemetry Sensor ID: EN-XAX-OKU550-5
- PLC Hex Register Address: DB28.DBW_0x9A08
- Nominal Operating Range: 27 - 73 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 103 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU550-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU550-LIN-COIL-3905
- Overhaul Seal & Consumable Kit: KIT-OKU550-XAX-5
- Critical Fastener Torque Value: 205 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U550 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU550-XAX-5 at PLC register 0x9A08.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Okuma-Multus-U550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU550-LIN-COIL-3905) with seal kit (KIT-OKU550-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 205 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU550-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U550, technician must log: "Resolved ERR-OKU550-XAX-5 on Okuma-Multus-U550 by installing PN-OKU550-LIN-COIL-3905 & kit KIT-OKU550-XAX-5. Fasteners torqued to 205 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU550-YAX-6

Target Machine Model: Okuma-Multus-U550 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U550 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Okuma-Multus-U550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U550 (OSP-P500A Aerospace)
- Telemetry Sensor ID: EN-YAX-OKU550-6
- PLC Hex Register Address: DB28.DBW_0x9A10
- Nominal Operating Range: 30 - 73 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 103 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU550-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU550-BSW-NUT-3906
- Overhaul Seal & Consumable Kit: KIT-OKU550-YAX-6
- Critical Fastener Torque Value: 24 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U550 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU550-YAX-6 at PLC register 0x9A10.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Okuma-Multus-U550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU550-BSW-NUT-3906) with seal kit (KIT-OKU550-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 24 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU550-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U550, technician must log: "Resolved ERR-OKU550-YAX-6 on Okuma-Multus-U550 by installing PN-OKU550-BSW-NUT-3906 & kit KIT-OKU550-YAX-6. Fasteners torqued to 24 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU550-ZAX-7

Target Machine Model: Okuma-Multus-U550 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U550 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U550 (OSP-P500A Aerospace)
- Telemetry Sensor ID: PR-ZAX-OKU550-7
- PLC Hex Register Address: DB28.DBW_0x9A18
- Nominal Operating Range: 33 - 73 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 103 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU550-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU550-CYL-SEAL-3907
- Overhaul Seal & Consumable Kit: KIT-OKU550-ZAX-7
- Critical Fastener Torque Value: 33 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U550 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU550-ZAX-7 at PLC register 0x9A18.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Okuma-Multus-U550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU550-CYL-SEAL-3907) with seal kit (KIT-OKU550-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 33 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU550-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U550, technician must log: "Resolved ERR-OKU550-ZAX-7 on Okuma-Multus-U550 by installing PN-OKU550-CYL-SEAL-3907 & kit KIT-OKU550-ZAX-7. Fasteners torqued to 33 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU550-BAX-8

Target Machine Model: Okuma-Multus-U550 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U550 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Okuma-Multus-U550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U550 (OSP-P500A Aerospace)
- Telemetry Sensor ID: EN-BAX-OKU550-8
- PLC Hex Register Address: DB28.DBW_0x9A20
- Nominal Operating Range: 36 - 73 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 103 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU550-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU550-TRQ-STAT-3908
- Overhaul Seal & Consumable Kit: KIT-OKU550-BAX-8
- Critical Fastener Torque Value: 42 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U550 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU550-BAX-8 at PLC register 0x9A20.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Okuma-Multus-U550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU550-TRQ-STAT-3908) with seal kit (KIT-OKU550-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 42 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU550-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U550, technician must log: "Resolved ERR-OKU550-BAX-8 on Okuma-Multus-U550 by installing PN-OKU550-TRQ-STAT-3908 & kit KIT-OKU550-BAX-8. Fasteners torqued to 42 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU550-CAX-9

Target Machine Model: Okuma-Multus-U550 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U550 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U550 (OSP-P500A Aerospace)
- Telemetry Sensor ID: PR-CAX-OKU550-9
- PLC Hex Register Address: DB28.DBW_0x9A28
- Nominal Operating Range: 39 - 73 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 103 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU550-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU550-BRK-DISC-3909
- Overhaul Seal & Consumable Kit: KIT-OKU550-CAX-9
- Critical Fastener Torque Value: 51 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U550 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU550-CAX-9 at PLC register 0x9A28.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Okuma-Multus-U550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU550-BRK-DISC-3909) with seal kit (KIT-OKU550-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 51 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU550-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U550, technician must log: "Resolved ERR-OKU550-CAX-9 on Okuma-Multus-U550 by installing PN-OKU550-BRK-DISC-3909 & kit KIT-OKU550-CAX-9. Fasteners torqued to 51 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU550-OPT-10

Target Machine Model: Okuma-Multus-U550 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U550 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Okuma-Multus-U550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U550 (OSP-P500A Aerospace)
- Telemetry Sensor ID: OP-SCL-OKU550-10
- PLC Hex Register Address: DB28.DBW_0x9A30
- Nominal Operating Range: 42 - 73 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 103 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU550-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU550-GLASS-RD-3910
- Overhaul Seal & Consumable Kit: KIT-OKU550-OPT-10
- Critical Fastener Torque Value: 60 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U550 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU550-OPT-10 at PLC register 0x9A30.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Okuma-Multus-U550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU550-GLASS-RD-3910) with seal kit (KIT-OKU550-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 60 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU550-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U550, technician must log: "Resolved ERR-OKU550-OPT-10 on Okuma-Multus-U550 by installing PN-OKU550-GLASS-RD-3910 & kit KIT-OKU550-OPT-10. Fasteners torqued to 60 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU550-PLC-11

Target Machine Model: Okuma-Multus-U550 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U550 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U550 (OSP-P500A Aerospace)
- Telemetry Sensor ID: VT-DRV-OKU550-11
- PLC Hex Register Address: DB28.DBW_0x9A38
- Nominal Operating Range: 45 - 73 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 103 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU550-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU550-IGBT-MOD-3911
- Overhaul Seal & Consumable Kit: KIT-OKU550-PLC-11
- Critical Fastener Torque Value: 69 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U550 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU550-PLC-11 at PLC register 0x9A38.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Okuma-Multus-U550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU550-IGBT-MOD-3911) with seal kit (KIT-OKU550-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 69 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU550-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U550, technician must log: "Resolved ERR-OKU550-PLC-11 on Okuma-Multus-U550 by installing PN-OKU550-IGBT-MOD-3911 & kit KIT-OKU550-PLC-11. Fasteners torqued to 69 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU550-LUB-12

Target Machine Model: Okuma-Multus-U550 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U550 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Okuma-Multus-U550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U550 (OSP-P500A Aerospace)
- Telemetry Sensor ID: FL-LUB-OKU550-12
- PLC Hex Register Address: DB28.DBW_0x9A40
- Nominal Operating Range: 48 - 73 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 103 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU550-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU550-INJ-MET-3912
- Overhaul Seal & Consumable Kit: KIT-OKU550-LUB-12
- Critical Fastener Torque Value: 78 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U550 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU550-LUB-12 at PLC register 0x9A40.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Okuma-Multus-U550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU550-INJ-MET-3912) with seal kit (KIT-OKU550-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 78 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU550-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U550, technician must log: "Resolved ERR-OKU550-LUB-12 on Okuma-Multus-U550 by installing PN-OKU550-INJ-MET-3912 & kit KIT-OKU550-LUB-12. Fasteners torqued to 78 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU550-VAC-13

Target Machine Model: Okuma-Multus-U550 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U550 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U550 (OSP-P500A Aerospace)
- Telemetry Sensor ID: VC-MNF-OKU550-13
- PLC Hex Register Address: DB28.DBW_0x9A48
- Nominal Operating Range: 51 - 73 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 103 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU550-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU550-GASK-VIT-3913
- Overhaul Seal & Consumable Kit: KIT-OKU550-VAC-13
- Critical Fastener Torque Value: 87 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U550 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU550-VAC-13 at PLC register 0x9A48.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Okuma-Multus-U550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU550-GASK-VIT-3913) with seal kit (KIT-OKU550-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 87 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU550-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U550, technician must log: "Resolved ERR-OKU550-VAC-13 on Okuma-Multus-U550 by installing PN-OKU550-GASK-VIT-3913 & kit KIT-OKU550-VAC-13. Fasteners torqued to 87 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU550-THM-14

Target Machine Model: Okuma-Multus-U550 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U550 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Okuma-Multus-U550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U550 (OSP-P500A Aerospace)
- Telemetry Sensor ID: TM-CHL-OKU550-14
- PLC Hex Register Address: DB28.DBW_0x9A50
- Nominal Operating Range: 54 - 73 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 103 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU550-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU550-HEX-CORE-3914
- Overhaul Seal & Consumable Kit: KIT-OKU550-THM-14
- Critical Fastener Torque Value: 96 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U550 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU550-THM-14 at PLC register 0x9A50.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Okuma-Multus-U550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU550-HEX-CORE-3914) with seal kit (KIT-OKU550-THM-14).

Step 6: Torque retaining bolts in cross sequence to 96 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU550-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U550, technician must log: "Resolved ERR-OKU550-THM-14 on Okuma-Multus-U550 by installing PN-OKU550-HEX-CORE-3914 & kit KIT-OKU550-THM-14. Fasteners torqued to 96 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU600-SPN-1

Target Machine Model: Okuma-Multus-U600 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U600 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Okuma-Multus-U600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U600 (OSP-P500A Aerospace)
- Telemetry Sensor ID: PT-SPN-OKU600-1
- PLC Hex Register Address: DB29.DBW_0x9D08
- Nominal Operating Range: 15 - 74 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 104 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU600-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU600-BRG-CER-4001
- Overhaul Seal & Consumable Kit: KIT-OKU600-SPN-1
- Critical Fastener Torque Value: 119 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U600 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU600-SPN-1 at PLC register 0x9D08.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Okuma-Multus-U600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU600-BRG-CER-4001) with seal kit (KIT-OKU600-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 119 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU600-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U600, technician must log: "Resolved ERR-OKU600-SPN-1 on Okuma-Multus-U600 by installing PN-OKU600-BRG-CER-4001 & kit KIT-OKU600-SPN-1. Fasteners torqued to 119 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU600-ATC-2

Target Machine Model: Okuma-Multus-U600 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U600 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Okuma-Multus-U600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U600 (OSP-P500A Aerospace)
- Telemetry Sensor ID: PX-ATC-OKU600-2
- PLC Hex Register Address: DB29.DBW_0x9D10
- Nominal Operating Range: 18 - 74 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 104 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU600-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU600-CAM-GRIP-4002
- Overhaul Seal & Consumable Kit: KIT-OKU600-ATC-2
- Critical Fastener Torque Value: 128 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U600 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU600-ATC-2 at PLC register 0x9D10.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Okuma-Multus-U600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU600-CAM-GRIP-4002) with seal kit (KIT-OKU600-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 128 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU600-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U600, technician must log: "Resolved ERR-OKU600-ATC-2 on Okuma-Multus-U600 by installing PN-OKU600-CAM-GRIP-4002 & kit KIT-OKU600-ATC-2. Fasteners torqued to 128 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU600-HYD-3

Target Machine Model: Okuma-Multus-U600 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U600 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Okuma-Multus-U600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U600 (OSP-P500A Aerospace)
- Telemetry Sensor ID: PR-HYD-OKU600-3
- PLC Hex Register Address: DB29.DBW_0x9D18
- Nominal Operating Range: 21 - 74 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 104 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU600-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU600-VLV-PROP-4003
- Overhaul Seal & Consumable Kit: KIT-OKU600-HYD-3
- Critical Fastener Torque Value: 137 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U600 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU600-HYD-3 at PLC register 0x9D18.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Okuma-Multus-U600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU600-VLV-PROP-4003) with seal kit (KIT-OKU600-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 137 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU600-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U600, technician must log: "Resolved ERR-OKU600-HYD-3 on Okuma-Multus-U600 by installing PN-OKU600-VLV-PROP-4003 & kit KIT-OKU600-HYD-3. Fasteners torqued to 137 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU600-CLN-4

Target Machine Model: Okuma-Multus-U600 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U600 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Okuma-Multus-U600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U600 (OSP-P500A Aerospace)
- Telemetry Sensor ID: FL-CLN-OKU600-4
- PLC Hex Register Address: DB29.DBW_0x9D20
- Nominal Operating Range: 24 - 74 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 104 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU600-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU600-PMP-PLG-4004
- Overhaul Seal & Consumable Kit: KIT-OKU600-CLN-4
- Critical Fastener Torque Value: 146 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U600 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU600-CLN-4 at PLC register 0x9D20.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Okuma-Multus-U600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU600-PMP-PLG-4004) with seal kit (KIT-OKU600-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 146 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU600-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U600, technician must log: "Resolved ERR-OKU600-CLN-4 on Okuma-Multus-U600 by installing PN-OKU600-PMP-PLG-4004 & kit KIT-OKU600-CLN-4. Fasteners torqued to 146 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU600-XAX-5

Target Machine Model: Okuma-Multus-U600 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U600 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U600 (OSP-P500A Aerospace)
- Telemetry Sensor ID: EN-XAX-OKU600-5
- PLC Hex Register Address: DB29.DBW_0x9D28
- Nominal Operating Range: 27 - 74 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 104 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU600-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU600-LIN-COIL-4005
- Overhaul Seal & Consumable Kit: KIT-OKU600-XAX-5
- Critical Fastener Torque Value: 155 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U600 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU600-XAX-5 at PLC register 0x9D28.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Okuma-Multus-U600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU600-LIN-COIL-4005) with seal kit (KIT-OKU600-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 155 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU600-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U600, technician must log: "Resolved ERR-OKU600-XAX-5 on Okuma-Multus-U600 by installing PN-OKU600-LIN-COIL-4005 & kit KIT-OKU600-XAX-5. Fasteners torqued to 155 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU600-YAX-6

Target Machine Model: Okuma-Multus-U600 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U600 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Okuma-Multus-U600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U600 (OSP-P500A Aerospace)
- Telemetry Sensor ID: EN-YAX-OKU600-6
- PLC Hex Register Address: DB29.DBW_0x9D30
- Nominal Operating Range: 30 - 74 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 104 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU600-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU600-BSW-NUT-4006
- Overhaul Seal & Consumable Kit: KIT-OKU600-YAX-6
- Critical Fastener Torque Value: 164 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U600 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU600-YAX-6 at PLC register 0x9D30.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Okuma-Multus-U600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU600-BSW-NUT-4006) with seal kit (KIT-OKU600-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 164 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU600-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U600, technician must log: "Resolved ERR-OKU600-YAX-6 on Okuma-Multus-U600 by installing PN-OKU600-BSW-NUT-4006 & kit KIT-OKU600-YAX-6. Fasteners torqued to 164 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU600-ZAX-7

Target Machine Model: Okuma-Multus-U600 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U600 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U600 (OSP-P500A Aerospace)
- Telemetry Sensor ID: PR-ZAX-OKU600-7
- PLC Hex Register Address: DB29.DBW_0x9D38
- Nominal Operating Range: 33 - 74 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 104 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU600-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU600-CYL-SEAL-4007
- Overhaul Seal & Consumable Kit: KIT-OKU600-ZAX-7
- Critical Fastener Torque Value: 173 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U600 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU600-ZAX-7 at PLC register 0x9D38.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Okuma-Multus-U600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU600-CYL-SEAL-4007) with seal kit (KIT-OKU600-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 173 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU600-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U600, technician must log: "Resolved ERR-OKU600-ZAX-7 on Okuma-Multus-U600 by installing PN-OKU600-CYL-SEAL-4007 & kit KIT-OKU600-ZAX-7. Fasteners torqued to 173 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU600-BAX-8

Target Machine Model: Okuma-Multus-U600 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U600 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Okuma-Multus-U600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U600 (OSP-P500A Aerospace)
- Telemetry Sensor ID: EN-BAX-OKU600-8
- PLC Hex Register Address: DB29.DBW_0x9D40
- Nominal Operating Range: 36 - 74 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 104 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU600-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU600-TRQ-STAT-4008
- Overhaul Seal & Consumable Kit: KIT-OKU600-BAX-8
- Critical Fastener Torque Value: 182 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U600 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU600-BAX-8 at PLC register 0x9D40.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Okuma-Multus-U600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU600-TRQ-STAT-4008) with seal kit (KIT-OKU600-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 182 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU600-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U600, technician must log: "Resolved ERR-OKU600-BAX-8 on Okuma-Multus-U600 by installing PN-OKU600-TRQ-STAT-4008 & kit KIT-OKU600-BAX-8. Fasteners torqued to 182 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU600-CAX-9

Target Machine Model: Okuma-Multus-U600 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U600 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U600 (OSP-P500A Aerospace)
- Telemetry Sensor ID: PR-CAX-OKU600-9
- PLC Hex Register Address: DB29.DBW_0x9D48
- Nominal Operating Range: 39 - 74 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 104 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU600-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU600-BRK-DISC-4009
- Overhaul Seal & Consumable Kit: KIT-OKU600-CAX-9
- Critical Fastener Torque Value: 191 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U600 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU600-CAX-9 at PLC register 0x9D48.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Okuma-Multus-U600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU600-BRK-DISC-4009) with seal kit (KIT-OKU600-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 191 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU600-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U600, technician must log: "Resolved ERR-OKU600-CAX-9 on Okuma-Multus-U600 by installing PN-OKU600-BRK-DISC-4009 & kit KIT-OKU600-CAX-9. Fasteners torqued to 191 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU600-OPT-10

Target Machine Model: Okuma-Multus-U600 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U600 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Okuma-Multus-U600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U600 (OSP-P500A Aerospace)
- Telemetry Sensor ID: OP-SCL-OKU600-10
- PLC Hex Register Address: DB29.DBW_0x9D50
- Nominal Operating Range: 42 - 74 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 104 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU600-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU600-GLASS-RD-4010
- Overhaul Seal & Consumable Kit: KIT-OKU600-OPT-10
- Critical Fastener Torque Value: 200 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U600 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU600-OPT-10 at PLC register 0x9D50.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Okuma-Multus-U600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU600-GLASS-RD-4010) with seal kit (KIT-OKU600-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 200 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU600-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U600, technician must log: "Resolved ERR-OKU600-OPT-10 on Okuma-Multus-U600 by installing PN-OKU600-GLASS-RD-4010 & kit KIT-OKU600-OPT-10. Fasteners torqued to 200 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU600-PLC-11

Target Machine Model: Okuma-Multus-U600 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U600 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U600 (OSP-P500A Aerospace)
- Telemetry Sensor ID: VT-DRV-OKU600-11
- PLC Hex Register Address: DB29.DBW_0x9D58
- Nominal Operating Range: 45 - 74 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 104 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU600-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU600-IGBT-MOD-4011
- Overhaul Seal & Consumable Kit: KIT-OKU600-PLC-11
- Critical Fastener Torque Value: 209 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U600 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU600-PLC-11 at PLC register 0x9D58.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Okuma-Multus-U600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU600-IGBT-MOD-4011) with seal kit (KIT-OKU600-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 209 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU600-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U600, technician must log: "Resolved ERR-OKU600-PLC-11 on Okuma-Multus-U600 by installing PN-OKU600-IGBT-MOD-4011 & kit KIT-OKU600-PLC-11. Fasteners torqued to 209 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU600-LUB-12

Target Machine Model: Okuma-Multus-U600 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U600 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Okuma-Multus-U600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U600 (OSP-P500A Aerospace)
- Telemetry Sensor ID: FL-LUB-OKU600-12
- PLC Hex Register Address: DB29.DBW_0x9D60
- Nominal Operating Range: 48 - 74 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 104 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU600-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU600-INJ-MET-4012
- Overhaul Seal & Consumable Kit: KIT-OKU600-LUB-12
- Critical Fastener Torque Value: 28 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U600 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU600-LUB-12 at PLC register 0x9D60.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Okuma-Multus-U600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU600-INJ-MET-4012) with seal kit (KIT-OKU600-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 28 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU600-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U600, technician must log: "Resolved ERR-OKU600-LUB-12 on Okuma-Multus-U600 by installing PN-OKU600-INJ-MET-4012 & kit KIT-OKU600-LUB-12. Fasteners torqued to 28 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU600-VAC-13

Target Machine Model: Okuma-Multus-U600 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U600 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U600 (OSP-P500A Aerospace)
- Telemetry Sensor ID: VC-MNF-OKU600-13
- PLC Hex Register Address: DB29.DBW_0x9D68
- Nominal Operating Range: 51 - 74 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 104 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU600-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU600-GASK-VIT-4013
- Overhaul Seal & Consumable Kit: KIT-OKU600-VAC-13
- Critical Fastener Torque Value: 37 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U600 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU600-VAC-13 at PLC register 0x9D68.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Okuma-Multus-U600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU600-GASK-VIT-4013) with seal kit (KIT-OKU600-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 37 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU600-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U600, technician must log: "Resolved ERR-OKU600-VAC-13 on Okuma-Multus-U600 by installing PN-OKU600-GASK-VIT-4013 & kit KIT-OKU600-VAC-13. Fasteners torqued to 37 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-OKU600-THM-14

Target Machine Model: Okuma-Multus-U600 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Okuma-Multus-U600 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Okuma-Multus-U600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Okuma-Multus-U600 (OSP-P500A Aerospace)
- Telemetry Sensor ID: TM-CHL-OKU600-14
- PLC Hex Register Address: DB29.DBW_0x9D70
- Nominal Operating Range: 54 - 74 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 104 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-OKU600-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-OKU600-HEX-CORE-4014
- Overhaul Seal & Consumable Kit: KIT-OKU600-THM-14
- Critical Fastener Torque Value: 46 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Okuma-Multus-U600 (OSP-P500A Aerospace) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to OSP-P500A Aerospace and verify fault latch ERR-OKU600-THM-14 at PLC register 0x9D70.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Okuma-Multus-U600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-OKU600-HEX-CORE-4014) with seal kit (KIT-OKU600-THM-14).

Step 6: Torque retaining bolts in cross sequence to 46 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-OKU600-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Okuma-Multus-U600, technician must log: "Resolved ERR-OKU600-THM-14 on Okuma-Multus-U600 by installing PN-OKU600-HEX-CORE-4014 & kit KIT-OKU600-THM-14. Fasteners torqued to 46 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS150-SPN-1

Target Machine Model: EOS-Additive-M150 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M150 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on EOS-Additive-M150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M150 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: PT-SPN-EOS150-1
- PLC Hex Register Address: DB30.DBW_0xA028
- Nominal Operating Range: 15 - 75 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 105 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS150-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS150-BRG-CER-4101
- Overhaul Seal & Consumable Kit: KIT-EOS150-SPN-1
- Critical Fastener Torque Value: 69 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M150 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS150-SPN-1 at PLC register 0xA028.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on EOS-Additive-M150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS150-BRG-CER-4101) with seal kit (KIT-EOS150-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 69 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS150-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M150, technician must log: "Resolved ERR-EOS150-SPN-1 on EOS-Additive-M150 by installing PN-EOS150-BRG-CER-4101 & kit KIT-EOS150-SPN-1. Fasteners torqued to 69 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS150-ATC-2

Target Machine Model: EOS-Additive-M150 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M150 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on EOS-Additive-M150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M150 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: PX-ATC-EOS150-2
- PLC Hex Register Address: DB30.DBW_0xA030
- Nominal Operating Range: 18 - 75 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 105 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS150-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS150-CAM-GRIP-4102
- Overhaul Seal & Consumable Kit: KIT-EOS150-ATC-2
- Critical Fastener Torque Value: 78 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M150 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS150-ATC-2 at PLC register 0xA030.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on EOS-Additive-M150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS150-CAM-GRIP-4102) with seal kit (KIT-EOS150-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 78 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS150-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M150, technician must log: "Resolved ERR-EOS150-ATC-2 on EOS-Additive-M150 by installing PN-EOS150-CAM-GRIP-4102 & kit KIT-EOS150-ATC-2. Fasteners torqued to 78 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS150-HYD-3

Target Machine Model: EOS-Additive-M150 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M150 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on EOS-Additive-M150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M150 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: PR-HYD-EOS150-3
- PLC Hex Register Address: DB30.DBW_0xA038
- Nominal Operating Range: 21 - 75 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 105 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS150-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS150-VLV-PROP-4103
- Overhaul Seal & Consumable Kit: KIT-EOS150-HYD-3
- Critical Fastener Torque Value: 87 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M150 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS150-HYD-3 at PLC register 0xA038.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on EOS-Additive-M150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS150-VLV-PROP-4103) with seal kit (KIT-EOS150-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 87 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS150-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M150, technician must log: "Resolved ERR-EOS150-HYD-3 on EOS-Additive-M150 by installing PN-EOS150-VLV-PROP-4103 & kit KIT-EOS150-HYD-3. Fasteners torqued to 87 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS150-CLN-4

Target Machine Model: EOS-Additive-M150 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M150 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on EOS-Additive-M150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M150 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: FL-CLN-EOS150-4
- PLC Hex Register Address: DB30.DBW_0xA040
- Nominal Operating Range: 24 - 75 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 105 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS150-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS150-PMP-PLG-4104
- Overhaul Seal & Consumable Kit: KIT-EOS150-CLN-4
- Critical Fastener Torque Value: 96 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M150 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS150-CLN-4 at PLC register 0xA040.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on EOS-Additive-M150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS150-PMP-PLG-4104) with seal kit (KIT-EOS150-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 96 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS150-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M150, technician must log: "Resolved ERR-EOS150-CLN-4 on EOS-Additive-M150 by installing PN-EOS150-PMP-PLG-4104 & kit KIT-EOS150-CLN-4. Fasteners torqued to 96 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS150-XAX-5

Target Machine Model: EOS-Additive-M150 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M150 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M150 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: EN-XAX-EOS150-5
- PLC Hex Register Address: DB30.DBW_0xA048
- Nominal Operating Range: 27 - 75 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 105 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS150-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS150-LIN-COIL-4105
- Overhaul Seal & Consumable Kit: KIT-EOS150-XAX-5
- Critical Fastener Torque Value: 105 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M150 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS150-XAX-5 at PLC register 0xA048.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on EOS-Additive-M150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS150-LIN-COIL-4105) with seal kit (KIT-EOS150-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 105 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS150-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M150, technician must log: "Resolved ERR-EOS150-XAX-5 on EOS-Additive-M150 by installing PN-EOS150-LIN-COIL-4105 & kit KIT-EOS150-XAX-5. Fasteners torqued to 105 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS150-YAX-6

Target Machine Model: EOS-Additive-M150 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M150 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on EOS-Additive-M150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M150 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: EN-YAX-EOS150-6
- PLC Hex Register Address: DB30.DBW_0xA050
- Nominal Operating Range: 30 - 75 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 105 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS150-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS150-BSW-NUT-4106
- Overhaul Seal & Consumable Kit: KIT-EOS150-YAX-6
- Critical Fastener Torque Value: 114 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M150 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS150-YAX-6 at PLC register 0xA050.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on EOS-Additive-M150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS150-BSW-NUT-4106) with seal kit (KIT-EOS150-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 114 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS150-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M150, technician must log: "Resolved ERR-EOS150-YAX-6 on EOS-Additive-M150 by installing PN-EOS150-BSW-NUT-4106 & kit KIT-EOS150-YAX-6. Fasteners torqued to 114 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS150-ZAX-7

Target Machine Model: EOS-Additive-M150 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M150 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M150 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: PR-ZAX-EOS150-7
- PLC Hex Register Address: DB30.DBW_0xA058
- Nominal Operating Range: 33 - 75 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 105 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS150-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS150-CYL-SEAL-4107
- Overhaul Seal & Consumable Kit: KIT-EOS150-ZAX-7
- Critical Fastener Torque Value: 123 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M150 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS150-ZAX-7 at PLC register 0xA058.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on EOS-Additive-M150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS150-CYL-SEAL-4107) with seal kit (KIT-EOS150-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 123 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS150-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M150, technician must log: "Resolved ERR-EOS150-ZAX-7 on EOS-Additive-M150 by installing PN-EOS150-CYL-SEAL-4107 & kit KIT-EOS150-ZAX-7. Fasteners torqued to 123 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS150-BAX-8

Target Machine Model: EOS-Additive-M150 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M150 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on EOS-Additive-M150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M150 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: EN-BAX-EOS150-8
- PLC Hex Register Address: DB30.DBW_0xA060
- Nominal Operating Range: 36 - 75 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 105 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS150-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS150-TRQ-STAT-4108
- Overhaul Seal & Consumable Kit: KIT-EOS150-BAX-8
- Critical Fastener Torque Value: 132 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M150 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS150-BAX-8 at PLC register 0xA060.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on EOS-Additive-M150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS150-TRQ-STAT-4108) with seal kit (KIT-EOS150-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 132 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS150-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M150, technician must log: "Resolved ERR-EOS150-BAX-8 on EOS-Additive-M150 by installing PN-EOS150-TRQ-STAT-4108 & kit KIT-EOS150-BAX-8. Fasteners torqued to 132 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS150-CAX-9

Target Machine Model: EOS-Additive-M150 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M150 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M150 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: PR-CAX-EOS150-9
- PLC Hex Register Address: DB30.DBW_0xA068
- Nominal Operating Range: 39 - 75 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 105 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS150-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS150-BRK-DISC-4109
- Overhaul Seal & Consumable Kit: KIT-EOS150-CAX-9
- Critical Fastener Torque Value: 141 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M150 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS150-CAX-9 at PLC register 0xA068.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on EOS-Additive-M150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS150-BRK-DISC-4109) with seal kit (KIT-EOS150-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 141 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS150-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M150, technician must log: "Resolved ERR-EOS150-CAX-9 on EOS-Additive-M150 by installing PN-EOS150-BRK-DISC-4109 & kit KIT-EOS150-CAX-9. Fasteners torqued to 141 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS150-OPT-10

Target Machine Model: EOS-Additive-M150 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M150 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on EOS-Additive-M150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M150 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: OP-SCL-EOS150-10
- PLC Hex Register Address: DB30.DBW_0xA070
- Nominal Operating Range: 42 - 75 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 105 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS150-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS150-GLASS-RD-4110
- Overhaul Seal & Consumable Kit: KIT-EOS150-OPT-10
- Critical Fastener Torque Value: 150 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M150 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS150-OPT-10 at PLC register 0xA070.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on EOS-Additive-M150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS150-GLASS-RD-4110) with seal kit (KIT-EOS150-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 150 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS150-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M150, technician must log: "Resolved ERR-EOS150-OPT-10 on EOS-Additive-M150 by installing PN-EOS150-GLASS-RD-4110 & kit KIT-EOS150-OPT-10. Fasteners torqued to 150 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS150-PLC-11

Target Machine Model: EOS-Additive-M150 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M150 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M150 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: VT-DRV-EOS150-11
- PLC Hex Register Address: DB30.DBW_0xA078
- Nominal Operating Range: 45 - 75 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 105 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS150-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS150-IGBT-MOD-4111
- Overhaul Seal & Consumable Kit: KIT-EOS150-PLC-11
- Critical Fastener Torque Value: 159 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M150 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS150-PLC-11 at PLC register 0xA078.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on EOS-Additive-M150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS150-IGBT-MOD-4111) with seal kit (KIT-EOS150-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 159 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS150-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M150, technician must log: "Resolved ERR-EOS150-PLC-11 on EOS-Additive-M150 by installing PN-EOS150-IGBT-MOD-4111 & kit KIT-EOS150-PLC-11. Fasteners torqued to 159 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS150-LUB-12

Target Machine Model: EOS-Additive-M150 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M150 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on EOS-Additive-M150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M150 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: FL-LUB-EOS150-12
- PLC Hex Register Address: DB30.DBW_0xA080
- Nominal Operating Range: 48 - 75 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 105 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS150-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS150-INJ-MET-4112
- Overhaul Seal & Consumable Kit: KIT-EOS150-LUB-12
- Critical Fastener Torque Value: 168 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M150 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS150-LUB-12 at PLC register 0xA080.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on EOS-Additive-M150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS150-INJ-MET-4112) with seal kit (KIT-EOS150-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 168 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS150-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M150, technician must log: "Resolved ERR-EOS150-LUB-12 on EOS-Additive-M150 by installing PN-EOS150-INJ-MET-4112 & kit KIT-EOS150-LUB-12. Fasteners torqued to 168 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS150-VAC-13

Target Machine Model: EOS-Additive-M150 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M150 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M150 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: VC-MNF-EOS150-13
- PLC Hex Register Address: DB30.DBW_0xA088
- Nominal Operating Range: 51 - 75 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 105 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS150-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS150-GASK-VIT-4113
- Overhaul Seal & Consumable Kit: KIT-EOS150-VAC-13
- Critical Fastener Torque Value: 177 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M150 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS150-VAC-13 at PLC register 0xA088.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on EOS-Additive-M150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS150-GASK-VIT-4113) with seal kit (KIT-EOS150-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 177 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS150-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M150, technician must log: "Resolved ERR-EOS150-VAC-13 on EOS-Additive-M150 by installing PN-EOS150-GASK-VIT-4113 & kit KIT-EOS150-VAC-13. Fasteners torqued to 177 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS150-THM-14

Target Machine Model: EOS-Additive-M150 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M150 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on EOS-Additive-M150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M150 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: TM-CHL-EOS150-14
- PLC Hex Register Address: DB30.DBW_0xA090
- Nominal Operating Range: 54 - 75 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 105 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS150-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS150-HEX-CORE-4114
- Overhaul Seal & Consumable Kit: KIT-EOS150-THM-14
- Critical Fastener Torque Value: 186 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M150 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS150-THM-14 at PLC register 0xA090.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on EOS-Additive-M150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS150-HEX-CORE-4114) with seal kit (KIT-EOS150-THM-14).

Step 6: Torque retaining bolts in cross sequence to 186 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS150-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M150, technician must log: "Resolved ERR-EOS150-THM-14 on EOS-Additive-M150 by installing PN-EOS150-HEX-CORE-4114 & kit KIT-EOS150-THM-14. Fasteners torqued to 186 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS200-SPN-1

Target Machine Model: EOS-Additive-M200 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M200 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on EOS-Additive-M200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M200 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: PT-SPN-EOS200-1
- PLC Hex Register Address: DB31.DBW_0xA348
- Nominal Operating Range: 15 - 76 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 106 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS200-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS200-BRG-CER-4201
- Overhaul Seal & Consumable Kit: KIT-EOS200-SPN-1
- Critical Fastener Torque Value: 209 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M200 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS200-SPN-1 at PLC register 0xA348.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on EOS-Additive-M200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS200-BRG-CER-4201) with seal kit (KIT-EOS200-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 209 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS200-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M200, technician must log: "Resolved ERR-EOS200-SPN-1 on EOS-Additive-M200 by installing PN-EOS200-BRG-CER-4201 & kit KIT-EOS200-SPN-1. Fasteners torqued to 209 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS200-ATC-2

Target Machine Model: EOS-Additive-M200 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M200 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on EOS-Additive-M200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M200 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: PX-ATC-EOS200-2
- PLC Hex Register Address: DB31.DBW_0xA350
- Nominal Operating Range: 18 - 76 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 106 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS200-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS200-CAM-GRIP-4202
- Overhaul Seal & Consumable Kit: KIT-EOS200-ATC-2
- Critical Fastener Torque Value: 28 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M200 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS200-ATC-2 at PLC register 0xA350.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on EOS-Additive-M200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS200-CAM-GRIP-4202) with seal kit (KIT-EOS200-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 28 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS200-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M200, technician must log: "Resolved ERR-EOS200-ATC-2 on EOS-Additive-M200 by installing PN-EOS200-CAM-GRIP-4202 & kit KIT-EOS200-ATC-2. Fasteners torqued to 28 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS200-HYD-3

Target Machine Model: EOS-Additive-M200 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M200 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on EOS-Additive-M200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M200 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: PR-HYD-EOS200-3
- PLC Hex Register Address: DB31.DBW_0xA358
- Nominal Operating Range: 21 - 76 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 106 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS200-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS200-VLV-PROP-4203
- Overhaul Seal & Consumable Kit: KIT-EOS200-HYD-3
- Critical Fastener Torque Value: 37 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M200 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS200-HYD-3 at PLC register 0xA358.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on EOS-Additive-M200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS200-VLV-PROP-4203) with seal kit (KIT-EOS200-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 37 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS200-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M200, technician must log: "Resolved ERR-EOS200-HYD-3 on EOS-Additive-M200 by installing PN-EOS200-VLV-PROP-4203 & kit KIT-EOS200-HYD-3. Fasteners torqued to 37 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS200-CLN-4

Target Machine Model: EOS-Additive-M200 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M200 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on EOS-Additive-M200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M200 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: FL-CLN-EOS200-4
- PLC Hex Register Address: DB31.DBW_0xA360
- Nominal Operating Range: 24 - 76 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 106 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS200-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS200-PMP-PLG-4204
- Overhaul Seal & Consumable Kit: KIT-EOS200-CLN-4
- Critical Fastener Torque Value: 46 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M200 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS200-CLN-4 at PLC register 0xA360.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on EOS-Additive-M200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS200-PMP-PLG-4204) with seal kit (KIT-EOS200-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 46 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS200-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M200, technician must log: "Resolved ERR-EOS200-CLN-4 on EOS-Additive-M200 by installing PN-EOS200-PMP-PLG-4204 & kit KIT-EOS200-CLN-4. Fasteners torqued to 46 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS200-XAX-5

Target Machine Model: EOS-Additive-M200 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M200 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M200 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: EN-XAX-EOS200-5
- PLC Hex Register Address: DB31.DBW_0xA368
- Nominal Operating Range: 27 - 76 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 106 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS200-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS200-LIN-COIL-4205
- Overhaul Seal & Consumable Kit: KIT-EOS200-XAX-5
- Critical Fastener Torque Value: 55 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M200 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS200-XAX-5 at PLC register 0xA368.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on EOS-Additive-M200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS200-LIN-COIL-4205) with seal kit (KIT-EOS200-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 55 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS200-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M200, technician must log: "Resolved ERR-EOS200-XAX-5 on EOS-Additive-M200 by installing PN-EOS200-LIN-COIL-4205 & kit KIT-EOS200-XAX-5. Fasteners torqued to 55 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS200-YAX-6

Target Machine Model: EOS-Additive-M200 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M200 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on EOS-Additive-M200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M200 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: EN-YAX-EOS200-6
- PLC Hex Register Address: DB31.DBW_0xA370
- Nominal Operating Range: 30 - 76 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 106 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS200-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS200-BSW-NUT-4206
- Overhaul Seal & Consumable Kit: KIT-EOS200-YAX-6
- Critical Fastener Torque Value: 64 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M200 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS200-YAX-6 at PLC register 0xA370.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on EOS-Additive-M200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS200-BSW-NUT-4206) with seal kit (KIT-EOS200-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 64 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS200-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M200, technician must log: "Resolved ERR-EOS200-YAX-6 on EOS-Additive-M200 by installing PN-EOS200-BSW-NUT-4206 & kit KIT-EOS200-YAX-6. Fasteners torqued to 64 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS200-ZAX-7

Target Machine Model: EOS-Additive-M200 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M200 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M200 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: PR-ZAX-EOS200-7
- PLC Hex Register Address: DB31.DBW_0xA378
- Nominal Operating Range: 33 - 76 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 106 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS200-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS200-CYL-SEAL-4207
- Overhaul Seal & Consumable Kit: KIT-EOS200-ZAX-7
- Critical Fastener Torque Value: 73 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M200 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS200-ZAX-7 at PLC register 0xA378.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on EOS-Additive-M200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS200-CYL-SEAL-4207) with seal kit (KIT-EOS200-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 73 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS200-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M200, technician must log: "Resolved ERR-EOS200-ZAX-7 on EOS-Additive-M200 by installing PN-EOS200-CYL-SEAL-4207 & kit KIT-EOS200-ZAX-7. Fasteners torqued to 73 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS200-BAX-8

Target Machine Model: EOS-Additive-M200 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M200 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on EOS-Additive-M200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M200 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: EN-BAX-EOS200-8
- PLC Hex Register Address: DB31.DBW_0xA380
- Nominal Operating Range: 36 - 76 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 106 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS200-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS200-TRQ-STAT-4208
- Overhaul Seal & Consumable Kit: KIT-EOS200-BAX-8
- Critical Fastener Torque Value: 82 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M200 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS200-BAX-8 at PLC register 0xA380.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on EOS-Additive-M200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS200-TRQ-STAT-4208) with seal kit (KIT-EOS200-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 82 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS200-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M200, technician must log: "Resolved ERR-EOS200-BAX-8 on EOS-Additive-M200 by installing PN-EOS200-TRQ-STAT-4208 & kit KIT-EOS200-BAX-8. Fasteners torqued to 82 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS200-CAX-9

Target Machine Model: EOS-Additive-M200 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M200 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M200 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: PR-CAX-EOS200-9
- PLC Hex Register Address: DB31.DBW_0xA388
- Nominal Operating Range: 39 - 76 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 106 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS200-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS200-BRK-DISC-4209
- Overhaul Seal & Consumable Kit: KIT-EOS200-CAX-9
- Critical Fastener Torque Value: 91 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M200 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS200-CAX-9 at PLC register 0xA388.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on EOS-Additive-M200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS200-BRK-DISC-4209) with seal kit (KIT-EOS200-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 91 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS200-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M200, technician must log: "Resolved ERR-EOS200-CAX-9 on EOS-Additive-M200 by installing PN-EOS200-BRK-DISC-4209 & kit KIT-EOS200-CAX-9. Fasteners torqued to 91 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS200-OPT-10

Target Machine Model: EOS-Additive-M200 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M200 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on EOS-Additive-M200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M200 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: OP-SCL-EOS200-10
- PLC Hex Register Address: DB31.DBW_0xA390
- Nominal Operating Range: 42 - 76 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 106 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS200-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS200-GLASS-RD-4210
- Overhaul Seal & Consumable Kit: KIT-EOS200-OPT-10
- Critical Fastener Torque Value: 100 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M200 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS200-OPT-10 at PLC register 0xA390.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on EOS-Additive-M200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS200-GLASS-RD-4210) with seal kit (KIT-EOS200-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 100 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS200-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M200, technician must log: "Resolved ERR-EOS200-OPT-10 on EOS-Additive-M200 by installing PN-EOS200-GLASS-RD-4210 & kit KIT-EOS200-OPT-10. Fasteners torqued to 100 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS200-PLC-11

Target Machine Model: EOS-Additive-M200 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M200 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M200 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: VT-DRV-EOS200-11
- PLC Hex Register Address: DB31.DBW_0xA398
- Nominal Operating Range: 45 - 76 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 106 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS200-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS200-IGBT-MOD-4211
- Overhaul Seal & Consumable Kit: KIT-EOS200-PLC-11
- Critical Fastener Torque Value: 109 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M200 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS200-PLC-11 at PLC register 0xA398.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on EOS-Additive-M200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS200-IGBT-MOD-4211) with seal kit (KIT-EOS200-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 109 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS200-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M200, technician must log: "Resolved ERR-EOS200-PLC-11 on EOS-Additive-M200 by installing PN-EOS200-IGBT-MOD-4211 & kit KIT-EOS200-PLC-11. Fasteners torqued to 109 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS200-LUB-12

Target Machine Model: EOS-Additive-M200 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M200 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on EOS-Additive-M200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M200 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: FL-LUB-EOS200-12
- PLC Hex Register Address: DB31.DBW_0xA3A0
- Nominal Operating Range: 48 - 76 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 106 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS200-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS200-INJ-MET-4212
- Overhaul Seal & Consumable Kit: KIT-EOS200-LUB-12
- Critical Fastener Torque Value: 118 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M200 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS200-LUB-12 at PLC register 0xA3A0.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on EOS-Additive-M200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS200-INJ-MET-4212) with seal kit (KIT-EOS200-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 118 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS200-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M200, technician must log: "Resolved ERR-EOS200-LUB-12 on EOS-Additive-M200 by installing PN-EOS200-INJ-MET-4212 & kit KIT-EOS200-LUB-12. Fasteners torqued to 118 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS200-VAC-13

Target Machine Model: EOS-Additive-M200 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M200 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M200 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: VC-MNF-EOS200-13
- PLC Hex Register Address: DB31.DBW_0xA3A8
- Nominal Operating Range: 51 - 76 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 106 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS200-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS200-GASK-VIT-4213
- Overhaul Seal & Consumable Kit: KIT-EOS200-VAC-13
- Critical Fastener Torque Value: 127 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M200 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS200-VAC-13 at PLC register 0xA3A8.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on EOS-Additive-M200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS200-GASK-VIT-4213) with seal kit (KIT-EOS200-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 127 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS200-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M200, technician must log: "Resolved ERR-EOS200-VAC-13 on EOS-Additive-M200 by installing PN-EOS200-GASK-VIT-4213 & kit KIT-EOS200-VAC-13. Fasteners torqued to 127 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS200-THM-14

Target Machine Model: EOS-Additive-M200 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M200 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on EOS-Additive-M200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M200 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: TM-CHL-EOS200-14
- PLC Hex Register Address: DB31.DBW_0xA3B0
- Nominal Operating Range: 54 - 76 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 106 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS200-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS200-HEX-CORE-4214
- Overhaul Seal & Consumable Kit: KIT-EOS200-THM-14
- Critical Fastener Torque Value: 136 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M200 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS200-THM-14 at PLC register 0xA3B0.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on EOS-Additive-M200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS200-HEX-CORE-4214) with seal kit (KIT-EOS200-THM-14).

Step 6: Torque retaining bolts in cross sequence to 136 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS200-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M200, technician must log: "Resolved ERR-EOS200-THM-14 on EOS-Additive-M200 by installing PN-EOS200-HEX-CORE-4214 & kit KIT-EOS200-THM-14. Fasteners torqued to 136 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS250-SPN-1

Target Machine Model: EOS-Additive-M250 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M250 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on EOS-Additive-M250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M250 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: PT-SPN-EOS250-1
- PLC Hex Register Address: DB32.DBW_0xA668
- Nominal Operating Range: 15 - 77 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 107 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS250-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS250-BRG-CER-4301
- Overhaul Seal & Consumable Kit: KIT-EOS250-SPN-1
- Critical Fastener Torque Value: 159 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M250 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS250-SPN-1 at PLC register 0xA668.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on EOS-Additive-M250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS250-BRG-CER-4301) with seal kit (KIT-EOS250-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 159 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS250-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M250, technician must log: "Resolved ERR-EOS250-SPN-1 on EOS-Additive-M250 by installing PN-EOS250-BRG-CER-4301 & kit KIT-EOS250-SPN-1. Fasteners torqued to 159 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS250-ATC-2

Target Machine Model: EOS-Additive-M250 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M250 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on EOS-Additive-M250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M250 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: PX-ATC-EOS250-2
- PLC Hex Register Address: DB32.DBW_0xA670
- Nominal Operating Range: 18 - 77 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 107 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS250-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS250-CAM-GRIP-4302
- Overhaul Seal & Consumable Kit: KIT-EOS250-ATC-2
- Critical Fastener Torque Value: 168 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M250 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS250-ATC-2 at PLC register 0xA670.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on EOS-Additive-M250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS250-CAM-GRIP-4302) with seal kit (KIT-EOS250-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 168 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS250-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M250, technician must log: "Resolved ERR-EOS250-ATC-2 on EOS-Additive-M250 by installing PN-EOS250-CAM-GRIP-4302 & kit KIT-EOS250-ATC-2. Fasteners torqued to 168 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS250-HYD-3

Target Machine Model: EOS-Additive-M250 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M250 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on EOS-Additive-M250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M250 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: PR-HYD-EOS250-3
- PLC Hex Register Address: DB32.DBW_0xA678
- Nominal Operating Range: 21 - 77 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 107 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS250-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS250-VLV-PROP-4303
- Overhaul Seal & Consumable Kit: KIT-EOS250-HYD-3
- Critical Fastener Torque Value: 177 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M250 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS250-HYD-3 at PLC register 0xA678.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on EOS-Additive-M250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS250-VLV-PROP-4303) with seal kit (KIT-EOS250-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 177 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS250-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M250, technician must log: "Resolved ERR-EOS250-HYD-3 on EOS-Additive-M250 by installing PN-EOS250-VLV-PROP-4303 & kit KIT-EOS250-HYD-3. Fasteners torqued to 177 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS250-CLN-4

Target Machine Model: EOS-Additive-M250 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M250 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on EOS-Additive-M250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M250 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: FL-CLN-EOS250-4
- PLC Hex Register Address: DB32.DBW_0xA680
- Nominal Operating Range: 24 - 77 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 107 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS250-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS250-PMP-PLG-4304
- Overhaul Seal & Consumable Kit: KIT-EOS250-CLN-4
- Critical Fastener Torque Value: 186 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M250 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS250-CLN-4 at PLC register 0xA680.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on EOS-Additive-M250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS250-PMP-PLG-4304) with seal kit (KIT-EOS250-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 186 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS250-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M250, technician must log: "Resolved ERR-EOS250-CLN-4 on EOS-Additive-M250 by installing PN-EOS250-PMP-PLG-4304 & kit KIT-EOS250-CLN-4. Fasteners torqued to 186 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS250-XAX-5

Target Machine Model: EOS-Additive-M250 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M250 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M250 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: EN-XAX-EOS250-5
- PLC Hex Register Address: DB32.DBW_0xA688
- Nominal Operating Range: 27 - 77 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 107 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS250-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS250-LIN-COIL-4305
- Overhaul Seal & Consumable Kit: KIT-EOS250-XAX-5
- Critical Fastener Torque Value: 195 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M250 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS250-XAX-5 at PLC register 0xA688.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on EOS-Additive-M250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS250-LIN-COIL-4305) with seal kit (KIT-EOS250-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 195 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS250-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M250, technician must log: "Resolved ERR-EOS250-XAX-5 on EOS-Additive-M250 by installing PN-EOS250-LIN-COIL-4305 & kit KIT-EOS250-XAX-5. Fasteners torqued to 195 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS250-YAX-6

Target Machine Model: EOS-Additive-M250 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M250 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on EOS-Additive-M250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M250 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: EN-YAX-EOS250-6
- PLC Hex Register Address: DB32.DBW_0xA690
- Nominal Operating Range: 30 - 77 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 107 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS250-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS250-BSW-NUT-4306
- Overhaul Seal & Consumable Kit: KIT-EOS250-YAX-6
- Critical Fastener Torque Value: 204 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M250 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS250-YAX-6 at PLC register 0xA690.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on EOS-Additive-M250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS250-BSW-NUT-4306) with seal kit (KIT-EOS250-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 204 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS250-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M250, technician must log: "Resolved ERR-EOS250-YAX-6 on EOS-Additive-M250 by installing PN-EOS250-BSW-NUT-4306 & kit KIT-EOS250-YAX-6. Fasteners torqued to 204 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS250-ZAX-7

Target Machine Model: EOS-Additive-M250 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M250 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M250 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: PR-ZAX-EOS250-7
- PLC Hex Register Address: DB32.DBW_0xA698
- Nominal Operating Range: 33 - 77 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 107 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS250-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS250-CYL-SEAL-4307
- Overhaul Seal & Consumable Kit: KIT-EOS250-ZAX-7
- Critical Fastener Torque Value: 23 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M250 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS250-ZAX-7 at PLC register 0xA698.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on EOS-Additive-M250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS250-CYL-SEAL-4307) with seal kit (KIT-EOS250-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 23 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS250-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M250, technician must log: "Resolved ERR-EOS250-ZAX-7 on EOS-Additive-M250 by installing PN-EOS250-CYL-SEAL-4307 & kit KIT-EOS250-ZAX-7. Fasteners torqued to 23 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS250-BAX-8

Target Machine Model: EOS-Additive-M250 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M250 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on EOS-Additive-M250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M250 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: EN-BAX-EOS250-8
- PLC Hex Register Address: DB32.DBW_0xA6A0
- Nominal Operating Range: 36 - 77 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 107 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS250-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS250-TRQ-STAT-4308
- Overhaul Seal & Consumable Kit: KIT-EOS250-BAX-8
- Critical Fastener Torque Value: 32 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M250 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS250-BAX-8 at PLC register 0xA6A0.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on EOS-Additive-M250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS250-TRQ-STAT-4308) with seal kit (KIT-EOS250-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 32 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS250-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M250, technician must log: "Resolved ERR-EOS250-BAX-8 on EOS-Additive-M250 by installing PN-EOS250-TRQ-STAT-4308 & kit KIT-EOS250-BAX-8. Fasteners torqued to 32 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS250-CAX-9

Target Machine Model: EOS-Additive-M250 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M250 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M250 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: PR-CAX-EOS250-9
- PLC Hex Register Address: DB32.DBW_0xA6A8
- Nominal Operating Range: 39 - 77 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 107 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS250-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS250-BRK-DISC-4309
- Overhaul Seal & Consumable Kit: KIT-EOS250-CAX-9
- Critical Fastener Torque Value: 41 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M250 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS250-CAX-9 at PLC register 0xA6A8.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on EOS-Additive-M250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS250-BRK-DISC-4309) with seal kit (KIT-EOS250-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 41 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS250-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M250, technician must log: "Resolved ERR-EOS250-CAX-9 on EOS-Additive-M250 by installing PN-EOS250-BRK-DISC-4309 & kit KIT-EOS250-CAX-9. Fasteners torqued to 41 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS250-OPT-10

Target Machine Model: EOS-Additive-M250 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M250 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on EOS-Additive-M250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M250 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: OP-SCL-EOS250-10
- PLC Hex Register Address: DB32.DBW_0xA6B0
- Nominal Operating Range: 42 - 77 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 107 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS250-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS250-GLASS-RD-4310
- Overhaul Seal & Consumable Kit: KIT-EOS250-OPT-10
- Critical Fastener Torque Value: 50 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M250 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS250-OPT-10 at PLC register 0xA6B0.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on EOS-Additive-M250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS250-GLASS-RD-4310) with seal kit (KIT-EOS250-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 50 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS250-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M250, technician must log: "Resolved ERR-EOS250-OPT-10 on EOS-Additive-M250 by installing PN-EOS250-GLASS-RD-4310 & kit KIT-EOS250-OPT-10. Fasteners torqued to 50 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS250-PLC-11

Target Machine Model: EOS-Additive-M250 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M250 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M250 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: VT-DRV-EOS250-11
- PLC Hex Register Address: DB32.DBW_0xA6B8
- Nominal Operating Range: 45 - 77 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 107 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS250-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS250-IGBT-MOD-4311
- Overhaul Seal & Consumable Kit: KIT-EOS250-PLC-11
- Critical Fastener Torque Value: 59 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M250 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS250-PLC-11 at PLC register 0xA6B8.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on EOS-Additive-M250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS250-IGBT-MOD-4311) with seal kit (KIT-EOS250-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 59 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS250-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M250, technician must log: "Resolved ERR-EOS250-PLC-11 on EOS-Additive-M250 by installing PN-EOS250-IGBT-MOD-4311 & kit KIT-EOS250-PLC-11. Fasteners torqued to 59 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS250-LUB-12

Target Machine Model: EOS-Additive-M250 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M250 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on EOS-Additive-M250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M250 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: FL-LUB-EOS250-12
- PLC Hex Register Address: DB32.DBW_0xA6C0
- Nominal Operating Range: 48 - 77 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 107 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS250-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS250-INJ-MET-4312
- Overhaul Seal & Consumable Kit: KIT-EOS250-LUB-12
- Critical Fastener Torque Value: 68 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M250 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS250-LUB-12 at PLC register 0xA6C0.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on EOS-Additive-M250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS250-INJ-MET-4312) with seal kit (KIT-EOS250-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 68 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS250-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M250, technician must log: "Resolved ERR-EOS250-LUB-12 on EOS-Additive-M250 by installing PN-EOS250-INJ-MET-4312 & kit KIT-EOS250-LUB-12. Fasteners torqued to 68 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS250-VAC-13

Target Machine Model: EOS-Additive-M250 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M250 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M250 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: VC-MNF-EOS250-13
- PLC Hex Register Address: DB32.DBW_0xA6C8
- Nominal Operating Range: 51 - 77 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 107 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS250-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS250-GASK-VIT-4313
- Overhaul Seal & Consumable Kit: KIT-EOS250-VAC-13
- Critical Fastener Torque Value: 77 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M250 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS250-VAC-13 at PLC register 0xA6C8.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on EOS-Additive-M250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS250-GASK-VIT-4313) with seal kit (KIT-EOS250-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 77 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS250-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M250, technician must log: "Resolved ERR-EOS250-VAC-13 on EOS-Additive-M250 by installing PN-EOS250-GASK-VIT-4313 & kit KIT-EOS250-VAC-13. Fasteners torqued to 77 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS250-THM-14

Target Machine Model: EOS-Additive-M250 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M250 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on EOS-Additive-M250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M250 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: TM-CHL-EOS250-14
- PLC Hex Register Address: DB32.DBW_0xA6D0
- Nominal Operating Range: 54 - 77 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 107 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS250-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS250-HEX-CORE-4314
- Overhaul Seal & Consumable Kit: KIT-EOS250-THM-14
- Critical Fastener Torque Value: 86 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M250 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS250-THM-14 at PLC register 0xA6D0.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on EOS-Additive-M250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS250-HEX-CORE-4314) with seal kit (KIT-EOS250-THM-14).

Step 6: Torque retaining bolts in cross sequence to 86 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS250-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M250, technician must log: "Resolved ERR-EOS250-THM-14 on EOS-Additive-M250 by installing PN-EOS250-HEX-CORE-4314 & kit KIT-EOS250-THM-14. Fasteners torqued to 86 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS300-SPN-1

Target Machine Model: EOS-Additive-M300 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M300 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on EOS-Additive-M300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M300 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: PT-SPN-EOS300-1
- PLC Hex Register Address: DB33.DBW_0xA988
- Nominal Operating Range: 15 - 78 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 108 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS300-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS300-BRG-CER-4401
- Overhaul Seal & Consumable Kit: KIT-EOS300-SPN-1
- Critical Fastener Torque Value: 109 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M300 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS300-SPN-1 at PLC register 0xA988.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on EOS-Additive-M300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS300-BRG-CER-4401) with seal kit (KIT-EOS300-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 109 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS300-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M300, technician must log: "Resolved ERR-EOS300-SPN-1 on EOS-Additive-M300 by installing PN-EOS300-BRG-CER-4401 & kit KIT-EOS300-SPN-1. Fasteners torqued to 109 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS300-ATC-2

Target Machine Model: EOS-Additive-M300 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M300 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on EOS-Additive-M300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M300 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: PX-ATC-EOS300-2
- PLC Hex Register Address: DB33.DBW_0xA990
- Nominal Operating Range: 18 - 78 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 108 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS300-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS300-CAM-GRIP-4402
- Overhaul Seal & Consumable Kit: KIT-EOS300-ATC-2
- Critical Fastener Torque Value: 118 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M300 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS300-ATC-2 at PLC register 0xA990.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on EOS-Additive-M300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS300-CAM-GRIP-4402) with seal kit (KIT-EOS300-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 118 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS300-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M300, technician must log: "Resolved ERR-EOS300-ATC-2 on EOS-Additive-M300 by installing PN-EOS300-CAM-GRIP-4402 & kit KIT-EOS300-ATC-2. Fasteners torqued to 118 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS300-HYD-3

Target Machine Model: EOS-Additive-M300 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M300 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on EOS-Additive-M300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M300 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: PR-HYD-EOS300-3
- PLC Hex Register Address: DB33.DBW_0xA998
- Nominal Operating Range: 21 - 78 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 108 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS300-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS300-VLV-PROP-4403
- Overhaul Seal & Consumable Kit: KIT-EOS300-HYD-3
- Critical Fastener Torque Value: 127 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M300 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS300-HYD-3 at PLC register 0xA998.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on EOS-Additive-M300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS300-VLV-PROP-4403) with seal kit (KIT-EOS300-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 127 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS300-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M300, technician must log: "Resolved ERR-EOS300-HYD-3 on EOS-Additive-M300 by installing PN-EOS300-VLV-PROP-4403 & kit KIT-EOS300-HYD-3. Fasteners torqued to 127 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS300-CLN-4

Target Machine Model: EOS-Additive-M300 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M300 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on EOS-Additive-M300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M300 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: FL-CLN-EOS300-4
- PLC Hex Register Address: DB33.DBW_0xA9A0
- Nominal Operating Range: 24 - 78 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 108 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS300-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS300-PMP-PLG-4404
- Overhaul Seal & Consumable Kit: KIT-EOS300-CLN-4
- Critical Fastener Torque Value: 136 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M300 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS300-CLN-4 at PLC register 0xA9A0.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on EOS-Additive-M300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS300-PMP-PLG-4404) with seal kit (KIT-EOS300-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 136 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS300-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M300, technician must log: "Resolved ERR-EOS300-CLN-4 on EOS-Additive-M300 by installing PN-EOS300-PMP-PLG-4404 & kit KIT-EOS300-CLN-4. Fasteners torqued to 136 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS300-XAX-5

Target Machine Model: EOS-Additive-M300 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M300 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M300 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: EN-XAX-EOS300-5
- PLC Hex Register Address: DB33.DBW_0xA9A8
- Nominal Operating Range: 27 - 78 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 108 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS300-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS300-LIN-COIL-4405
- Overhaul Seal & Consumable Kit: KIT-EOS300-XAX-5
- Critical Fastener Torque Value: 145 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M300 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS300-XAX-5 at PLC register 0xA9A8.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on EOS-Additive-M300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS300-LIN-COIL-4405) with seal kit (KIT-EOS300-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 145 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS300-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M300, technician must log: "Resolved ERR-EOS300-XAX-5 on EOS-Additive-M300 by installing PN-EOS300-LIN-COIL-4405 & kit KIT-EOS300-XAX-5. Fasteners torqued to 145 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS300-YAX-6

Target Machine Model: EOS-Additive-M300 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M300 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on EOS-Additive-M300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M300 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: EN-YAX-EOS300-6
- PLC Hex Register Address: DB33.DBW_0xA9B0
- Nominal Operating Range: 30 - 78 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 108 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS300-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS300-BSW-NUT-4406
- Overhaul Seal & Consumable Kit: KIT-EOS300-YAX-6
- Critical Fastener Torque Value: 154 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M300 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS300-YAX-6 at PLC register 0xA9B0.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on EOS-Additive-M300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS300-BSW-NUT-4406) with seal kit (KIT-EOS300-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 154 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS300-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M300, technician must log: "Resolved ERR-EOS300-YAX-6 on EOS-Additive-M300 by installing PN-EOS300-BSW-NUT-4406 & kit KIT-EOS300-YAX-6. Fasteners torqued to 154 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS300-ZAX-7

Target Machine Model: EOS-Additive-M300 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M300 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M300 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: PR-ZAX-EOS300-7
- PLC Hex Register Address: DB33.DBW_0xA9B8
- Nominal Operating Range: 33 - 78 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 108 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS300-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS300-CYL-SEAL-4407
- Overhaul Seal & Consumable Kit: KIT-EOS300-ZAX-7
- Critical Fastener Torque Value: 163 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M300 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS300-ZAX-7 at PLC register 0xA9B8.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on EOS-Additive-M300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS300-CYL-SEAL-4407) with seal kit (KIT-EOS300-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 163 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS300-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M300, technician must log: "Resolved ERR-EOS300-ZAX-7 on EOS-Additive-M300 by installing PN-EOS300-CYL-SEAL-4407 & kit KIT-EOS300-ZAX-7. Fasteners torqued to 163 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS300-BAX-8

Target Machine Model: EOS-Additive-M300 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M300 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on EOS-Additive-M300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M300 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: EN-BAX-EOS300-8
- PLC Hex Register Address: DB33.DBW_0xA9C0
- Nominal Operating Range: 36 - 78 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 108 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS300-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS300-TRQ-STAT-4408
- Overhaul Seal & Consumable Kit: KIT-EOS300-BAX-8
- Critical Fastener Torque Value: 172 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M300 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS300-BAX-8 at PLC register 0xA9C0.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on EOS-Additive-M300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS300-TRQ-STAT-4408) with seal kit (KIT-EOS300-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 172 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS300-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M300, technician must log: "Resolved ERR-EOS300-BAX-8 on EOS-Additive-M300 by installing PN-EOS300-TRQ-STAT-4408 & kit KIT-EOS300-BAX-8. Fasteners torqued to 172 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS300-CAX-9

Target Machine Model: EOS-Additive-M300 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M300 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M300 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: PR-CAX-EOS300-9
- PLC Hex Register Address: DB33.DBW_0xA9C8
- Nominal Operating Range: 39 - 78 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 108 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS300-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS300-BRK-DISC-4409
- Overhaul Seal & Consumable Kit: KIT-EOS300-CAX-9
- Critical Fastener Torque Value: 181 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M300 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS300-CAX-9 at PLC register 0xA9C8.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on EOS-Additive-M300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS300-BRK-DISC-4409) with seal kit (KIT-EOS300-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 181 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS300-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M300, technician must log: "Resolved ERR-EOS300-CAX-9 on EOS-Additive-M300 by installing PN-EOS300-BRK-DISC-4409 & kit KIT-EOS300-CAX-9. Fasteners torqued to 181 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS300-OPT-10

Target Machine Model: EOS-Additive-M300 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M300 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on EOS-Additive-M300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M300 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: OP-SCL-EOS300-10
- PLC Hex Register Address: DB33.DBW_0xA9D0
- Nominal Operating Range: 42 - 78 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 108 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS300-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS300-GLASS-RD-4410
- Overhaul Seal & Consumable Kit: KIT-EOS300-OPT-10
- Critical Fastener Torque Value: 190 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M300 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS300-OPT-10 at PLC register 0xA9D0.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on EOS-Additive-M300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS300-GLASS-RD-4410) with seal kit (KIT-EOS300-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 190 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS300-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M300, technician must log: "Resolved ERR-EOS300-OPT-10 on EOS-Additive-M300 by installing PN-EOS300-GLASS-RD-4410 & kit KIT-EOS300-OPT-10. Fasteners torqued to 190 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS300-PLC-11

Target Machine Model: EOS-Additive-M300 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M300 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M300 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: VT-DRV-EOS300-11
- PLC Hex Register Address: DB33.DBW_0xA9D8
- Nominal Operating Range: 45 - 78 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 108 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS300-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS300-IGBT-MOD-4411
- Overhaul Seal & Consumable Kit: KIT-EOS300-PLC-11
- Critical Fastener Torque Value: 199 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M300 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS300-PLC-11 at PLC register 0xA9D8.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on EOS-Additive-M300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS300-IGBT-MOD-4411) with seal kit (KIT-EOS300-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 199 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS300-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M300, technician must log: "Resolved ERR-EOS300-PLC-11 on EOS-Additive-M300 by installing PN-EOS300-IGBT-MOD-4411 & kit KIT-EOS300-PLC-11. Fasteners torqued to 199 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS300-LUB-12

Target Machine Model: EOS-Additive-M300 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M300 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on EOS-Additive-M300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M300 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: FL-LUB-EOS300-12
- PLC Hex Register Address: DB33.DBW_0xA9E0
- Nominal Operating Range: 48 - 78 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 108 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS300-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS300-INJ-MET-4412
- Overhaul Seal & Consumable Kit: KIT-EOS300-LUB-12
- Critical Fastener Torque Value: 208 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M300 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS300-LUB-12 at PLC register 0xA9E0.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on EOS-Additive-M300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS300-INJ-MET-4412) with seal kit (KIT-EOS300-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 208 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS300-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M300, technician must log: "Resolved ERR-EOS300-LUB-12 on EOS-Additive-M300 by installing PN-EOS300-INJ-MET-4412 & kit KIT-EOS300-LUB-12. Fasteners torqued to 208 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS300-VAC-13

Target Machine Model: EOS-Additive-M300 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M300 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M300 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: VC-MNF-EOS300-13
- PLC Hex Register Address: DB33.DBW_0xA9E8
- Nominal Operating Range: 51 - 78 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 108 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS300-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS300-GASK-VIT-4413
- Overhaul Seal & Consumable Kit: KIT-EOS300-VAC-13
- Critical Fastener Torque Value: 27 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M300 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS300-VAC-13 at PLC register 0xA9E8.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on EOS-Additive-M300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS300-GASK-VIT-4413) with seal kit (KIT-EOS300-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 27 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS300-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M300, technician must log: "Resolved ERR-EOS300-VAC-13 on EOS-Additive-M300 by installing PN-EOS300-GASK-VIT-4413 & kit KIT-EOS300-VAC-13. Fasteners torqued to 27 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS300-THM-14

Target Machine Model: EOS-Additive-M300 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M300 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on EOS-Additive-M300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M300 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: TM-CHL-EOS300-14
- PLC Hex Register Address: DB33.DBW_0xA9F0
- Nominal Operating Range: 54 - 78 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 108 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS300-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS300-HEX-CORE-4414
- Overhaul Seal & Consumable Kit: KIT-EOS300-THM-14
- Critical Fastener Torque Value: 36 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M300 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS300-THM-14 at PLC register 0xA9F0.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on EOS-Additive-M300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS300-HEX-CORE-4414) with seal kit (KIT-EOS300-THM-14).

Step 6: Torque retaining bolts in cross sequence to 36 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS300-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M300, technician must log: "Resolved ERR-EOS300-THM-14 on EOS-Additive-M300 by installing PN-EOS300-HEX-CORE-4414 & kit KIT-EOS300-THM-14. Fasteners torqued to 36 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS350-SPN-1

Target Machine Model: EOS-Additive-M350 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M350 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on EOS-Additive-M350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M350 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: PT-SPN-EOS350-1
- PLC Hex Register Address: DB34.DBW_0xACA8
- Nominal Operating Range: 15 - 79 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 109 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS350-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS350-BRG-CER-4501
- Overhaul Seal & Consumable Kit: KIT-EOS350-SPN-1
- Critical Fastener Torque Value: 59 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M350 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS350-SPN-1 at PLC register 0xACA8.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on EOS-Additive-M350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS350-BRG-CER-4501) with seal kit (KIT-EOS350-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 59 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS350-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M350, technician must log: "Resolved ERR-EOS350-SPN-1 on EOS-Additive-M350 by installing PN-EOS350-BRG-CER-4501 & kit KIT-EOS350-SPN-1. Fasteners torqued to 59 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS350-ATC-2

Target Machine Model: EOS-Additive-M350 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M350 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on EOS-Additive-M350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M350 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: PX-ATC-EOS350-2
- PLC Hex Register Address: DB34.DBW_0xACB0
- Nominal Operating Range: 18 - 79 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 109 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS350-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS350-CAM-GRIP-4502
- Overhaul Seal & Consumable Kit: KIT-EOS350-ATC-2
- Critical Fastener Torque Value: 68 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M350 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS350-ATC-2 at PLC register 0xACB0.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on EOS-Additive-M350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS350-CAM-GRIP-4502) with seal kit (KIT-EOS350-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 68 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS350-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M350, technician must log: "Resolved ERR-EOS350-ATC-2 on EOS-Additive-M350 by installing PN-EOS350-CAM-GRIP-4502 & kit KIT-EOS350-ATC-2. Fasteners torqued to 68 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS350-HYD-3

Target Machine Model: EOS-Additive-M350 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M350 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on EOS-Additive-M350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M350 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: PR-HYD-EOS350-3
- PLC Hex Register Address: DB34.DBW_0xACB8
- Nominal Operating Range: 21 - 79 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 109 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS350-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS350-VLV-PROP-4503
- Overhaul Seal & Consumable Kit: KIT-EOS350-HYD-3
- Critical Fastener Torque Value: 77 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M350 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS350-HYD-3 at PLC register 0xACB8.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on EOS-Additive-M350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS350-VLV-PROP-4503) with seal kit (KIT-EOS350-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 77 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS350-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M350, technician must log: "Resolved ERR-EOS350-HYD-3 on EOS-Additive-M350 by installing PN-EOS350-VLV-PROP-4503 & kit KIT-EOS350-HYD-3. Fasteners torqued to 77 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS350-CLN-4

Target Machine Model: EOS-Additive-M350 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M350 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on EOS-Additive-M350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M350 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: FL-CLN-EOS350-4
- PLC Hex Register Address: DB34.DBW_0xACC0
- Nominal Operating Range: 24 - 79 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 109 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS350-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS350-PMP-PLG-4504
- Overhaul Seal & Consumable Kit: KIT-EOS350-CLN-4
- Critical Fastener Torque Value: 86 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M350 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS350-CLN-4 at PLC register 0xACC0.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on EOS-Additive-M350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS350-PMP-PLG-4504) with seal kit (KIT-EOS350-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 86 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS350-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M350, technician must log: "Resolved ERR-EOS350-CLN-4 on EOS-Additive-M350 by installing PN-EOS350-PMP-PLG-4504 & kit KIT-EOS350-CLN-4. Fasteners torqued to 86 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS350-XAX-5

Target Machine Model: EOS-Additive-M350 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M350 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M350 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: EN-XAX-EOS350-5
- PLC Hex Register Address: DB34.DBW_0xACC8
- Nominal Operating Range: 27 - 79 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 109 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS350-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS350-LIN-COIL-4505
- Overhaul Seal & Consumable Kit: KIT-EOS350-XAX-5
- Critical Fastener Torque Value: 95 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M350 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS350-XAX-5 at PLC register 0xACC8.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on EOS-Additive-M350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS350-LIN-COIL-4505) with seal kit (KIT-EOS350-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 95 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS350-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M350, technician must log: "Resolved ERR-EOS350-XAX-5 on EOS-Additive-M350 by installing PN-EOS350-LIN-COIL-4505 & kit KIT-EOS350-XAX-5. Fasteners torqued to 95 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS350-YAX-6

Target Machine Model: EOS-Additive-M350 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M350 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on EOS-Additive-M350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M350 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: EN-YAX-EOS350-6
- PLC Hex Register Address: DB34.DBW_0xACD0
- Nominal Operating Range: 30 - 79 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 109 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS350-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS350-BSW-NUT-4506
- Overhaul Seal & Consumable Kit: KIT-EOS350-YAX-6
- Critical Fastener Torque Value: 104 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M350 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS350-YAX-6 at PLC register 0xACD0.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on EOS-Additive-M350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS350-BSW-NUT-4506) with seal kit (KIT-EOS350-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 104 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS350-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M350, technician must log: "Resolved ERR-EOS350-YAX-6 on EOS-Additive-M350 by installing PN-EOS350-BSW-NUT-4506 & kit KIT-EOS350-YAX-6. Fasteners torqued to 104 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS350-ZAX-7

Target Machine Model: EOS-Additive-M350 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M350 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M350 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: PR-ZAX-EOS350-7
- PLC Hex Register Address: DB34.DBW_0xACD8
- Nominal Operating Range: 33 - 79 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 109 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS350-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS350-CYL-SEAL-4507
- Overhaul Seal & Consumable Kit: KIT-EOS350-ZAX-7
- Critical Fastener Torque Value: 113 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M350 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS350-ZAX-7 at PLC register 0xACD8.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on EOS-Additive-M350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS350-CYL-SEAL-4507) with seal kit (KIT-EOS350-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 113 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS350-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M350, technician must log: "Resolved ERR-EOS350-ZAX-7 on EOS-Additive-M350 by installing PN-EOS350-CYL-SEAL-4507 & kit KIT-EOS350-ZAX-7. Fasteners torqued to 113 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS350-BAX-8

Target Machine Model: EOS-Additive-M350 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M350 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on EOS-Additive-M350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M350 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: EN-BAX-EOS350-8
- PLC Hex Register Address: DB34.DBW_0xACE0
- Nominal Operating Range: 36 - 79 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 109 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS350-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS350-TRQ-STAT-4508
- Overhaul Seal & Consumable Kit: KIT-EOS350-BAX-8
- Critical Fastener Torque Value: 122 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M350 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS350-BAX-8 at PLC register 0xACE0.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on EOS-Additive-M350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS350-TRQ-STAT-4508) with seal kit (KIT-EOS350-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 122 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS350-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M350, technician must log: "Resolved ERR-EOS350-BAX-8 on EOS-Additive-M350 by installing PN-EOS350-TRQ-STAT-4508 & kit KIT-EOS350-BAX-8. Fasteners torqued to 122 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS350-CAX-9

Target Machine Model: EOS-Additive-M350 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M350 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M350 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: PR-CAX-EOS350-9
- PLC Hex Register Address: DB34.DBW_0xACE8
- Nominal Operating Range: 39 - 79 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 109 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS350-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS350-BRK-DISC-4509
- Overhaul Seal & Consumable Kit: KIT-EOS350-CAX-9
- Critical Fastener Torque Value: 131 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M350 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS350-CAX-9 at PLC register 0xACE8.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on EOS-Additive-M350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS350-BRK-DISC-4509) with seal kit (KIT-EOS350-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 131 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS350-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M350, technician must log: "Resolved ERR-EOS350-CAX-9 on EOS-Additive-M350 by installing PN-EOS350-BRK-DISC-4509 & kit KIT-EOS350-CAX-9. Fasteners torqued to 131 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS350-OPT-10

Target Machine Model: EOS-Additive-M350 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M350 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on EOS-Additive-M350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M350 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: OP-SCL-EOS350-10
- PLC Hex Register Address: DB34.DBW_0xACF0
- Nominal Operating Range: 42 - 79 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 109 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS350-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS350-GLASS-RD-4510
- Overhaul Seal & Consumable Kit: KIT-EOS350-OPT-10
- Critical Fastener Torque Value: 140 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M350 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS350-OPT-10 at PLC register 0xACF0.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on EOS-Additive-M350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS350-GLASS-RD-4510) with seal kit (KIT-EOS350-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 140 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS350-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M350, technician must log: "Resolved ERR-EOS350-OPT-10 on EOS-Additive-M350 by installing PN-EOS350-GLASS-RD-4510 & kit KIT-EOS350-OPT-10. Fasteners torqued to 140 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS350-PLC-11

Target Machine Model: EOS-Additive-M350 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M350 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M350 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: VT-DRV-EOS350-11
- PLC Hex Register Address: DB34.DBW_0xACF8
- Nominal Operating Range: 45 - 79 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 109 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS350-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS350-IGBT-MOD-4511
- Overhaul Seal & Consumable Kit: KIT-EOS350-PLC-11
- Critical Fastener Torque Value: 149 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M350 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS350-PLC-11 at PLC register 0xACF8.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on EOS-Additive-M350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS350-IGBT-MOD-4511) with seal kit (KIT-EOS350-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 149 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS350-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M350, technician must log: "Resolved ERR-EOS350-PLC-11 on EOS-Additive-M350 by installing PN-EOS350-IGBT-MOD-4511 & kit KIT-EOS350-PLC-11. Fasteners torqued to 149 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS350-LUB-12

Target Machine Model: EOS-Additive-M350 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M350 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on EOS-Additive-M350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M350 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: FL-LUB-EOS350-12
- PLC Hex Register Address: DB34.DBW_0xAD00
- Nominal Operating Range: 48 - 79 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 109 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS350-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS350-INJ-MET-4512
- Overhaul Seal & Consumable Kit: KIT-EOS350-LUB-12
- Critical Fastener Torque Value: 158 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M350 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS350-LUB-12 at PLC register 0xAD00.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on EOS-Additive-M350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS350-INJ-MET-4512) with seal kit (KIT-EOS350-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 158 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS350-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M350, technician must log: "Resolved ERR-EOS350-LUB-12 on EOS-Additive-M350 by installing PN-EOS350-INJ-MET-4512 & kit KIT-EOS350-LUB-12. Fasteners torqued to 158 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS350-VAC-13

Target Machine Model: EOS-Additive-M350 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M350 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M350 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: VC-MNF-EOS350-13
- PLC Hex Register Address: DB34.DBW_0xAD08
- Nominal Operating Range: 51 - 79 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 109 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS350-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS350-GASK-VIT-4513
- Overhaul Seal & Consumable Kit: KIT-EOS350-VAC-13
- Critical Fastener Torque Value: 167 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M350 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS350-VAC-13 at PLC register 0xAD08.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on EOS-Additive-M350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS350-GASK-VIT-4513) with seal kit (KIT-EOS350-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 167 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS350-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M350, technician must log: "Resolved ERR-EOS350-VAC-13 on EOS-Additive-M350 by installing PN-EOS350-GASK-VIT-4513 & kit KIT-EOS350-VAC-13. Fasteners torqued to 167 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS350-THM-14

Target Machine Model: EOS-Additive-M350 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M350 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on EOS-Additive-M350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M350 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: TM-CHL-EOS350-14
- PLC Hex Register Address: DB34.DBW_0xAD10
- Nominal Operating Range: 54 - 79 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 109 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS350-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS350-HEX-CORE-4514
- Overhaul Seal & Consumable Kit: KIT-EOS350-THM-14
- Critical Fastener Torque Value: 176 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M350 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS350-THM-14 at PLC register 0xAD10.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on EOS-Additive-M350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS350-HEX-CORE-4514) with seal kit (KIT-EOS350-THM-14).

Step 6: Torque retaining bolts in cross sequence to 176 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS350-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M350, technician must log: "Resolved ERR-EOS350-THM-14 on EOS-Additive-M350 by installing PN-EOS350-HEX-CORE-4514 & kit KIT-EOS350-THM-14. Fasteners torqued to 176 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS400-SPN-1

Target Machine Model: EOS-Additive-M400 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M400 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on EOS-Additive-M400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M400 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: PT-SPN-EOS400-1
- PLC Hex Register Address: DB20.DBW_0xAFC8
- Nominal Operating Range: 15 - 80 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 110 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS400-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS400-BRG-CER-4601
- Overhaul Seal & Consumable Kit: KIT-EOS400-SPN-1
- Critical Fastener Torque Value: 199 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M400 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS400-SPN-1 at PLC register 0xAFC8.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on EOS-Additive-M400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS400-BRG-CER-4601) with seal kit (KIT-EOS400-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 199 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS400-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M400, technician must log: "Resolved ERR-EOS400-SPN-1 on EOS-Additive-M400 by installing PN-EOS400-BRG-CER-4601 & kit KIT-EOS400-SPN-1. Fasteners torqued to 199 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS400-ATC-2

Target Machine Model: EOS-Additive-M400 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M400 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on EOS-Additive-M400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M400 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: PX-ATC-EOS400-2
- PLC Hex Register Address: DB20.DBW_0xAFD0
- Nominal Operating Range: 18 - 80 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 110 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS400-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS400-CAM-GRIP-4602
- Overhaul Seal & Consumable Kit: KIT-EOS400-ATC-2
- Critical Fastener Torque Value: 208 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M400 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS400-ATC-2 at PLC register 0xAFD0.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on EOS-Additive-M400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS400-CAM-GRIP-4602) with seal kit (KIT-EOS400-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 208 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS400-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M400, technician must log: "Resolved ERR-EOS400-ATC-2 on EOS-Additive-M400 by installing PN-EOS400-CAM-GRIP-4602 & kit KIT-EOS400-ATC-2. Fasteners torqued to 208 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS400-HYD-3

Target Machine Model: EOS-Additive-M400 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M400 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on EOS-Additive-M400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M400 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: PR-HYD-EOS400-3
- PLC Hex Register Address: DB20.DBW_0xAFD8
- Nominal Operating Range: 21 - 80 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 110 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS400-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS400-VLV-PROP-4603
- Overhaul Seal & Consumable Kit: KIT-EOS400-HYD-3
- Critical Fastener Torque Value: 27 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M400 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS400-HYD-3 at PLC register 0xAFD8.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on EOS-Additive-M400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS400-VLV-PROP-4603) with seal kit (KIT-EOS400-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 27 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS400-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M400, technician must log: "Resolved ERR-EOS400-HYD-3 on EOS-Additive-M400 by installing PN-EOS400-VLV-PROP-4603 & kit KIT-EOS400-HYD-3. Fasteners torqued to 27 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS400-CLN-4

Target Machine Model: EOS-Additive-M400 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M400 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on EOS-Additive-M400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M400 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: FL-CLN-EOS400-4
- PLC Hex Register Address: DB20.DBW_0xAFE0
- Nominal Operating Range: 24 - 80 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 110 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS400-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS400-PMP-PLG-4604
- Overhaul Seal & Consumable Kit: KIT-EOS400-CLN-4
- Critical Fastener Torque Value: 36 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M400 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS400-CLN-4 at PLC register 0xAFE0.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on EOS-Additive-M400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS400-PMP-PLG-4604) with seal kit (KIT-EOS400-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 36 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS400-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M400, technician must log: "Resolved ERR-EOS400-CLN-4 on EOS-Additive-M400 by installing PN-EOS400-PMP-PLG-4604 & kit KIT-EOS400-CLN-4. Fasteners torqued to 36 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS400-XAX-5

Target Machine Model: EOS-Additive-M400 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M400 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M400 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: EN-XAX-EOS400-5
- PLC Hex Register Address: DB20.DBW_0xAFE8
- Nominal Operating Range: 27 - 80 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 110 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS400-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS400-LIN-COIL-4605
- Overhaul Seal & Consumable Kit: KIT-EOS400-XAX-5
- Critical Fastener Torque Value: 45 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M400 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS400-XAX-5 at PLC register 0xAFE8.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on EOS-Additive-M400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS400-LIN-COIL-4605) with seal kit (KIT-EOS400-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 45 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS400-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M400, technician must log: "Resolved ERR-EOS400-XAX-5 on EOS-Additive-M400 by installing PN-EOS400-LIN-COIL-4605 & kit KIT-EOS400-XAX-5. Fasteners torqued to 45 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS400-YAX-6

Target Machine Model: EOS-Additive-M400 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M400 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on EOS-Additive-M400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M400 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: EN-YAX-EOS400-6
- PLC Hex Register Address: DB20.DBW_0xAFF0
- Nominal Operating Range: 30 - 80 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 110 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS400-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS400-BSW-NUT-4606
- Overhaul Seal & Consumable Kit: KIT-EOS400-YAX-6
- Critical Fastener Torque Value: 54 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M400 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS400-YAX-6 at PLC register 0xAFF0.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on EOS-Additive-M400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS400-BSW-NUT-4606) with seal kit (KIT-EOS400-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 54 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS400-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M400, technician must log: "Resolved ERR-EOS400-YAX-6 on EOS-Additive-M400 by installing PN-EOS400-BSW-NUT-4606 & kit KIT-EOS400-YAX-6. Fasteners torqued to 54 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS400-ZAX-7

Target Machine Model: EOS-Additive-M400 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M400 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M400 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: PR-ZAX-EOS400-7
- PLC Hex Register Address: DB20.DBW_0xAFF8
- Nominal Operating Range: 33 - 80 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 110 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS400-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS400-CYL-SEAL-4607
- Overhaul Seal & Consumable Kit: KIT-EOS400-ZAX-7
- Critical Fastener Torque Value: 63 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M400 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS400-ZAX-7 at PLC register 0xAFF8.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on EOS-Additive-M400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS400-CYL-SEAL-4607) with seal kit (KIT-EOS400-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 63 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS400-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M400, technician must log: "Resolved ERR-EOS400-ZAX-7 on EOS-Additive-M400 by installing PN-EOS400-CYL-SEAL-4607 & kit KIT-EOS400-ZAX-7. Fasteners torqued to 63 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS400-BAX-8

Target Machine Model: EOS-Additive-M400 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M400 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on EOS-Additive-M400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M400 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: EN-BAX-EOS400-8
- PLC Hex Register Address: DB20.DBW_0xB000
- Nominal Operating Range: 36 - 80 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 110 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS400-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS400-TRQ-STAT-4608
- Overhaul Seal & Consumable Kit: KIT-EOS400-BAX-8
- Critical Fastener Torque Value: 72 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M400 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS400-BAX-8 at PLC register 0xB000.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on EOS-Additive-M400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS400-TRQ-STAT-4608) with seal kit (KIT-EOS400-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 72 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS400-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M400, technician must log: "Resolved ERR-EOS400-BAX-8 on EOS-Additive-M400 by installing PN-EOS400-TRQ-STAT-4608 & kit KIT-EOS400-BAX-8. Fasteners torqued to 72 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS400-CAX-9

Target Machine Model: EOS-Additive-M400 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M400 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M400 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: PR-CAX-EOS400-9
- PLC Hex Register Address: DB20.DBW_0xB008
- Nominal Operating Range: 39 - 80 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 110 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS400-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS400-BRK-DISC-4609
- Overhaul Seal & Consumable Kit: KIT-EOS400-CAX-9
- Critical Fastener Torque Value: 81 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M400 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS400-CAX-9 at PLC register 0xB008.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on EOS-Additive-M400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS400-BRK-DISC-4609) with seal kit (KIT-EOS400-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 81 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS400-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M400, technician must log: "Resolved ERR-EOS400-CAX-9 on EOS-Additive-M400 by installing PN-EOS400-BRK-DISC-4609 & kit KIT-EOS400-CAX-9. Fasteners torqued to 81 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS400-OPT-10

Target Machine Model: EOS-Additive-M400 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M400 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on EOS-Additive-M400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M400 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: OP-SCL-EOS400-10
- PLC Hex Register Address: DB20.DBW_0xB010
- Nominal Operating Range: 42 - 80 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 110 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS400-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS400-GLASS-RD-4610
- Overhaul Seal & Consumable Kit: KIT-EOS400-OPT-10
- Critical Fastener Torque Value: 90 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M400 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS400-OPT-10 at PLC register 0xB010.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on EOS-Additive-M400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS400-GLASS-RD-4610) with seal kit (KIT-EOS400-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 90 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS400-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M400, technician must log: "Resolved ERR-EOS400-OPT-10 on EOS-Additive-M400 by installing PN-EOS400-GLASS-RD-4610 & kit KIT-EOS400-OPT-10. Fasteners torqued to 90 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS400-PLC-11

Target Machine Model: EOS-Additive-M400 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M400 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M400 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: VT-DRV-EOS400-11
- PLC Hex Register Address: DB20.DBW_0xB018
- Nominal Operating Range: 45 - 80 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 110 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS400-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS400-IGBT-MOD-4611
- Overhaul Seal & Consumable Kit: KIT-EOS400-PLC-11
- Critical Fastener Torque Value: 99 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M400 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS400-PLC-11 at PLC register 0xB018.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on EOS-Additive-M400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS400-IGBT-MOD-4611) with seal kit (KIT-EOS400-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 99 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS400-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M400, technician must log: "Resolved ERR-EOS400-PLC-11 on EOS-Additive-M400 by installing PN-EOS400-IGBT-MOD-4611 & kit KIT-EOS400-PLC-11. Fasteners torqued to 99 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS400-LUB-12

Target Machine Model: EOS-Additive-M400 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M400 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on EOS-Additive-M400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M400 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: FL-LUB-EOS400-12
- PLC Hex Register Address: DB20.DBW_0xB020
- Nominal Operating Range: 48 - 80 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 110 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS400-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS400-INJ-MET-4612
- Overhaul Seal & Consumable Kit: KIT-EOS400-LUB-12
- Critical Fastener Torque Value: 108 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M400 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS400-LUB-12 at PLC register 0xB020.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on EOS-Additive-M400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS400-INJ-MET-4612) with seal kit (KIT-EOS400-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 108 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS400-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M400, technician must log: "Resolved ERR-EOS400-LUB-12 on EOS-Additive-M400 by installing PN-EOS400-INJ-MET-4612 & kit KIT-EOS400-LUB-12. Fasteners torqued to 108 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS400-VAC-13

Target Machine Model: EOS-Additive-M400 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M400 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M400 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: VC-MNF-EOS400-13
- PLC Hex Register Address: DB20.DBW_0xB028
- Nominal Operating Range: 51 - 80 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 110 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS400-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS400-GASK-VIT-4613
- Overhaul Seal & Consumable Kit: KIT-EOS400-VAC-13
- Critical Fastener Torque Value: 117 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M400 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS400-VAC-13 at PLC register 0xB028.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on EOS-Additive-M400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS400-GASK-VIT-4613) with seal kit (KIT-EOS400-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 117 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS400-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M400, technician must log: "Resolved ERR-EOS400-VAC-13 on EOS-Additive-M400 by installing PN-EOS400-GASK-VIT-4613 & kit KIT-EOS400-VAC-13. Fasteners torqued to 117 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS400-THM-14

Target Machine Model: EOS-Additive-M400 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M400 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on EOS-Additive-M400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M400 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: TM-CHL-EOS400-14
- PLC Hex Register Address: DB20.DBW_0xB030
- Nominal Operating Range: 54 - 80 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 110 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS400-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS400-HEX-CORE-4614
- Overhaul Seal & Consumable Kit: KIT-EOS400-THM-14
- Critical Fastener Torque Value: 126 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M400 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS400-THM-14 at PLC register 0xB030.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on EOS-Additive-M400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS400-HEX-CORE-4614) with seal kit (KIT-EOS400-THM-14).

Step 6: Torque retaining bolts in cross sequence to 126 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS400-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M400, technician must log: "Resolved ERR-EOS400-THM-14 on EOS-Additive-M400 by installing PN-EOS400-HEX-CORE-4614 & kit KIT-EOS400-THM-14. Fasteners torqued to 126 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS450-SPN-1

Target Machine Model: EOS-Additive-M450 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M450 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on EOS-Additive-M450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M450 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: PT-SPN-EOS450-1
- PLC Hex Register Address: DB21.DBW_0xB2E8
- Nominal Operating Range: 15 - 81 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 111 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS450-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS450-BRG-CER-4701
- Overhaul Seal & Consumable Kit: KIT-EOS450-SPN-1
- Critical Fastener Torque Value: 149 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M450 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS450-SPN-1 at PLC register 0xB2E8.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on EOS-Additive-M450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS450-BRG-CER-4701) with seal kit (KIT-EOS450-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 149 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS450-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M450, technician must log: "Resolved ERR-EOS450-SPN-1 on EOS-Additive-M450 by installing PN-EOS450-BRG-CER-4701 & kit KIT-EOS450-SPN-1. Fasteners torqued to 149 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS450-ATC-2

Target Machine Model: EOS-Additive-M450 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M450 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on EOS-Additive-M450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M450 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: PX-ATC-EOS450-2
- PLC Hex Register Address: DB21.DBW_0xB2F0
- Nominal Operating Range: 18 - 81 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 111 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS450-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS450-CAM-GRIP-4702
- Overhaul Seal & Consumable Kit: KIT-EOS450-ATC-2
- Critical Fastener Torque Value: 158 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M450 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS450-ATC-2 at PLC register 0xB2F0.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on EOS-Additive-M450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS450-CAM-GRIP-4702) with seal kit (KIT-EOS450-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 158 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS450-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M450, technician must log: "Resolved ERR-EOS450-ATC-2 on EOS-Additive-M450 by installing PN-EOS450-CAM-GRIP-4702 & kit KIT-EOS450-ATC-2. Fasteners torqued to 158 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS450-HYD-3

Target Machine Model: EOS-Additive-M450 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M450 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on EOS-Additive-M450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M450 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: PR-HYD-EOS450-3
- PLC Hex Register Address: DB21.DBW_0xB2F8
- Nominal Operating Range: 21 - 81 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 111 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS450-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS450-VLV-PROP-4703
- Overhaul Seal & Consumable Kit: KIT-EOS450-HYD-3
- Critical Fastener Torque Value: 167 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M450 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS450-HYD-3 at PLC register 0xB2F8.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on EOS-Additive-M450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS450-VLV-PROP-4703) with seal kit (KIT-EOS450-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 167 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS450-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M450, technician must log: "Resolved ERR-EOS450-HYD-3 on EOS-Additive-M450 by installing PN-EOS450-VLV-PROP-4703 & kit KIT-EOS450-HYD-3. Fasteners torqued to 167 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS450-CLN-4

Target Machine Model: EOS-Additive-M450 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M450 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on EOS-Additive-M450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M450 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: FL-CLN-EOS450-4
- PLC Hex Register Address: DB21.DBW_0xB300
- Nominal Operating Range: 24 - 81 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 111 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS450-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS450-PMP-PLG-4704
- Overhaul Seal & Consumable Kit: KIT-EOS450-CLN-4
- Critical Fastener Torque Value: 176 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M450 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS450-CLN-4 at PLC register 0xB300.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on EOS-Additive-M450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS450-PMP-PLG-4704) with seal kit (KIT-EOS450-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 176 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS450-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M450, technician must log: "Resolved ERR-EOS450-CLN-4 on EOS-Additive-M450 by installing PN-EOS450-PMP-PLG-4704 & kit KIT-EOS450-CLN-4. Fasteners torqued to 176 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS450-XAX-5

Target Machine Model: EOS-Additive-M450 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M450 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M450 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: EN-XAX-EOS450-5
- PLC Hex Register Address: DB21.DBW_0xB308
- Nominal Operating Range: 27 - 81 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 111 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS450-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS450-LIN-COIL-4705
- Overhaul Seal & Consumable Kit: KIT-EOS450-XAX-5
- Critical Fastener Torque Value: 185 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M450 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS450-XAX-5 at PLC register 0xB308.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on EOS-Additive-M450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS450-LIN-COIL-4705) with seal kit (KIT-EOS450-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 185 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS450-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M450, technician must log: "Resolved ERR-EOS450-XAX-5 on EOS-Additive-M450 by installing PN-EOS450-LIN-COIL-4705 & kit KIT-EOS450-XAX-5. Fasteners torqued to 185 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS450-YAX-6

Target Machine Model: EOS-Additive-M450 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M450 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on EOS-Additive-M450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M450 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: EN-YAX-EOS450-6
- PLC Hex Register Address: DB21.DBW_0xB310
- Nominal Operating Range: 30 - 81 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 111 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS450-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS450-BSW-NUT-4706
- Overhaul Seal & Consumable Kit: KIT-EOS450-YAX-6
- Critical Fastener Torque Value: 194 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M450 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS450-YAX-6 at PLC register 0xB310.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on EOS-Additive-M450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS450-BSW-NUT-4706) with seal kit (KIT-EOS450-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 194 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS450-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M450, technician must log: "Resolved ERR-EOS450-YAX-6 on EOS-Additive-M450 by installing PN-EOS450-BSW-NUT-4706 & kit KIT-EOS450-YAX-6. Fasteners torqued to 194 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS450-ZAX-7

Target Machine Model: EOS-Additive-M450 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M450 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M450 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: PR-ZAX-EOS450-7
- PLC Hex Register Address: DB21.DBW_0xB318
- Nominal Operating Range: 33 - 81 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 111 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS450-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS450-CYL-SEAL-4707
- Overhaul Seal & Consumable Kit: KIT-EOS450-ZAX-7
- Critical Fastener Torque Value: 203 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M450 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS450-ZAX-7 at PLC register 0xB318.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on EOS-Additive-M450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS450-CYL-SEAL-4707) with seal kit (KIT-EOS450-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 203 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS450-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M450, technician must log: "Resolved ERR-EOS450-ZAX-7 on EOS-Additive-M450 by installing PN-EOS450-CYL-SEAL-4707 & kit KIT-EOS450-ZAX-7. Fasteners torqued to 203 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS450-BAX-8

Target Machine Model: EOS-Additive-M450 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M450 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on EOS-Additive-M450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M450 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: EN-BAX-EOS450-8
- PLC Hex Register Address: DB21.DBW_0xB320
- Nominal Operating Range: 36 - 81 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 111 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS450-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS450-TRQ-STAT-4708
- Overhaul Seal & Consumable Kit: KIT-EOS450-BAX-8
- Critical Fastener Torque Value: 22 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M450 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS450-BAX-8 at PLC register 0xB320.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on EOS-Additive-M450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS450-TRQ-STAT-4708) with seal kit (KIT-EOS450-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 22 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS450-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M450, technician must log: "Resolved ERR-EOS450-BAX-8 on EOS-Additive-M450 by installing PN-EOS450-TRQ-STAT-4708 & kit KIT-EOS450-BAX-8. Fasteners torqued to 22 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS450-CAX-9

Target Machine Model: EOS-Additive-M450 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M450 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M450 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: PR-CAX-EOS450-9
- PLC Hex Register Address: DB21.DBW_0xB328
- Nominal Operating Range: 39 - 81 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 111 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS450-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS450-BRK-DISC-4709
- Overhaul Seal & Consumable Kit: KIT-EOS450-CAX-9
- Critical Fastener Torque Value: 31 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M450 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS450-CAX-9 at PLC register 0xB328.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on EOS-Additive-M450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS450-BRK-DISC-4709) with seal kit (KIT-EOS450-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 31 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS450-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M450, technician must log: "Resolved ERR-EOS450-CAX-9 on EOS-Additive-M450 by installing PN-EOS450-BRK-DISC-4709 & kit KIT-EOS450-CAX-9. Fasteners torqued to 31 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS450-OPT-10

Target Machine Model: EOS-Additive-M450 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M450 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on EOS-Additive-M450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M450 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: OP-SCL-EOS450-10
- PLC Hex Register Address: DB21.DBW_0xB330
- Nominal Operating Range: 42 - 81 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 111 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS450-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS450-GLASS-RD-4710
- Overhaul Seal & Consumable Kit: KIT-EOS450-OPT-10
- Critical Fastener Torque Value: 40 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M450 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS450-OPT-10 at PLC register 0xB330.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on EOS-Additive-M450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS450-GLASS-RD-4710) with seal kit (KIT-EOS450-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 40 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS450-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M450, technician must log: "Resolved ERR-EOS450-OPT-10 on EOS-Additive-M450 by installing PN-EOS450-GLASS-RD-4710 & kit KIT-EOS450-OPT-10. Fasteners torqued to 40 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS450-PLC-11

Target Machine Model: EOS-Additive-M450 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M450 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M450 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: VT-DRV-EOS450-11
- PLC Hex Register Address: DB21.DBW_0xB338
- Nominal Operating Range: 45 - 81 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 111 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS450-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS450-IGBT-MOD-4711
- Overhaul Seal & Consumable Kit: KIT-EOS450-PLC-11
- Critical Fastener Torque Value: 49 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M450 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS450-PLC-11 at PLC register 0xB338.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on EOS-Additive-M450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS450-IGBT-MOD-4711) with seal kit (KIT-EOS450-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 49 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS450-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M450, technician must log: "Resolved ERR-EOS450-PLC-11 on EOS-Additive-M450 by installing PN-EOS450-IGBT-MOD-4711 & kit KIT-EOS450-PLC-11. Fasteners torqued to 49 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS450-LUB-12

Target Machine Model: EOS-Additive-M450 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M450 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on EOS-Additive-M450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M450 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: FL-LUB-EOS450-12
- PLC Hex Register Address: DB21.DBW_0xB340
- Nominal Operating Range: 48 - 81 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 111 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS450-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS450-INJ-MET-4712
- Overhaul Seal & Consumable Kit: KIT-EOS450-LUB-12
- Critical Fastener Torque Value: 58 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M450 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS450-LUB-12 at PLC register 0xB340.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on EOS-Additive-M450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS450-INJ-MET-4712) with seal kit (KIT-EOS450-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 58 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS450-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M450, technician must log: "Resolved ERR-EOS450-LUB-12 on EOS-Additive-M450 by installing PN-EOS450-INJ-MET-4712 & kit KIT-EOS450-LUB-12. Fasteners torqued to 58 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS450-VAC-13

Target Machine Model: EOS-Additive-M450 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M450 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M450 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: VC-MNF-EOS450-13
- PLC Hex Register Address: DB21.DBW_0xB348
- Nominal Operating Range: 51 - 81 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 111 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS450-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS450-GASK-VIT-4713
- Overhaul Seal & Consumable Kit: KIT-EOS450-VAC-13
- Critical Fastener Torque Value: 67 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M450 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS450-VAC-13 at PLC register 0xB348.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on EOS-Additive-M450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS450-GASK-VIT-4713) with seal kit (KIT-EOS450-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 67 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS450-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M450, technician must log: "Resolved ERR-EOS450-VAC-13 on EOS-Additive-M450 by installing PN-EOS450-GASK-VIT-4713 & kit KIT-EOS450-VAC-13. Fasteners torqued to 67 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS450-THM-14

Target Machine Model: EOS-Additive-M450 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M450 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on EOS-Additive-M450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M450 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: TM-CHL-EOS450-14
- PLC Hex Register Address: DB21.DBW_0xB350
- Nominal Operating Range: 54 - 81 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 111 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS450-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS450-HEX-CORE-4714
- Overhaul Seal & Consumable Kit: KIT-EOS450-THM-14
- Critical Fastener Torque Value: 76 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M450 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS450-THM-14 at PLC register 0xB350.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on EOS-Additive-M450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS450-HEX-CORE-4714) with seal kit (KIT-EOS450-THM-14).

Step 6: Torque retaining bolts in cross sequence to 76 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS450-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M450, technician must log: "Resolved ERR-EOS450-THM-14 on EOS-Additive-M450 by installing PN-EOS450-HEX-CORE-4714 & kit KIT-EOS450-THM-14. Fasteners torqued to 76 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS500-SPN-1

Target Machine Model: EOS-Additive-M500 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M500 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on EOS-Additive-M500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M500 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: PT-SPN-EOS500-1
- PLC Hex Register Address: DB22.DBW_0xB608
- Nominal Operating Range: 15 - 82 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 112 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS500-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS500-BRG-CER-4801
- Overhaul Seal & Consumable Kit: KIT-EOS500-SPN-1
- Critical Fastener Torque Value: 99 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M500 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS500-SPN-1 at PLC register 0xB608.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on EOS-Additive-M500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS500-BRG-CER-4801) with seal kit (KIT-EOS500-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 99 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS500-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M500, technician must log: "Resolved ERR-EOS500-SPN-1 on EOS-Additive-M500 by installing PN-EOS500-BRG-CER-4801 & kit KIT-EOS500-SPN-1. Fasteners torqued to 99 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS500-ATC-2

Target Machine Model: EOS-Additive-M500 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M500 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on EOS-Additive-M500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M500 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: PX-ATC-EOS500-2
- PLC Hex Register Address: DB22.DBW_0xB610
- Nominal Operating Range: 18 - 82 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 112 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS500-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS500-CAM-GRIP-4802
- Overhaul Seal & Consumable Kit: KIT-EOS500-ATC-2
- Critical Fastener Torque Value: 108 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M500 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS500-ATC-2 at PLC register 0xB610.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on EOS-Additive-M500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS500-CAM-GRIP-4802) with seal kit (KIT-EOS500-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 108 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS500-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M500, technician must log: "Resolved ERR-EOS500-ATC-2 on EOS-Additive-M500 by installing PN-EOS500-CAM-GRIP-4802 & kit KIT-EOS500-ATC-2. Fasteners torqued to 108 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS500-HYD-3

Target Machine Model: EOS-Additive-M500 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M500 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on EOS-Additive-M500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M500 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: PR-HYD-EOS500-3
- PLC Hex Register Address: DB22.DBW_0xB618
- Nominal Operating Range: 21 - 82 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 112 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS500-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS500-VLV-PROP-4803
- Overhaul Seal & Consumable Kit: KIT-EOS500-HYD-3
- Critical Fastener Torque Value: 117 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M500 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS500-HYD-3 at PLC register 0xB618.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on EOS-Additive-M500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS500-VLV-PROP-4803) with seal kit (KIT-EOS500-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 117 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS500-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M500, technician must log: "Resolved ERR-EOS500-HYD-3 on EOS-Additive-M500 by installing PN-EOS500-VLV-PROP-4803 & kit KIT-EOS500-HYD-3. Fasteners torqued to 117 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS500-CLN-4

Target Machine Model: EOS-Additive-M500 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M500 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on EOS-Additive-M500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M500 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: FL-CLN-EOS500-4
- PLC Hex Register Address: DB22.DBW_0xB620
- Nominal Operating Range: 24 - 82 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 112 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS500-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS500-PMP-PLG-4804
- Overhaul Seal & Consumable Kit: KIT-EOS500-CLN-4
- Critical Fastener Torque Value: 126 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M500 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS500-CLN-4 at PLC register 0xB620.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on EOS-Additive-M500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS500-PMP-PLG-4804) with seal kit (KIT-EOS500-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 126 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS500-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M500, technician must log: "Resolved ERR-EOS500-CLN-4 on EOS-Additive-M500 by installing PN-EOS500-PMP-PLG-4804 & kit KIT-EOS500-CLN-4. Fasteners torqued to 126 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS500-XAX-5

Target Machine Model: EOS-Additive-M500 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M500 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M500 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: EN-XAX-EOS500-5
- PLC Hex Register Address: DB22.DBW_0xB628
- Nominal Operating Range: 27 - 82 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 112 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS500-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS500-LIN-COIL-4805
- Overhaul Seal & Consumable Kit: KIT-EOS500-XAX-5
- Critical Fastener Torque Value: 135 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M500 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS500-XAX-5 at PLC register 0xB628.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on EOS-Additive-M500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS500-LIN-COIL-4805) with seal kit (KIT-EOS500-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 135 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS500-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M500, technician must log: "Resolved ERR-EOS500-XAX-5 on EOS-Additive-M500 by installing PN-EOS500-LIN-COIL-4805 & kit KIT-EOS500-XAX-5. Fasteners torqued to 135 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS500-YAX-6

Target Machine Model: EOS-Additive-M500 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M500 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on EOS-Additive-M500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M500 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: EN-YAX-EOS500-6
- PLC Hex Register Address: DB22.DBW_0xB630
- Nominal Operating Range: 30 - 82 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 112 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS500-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS500-BSW-NUT-4806
- Overhaul Seal & Consumable Kit: KIT-EOS500-YAX-6
- Critical Fastener Torque Value: 144 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M500 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS500-YAX-6 at PLC register 0xB630.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on EOS-Additive-M500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS500-BSW-NUT-4806) with seal kit (KIT-EOS500-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 144 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS500-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M500, technician must log: "Resolved ERR-EOS500-YAX-6 on EOS-Additive-M500 by installing PN-EOS500-BSW-NUT-4806 & kit KIT-EOS500-YAX-6. Fasteners torqued to 144 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS500-ZAX-7

Target Machine Model: EOS-Additive-M500 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M500 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M500 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: PR-ZAX-EOS500-7
- PLC Hex Register Address: DB22.DBW_0xB638
- Nominal Operating Range: 33 - 82 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 112 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS500-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS500-CYL-SEAL-4807
- Overhaul Seal & Consumable Kit: KIT-EOS500-ZAX-7
- Critical Fastener Torque Value: 153 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M500 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS500-ZAX-7 at PLC register 0xB638.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on EOS-Additive-M500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS500-CYL-SEAL-4807) with seal kit (KIT-EOS500-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 153 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS500-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M500, technician must log: "Resolved ERR-EOS500-ZAX-7 on EOS-Additive-M500 by installing PN-EOS500-CYL-SEAL-4807 & kit KIT-EOS500-ZAX-7. Fasteners torqued to 153 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS500-BAX-8

Target Machine Model: EOS-Additive-M500 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M500 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on EOS-Additive-M500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M500 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: EN-BAX-EOS500-8
- PLC Hex Register Address: DB22.DBW_0xB640
- Nominal Operating Range: 36 - 82 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 112 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS500-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS500-TRQ-STAT-4808
- Overhaul Seal & Consumable Kit: KIT-EOS500-BAX-8
- Critical Fastener Torque Value: 162 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M500 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS500-BAX-8 at PLC register 0xB640.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on EOS-Additive-M500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS500-TRQ-STAT-4808) with seal kit (KIT-EOS500-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 162 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS500-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M500, technician must log: "Resolved ERR-EOS500-BAX-8 on EOS-Additive-M500 by installing PN-EOS500-TRQ-STAT-4808 & kit KIT-EOS500-BAX-8. Fasteners torqued to 162 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS500-CAX-9

Target Machine Model: EOS-Additive-M500 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M500 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M500 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: PR-CAX-EOS500-9
- PLC Hex Register Address: DB22.DBW_0xB648
- Nominal Operating Range: 39 - 82 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 112 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS500-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS500-BRK-DISC-4809
- Overhaul Seal & Consumable Kit: KIT-EOS500-CAX-9
- Critical Fastener Torque Value: 171 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M500 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS500-CAX-9 at PLC register 0xB648.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on EOS-Additive-M500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS500-BRK-DISC-4809) with seal kit (KIT-EOS500-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 171 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS500-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M500, technician must log: "Resolved ERR-EOS500-CAX-9 on EOS-Additive-M500 by installing PN-EOS500-BRK-DISC-4809 & kit KIT-EOS500-CAX-9. Fasteners torqued to 171 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS500-OPT-10

Target Machine Model: EOS-Additive-M500 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M500 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on EOS-Additive-M500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M500 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: OP-SCL-EOS500-10
- PLC Hex Register Address: DB22.DBW_0xB650
- Nominal Operating Range: 42 - 82 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 112 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS500-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS500-GLASS-RD-4810
- Overhaul Seal & Consumable Kit: KIT-EOS500-OPT-10
- Critical Fastener Torque Value: 180 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M500 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS500-OPT-10 at PLC register 0xB650.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on EOS-Additive-M500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS500-GLASS-RD-4810) with seal kit (KIT-EOS500-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 180 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS500-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M500, technician must log: "Resolved ERR-EOS500-OPT-10 on EOS-Additive-M500 by installing PN-EOS500-GLASS-RD-4810 & kit KIT-EOS500-OPT-10. Fasteners torqued to 180 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS500-PLC-11

Target Machine Model: EOS-Additive-M500 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M500 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M500 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: VT-DRV-EOS500-11
- PLC Hex Register Address: DB22.DBW_0xB658
- Nominal Operating Range: 45 - 82 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 112 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS500-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS500-IGBT-MOD-4811
- Overhaul Seal & Consumable Kit: KIT-EOS500-PLC-11
- Critical Fastener Torque Value: 189 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M500 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS500-PLC-11 at PLC register 0xB658.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on EOS-Additive-M500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS500-IGBT-MOD-4811) with seal kit (KIT-EOS500-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 189 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS500-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M500, technician must log: "Resolved ERR-EOS500-PLC-11 on EOS-Additive-M500 by installing PN-EOS500-IGBT-MOD-4811 & kit KIT-EOS500-PLC-11. Fasteners torqued to 189 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS500-LUB-12

Target Machine Model: EOS-Additive-M500 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M500 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on EOS-Additive-M500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M500 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: FL-LUB-EOS500-12
- PLC Hex Register Address: DB22.DBW_0xB660
- Nominal Operating Range: 48 - 82 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 112 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS500-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS500-INJ-MET-4812
- Overhaul Seal & Consumable Kit: KIT-EOS500-LUB-12
- Critical Fastener Torque Value: 198 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M500 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS500-LUB-12 at PLC register 0xB660.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on EOS-Additive-M500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS500-INJ-MET-4812) with seal kit (KIT-EOS500-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 198 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS500-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M500, technician must log: "Resolved ERR-EOS500-LUB-12 on EOS-Additive-M500 by installing PN-EOS500-INJ-MET-4812 & kit KIT-EOS500-LUB-12. Fasteners torqued to 198 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS500-VAC-13

Target Machine Model: EOS-Additive-M500 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M500 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M500 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: VC-MNF-EOS500-13
- PLC Hex Register Address: DB22.DBW_0xB668
- Nominal Operating Range: 51 - 82 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 112 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS500-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS500-GASK-VIT-4813
- Overhaul Seal & Consumable Kit: KIT-EOS500-VAC-13
- Critical Fastener Torque Value: 207 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M500 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS500-VAC-13 at PLC register 0xB668.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on EOS-Additive-M500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS500-GASK-VIT-4813) with seal kit (KIT-EOS500-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 207 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS500-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M500, technician must log: "Resolved ERR-EOS500-VAC-13 on EOS-Additive-M500 by installing PN-EOS500-GASK-VIT-4813 & kit KIT-EOS500-VAC-13. Fasteners torqued to 207 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS500-THM-14

Target Machine Model: EOS-Additive-M500 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M500 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on EOS-Additive-M500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M500 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: TM-CHL-EOS500-14
- PLC Hex Register Address: DB22.DBW_0xB670
- Nominal Operating Range: 54 - 82 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 112 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS500-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS500-HEX-CORE-4814
- Overhaul Seal & Consumable Kit: KIT-EOS500-THM-14
- Critical Fastener Torque Value: 26 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M500 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS500-THM-14 at PLC register 0xB670.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on EOS-Additive-M500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS500-HEX-CORE-4814) with seal kit (KIT-EOS500-THM-14).

Step 6: Torque retaining bolts in cross sequence to 26 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS500-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M500, technician must log: "Resolved ERR-EOS500-THM-14 on EOS-Additive-M500 by installing PN-EOS500-HEX-CORE-4814 & kit KIT-EOS500-THM-14. Fasteners torqued to 26 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS550-SPN-1

Target Machine Model: EOS-Additive-M550 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M550 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on EOS-Additive-M550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M550 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: PT-SPN-EOS550-1
- PLC Hex Register Address: DB23.DBW_0xB928
- Nominal Operating Range: 15 - 83 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 113 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS550-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS550-BRG-CER-4901
- Overhaul Seal & Consumable Kit: KIT-EOS550-SPN-1
- Critical Fastener Torque Value: 49 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M550 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS550-SPN-1 at PLC register 0xB928.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on EOS-Additive-M550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS550-BRG-CER-4901) with seal kit (KIT-EOS550-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 49 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS550-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M550, technician must log: "Resolved ERR-EOS550-SPN-1 on EOS-Additive-M550 by installing PN-EOS550-BRG-CER-4901 & kit KIT-EOS550-SPN-1. Fasteners torqued to 49 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS550-ATC-2

Target Machine Model: EOS-Additive-M550 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M550 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on EOS-Additive-M550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M550 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: PX-ATC-EOS550-2
- PLC Hex Register Address: DB23.DBW_0xB930
- Nominal Operating Range: 18 - 83 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 113 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS550-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS550-CAM-GRIP-4902
- Overhaul Seal & Consumable Kit: KIT-EOS550-ATC-2
- Critical Fastener Torque Value: 58 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M550 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS550-ATC-2 at PLC register 0xB930.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on EOS-Additive-M550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS550-CAM-GRIP-4902) with seal kit (KIT-EOS550-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 58 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS550-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M550, technician must log: "Resolved ERR-EOS550-ATC-2 on EOS-Additive-M550 by installing PN-EOS550-CAM-GRIP-4902 & kit KIT-EOS550-ATC-2. Fasteners torqued to 58 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS550-HYD-3

Target Machine Model: EOS-Additive-M550 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M550 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on EOS-Additive-M550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M550 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: PR-HYD-EOS550-3
- PLC Hex Register Address: DB23.DBW_0xB938
- Nominal Operating Range: 21 - 83 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 113 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS550-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS550-VLV-PROP-4903
- Overhaul Seal & Consumable Kit: KIT-EOS550-HYD-3
- Critical Fastener Torque Value: 67 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M550 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS550-HYD-3 at PLC register 0xB938.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on EOS-Additive-M550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS550-VLV-PROP-4903) with seal kit (KIT-EOS550-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 67 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS550-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M550, technician must log: "Resolved ERR-EOS550-HYD-3 on EOS-Additive-M550 by installing PN-EOS550-VLV-PROP-4903 & kit KIT-EOS550-HYD-3. Fasteners torqued to 67 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS550-CLN-4

Target Machine Model: EOS-Additive-M550 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M550 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on EOS-Additive-M550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M550 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: FL-CLN-EOS550-4
- PLC Hex Register Address: DB23.DBW_0xB940
- Nominal Operating Range: 24 - 83 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 113 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS550-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS550-PMP-PLG-4904
- Overhaul Seal & Consumable Kit: KIT-EOS550-CLN-4
- Critical Fastener Torque Value: 76 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M550 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS550-CLN-4 at PLC register 0xB940.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on EOS-Additive-M550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS550-PMP-PLG-4904) with seal kit (KIT-EOS550-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 76 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS550-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M550, technician must log: "Resolved ERR-EOS550-CLN-4 on EOS-Additive-M550 by installing PN-EOS550-PMP-PLG-4904 & kit KIT-EOS550-CLN-4. Fasteners torqued to 76 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS550-XAX-5

Target Machine Model: EOS-Additive-M550 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M550 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M550 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: EN-XAX-EOS550-5
- PLC Hex Register Address: DB23.DBW_0xB948
- Nominal Operating Range: 27 - 83 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 113 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS550-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS550-LIN-COIL-4905
- Overhaul Seal & Consumable Kit: KIT-EOS550-XAX-5
- Critical Fastener Torque Value: 85 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M550 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS550-XAX-5 at PLC register 0xB948.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on EOS-Additive-M550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS550-LIN-COIL-4905) with seal kit (KIT-EOS550-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 85 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS550-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M550, technician must log: "Resolved ERR-EOS550-XAX-5 on EOS-Additive-M550 by installing PN-EOS550-LIN-COIL-4905 & kit KIT-EOS550-XAX-5. Fasteners torqued to 85 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS550-YAX-6

Target Machine Model: EOS-Additive-M550 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M550 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on EOS-Additive-M550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M550 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: EN-YAX-EOS550-6
- PLC Hex Register Address: DB23.DBW_0xB950
- Nominal Operating Range: 30 - 83 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 113 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS550-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS550-BSW-NUT-4906
- Overhaul Seal & Consumable Kit: KIT-EOS550-YAX-6
- Critical Fastener Torque Value: 94 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M550 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS550-YAX-6 at PLC register 0xB950.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on EOS-Additive-M550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS550-BSW-NUT-4906) with seal kit (KIT-EOS550-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 94 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS550-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M550, technician must log: "Resolved ERR-EOS550-YAX-6 on EOS-Additive-M550 by installing PN-EOS550-BSW-NUT-4906 & kit KIT-EOS550-YAX-6. Fasteners torqued to 94 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS550-ZAX-7

Target Machine Model: EOS-Additive-M550 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M550 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M550 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: PR-ZAX-EOS550-7
- PLC Hex Register Address: DB23.DBW_0xB958
- Nominal Operating Range: 33 - 83 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 113 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS550-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS550-CYL-SEAL-4907
- Overhaul Seal & Consumable Kit: KIT-EOS550-ZAX-7
- Critical Fastener Torque Value: 103 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M550 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS550-ZAX-7 at PLC register 0xB958.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on EOS-Additive-M550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS550-CYL-SEAL-4907) with seal kit (KIT-EOS550-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 103 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS550-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M550, technician must log: "Resolved ERR-EOS550-ZAX-7 on EOS-Additive-M550 by installing PN-EOS550-CYL-SEAL-4907 & kit KIT-EOS550-ZAX-7. Fasteners torqued to 103 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS550-BAX-8

Target Machine Model: EOS-Additive-M550 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M550 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on EOS-Additive-M550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M550 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: EN-BAX-EOS550-8
- PLC Hex Register Address: DB23.DBW_0xB960
- Nominal Operating Range: 36 - 83 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 113 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS550-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS550-TRQ-STAT-4908
- Overhaul Seal & Consumable Kit: KIT-EOS550-BAX-8
- Critical Fastener Torque Value: 112 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M550 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS550-BAX-8 at PLC register 0xB960.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on EOS-Additive-M550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS550-TRQ-STAT-4908) with seal kit (KIT-EOS550-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 112 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS550-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M550, technician must log: "Resolved ERR-EOS550-BAX-8 on EOS-Additive-M550 by installing PN-EOS550-TRQ-STAT-4908 & kit KIT-EOS550-BAX-8. Fasteners torqued to 112 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS550-CAX-9

Target Machine Model: EOS-Additive-M550 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M550 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M550 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: PR-CAX-EOS550-9
- PLC Hex Register Address: DB23.DBW_0xB968
- Nominal Operating Range: 39 - 83 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 113 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS550-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS550-BRK-DISC-4909
- Overhaul Seal & Consumable Kit: KIT-EOS550-CAX-9
- Critical Fastener Torque Value: 121 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M550 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS550-CAX-9 at PLC register 0xB968.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on EOS-Additive-M550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS550-BRK-DISC-4909) with seal kit (KIT-EOS550-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 121 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS550-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M550, technician must log: "Resolved ERR-EOS550-CAX-9 on EOS-Additive-M550 by installing PN-EOS550-BRK-DISC-4909 & kit KIT-EOS550-CAX-9. Fasteners torqued to 121 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS550-OPT-10

Target Machine Model: EOS-Additive-M550 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M550 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on EOS-Additive-M550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M550 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: OP-SCL-EOS550-10
- PLC Hex Register Address: DB23.DBW_0xB970
- Nominal Operating Range: 42 - 83 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 113 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS550-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS550-GLASS-RD-4910
- Overhaul Seal & Consumable Kit: KIT-EOS550-OPT-10
- Critical Fastener Torque Value: 130 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M550 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS550-OPT-10 at PLC register 0xB970.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on EOS-Additive-M550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS550-GLASS-RD-4910) with seal kit (KIT-EOS550-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 130 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS550-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M550, technician must log: "Resolved ERR-EOS550-OPT-10 on EOS-Additive-M550 by installing PN-EOS550-GLASS-RD-4910 & kit KIT-EOS550-OPT-10. Fasteners torqued to 130 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS550-PLC-11

Target Machine Model: EOS-Additive-M550 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M550 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M550 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: VT-DRV-EOS550-11
- PLC Hex Register Address: DB23.DBW_0xB978
- Nominal Operating Range: 45 - 83 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 113 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS550-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS550-IGBT-MOD-4911
- Overhaul Seal & Consumable Kit: KIT-EOS550-PLC-11
- Critical Fastener Torque Value: 139 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M550 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS550-PLC-11 at PLC register 0xB978.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on EOS-Additive-M550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS550-IGBT-MOD-4911) with seal kit (KIT-EOS550-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 139 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS550-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M550, technician must log: "Resolved ERR-EOS550-PLC-11 on EOS-Additive-M550 by installing PN-EOS550-IGBT-MOD-4911 & kit KIT-EOS550-PLC-11. Fasteners torqued to 139 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS550-LUB-12

Target Machine Model: EOS-Additive-M550 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M550 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on EOS-Additive-M550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M550 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: FL-LUB-EOS550-12
- PLC Hex Register Address: DB23.DBW_0xB980
- Nominal Operating Range: 48 - 83 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 113 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS550-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS550-INJ-MET-4912
- Overhaul Seal & Consumable Kit: KIT-EOS550-LUB-12
- Critical Fastener Torque Value: 148 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M550 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS550-LUB-12 at PLC register 0xB980.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on EOS-Additive-M550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS550-INJ-MET-4912) with seal kit (KIT-EOS550-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 148 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS550-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M550, technician must log: "Resolved ERR-EOS550-LUB-12 on EOS-Additive-M550 by installing PN-EOS550-INJ-MET-4912 & kit KIT-EOS550-LUB-12. Fasteners torqued to 148 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS550-VAC-13

Target Machine Model: EOS-Additive-M550 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M550 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M550 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: VC-MNF-EOS550-13
- PLC Hex Register Address: DB23.DBW_0xB988
- Nominal Operating Range: 51 - 83 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 113 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS550-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS550-GASK-VIT-4913
- Overhaul Seal & Consumable Kit: KIT-EOS550-VAC-13
- Critical Fastener Torque Value: 157 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M550 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS550-VAC-13 at PLC register 0xB988.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on EOS-Additive-M550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS550-GASK-VIT-4913) with seal kit (KIT-EOS550-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 157 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS550-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M550, technician must log: "Resolved ERR-EOS550-VAC-13 on EOS-Additive-M550 by installing PN-EOS550-GASK-VIT-4913 & kit KIT-EOS550-VAC-13. Fasteners torqued to 157 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS550-THM-14

Target Machine Model: EOS-Additive-M550 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M550 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on EOS-Additive-M550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M550 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: TM-CHL-EOS550-14
- PLC Hex Register Address: DB23.DBW_0xB990
- Nominal Operating Range: 54 - 83 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 113 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS550-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS550-HEX-CORE-4914
- Overhaul Seal & Consumable Kit: KIT-EOS550-THM-14
- Critical Fastener Torque Value: 166 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M550 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS550-THM-14 at PLC register 0xB990.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on EOS-Additive-M550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS550-HEX-CORE-4914) with seal kit (KIT-EOS550-THM-14).

Step 6: Torque retaining bolts in cross sequence to 166 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS550-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M550, technician must log: "Resolved ERR-EOS550-THM-14 on EOS-Additive-M550 by installing PN-EOS550-HEX-CORE-4914 & kit KIT-EOS550-THM-14. Fasteners torqued to 166 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS600-SPN-1

Target Machine Model: EOS-Additive-M600 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M600 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on EOS-Additive-M600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M600 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: PT-SPN-EOS600-1
- PLC Hex Register Address: DB24.DBW_0xBC48
- Nominal Operating Range: 15 - 84 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 114 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS600-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS600-BRG-CER-5001
- Overhaul Seal & Consumable Kit: KIT-EOS600-SPN-1
- Critical Fastener Torque Value: 189 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M600 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS600-SPN-1 at PLC register 0xBC48.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on EOS-Additive-M600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS600-BRG-CER-5001) with seal kit (KIT-EOS600-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 189 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS600-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M600, technician must log: "Resolved ERR-EOS600-SPN-1 on EOS-Additive-M600 by installing PN-EOS600-BRG-CER-5001 & kit KIT-EOS600-SPN-1. Fasteners torqued to 189 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS600-ATC-2

Target Machine Model: EOS-Additive-M600 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M600 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on EOS-Additive-M600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M600 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: PX-ATC-EOS600-2
- PLC Hex Register Address: DB24.DBW_0xBC50
- Nominal Operating Range: 18 - 84 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 114 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS600-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS600-CAM-GRIP-5002
- Overhaul Seal & Consumable Kit: KIT-EOS600-ATC-2
- Critical Fastener Torque Value: 198 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M600 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS600-ATC-2 at PLC register 0xBC50.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on EOS-Additive-M600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS600-CAM-GRIP-5002) with seal kit (KIT-EOS600-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 198 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS600-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M600, technician must log: "Resolved ERR-EOS600-ATC-2 on EOS-Additive-M600 by installing PN-EOS600-CAM-GRIP-5002 & kit KIT-EOS600-ATC-2. Fasteners torqued to 198 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS600-HYD-3

Target Machine Model: EOS-Additive-M600 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M600 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on EOS-Additive-M600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M600 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: PR-HYD-EOS600-3
- PLC Hex Register Address: DB24.DBW_0xBC58
- Nominal Operating Range: 21 - 84 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 114 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS600-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS600-VLV-PROP-5003
- Overhaul Seal & Consumable Kit: KIT-EOS600-HYD-3
- Critical Fastener Torque Value: 207 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M600 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS600-HYD-3 at PLC register 0xBC58.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on EOS-Additive-M600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS600-VLV-PROP-5003) with seal kit (KIT-EOS600-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 207 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS600-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M600, technician must log: "Resolved ERR-EOS600-HYD-3 on EOS-Additive-M600 by installing PN-EOS600-VLV-PROP-5003 & kit KIT-EOS600-HYD-3. Fasteners torqued to 207 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS600-CLN-4

Target Machine Model: EOS-Additive-M600 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M600 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on EOS-Additive-M600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M600 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: FL-CLN-EOS600-4
- PLC Hex Register Address: DB24.DBW_0xBC60
- Nominal Operating Range: 24 - 84 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 114 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS600-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS600-PMP-PLG-5004
- Overhaul Seal & Consumable Kit: KIT-EOS600-CLN-4
- Critical Fastener Torque Value: 26 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M600 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS600-CLN-4 at PLC register 0xBC60.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on EOS-Additive-M600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS600-PMP-PLG-5004) with seal kit (KIT-EOS600-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 26 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS600-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M600, technician must log: "Resolved ERR-EOS600-CLN-4 on EOS-Additive-M600 by installing PN-EOS600-PMP-PLG-5004 & kit KIT-EOS600-CLN-4. Fasteners torqued to 26 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS600-XAX-5

Target Machine Model: EOS-Additive-M600 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M600 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M600 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: EN-XAX-EOS600-5
- PLC Hex Register Address: DB24.DBW_0xBC68
- Nominal Operating Range: 27 - 84 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 114 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS600-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS600-LIN-COIL-5005
- Overhaul Seal & Consumable Kit: KIT-EOS600-XAX-5
- Critical Fastener Torque Value: 35 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M600 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS600-XAX-5 at PLC register 0xBC68.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on EOS-Additive-M600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS600-LIN-COIL-5005) with seal kit (KIT-EOS600-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 35 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS600-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M600, technician must log: "Resolved ERR-EOS600-XAX-5 on EOS-Additive-M600 by installing PN-EOS600-LIN-COIL-5005 & kit KIT-EOS600-XAX-5. Fasteners torqued to 35 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS600-YAX-6

Target Machine Model: EOS-Additive-M600 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M600 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on EOS-Additive-M600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M600 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: EN-YAX-EOS600-6
- PLC Hex Register Address: DB24.DBW_0xBC70
- Nominal Operating Range: 30 - 84 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 114 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS600-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS600-BSW-NUT-5006
- Overhaul Seal & Consumable Kit: KIT-EOS600-YAX-6
- Critical Fastener Torque Value: 44 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M600 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS600-YAX-6 at PLC register 0xBC70.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on EOS-Additive-M600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS600-BSW-NUT-5006) with seal kit (KIT-EOS600-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 44 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS600-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M600, technician must log: "Resolved ERR-EOS600-YAX-6 on EOS-Additive-M600 by installing PN-EOS600-BSW-NUT-5006 & kit KIT-EOS600-YAX-6. Fasteners torqued to 44 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS600-ZAX-7

Target Machine Model: EOS-Additive-M600 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M600 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M600 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: PR-ZAX-EOS600-7
- PLC Hex Register Address: DB24.DBW_0xBC78
- Nominal Operating Range: 33 - 84 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 114 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS600-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS600-CYL-SEAL-5007
- Overhaul Seal & Consumable Kit: KIT-EOS600-ZAX-7
- Critical Fastener Torque Value: 53 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M600 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS600-ZAX-7 at PLC register 0xBC78.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on EOS-Additive-M600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS600-CYL-SEAL-5007) with seal kit (KIT-EOS600-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 53 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS600-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M600, technician must log: "Resolved ERR-EOS600-ZAX-7 on EOS-Additive-M600 by installing PN-EOS600-CYL-SEAL-5007 & kit KIT-EOS600-ZAX-7. Fasteners torqued to 53 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS600-BAX-8

Target Machine Model: EOS-Additive-M600 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M600 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on EOS-Additive-M600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M600 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: EN-BAX-EOS600-8
- PLC Hex Register Address: DB24.DBW_0xBC80
- Nominal Operating Range: 36 - 84 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 114 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS600-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS600-TRQ-STAT-5008
- Overhaul Seal & Consumable Kit: KIT-EOS600-BAX-8
- Critical Fastener Torque Value: 62 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M600 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS600-BAX-8 at PLC register 0xBC80.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on EOS-Additive-M600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS600-TRQ-STAT-5008) with seal kit (KIT-EOS600-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 62 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS600-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M600, technician must log: "Resolved ERR-EOS600-BAX-8 on EOS-Additive-M600 by installing PN-EOS600-TRQ-STAT-5008 & kit KIT-EOS600-BAX-8. Fasteners torqued to 62 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS600-CAX-9

Target Machine Model: EOS-Additive-M600 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M600 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M600 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: PR-CAX-EOS600-9
- PLC Hex Register Address: DB24.DBW_0xBC88
- Nominal Operating Range: 39 - 84 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 114 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS600-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS600-BRK-DISC-5009
- Overhaul Seal & Consumable Kit: KIT-EOS600-CAX-9
- Critical Fastener Torque Value: 71 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M600 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS600-CAX-9 at PLC register 0xBC88.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on EOS-Additive-M600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS600-BRK-DISC-5009) with seal kit (KIT-EOS600-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 71 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS600-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M600, technician must log: "Resolved ERR-EOS600-CAX-9 on EOS-Additive-M600 by installing PN-EOS600-BRK-DISC-5009 & kit KIT-EOS600-CAX-9. Fasteners torqued to 71 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS600-OPT-10

Target Machine Model: EOS-Additive-M600 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M600 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on EOS-Additive-M600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M600 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: OP-SCL-EOS600-10
- PLC Hex Register Address: DB24.DBW_0xBC90
- Nominal Operating Range: 42 - 84 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 114 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS600-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS600-GLASS-RD-5010
- Overhaul Seal & Consumable Kit: KIT-EOS600-OPT-10
- Critical Fastener Torque Value: 80 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M600 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS600-OPT-10 at PLC register 0xBC90.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on EOS-Additive-M600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS600-GLASS-RD-5010) with seal kit (KIT-EOS600-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 80 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS600-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M600, technician must log: "Resolved ERR-EOS600-OPT-10 on EOS-Additive-M600 by installing PN-EOS600-GLASS-RD-5010 & kit KIT-EOS600-OPT-10. Fasteners torqued to 80 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS600-PLC-11

Target Machine Model: EOS-Additive-M600 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M600 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M600 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: VT-DRV-EOS600-11
- PLC Hex Register Address: DB24.DBW_0xBC98
- Nominal Operating Range: 45 - 84 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 114 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS600-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS600-IGBT-MOD-5011
- Overhaul Seal & Consumable Kit: KIT-EOS600-PLC-11
- Critical Fastener Torque Value: 89 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M600 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS600-PLC-11 at PLC register 0xBC98.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on EOS-Additive-M600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS600-IGBT-MOD-5011) with seal kit (KIT-EOS600-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 89 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS600-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M600, technician must log: "Resolved ERR-EOS600-PLC-11 on EOS-Additive-M600 by installing PN-EOS600-IGBT-MOD-5011 & kit KIT-EOS600-PLC-11. Fasteners torqued to 89 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS600-LUB-12

Target Machine Model: EOS-Additive-M600 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M600 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on EOS-Additive-M600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M600 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: FL-LUB-EOS600-12
- PLC Hex Register Address: DB24.DBW_0xBCA0
- Nominal Operating Range: 48 - 84 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 114 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS600-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS600-INJ-MET-5012
- Overhaul Seal & Consumable Kit: KIT-EOS600-LUB-12
- Critical Fastener Torque Value: 98 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M600 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS600-LUB-12 at PLC register 0xBCA0.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on EOS-Additive-M600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS600-INJ-MET-5012) with seal kit (KIT-EOS600-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 98 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS600-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M600, technician must log: "Resolved ERR-EOS600-LUB-12 on EOS-Additive-M600 by installing PN-EOS600-INJ-MET-5012 & kit KIT-EOS600-LUB-12. Fasteners torqued to 98 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS600-VAC-13

Target Machine Model: EOS-Additive-M600 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M600 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M600 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: VC-MNF-EOS600-13
- PLC Hex Register Address: DB24.DBW_0xBCA8
- Nominal Operating Range: 51 - 84 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 114 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS600-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS600-GASK-VIT-5013
- Overhaul Seal & Consumable Kit: KIT-EOS600-VAC-13
- Critical Fastener Torque Value: 107 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M600 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS600-VAC-13 at PLC register 0xBCA8.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on EOS-Additive-M600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS600-GASK-VIT-5013) with seal kit (KIT-EOS600-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 107 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS600-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M600, technician must log: "Resolved ERR-EOS600-VAC-13 on EOS-Additive-M600 by installing PN-EOS600-GASK-VIT-5013 & kit KIT-EOS600-VAC-13. Fasteners torqued to 107 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-EOS600-THM-14

Target Machine Model: EOS-Additive-M600 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on EOS-Additive-M600 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on EOS-Additive-M600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: EOS-Additive-M600 (EOSYSTEM Dual-PLC)
- Telemetry Sensor ID: TM-CHL-EOS600-14
- PLC Hex Register Address: DB24.DBW_0xBCB0
- Nominal Operating Range: 54 - 84 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 114 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-EOS600-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-EOS600-HEX-CORE-5014
- Overhaul Seal & Consumable Kit: KIT-EOS600-THM-14
- Critical Fastener Torque Value: 116 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on EOS-Additive-M600 (EOSYSTEM Dual-PLC) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to EOSYSTEM Dual-PLC and verify fault latch ERR-EOS600-THM-14 at PLC register 0xBCB0.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on EOS-Additive-M600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-EOS600-HEX-CORE-5014) with seal kit (KIT-EOS600-THM-14).

Step 6: Torque retaining bolts in cross sequence to 116 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-EOS600-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on EOS-Additive-M600, technician must log: "Resolved ERR-EOS600-THM-14 on EOS-Additive-M600 by installing PN-EOS600-HEX-CORE-5014 & kit KIT-EOS600-THM-14. Fasteners torqued to 116 Nm; alignment verified at ±0.0060 mm."

CNC/PLC SERVO PARAMETER DICTIONARY — Mazak-Integrex-i400 (BANK 1)

Machine Model: Mazak-Integrex-i400 | Controller: MAZATROL SmoothAi. Use these parameter registers when diagnosing servo following errors or thermal drift on Mazak-Integrex-i400.

PRM_42621 [Mazak-Integrex-i400] — ATC_LOOP_GAIN_1 | Default: 29.900 | Linked Fault: ERR-MAZ400-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Mazak-Integrex-i400.

PRM_42622 [Mazak-Integrex-i400] — HYD_LOOP_GAIN_2 | Default: 59.800 | Linked Fault: ERR-MAZ400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i400.

PRM_42623 [Mazak-Integrex-i400] — CLN_LOOP_GAIN_3 | Default: 89.700 | Linked Fault: ERR-MAZ400-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Mazak-Integrex-i400.

PRM_42624 [Mazak-Integrex-i400] — XAX_LOOP_GAIN_4 | Default: 119.600 | Linked Fault: ERR-MAZ400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i400.

PRM_42625 [Mazak-Integrex-i400] — YAX_LOOP_GAIN_5 | Default: 149.500 | Linked Fault: ERR-MAZ400-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Mazak-Integrex-i400.

PRM_42626 [Mazak-Integrex-i400] — ZAX_LOOP_GAIN_6 | Default: 179.400 | Linked Fault: ERR-MAZ400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i400.

PRM_42627 [Mazak-Integrex-i400] — BAX_LOOP_GAIN_7 | Default: 209.300 | Linked Fault: ERR-MAZ400-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Mazak-Integrex-i400.

PRM_42628 [Mazak-Integrex-i400] — CAX_LOOP_GAIN_8 | Default: 239.200 | Linked Fault: ERR-MAZ400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i400.

PRM_42629 [Mazak-Integrex-i400] — OPT_LOOP_GAIN_9 | Default: 19.100 | Linked Fault: ERR-MAZ400-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Mazak-Integrex-i400.

PRM_42630 [Mazak-Integrex-i400] — PLC_LOOP_GAIN_10 | Default: 49.000 | Linked Fault: ERR-MAZ400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i400.

PRM_42631 [Mazak-Integrex-i400] — LUB_LOOP_GAIN_11 | Default: 78.900 | Linked Fault: ERR-MAZ400-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Mazak-Integrex-i400.

PRM_42632 [Mazak-Integrex-i400] — VAC_LOOP_GAIN_12 | Default: 108.800 | Linked Fault: ERR-MAZ400-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i400.

PRM_42633 [Mazak-Integrex-i400] — THM_LOOP_GAIN_13 | Default: 138.700 | Linked Fault: ERR-MAZ400-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Mazak-Integrex-i400.

PRM_42634 [Mazak-Integrex-i400] — SPN_LOOP_GAIN_14 | Default: 168.600 | Linked Fault: ERR-MAZ400-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i400.

PRM_42635 [Mazak-Integrex-i400] — ATC_LOOP_GAIN_15 | Default: 198.500 | Linked Fault: ERR-MAZ400-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Mazak-Integrex-i400.

PRM_42636 [Mazak-Integrex-i400] — HYD_LOOP_GAIN_16 | Default: 228.400 | Linked Fault: ERR-MAZ400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i400.

CNC/PLC SERVO PARAMETER DICTIONARY — Mazak-Integrex-i400 (BANK 2)

Machine Model: Mazak-Integrex-i400 | Controller: MAZATROL SmoothAi. Use these parameter registers when diagnosing servo following errors or thermal drift on Mazak-Integrex-i400.

PRM_42641 [Mazak-Integrex-i400] — HYD_LOOP_GAIN_1 | Default: 59.800 | Linked Fault: ERR-MAZ400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i400.

PRM_42642 [Mazak-Integrex-i400] — XAX_LOOP_GAIN_2 | Default: 119.600 | Linked Fault: ERR-MAZ400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i400.

PRM_42643 [Mazak-Integrex-i400] — ZAX_LOOP_GAIN_3 | Default: 179.400 | Linked Fault: ERR-MAZ400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i400.

PRM_42644 [Mazak-Integrex-i400] — CAX_LOOP_GAIN_4 | Default: 239.200 | Linked Fault: ERR-MAZ400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i400.

PRM_42645 [Mazak-Integrex-i400] — PLC_LOOP_GAIN_5 | Default: 49.000 | Linked Fault: ERR-MAZ400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i400.

PRM_42646 [Mazak-Integrex-i400] — VAC_LOOP_GAIN_6 | Default: 108.800 | Linked Fault: ERR-MAZ400-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i400.

PRM_42647 [Mazak-Integrex-i400] — SPN_LOOP_GAIN_7 | Default: 168.600 | Linked Fault: ERR-MAZ400-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i400.

PRM_42648 [Mazak-Integrex-i400] — HYD_LOOP_GAIN_8 | Default: 228.400 | Linked Fault: ERR-MAZ400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i400.

PRM_42649 [Mazak-Integrex-i400] — XAX_LOOP_GAIN_9 | Default: 38.200 | Linked Fault: ERR-MAZ400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i400.

PRM_42650 [Mazak-Integrex-i400] — ZAX_LOOP_GAIN_10 | Default: 98.000 | Linked Fault: ERR-MAZ400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i400.

PRM_42651 [Mazak-Integrex-i400] — CAX_LOOP_GAIN_11 | Default: 157.800 | Linked Fault: ERR-MAZ400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i400.

PRM_42652 [Mazak-Integrex-i400] — PLC_LOOP_GAIN_12 | Default: 217.600 | Linked Fault: ERR-MAZ400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i400.

PRM_42653 [Mazak-Integrex-i400] — VAC_LOOP_GAIN_13 | Default: 27.400 | Linked Fault: ERR-MAZ400-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i400.

PRM_42654 [Mazak-Integrex-i400] — SPN_LOOP_GAIN_14 | Default: 87.200 | Linked Fault: ERR-MAZ400-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i400.

PRM_42655 [Mazak-Integrex-i400] — HYD_LOOP_GAIN_15 | Default: 147.000 | Linked Fault: ERR-MAZ400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i400.

PRM_42656 [Mazak-Integrex-i400] — XAX_LOOP_GAIN_16 | Default: 206.800 | Linked Fault: ERR-MAZ400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i400.

CNC/PLC SERVO PARAMETER DICTIONARY — Mazak-Integrex-i400 (BANK 3)

Machine Model: Mazak-Integrex-i400 | Controller: MAZATROL SmoothAi. Use these parameter registers when diagnosing servo following errors or thermal drift on Mazak-Integrex-i400.

PRM_42661 [Mazak-Integrex-i400] — CLN_LOOP_GAIN_1 | Default: 89.700 | Linked Fault: ERR-MAZ400-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Mazak-Integrex-i400.

PRM_42662 [Mazak-Integrex-i400] — ZAX_LOOP_GAIN_2 | Default: 179.400 | Linked Fault: ERR-MAZ400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i400.

PRM_42663 [Mazak-Integrex-i400] — OPT_LOOP_GAIN_3 | Default: 19.100 | Linked Fault: ERR-MAZ400-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Mazak-Integrex-i400.

PRM_42664 [Mazak-Integrex-i400] — VAC_LOOP_GAIN_4 | Default: 108.800 | Linked Fault: ERR-MAZ400-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i400.

PRM_42665 [Mazak-Integrex-i400] — ATC_LOOP_GAIN_5 | Default: 198.500 | Linked Fault: ERR-MAZ400-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Mazak-Integrex-i400.

PRM_42666 [Mazak-Integrex-i400] — XAX_LOOP_GAIN_6 | Default: 38.200 | Linked Fault: ERR-MAZ400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i400.

PRM_42667 [Mazak-Integrex-i400] — BAX_LOOP_GAIN_7 | Default: 127.900 | Linked Fault: ERR-MAZ400-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Mazak-Integrex-i400.

PRM_42668 [Mazak-Integrex-i400] — PLC_LOOP_GAIN_8 | Default: 217.600 | Linked Fault: ERR-MAZ400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i400.

PRM_42669 [Mazak-Integrex-i400] — THM_LOOP_GAIN_9 | Default: 57.300 | Linked Fault: ERR-MAZ400-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Mazak-Integrex-i400.

PRM_42670 [Mazak-Integrex-i400] — HYD_LOOP_GAIN_10 | Default: 147.000 | Linked Fault: ERR-MAZ400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i400.

PRM_42671 [Mazak-Integrex-i400] — YAX_LOOP_GAIN_11 | Default: 236.700 | Linked Fault: ERR-MAZ400-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Mazak-Integrex-i400.

PRM_42672 [Mazak-Integrex-i400] — CAX_LOOP_GAIN_12 | Default: 76.400 | Linked Fault: ERR-MAZ400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i400.

PRM_42673 [Mazak-Integrex-i400] — LUB_LOOP_GAIN_13 | Default: 166.100 | Linked Fault: ERR-MAZ400-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Mazak-Integrex-i400.

PRM_42674 [Mazak-Integrex-i400] — SPN_LOOP_GAIN_14 | Default: 5.800 | Linked Fault: ERR-MAZ400-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i400.

PRM_42675 [Mazak-Integrex-i400] — CLN_LOOP_GAIN_15 | Default: 95.500 | Linked Fault: ERR-MAZ400-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Mazak-Integrex-i400.

PRM_42676 [Mazak-Integrex-i400] — ZAX_LOOP_GAIN_16 | Default: 185.200 | Linked Fault: ERR-MAZ400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i400.

CNC/PLC SERVO PARAMETER DICTIONARY — Mazak-Integrex-i400 (BANK 4)

Machine Model: Mazak-Integrex-i400 | Controller: MAZATROL SmoothAi. Use these parameter registers when diagnosing servo following errors or thermal drift on Mazak-Integrex-i400.

PRM_42681 [Mazak-Integrex-i400] — XAX_LOOP_GAIN_1 | Default: 119.600 | Linked Fault: ERR-MAZ400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i400.

PRM_42682 [Mazak-Integrex-i400] — CAX_LOOP_GAIN_2 | Default: 239.200 | Linked Fault: ERR-MAZ400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i400.

PRM_42683 [Mazak-Integrex-i400] — VAC_LOOP_GAIN_3 | Default: 108.800 | Linked Fault: ERR-MAZ400-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i400.

PRM_42684 [Mazak-Integrex-i400] — HYD_LOOP_GAIN_4 | Default: 228.400 | Linked Fault: ERR-MAZ400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i400.

PRM_42685 [Mazak-Integrex-i400] — ZAX_LOOP_GAIN_5 | Default: 98.000 | Linked Fault: ERR-MAZ400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i400.

PRM_42686 [Mazak-Integrex-i400] — PLC_LOOP_GAIN_6 | Default: 217.600 | Linked Fault: ERR-MAZ400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i400.

PRM_42687 [Mazak-Integrex-i400] — SPN_LOOP_GAIN_7 | Default: 87.200 | Linked Fault: ERR-MAZ400-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i400.

PRM_42688 [Mazak-Integrex-i400] — XAX_LOOP_GAIN_8 | Default: 206.800 | Linked Fault: ERR-MAZ400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i400.

PRM_42689 [Mazak-Integrex-i400] — CAX_LOOP_GAIN_9 | Default: 76.400 | Linked Fault: ERR-MAZ400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i400.

PRM_42690 [Mazak-Integrex-i400] — VAC_LOOP_GAIN_10 | Default: 196.000 | Linked Fault: ERR-MAZ400-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i400.

PRM_42691 [Mazak-Integrex-i400] — HYD_LOOP_GAIN_11 | Default: 65.600 | Linked Fault: ERR-MAZ400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i400.

PRM_42692 [Mazak-Integrex-i400] — ZAX_LOOP_GAIN_12 | Default: 185.200 | Linked Fault: ERR-MAZ400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i400.

PRM_42693 [Mazak-Integrex-i400] — PLC_LOOP_GAIN_13 | Default: 54.800 | Linked Fault: ERR-MAZ400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i400.

PRM_42694 [Mazak-Integrex-i400] — SPN_LOOP_GAIN_14 | Default: 174.400 | Linked Fault: ERR-MAZ400-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i400.

PRM_42695 [Mazak-Integrex-i400] — XAX_LOOP_GAIN_15 | Default: 44.000 | Linked Fault: ERR-MAZ400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i400.

PRM_42696 [Mazak-Integrex-i400] — CAX_LOOP_GAIN_16 | Default: 163.600 | Linked Fault: ERR-MAZ400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i400.

CNC/PLC SERVO PARAMETER DICTIONARY — Mazak-Integrex-i400 (BANK 5)

Machine Model: Mazak-Integrex-i400 | Controller: MAZATROL SmoothAi. Use these parameter registers when diagnosing servo following errors or thermal drift on Mazak-Integrex-i400.

PRM_42701 [Mazak-Integrex-i400] — YAX_LOOP_GAIN_1 | Default: 149.500 | Linked Fault: ERR-MAZ400-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Mazak-Integrex-i400.

PRM_42702 [Mazak-Integrex-i400] — PLC_LOOP_GAIN_2 | Default: 49.000 | Linked Fault: ERR-MAZ400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i400.

PRM_42703 [Mazak-Integrex-i400] — ATC_LOOP_GAIN_3 | Default: 198.500 | Linked Fault: ERR-MAZ400-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Mazak-Integrex-i400.

PRM_42704 [Mazak-Integrex-i400] — ZAX_LOOP_GAIN_4 | Default: 98.000 | Linked Fault: ERR-MAZ400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i400.

PRM_42705 [Mazak-Integrex-i400] — LUB_LOOP_GAIN_5 | Default: 247.500 | Linked Fault: ERR-MAZ400-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Mazak-Integrex-i400.

PRM_42706 [Mazak-Integrex-i400] — HYD_LOOP_GAIN_6 | Default: 147.000 | Linked Fault: ERR-MAZ400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i400.

PRM_42707 [Mazak-Integrex-i400] — BAX_LOOP_GAIN_7 | Default: 46.500 | Linked Fault: ERR-MAZ400-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Mazak-Integrex-i400.

PRM_42708 [Mazak-Integrex-i400] — VAC_LOOP_GAIN_8 | Default: 196.000 | Linked Fault: ERR-MAZ400-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i400.

PRM_42709 [Mazak-Integrex-i400] — CLN_LOOP_GAIN_9 | Default: 95.500 | Linked Fault: ERR-MAZ400-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Mazak-Integrex-i400.

PRM_42710 [Mazak-Integrex-i400] — CAX_LOOP_GAIN_10 | Default: 245.000 | Linked Fault: ERR-MAZ400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i400.

PRM_42711 [Mazak-Integrex-i400] — THM_LOOP_GAIN_11 | Default: 144.500 | Linked Fault: ERR-MAZ400-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Mazak-Integrex-i400.

PRM_42712 [Mazak-Integrex-i400] — XAX_LOOP_GAIN_12 | Default: 44.000 | Linked Fault: ERR-MAZ400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i400.

PRM_42713 [Mazak-Integrex-i400] — OPT_LOOP_GAIN_13 | Default: 193.500 | Linked Fault: ERR-MAZ400-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Mazak-Integrex-i400.

PRM_42714 [Mazak-Integrex-i400] — SPN_LOOP_GAIN_14 | Default: 93.000 | Linked Fault: ERR-MAZ400-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i400.

PRM_42715 [Mazak-Integrex-i400] — YAX_LOOP_GAIN_15 | Default: 242.500 | Linked Fault: ERR-MAZ400-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Mazak-Integrex-i400.

PRM_42716 [Mazak-Integrex-i400] — PLC_LOOP_GAIN_16 | Default: 142.000 | Linked Fault: ERR-MAZ400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i400.

CNC/PLC SERVO PARAMETER DICTIONARY — Mazak-Integrex-i450 (BANK 1)

Machine Model: Mazak-Integrex-i450 | Controller: MAZATROL SmoothAi. Use these parameter registers when diagnosing servo following errors or thermal drift on Mazak-Integrex-i450.

PRM_42721 [Mazak-Integrex-i450] — ATC_LOOP_GAIN_1 | Default: 31.050 | Linked Fault: ERR-MAZ450-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Mazak-Integrex-i450.

PRM_42722 [Mazak-Integrex-i450] — HYD_LOOP_GAIN_2 | Default: 62.100 | Linked Fault: ERR-MAZ450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i450.

PRM_42723 [Mazak-Integrex-i450] — CLN_LOOP_GAIN_3 | Default: 93.150 | Linked Fault: ERR-MAZ450-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Mazak-Integrex-i450.

PRM_42724 [Mazak-Integrex-i450] — XAX_LOOP_GAIN_4 | Default: 124.200 | Linked Fault: ERR-MAZ450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i450.

PRM_42725 [Mazak-Integrex-i450] — YAX_LOOP_GAIN_5 | Default: 155.250 | Linked Fault: ERR-MAZ450-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Mazak-Integrex-i450.

PRM_42726 [Mazak-Integrex-i450] — ZAX_LOOP_GAIN_6 | Default: 186.300 | Linked Fault: ERR-MAZ450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i450.

PRM_42727 [Mazak-Integrex-i450] — BAX_LOOP_GAIN_7 | Default: 217.350 | Linked Fault: ERR-MAZ450-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Mazak-Integrex-i450.

PRM_42728 [Mazak-Integrex-i450] — CAX_LOOP_GAIN_8 | Default: 248.400 | Linked Fault: ERR-MAZ450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i450.

PRM_42729 [Mazak-Integrex-i450] — OPT_LOOP_GAIN_9 | Default: 29.450 | Linked Fault: ERR-MAZ450-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Mazak-Integrex-i450.

PRM_42730 [Mazak-Integrex-i450] — PLC_LOOP_GAIN_10 | Default: 60.500 | Linked Fault: ERR-MAZ450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i450.

PRM_42731 [Mazak-Integrex-i450] — LUB_LOOP_GAIN_11 | Default: 91.550 | Linked Fault: ERR-MAZ450-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Mazak-Integrex-i450.

PRM_42732 [Mazak-Integrex-i450] — VAC_LOOP_GAIN_12 | Default: 122.600 | Linked Fault: ERR-MAZ450-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i450.

PRM_42733 [Mazak-Integrex-i450] — THM_LOOP_GAIN_13 | Default: 153.650 | Linked Fault: ERR-MAZ450-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Mazak-Integrex-i450.

PRM_42734 [Mazak-Integrex-i450] — SPN_LOOP_GAIN_14 | Default: 184.700 | Linked Fault: ERR-MAZ450-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i450.

PRM_42735 [Mazak-Integrex-i450] — ATC_LOOP_GAIN_15 | Default: 215.750 | Linked Fault: ERR-MAZ450-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Mazak-Integrex-i450.

PRM_42736 [Mazak-Integrex-i450] — HYD_LOOP_GAIN_16 | Default: 246.800 | Linked Fault: ERR-MAZ450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i450.

CNC/PLC SERVO PARAMETER DICTIONARY — Mazak-Integrex-i450 (BANK 2)

Machine Model: Mazak-Integrex-i450 | Controller: MAZATROL SmoothAi. Use these parameter registers when diagnosing servo following errors or thermal drift on Mazak-Integrex-i450.

PRM_42741 [Mazak-Integrex-i450] — HYD_LOOP_GAIN_1 | Default: 62.100 | Linked Fault: ERR-MAZ450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i450.

PRM_42742 [Mazak-Integrex-i450] — XAX_LOOP_GAIN_2 | Default: 124.200 | Linked Fault: ERR-MAZ450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i450.

PRM_42743 [Mazak-Integrex-i450] — ZAX_LOOP_GAIN_3 | Default: 186.300 | Linked Fault: ERR-MAZ450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i450.

PRM_42744 [Mazak-Integrex-i450] — CAX_LOOP_GAIN_4 | Default: 248.400 | Linked Fault: ERR-MAZ450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i450.

PRM_42745 [Mazak-Integrex-i450] — PLC_LOOP_GAIN_5 | Default: 60.500 | Linked Fault: ERR-MAZ450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i450.

PRM_42746 [Mazak-Integrex-i450] — VAC_LOOP_GAIN_6 | Default: 122.600 | Linked Fault: ERR-MAZ450-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i450.

PRM_42747 [Mazak-Integrex-i450] — SPN_LOOP_GAIN_7 | Default: 184.700 | Linked Fault: ERR-MAZ450-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i450.

PRM_42748 [Mazak-Integrex-i450] — HYD_LOOP_GAIN_8 | Default: 246.800 | Linked Fault: ERR-MAZ450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i450.

PRM_42749 [Mazak-Integrex-i450] — XAX_LOOP_GAIN_9 | Default: 58.900 | Linked Fault: ERR-MAZ450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i450.

PRM_42750 [Mazak-Integrex-i450] — ZAX_LOOP_GAIN_10 | Default: 121.000 | Linked Fault: ERR-MAZ450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i450.

PRM_42751 [Mazak-Integrex-i450] — CAX_LOOP_GAIN_11 | Default: 183.100 | Linked Fault: ERR-MAZ450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i450.

PRM_42752 [Mazak-Integrex-i450] — PLC_LOOP_GAIN_12 | Default: 245.200 | Linked Fault: ERR-MAZ450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i450.

PRM_42753 [Mazak-Integrex-i450] — VAC_LOOP_GAIN_13 | Default: 57.300 | Linked Fault: ERR-MAZ450-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i450.

PRM_42754 [Mazak-Integrex-i450] — SPN_LOOP_GAIN_14 | Default: 119.400 | Linked Fault: ERR-MAZ450-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i450.

PRM_42755 [Mazak-Integrex-i450] — HYD_LOOP_GAIN_15 | Default: 181.500 | Linked Fault: ERR-MAZ450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i450.

PRM_42756 [Mazak-Integrex-i450] — XAX_LOOP_GAIN_16 | Default: 243.600 | Linked Fault: ERR-MAZ450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i450.

CNC/PLC SERVO PARAMETER DICTIONARY — Mazak-Integrex-i450 (BANK 3)

Machine Model: Mazak-Integrex-i450 | Controller: MAZATROL SmoothAi. Use these parameter registers when diagnosing servo following errors or thermal drift on Mazak-Integrex-i450.

PRM_42761 [Mazak-Integrex-i450] — CLN_LOOP_GAIN_1 | Default: 93.150 | Linked Fault: ERR-MAZ450-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Mazak-Integrex-i450.

PRM_42762 [Mazak-Integrex-i450] — ZAX_LOOP_GAIN_2 | Default: 186.300 | Linked Fault: ERR-MAZ450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i450.

PRM_42763 [Mazak-Integrex-i450] — OPT_LOOP_GAIN_3 | Default: 29.450 | Linked Fault: ERR-MAZ450-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Mazak-Integrex-i450.

PRM_42764 [Mazak-Integrex-i450] — VAC_LOOP_GAIN_4 | Default: 122.600 | Linked Fault: ERR-MAZ450-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i450.

PRM_42765 [Mazak-Integrex-i450] — ATC_LOOP_GAIN_5 | Default: 215.750 | Linked Fault: ERR-MAZ450-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Mazak-Integrex-i450.

PRM_42766 [Mazak-Integrex-i450] — XAX_LOOP_GAIN_6 | Default: 58.900 | Linked Fault: ERR-MAZ450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i450.

PRM_42767 [Mazak-Integrex-i450] — BAX_LOOP_GAIN_7 | Default: 152.050 | Linked Fault: ERR-MAZ450-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Mazak-Integrex-i450.

PRM_42768 [Mazak-Integrex-i450] — PLC_LOOP_GAIN_8 | Default: 245.200 | Linked Fault: ERR-MAZ450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i450.

PRM_42769 [Mazak-Integrex-i450] — THM_LOOP_GAIN_9 | Default: 88.350 | Linked Fault: ERR-MAZ450-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Mazak-Integrex-i450.

PRM_42770 [Mazak-Integrex-i450] — HYD_LOOP_GAIN_10 | Default: 181.500 | Linked Fault: ERR-MAZ450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i450.

PRM_42771 [Mazak-Integrex-i450] — YAX_LOOP_GAIN_11 | Default: 24.650 | Linked Fault: ERR-MAZ450-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Mazak-Integrex-i450.

PRM_42772 [Mazak-Integrex-i450] — CAX_LOOP_GAIN_12 | Default: 117.800 | Linked Fault: ERR-MAZ450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i450.

PRM_42773 [Mazak-Integrex-i450] — LUB_LOOP_GAIN_13 | Default: 210.950 | Linked Fault: ERR-MAZ450-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Mazak-Integrex-i450.

PRM_42774 [Mazak-Integrex-i450] — SPN_LOOP_GAIN_14 | Default: 54.100 | Linked Fault: ERR-MAZ450-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i450.

PRM_42775 [Mazak-Integrex-i450] — CLN_LOOP_GAIN_15 | Default: 147.250 | Linked Fault: ERR-MAZ450-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Mazak-Integrex-i450.

PRM_42776 [Mazak-Integrex-i450] — ZAX_LOOP_GAIN_16 | Default: 240.400 | Linked Fault: ERR-MAZ450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i450.

CNC/PLC SERVO PARAMETER DICTIONARY — Mazak-Integrex-i450 (BANK 4)

Machine Model: Mazak-Integrex-i450 | Controller: MAZATROL SmoothAi. Use these parameter registers when diagnosing servo following errors or thermal drift on Mazak-Integrex-i450.

PRM_42781 [Mazak-Integrex-i450] — XAX_LOOP_GAIN_1 | Default: 124.200 | Linked Fault: ERR-MAZ450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i450.

PRM_42782 [Mazak-Integrex-i450] — CAX_LOOP_GAIN_2 | Default: 248.400 | Linked Fault: ERR-MAZ450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i450.

PRM_42783 [Mazak-Integrex-i450] — VAC_LOOP_GAIN_3 | Default: 122.600 | Linked Fault: ERR-MAZ450-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i450.

PRM_42784 [Mazak-Integrex-i450] — HYD_LOOP_GAIN_4 | Default: 246.800 | Linked Fault: ERR-MAZ450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i450.

PRM_42785 [Mazak-Integrex-i450] — ZAX_LOOP_GAIN_5 | Default: 121.000 | Linked Fault: ERR-MAZ450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i450.

PRM_42786 [Mazak-Integrex-i450] — PLC_LOOP_GAIN_6 | Default: 245.200 | Linked Fault: ERR-MAZ450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i450.

PRM_42787 [Mazak-Integrex-i450] — SPN_LOOP_GAIN_7 | Default: 119.400 | Linked Fault: ERR-MAZ450-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i450.

PRM_42788 [Mazak-Integrex-i450] — XAX_LOOP_GAIN_8 | Default: 243.600 | Linked Fault: ERR-MAZ450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i450.

PRM_42789 [Mazak-Integrex-i450] — CAX_LOOP_GAIN_9 | Default: 117.800 | Linked Fault: ERR-MAZ450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i450.

PRM_42790 [Mazak-Integrex-i450] — VAC_LOOP_GAIN_10 | Default: 242.000 | Linked Fault: ERR-MAZ450-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i450.

PRM_42791 [Mazak-Integrex-i450] — HYD_LOOP_GAIN_11 | Default: 116.200 | Linked Fault: ERR-MAZ450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i450.

PRM_42792 [Mazak-Integrex-i450] — ZAX_LOOP_GAIN_12 | Default: 240.400 | Linked Fault: ERR-MAZ450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i450.

PRM_42793 [Mazak-Integrex-i450] — PLC_LOOP_GAIN_13 | Default: 114.600 | Linked Fault: ERR-MAZ450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i450.

PRM_42794 [Mazak-Integrex-i450] — SPN_LOOP_GAIN_14 | Default: 238.800 | Linked Fault: ERR-MAZ450-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i450.

PRM_42795 [Mazak-Integrex-i450] — XAX_LOOP_GAIN_15 | Default: 113.000 | Linked Fault: ERR-MAZ450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i450.

PRM_42796 [Mazak-Integrex-i450] — CAX_LOOP_GAIN_16 | Default: 237.200 | Linked Fault: ERR-MAZ450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i450.

CNC/PLC SERVO PARAMETER DICTIONARY — Mazak-Integrex-i450 (BANK 5)

Machine Model: Mazak-Integrex-i450 | Controller: MAZATROL SmoothAi. Use these parameter registers when diagnosing servo following errors or thermal drift on Mazak-Integrex-i450.

PRM_42801 [Mazak-Integrex-i450] — YAX_LOOP_GAIN_1 | Default: 155.250 | Linked Fault: ERR-MAZ450-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Mazak-Integrex-i450.

PRM_42802 [Mazak-Integrex-i450] — PLC_LOOP_GAIN_2 | Default: 60.500 | Linked Fault: ERR-MAZ450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i450.

PRM_42803 [Mazak-Integrex-i450] — ATC_LOOP_GAIN_3 | Default: 215.750 | Linked Fault: ERR-MAZ450-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Mazak-Integrex-i450.

PRM_42804 [Mazak-Integrex-i450] — ZAX_LOOP_GAIN_4 | Default: 121.000 | Linked Fault: ERR-MAZ450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i450.

PRM_42805 [Mazak-Integrex-i450] — LUB_LOOP_GAIN_5 | Default: 26.250 | Linked Fault: ERR-MAZ450-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Mazak-Integrex-i450.

PRM_42806 [Mazak-Integrex-i450] — HYD_LOOP_GAIN_6 | Default: 181.500 | Linked Fault: ERR-MAZ450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i450.

PRM_42807 [Mazak-Integrex-i450] — BAX_LOOP_GAIN_7 | Default: 86.750 | Linked Fault: ERR-MAZ450-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Mazak-Integrex-i450.

PRM_42808 [Mazak-Integrex-i450] — VAC_LOOP_GAIN_8 | Default: 242.000 | Linked Fault: ERR-MAZ450-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i450.

PRM_42809 [Mazak-Integrex-i450] — CLN_LOOP_GAIN_9 | Default: 147.250 | Linked Fault: ERR-MAZ450-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Mazak-Integrex-i450.

PRM_42810 [Mazak-Integrex-i450] — CAX_LOOP_GAIN_10 | Default: 52.500 | Linked Fault: ERR-MAZ450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i450.

PRM_42811 [Mazak-Integrex-i450] — THM_LOOP_GAIN_11 | Default: 207.750 | Linked Fault: ERR-MAZ450-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Mazak-Integrex-i450.

PRM_42812 [Mazak-Integrex-i450] — XAX_LOOP_GAIN_12 | Default: 113.000 | Linked Fault: ERR-MAZ450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i450.

PRM_42813 [Mazak-Integrex-i450] — OPT_LOOP_GAIN_13 | Default: 18.250 | Linked Fault: ERR-MAZ450-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Mazak-Integrex-i450.

PRM_42814 [Mazak-Integrex-i450] — SPN_LOOP_GAIN_14 | Default: 173.500 | Linked Fault: ERR-MAZ450-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i450.

PRM_42815 [Mazak-Integrex-i450] — YAX_LOOP_GAIN_15 | Default: 78.750 | Linked Fault: ERR-MAZ450-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Mazak-Integrex-i450.

PRM_42816 [Mazak-Integrex-i450] — PLC_LOOP_GAIN_16 | Default: 234.000 | Linked Fault: ERR-MAZ450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i450.

CNC/PLC SERVO PARAMETER DICTIONARY — Mazak-Integrex-i500 (BANK 1)

Machine Model: Mazak-Integrex-i500 | Controller: MAZATROL SmoothAi. Use these parameter registers when diagnosing servo following errors or thermal drift on Mazak-Integrex-i500.

PRM_42821 [Mazak-Integrex-i500] — ATC_LOOP_GAIN_1 | Default: 32.200 | Linked Fault: ERR-MAZ500-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Mazak-Integrex-i500.

PRM_42822 [Mazak-Integrex-i500] — HYD_LOOP_GAIN_2 | Default: 64.400 | Linked Fault: ERR-MAZ500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i500.

PRM_42823 [Mazak-Integrex-i500] — CLN_LOOP_GAIN_3 | Default: 96.600 | Linked Fault: ERR-MAZ500-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Mazak-Integrex-i500.

PRM_42824 [Mazak-Integrex-i500] — XAX_LOOP_GAIN_4 | Default: 128.800 | Linked Fault: ERR-MAZ500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i500.

PRM_42825 [Mazak-Integrex-i500] — YAX_LOOP_GAIN_5 | Default: 161.000 | Linked Fault: ERR-MAZ500-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Mazak-Integrex-i500.

PRM_42826 [Mazak-Integrex-i500] — ZAX_LOOP_GAIN_6 | Default: 193.200 | Linked Fault: ERR-MAZ500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i500.

PRM_42827 [Mazak-Integrex-i500] — BAX_LOOP_GAIN_7 | Default: 225.400 | Linked Fault: ERR-MAZ500-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Mazak-Integrex-i500.

PRM_42828 [Mazak-Integrex-i500] — CAX_LOOP_GAIN_8 | Default: 7.600 | Linked Fault: ERR-MAZ500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i500.

PRM_42829 [Mazak-Integrex-i500] — OPT_LOOP_GAIN_9 | Default: 39.800 | Linked Fault: ERR-MAZ500-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Mazak-Integrex-i500.

PRM_42830 [Mazak-Integrex-i500] — PLC_LOOP_GAIN_10 | Default: 72.000 | Linked Fault: ERR-MAZ500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i500.

PRM_42831 [Mazak-Integrex-i500] — LUB_LOOP_GAIN_11 | Default: 104.200 | Linked Fault: ERR-MAZ500-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Mazak-Integrex-i500.

PRM_42832 [Mazak-Integrex-i500] — VAC_LOOP_GAIN_12 | Default: 136.400 | Linked Fault: ERR-MAZ500-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i500.

PRM_42833 [Mazak-Integrex-i500] — THM_LOOP_GAIN_13 | Default: 168.600 | Linked Fault: ERR-MAZ500-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Mazak-Integrex-i500.

PRM_42834 [Mazak-Integrex-i500] — SPN_LOOP_GAIN_14 | Default: 200.800 | Linked Fault: ERR-MAZ500-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i500.

PRM_42835 [Mazak-Integrex-i500] — ATC_LOOP_GAIN_15 | Default: 233.000 | Linked Fault: ERR-MAZ500-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Mazak-Integrex-i500.

PRM_42836 [Mazak-Integrex-i500] — HYD_LOOP_GAIN_16 | Default: 15.200 | Linked Fault: ERR-MAZ500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i500.

CNC/PLC SERVO PARAMETER DICTIONARY — Mazak-Integrex-i500 (BANK 2)

Machine Model: Mazak-Integrex-i500 | Controller: MAZATROL SmoothAi. Use these parameter registers when diagnosing servo following errors or thermal drift on Mazak-Integrex-i500.

PRM_42841 [Mazak-Integrex-i500] — HYD_LOOP_GAIN_1 | Default: 64.400 | Linked Fault: ERR-MAZ500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i500.

PRM_42842 [Mazak-Integrex-i500] — XAX_LOOP_GAIN_2 | Default: 128.800 | Linked Fault: ERR-MAZ500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i500.

PRM_42843 [Mazak-Integrex-i500] — ZAX_LOOP_GAIN_3 | Default: 193.200 | Linked Fault: ERR-MAZ500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i500.

PRM_42844 [Mazak-Integrex-i500] — CAX_LOOP_GAIN_4 | Default: 7.600 | Linked Fault: ERR-MAZ500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i500.

PRM_42845 [Mazak-Integrex-i500] — PLC_LOOP_GAIN_5 | Default: 72.000 | Linked Fault: ERR-MAZ500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i500.

PRM_42846 [Mazak-Integrex-i500] — VAC_LOOP_GAIN_6 | Default: 136.400 | Linked Fault: ERR-MAZ500-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i500.

PRM_42847 [Mazak-Integrex-i500] — SPN_LOOP_GAIN_7 | Default: 200.800 | Linked Fault: ERR-MAZ500-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i500.

PRM_42848 [Mazak-Integrex-i500] — HYD_LOOP_GAIN_8 | Default: 15.200 | Linked Fault: ERR-MAZ500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i500.

PRM_42849 [Mazak-Integrex-i500] — XAX_LOOP_GAIN_9 | Default: 79.600 | Linked Fault: ERR-MAZ500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i500.

PRM_42850 [Mazak-Integrex-i500] — ZAX_LOOP_GAIN_10 | Default: 144.000 | Linked Fault: ERR-MAZ500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i500.

PRM_42851 [Mazak-Integrex-i500] — CAX_LOOP_GAIN_11 | Default: 208.400 | Linked Fault: ERR-MAZ500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i500.

PRM_42852 [Mazak-Integrex-i500] — PLC_LOOP_GAIN_12 | Default: 22.800 | Linked Fault: ERR-MAZ500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i500.

PRM_42853 [Mazak-Integrex-i500] — VAC_LOOP_GAIN_13 | Default: 87.200 | Linked Fault: ERR-MAZ500-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i500.

PRM_42854 [Mazak-Integrex-i500] — SPN_LOOP_GAIN_14 | Default: 151.600 | Linked Fault: ERR-MAZ500-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i500.

PRM_42855 [Mazak-Integrex-i500] — HYD_LOOP_GAIN_15 | Default: 216.000 | Linked Fault: ERR-MAZ500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i500.

PRM_42856 [Mazak-Integrex-i500] — XAX_LOOP_GAIN_16 | Default: 30.400 | Linked Fault: ERR-MAZ500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i500.

CNC/PLC SERVO PARAMETER DICTIONARY — Mazak-Integrex-i500 (BANK 3)

Machine Model: Mazak-Integrex-i500 | Controller: MAZATROL SmoothAi. Use these parameter registers when diagnosing servo following errors or thermal drift on Mazak-Integrex-i500.

PRM_42861 [Mazak-Integrex-i500] — CLN_LOOP_GAIN_1 | Default: 96.600 | Linked Fault: ERR-MAZ500-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Mazak-Integrex-i500.

PRM_42862 [Mazak-Integrex-i500] — ZAX_LOOP_GAIN_2 | Default: 193.200 | Linked Fault: ERR-MAZ500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i500.

PRM_42863 [Mazak-Integrex-i500] — OPT_LOOP_GAIN_3 | Default: 39.800 | Linked Fault: ERR-MAZ500-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Mazak-Integrex-i500.

PRM_42864 [Mazak-Integrex-i500] — VAC_LOOP_GAIN_4 | Default: 136.400 | Linked Fault: ERR-MAZ500-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i500.

PRM_42865 [Mazak-Integrex-i500] — ATC_LOOP_GAIN_5 | Default: 233.000 | Linked Fault: ERR-MAZ500-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Mazak-Integrex-i500.

PRM_42866 [Mazak-Integrex-i500] — XAX_LOOP_GAIN_6 | Default: 79.600 | Linked Fault: ERR-MAZ500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i500.

PRM_42867 [Mazak-Integrex-i500] — BAX_LOOP_GAIN_7 | Default: 176.200 | Linked Fault: ERR-MAZ500-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Mazak-Integrex-i500.

PRM_42868 [Mazak-Integrex-i500] — PLC_LOOP_GAIN_8 | Default: 22.800 | Linked Fault: ERR-MAZ500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i500.

PRM_42869 [Mazak-Integrex-i500] — THM_LOOP_GAIN_9 | Default: 119.400 | Linked Fault: ERR-MAZ500-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Mazak-Integrex-i500.

PRM_42870 [Mazak-Integrex-i500] — HYD_LOOP_GAIN_10 | Default: 216.000 | Linked Fault: ERR-MAZ500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i500.

PRM_42871 [Mazak-Integrex-i500] — YAX_LOOP_GAIN_11 | Default: 62.600 | Linked Fault: ERR-MAZ500-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Mazak-Integrex-i500.

PRM_42872 [Mazak-Integrex-i500] — CAX_LOOP_GAIN_12 | Default: 159.200 | Linked Fault: ERR-MAZ500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i500.

PRM_42873 [Mazak-Integrex-i500] — LUB_LOOP_GAIN_13 | Default: 5.800 | Linked Fault: ERR-MAZ500-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Mazak-Integrex-i500.

PRM_42874 [Mazak-Integrex-i500] — SPN_LOOP_GAIN_14 | Default: 102.400 | Linked Fault: ERR-MAZ500-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i500.

PRM_42875 [Mazak-Integrex-i500] — CLN_LOOP_GAIN_15 | Default: 199.000 | Linked Fault: ERR-MAZ500-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Mazak-Integrex-i500.

PRM_42876 [Mazak-Integrex-i500] — ZAX_LOOP_GAIN_16 | Default: 45.600 | Linked Fault: ERR-MAZ500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i500.

CNC/PLC SERVO PARAMETER DICTIONARY — Mazak-Integrex-i500 (BANK 4)

Machine Model: Mazak-Integrex-i500 | Controller: MAZATROL SmoothAi. Use these parameter registers when diagnosing servo following errors or thermal drift on Mazak-Integrex-i500.

PRM_42881 [Mazak-Integrex-i500] — XAX_LOOP_GAIN_1 | Default: 128.800 | Linked Fault: ERR-MAZ500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i500.

PRM_42882 [Mazak-Integrex-i500] — CAX_LOOP_GAIN_2 | Default: 7.600 | Linked Fault: ERR-MAZ500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i500.

PRM_42883 [Mazak-Integrex-i500] — VAC_LOOP_GAIN_3 | Default: 136.400 | Linked Fault: ERR-MAZ500-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i500.

PRM_42884 [Mazak-Integrex-i500] — HYD_LOOP_GAIN_4 | Default: 15.200 | Linked Fault: ERR-MAZ500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i500.

PRM_42885 [Mazak-Integrex-i500] — ZAX_LOOP_GAIN_5 | Default: 144.000 | Linked Fault: ERR-MAZ500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i500.

PRM_42886 [Mazak-Integrex-i500] — PLC_LOOP_GAIN_6 | Default: 22.800 | Linked Fault: ERR-MAZ500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i500.

PRM_42887 [Mazak-Integrex-i500] — SPN_LOOP_GAIN_7 | Default: 151.600 | Linked Fault: ERR-MAZ500-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i500.

PRM_42888 [Mazak-Integrex-i500] — XAX_LOOP_GAIN_8 | Default: 30.400 | Linked Fault: ERR-MAZ500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i500.

PRM_42889 [Mazak-Integrex-i500] — CAX_LOOP_GAIN_9 | Default: 159.200 | Linked Fault: ERR-MAZ500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i500.

PRM_42890 [Mazak-Integrex-i500] — VAC_LOOP_GAIN_10 | Default: 38.000 | Linked Fault: ERR-MAZ500-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i500.

PRM_42891 [Mazak-Integrex-i500] — HYD_LOOP_GAIN_11 | Default: 166.800 | Linked Fault: ERR-MAZ500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i500.

PRM_42892 [Mazak-Integrex-i500] — ZAX_LOOP_GAIN_12 | Default: 45.600 | Linked Fault: ERR-MAZ500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i500.

PRM_42893 [Mazak-Integrex-i500] — PLC_LOOP_GAIN_13 | Default: 174.400 | Linked Fault: ERR-MAZ500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i500.

PRM_42894 [Mazak-Integrex-i500] — SPN_LOOP_GAIN_14 | Default: 53.200 | Linked Fault: ERR-MAZ500-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i500.

PRM_42895 [Mazak-Integrex-i500] — XAX_LOOP_GAIN_15 | Default: 182.000 | Linked Fault: ERR-MAZ500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i500.

PRM_42896 [Mazak-Integrex-i500] — CAX_LOOP_GAIN_16 | Default: 60.800 | Linked Fault: ERR-MAZ500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i500.

CNC/PLC SERVO PARAMETER DICTIONARY — Mazak-Integrex-i500 (BANK 5)

Machine Model: Mazak-Integrex-i500 | Controller: MAZATROL SmoothAi. Use these parameter registers when diagnosing servo following errors or thermal drift on Mazak-Integrex-i500.

PRM_42901 [Mazak-Integrex-i500] — YAX_LOOP_GAIN_1 | Default: 161.000 | Linked Fault: ERR-MAZ500-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Mazak-Integrex-i500.

PRM_42902 [Mazak-Integrex-i500] — PLC_LOOP_GAIN_2 | Default: 72.000 | Linked Fault: ERR-MAZ500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i500.

PRM_42903 [Mazak-Integrex-i500] — ATC_LOOP_GAIN_3 | Default: 233.000 | Linked Fault: ERR-MAZ500-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Mazak-Integrex-i500.

PRM_42904 [Mazak-Integrex-i500] — ZAX_LOOP_GAIN_4 | Default: 144.000 | Linked Fault: ERR-MAZ500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i500.

PRM_42905 [Mazak-Integrex-i500] — LUB_LOOP_GAIN_5 | Default: 55.000 | Linked Fault: ERR-MAZ500-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Mazak-Integrex-i500.

PRM_42906 [Mazak-Integrex-i500] — HYD_LOOP_GAIN_6 | Default: 216.000 | Linked Fault: ERR-MAZ500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i500.

PRM_42907 [Mazak-Integrex-i500] — BAX_LOOP_GAIN_7 | Default: 127.000 | Linked Fault: ERR-MAZ500-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Mazak-Integrex-i500.

PRM_42908 [Mazak-Integrex-i500] — VAC_LOOP_GAIN_8 | Default: 38.000 | Linked Fault: ERR-MAZ500-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i500.

PRM_42909 [Mazak-Integrex-i500] — CLN_LOOP_GAIN_9 | Default: 199.000 | Linked Fault: ERR-MAZ500-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Mazak-Integrex-i500.

PRM_42910 [Mazak-Integrex-i500] — CAX_LOOP_GAIN_10 | Default: 110.000 | Linked Fault: ERR-MAZ500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i500.

PRM_42911 [Mazak-Integrex-i500] — THM_LOOP_GAIN_11 | Default: 21.000 | Linked Fault: ERR-MAZ500-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Mazak-Integrex-i500.

PRM_42912 [Mazak-Integrex-i500] — XAX_LOOP_GAIN_12 | Default: 182.000 | Linked Fault: ERR-MAZ500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i500.

PRM_42913 [Mazak-Integrex-i500] — OPT_LOOP_GAIN_13 | Default: 93.000 | Linked Fault: ERR-MAZ500-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Mazak-Integrex-i500.

PRM_42914 [Mazak-Integrex-i500] — SPN_LOOP_GAIN_14 | Default: 4.000 | Linked Fault: ERR-MAZ500-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i500.

PRM_42915 [Mazak-Integrex-i500] — YAX_LOOP_GAIN_15 | Default: 165.000 | Linked Fault: ERR-MAZ500-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Mazak-Integrex-i500.

PRM_42916 [Mazak-Integrex-i500] — PLC_LOOP_GAIN_16 | Default: 76.000 | Linked Fault: ERR-MAZ500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i500.

CNC/PLC SERVO PARAMETER DICTIONARY — Mazak-Integrex-i550 (BANK 1)

Machine Model: Mazak-Integrex-i550 | Controller: MAZATROL SmoothAi. Use these parameter registers when diagnosing servo following errors or thermal drift on Mazak-Integrex-i550.

PRM_42921 [Mazak-Integrex-i550] — ATC_LOOP_GAIN_1 | Default: 33.350 | Linked Fault: ERR-MAZ550-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Mazak-Integrex-i550.

PRM_42922 [Mazak-Integrex-i550] — HYD_LOOP_GAIN_2 | Default: 66.700 | Linked Fault: ERR-MAZ550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i550.

PRM_42923 [Mazak-Integrex-i550] — CLN_LOOP_GAIN_3 | Default: 100.050 | Linked Fault: ERR-MAZ550-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Mazak-Integrex-i550.

PRM_42924 [Mazak-Integrex-i550] — XAX_LOOP_GAIN_4 | Default: 133.400 | Linked Fault: ERR-MAZ550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i550.

PRM_42925 [Mazak-Integrex-i550] — YAX_LOOP_GAIN_5 | Default: 166.750 | Linked Fault: ERR-MAZ550-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Mazak-Integrex-i550.

PRM_42926 [Mazak-Integrex-i550] — ZAX_LOOP_GAIN_6 | Default: 200.100 | Linked Fault: ERR-MAZ550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i550.

PRM_42927 [Mazak-Integrex-i550] — BAX_LOOP_GAIN_7 | Default: 233.450 | Linked Fault: ERR-MAZ550-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Mazak-Integrex-i550.

PRM_42928 [Mazak-Integrex-i550] — CAX_LOOP_GAIN_8 | Default: 16.800 | Linked Fault: ERR-MAZ550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i550.

PRM_42929 [Mazak-Integrex-i550] — OPT_LOOP_GAIN_9 | Default: 50.150 | Linked Fault: ERR-MAZ550-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Mazak-Integrex-i550.

PRM_42930 [Mazak-Integrex-i550] — PLC_LOOP_GAIN_10 | Default: 83.500 | Linked Fault: ERR-MAZ550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i550.

PRM_42931 [Mazak-Integrex-i550] — LUB_LOOP_GAIN_11 | Default: 116.850 | Linked Fault: ERR-MAZ550-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Mazak-Integrex-i550.

PRM_42932 [Mazak-Integrex-i550] — VAC_LOOP_GAIN_12 | Default: 150.200 | Linked Fault: ERR-MAZ550-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i550.

PRM_42933 [Mazak-Integrex-i550] — THM_LOOP_GAIN_13 | Default: 183.550 | Linked Fault: ERR-MAZ550-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Mazak-Integrex-i550.

PRM_42934 [Mazak-Integrex-i550] — SPN_LOOP_GAIN_14 | Default: 216.900 | Linked Fault: ERR-MAZ550-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i550.

PRM_42935 [Mazak-Integrex-i550] — ATC_LOOP_GAIN_15 | Default: 0.250 | Linked Fault: ERR-MAZ550-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Mazak-Integrex-i550.

PRM_42936 [Mazak-Integrex-i550] — HYD_LOOP_GAIN_16 | Default: 33.600 | Linked Fault: ERR-MAZ550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i550.

CNC/PLC SERVO PARAMETER DICTIONARY — Mazak-Integrex-i550 (BANK 2)

Machine Model: Mazak-Integrex-i550 | Controller: MAZATROL SmoothAi. Use these parameter registers when diagnosing servo following errors or thermal drift on Mazak-Integrex-i550.

PRM_42941 [Mazak-Integrex-i550] — HYD_LOOP_GAIN_1 | Default: 66.700 | Linked Fault: ERR-MAZ550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i550.

PRM_42942 [Mazak-Integrex-i550] — XAX_LOOP_GAIN_2 | Default: 133.400 | Linked Fault: ERR-MAZ550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i550.

PRM_42943 [Mazak-Integrex-i550] — ZAX_LOOP_GAIN_3 | Default: 200.100 | Linked Fault: ERR-MAZ550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i550.

PRM_42944 [Mazak-Integrex-i550] — CAX_LOOP_GAIN_4 | Default: 16.800 | Linked Fault: ERR-MAZ550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i550.

PRM_42945 [Mazak-Integrex-i550] — PLC_LOOP_GAIN_5 | Default: 83.500 | Linked Fault: ERR-MAZ550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i550.

PRM_42946 [Mazak-Integrex-i550] — VAC_LOOP_GAIN_6 | Default: 150.200 | Linked Fault: ERR-MAZ550-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i550.

PRM_42947 [Mazak-Integrex-i550] — SPN_LOOP_GAIN_7 | Default: 216.900 | Linked Fault: ERR-MAZ550-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i550.

PRM_42948 [Mazak-Integrex-i550] — HYD_LOOP_GAIN_8 | Default: 33.600 | Linked Fault: ERR-MAZ550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i550.

PRM_42949 [Mazak-Integrex-i550] — XAX_LOOP_GAIN_9 | Default: 100.300 | Linked Fault: ERR-MAZ550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i550.

PRM_42950 [Mazak-Integrex-i550] — ZAX_LOOP_GAIN_10 | Default: 167.000 | Linked Fault: ERR-MAZ550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i550.

PRM_42951 [Mazak-Integrex-i550] — CAX_LOOP_GAIN_11 | Default: 233.700 | Linked Fault: ERR-MAZ550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i550.

PRM_42952 [Mazak-Integrex-i550] — PLC_LOOP_GAIN_12 | Default: 50.400 | Linked Fault: ERR-MAZ550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i550.

PRM_42953 [Mazak-Integrex-i550] — VAC_LOOP_GAIN_13 | Default: 117.100 | Linked Fault: ERR-MAZ550-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i550.

PRM_42954 [Mazak-Integrex-i550] — SPN_LOOP_GAIN_14 | Default: 183.800 | Linked Fault: ERR-MAZ550-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i550.

PRM_42955 [Mazak-Integrex-i550] — HYD_LOOP_GAIN_15 | Default: 0.500 | Linked Fault: ERR-MAZ550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i550.

PRM_42956 [Mazak-Integrex-i550] — XAX_LOOP_GAIN_16 | Default: 67.200 | Linked Fault: ERR-MAZ550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i550.

CNC/PLC SERVO PARAMETER DICTIONARY — Mazak-Integrex-i550 (BANK 3)

Machine Model: Mazak-Integrex-i550 | Controller: MAZATROL SmoothAi. Use these parameter registers when diagnosing servo following errors or thermal drift on Mazak-Integrex-i550.

PRM_42961 [Mazak-Integrex-i550] — CLN_LOOP_GAIN_1 | Default: 100.050 | Linked Fault: ERR-MAZ550-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Mazak-Integrex-i550.

PRM_42962 [Mazak-Integrex-i550] — ZAX_LOOP_GAIN_2 | Default: 200.100 | Linked Fault: ERR-MAZ550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i550.

PRM_42963 [Mazak-Integrex-i550] — OPT_LOOP_GAIN_3 | Default: 50.150 | Linked Fault: ERR-MAZ550-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Mazak-Integrex-i550.

PRM_42964 [Mazak-Integrex-i550] — VAC_LOOP_GAIN_4 | Default: 150.200 | Linked Fault: ERR-MAZ550-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i550.

PRM_42965 [Mazak-Integrex-i550] — ATC_LOOP_GAIN_5 | Default: 0.250 | Linked Fault: ERR-MAZ550-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Mazak-Integrex-i550.

PRM_42966 [Mazak-Integrex-i550] — XAX_LOOP_GAIN_6 | Default: 100.300 | Linked Fault: ERR-MAZ550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i550.

PRM_42967 [Mazak-Integrex-i550] — BAX_LOOP_GAIN_7 | Default: 200.350 | Linked Fault: ERR-MAZ550-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Mazak-Integrex-i550.

PRM_42968 [Mazak-Integrex-i550] — PLC_LOOP_GAIN_8 | Default: 50.400 | Linked Fault: ERR-MAZ550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i550.

PRM_42969 [Mazak-Integrex-i550] — THM_LOOP_GAIN_9 | Default: 150.450 | Linked Fault: ERR-MAZ550-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Mazak-Integrex-i550.

PRM_42970 [Mazak-Integrex-i550] — HYD_LOOP_GAIN_10 | Default: 0.500 | Linked Fault: ERR-MAZ550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i550.

PRM_42971 [Mazak-Integrex-i550] — YAX_LOOP_GAIN_11 | Default: 100.550 | Linked Fault: ERR-MAZ550-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Mazak-Integrex-i550.

PRM_42972 [Mazak-Integrex-i550] — CAX_LOOP_GAIN_12 | Default: 200.600 | Linked Fault: ERR-MAZ550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i550.

PRM_42973 [Mazak-Integrex-i550] — LUB_LOOP_GAIN_13 | Default: 50.650 | Linked Fault: ERR-MAZ550-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Mazak-Integrex-i550.

PRM_42974 [Mazak-Integrex-i550] — SPN_LOOP_GAIN_14 | Default: 150.700 | Linked Fault: ERR-MAZ550-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i550.

PRM_42975 [Mazak-Integrex-i550] — CLN_LOOP_GAIN_15 | Default: 0.750 | Linked Fault: ERR-MAZ550-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Mazak-Integrex-i550.

PRM_42976 [Mazak-Integrex-i550] — ZAX_LOOP_GAIN_16 | Default: 100.800 | Linked Fault: ERR-MAZ550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i550.

CNC/PLC SERVO PARAMETER DICTIONARY — Mazak-Integrex-i550 (BANK 4)

Machine Model: Mazak-Integrex-i550 | Controller: MAZATROL SmoothAi. Use these parameter registers when diagnosing servo following errors or thermal drift on Mazak-Integrex-i550.

PRM_42981 [Mazak-Integrex-i550] — XAX_LOOP_GAIN_1 | Default: 133.400 | Linked Fault: ERR-MAZ550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i550.

PRM_42982 [Mazak-Integrex-i550] — CAX_LOOP_GAIN_2 | Default: 16.800 | Linked Fault: ERR-MAZ550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i550.

PRM_42983 [Mazak-Integrex-i550] — VAC_LOOP_GAIN_3 | Default: 150.200 | Linked Fault: ERR-MAZ550-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i550.

PRM_42984 [Mazak-Integrex-i550] — HYD_LOOP_GAIN_4 | Default: 33.600 | Linked Fault: ERR-MAZ550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i550.

PRM_42985 [Mazak-Integrex-i550] — ZAX_LOOP_GAIN_5 | Default: 167.000 | Linked Fault: ERR-MAZ550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i550.

PRM_42986 [Mazak-Integrex-i550] — PLC_LOOP_GAIN_6 | Default: 50.400 | Linked Fault: ERR-MAZ550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i550.

PRM_42987 [Mazak-Integrex-i550] — SPN_LOOP_GAIN_7 | Default: 183.800 | Linked Fault: ERR-MAZ550-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i550.

PRM_42988 [Mazak-Integrex-i550] — XAX_LOOP_GAIN_8 | Default: 67.200 | Linked Fault: ERR-MAZ550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i550.

PRM_42989 [Mazak-Integrex-i550] — CAX_LOOP_GAIN_9 | Default: 200.600 | Linked Fault: ERR-MAZ550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i550.

PRM_42990 [Mazak-Integrex-i550] — VAC_LOOP_GAIN_10 | Default: 84.000 | Linked Fault: ERR-MAZ550-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i550.

PRM_42991 [Mazak-Integrex-i550] — HYD_LOOP_GAIN_11 | Default: 217.400 | Linked Fault: ERR-MAZ550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i550.

PRM_42992 [Mazak-Integrex-i550] — ZAX_LOOP_GAIN_12 | Default: 100.800 | Linked Fault: ERR-MAZ550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i550.

PRM_42993 [Mazak-Integrex-i550] — PLC_LOOP_GAIN_13 | Default: 234.200 | Linked Fault: ERR-MAZ550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i550.

PRM_42994 [Mazak-Integrex-i550] — SPN_LOOP_GAIN_14 | Default: 117.600 | Linked Fault: ERR-MAZ550-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i550.

PRM_42995 [Mazak-Integrex-i550] — XAX_LOOP_GAIN_15 | Default: 1.000 | Linked Fault: ERR-MAZ550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i550.

PRM_42996 [Mazak-Integrex-i550] — CAX_LOOP_GAIN_16 | Default: 134.400 | Linked Fault: ERR-MAZ550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i550.

CNC/PLC SERVO PARAMETER DICTIONARY — Mazak-Integrex-i550 (BANK 5)

Machine Model: Mazak-Integrex-i550 | Controller: MAZATROL SmoothAi. Use these parameter registers when diagnosing servo following errors or thermal drift on Mazak-Integrex-i550.

PRM_43001 [Mazak-Integrex-i550] — YAX_LOOP_GAIN_1 | Default: 166.750 | Linked Fault: ERR-MAZ550-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Mazak-Integrex-i550.

PRM_43002 [Mazak-Integrex-i550] — PLC_LOOP_GAIN_2 | Default: 83.500 | Linked Fault: ERR-MAZ550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i550.

PRM_43003 [Mazak-Integrex-i550] — ATC_LOOP_GAIN_3 | Default: 0.250 | Linked Fault: ERR-MAZ550-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Mazak-Integrex-i550.

PRM_43004 [Mazak-Integrex-i550] — ZAX_LOOP_GAIN_4 | Default: 167.000 | Linked Fault: ERR-MAZ550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i550.

PRM_43005 [Mazak-Integrex-i550] — LUB_LOOP_GAIN_5 | Default: 83.750 | Linked Fault: ERR-MAZ550-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Mazak-Integrex-i550.

PRM_43006 [Mazak-Integrex-i550] — HYD_LOOP_GAIN_6 | Default: 0.500 | Linked Fault: ERR-MAZ550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i550.

PRM_43007 [Mazak-Integrex-i550] — BAX_LOOP_GAIN_7 | Default: 167.250 | Linked Fault: ERR-MAZ550-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Mazak-Integrex-i550.

PRM_43008 [Mazak-Integrex-i550] — VAC_LOOP_GAIN_8 | Default: 84.000 | Linked Fault: ERR-MAZ550-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i550.

PRM_43009 [Mazak-Integrex-i550] — CLN_LOOP_GAIN_9 | Default: 0.750 | Linked Fault: ERR-MAZ550-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Mazak-Integrex-i550.

PRM_43010 [Mazak-Integrex-i550] — CAX_LOOP_GAIN_10 | Default: 167.500 | Linked Fault: ERR-MAZ550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i550.

PRM_43011 [Mazak-Integrex-i550] — THM_LOOP_GAIN_11 | Default: 84.250 | Linked Fault: ERR-MAZ550-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Mazak-Integrex-i550.

PRM_43012 [Mazak-Integrex-i550] — XAX_LOOP_GAIN_12 | Default: 1.000 | Linked Fault: ERR-MAZ550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i550.

PRM_43013 [Mazak-Integrex-i550] — OPT_LOOP_GAIN_13 | Default: 167.750 | Linked Fault: ERR-MAZ550-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Mazak-Integrex-i550.

PRM_43014 [Mazak-Integrex-i550] — SPN_LOOP_GAIN_14 | Default: 84.500 | Linked Fault: ERR-MAZ550-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i550.

PRM_43015 [Mazak-Integrex-i550] — YAX_LOOP_GAIN_15 | Default: 1.250 | Linked Fault: ERR-MAZ550-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Mazak-Integrex-i550.

PRM_43016 [Mazak-Integrex-i550] — PLC_LOOP_GAIN_16 | Default: 168.000 | Linked Fault: ERR-MAZ550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i550.

CNC/PLC SERVO PARAMETER DICTIONARY — Mazak-Integrex-i600 (BANK 1)

Machine Model: Mazak-Integrex-i600 | Controller: MAZATROL SmoothAi. Use these parameter registers when diagnosing servo following errors or thermal drift on Mazak-Integrex-i600.

PRM_43021 [Mazak-Integrex-i600] — ATC_LOOP_GAIN_1 | Default: 34.500 | Linked Fault: ERR-MAZ600-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Mazak-Integrex-i600.

PRM_43022 [Mazak-Integrex-i600] — HYD_LOOP_GAIN_2 | Default: 69.000 | Linked Fault: ERR-MAZ600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i600.

PRM_43023 [Mazak-Integrex-i600] — CLN_LOOP_GAIN_3 | Default: 103.500 | Linked Fault: ERR-MAZ600-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Mazak-Integrex-i600.

PRM_43024 [Mazak-Integrex-i600] — XAX_LOOP_GAIN_4 | Default: 138.000 | Linked Fault: ERR-MAZ600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i600.

PRM_43025 [Mazak-Integrex-i600] — YAX_LOOP_GAIN_5 | Default: 172.500 | Linked Fault: ERR-MAZ600-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Mazak-Integrex-i600.

PRM_43026 [Mazak-Integrex-i600] — ZAX_LOOP_GAIN_6 | Default: 207.000 | Linked Fault: ERR-MAZ600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i600.

PRM_43027 [Mazak-Integrex-i600] — BAX_LOOP_GAIN_7 | Default: 241.500 | Linked Fault: ERR-MAZ600-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Mazak-Integrex-i600.

PRM_43028 [Mazak-Integrex-i600] — CAX_LOOP_GAIN_8 | Default: 26.000 | Linked Fault: ERR-MAZ600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i600.

PRM_43029 [Mazak-Integrex-i600] — OPT_LOOP_GAIN_9 | Default: 60.500 | Linked Fault: ERR-MAZ600-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Mazak-Integrex-i600.

PRM_43030 [Mazak-Integrex-i600] — PLC_LOOP_GAIN_10 | Default: 95.000 | Linked Fault: ERR-MAZ600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i600.

PRM_43031 [Mazak-Integrex-i600] — LUB_LOOP_GAIN_11 | Default: 129.500 | Linked Fault: ERR-MAZ600-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Mazak-Integrex-i600.

PRM_43032 [Mazak-Integrex-i600] — VAC_LOOP_GAIN_12 | Default: 164.000 | Linked Fault: ERR-MAZ600-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i600.

PRM_43033 [Mazak-Integrex-i600] — THM_LOOP_GAIN_13 | Default: 198.500 | Linked Fault: ERR-MAZ600-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Mazak-Integrex-i600.

PRM_43034 [Mazak-Integrex-i600] — SPN_LOOP_GAIN_14 | Default: 233.000 | Linked Fault: ERR-MAZ600-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i600.

PRM_43035 [Mazak-Integrex-i600] — ATC_LOOP_GAIN_15 | Default: 17.500 | Linked Fault: ERR-MAZ600-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Mazak-Integrex-i600.

PRM_43036 [Mazak-Integrex-i600] — HYD_LOOP_GAIN_16 | Default: 52.000 | Linked Fault: ERR-MAZ600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i600.

CNC/PLC SERVO PARAMETER DICTIONARY — Mazak-Integrex-i600 (BANK 2)

Machine Model: Mazak-Integrex-i600 | Controller: MAZATROL SmoothAi. Use these parameter registers when diagnosing servo following errors or thermal drift on Mazak-Integrex-i600.

PRM_43041 [Mazak-Integrex-i600] — HYD_LOOP_GAIN_1 | Default: 69.000 | Linked Fault: ERR-MAZ600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i600.

PRM_43042 [Mazak-Integrex-i600] — XAX_LOOP_GAIN_2 | Default: 138.000 | Linked Fault: ERR-MAZ600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i600.

PRM_43043 [Mazak-Integrex-i600] — ZAX_LOOP_GAIN_3 | Default: 207.000 | Linked Fault: ERR-MAZ600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i600.

PRM_43044 [Mazak-Integrex-i600] — CAX_LOOP_GAIN_4 | Default: 26.000 | Linked Fault: ERR-MAZ600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i600.

PRM_43045 [Mazak-Integrex-i600] — PLC_LOOP_GAIN_5 | Default: 95.000 | Linked Fault: ERR-MAZ600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i600.

PRM_43046 [Mazak-Integrex-i600] — VAC_LOOP_GAIN_6 | Default: 164.000 | Linked Fault: ERR-MAZ600-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i600.

PRM_43047 [Mazak-Integrex-i600] — SPN_LOOP_GAIN_7 | Default: 233.000 | Linked Fault: ERR-MAZ600-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i600.

PRM_43048 [Mazak-Integrex-i600] — HYD_LOOP_GAIN_8 | Default: 52.000 | Linked Fault: ERR-MAZ600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i600.

PRM_43049 [Mazak-Integrex-i600] — XAX_LOOP_GAIN_9 | Default: 121.000 | Linked Fault: ERR-MAZ600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i600.

PRM_43050 [Mazak-Integrex-i600] — ZAX_LOOP_GAIN_10 | Default: 190.000 | Linked Fault: ERR-MAZ600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i600.

PRM_43051 [Mazak-Integrex-i600] — CAX_LOOP_GAIN_11 | Default: 9.000 | Linked Fault: ERR-MAZ600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i600.

PRM_43052 [Mazak-Integrex-i600] — PLC_LOOP_GAIN_12 | Default: 78.000 | Linked Fault: ERR-MAZ600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i600.

PRM_43053 [Mazak-Integrex-i600] — VAC_LOOP_GAIN_13 | Default: 147.000 | Linked Fault: ERR-MAZ600-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i600.

PRM_43054 [Mazak-Integrex-i600] — SPN_LOOP_GAIN_14 | Default: 216.000 | Linked Fault: ERR-MAZ600-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i600.

PRM_43055 [Mazak-Integrex-i600] — HYD_LOOP_GAIN_15 | Default: 35.000 | Linked Fault: ERR-MAZ600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i600.

PRM_43056 [Mazak-Integrex-i600] — XAX_LOOP_GAIN_16 | Default: 104.000 | Linked Fault: ERR-MAZ600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i600.

CNC/PLC SERVO PARAMETER DICTIONARY — Mazak-Integrex-i600 (BANK 3)

Machine Model: Mazak-Integrex-i600 | Controller: MAZATROL SmoothAi. Use these parameter registers when diagnosing servo following errors or thermal drift on Mazak-Integrex-i600.

PRM_43061 [Mazak-Integrex-i600] — CLN_LOOP_GAIN_1 | Default: 103.500 | Linked Fault: ERR-MAZ600-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Mazak-Integrex-i600.

PRM_43062 [Mazak-Integrex-i600] — ZAX_LOOP_GAIN_2 | Default: 207.000 | Linked Fault: ERR-MAZ600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i600.

PRM_43063 [Mazak-Integrex-i600] — OPT_LOOP_GAIN_3 | Default: 60.500 | Linked Fault: ERR-MAZ600-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Mazak-Integrex-i600.

PRM_43064 [Mazak-Integrex-i600] — VAC_LOOP_GAIN_4 | Default: 164.000 | Linked Fault: ERR-MAZ600-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i600.

PRM_43065 [Mazak-Integrex-i600] — ATC_LOOP_GAIN_5 | Default: 17.500 | Linked Fault: ERR-MAZ600-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Mazak-Integrex-i600.

PRM_43066 [Mazak-Integrex-i600] — XAX_LOOP_GAIN_6 | Default: 121.000 | Linked Fault: ERR-MAZ600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i600.

PRM_43067 [Mazak-Integrex-i600] — BAX_LOOP_GAIN_7 | Default: 224.500 | Linked Fault: ERR-MAZ600-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Mazak-Integrex-i600.

PRM_43068 [Mazak-Integrex-i600] — PLC_LOOP_GAIN_8 | Default: 78.000 | Linked Fault: ERR-MAZ600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i600.

PRM_43069 [Mazak-Integrex-i600] — THM_LOOP_GAIN_9 | Default: 181.500 | Linked Fault: ERR-MAZ600-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Mazak-Integrex-i600.

PRM_43070 [Mazak-Integrex-i600] — HYD_LOOP_GAIN_10 | Default: 35.000 | Linked Fault: ERR-MAZ600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i600.

PRM_43071 [Mazak-Integrex-i600] — YAX_LOOP_GAIN_11 | Default: 138.500 | Linked Fault: ERR-MAZ600-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Mazak-Integrex-i600.

PRM_43072 [Mazak-Integrex-i600] — CAX_LOOP_GAIN_12 | Default: 242.000 | Linked Fault: ERR-MAZ600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i600.

PRM_43073 [Mazak-Integrex-i600] — LUB_LOOP_GAIN_13 | Default: 95.500 | Linked Fault: ERR-MAZ600-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Mazak-Integrex-i600.

PRM_43074 [Mazak-Integrex-i600] — SPN_LOOP_GAIN_14 | Default: 199.000 | Linked Fault: ERR-MAZ600-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i600.

PRM_43075 [Mazak-Integrex-i600] — CLN_LOOP_GAIN_15 | Default: 52.500 | Linked Fault: ERR-MAZ600-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Mazak-Integrex-i600.

PRM_43076 [Mazak-Integrex-i600] — ZAX_LOOP_GAIN_16 | Default: 156.000 | Linked Fault: ERR-MAZ600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i600.

CNC/PLC SERVO PARAMETER DICTIONARY — Mazak-Integrex-i600 (BANK 4)

Machine Model: Mazak-Integrex-i600 | Controller: MAZATROL SmoothAi. Use these parameter registers when diagnosing servo following errors or thermal drift on Mazak-Integrex-i600.

PRM_43081 [Mazak-Integrex-i600] — XAX_LOOP_GAIN_1 | Default: 138.000 | Linked Fault: ERR-MAZ600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i600.

PRM_43082 [Mazak-Integrex-i600] — CAX_LOOP_GAIN_2 | Default: 26.000 | Linked Fault: ERR-MAZ600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i600.

PRM_43083 [Mazak-Integrex-i600] — VAC_LOOP_GAIN_3 | Default: 164.000 | Linked Fault: ERR-MAZ600-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i600.

PRM_43084 [Mazak-Integrex-i600] — HYD_LOOP_GAIN_4 | Default: 52.000 | Linked Fault: ERR-MAZ600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i600.

PRM_43085 [Mazak-Integrex-i600] — ZAX_LOOP_GAIN_5 | Default: 190.000 | Linked Fault: ERR-MAZ600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i600.

PRM_43086 [Mazak-Integrex-i600] — PLC_LOOP_GAIN_6 | Default: 78.000 | Linked Fault: ERR-MAZ600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i600.

PRM_43087 [Mazak-Integrex-i600] — SPN_LOOP_GAIN_7 | Default: 216.000 | Linked Fault: ERR-MAZ600-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i600.

PRM_43088 [Mazak-Integrex-i600] — XAX_LOOP_GAIN_8 | Default: 104.000 | Linked Fault: ERR-MAZ600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i600.

PRM_43089 [Mazak-Integrex-i600] — CAX_LOOP_GAIN_9 | Default: 242.000 | Linked Fault: ERR-MAZ600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i600.

PRM_43090 [Mazak-Integrex-i600] — VAC_LOOP_GAIN_10 | Default: 130.000 | Linked Fault: ERR-MAZ600-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i600.

PRM_43091 [Mazak-Integrex-i600] — HYD_LOOP_GAIN_11 | Default: 18.000 | Linked Fault: ERR-MAZ600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i600.

PRM_43092 [Mazak-Integrex-i600] — ZAX_LOOP_GAIN_12 | Default: 156.000 | Linked Fault: ERR-MAZ600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i600.

PRM_43093 [Mazak-Integrex-i600] — PLC_LOOP_GAIN_13 | Default: 44.000 | Linked Fault: ERR-MAZ600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i600.

PRM_43094 [Mazak-Integrex-i600] — SPN_LOOP_GAIN_14 | Default: 182.000 | Linked Fault: ERR-MAZ600-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i600.

PRM_43095 [Mazak-Integrex-i600] — XAX_LOOP_GAIN_15 | Default: 70.000 | Linked Fault: ERR-MAZ600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i600.

PRM_43096 [Mazak-Integrex-i600] — CAX_LOOP_GAIN_16 | Default: 208.000 | Linked Fault: ERR-MAZ600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i600.

CNC/PLC SERVO PARAMETER DICTIONARY — Mazak-Integrex-i600 (BANK 5)

Machine Model: Mazak-Integrex-i600 | Controller: MAZATROL SmoothAi. Use these parameter registers when diagnosing servo following errors or thermal drift on Mazak-Integrex-i600.

PRM_43101 [Mazak-Integrex-i600] — YAX_LOOP_GAIN_1 | Default: 172.500 | Linked Fault: ERR-MAZ600-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Mazak-Integrex-i600.

PRM_43102 [Mazak-Integrex-i600] — PLC_LOOP_GAIN_2 | Default: 95.000 | Linked Fault: ERR-MAZ600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i600.

PRM_43103 [Mazak-Integrex-i600] — ATC_LOOP_GAIN_3 | Default: 17.500 | Linked Fault: ERR-MAZ600-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Mazak-Integrex-i600.

PRM_43104 [Mazak-Integrex-i600] — ZAX_LOOP_GAIN_4 | Default: 190.000 | Linked Fault: ERR-MAZ600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i600.

PRM_43105 [Mazak-Integrex-i600] — LUB_LOOP_GAIN_5 | Default: 112.500 | Linked Fault: ERR-MAZ600-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Mazak-Integrex-i600.

PRM_43106 [Mazak-Integrex-i600] — HYD_LOOP_GAIN_6 | Default: 35.000 | Linked Fault: ERR-MAZ600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i600.

PRM_43107 [Mazak-Integrex-i600] — BAX_LOOP_GAIN_7 | Default: 207.500 | Linked Fault: ERR-MAZ600-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Mazak-Integrex-i600.

PRM_43108 [Mazak-Integrex-i600] — VAC_LOOP_GAIN_8 | Default: 130.000 | Linked Fault: ERR-MAZ600-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i600.

PRM_43109 [Mazak-Integrex-i600] — CLN_LOOP_GAIN_9 | Default: 52.500 | Linked Fault: ERR-MAZ600-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Mazak-Integrex-i600.

PRM_43110 [Mazak-Integrex-i600] — CAX_LOOP_GAIN_10 | Default: 225.000 | Linked Fault: ERR-MAZ600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i600.

PRM_43111 [Mazak-Integrex-i600] — THM_LOOP_GAIN_11 | Default: 147.500 | Linked Fault: ERR-MAZ600-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Mazak-Integrex-i600.

PRM_43112 [Mazak-Integrex-i600] — XAX_LOOP_GAIN_12 | Default: 70.000 | Linked Fault: ERR-MAZ600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i600.

PRM_43113 [Mazak-Integrex-i600] — OPT_LOOP_GAIN_13 | Default: 242.500 | Linked Fault: ERR-MAZ600-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Mazak-Integrex-i600.

PRM_43114 [Mazak-Integrex-i600] — SPN_LOOP_GAIN_14 | Default: 165.000 | Linked Fault: ERR-MAZ600-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i600.

PRM_43115 [Mazak-Integrex-i600] — YAX_LOOP_GAIN_15 | Default: 87.500 | Linked Fault: ERR-MAZ600-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Mazak-Integrex-i600.

PRM_43116 [Mazak-Integrex-i600] — PLC_LOOP_GAIN_16 | Default: 10.000 | Linked Fault: ERR-MAZ600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i600.

CNC/PLC SERVO PARAMETER DICTIONARY — Okuma-Multus-U150 (BANK 1)

Machine Model: Okuma-Multus-U150 | Controller: OSP-P500A Aerospace. Use these parameter registers when diagnosing servo following errors or thermal drift on Okuma-Multus-U150.

PRM_43121 [Okuma-Multus-U150] — ATC_LOOP_GAIN_1 | Default: 35.650 | Linked Fault: ERR-OKU150-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Okuma-Multus-U150.

PRM_43122 [Okuma-Multus-U150] — HYD_LOOP_GAIN_2 | Default: 71.300 | Linked Fault: ERR-OKU150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U150.

PRM_43123 [Okuma-Multus-U150] — CLN_LOOP_GAIN_3 | Default: 106.950 | Linked Fault: ERR-OKU150-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Okuma-Multus-U150.

PRM_43124 [Okuma-Multus-U150] — XAX_LOOP_GAIN_4 | Default: 142.600 | Linked Fault: ERR-OKU150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U150.

PRM_43125 [Okuma-Multus-U150] — YAX_LOOP_GAIN_5 | Default: 178.250 | Linked Fault: ERR-OKU150-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Okuma-Multus-U150.

PRM_43126 [Okuma-Multus-U150] — ZAX_LOOP_GAIN_6 | Default: 213.900 | Linked Fault: ERR-OKU150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U150.

PRM_43127 [Okuma-Multus-U150] — BAX_LOOP_GAIN_7 | Default: 249.550 | Linked Fault: ERR-OKU150-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Okuma-Multus-U150.

PRM_43128 [Okuma-Multus-U150] — CAX_LOOP_GAIN_8 | Default: 35.200 | Linked Fault: ERR-OKU150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U150.

PRM_43129 [Okuma-Multus-U150] — OPT_LOOP_GAIN_9 | Default: 70.850 | Linked Fault: ERR-OKU150-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Okuma-Multus-U150.

PRM_43130 [Okuma-Multus-U150] — PLC_LOOP_GAIN_10 | Default: 106.500 | Linked Fault: ERR-OKU150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U150.

PRM_43131 [Okuma-Multus-U150] — LUB_LOOP_GAIN_11 | Default: 142.150 | Linked Fault: ERR-OKU150-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Okuma-Multus-U150.

PRM_43132 [Okuma-Multus-U150] — VAC_LOOP_GAIN_12 | Default: 177.800 | Linked Fault: ERR-OKU150-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U150.

PRM_43133 [Okuma-Multus-U150] — THM_LOOP_GAIN_13 | Default: 213.450 | Linked Fault: ERR-OKU150-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Okuma-Multus-U150.

PRM_43134 [Okuma-Multus-U150] — SPN_LOOP_GAIN_14 | Default: 249.100 | Linked Fault: ERR-OKU150-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Okuma-Multus-U150.

PRM_43135 [Okuma-Multus-U150] — ATC_LOOP_GAIN_15 | Default: 34.750 | Linked Fault: ERR-OKU150-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Okuma-Multus-U150.

PRM_43136 [Okuma-Multus-U150] — HYD_LOOP_GAIN_16 | Default: 70.400 | Linked Fault: ERR-OKU150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U150.

CNC/PLC SERVO PARAMETER DICTIONARY — Okuma-Multus-U150 (BANK 2)

Machine Model: Okuma-Multus-U150 | Controller: OSP-P500A Aerospace. Use these parameter registers when diagnosing servo following errors or thermal drift on Okuma-Multus-U150.

PRM_43141 [Okuma-Multus-U150] — HYD_LOOP_GAIN_1 | Default: 71.300 | Linked Fault: ERR-OKU150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U150.

PRM_43142 [Okuma-Multus-U150] — XAX_LOOP_GAIN_2 | Default: 142.600 | Linked Fault: ERR-OKU150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U150.

PRM_43143 [Okuma-Multus-U150] — ZAX_LOOP_GAIN_3 | Default: 213.900 | Linked Fault: ERR-OKU150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U150.

PRM_43144 [Okuma-Multus-U150] — CAX_LOOP_GAIN_4 | Default: 35.200 | Linked Fault: ERR-OKU150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U150.

PRM_43145 [Okuma-Multus-U150] — PLC_LOOP_GAIN_5 | Default: 106.500 | Linked Fault: ERR-OKU150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U150.

PRM_43146 [Okuma-Multus-U150] — VAC_LOOP_GAIN_6 | Default: 177.800 | Linked Fault: ERR-OKU150-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U150.

PRM_43147 [Okuma-Multus-U150] — SPN_LOOP_GAIN_7 | Default: 249.100 | Linked Fault: ERR-OKU150-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Okuma-Multus-U150.

PRM_43148 [Okuma-Multus-U150] — HYD_LOOP_GAIN_8 | Default: 70.400 | Linked Fault: ERR-OKU150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U150.

PRM_43149 [Okuma-Multus-U150] — XAX_LOOP_GAIN_9 | Default: 141.700 | Linked Fault: ERR-OKU150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U150.

PRM_43150 [Okuma-Multus-U150] — ZAX_LOOP_GAIN_10 | Default: 213.000 | Linked Fault: ERR-OKU150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U150.

PRM_43151 [Okuma-Multus-U150] — CAX_LOOP_GAIN_11 | Default: 34.300 | Linked Fault: ERR-OKU150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U150.

PRM_43152 [Okuma-Multus-U150] — PLC_LOOP_GAIN_12 | Default: 105.600 | Linked Fault: ERR-OKU150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U150.

PRM_43153 [Okuma-Multus-U150] — VAC_LOOP_GAIN_13 | Default: 176.900 | Linked Fault: ERR-OKU150-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U150.

PRM_43154 [Okuma-Multus-U150] — SPN_LOOP_GAIN_14 | Default: 248.200 | Linked Fault: ERR-OKU150-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Okuma-Multus-U150.

PRM_43155 [Okuma-Multus-U150] — HYD_LOOP_GAIN_15 | Default: 69.500 | Linked Fault: ERR-OKU150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U150.

PRM_43156 [Okuma-Multus-U150] — XAX_LOOP_GAIN_16 | Default: 140.800 | Linked Fault: ERR-OKU150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U150.

CNC/PLC SERVO PARAMETER DICTIONARY — Okuma-Multus-U150 (BANK 3)

Machine Model: Okuma-Multus-U150 | Controller: OSP-P500A Aerospace. Use these parameter registers when diagnosing servo following errors or thermal drift on Okuma-Multus-U150.

PRM_43161 [Okuma-Multus-U150] — CLN_LOOP_GAIN_1 | Default: 106.950 | Linked Fault: ERR-OKU150-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Okuma-Multus-U150.

PRM_43162 [Okuma-Multus-U150] — ZAX_LOOP_GAIN_2 | Default: 213.900 | Linked Fault: ERR-OKU150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U150.

PRM_43163 [Okuma-Multus-U150] — OPT_LOOP_GAIN_3 | Default: 70.850 | Linked Fault: ERR-OKU150-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Okuma-Multus-U150.

PRM_43164 [Okuma-Multus-U150] — VAC_LOOP_GAIN_4 | Default: 177.800 | Linked Fault: ERR-OKU150-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U150.

PRM_43165 [Okuma-Multus-U150] — ATC_LOOP_GAIN_5 | Default: 34.750 | Linked Fault: ERR-OKU150-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Okuma-Multus-U150.

PRM_43166 [Okuma-Multus-U150] — XAX_LOOP_GAIN_6 | Default: 141.700 | Linked Fault: ERR-OKU150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U150.

PRM_43167 [Okuma-Multus-U150] — BAX_LOOP_GAIN_7 | Default: 248.650 | Linked Fault: ERR-OKU150-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Okuma-Multus-U150.

PRM_43168 [Okuma-Multus-U150] — PLC_LOOP_GAIN_8 | Default: 105.600 | Linked Fault: ERR-OKU150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U150.

PRM_43169 [Okuma-Multus-U150] — THM_LOOP_GAIN_9 | Default: 212.550 | Linked Fault: ERR-OKU150-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Okuma-Multus-U150.

PRM_43170 [Okuma-Multus-U150] — HYD_LOOP_GAIN_10 | Default: 69.500 | Linked Fault: ERR-OKU150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U150.

PRM_43171 [Okuma-Multus-U150] — YAX_LOOP_GAIN_11 | Default: 176.450 | Linked Fault: ERR-OKU150-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Okuma-Multus-U150.

PRM_43172 [Okuma-Multus-U150] — CAX_LOOP_GAIN_12 | Default: 33.400 | Linked Fault: ERR-OKU150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U150.

PRM_43173 [Okuma-Multus-U150] — LUB_LOOP_GAIN_13 | Default: 140.350 | Linked Fault: ERR-OKU150-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Okuma-Multus-U150.

PRM_43174 [Okuma-Multus-U150] — SPN_LOOP_GAIN_14 | Default: 247.300 | Linked Fault: ERR-OKU150-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Okuma-Multus-U150.

PRM_43175 [Okuma-Multus-U150] — CLN_LOOP_GAIN_15 | Default: 104.250 | Linked Fault: ERR-OKU150-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Okuma-Multus-U150.

PRM_43176 [Okuma-Multus-U150] — ZAX_LOOP_GAIN_16 | Default: 211.200 | Linked Fault: ERR-OKU150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U150.

CNC/PLC SERVO PARAMETER DICTIONARY — Okuma-Multus-U150 (BANK 4)

Machine Model: Okuma-Multus-U150 | Controller: OSP-P500A Aerospace. Use these parameter registers when diagnosing servo following errors or thermal drift on Okuma-Multus-U150.

PRM_43181 [Okuma-Multus-U150] — XAX_LOOP_GAIN_1 | Default: 142.600 | Linked Fault: ERR-OKU150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U150.

PRM_43182 [Okuma-Multus-U150] — CAX_LOOP_GAIN_2 | Default: 35.200 | Linked Fault: ERR-OKU150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U150.

PRM_43183 [Okuma-Multus-U150] — VAC_LOOP_GAIN_3 | Default: 177.800 | Linked Fault: ERR-OKU150-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U150.

PRM_43184 [Okuma-Multus-U150] — HYD_LOOP_GAIN_4 | Default: 70.400 | Linked Fault: ERR-OKU150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U150.

PRM_43185 [Okuma-Multus-U150] — ZAX_LOOP_GAIN_5 | Default: 213.000 | Linked Fault: ERR-OKU150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U150.

PRM_43186 [Okuma-Multus-U150] — PLC_LOOP_GAIN_6 | Default: 105.600 | Linked Fault: ERR-OKU150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U150.

PRM_43187 [Okuma-Multus-U150] — SPN_LOOP_GAIN_7 | Default: 248.200 | Linked Fault: ERR-OKU150-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Okuma-Multus-U150.

PRM_43188 [Okuma-Multus-U150] — XAX_LOOP_GAIN_8 | Default: 140.800 | Linked Fault: ERR-OKU150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U150.

PRM_43189 [Okuma-Multus-U150] — CAX_LOOP_GAIN_9 | Default: 33.400 | Linked Fault: ERR-OKU150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U150.

PRM_43190 [Okuma-Multus-U150] — VAC_LOOP_GAIN_10 | Default: 176.000 | Linked Fault: ERR-OKU150-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U150.

PRM_43191 [Okuma-Multus-U150] — HYD_LOOP_GAIN_11 | Default: 68.600 | Linked Fault: ERR-OKU150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U150.

PRM_43192 [Okuma-Multus-U150] — ZAX_LOOP_GAIN_12 | Default: 211.200 | Linked Fault: ERR-OKU150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U150.

PRM_43193 [Okuma-Multus-U150] — PLC_LOOP_GAIN_13 | Default: 103.800 | Linked Fault: ERR-OKU150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U150.

PRM_43194 [Okuma-Multus-U150] — SPN_LOOP_GAIN_14 | Default: 246.400 | Linked Fault: ERR-OKU150-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Okuma-Multus-U150.

PRM_43195 [Okuma-Multus-U150] — XAX_LOOP_GAIN_15 | Default: 139.000 | Linked Fault: ERR-OKU150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U150.

PRM_43196 [Okuma-Multus-U150] — CAX_LOOP_GAIN_16 | Default: 31.600 | Linked Fault: ERR-OKU150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U150.

CNC/PLC SERVO PARAMETER DICTIONARY — Okuma-Multus-U150 (BANK 5)

Machine Model: Okuma-Multus-U150 | Controller: OSP-P500A Aerospace. Use these parameter registers when diagnosing servo following errors or thermal drift on Okuma-Multus-U150.

PRM_43201 [Okuma-Multus-U150] — YAX_LOOP_GAIN_1 | Default: 178.250 | Linked Fault: ERR-OKU150-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Okuma-Multus-U150.

PRM_43202 [Okuma-Multus-U150] — PLC_LOOP_GAIN_2 | Default: 106.500 | Linked Fault: ERR-OKU150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U150.

PRM_43203 [Okuma-Multus-U150] — ATC_LOOP_GAIN_3 | Default: 34.750 | Linked Fault: ERR-OKU150-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Okuma-Multus-U150.

PRM_43204 [Okuma-Multus-U150] — ZAX_LOOP_GAIN_4 | Default: 213.000 | Linked Fault: ERR-OKU150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U150.

PRM_43205 [Okuma-Multus-U150] — LUB_LOOP_GAIN_5 | Default: 141.250 | Linked Fault: ERR-OKU150-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Okuma-Multus-U150.

PRM_43206 [Okuma-Multus-U150] — HYD_LOOP_GAIN_6 | Default: 69.500 | Linked Fault: ERR-OKU150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U150.

PRM_43207 [Okuma-Multus-U150] — BAX_LOOP_GAIN_7 | Default: 247.750 | Linked Fault: ERR-OKU150-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Okuma-Multus-U150.

PRM_43208 [Okuma-Multus-U150] — VAC_LOOP_GAIN_8 | Default: 176.000 | Linked Fault: ERR-OKU150-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U150.

PRM_43209 [Okuma-Multus-U150] — CLN_LOOP_GAIN_9 | Default: 104.250 | Linked Fault: ERR-OKU150-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Okuma-Multus-U150.

PRM_43210 [Okuma-Multus-U150] — CAX_LOOP_GAIN_10 | Default: 32.500 | Linked Fault: ERR-OKU150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U150.

PRM_43211 [Okuma-Multus-U150] — THM_LOOP_GAIN_11 | Default: 210.750 | Linked Fault: ERR-OKU150-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Okuma-Multus-U150.

PRM_43212 [Okuma-Multus-U150] — XAX_LOOP_GAIN_12 | Default: 139.000 | Linked Fault: ERR-OKU150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U150.

PRM_43213 [Okuma-Multus-U150] — OPT_LOOP_GAIN_13 | Default: 67.250 | Linked Fault: ERR-OKU150-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Okuma-Multus-U150.

PRM_43214 [Okuma-Multus-U150] — SPN_LOOP_GAIN_14 | Default: 245.500 | Linked Fault: ERR-OKU150-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Okuma-Multus-U150.

PRM_43215 [Okuma-Multus-U150] — YAX_LOOP_GAIN_15 | Default: 173.750 | Linked Fault: ERR-OKU150-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Okuma-Multus-U150.

PRM_43216 [Okuma-Multus-U150] — PLC_LOOP_GAIN_16 | Default: 102.000 | Linked Fault: ERR-OKU150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U150.

CNC/PLC SERVO PARAMETER DICTIONARY — Okuma-Multus-U200 (BANK 1)

Machine Model: Okuma-Multus-U200 | Controller: OSP-P500A Aerospace. Use these parameter registers when diagnosing servo following errors or thermal drift on Okuma-Multus-U200.

PRM_43221 [Okuma-Multus-U200] — ATC_LOOP_GAIN_1 | Default: 36.800 | Linked Fault: ERR-OKU200-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Okuma-Multus-U200.

PRM_43222 [Okuma-Multus-U200] — HYD_LOOP_GAIN_2 | Default: 73.600 | Linked Fault: ERR-OKU200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U200.

PRM_43223 [Okuma-Multus-U200] — CLN_LOOP_GAIN_3 | Default: 110.400 | Linked Fault: ERR-OKU200-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Okuma-Multus-U200.

PRM_43224 [Okuma-Multus-U200] — XAX_LOOP_GAIN_4 | Default: 147.200 | Linked Fault: ERR-OKU200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U200.

PRM_43225 [Okuma-Multus-U200] — YAX_LOOP_GAIN_5 | Default: 184.000 | Linked Fault: ERR-OKU200-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Okuma-Multus-U200.

PRM_43226 [Okuma-Multus-U200] — ZAX_LOOP_GAIN_6 | Default: 220.800 | Linked Fault: ERR-OKU200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U200.

PRM_43227 [Okuma-Multus-U200] — BAX_LOOP_GAIN_7 | Default: 7.600 | Linked Fault: ERR-OKU200-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Okuma-Multus-U200.

PRM_43228 [Okuma-Multus-U200] — CAX_LOOP_GAIN_8 | Default: 44.400 | Linked Fault: ERR-OKU200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U200.

PRM_43229 [Okuma-Multus-U200] — OPT_LOOP_GAIN_9 | Default: 81.200 | Linked Fault: ERR-OKU200-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Okuma-Multus-U200.

PRM_43230 [Okuma-Multus-U200] — PLC_LOOP_GAIN_10 | Default: 118.000 | Linked Fault: ERR-OKU200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U200.

PRM_43231 [Okuma-Multus-U200] — LUB_LOOP_GAIN_11 | Default: 154.800 | Linked Fault: ERR-OKU200-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Okuma-Multus-U200.

PRM_43232 [Okuma-Multus-U200] — VAC_LOOP_GAIN_12 | Default: 191.600 | Linked Fault: ERR-OKU200-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U200.

PRM_43233 [Okuma-Multus-U200] — THM_LOOP_GAIN_13 | Default: 228.400 | Linked Fault: ERR-OKU200-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Okuma-Multus-U200.

PRM_43234 [Okuma-Multus-U200] — SPN_LOOP_GAIN_14 | Default: 15.200 | Linked Fault: ERR-OKU200-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Okuma-Multus-U200.

PRM_43235 [Okuma-Multus-U200] — ATC_LOOP_GAIN_15 | Default: 52.000 | Linked Fault: ERR-OKU200-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Okuma-Multus-U200.

PRM_43236 [Okuma-Multus-U200] — HYD_LOOP_GAIN_16 | Default: 88.800 | Linked Fault: ERR-OKU200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U200.

CNC/PLC SERVO PARAMETER DICTIONARY — Okuma-Multus-U200 (BANK 2)

Machine Model: Okuma-Multus-U200 | Controller: OSP-P500A Aerospace. Use these parameter registers when diagnosing servo following errors or thermal drift on Okuma-Multus-U200.

PRM_43241 [Okuma-Multus-U200] — HYD_LOOP_GAIN_1 | Default: 73.600 | Linked Fault: ERR-OKU200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U200.

PRM_43242 [Okuma-Multus-U200] — XAX_LOOP_GAIN_2 | Default: 147.200 | Linked Fault: ERR-OKU200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U200.

PRM_43243 [Okuma-Multus-U200] — ZAX_LOOP_GAIN_3 | Default: 220.800 | Linked Fault: ERR-OKU200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U200.

PRM_43244 [Okuma-Multus-U200] — CAX_LOOP_GAIN_4 | Default: 44.400 | Linked Fault: ERR-OKU200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U200.

PRM_43245 [Okuma-Multus-U200] — PLC_LOOP_GAIN_5 | Default: 118.000 | Linked Fault: ERR-OKU200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U200.

PRM_43246 [Okuma-Multus-U200] — VAC_LOOP_GAIN_6 | Default: 191.600 | Linked Fault: ERR-OKU200-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U200.

PRM_43247 [Okuma-Multus-U200] — SPN_LOOP_GAIN_7 | Default: 15.200 | Linked Fault: ERR-OKU200-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Okuma-Multus-U200.

PRM_43248 [Okuma-Multus-U200] — HYD_LOOP_GAIN_8 | Default: 88.800 | Linked Fault: ERR-OKU200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U200.

PRM_43249 [Okuma-Multus-U200] — XAX_LOOP_GAIN_9 | Default: 162.400 | Linked Fault: ERR-OKU200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U200.

PRM_43250 [Okuma-Multus-U200] — ZAX_LOOP_GAIN_10 | Default: 236.000 | Linked Fault: ERR-OKU200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U200.

PRM_43251 [Okuma-Multus-U200] — CAX_LOOP_GAIN_11 | Default: 59.600 | Linked Fault: ERR-OKU200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U200.

PRM_43252 [Okuma-Multus-U200] — PLC_LOOP_GAIN_12 | Default: 133.200 | Linked Fault: ERR-OKU200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U200.

PRM_43253 [Okuma-Multus-U200] — VAC_LOOP_GAIN_13 | Default: 206.800 | Linked Fault: ERR-OKU200-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U200.

PRM_43254 [Okuma-Multus-U200] — SPN_LOOP_GAIN_14 | Default: 30.400 | Linked Fault: ERR-OKU200-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Okuma-Multus-U200.

PRM_43255 [Okuma-Multus-U200] — HYD_LOOP_GAIN_15 | Default: 104.000 | Linked Fault: ERR-OKU200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U200.

PRM_43256 [Okuma-Multus-U200] — XAX_LOOP_GAIN_16 | Default: 177.600 | Linked Fault: ERR-OKU200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U200.

CNC/PLC SERVO PARAMETER DICTIONARY — Okuma-Multus-U200 (BANK 3)

Machine Model: Okuma-Multus-U200 | Controller: OSP-P500A Aerospace. Use these parameter registers when diagnosing servo following errors or thermal drift on Okuma-Multus-U200.

PRM_43261 [Okuma-Multus-U200] — CLN_LOOP_GAIN_1 | Default: 110.400 | Linked Fault: ERR-OKU200-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Okuma-Multus-U200.

PRM_43262 [Okuma-Multus-U200] — ZAX_LOOP_GAIN_2 | Default: 220.800 | Linked Fault: ERR-OKU200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U200.

PRM_43263 [Okuma-Multus-U200] — OPT_LOOP_GAIN_3 | Default: 81.200 | Linked Fault: ERR-OKU200-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Okuma-Multus-U200.

PRM_43264 [Okuma-Multus-U200] — VAC_LOOP_GAIN_4 | Default: 191.600 | Linked Fault: ERR-OKU200-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U200.

PRM_43265 [Okuma-Multus-U200] — ATC_LOOP_GAIN_5 | Default: 52.000 | Linked Fault: ERR-OKU200-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Okuma-Multus-U200.

PRM_43266 [Okuma-Multus-U200] — XAX_LOOP_GAIN_6 | Default: 162.400 | Linked Fault: ERR-OKU200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U200.

PRM_43267 [Okuma-Multus-U200] — BAX_LOOP_GAIN_7 | Default: 22.800 | Linked Fault: ERR-OKU200-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Okuma-Multus-U200.

PRM_43268 [Okuma-Multus-U200] — PLC_LOOP_GAIN_8 | Default: 133.200 | Linked Fault: ERR-OKU200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U200.

PRM_43269 [Okuma-Multus-U200] — THM_LOOP_GAIN_9 | Default: 243.600 | Linked Fault: ERR-OKU200-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Okuma-Multus-U200.

PRM_43270 [Okuma-Multus-U200] — HYD_LOOP_GAIN_10 | Default: 104.000 | Linked Fault: ERR-OKU200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U200.

PRM_43271 [Okuma-Multus-U200] — YAX_LOOP_GAIN_11 | Default: 214.400 | Linked Fault: ERR-OKU200-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Okuma-Multus-U200.

PRM_43272 [Okuma-Multus-U200] — CAX_LOOP_GAIN_12 | Default: 74.800 | Linked Fault: ERR-OKU200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U200.

PRM_43273 [Okuma-Multus-U200] — LUB_LOOP_GAIN_13 | Default: 185.200 | Linked Fault: ERR-OKU200-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Okuma-Multus-U200.

PRM_43274 [Okuma-Multus-U200] — SPN_LOOP_GAIN_14 | Default: 45.600 | Linked Fault: ERR-OKU200-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Okuma-Multus-U200.

PRM_43275 [Okuma-Multus-U200] — CLN_LOOP_GAIN_15 | Default: 156.000 | Linked Fault: ERR-OKU200-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Okuma-Multus-U200.

PRM_43276 [Okuma-Multus-U200] — ZAX_LOOP_GAIN_16 | Default: 16.400 | Linked Fault: ERR-OKU200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U200.

CNC/PLC SERVO PARAMETER DICTIONARY — Okuma-Multus-U200 (BANK 4)

Machine Model: Okuma-Multus-U200 | Controller: OSP-P500A Aerospace. Use these parameter registers when diagnosing servo following errors or thermal drift on Okuma-Multus-U200.

PRM_43281 [Okuma-Multus-U200] — XAX_LOOP_GAIN_1 | Default: 147.200 | Linked Fault: ERR-OKU200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U200.

PRM_43282 [Okuma-Multus-U200] — CAX_LOOP_GAIN_2 | Default: 44.400 | Linked Fault: ERR-OKU200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U200.

PRM_43283 [Okuma-Multus-U200] — VAC_LOOP_GAIN_3 | Default: 191.600 | Linked Fault: ERR-OKU200-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U200.

PRM_43284 [Okuma-Multus-U200] — HYD_LOOP_GAIN_4 | Default: 88.800 | Linked Fault: ERR-OKU200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U200.

PRM_43285 [Okuma-Multus-U200] — ZAX_LOOP_GAIN_5 | Default: 236.000 | Linked Fault: ERR-OKU200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U200.

PRM_43286 [Okuma-Multus-U200] — PLC_LOOP_GAIN_6 | Default: 133.200 | Linked Fault: ERR-OKU200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U200.

PRM_43287 [Okuma-Multus-U200] — SPN_LOOP_GAIN_7 | Default: 30.400 | Linked Fault: ERR-OKU200-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Okuma-Multus-U200.

PRM_43288 [Okuma-Multus-U200] — XAX_LOOP_GAIN_8 | Default: 177.600 | Linked Fault: ERR-OKU200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U200.

PRM_43289 [Okuma-Multus-U200] — CAX_LOOP_GAIN_9 | Default: 74.800 | Linked Fault: ERR-OKU200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U200.

PRM_43290 [Okuma-Multus-U200] — VAC_LOOP_GAIN_10 | Default: 222.000 | Linked Fault: ERR-OKU200-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U200.

PRM_43291 [Okuma-Multus-U200] — HYD_LOOP_GAIN_11 | Default: 119.200 | Linked Fault: ERR-OKU200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U200.

PRM_43292 [Okuma-Multus-U200] — ZAX_LOOP_GAIN_12 | Default: 16.400 | Linked Fault: ERR-OKU200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U200.

PRM_43293 [Okuma-Multus-U200] — PLC_LOOP_GAIN_13 | Default: 163.600 | Linked Fault: ERR-OKU200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U200.

PRM_43294 [Okuma-Multus-U200] — SPN_LOOP_GAIN_14 | Default: 60.800 | Linked Fault: ERR-OKU200-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Okuma-Multus-U200.

PRM_43295 [Okuma-Multus-U200] — XAX_LOOP_GAIN_15 | Default: 208.000 | Linked Fault: ERR-OKU200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U200.

PRM_43296 [Okuma-Multus-U200] — CAX_LOOP_GAIN_16 | Default: 105.200 | Linked Fault: ERR-OKU200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U200.

CNC/PLC SERVO PARAMETER DICTIONARY — Okuma-Multus-U200 (BANK 5)

Machine Model: Okuma-Multus-U200 | Controller: OSP-P500A Aerospace. Use these parameter registers when diagnosing servo following errors or thermal drift on Okuma-Multus-U200.

PRM_43301 [Okuma-Multus-U200] — YAX_LOOP_GAIN_1 | Default: 184.000 | Linked Fault: ERR-OKU200-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Okuma-Multus-U200.

PRM_43302 [Okuma-Multus-U200] — PLC_LOOP_GAIN_2 | Default: 118.000 | Linked Fault: ERR-OKU200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U200.

PRM_43303 [Okuma-Multus-U200] — ATC_LOOP_GAIN_3 | Default: 52.000 | Linked Fault: ERR-OKU200-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Okuma-Multus-U200.

PRM_43304 [Okuma-Multus-U200] — ZAX_LOOP_GAIN_4 | Default: 236.000 | Linked Fault: ERR-OKU200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U200.

PRM_43305 [Okuma-Multus-U200] — LUB_LOOP_GAIN_5 | Default: 170.000 | Linked Fault: ERR-OKU200-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Okuma-Multus-U200.

PRM_43306 [Okuma-Multus-U200] — HYD_LOOP_GAIN_6 | Default: 104.000 | Linked Fault: ERR-OKU200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U200.

PRM_43307 [Okuma-Multus-U200] — BAX_LOOP_GAIN_7 | Default: 38.000 | Linked Fault: ERR-OKU200-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Okuma-Multus-U200.

PRM_43308 [Okuma-Multus-U200] — VAC_LOOP_GAIN_8 | Default: 222.000 | Linked Fault: ERR-OKU200-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U200.

PRM_43309 [Okuma-Multus-U200] — CLN_LOOP_GAIN_9 | Default: 156.000 | Linked Fault: ERR-OKU200-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Okuma-Multus-U200.

PRM_43310 [Okuma-Multus-U200] — CAX_LOOP_GAIN_10 | Default: 90.000 | Linked Fault: ERR-OKU200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U200.

PRM_43311 [Okuma-Multus-U200] — THM_LOOP_GAIN_11 | Default: 24.000 | Linked Fault: ERR-OKU200-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Okuma-Multus-U200.

PRM_43312 [Okuma-Multus-U200] — XAX_LOOP_GAIN_12 | Default: 208.000 | Linked Fault: ERR-OKU200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U200.

PRM_43313 [Okuma-Multus-U200] — OPT_LOOP_GAIN_13 | Default: 142.000 | Linked Fault: ERR-OKU200-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Okuma-Multus-U200.

PRM_43314 [Okuma-Multus-U200] — SPN_LOOP_GAIN_14 | Default: 76.000 | Linked Fault: ERR-OKU200-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Okuma-Multus-U200.

PRM_43315 [Okuma-Multus-U200] — YAX_LOOP_GAIN_15 | Default: 10.000 | Linked Fault: ERR-OKU200-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Okuma-Multus-U200.

PRM_43316 [Okuma-Multus-U200] — PLC_LOOP_GAIN_16 | Default: 194.000 | Linked Fault: ERR-OKU200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U200.

CNC/PLC SERVO PARAMETER DICTIONARY — Okuma-Multus-U250 (BANK 1)

Machine Model: Okuma-Multus-U250 | Controller: OSP-P500A Aerospace. Use these parameter registers when diagnosing servo following errors or thermal drift on Okuma-Multus-U250.

PRM_43321 [Okuma-Multus-U250] — ATC_LOOP_GAIN_1 | Default: 37.950 | Linked Fault: ERR-OKU250-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Okuma-Multus-U250.

PRM_43322 [Okuma-Multus-U250] — HYD_LOOP_GAIN_2 | Default: 75.900 | Linked Fault: ERR-OKU250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U250.

PRM_43323 [Okuma-Multus-U250] — CLN_LOOP_GAIN_3 | Default: 113.850 | Linked Fault: ERR-OKU250-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Okuma-Multus-U250.

PRM_43324 [Okuma-Multus-U250] — XAX_LOOP_GAIN_4 | Default: 151.800 | Linked Fault: ERR-OKU250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U250.

PRM_43325 [Okuma-Multus-U250] — YAX_LOOP_GAIN_5 | Default: 189.750 | Linked Fault: ERR-OKU250-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Okuma-Multus-U250.

PRM_43326 [Okuma-Multus-U250] — ZAX_LOOP_GAIN_6 | Default: 227.700 | Linked Fault: ERR-OKU250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U250.

PRM_43327 [Okuma-Multus-U250] — BAX_LOOP_GAIN_7 | Default: 15.650 | Linked Fault: ERR-OKU250-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Okuma-Multus-U250.

PRM_43328 [Okuma-Multus-U250] — CAX_LOOP_GAIN_8 | Default: 53.600 | Linked Fault: ERR-OKU250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U250.

PRM_43329 [Okuma-Multus-U250] — OPT_LOOP_GAIN_9 | Default: 91.550 | Linked Fault: ERR-OKU250-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Okuma-Multus-U250.

PRM_43330 [Okuma-Multus-U250] — PLC_LOOP_GAIN_10 | Default: 129.500 | Linked Fault: ERR-OKU250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U250.

PRM_43331 [Okuma-Multus-U250] — LUB_LOOP_GAIN_11 | Default: 167.450 | Linked Fault: ERR-OKU250-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Okuma-Multus-U250.

PRM_43332 [Okuma-Multus-U250] — VAC_LOOP_GAIN_12 | Default: 205.400 | Linked Fault: ERR-OKU250-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U250.

PRM_43333 [Okuma-Multus-U250] — THM_LOOP_GAIN_13 | Default: 243.350 | Linked Fault: ERR-OKU250-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Okuma-Multus-U250.

PRM_43334 [Okuma-Multus-U250] — SPN_LOOP_GAIN_14 | Default: 31.300 | Linked Fault: ERR-OKU250-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Okuma-Multus-U250.

PRM_43335 [Okuma-Multus-U250] — ATC_LOOP_GAIN_15 | Default: 69.250 | Linked Fault: ERR-OKU250-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Okuma-Multus-U250.

PRM_43336 [Okuma-Multus-U250] — HYD_LOOP_GAIN_16 | Default: 107.200 | Linked Fault: ERR-OKU250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U250.

CNC/PLC SERVO PARAMETER DICTIONARY — Okuma-Multus-U250 (BANK 2)

Machine Model: Okuma-Multus-U250 | Controller: OSP-P500A Aerospace. Use these parameter registers when diagnosing servo following errors or thermal drift on Okuma-Multus-U250.

PRM_43341 [Okuma-Multus-U250] — HYD_LOOP_GAIN_1 | Default: 75.900 | Linked Fault: ERR-OKU250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U250.

PRM_43342 [Okuma-Multus-U250] — XAX_LOOP_GAIN_2 | Default: 151.800 | Linked Fault: ERR-OKU250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U250.

PRM_43343 [Okuma-Multus-U250] — ZAX_LOOP_GAIN_3 | Default: 227.700 | Linked Fault: ERR-OKU250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U250.

PRM_43344 [Okuma-Multus-U250] — CAX_LOOP_GAIN_4 | Default: 53.600 | Linked Fault: ERR-OKU250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U250.

PRM_43345 [Okuma-Multus-U250] — PLC_LOOP_GAIN_5 | Default: 129.500 | Linked Fault: ERR-OKU250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U250.

PRM_43346 [Okuma-Multus-U250] — VAC_LOOP_GAIN_6 | Default: 205.400 | Linked Fault: ERR-OKU250-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U250.

PRM_43347 [Okuma-Multus-U250] — SPN_LOOP_GAIN_7 | Default: 31.300 | Linked Fault: ERR-OKU250-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Okuma-Multus-U250.

PRM_43348 [Okuma-Multus-U250] — HYD_LOOP_GAIN_8 | Default: 107.200 | Linked Fault: ERR-OKU250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U250.

PRM_43349 [Okuma-Multus-U250] — XAX_LOOP_GAIN_9 | Default: 183.100 | Linked Fault: ERR-OKU250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U250.

PRM_43350 [Okuma-Multus-U250] — ZAX_LOOP_GAIN_10 | Default: 9.000 | Linked Fault: ERR-OKU250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U250.

PRM_43351 [Okuma-Multus-U250] — CAX_LOOP_GAIN_11 | Default: 84.900 | Linked Fault: ERR-OKU250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U250.

PRM_43352 [Okuma-Multus-U250] — PLC_LOOP_GAIN_12 | Default: 160.800 | Linked Fault: ERR-OKU250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U250.

PRM_43353 [Okuma-Multus-U250] — VAC_LOOP_GAIN_13 | Default: 236.700 | Linked Fault: ERR-OKU250-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U250.

PRM_43354 [Okuma-Multus-U250] — SPN_LOOP_GAIN_14 | Default: 62.600 | Linked Fault: ERR-OKU250-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Okuma-Multus-U250.

PRM_43355 [Okuma-Multus-U250] — HYD_LOOP_GAIN_15 | Default: 138.500 | Linked Fault: ERR-OKU250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U250.

PRM_43356 [Okuma-Multus-U250] — XAX_LOOP_GAIN_16 | Default: 214.400 | Linked Fault: ERR-OKU250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U250.

CNC/PLC SERVO PARAMETER DICTIONARY — Okuma-Multus-U250 (BANK 3)

Machine Model: Okuma-Multus-U250 | Controller: OSP-P500A Aerospace. Use these parameter registers when diagnosing servo following errors or thermal drift on Okuma-Multus-U250.

PRM_43361 [Okuma-Multus-U250] — CLN_LOOP_GAIN_1 | Default: 113.850 | Linked Fault: ERR-OKU250-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Okuma-Multus-U250.

PRM_43362 [Okuma-Multus-U250] — ZAX_LOOP_GAIN_2 | Default: 227.700 | Linked Fault: ERR-OKU250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U250.

PRM_43363 [Okuma-Multus-U250] — OPT_LOOP_GAIN_3 | Default: 91.550 | Linked Fault: ERR-OKU250-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Okuma-Multus-U250.

PRM_43364 [Okuma-Multus-U250] — VAC_LOOP_GAIN_4 | Default: 205.400 | Linked Fault: ERR-OKU250-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U250.

PRM_43365 [Okuma-Multus-U250] — ATC_LOOP_GAIN_5 | Default: 69.250 | Linked Fault: ERR-OKU250-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Okuma-Multus-U250.

PRM_43366 [Okuma-Multus-U250] — XAX_LOOP_GAIN_6 | Default: 183.100 | Linked Fault: ERR-OKU250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U250.

PRM_43367 [Okuma-Multus-U250] — BAX_LOOP_GAIN_7 | Default: 46.950 | Linked Fault: ERR-OKU250-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Okuma-Multus-U250.

PRM_43368 [Okuma-Multus-U250] — PLC_LOOP_GAIN_8 | Default: 160.800 | Linked Fault: ERR-OKU250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U250.

PRM_43369 [Okuma-Multus-U250] — THM_LOOP_GAIN_9 | Default: 24.650 | Linked Fault: ERR-OKU250-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Okuma-Multus-U250.

PRM_43370 [Okuma-Multus-U250] — HYD_LOOP_GAIN_10 | Default: 138.500 | Linked Fault: ERR-OKU250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U250.

PRM_43371 [Okuma-Multus-U250] — YAX_LOOP_GAIN_11 | Default: 2.350 | Linked Fault: ERR-OKU250-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Okuma-Multus-U250.

PRM_43372 [Okuma-Multus-U250] — CAX_LOOP_GAIN_12 | Default: 116.200 | Linked Fault: ERR-OKU250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U250.

PRM_43373 [Okuma-Multus-U250] — LUB_LOOP_GAIN_13 | Default: 230.050 | Linked Fault: ERR-OKU250-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Okuma-Multus-U250.

PRM_43374 [Okuma-Multus-U250] — SPN_LOOP_GAIN_14 | Default: 93.900 | Linked Fault: ERR-OKU250-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Okuma-Multus-U250.

PRM_43375 [Okuma-Multus-U250] — CLN_LOOP_GAIN_15 | Default: 207.750 | Linked Fault: ERR-OKU250-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Okuma-Multus-U250.

PRM_43376 [Okuma-Multus-U250] — ZAX_LOOP_GAIN_16 | Default: 71.600 | Linked Fault: ERR-OKU250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U250.

CNC/PLC SERVO PARAMETER DICTIONARY — Okuma-Multus-U250 (BANK 4)

Machine Model: Okuma-Multus-U250 | Controller: OSP-P500A Aerospace. Use these parameter registers when diagnosing servo following errors or thermal drift on Okuma-Multus-U250.

PRM_43381 [Okuma-Multus-U250] — XAX_LOOP_GAIN_1 | Default: 151.800 | Linked Fault: ERR-OKU250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U250.

PRM_43382 [Okuma-Multus-U250] — CAX_LOOP_GAIN_2 | Default: 53.600 | Linked Fault: ERR-OKU250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U250.

PRM_43383 [Okuma-Multus-U250] — VAC_LOOP_GAIN_3 | Default: 205.400 | Linked Fault: ERR-OKU250-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U250.

PRM_43384 [Okuma-Multus-U250] — HYD_LOOP_GAIN_4 | Default: 107.200 | Linked Fault: ERR-OKU250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U250.

PRM_43385 [Okuma-Multus-U250] — ZAX_LOOP_GAIN_5 | Default: 9.000 | Linked Fault: ERR-OKU250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U250.

PRM_43386 [Okuma-Multus-U250] — PLC_LOOP_GAIN_6 | Default: 160.800 | Linked Fault: ERR-OKU250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U250.

PRM_43387 [Okuma-Multus-U250] — SPN_LOOP_GAIN_7 | Default: 62.600 | Linked Fault: ERR-OKU250-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Okuma-Multus-U250.

PRM_43388 [Okuma-Multus-U250] — XAX_LOOP_GAIN_8 | Default: 214.400 | Linked Fault: ERR-OKU250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U250.

PRM_43389 [Okuma-Multus-U250] — CAX_LOOP_GAIN_9 | Default: 116.200 | Linked Fault: ERR-OKU250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U250.

PRM_43390 [Okuma-Multus-U250] — VAC_LOOP_GAIN_10 | Default: 18.000 | Linked Fault: ERR-OKU250-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U250.

PRM_43391 [Okuma-Multus-U250] — HYD_LOOP_GAIN_11 | Default: 169.800 | Linked Fault: ERR-OKU250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U250.

PRM_43392 [Okuma-Multus-U250] — ZAX_LOOP_GAIN_12 | Default: 71.600 | Linked Fault: ERR-OKU250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U250.

PRM_43393 [Okuma-Multus-U250] — PLC_LOOP_GAIN_13 | Default: 223.400 | Linked Fault: ERR-OKU250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U250.

PRM_43394 [Okuma-Multus-U250] — SPN_LOOP_GAIN_14 | Default: 125.200 | Linked Fault: ERR-OKU250-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Okuma-Multus-U250.

PRM_43395 [Okuma-Multus-U250] — XAX_LOOP_GAIN_15 | Default: 27.000 | Linked Fault: ERR-OKU250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U250.

PRM_43396 [Okuma-Multus-U250] — CAX_LOOP_GAIN_16 | Default: 178.800 | Linked Fault: ERR-OKU250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U250.

CNC/PLC SERVO PARAMETER DICTIONARY — Okuma-Multus-U250 (BANK 5)

Machine Model: Okuma-Multus-U250 | Controller: OSP-P500A Aerospace. Use these parameter registers when diagnosing servo following errors or thermal drift on Okuma-Multus-U250.

PRM_43401 [Okuma-Multus-U250] — YAX_LOOP_GAIN_1 | Default: 189.750 | Linked Fault: ERR-OKU250-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Okuma-Multus-U250.

PRM_43402 [Okuma-Multus-U250] — PLC_LOOP_GAIN_2 | Default: 129.500 | Linked Fault: ERR-OKU250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U250.

PRM_43403 [Okuma-Multus-U250] — ATC_LOOP_GAIN_3 | Default: 69.250 | Linked Fault: ERR-OKU250-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Okuma-Multus-U250.

PRM_43404 [Okuma-Multus-U250] — ZAX_LOOP_GAIN_4 | Default: 9.000 | Linked Fault: ERR-OKU250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U250.

PRM_43405 [Okuma-Multus-U250] — LUB_LOOP_GAIN_5 | Default: 198.750 | Linked Fault: ERR-OKU250-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Okuma-Multus-U250.

PRM_43406 [Okuma-Multus-U250] — HYD_LOOP_GAIN_6 | Default: 138.500 | Linked Fault: ERR-OKU250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U250.

PRM_43407 [Okuma-Multus-U250] — BAX_LOOP_GAIN_7 | Default: 78.250 | Linked Fault: ERR-OKU250-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Okuma-Multus-U250.

PRM_43408 [Okuma-Multus-U250] — VAC_LOOP_GAIN_8 | Default: 18.000 | Linked Fault: ERR-OKU250-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U250.

PRM_43409 [Okuma-Multus-U250] — CLN_LOOP_GAIN_9 | Default: 207.750 | Linked Fault: ERR-OKU250-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Okuma-Multus-U250.

PRM_43410 [Okuma-Multus-U250] — CAX_LOOP_GAIN_10 | Default: 147.500 | Linked Fault: ERR-OKU250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U250.

PRM_43411 [Okuma-Multus-U250] — THM_LOOP_GAIN_11 | Default: 87.250 | Linked Fault: ERR-OKU250-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Okuma-Multus-U250.

PRM_43412 [Okuma-Multus-U250] — XAX_LOOP_GAIN_12 | Default: 27.000 | Linked Fault: ERR-OKU250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U250.

PRM_43413 [Okuma-Multus-U250] — OPT_LOOP_GAIN_13 | Default: 216.750 | Linked Fault: ERR-OKU250-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Okuma-Multus-U250.

PRM_43414 [Okuma-Multus-U250] — SPN_LOOP_GAIN_14 | Default: 156.500 | Linked Fault: ERR-OKU250-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Okuma-Multus-U250.

PRM_43415 [Okuma-Multus-U250] — YAX_LOOP_GAIN_15 | Default: 96.250 | Linked Fault: ERR-OKU250-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Okuma-Multus-U250.

PRM_43416 [Okuma-Multus-U250] — PLC_LOOP_GAIN_16 | Default: 36.000 | Linked Fault: ERR-OKU250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U250.

CNC/PLC SERVO PARAMETER DICTIONARY — Okuma-Multus-U300 (BANK 1)

Machine Model: Okuma-Multus-U300 | Controller: OSP-P500A Aerospace. Use these parameter registers when diagnosing servo following errors or thermal drift on Okuma-Multus-U300.

PRM_43421 [Okuma-Multus-U300] — ATC_LOOP_GAIN_1 | Default: 39.100 | Linked Fault: ERR-OKU300-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Okuma-Multus-U300.

PRM_43422 [Okuma-Multus-U300] — HYD_LOOP_GAIN_2 | Default: 78.200 | Linked Fault: ERR-OKU300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U300.

PRM_43423 [Okuma-Multus-U300] — CLN_LOOP_GAIN_3 | Default: 117.300 | Linked Fault: ERR-OKU300-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Okuma-Multus-U300.

PRM_43424 [Okuma-Multus-U300] — XAX_LOOP_GAIN_4 | Default: 156.400 | Linked Fault: ERR-OKU300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U300.

PRM_43425 [Okuma-Multus-U300] — YAX_LOOP_GAIN_5 | Default: 195.500 | Linked Fault: ERR-OKU300-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Okuma-Multus-U300.

PRM_43426 [Okuma-Multus-U300] — ZAX_LOOP_GAIN_6 | Default: 234.600 | Linked Fault: ERR-OKU300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U300.

PRM_43427 [Okuma-Multus-U300] — BAX_LOOP_GAIN_7 | Default: 23.700 | Linked Fault: ERR-OKU300-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Okuma-Multus-U300.

PRM_43428 [Okuma-Multus-U300] — CAX_LOOP_GAIN_8 | Default: 62.800 | Linked Fault: ERR-OKU300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U300.

PRM_43429 [Okuma-Multus-U300] — OPT_LOOP_GAIN_9 | Default: 101.900 | Linked Fault: ERR-OKU300-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Okuma-Multus-U300.

PRM_43430 [Okuma-Multus-U300] — PLC_LOOP_GAIN_10 | Default: 141.000 | Linked Fault: ERR-OKU300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U300.

PRM_43431 [Okuma-Multus-U300] — LUB_LOOP_GAIN_11 | Default: 180.100 | Linked Fault: ERR-OKU300-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Okuma-Multus-U300.

PRM_43432 [Okuma-Multus-U300] — VAC_LOOP_GAIN_12 | Default: 219.200 | Linked Fault: ERR-OKU300-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U300.

PRM_43433 [Okuma-Multus-U300] — THM_LOOP_GAIN_13 | Default: 8.300 | Linked Fault: ERR-OKU300-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Okuma-Multus-U300.

PRM_43434 [Okuma-Multus-U300] — SPN_LOOP_GAIN_14 | Default: 47.400 | Linked Fault: ERR-OKU300-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Okuma-Multus-U300.

PRM_43435 [Okuma-Multus-U300] — ATC_LOOP_GAIN_15 | Default: 86.500 | Linked Fault: ERR-OKU300-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Okuma-Multus-U300.

PRM_43436 [Okuma-Multus-U300] — HYD_LOOP_GAIN_16 | Default: 125.600 | Linked Fault: ERR-OKU300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U300.

CNC/PLC SERVO PARAMETER DICTIONARY — Okuma-Multus-U300 (BANK 2)

Machine Model: Okuma-Multus-U300 | Controller: OSP-P500A Aerospace. Use these parameter registers when diagnosing servo following errors or thermal drift on Okuma-Multus-U300.

PRM_43441 [Okuma-Multus-U300] — HYD_LOOP_GAIN_1 | Default: 78.200 | Linked Fault: ERR-OKU300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U300.

PRM_43442 [Okuma-Multus-U300] — XAX_LOOP_GAIN_2 | Default: 156.400 | Linked Fault: ERR-OKU300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U300.

PRM_43443 [Okuma-Multus-U300] — ZAX_LOOP_GAIN_3 | Default: 234.600 | Linked Fault: ERR-OKU300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U300.

PRM_43444 [Okuma-Multus-U300] — CAX_LOOP_GAIN_4 | Default: 62.800 | Linked Fault: ERR-OKU300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U300.

PRM_43445 [Okuma-Multus-U300] — PLC_LOOP_GAIN_5 | Default: 141.000 | Linked Fault: ERR-OKU300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U300.

PRM_43446 [Okuma-Multus-U300] — VAC_LOOP_GAIN_6 | Default: 219.200 | Linked Fault: ERR-OKU300-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U300.

PRM_43447 [Okuma-Multus-U300] — SPN_LOOP_GAIN_7 | Default: 47.400 | Linked Fault: ERR-OKU300-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Okuma-Multus-U300.

PRM_43448 [Okuma-Multus-U300] — HYD_LOOP_GAIN_8 | Default: 125.600 | Linked Fault: ERR-OKU300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U300.

PRM_43449 [Okuma-Multus-U300] — XAX_LOOP_GAIN_9 | Default: 203.800 | Linked Fault: ERR-OKU300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U300.

PRM_43450 [Okuma-Multus-U300] — ZAX_LOOP_GAIN_10 | Default: 32.000 | Linked Fault: ERR-OKU300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U300.

PRM_43451 [Okuma-Multus-U300] — CAX_LOOP_GAIN_11 | Default: 110.200 | Linked Fault: ERR-OKU300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U300.

PRM_43452 [Okuma-Multus-U300] — PLC_LOOP_GAIN_12 | Default: 188.400 | Linked Fault: ERR-OKU300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U300.

PRM_43453 [Okuma-Multus-U300] — VAC_LOOP_GAIN_13 | Default: 16.600 | Linked Fault: ERR-OKU300-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U300.

PRM_43454 [Okuma-Multus-U300] — SPN_LOOP_GAIN_14 | Default: 94.800 | Linked Fault: ERR-OKU300-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Okuma-Multus-U300.

PRM_43455 [Okuma-Multus-U300] — HYD_LOOP_GAIN_15 | Default: 173.000 | Linked Fault: ERR-OKU300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U300.

PRM_43456 [Okuma-Multus-U300] — XAX_LOOP_GAIN_16 | Default: 1.200 | Linked Fault: ERR-OKU300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U300.

CNC/PLC SERVO PARAMETER DICTIONARY — Okuma-Multus-U300 (BANK 3)

Machine Model: Okuma-Multus-U300 | Controller: OSP-P500A Aerospace. Use these parameter registers when diagnosing servo following errors or thermal drift on Okuma-Multus-U300.

PRM_43461 [Okuma-Multus-U300] — CLN_LOOP_GAIN_1 | Default: 117.300 | Linked Fault: ERR-OKU300-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Okuma-Multus-U300.

PRM_43462 [Okuma-Multus-U300] — ZAX_LOOP_GAIN_2 | Default: 234.600 | Linked Fault: ERR-OKU300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U300.

PRM_43463 [Okuma-Multus-U300] — OPT_LOOP_GAIN_3 | Default: 101.900 | Linked Fault: ERR-OKU300-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Okuma-Multus-U300.

PRM_43464 [Okuma-Multus-U300] — VAC_LOOP_GAIN_4 | Default: 219.200 | Linked Fault: ERR-OKU300-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U300.

PRM_43465 [Okuma-Multus-U300] — ATC_LOOP_GAIN_5 | Default: 86.500 | Linked Fault: ERR-OKU300-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Okuma-Multus-U300.

PRM_43466 [Okuma-Multus-U300] — XAX_LOOP_GAIN_6 | Default: 203.800 | Linked Fault: ERR-OKU300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U300.

PRM_43467 [Okuma-Multus-U300] — BAX_LOOP_GAIN_7 | Default: 71.100 | Linked Fault: ERR-OKU300-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Okuma-Multus-U300.

PRM_43468 [Okuma-Multus-U300] — PLC_LOOP_GAIN_8 | Default: 188.400 | Linked Fault: ERR-OKU300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U300.

PRM_43469 [Okuma-Multus-U300] — THM_LOOP_GAIN_9 | Default: 55.700 | Linked Fault: ERR-OKU300-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Okuma-Multus-U300.

PRM_43470 [Okuma-Multus-U300] — HYD_LOOP_GAIN_10 | Default: 173.000 | Linked Fault: ERR-OKU300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U300.

PRM_43471 [Okuma-Multus-U300] — YAX_LOOP_GAIN_11 | Default: 40.300 | Linked Fault: ERR-OKU300-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Okuma-Multus-U300.

PRM_43472 [Okuma-Multus-U300] — CAX_LOOP_GAIN_12 | Default: 157.600 | Linked Fault: ERR-OKU300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U300.

PRM_43473 [Okuma-Multus-U300] — LUB_LOOP_GAIN_13 | Default: 24.900 | Linked Fault: ERR-OKU300-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Okuma-Multus-U300.

PRM_43474 [Okuma-Multus-U300] — SPN_LOOP_GAIN_14 | Default: 142.200 | Linked Fault: ERR-OKU300-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Okuma-Multus-U300.

PRM_43475 [Okuma-Multus-U300] — CLN_LOOP_GAIN_15 | Default: 9.500 | Linked Fault: ERR-OKU300-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Okuma-Multus-U300.

PRM_43476 [Okuma-Multus-U300] — ZAX_LOOP_GAIN_16 | Default: 126.800 | Linked Fault: ERR-OKU300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U300.

CNC/PLC SERVO PARAMETER DICTIONARY — Okuma-Multus-U300 (BANK 4)

Machine Model: Okuma-Multus-U300 | Controller: OSP-P500A Aerospace. Use these parameter registers when diagnosing servo following errors or thermal drift on Okuma-Multus-U300.

PRM_43481 [Okuma-Multus-U300] — XAX_LOOP_GAIN_1 | Default: 156.400 | Linked Fault: ERR-OKU300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U300.

PRM_43482 [Okuma-Multus-U300] — CAX_LOOP_GAIN_2 | Default: 62.800 | Linked Fault: ERR-OKU300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U300.

PRM_43483 [Okuma-Multus-U300] — VAC_LOOP_GAIN_3 | Default: 219.200 | Linked Fault: ERR-OKU300-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U300.

PRM_43484 [Okuma-Multus-U300] — HYD_LOOP_GAIN_4 | Default: 125.600 | Linked Fault: ERR-OKU300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U300.

PRM_43485 [Okuma-Multus-U300] — ZAX_LOOP_GAIN_5 | Default: 32.000 | Linked Fault: ERR-OKU300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U300.

PRM_43486 [Okuma-Multus-U300] — PLC_LOOP_GAIN_6 | Default: 188.400 | Linked Fault: ERR-OKU300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U300.

PRM_43487 [Okuma-Multus-U300] — SPN_LOOP_GAIN_7 | Default: 94.800 | Linked Fault: ERR-OKU300-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Okuma-Multus-U300.

PRM_43488 [Okuma-Multus-U300] — XAX_LOOP_GAIN_8 | Default: 1.200 | Linked Fault: ERR-OKU300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U300.

PRM_43489 [Okuma-Multus-U300] — CAX_LOOP_GAIN_9 | Default: 157.600 | Linked Fault: ERR-OKU300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U300.

PRM_43490 [Okuma-Multus-U300] — VAC_LOOP_GAIN_10 | Default: 64.000 | Linked Fault: ERR-OKU300-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U300.

PRM_43491 [Okuma-Multus-U300] — HYD_LOOP_GAIN_11 | Default: 220.400 | Linked Fault: ERR-OKU300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U300.

PRM_43492 [Okuma-Multus-U300] — ZAX_LOOP_GAIN_12 | Default: 126.800 | Linked Fault: ERR-OKU300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U300.

PRM_43493 [Okuma-Multus-U300] — PLC_LOOP_GAIN_13 | Default: 33.200 | Linked Fault: ERR-OKU300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U300.

PRM_43494 [Okuma-Multus-U300] — SPN_LOOP_GAIN_14 | Default: 189.600 | Linked Fault: ERR-OKU300-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Okuma-Multus-U300.

PRM_43495 [Okuma-Multus-U300] — XAX_LOOP_GAIN_15 | Default: 96.000 | Linked Fault: ERR-OKU300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U300.

PRM_43496 [Okuma-Multus-U300] — CAX_LOOP_GAIN_16 | Default: 2.400 | Linked Fault: ERR-OKU300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U300.

CNC/PLC SERVO PARAMETER DICTIONARY — Okuma-Multus-U300 (BANK 5)

Machine Model: Okuma-Multus-U300 | Controller: OSP-P500A Aerospace. Use these parameter registers when diagnosing servo following errors or thermal drift on Okuma-Multus-U300.

PRM_43501 [Okuma-Multus-U300] — YAX_LOOP_GAIN_1 | Default: 195.500 | Linked Fault: ERR-OKU300-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Okuma-Multus-U300.

PRM_43502 [Okuma-Multus-U300] — PLC_LOOP_GAIN_2 | Default: 141.000 | Linked Fault: ERR-OKU300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U300.

PRM_43503 [Okuma-Multus-U300] — ATC_LOOP_GAIN_3 | Default: 86.500 | Linked Fault: ERR-OKU300-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Okuma-Multus-U300.

PRM_43504 [Okuma-Multus-U300] — ZAX_LOOP_GAIN_4 | Default: 32.000 | Linked Fault: ERR-OKU300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U300.

PRM_43505 [Okuma-Multus-U300] — LUB_LOOP_GAIN_5 | Default: 227.500 | Linked Fault: ERR-OKU300-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Okuma-Multus-U300.

PRM_43506 [Okuma-Multus-U300] — HYD_LOOP_GAIN_6 | Default: 173.000 | Linked Fault: ERR-OKU300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U300.

PRM_43507 [Okuma-Multus-U300] — BAX_LOOP_GAIN_7 | Default: 118.500 | Linked Fault: ERR-OKU300-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Okuma-Multus-U300.

PRM_43508 [Okuma-Multus-U300] — VAC_LOOP_GAIN_8 | Default: 64.000 | Linked Fault: ERR-OKU300-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U300.

PRM_43509 [Okuma-Multus-U300] — CLN_LOOP_GAIN_9 | Default: 9.500 | Linked Fault: ERR-OKU300-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Okuma-Multus-U300.

PRM_43510 [Okuma-Multus-U300] — CAX_LOOP_GAIN_10 | Default: 205.000 | Linked Fault: ERR-OKU300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U300.

PRM_43511 [Okuma-Multus-U300] — THM_LOOP_GAIN_11 | Default: 150.500 | Linked Fault: ERR-OKU300-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Okuma-Multus-U300.

PRM_43512 [Okuma-Multus-U300] — XAX_LOOP_GAIN_12 | Default: 96.000 | Linked Fault: ERR-OKU300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U300.

PRM_43513 [Okuma-Multus-U300] — OPT_LOOP_GAIN_13 | Default: 41.500 | Linked Fault: ERR-OKU300-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Okuma-Multus-U300.

PRM_43514 [Okuma-Multus-U300] — SPN_LOOP_GAIN_14 | Default: 237.000 | Linked Fault: ERR-OKU300-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Okuma-Multus-U300.

PRM_43515 [Okuma-Multus-U300] — YAX_LOOP_GAIN_15 | Default: 182.500 | Linked Fault: ERR-OKU300-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Okuma-Multus-U300.

PRM_43516 [Okuma-Multus-U300] — PLC_LOOP_GAIN_16 | Default: 128.000 | Linked Fault: ERR-OKU300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U300.

CNC/PLC SERVO PARAMETER DICTIONARY — Okuma-Multus-U350 (BANK 1)

Machine Model: Okuma-Multus-U350 | Controller: OSP-P500A Aerospace. Use these parameter registers when diagnosing servo following errors or thermal drift on Okuma-Multus-U350.

PRM_43521 [Okuma-Multus-U350] — ATC_LOOP_GAIN_1 | Default: 40.250 | Linked Fault: ERR-OKU350-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Okuma-Multus-U350.

PRM_43522 [Okuma-Multus-U350] — HYD_LOOP_GAIN_2 | Default: 80.500 | Linked Fault: ERR-OKU350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U350.

PRM_43523 [Okuma-Multus-U350] — CLN_LOOP_GAIN_3 | Default: 120.750 | Linked Fault: ERR-OKU350-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Okuma-Multus-U350.

PRM_43524 [Okuma-Multus-U350] — XAX_LOOP_GAIN_4 | Default: 161.000 | Linked Fault: ERR-OKU350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U350.

PRM_43525 [Okuma-Multus-U350] — YAX_LOOP_GAIN_5 | Default: 201.250 | Linked Fault: ERR-OKU350-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Okuma-Multus-U350.

PRM_43526 [Okuma-Multus-U350] — ZAX_LOOP_GAIN_6 | Default: 241.500 | Linked Fault: ERR-OKU350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U350.

PRM_43527 [Okuma-Multus-U350] — BAX_LOOP_GAIN_7 | Default: 31.750 | Linked Fault: ERR-OKU350-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Okuma-Multus-U350.

PRM_43528 [Okuma-Multus-U350] — CAX_LOOP_GAIN_8 | Default: 72.000 | Linked Fault: ERR-OKU350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U350.

PRM_43529 [Okuma-Multus-U350] — OPT_LOOP_GAIN_9 | Default: 112.250 | Linked Fault: ERR-OKU350-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Okuma-Multus-U350.

PRM_43530 [Okuma-Multus-U350] — PLC_LOOP_GAIN_10 | Default: 152.500 | Linked Fault: ERR-OKU350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U350.

PRM_43531 [Okuma-Multus-U350] — LUB_LOOP_GAIN_11 | Default: 192.750 | Linked Fault: ERR-OKU350-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Okuma-Multus-U350.

PRM_43532 [Okuma-Multus-U350] — VAC_LOOP_GAIN_12 | Default: 233.000 | Linked Fault: ERR-OKU350-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U350.

PRM_43533 [Okuma-Multus-U350] — THM_LOOP_GAIN_13 | Default: 23.250 | Linked Fault: ERR-OKU350-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Okuma-Multus-U350.

PRM_43534 [Okuma-Multus-U350] — SPN_LOOP_GAIN_14 | Default: 63.500 | Linked Fault: ERR-OKU350-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Okuma-Multus-U350.

PRM_43535 [Okuma-Multus-U350] — ATC_LOOP_GAIN_15 | Default: 103.750 | Linked Fault: ERR-OKU350-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Okuma-Multus-U350.

PRM_43536 [Okuma-Multus-U350] — HYD_LOOP_GAIN_16 | Default: 144.000 | Linked Fault: ERR-OKU350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U350.

CNC/PLC SERVO PARAMETER DICTIONARY — Okuma-Multus-U350 (BANK 2)

Machine Model: Okuma-Multus-U350 | Controller: OSP-P500A Aerospace. Use these parameter registers when diagnosing servo following errors or thermal drift on Okuma-Multus-U350.

PRM_43541 [Okuma-Multus-U350] — HYD_LOOP_GAIN_1 | Default: 80.500 | Linked Fault: ERR-OKU350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U350.

PRM_43542 [Okuma-Multus-U350] — XAX_LOOP_GAIN_2 | Default: 161.000 | Linked Fault: ERR-OKU350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U350.

PRM_43543 [Okuma-Multus-U350] — ZAX_LOOP_GAIN_3 | Default: 241.500 | Linked Fault: ERR-OKU350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U350.

PRM_43544 [Okuma-Multus-U350] — CAX_LOOP_GAIN_4 | Default: 72.000 | Linked Fault: ERR-OKU350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U350.

PRM_43545 [Okuma-Multus-U350] — PLC_LOOP_GAIN_5 | Default: 152.500 | Linked Fault: ERR-OKU350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U350.

PRM_43546 [Okuma-Multus-U350] — VAC_LOOP_GAIN_6 | Default: 233.000 | Linked Fault: ERR-OKU350-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U350.

PRM_43547 [Okuma-Multus-U350] — SPN_LOOP_GAIN_7 | Default: 63.500 | Linked Fault: ERR-OKU350-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Okuma-Multus-U350.

PRM_43548 [Okuma-Multus-U350] — HYD_LOOP_GAIN_8 | Default: 144.000 | Linked Fault: ERR-OKU350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U350.

PRM_43549 [Okuma-Multus-U350] — XAX_LOOP_GAIN_9 | Default: 224.500 | Linked Fault: ERR-OKU350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U350.

PRM_43550 [Okuma-Multus-U350] — ZAX_LOOP_GAIN_10 | Default: 55.000 | Linked Fault: ERR-OKU350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U350.

PRM_43551 [Okuma-Multus-U350] — CAX_LOOP_GAIN_11 | Default: 135.500 | Linked Fault: ERR-OKU350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U350.

PRM_43552 [Okuma-Multus-U350] — PLC_LOOP_GAIN_12 | Default: 216.000 | Linked Fault: ERR-OKU350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U350.

PRM_43553 [Okuma-Multus-U350] — VAC_LOOP_GAIN_13 | Default: 46.500 | Linked Fault: ERR-OKU350-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U350.

PRM_43554 [Okuma-Multus-U350] — SPN_LOOP_GAIN_14 | Default: 127.000 | Linked Fault: ERR-OKU350-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Okuma-Multus-U350.

PRM_43555 [Okuma-Multus-U350] — HYD_LOOP_GAIN_15 | Default: 207.500 | Linked Fault: ERR-OKU350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U350.

PRM_43556 [Okuma-Multus-U350] — XAX_LOOP_GAIN_16 | Default: 38.000 | Linked Fault: ERR-OKU350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U350.

CNC/PLC SERVO PARAMETER DICTIONARY — Okuma-Multus-U350 (BANK 3)

Machine Model: Okuma-Multus-U350 | Controller: OSP-P500A Aerospace. Use these parameter registers when diagnosing servo following errors or thermal drift on Okuma-Multus-U350.

PRM_43561 [Okuma-Multus-U350] — CLN_LOOP_GAIN_1 | Default: 120.750 | Linked Fault: ERR-OKU350-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Okuma-Multus-U350.

PRM_43562 [Okuma-Multus-U350] — ZAX_LOOP_GAIN_2 | Default: 241.500 | Linked Fault: ERR-OKU350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U350.

PRM_43563 [Okuma-Multus-U350] — OPT_LOOP_GAIN_3 | Default: 112.250 | Linked Fault: ERR-OKU350-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Okuma-Multus-U350.

PRM_43564 [Okuma-Multus-U350] — VAC_LOOP_GAIN_4 | Default: 233.000 | Linked Fault: ERR-OKU350-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U350.

PRM_43565 [Okuma-Multus-U350] — ATC_LOOP_GAIN_5 | Default: 103.750 | Linked Fault: ERR-OKU350-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Okuma-Multus-U350.

PRM_43566 [Okuma-Multus-U350] — XAX_LOOP_GAIN_6 | Default: 224.500 | Linked Fault: ERR-OKU350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U350.

PRM_43567 [Okuma-Multus-U350] — BAX_LOOP_GAIN_7 | Default: 95.250 | Linked Fault: ERR-OKU350-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Okuma-Multus-U350.

PRM_43568 [Okuma-Multus-U350] — PLC_LOOP_GAIN_8 | Default: 216.000 | Linked Fault: ERR-OKU350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U350.

PRM_43569 [Okuma-Multus-U350] — THM_LOOP_GAIN_9 | Default: 86.750 | Linked Fault: ERR-OKU350-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Okuma-Multus-U350.

PRM_43570 [Okuma-Multus-U350] — HYD_LOOP_GAIN_10 | Default: 207.500 | Linked Fault: ERR-OKU350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U350.

PRM_43571 [Okuma-Multus-U350] — YAX_LOOP_GAIN_11 | Default: 78.250 | Linked Fault: ERR-OKU350-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Okuma-Multus-U350.

PRM_43572 [Okuma-Multus-U350] — CAX_LOOP_GAIN_12 | Default: 199.000 | Linked Fault: ERR-OKU350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U350.

PRM_43573 [Okuma-Multus-U350] — LUB_LOOP_GAIN_13 | Default: 69.750 | Linked Fault: ERR-OKU350-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Okuma-Multus-U350.

PRM_43574 [Okuma-Multus-U350] — SPN_LOOP_GAIN_14 | Default: 190.500 | Linked Fault: ERR-OKU350-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Okuma-Multus-U350.

PRM_43575 [Okuma-Multus-U350] — CLN_LOOP_GAIN_15 | Default: 61.250 | Linked Fault: ERR-OKU350-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Okuma-Multus-U350.

PRM_43576 [Okuma-Multus-U350] — ZAX_LOOP_GAIN_16 | Default: 182.000 | Linked Fault: ERR-OKU350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U350.

CNC/PLC SERVO PARAMETER DICTIONARY — Okuma-Multus-U350 (BANK 4)

Machine Model: Okuma-Multus-U350 | Controller: OSP-P500A Aerospace. Use these parameter registers when diagnosing servo following errors or thermal drift on Okuma-Multus-U350.

PRM_43581 [Okuma-Multus-U350] — XAX_LOOP_GAIN_1 | Default: 161.000 | Linked Fault: ERR-OKU350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U350.

PRM_43582 [Okuma-Multus-U350] — CAX_LOOP_GAIN_2 | Default: 72.000 | Linked Fault: ERR-OKU350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U350.

PRM_43583 [Okuma-Multus-U350] — VAC_LOOP_GAIN_3 | Default: 233.000 | Linked Fault: ERR-OKU350-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U350.

PRM_43584 [Okuma-Multus-U350] — HYD_LOOP_GAIN_4 | Default: 144.000 | Linked Fault: ERR-OKU350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U350.

PRM_43585 [Okuma-Multus-U350] — ZAX_LOOP_GAIN_5 | Default: 55.000 | Linked Fault: ERR-OKU350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U350.

PRM_43586 [Okuma-Multus-U350] — PLC_LOOP_GAIN_6 | Default: 216.000 | Linked Fault: ERR-OKU350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U350.

PRM_43587 [Okuma-Multus-U350] — SPN_LOOP_GAIN_7 | Default: 127.000 | Linked Fault: ERR-OKU350-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Okuma-Multus-U350.

PRM_43588 [Okuma-Multus-U350] — XAX_LOOP_GAIN_8 | Default: 38.000 | Linked Fault: ERR-OKU350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U350.

PRM_43589 [Okuma-Multus-U350] — CAX_LOOP_GAIN_9 | Default: 199.000 | Linked Fault: ERR-OKU350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U350.

PRM_43590 [Okuma-Multus-U350] — VAC_LOOP_GAIN_10 | Default: 110.000 | Linked Fault: ERR-OKU350-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U350.

PRM_43591 [Okuma-Multus-U350] — HYD_LOOP_GAIN_11 | Default: 21.000 | Linked Fault: ERR-OKU350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U350.

PRM_43592 [Okuma-Multus-U350] — ZAX_LOOP_GAIN_12 | Default: 182.000 | Linked Fault: ERR-OKU350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U350.

PRM_43593 [Okuma-Multus-U350] — PLC_LOOP_GAIN_13 | Default: 93.000 | Linked Fault: ERR-OKU350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U350.

PRM_43594 [Okuma-Multus-U350] — SPN_LOOP_GAIN_14 | Default: 4.000 | Linked Fault: ERR-OKU350-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Okuma-Multus-U350.

PRM_43595 [Okuma-Multus-U350] — XAX_LOOP_GAIN_15 | Default: 165.000 | Linked Fault: ERR-OKU350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U350.

PRM_43596 [Okuma-Multus-U350] — CAX_LOOP_GAIN_16 | Default: 76.000 | Linked Fault: ERR-OKU350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U350.

CNC/PLC SERVO PARAMETER DICTIONARY — Okuma-Multus-U350 (BANK 5)

Machine Model: Okuma-Multus-U350 | Controller: OSP-P500A Aerospace. Use these parameter registers when diagnosing servo following errors or thermal drift on Okuma-Multus-U350.

PRM_43601 [Okuma-Multus-U350] — YAX_LOOP_GAIN_1 | Default: 201.250 | Linked Fault: ERR-OKU350-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Okuma-Multus-U350.

PRM_43602 [Okuma-Multus-U350] — PLC_LOOP_GAIN_2 | Default: 152.500 | Linked Fault: ERR-OKU350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U350.

PRM_43603 [Okuma-Multus-U350] — ATC_LOOP_GAIN_3 | Default: 103.750 | Linked Fault: ERR-OKU350-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Okuma-Multus-U350.

PRM_43604 [Okuma-Multus-U350] — ZAX_LOOP_GAIN_4 | Default: 55.000 | Linked Fault: ERR-OKU350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U350.

PRM_43605 [Okuma-Multus-U350] — LUB_LOOP_GAIN_5 | Default: 6.250 | Linked Fault: ERR-OKU350-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Okuma-Multus-U350.

PRM_43606 [Okuma-Multus-U350] — HYD_LOOP_GAIN_6 | Default: 207.500 | Linked Fault: ERR-OKU350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U350.

PRM_43607 [Okuma-Multus-U350] — BAX_LOOP_GAIN_7 | Default: 158.750 | Linked Fault: ERR-OKU350-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Okuma-Multus-U350.

PRM_43608 [Okuma-Multus-U350] — VAC_LOOP_GAIN_8 | Default: 110.000 | Linked Fault: ERR-OKU350-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U350.

PRM_43609 [Okuma-Multus-U350] — CLN_LOOP_GAIN_9 | Default: 61.250 | Linked Fault: ERR-OKU350-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Okuma-Multus-U350.

PRM_43610 [Okuma-Multus-U350] — CAX_LOOP_GAIN_10 | Default: 12.500 | Linked Fault: ERR-OKU350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U350.

PRM_43611 [Okuma-Multus-U350] — THM_LOOP_GAIN_11 | Default: 213.750 | Linked Fault: ERR-OKU350-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Okuma-Multus-U350.

PRM_43612 [Okuma-Multus-U350] — XAX_LOOP_GAIN_12 | Default: 165.000 | Linked Fault: ERR-OKU350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U350.

PRM_43613 [Okuma-Multus-U350] — OPT_LOOP_GAIN_13 | Default: 116.250 | Linked Fault: ERR-OKU350-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Okuma-Multus-U350.

PRM_43614 [Okuma-Multus-U350] — SPN_LOOP_GAIN_14 | Default: 67.500 | Linked Fault: ERR-OKU350-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Okuma-Multus-U350.

PRM_43615 [Okuma-Multus-U350] — YAX_LOOP_GAIN_15 | Default: 18.750 | Linked Fault: ERR-OKU350-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Okuma-Multus-U350.

PRM_43616 [Okuma-Multus-U350] — PLC_LOOP_GAIN_16 | Default: 220.000 | Linked Fault: ERR-OKU350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U350.

CNC/PLC SERVO PARAMETER DICTIONARY — Okuma-Multus-U400 (BANK 1)

Machine Model: Okuma-Multus-U400 | Controller: OSP-P500A Aerospace. Use these parameter registers when diagnosing servo following errors or thermal drift on Okuma-Multus-U400.

PRM_43621 [Okuma-Multus-U400] — ATC_LOOP_GAIN_1 | Default: 41.400 | Linked Fault: ERR-OKU400-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Okuma-Multus-U400.

PRM_43622 [Okuma-Multus-U400] — HYD_LOOP_GAIN_2 | Default: 82.800 | Linked Fault: ERR-OKU400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U400.

PRM_43623 [Okuma-Multus-U400] — CLN_LOOP_GAIN_3 | Default: 124.200 | Linked Fault: ERR-OKU400-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Okuma-Multus-U400.

PRM_43624 [Okuma-Multus-U400] — XAX_LOOP_GAIN_4 | Default: 165.600 | Linked Fault: ERR-OKU400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U400.

PRM_43625 [Okuma-Multus-U400] — YAX_LOOP_GAIN_5 | Default: 207.000 | Linked Fault: ERR-OKU400-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Okuma-Multus-U400.

PRM_43626 [Okuma-Multus-U400] — ZAX_LOOP_GAIN_6 | Default: 248.400 | Linked Fault: ERR-OKU400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U400.

PRM_43627 [Okuma-Multus-U400] — BAX_LOOP_GAIN_7 | Default: 39.800 | Linked Fault: ERR-OKU400-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Okuma-Multus-U400.

PRM_43628 [Okuma-Multus-U400] — CAX_LOOP_GAIN_8 | Default: 81.200 | Linked Fault: ERR-OKU400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U400.

PRM_43629 [Okuma-Multus-U400] — OPT_LOOP_GAIN_9 | Default: 122.600 | Linked Fault: ERR-OKU400-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Okuma-Multus-U400.

PRM_43630 [Okuma-Multus-U400] — PLC_LOOP_GAIN_10 | Default: 164.000 | Linked Fault: ERR-OKU400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U400.

PRM_43631 [Okuma-Multus-U400] — LUB_LOOP_GAIN_11 | Default: 205.400 | Linked Fault: ERR-OKU400-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Okuma-Multus-U400.

PRM_43632 [Okuma-Multus-U400] — VAC_LOOP_GAIN_12 | Default: 246.800 | Linked Fault: ERR-OKU400-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U400.

PRM_43633 [Okuma-Multus-U400] — THM_LOOP_GAIN_13 | Default: 38.200 | Linked Fault: ERR-OKU400-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Okuma-Multus-U400.

PRM_43634 [Okuma-Multus-U400] — SPN_LOOP_GAIN_14 | Default: 79.600 | Linked Fault: ERR-OKU400-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Okuma-Multus-U400.

PRM_43635 [Okuma-Multus-U400] — ATC_LOOP_GAIN_15 | Default: 121.000 | Linked Fault: ERR-OKU400-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Okuma-Multus-U400.

PRM_43636 [Okuma-Multus-U400] — HYD_LOOP_GAIN_16 | Default: 162.400 | Linked Fault: ERR-OKU400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U400.

CNC/PLC SERVO PARAMETER DICTIONARY — Okuma-Multus-U400 (BANK 2)

Machine Model: Okuma-Multus-U400 | Controller: OSP-P500A Aerospace. Use these parameter registers when diagnosing servo following errors or thermal drift on Okuma-Multus-U400.

PRM_43641 [Okuma-Multus-U400] — HYD_LOOP_GAIN_1 | Default: 82.800 | Linked Fault: ERR-OKU400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U400.

PRM_43642 [Okuma-Multus-U400] — XAX_LOOP_GAIN_2 | Default: 165.600 | Linked Fault: ERR-OKU400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U400.

PRM_43643 [Okuma-Multus-U400] — ZAX_LOOP_GAIN_3 | Default: 248.400 | Linked Fault: ERR-OKU400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U400.

PRM_43644 [Okuma-Multus-U400] — CAX_LOOP_GAIN_4 | Default: 81.200 | Linked Fault: ERR-OKU400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U400.

PRM_43645 [Okuma-Multus-U400] — PLC_LOOP_GAIN_5 | Default: 164.000 | Linked Fault: ERR-OKU400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U400.

PRM_43646 [Okuma-Multus-U400] — VAC_LOOP_GAIN_6 | Default: 246.800 | Linked Fault: ERR-OKU400-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U400.

PRM_43647 [Okuma-Multus-U400] — SPN_LOOP_GAIN_7 | Default: 79.600 | Linked Fault: ERR-OKU400-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Okuma-Multus-U400.

PRM_43648 [Okuma-Multus-U400] — HYD_LOOP_GAIN_8 | Default: 162.400 | Linked Fault: ERR-OKU400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U400.

PRM_43649 [Okuma-Multus-U400] — XAX_LOOP_GAIN_9 | Default: 245.200 | Linked Fault: ERR-OKU400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U400.

PRM_43650 [Okuma-Multus-U400] — ZAX_LOOP_GAIN_10 | Default: 78.000 | Linked Fault: ERR-OKU400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U400.

PRM_43651 [Okuma-Multus-U400] — CAX_LOOP_GAIN_11 | Default: 160.800 | Linked Fault: ERR-OKU400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U400.

PRM_43652 [Okuma-Multus-U400] — PLC_LOOP_GAIN_12 | Default: 243.600 | Linked Fault: ERR-OKU400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U400.

PRM_43653 [Okuma-Multus-U400] — VAC_LOOP_GAIN_13 | Default: 76.400 | Linked Fault: ERR-OKU400-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U400.

PRM_43654 [Okuma-Multus-U400] — SPN_LOOP_GAIN_14 | Default: 159.200 | Linked Fault: ERR-OKU400-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Okuma-Multus-U400.

PRM_43655 [Okuma-Multus-U400] — HYD_LOOP_GAIN_15 | Default: 242.000 | Linked Fault: ERR-OKU400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U400.

PRM_43656 [Okuma-Multus-U400] — XAX_LOOP_GAIN_16 | Default: 74.800 | Linked Fault: ERR-OKU400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U400.

CNC/PLC SERVO PARAMETER DICTIONARY — Okuma-Multus-U400 (BANK 3)

Machine Model: Okuma-Multus-U400 | Controller: OSP-P500A Aerospace. Use these parameter registers when diagnosing servo following errors or thermal drift on Okuma-Multus-U400.

PRM_43661 [Okuma-Multus-U400] — CLN_LOOP_GAIN_1 | Default: 124.200 | Linked Fault: ERR-OKU400-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Okuma-Multus-U400.

PRM_43662 [Okuma-Multus-U400] — ZAX_LOOP_GAIN_2 | Default: 248.400 | Linked Fault: ERR-OKU400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U400.

PRM_43663 [Okuma-Multus-U400] — OPT_LOOP_GAIN_3 | Default: 122.600 | Linked Fault: ERR-OKU400-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Okuma-Multus-U400.

PRM_43664 [Okuma-Multus-U400] — VAC_LOOP_GAIN_4 | Default: 246.800 | Linked Fault: ERR-OKU400-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U400.

PRM_43665 [Okuma-Multus-U400] — ATC_LOOP_GAIN_5 | Default: 121.000 | Linked Fault: ERR-OKU400-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Okuma-Multus-U400.

PRM_43666 [Okuma-Multus-U400] — XAX_LOOP_GAIN_6 | Default: 245.200 | Linked Fault: ERR-OKU400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U400.

PRM_43667 [Okuma-Multus-U400] — BAX_LOOP_GAIN_7 | Default: 119.400 | Linked Fault: ERR-OKU400-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Okuma-Multus-U400.

PRM_43668 [Okuma-Multus-U400] — PLC_LOOP_GAIN_8 | Default: 243.600 | Linked Fault: ERR-OKU400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U400.

PRM_43669 [Okuma-Multus-U400] — THM_LOOP_GAIN_9 | Default: 117.800 | Linked Fault: ERR-OKU400-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Okuma-Multus-U400.

PRM_43670 [Okuma-Multus-U400] — HYD_LOOP_GAIN_10 | Default: 242.000 | Linked Fault: ERR-OKU400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U400.

PRM_43671 [Okuma-Multus-U400] — YAX_LOOP_GAIN_11 | Default: 116.200 | Linked Fault: ERR-OKU400-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Okuma-Multus-U400.

PRM_43672 [Okuma-Multus-U400] — CAX_LOOP_GAIN_12 | Default: 240.400 | Linked Fault: ERR-OKU400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U400.

PRM_43673 [Okuma-Multus-U400] — LUB_LOOP_GAIN_13 | Default: 114.600 | Linked Fault: ERR-OKU400-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Okuma-Multus-U400.

PRM_43674 [Okuma-Multus-U400] — SPN_LOOP_GAIN_14 | Default: 238.800 | Linked Fault: ERR-OKU400-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Okuma-Multus-U400.

PRM_43675 [Okuma-Multus-U400] — CLN_LOOP_GAIN_15 | Default: 113.000 | Linked Fault: ERR-OKU400-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Okuma-Multus-U400.

PRM_43676 [Okuma-Multus-U400] — ZAX_LOOP_GAIN_16 | Default: 237.200 | Linked Fault: ERR-OKU400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U400.

CNC/PLC SERVO PARAMETER DICTIONARY — Okuma-Multus-U400 (BANK 4)

Machine Model: Okuma-Multus-U400 | Controller: OSP-P500A Aerospace. Use these parameter registers when diagnosing servo following errors or thermal drift on Okuma-Multus-U400.

PRM_43681 [Okuma-Multus-U400] — XAX_LOOP_GAIN_1 | Default: 165.600 | Linked Fault: ERR-OKU400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U400.

PRM_43682 [Okuma-Multus-U400] — CAX_LOOP_GAIN_2 | Default: 81.200 | Linked Fault: ERR-OKU400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U400.

PRM_43683 [Okuma-Multus-U400] — VAC_LOOP_GAIN_3 | Default: 246.800 | Linked Fault: ERR-OKU400-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U400.

PRM_43684 [Okuma-Multus-U400] — HYD_LOOP_GAIN_4 | Default: 162.400 | Linked Fault: ERR-OKU400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U400.

PRM_43685 [Okuma-Multus-U400] — ZAX_LOOP_GAIN_5 | Default: 78.000 | Linked Fault: ERR-OKU400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U400.

PRM_43686 [Okuma-Multus-U400] — PLC_LOOP_GAIN_6 | Default: 243.600 | Linked Fault: ERR-OKU400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U400.

PRM_43687 [Okuma-Multus-U400] — SPN_LOOP_GAIN_7 | Default: 159.200 | Linked Fault: ERR-OKU400-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Okuma-Multus-U400.

PRM_43688 [Okuma-Multus-U400] — XAX_LOOP_GAIN_8 | Default: 74.800 | Linked Fault: ERR-OKU400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U400.

PRM_43689 [Okuma-Multus-U400] — CAX_LOOP_GAIN_9 | Default: 240.400 | Linked Fault: ERR-OKU400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U400.

PRM_43690 [Okuma-Multus-U400] — VAC_LOOP_GAIN_10 | Default: 156.000 | Linked Fault: ERR-OKU400-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U400.

PRM_43691 [Okuma-Multus-U400] — HYD_LOOP_GAIN_11 | Default: 71.600 | Linked Fault: ERR-OKU400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U400.

PRM_43692 [Okuma-Multus-U400] — ZAX_LOOP_GAIN_12 | Default: 237.200 | Linked Fault: ERR-OKU400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U400.

PRM_43693 [Okuma-Multus-U400] — PLC_LOOP_GAIN_13 | Default: 152.800 | Linked Fault: ERR-OKU400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U400.

PRM_43694 [Okuma-Multus-U400] — SPN_LOOP_GAIN_14 | Default: 68.400 | Linked Fault: ERR-OKU400-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Okuma-Multus-U400.

PRM_43695 [Okuma-Multus-U400] — XAX_LOOP_GAIN_15 | Default: 234.000 | Linked Fault: ERR-OKU400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U400.

PRM_43696 [Okuma-Multus-U400] — CAX_LOOP_GAIN_16 | Default: 149.600 | Linked Fault: ERR-OKU400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U400.

CNC/PLC SERVO PARAMETER DICTIONARY — Okuma-Multus-U400 (BANK 5)

Machine Model: Okuma-Multus-U400 | Controller: OSP-P500A Aerospace. Use these parameter registers when diagnosing servo following errors or thermal drift on Okuma-Multus-U400.

PRM_43701 [Okuma-Multus-U400] — YAX_LOOP_GAIN_1 | Default: 207.000 | Linked Fault: ERR-OKU400-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Okuma-Multus-U400.

PRM_43702 [Okuma-Multus-U400] — PLC_LOOP_GAIN_2 | Default: 164.000 | Linked Fault: ERR-OKU400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U400.

PRM_43703 [Okuma-Multus-U400] — ATC_LOOP_GAIN_3 | Default: 121.000 | Linked Fault: ERR-OKU400-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Okuma-Multus-U400.

PRM_43704 [Okuma-Multus-U400] — ZAX_LOOP_GAIN_4 | Default: 78.000 | Linked Fault: ERR-OKU400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U400.

PRM_43705 [Okuma-Multus-U400] — LUB_LOOP_GAIN_5 | Default: 35.000 | Linked Fault: ERR-OKU400-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Okuma-Multus-U400.

PRM_43706 [Okuma-Multus-U400] — HYD_LOOP_GAIN_6 | Default: 242.000 | Linked Fault: ERR-OKU400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U400.

PRM_43707 [Okuma-Multus-U400] — BAX_LOOP_GAIN_7 | Default: 199.000 | Linked Fault: ERR-OKU400-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Okuma-Multus-U400.

PRM_43708 [Okuma-Multus-U400] — VAC_LOOP_GAIN_8 | Default: 156.000 | Linked Fault: ERR-OKU400-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U400.

PRM_43709 [Okuma-Multus-U400] — CLN_LOOP_GAIN_9 | Default: 113.000 | Linked Fault: ERR-OKU400-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Okuma-Multus-U400.

PRM_43710 [Okuma-Multus-U400] — CAX_LOOP_GAIN_10 | Default: 70.000 | Linked Fault: ERR-OKU400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U400.

PRM_43711 [Okuma-Multus-U400] — THM_LOOP_GAIN_11 | Default: 27.000 | Linked Fault: ERR-OKU400-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Okuma-Multus-U400.

PRM_43712 [Okuma-Multus-U400] — XAX_LOOP_GAIN_12 | Default: 234.000 | Linked Fault: ERR-OKU400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U400.

PRM_43713 [Okuma-Multus-U400] — OPT_LOOP_GAIN_13 | Default: 191.000 | Linked Fault: ERR-OKU400-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Okuma-Multus-U400.

PRM_43714 [Okuma-Multus-U400] — SPN_LOOP_GAIN_14 | Default: 148.000 | Linked Fault: ERR-OKU400-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Okuma-Multus-U400.

PRM_43715 [Okuma-Multus-U400] — YAX_LOOP_GAIN_15 | Default: 105.000 | Linked Fault: ERR-OKU400-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Okuma-Multus-U400.

PRM_43716 [Okuma-Multus-U400] — PLC_LOOP_GAIN_16 | Default: 62.000 | Linked Fault: ERR-OKU400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U400.

CNC/PLC SERVO PARAMETER DICTIONARY — Okuma-Multus-U450 (BANK 1)

Machine Model: Okuma-Multus-U450 | Controller: OSP-P500A Aerospace. Use these parameter registers when diagnosing servo following errors or thermal drift on Okuma-Multus-U450.

PRM_43721 [Okuma-Multus-U450] — ATC_LOOP_GAIN_1 | Default: 42.550 | Linked Fault: ERR-OKU450-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Okuma-Multus-U450.

PRM_43722 [Okuma-Multus-U450] — HYD_LOOP_GAIN_2 | Default: 85.100 | Linked Fault: ERR-OKU450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U450.

PRM_43723 [Okuma-Multus-U450] — CLN_LOOP_GAIN_3 | Default: 127.650 | Linked Fault: ERR-OKU450-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Okuma-Multus-U450.

PRM_43724 [Okuma-Multus-U450] — XAX_LOOP_GAIN_4 | Default: 170.200 | Linked Fault: ERR-OKU450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U450.

PRM_43725 [Okuma-Multus-U450] — YAX_LOOP_GAIN_5 | Default: 212.750 | Linked Fault: ERR-OKU450-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Okuma-Multus-U450.

PRM_43726 [Okuma-Multus-U450] — ZAX_LOOP_GAIN_6 | Default: 5.300 | Linked Fault: ERR-OKU450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U450.

PRM_43727 [Okuma-Multus-U450] — BAX_LOOP_GAIN_7 | Default: 47.850 | Linked Fault: ERR-OKU450-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Okuma-Multus-U450.

PRM_43728 [Okuma-Multus-U450] — CAX_LOOP_GAIN_8 | Default: 90.400 | Linked Fault: ERR-OKU450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U450.

PRM_43729 [Okuma-Multus-U450] — OPT_LOOP_GAIN_9 | Default: 132.950 | Linked Fault: ERR-OKU450-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Okuma-Multus-U450.

PRM_43730 [Okuma-Multus-U450] — PLC_LOOP_GAIN_10 | Default: 175.500 | Linked Fault: ERR-OKU450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U450.

PRM_43731 [Okuma-Multus-U450] — LUB_LOOP_GAIN_11 | Default: 218.050 | Linked Fault: ERR-OKU450-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Okuma-Multus-U450.

PRM_43732 [Okuma-Multus-U450] — VAC_LOOP_GAIN_12 | Default: 10.600 | Linked Fault: ERR-OKU450-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U450.

PRM_43733 [Okuma-Multus-U450] — THM_LOOP_GAIN_13 | Default: 53.150 | Linked Fault: ERR-OKU450-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Okuma-Multus-U450.

PRM_43734 [Okuma-Multus-U450] — SPN_LOOP_GAIN_14 | Default: 95.700 | Linked Fault: ERR-OKU450-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Okuma-Multus-U450.

PRM_43735 [Okuma-Multus-U450] — ATC_LOOP_GAIN_15 | Default: 138.250 | Linked Fault: ERR-OKU450-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Okuma-Multus-U450.

PRM_43736 [Okuma-Multus-U450] — HYD_LOOP_GAIN_16 | Default: 180.800 | Linked Fault: ERR-OKU450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U450.

CNC/PLC SERVO PARAMETER DICTIONARY — Okuma-Multus-U450 (BANK 2)

Machine Model: Okuma-Multus-U450 | Controller: OSP-P500A Aerospace. Use these parameter registers when diagnosing servo following errors or thermal drift on Okuma-Multus-U450.

PRM_43741 [Okuma-Multus-U450] — HYD_LOOP_GAIN_1 | Default: 85.100 | Linked Fault: ERR-OKU450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U450.

PRM_43742 [Okuma-Multus-U450] — XAX_LOOP_GAIN_2 | Default: 170.200 | Linked Fault: ERR-OKU450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U450.

PRM_43743 [Okuma-Multus-U450] — ZAX_LOOP_GAIN_3 | Default: 5.300 | Linked Fault: ERR-OKU450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U450.

PRM_43744 [Okuma-Multus-U450] — CAX_LOOP_GAIN_4 | Default: 90.400 | Linked Fault: ERR-OKU450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U450.

PRM_43745 [Okuma-Multus-U450] — PLC_LOOP_GAIN_5 | Default: 175.500 | Linked Fault: ERR-OKU450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U450.

PRM_43746 [Okuma-Multus-U450] — VAC_LOOP_GAIN_6 | Default: 10.600 | Linked Fault: ERR-OKU450-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U450.

PRM_43747 [Okuma-Multus-U450] — SPN_LOOP_GAIN_7 | Default: 95.700 | Linked Fault: ERR-OKU450-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Okuma-Multus-U450.

PRM_43748 [Okuma-Multus-U450] — HYD_LOOP_GAIN_8 | Default: 180.800 | Linked Fault: ERR-OKU450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U450.

PRM_43749 [Okuma-Multus-U450] — XAX_LOOP_GAIN_9 | Default: 15.900 | Linked Fault: ERR-OKU450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U450.

PRM_43750 [Okuma-Multus-U450] — ZAX_LOOP_GAIN_10 | Default: 101.000 | Linked Fault: ERR-OKU450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U450.

PRM_43751 [Okuma-Multus-U450] — CAX_LOOP_GAIN_11 | Default: 186.100 | Linked Fault: ERR-OKU450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U450.

PRM_43752 [Okuma-Multus-U450] — PLC_LOOP_GAIN_12 | Default: 21.200 | Linked Fault: ERR-OKU450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U450.

PRM_43753 [Okuma-Multus-U450] — VAC_LOOP_GAIN_13 | Default: 106.300 | Linked Fault: ERR-OKU450-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U450.

PRM_43754 [Okuma-Multus-U450] — SPN_LOOP_GAIN_14 | Default: 191.400 | Linked Fault: ERR-OKU450-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Okuma-Multus-U450.

PRM_43755 [Okuma-Multus-U450] — HYD_LOOP_GAIN_15 | Default: 26.500 | Linked Fault: ERR-OKU450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U450.

PRM_43756 [Okuma-Multus-U450] — XAX_LOOP_GAIN_16 | Default: 111.600 | Linked Fault: ERR-OKU450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U450.

CNC/PLC SERVO PARAMETER DICTIONARY — Okuma-Multus-U450 (BANK 3)

Machine Model: Okuma-Multus-U450 | Controller: OSP-P500A Aerospace. Use these parameter registers when diagnosing servo following errors or thermal drift on Okuma-Multus-U450.

PRM_43761 [Okuma-Multus-U450] — CLN_LOOP_GAIN_1 | Default: 127.650 | Linked Fault: ERR-OKU450-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Okuma-Multus-U450.

PRM_43762 [Okuma-Multus-U450] — ZAX_LOOP_GAIN_2 | Default: 5.300 | Linked Fault: ERR-OKU450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U450.

PRM_43763 [Okuma-Multus-U450] — OPT_LOOP_GAIN_3 | Default: 132.950 | Linked Fault: ERR-OKU450-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Okuma-Multus-U450.

PRM_43764 [Okuma-Multus-U450] — VAC_LOOP_GAIN_4 | Default: 10.600 | Linked Fault: ERR-OKU450-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U450.

PRM_43765 [Okuma-Multus-U450] — ATC_LOOP_GAIN_5 | Default: 138.250 | Linked Fault: ERR-OKU450-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Okuma-Multus-U450.

PRM_43766 [Okuma-Multus-U450] — XAX_LOOP_GAIN_6 | Default: 15.900 | Linked Fault: ERR-OKU450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U450.

PRM_43767 [Okuma-Multus-U450] — BAX_LOOP_GAIN_7 | Default: 143.550 | Linked Fault: ERR-OKU450-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Okuma-Multus-U450.

PRM_43768 [Okuma-Multus-U450] — PLC_LOOP_GAIN_8 | Default: 21.200 | Linked Fault: ERR-OKU450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U450.

PRM_43769 [Okuma-Multus-U450] — THM_LOOP_GAIN_9 | Default: 148.850 | Linked Fault: ERR-OKU450-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Okuma-Multus-U450.

PRM_43770 [Okuma-Multus-U450] — HYD_LOOP_GAIN_10 | Default: 26.500 | Linked Fault: ERR-OKU450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U450.

PRM_43771 [Okuma-Multus-U450] — YAX_LOOP_GAIN_11 | Default: 154.150 | Linked Fault: ERR-OKU450-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Okuma-Multus-U450.

PRM_43772 [Okuma-Multus-U450] — CAX_LOOP_GAIN_12 | Default: 31.800 | Linked Fault: ERR-OKU450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U450.

PRM_43773 [Okuma-Multus-U450] — LUB_LOOP_GAIN_13 | Default: 159.450 | Linked Fault: ERR-OKU450-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Okuma-Multus-U450.

PRM_43774 [Okuma-Multus-U450] — SPN_LOOP_GAIN_14 | Default: 37.100 | Linked Fault: ERR-OKU450-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Okuma-Multus-U450.

PRM_43775 [Okuma-Multus-U450] — CLN_LOOP_GAIN_15 | Default: 164.750 | Linked Fault: ERR-OKU450-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Okuma-Multus-U450.

PRM_43776 [Okuma-Multus-U450] — ZAX_LOOP_GAIN_16 | Default: 42.400 | Linked Fault: ERR-OKU450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U450.

CNC/PLC SERVO PARAMETER DICTIONARY — Okuma-Multus-U450 (BANK 4)

Machine Model: Okuma-Multus-U450 | Controller: OSP-P500A Aerospace. Use these parameter registers when diagnosing servo following errors or thermal drift on Okuma-Multus-U450.

PRM_43781 [Okuma-Multus-U450] — XAX_LOOP_GAIN_1 | Default: 170.200 | Linked Fault: ERR-OKU450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U450.

PRM_43782 [Okuma-Multus-U450] — CAX_LOOP_GAIN_2 | Default: 90.400 | Linked Fault: ERR-OKU450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U450.

PRM_43783 [Okuma-Multus-U450] — VAC_LOOP_GAIN_3 | Default: 10.600 | Linked Fault: ERR-OKU450-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U450.

PRM_43784 [Okuma-Multus-U450] — HYD_LOOP_GAIN_4 | Default: 180.800 | Linked Fault: ERR-OKU450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U450.

PRM_43785 [Okuma-Multus-U450] — ZAX_LOOP_GAIN_5 | Default: 101.000 | Linked Fault: ERR-OKU450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U450.

PRM_43786 [Okuma-Multus-U450] — PLC_LOOP_GAIN_6 | Default: 21.200 | Linked Fault: ERR-OKU450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U450.

PRM_43787 [Okuma-Multus-U450] — SPN_LOOP_GAIN_7 | Default: 191.400 | Linked Fault: ERR-OKU450-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Okuma-Multus-U450.

PRM_43788 [Okuma-Multus-U450] — XAX_LOOP_GAIN_8 | Default: 111.600 | Linked Fault: ERR-OKU450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U450.

PRM_43789 [Okuma-Multus-U450] — CAX_LOOP_GAIN_9 | Default: 31.800 | Linked Fault: ERR-OKU450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U450.

PRM_43790 [Okuma-Multus-U450] — VAC_LOOP_GAIN_10 | Default: 202.000 | Linked Fault: ERR-OKU450-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U450.

PRM_43791 [Okuma-Multus-U450] — HYD_LOOP_GAIN_11 | Default: 122.200 | Linked Fault: ERR-OKU450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U450.

PRM_43792 [Okuma-Multus-U450] — ZAX_LOOP_GAIN_12 | Default: 42.400 | Linked Fault: ERR-OKU450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U450.

PRM_43793 [Okuma-Multus-U450] — PLC_LOOP_GAIN_13 | Default: 212.600 | Linked Fault: ERR-OKU450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U450.

PRM_43794 [Okuma-Multus-U450] — SPN_LOOP_GAIN_14 | Default: 132.800 | Linked Fault: ERR-OKU450-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Okuma-Multus-U450.

PRM_43795 [Okuma-Multus-U450] — XAX_LOOP_GAIN_15 | Default: 53.000 | Linked Fault: ERR-OKU450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U450.

PRM_43796 [Okuma-Multus-U450] — CAX_LOOP_GAIN_16 | Default: 223.200 | Linked Fault: ERR-OKU450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U450.

CNC/PLC SERVO PARAMETER DICTIONARY — Okuma-Multus-U450 (BANK 5)

Machine Model: Okuma-Multus-U450 | Controller: OSP-P500A Aerospace. Use these parameter registers when diagnosing servo following errors or thermal drift on Okuma-Multus-U450.

PRM_43801 [Okuma-Multus-U450] — YAX_LOOP_GAIN_1 | Default: 212.750 | Linked Fault: ERR-OKU450-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Okuma-Multus-U450.

PRM_43802 [Okuma-Multus-U450] — PLC_LOOP_GAIN_2 | Default: 175.500 | Linked Fault: ERR-OKU450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U450.

PRM_43803 [Okuma-Multus-U450] — ATC_LOOP_GAIN_3 | Default: 138.250 | Linked Fault: ERR-OKU450-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Okuma-Multus-U450.

PRM_43804 [Okuma-Multus-U450] — ZAX_LOOP_GAIN_4 | Default: 101.000 | Linked Fault: ERR-OKU450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U450.

PRM_43805 [Okuma-Multus-U450] — LUB_LOOP_GAIN_5 | Default: 63.750 | Linked Fault: ERR-OKU450-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Okuma-Multus-U450.

PRM_43806 [Okuma-Multus-U450] — HYD_LOOP_GAIN_6 | Default: 26.500 | Linked Fault: ERR-OKU450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U450.

PRM_43807 [Okuma-Multus-U450] — BAX_LOOP_GAIN_7 | Default: 239.250 | Linked Fault: ERR-OKU450-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Okuma-Multus-U450.

PRM_43808 [Okuma-Multus-U450] — VAC_LOOP_GAIN_8 | Default: 202.000 | Linked Fault: ERR-OKU450-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U450.

PRM_43809 [Okuma-Multus-U450] — CLN_LOOP_GAIN_9 | Default: 164.750 | Linked Fault: ERR-OKU450-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Okuma-Multus-U450.

PRM_43810 [Okuma-Multus-U450] — CAX_LOOP_GAIN_10 | Default: 127.500 | Linked Fault: ERR-OKU450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U450.

PRM_43811 [Okuma-Multus-U450] — THM_LOOP_GAIN_11 | Default: 90.250 | Linked Fault: ERR-OKU450-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Okuma-Multus-U450.

PRM_43812 [Okuma-Multus-U450] — XAX_LOOP_GAIN_12 | Default: 53.000 | Linked Fault: ERR-OKU450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U450.

PRM_43813 [Okuma-Multus-U450] — OPT_LOOP_GAIN_13 | Default: 15.750 | Linked Fault: ERR-OKU450-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Okuma-Multus-U450.

PRM_43814 [Okuma-Multus-U450] — SPN_LOOP_GAIN_14 | Default: 228.500 | Linked Fault: ERR-OKU450-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Okuma-Multus-U450.

PRM_43815 [Okuma-Multus-U450] — YAX_LOOP_GAIN_15 | Default: 191.250 | Linked Fault: ERR-OKU450-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Okuma-Multus-U450.

PRM_43816 [Okuma-Multus-U450] — PLC_LOOP_GAIN_16 | Default: 154.000 | Linked Fault: ERR-OKU450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U450.

CNC/PLC SERVO PARAMETER DICTIONARY — Okuma-Multus-U500 (BANK 1)

Machine Model: Okuma-Multus-U500 | Controller: OSP-P500A Aerospace. Use these parameter registers when diagnosing servo following errors or thermal drift on Okuma-Multus-U500.

PRM_43821 [Okuma-Multus-U500] — ATC_LOOP_GAIN_1 | Default: 43.700 | Linked Fault: ERR-OKU500-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Okuma-Multus-U500.

PRM_43822 [Okuma-Multus-U500] — HYD_LOOP_GAIN_2 | Default: 87.400 | Linked Fault: ERR-OKU500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U500.

PRM_43823 [Okuma-Multus-U500] — CLN_LOOP_GAIN_3 | Default: 131.100 | Linked Fault: ERR-OKU500-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Okuma-Multus-U500.

PRM_43824 [Okuma-Multus-U500] — XAX_LOOP_GAIN_4 | Default: 174.800 | Linked Fault: ERR-OKU500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U500.

PRM_43825 [Okuma-Multus-U500] — YAX_LOOP_GAIN_5 | Default: 218.500 | Linked Fault: ERR-OKU500-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Okuma-Multus-U500.

PRM_43826 [Okuma-Multus-U500] — ZAX_LOOP_GAIN_6 | Default: 12.200 | Linked Fault: ERR-OKU500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U500.

PRM_43827 [Okuma-Multus-U500] — BAX_LOOP_GAIN_7 | Default: 55.900 | Linked Fault: ERR-OKU500-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Okuma-Multus-U500.

PRM_43828 [Okuma-Multus-U500] — CAX_LOOP_GAIN_8 | Default: 99.600 | Linked Fault: ERR-OKU500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U500.

PRM_43829 [Okuma-Multus-U500] — OPT_LOOP_GAIN_9 | Default: 143.300 | Linked Fault: ERR-OKU500-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Okuma-Multus-U500.

PRM_43830 [Okuma-Multus-U500] — PLC_LOOP_GAIN_10 | Default: 187.000 | Linked Fault: ERR-OKU500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U500.

PRM_43831 [Okuma-Multus-U500] — LUB_LOOP_GAIN_11 | Default: 230.700 | Linked Fault: ERR-OKU500-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Okuma-Multus-U500.

PRM_43832 [Okuma-Multus-U500] — VAC_LOOP_GAIN_12 | Default: 24.400 | Linked Fault: ERR-OKU500-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U500.

PRM_43833 [Okuma-Multus-U500] — THM_LOOP_GAIN_13 | Default: 68.100 | Linked Fault: ERR-OKU500-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Okuma-Multus-U500.

PRM_43834 [Okuma-Multus-U500] — SPN_LOOP_GAIN_14 | Default: 111.800 | Linked Fault: ERR-OKU500-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Okuma-Multus-U500.

PRM_43835 [Okuma-Multus-U500] — ATC_LOOP_GAIN_15 | Default: 155.500 | Linked Fault: ERR-OKU500-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Okuma-Multus-U500.

PRM_43836 [Okuma-Multus-U500] — HYD_LOOP_GAIN_16 | Default: 199.200 | Linked Fault: ERR-OKU500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U500.

CNC/PLC SERVO PARAMETER DICTIONARY — Okuma-Multus-U500 (BANK 2)

Machine Model: Okuma-Multus-U500 | Controller: OSP-P500A Aerospace. Use these parameter registers when diagnosing servo following errors or thermal drift on Okuma-Multus-U500.

PRM_43841 [Okuma-Multus-U500] — HYD_LOOP_GAIN_1 | Default: 87.400 | Linked Fault: ERR-OKU500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U500.

PRM_43842 [Okuma-Multus-U500] — XAX_LOOP_GAIN_2 | Default: 174.800 | Linked Fault: ERR-OKU500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U500.

PRM_43843 [Okuma-Multus-U500] — ZAX_LOOP_GAIN_3 | Default: 12.200 | Linked Fault: ERR-OKU500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U500.

PRM_43844 [Okuma-Multus-U500] — CAX_LOOP_GAIN_4 | Default: 99.600 | Linked Fault: ERR-OKU500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U500.

PRM_43845 [Okuma-Multus-U500] — PLC_LOOP_GAIN_5 | Default: 187.000 | Linked Fault: ERR-OKU500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U500.

PRM_43846 [Okuma-Multus-U500] — VAC_LOOP_GAIN_6 | Default: 24.400 | Linked Fault: ERR-OKU500-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U500.

PRM_43847 [Okuma-Multus-U500] — SPN_LOOP_GAIN_7 | Default: 111.800 | Linked Fault: ERR-OKU500-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Okuma-Multus-U500.

PRM_43848 [Okuma-Multus-U500] — HYD_LOOP_GAIN_8 | Default: 199.200 | Linked Fault: ERR-OKU500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U500.

PRM_43849 [Okuma-Multus-U500] — XAX_LOOP_GAIN_9 | Default: 36.600 | Linked Fault: ERR-OKU500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U500.

PRM_43850 [Okuma-Multus-U500] — ZAX_LOOP_GAIN_10 | Default: 124.000 | Linked Fault: ERR-OKU500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U500.

PRM_43851 [Okuma-Multus-U500] — CAX_LOOP_GAIN_11 | Default: 211.400 | Linked Fault: ERR-OKU500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U500.

PRM_43852 [Okuma-Multus-U500] — PLC_LOOP_GAIN_12 | Default: 48.800 | Linked Fault: ERR-OKU500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U500.

PRM_43853 [Okuma-Multus-U500] — VAC_LOOP_GAIN_13 | Default: 136.200 | Linked Fault: ERR-OKU500-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U500.

PRM_43854 [Okuma-Multus-U500] — SPN_LOOP_GAIN_14 | Default: 223.600 | Linked Fault: ERR-OKU500-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Okuma-Multus-U500.

PRM_43855 [Okuma-Multus-U500] — HYD_LOOP_GAIN_15 | Default: 61.000 | Linked Fault: ERR-OKU500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U500.

PRM_43856 [Okuma-Multus-U500] — XAX_LOOP_GAIN_16 | Default: 148.400 | Linked Fault: ERR-OKU500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U500.

CNC/PLC SERVO PARAMETER DICTIONARY — Okuma-Multus-U500 (BANK 3)

Machine Model: Okuma-Multus-U500 | Controller: OSP-P500A Aerospace. Use these parameter registers when diagnosing servo following errors or thermal drift on Okuma-Multus-U500.

PRM_43861 [Okuma-Multus-U500] — CLN_LOOP_GAIN_1 | Default: 131.100 | Linked Fault: ERR-OKU500-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Okuma-Multus-U500.

PRM_43862 [Okuma-Multus-U500] — ZAX_LOOP_GAIN_2 | Default: 12.200 | Linked Fault: ERR-OKU500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U500.

PRM_43863 [Okuma-Multus-U500] — OPT_LOOP_GAIN_3 | Default: 143.300 | Linked Fault: ERR-OKU500-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Okuma-Multus-U500.

PRM_43864 [Okuma-Multus-U500] — VAC_LOOP_GAIN_4 | Default: 24.400 | Linked Fault: ERR-OKU500-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U500.

PRM_43865 [Okuma-Multus-U500] — ATC_LOOP_GAIN_5 | Default: 155.500 | Linked Fault: ERR-OKU500-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Okuma-Multus-U500.

PRM_43866 [Okuma-Multus-U500] — XAX_LOOP_GAIN_6 | Default: 36.600 | Linked Fault: ERR-OKU500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U500.

PRM_43867 [Okuma-Multus-U500] — BAX_LOOP_GAIN_7 | Default: 167.700 | Linked Fault: ERR-OKU500-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Okuma-Multus-U500.

PRM_43868 [Okuma-Multus-U500] — PLC_LOOP_GAIN_8 | Default: 48.800 | Linked Fault: ERR-OKU500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U500.

PRM_43869 [Okuma-Multus-U500] — THM_LOOP_GAIN_9 | Default: 179.900 | Linked Fault: ERR-OKU500-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Okuma-Multus-U500.

PRM_43870 [Okuma-Multus-U500] — HYD_LOOP_GAIN_10 | Default: 61.000 | Linked Fault: ERR-OKU500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U500.

PRM_43871 [Okuma-Multus-U500] — YAX_LOOP_GAIN_11 | Default: 192.100 | Linked Fault: ERR-OKU500-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Okuma-Multus-U500.

PRM_43872 [Okuma-Multus-U500] — CAX_LOOP_GAIN_12 | Default: 73.200 | Linked Fault: ERR-OKU500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U500.

PRM_43873 [Okuma-Multus-U500] — LUB_LOOP_GAIN_13 | Default: 204.300 | Linked Fault: ERR-OKU500-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Okuma-Multus-U500.

PRM_43874 [Okuma-Multus-U500] — SPN_LOOP_GAIN_14 | Default: 85.400 | Linked Fault: ERR-OKU500-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Okuma-Multus-U500.

PRM_43875 [Okuma-Multus-U500] — CLN_LOOP_GAIN_15 | Default: 216.500 | Linked Fault: ERR-OKU500-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Okuma-Multus-U500.

PRM_43876 [Okuma-Multus-U500] — ZAX_LOOP_GAIN_16 | Default: 97.600 | Linked Fault: ERR-OKU500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U500.

CNC/PLC SERVO PARAMETER DICTIONARY — Okuma-Multus-U500 (BANK 4)

Machine Model: Okuma-Multus-U500 | Controller: OSP-P500A Aerospace. Use these parameter registers when diagnosing servo following errors or thermal drift on Okuma-Multus-U500.

PRM_43881 [Okuma-Multus-U500] — XAX_LOOP_GAIN_1 | Default: 174.800 | Linked Fault: ERR-OKU500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U500.

PRM_43882 [Okuma-Multus-U500] — CAX_LOOP_GAIN_2 | Default: 99.600 | Linked Fault: ERR-OKU500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U500.

PRM_43883 [Okuma-Multus-U500] — VAC_LOOP_GAIN_3 | Default: 24.400 | Linked Fault: ERR-OKU500-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U500.

PRM_43884 [Okuma-Multus-U500] — HYD_LOOP_GAIN_4 | Default: 199.200 | Linked Fault: ERR-OKU500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U500.

PRM_43885 [Okuma-Multus-U500] — ZAX_LOOP_GAIN_5 | Default: 124.000 | Linked Fault: ERR-OKU500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U500.

PRM_43886 [Okuma-Multus-U500] — PLC_LOOP_GAIN_6 | Default: 48.800 | Linked Fault: ERR-OKU500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U500.

PRM_43887 [Okuma-Multus-U500] — SPN_LOOP_GAIN_7 | Default: 223.600 | Linked Fault: ERR-OKU500-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Okuma-Multus-U500.

PRM_43888 [Okuma-Multus-U500] — XAX_LOOP_GAIN_8 | Default: 148.400 | Linked Fault: ERR-OKU500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U500.

PRM_43889 [Okuma-Multus-U500] — CAX_LOOP_GAIN_9 | Default: 73.200 | Linked Fault: ERR-OKU500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U500.

PRM_43890 [Okuma-Multus-U500] — VAC_LOOP_GAIN_10 | Default: 248.000 | Linked Fault: ERR-OKU500-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U500.

PRM_43891 [Okuma-Multus-U500] — HYD_LOOP_GAIN_11 | Default: 172.800 | Linked Fault: ERR-OKU500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U500.

PRM_43892 [Okuma-Multus-U500] — ZAX_LOOP_GAIN_12 | Default: 97.600 | Linked Fault: ERR-OKU500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U500.

PRM_43893 [Okuma-Multus-U500] — PLC_LOOP_GAIN_13 | Default: 22.400 | Linked Fault: ERR-OKU500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U500.

PRM_43894 [Okuma-Multus-U500] — SPN_LOOP_GAIN_14 | Default: 197.200 | Linked Fault: ERR-OKU500-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Okuma-Multus-U500.

PRM_43895 [Okuma-Multus-U500] — XAX_LOOP_GAIN_15 | Default: 122.000 | Linked Fault: ERR-OKU500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U500.

PRM_43896 [Okuma-Multus-U500] — CAX_LOOP_GAIN_16 | Default: 46.800 | Linked Fault: ERR-OKU500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U500.

CNC/PLC SERVO PARAMETER DICTIONARY — Okuma-Multus-U500 (BANK 5)

Machine Model: Okuma-Multus-U500 | Controller: OSP-P500A Aerospace. Use these parameter registers when diagnosing servo following errors or thermal drift on Okuma-Multus-U500.

PRM_43901 [Okuma-Multus-U500] — YAX_LOOP_GAIN_1 | Default: 218.500 | Linked Fault: ERR-OKU500-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Okuma-Multus-U500.

PRM_43902 [Okuma-Multus-U500] — PLC_LOOP_GAIN_2 | Default: 187.000 | Linked Fault: ERR-OKU500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U500.

PRM_43903 [Okuma-Multus-U500] — ATC_LOOP_GAIN_3 | Default: 155.500 | Linked Fault: ERR-OKU500-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Okuma-Multus-U500.

PRM_43904 [Okuma-Multus-U500] — ZAX_LOOP_GAIN_4 | Default: 124.000 | Linked Fault: ERR-OKU500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U500.

PRM_43905 [Okuma-Multus-U500] — LUB_LOOP_GAIN_5 | Default: 92.500 | Linked Fault: ERR-OKU500-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Okuma-Multus-U500.

PRM_43906 [Okuma-Multus-U500] — HYD_LOOP_GAIN_6 | Default: 61.000 | Linked Fault: ERR-OKU500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U500.

PRM_43907 [Okuma-Multus-U500] — BAX_LOOP_GAIN_7 | Default: 29.500 | Linked Fault: ERR-OKU500-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Okuma-Multus-U500.

PRM_43908 [Okuma-Multus-U500] — VAC_LOOP_GAIN_8 | Default: 248.000 | Linked Fault: ERR-OKU500-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U500.

PRM_43909 [Okuma-Multus-U500] — CLN_LOOP_GAIN_9 | Default: 216.500 | Linked Fault: ERR-OKU500-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Okuma-Multus-U500.

PRM_43910 [Okuma-Multus-U500] — CAX_LOOP_GAIN_10 | Default: 185.000 | Linked Fault: ERR-OKU500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U500.

PRM_43911 [Okuma-Multus-U500] — THM_LOOP_GAIN_11 | Default: 153.500 | Linked Fault: ERR-OKU500-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Okuma-Multus-U500.

PRM_43912 [Okuma-Multus-U500] — XAX_LOOP_GAIN_12 | Default: 122.000 | Linked Fault: ERR-OKU500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U500.

PRM_43913 [Okuma-Multus-U500] — OPT_LOOP_GAIN_13 | Default: 90.500 | Linked Fault: ERR-OKU500-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Okuma-Multus-U500.

PRM_43914 [Okuma-Multus-U500] — SPN_LOOP_GAIN_14 | Default: 59.000 | Linked Fault: ERR-OKU500-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Okuma-Multus-U500.

PRM_43915 [Okuma-Multus-U500] — YAX_LOOP_GAIN_15 | Default: 27.500 | Linked Fault: ERR-OKU500-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Okuma-Multus-U500.

PRM_43916 [Okuma-Multus-U500] — PLC_LOOP_GAIN_16 | Default: 246.000 | Linked Fault: ERR-OKU500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U500.

CNC/PLC SERVO PARAMETER DICTIONARY — Okuma-Multus-U550 (BANK 1)

Machine Model: Okuma-Multus-U550 | Controller: OSP-P500A Aerospace. Use these parameter registers when diagnosing servo following errors or thermal drift on Okuma-Multus-U550.

PRM_43921 [Okuma-Multus-U550] — ATC_LOOP_GAIN_1 | Default: 44.850 | Linked Fault: ERR-OKU550-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Okuma-Multus-U550.

PRM_43922 [Okuma-Multus-U550] — HYD_LOOP_GAIN_2 | Default: 89.700 | Linked Fault: ERR-OKU550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U550.

PRM_43923 [Okuma-Multus-U550] — CLN_LOOP_GAIN_3 | Default: 134.550 | Linked Fault: ERR-OKU550-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Okuma-Multus-U550.

PRM_43924 [Okuma-Multus-U550] — XAX_LOOP_GAIN_4 | Default: 179.400 | Linked Fault: ERR-OKU550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U550.

PRM_43925 [Okuma-Multus-U550] — YAX_LOOP_GAIN_5 | Default: 224.250 | Linked Fault: ERR-OKU550-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Okuma-Multus-U550.

PRM_43926 [Okuma-Multus-U550] — ZAX_LOOP_GAIN_6 | Default: 19.100 | Linked Fault: ERR-OKU550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U550.

PRM_43927 [Okuma-Multus-U550] — BAX_LOOP_GAIN_7 | Default: 63.950 | Linked Fault: ERR-OKU550-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Okuma-Multus-U550.

PRM_43928 [Okuma-Multus-U550] — CAX_LOOP_GAIN_8 | Default: 108.800 | Linked Fault: ERR-OKU550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U550.

PRM_43929 [Okuma-Multus-U550] — OPT_LOOP_GAIN_9 | Default: 153.650 | Linked Fault: ERR-OKU550-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Okuma-Multus-U550.

PRM_43930 [Okuma-Multus-U550] — PLC_LOOP_GAIN_10 | Default: 198.500 | Linked Fault: ERR-OKU550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U550.

PRM_43931 [Okuma-Multus-U550] — LUB_LOOP_GAIN_11 | Default: 243.350 | Linked Fault: ERR-OKU550-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Okuma-Multus-U550.

PRM_43932 [Okuma-Multus-U550] — VAC_LOOP_GAIN_12 | Default: 38.200 | Linked Fault: ERR-OKU550-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U550.

PRM_43933 [Okuma-Multus-U550] — THM_LOOP_GAIN_13 | Default: 83.050 | Linked Fault: ERR-OKU550-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Okuma-Multus-U550.

PRM_43934 [Okuma-Multus-U550] — SPN_LOOP_GAIN_14 | Default: 127.900 | Linked Fault: ERR-OKU550-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Okuma-Multus-U550.

PRM_43935 [Okuma-Multus-U550] — ATC_LOOP_GAIN_15 | Default: 172.750 | Linked Fault: ERR-OKU550-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Okuma-Multus-U550.

PRM_43936 [Okuma-Multus-U550] — HYD_LOOP_GAIN_16 | Default: 217.600 | Linked Fault: ERR-OKU550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U550.

CNC/PLC SERVO PARAMETER DICTIONARY — Okuma-Multus-U550 (BANK 2)

Machine Model: Okuma-Multus-U550 | Controller: OSP-P500A Aerospace. Use these parameter registers when diagnosing servo following errors or thermal drift on Okuma-Multus-U550.

PRM_43941 [Okuma-Multus-U550] — HYD_LOOP_GAIN_1 | Default: 89.700 | Linked Fault: ERR-OKU550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U550.

PRM_43942 [Okuma-Multus-U550] — XAX_LOOP_GAIN_2 | Default: 179.400 | Linked Fault: ERR-OKU550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U550.

PRM_43943 [Okuma-Multus-U550] — ZAX_LOOP_GAIN_3 | Default: 19.100 | Linked Fault: ERR-OKU550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U550.

PRM_43944 [Okuma-Multus-U550] — CAX_LOOP_GAIN_4 | Default: 108.800 | Linked Fault: ERR-OKU550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U550.

PRM_43945 [Okuma-Multus-U550] — PLC_LOOP_GAIN_5 | Default: 198.500 | Linked Fault: ERR-OKU550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U550.

PRM_43946 [Okuma-Multus-U550] — VAC_LOOP_GAIN_6 | Default: 38.200 | Linked Fault: ERR-OKU550-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U550.

PRM_43947 [Okuma-Multus-U550] — SPN_LOOP_GAIN_7 | Default: 127.900 | Linked Fault: ERR-OKU550-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Okuma-Multus-U550.

PRM_43948 [Okuma-Multus-U550] — HYD_LOOP_GAIN_8 | Default: 217.600 | Linked Fault: ERR-OKU550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U550.

PRM_43949 [Okuma-Multus-U550] — XAX_LOOP_GAIN_9 | Default: 57.300 | Linked Fault: ERR-OKU550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U550.

PRM_43950 [Okuma-Multus-U550] — ZAX_LOOP_GAIN_10 | Default: 147.000 | Linked Fault: ERR-OKU550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U550.

PRM_43951 [Okuma-Multus-U550] — CAX_LOOP_GAIN_11 | Default: 236.700 | Linked Fault: ERR-OKU550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U550.

PRM_43952 [Okuma-Multus-U550] — PLC_LOOP_GAIN_12 | Default: 76.400 | Linked Fault: ERR-OKU550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U550.

PRM_43953 [Okuma-Multus-U550] — VAC_LOOP_GAIN_13 | Default: 166.100 | Linked Fault: ERR-OKU550-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U550.

PRM_43954 [Okuma-Multus-U550] — SPN_LOOP_GAIN_14 | Default: 5.800 | Linked Fault: ERR-OKU550-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Okuma-Multus-U550.

PRM_43955 [Okuma-Multus-U550] — HYD_LOOP_GAIN_15 | Default: 95.500 | Linked Fault: ERR-OKU550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U550.

PRM_43956 [Okuma-Multus-U550] — XAX_LOOP_GAIN_16 | Default: 185.200 | Linked Fault: ERR-OKU550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U550.

CNC/PLC SERVO PARAMETER DICTIONARY — Okuma-Multus-U550 (BANK 3)

Machine Model: Okuma-Multus-U550 | Controller: OSP-P500A Aerospace. Use these parameter registers when diagnosing servo following errors or thermal drift on Okuma-Multus-U550.

PRM_43961 [Okuma-Multus-U550] — CLN_LOOP_GAIN_1 | Default: 134.550 | Linked Fault: ERR-OKU550-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Okuma-Multus-U550.

PRM_43962 [Okuma-Multus-U550] — ZAX_LOOP_GAIN_2 | Default: 19.100 | Linked Fault: ERR-OKU550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U550.

PRM_43963 [Okuma-Multus-U550] — OPT_LOOP_GAIN_3 | Default: 153.650 | Linked Fault: ERR-OKU550-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Okuma-Multus-U550.

PRM_43964 [Okuma-Multus-U550] — VAC_LOOP_GAIN_4 | Default: 38.200 | Linked Fault: ERR-OKU550-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U550.

PRM_43965 [Okuma-Multus-U550] — ATC_LOOP_GAIN_5 | Default: 172.750 | Linked Fault: ERR-OKU550-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Okuma-Multus-U550.

PRM_43966 [Okuma-Multus-U550] — XAX_LOOP_GAIN_6 | Default: 57.300 | Linked Fault: ERR-OKU550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U550.

PRM_43967 [Okuma-Multus-U550] — BAX_LOOP_GAIN_7 | Default: 191.850 | Linked Fault: ERR-OKU550-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Okuma-Multus-U550.

PRM_43968 [Okuma-Multus-U550] — PLC_LOOP_GAIN_8 | Default: 76.400 | Linked Fault: ERR-OKU550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U550.

PRM_43969 [Okuma-Multus-U550] — THM_LOOP_GAIN_9 | Default: 210.950 | Linked Fault: ERR-OKU550-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Okuma-Multus-U550.

PRM_43970 [Okuma-Multus-U550] — HYD_LOOP_GAIN_10 | Default: 95.500 | Linked Fault: ERR-OKU550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U550.

PRM_43971 [Okuma-Multus-U550] — YAX_LOOP_GAIN_11 | Default: 230.050 | Linked Fault: ERR-OKU550-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Okuma-Multus-U550.

PRM_43972 [Okuma-Multus-U550] — CAX_LOOP_GAIN_12 | Default: 114.600 | Linked Fault: ERR-OKU550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U550.

PRM_43973 [Okuma-Multus-U550] — LUB_LOOP_GAIN_13 | Default: 249.150 | Linked Fault: ERR-OKU550-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Okuma-Multus-U550.

PRM_43974 [Okuma-Multus-U550] — SPN_LOOP_GAIN_14 | Default: 133.700 | Linked Fault: ERR-OKU550-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Okuma-Multus-U550.

PRM_43975 [Okuma-Multus-U550] — CLN_LOOP_GAIN_15 | Default: 18.250 | Linked Fault: ERR-OKU550-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Okuma-Multus-U550.

PRM_43976 [Okuma-Multus-U550] — ZAX_LOOP_GAIN_16 | Default: 152.800 | Linked Fault: ERR-OKU550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U550.

CNC/PLC SERVO PARAMETER DICTIONARY — Okuma-Multus-U550 (BANK 4)

Machine Model: Okuma-Multus-U550 | Controller: OSP-P500A Aerospace. Use these parameter registers when diagnosing servo following errors or thermal drift on Okuma-Multus-U550.

PRM_43981 [Okuma-Multus-U550] — XAX_LOOP_GAIN_1 | Default: 179.400 | Linked Fault: ERR-OKU550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U550.

PRM_43982 [Okuma-Multus-U550] — CAX_LOOP_GAIN_2 | Default: 108.800 | Linked Fault: ERR-OKU550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U550.

PRM_43983 [Okuma-Multus-U550] — VAC_LOOP_GAIN_3 | Default: 38.200 | Linked Fault: ERR-OKU550-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U550.

PRM_43984 [Okuma-Multus-U550] — HYD_LOOP_GAIN_4 | Default: 217.600 | Linked Fault: ERR-OKU550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U550.

PRM_43985 [Okuma-Multus-U550] — ZAX_LOOP_GAIN_5 | Default: 147.000 | Linked Fault: ERR-OKU550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U550.

PRM_43986 [Okuma-Multus-U550] — PLC_LOOP_GAIN_6 | Default: 76.400 | Linked Fault: ERR-OKU550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U550.

PRM_43987 [Okuma-Multus-U550] — SPN_LOOP_GAIN_7 | Default: 5.800 | Linked Fault: ERR-OKU550-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Okuma-Multus-U550.

PRM_43988 [Okuma-Multus-U550] — XAX_LOOP_GAIN_8 | Default: 185.200 | Linked Fault: ERR-OKU550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U550.

PRM_43989 [Okuma-Multus-U550] — CAX_LOOP_GAIN_9 | Default: 114.600 | Linked Fault: ERR-OKU550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U550.

PRM_43990 [Okuma-Multus-U550] — VAC_LOOP_GAIN_10 | Default: 44.000 | Linked Fault: ERR-OKU550-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U550.

PRM_43991 [Okuma-Multus-U550] — HYD_LOOP_GAIN_11 | Default: 223.400 | Linked Fault: ERR-OKU550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U550.

PRM_43992 [Okuma-Multus-U550] — ZAX_LOOP_GAIN_12 | Default: 152.800 | Linked Fault: ERR-OKU550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U550.

PRM_43993 [Okuma-Multus-U550] — PLC_LOOP_GAIN_13 | Default: 82.200 | Linked Fault: ERR-OKU550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U550.

PRM_43994 [Okuma-Multus-U550] — SPN_LOOP_GAIN_14 | Default: 11.600 | Linked Fault: ERR-OKU550-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Okuma-Multus-U550.

PRM_43995 [Okuma-Multus-U550] — XAX_LOOP_GAIN_15 | Default: 191.000 | Linked Fault: ERR-OKU550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U550.

PRM_43996 [Okuma-Multus-U550] — CAX_LOOP_GAIN_16 | Default: 120.400 | Linked Fault: ERR-OKU550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U550.

CNC/PLC SERVO PARAMETER DICTIONARY — Okuma-Multus-U550 (BANK 5)

Machine Model: Okuma-Multus-U550 | Controller: OSP-P500A Aerospace. Use these parameter registers when diagnosing servo following errors or thermal drift on Okuma-Multus-U550.

PRM_44001 [Okuma-Multus-U550] — YAX_LOOP_GAIN_1 | Default: 224.250 | Linked Fault: ERR-OKU550-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Okuma-Multus-U550.

PRM_44002 [Okuma-Multus-U550] — PLC_LOOP_GAIN_2 | Default: 198.500 | Linked Fault: ERR-OKU550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U550.

PRM_44003 [Okuma-Multus-U550] — ATC_LOOP_GAIN_3 | Default: 172.750 | Linked Fault: ERR-OKU550-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Okuma-Multus-U550.

PRM_44004 [Okuma-Multus-U550] — ZAX_LOOP_GAIN_4 | Default: 147.000 | Linked Fault: ERR-OKU550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U550.

PRM_44005 [Okuma-Multus-U550] — LUB_LOOP_GAIN_5 | Default: 121.250 | Linked Fault: ERR-OKU550-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Okuma-Multus-U550.

PRM_44006 [Okuma-Multus-U550] — HYD_LOOP_GAIN_6 | Default: 95.500 | Linked Fault: ERR-OKU550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U550.

PRM_44007 [Okuma-Multus-U550] — BAX_LOOP_GAIN_7 | Default: 69.750 | Linked Fault: ERR-OKU550-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Okuma-Multus-U550.

PRM_44008 [Okuma-Multus-U550] — VAC_LOOP_GAIN_8 | Default: 44.000 | Linked Fault: ERR-OKU550-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U550.

PRM_44009 [Okuma-Multus-U550] — CLN_LOOP_GAIN_9 | Default: 18.250 | Linked Fault: ERR-OKU550-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Okuma-Multus-U550.

PRM_44010 [Okuma-Multus-U550] — CAX_LOOP_GAIN_10 | Default: 242.500 | Linked Fault: ERR-OKU550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U550.

PRM_44011 [Okuma-Multus-U550] — THM_LOOP_GAIN_11 | Default: 216.750 | Linked Fault: ERR-OKU550-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Okuma-Multus-U550.

PRM_44012 [Okuma-Multus-U550] — XAX_LOOP_GAIN_12 | Default: 191.000 | Linked Fault: ERR-OKU550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U550.

PRM_44013 [Okuma-Multus-U550] — OPT_LOOP_GAIN_13 | Default: 165.250 | Linked Fault: ERR-OKU550-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Okuma-Multus-U550.

PRM_44014 [Okuma-Multus-U550] — SPN_LOOP_GAIN_14 | Default: 139.500 | Linked Fault: ERR-OKU550-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Okuma-Multus-U550.

PRM_44015 [Okuma-Multus-U550] — YAX_LOOP_GAIN_15 | Default: 113.750 | Linked Fault: ERR-OKU550-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Okuma-Multus-U550.

PRM_44016 [Okuma-Multus-U550] — PLC_LOOP_GAIN_16 | Default: 88.000 | Linked Fault: ERR-OKU550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U550.

CNC/PLC SERVO PARAMETER DICTIONARY — Okuma-Multus-U600 (BANK 1)

Machine Model: Okuma-Multus-U600 | Controller: OSP-P500A Aerospace. Use these parameter registers when diagnosing servo following errors or thermal drift on Okuma-Multus-U600.

PRM_44021 [Okuma-Multus-U600] — ATC_LOOP_GAIN_1 | Default: 46.000 | Linked Fault: ERR-OKU600-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Okuma-Multus-U600.

PRM_44022 [Okuma-Multus-U600] — HYD_LOOP_GAIN_2 | Default: 92.000 | Linked Fault: ERR-OKU600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U600.

PRM_44023 [Okuma-Multus-U600] — CLN_LOOP_GAIN_3 | Default: 138.000 | Linked Fault: ERR-OKU600-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Okuma-Multus-U600.

PRM_44024 [Okuma-Multus-U600] — XAX_LOOP_GAIN_4 | Default: 184.000 | Linked Fault: ERR-OKU600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U600.

PRM_44025 [Okuma-Multus-U600] — YAX_LOOP_GAIN_5 | Default: 230.000 | Linked Fault: ERR-OKU600-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Okuma-Multus-U600.

PRM_44026 [Okuma-Multus-U600] — ZAX_LOOP_GAIN_6 | Default: 26.000 | Linked Fault: ERR-OKU600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U600.

PRM_44027 [Okuma-Multus-U600] — BAX_LOOP_GAIN_7 | Default: 72.000 | Linked Fault: ERR-OKU600-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Okuma-Multus-U600.

PRM_44028 [Okuma-Multus-U600] — CAX_LOOP_GAIN_8 | Default: 118.000 | Linked Fault: ERR-OKU600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U600.

PRM_44029 [Okuma-Multus-U600] — OPT_LOOP_GAIN_9 | Default: 164.000 | Linked Fault: ERR-OKU600-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Okuma-Multus-U600.

PRM_44030 [Okuma-Multus-U600] — PLC_LOOP_GAIN_10 | Default: 210.000 | Linked Fault: ERR-OKU600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U600.

PRM_44031 [Okuma-Multus-U600] — LUB_LOOP_GAIN_11 | Default: 6.000 | Linked Fault: ERR-OKU600-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Okuma-Multus-U600.

PRM_44032 [Okuma-Multus-U600] — VAC_LOOP_GAIN_12 | Default: 52.000 | Linked Fault: ERR-OKU600-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U600.

PRM_44033 [Okuma-Multus-U600] — THM_LOOP_GAIN_13 | Default: 98.000 | Linked Fault: ERR-OKU600-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Okuma-Multus-U600.

PRM_44034 [Okuma-Multus-U600] — SPN_LOOP_GAIN_14 | Default: 144.000 | Linked Fault: ERR-OKU600-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Okuma-Multus-U600.

PRM_44035 [Okuma-Multus-U600] — ATC_LOOP_GAIN_15 | Default: 190.000 | Linked Fault: ERR-OKU600-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Okuma-Multus-U600.

PRM_44036 [Okuma-Multus-U600] — HYD_LOOP_GAIN_16 | Default: 236.000 | Linked Fault: ERR-OKU600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U600.

CNC/PLC SERVO PARAMETER DICTIONARY — Okuma-Multus-U600 (BANK 2)

Machine Model: Okuma-Multus-U600 | Controller: OSP-P500A Aerospace. Use these parameter registers when diagnosing servo following errors or thermal drift on Okuma-Multus-U600.

PRM_44041 [Okuma-Multus-U600] — HYD_LOOP_GAIN_1 | Default: 92.000 | Linked Fault: ERR-OKU600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U600.

PRM_44042 [Okuma-Multus-U600] — XAX_LOOP_GAIN_2 | Default: 184.000 | Linked Fault: ERR-OKU600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U600.

PRM_44043 [Okuma-Multus-U600] — ZAX_LOOP_GAIN_3 | Default: 26.000 | Linked Fault: ERR-OKU600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U600.

PRM_44044 [Okuma-Multus-U600] — CAX_LOOP_GAIN_4 | Default: 118.000 | Linked Fault: ERR-OKU600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U600.

PRM_44045 [Okuma-Multus-U600] — PLC_LOOP_GAIN_5 | Default: 210.000 | Linked Fault: ERR-OKU600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U600.

PRM_44046 [Okuma-Multus-U600] — VAC_LOOP_GAIN_6 | Default: 52.000 | Linked Fault: ERR-OKU600-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U600.

PRM_44047 [Okuma-Multus-U600] — SPN_LOOP_GAIN_7 | Default: 144.000 | Linked Fault: ERR-OKU600-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Okuma-Multus-U600.

PRM_44048 [Okuma-Multus-U600] — HYD_LOOP_GAIN_8 | Default: 236.000 | Linked Fault: ERR-OKU600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U600.

PRM_44049 [Okuma-Multus-U600] — XAX_LOOP_GAIN_9 | Default: 78.000 | Linked Fault: ERR-OKU600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U600.

PRM_44050 [Okuma-Multus-U600] — ZAX_LOOP_GAIN_10 | Default: 170.000 | Linked Fault: ERR-OKU600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U600.

PRM_44051 [Okuma-Multus-U600] — CAX_LOOP_GAIN_11 | Default: 12.000 | Linked Fault: ERR-OKU600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U600.

PRM_44052 [Okuma-Multus-U600] — PLC_LOOP_GAIN_12 | Default: 104.000 | Linked Fault: ERR-OKU600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U600.

PRM_44053 [Okuma-Multus-U600] — VAC_LOOP_GAIN_13 | Default: 196.000 | Linked Fault: ERR-OKU600-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U600.

PRM_44054 [Okuma-Multus-U600] — SPN_LOOP_GAIN_14 | Default: 38.000 | Linked Fault: ERR-OKU600-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Okuma-Multus-U600.

PRM_44055 [Okuma-Multus-U600] — HYD_LOOP_GAIN_15 | Default: 130.000 | Linked Fault: ERR-OKU600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U600.

PRM_44056 [Okuma-Multus-U600] — XAX_LOOP_GAIN_16 | Default: 222.000 | Linked Fault: ERR-OKU600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U600.

CNC/PLC SERVO PARAMETER DICTIONARY — Okuma-Multus-U600 (BANK 3)

Machine Model: Okuma-Multus-U600 | Controller: OSP-P500A Aerospace. Use these parameter registers when diagnosing servo following errors or thermal drift on Okuma-Multus-U600.

PRM_44061 [Okuma-Multus-U600] — CLN_LOOP_GAIN_1 | Default: 138.000 | Linked Fault: ERR-OKU600-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Okuma-Multus-U600.

PRM_44062 [Okuma-Multus-U600] — ZAX_LOOP_GAIN_2 | Default: 26.000 | Linked Fault: ERR-OKU600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U600.

PRM_44063 [Okuma-Multus-U600] — OPT_LOOP_GAIN_3 | Default: 164.000 | Linked Fault: ERR-OKU600-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Okuma-Multus-U600.

PRM_44064 [Okuma-Multus-U600] — VAC_LOOP_GAIN_4 | Default: 52.000 | Linked Fault: ERR-OKU600-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U600.

PRM_44065 [Okuma-Multus-U600] — ATC_LOOP_GAIN_5 | Default: 190.000 | Linked Fault: ERR-OKU600-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Okuma-Multus-U600.

PRM_44066 [Okuma-Multus-U600] — XAX_LOOP_GAIN_6 | Default: 78.000 | Linked Fault: ERR-OKU600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U600.

PRM_44067 [Okuma-Multus-U600] — BAX_LOOP_GAIN_7 | Default: 216.000 | Linked Fault: ERR-OKU600-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Okuma-Multus-U600.

PRM_44068 [Okuma-Multus-U600] — PLC_LOOP_GAIN_8 | Default: 104.000 | Linked Fault: ERR-OKU600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U600.

PRM_44069 [Okuma-Multus-U600] — THM_LOOP_GAIN_9 | Default: 242.000 | Linked Fault: ERR-OKU600-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Okuma-Multus-U600.

PRM_44070 [Okuma-Multus-U600] — HYD_LOOP_GAIN_10 | Default: 130.000 | Linked Fault: ERR-OKU600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U600.

PRM_44071 [Okuma-Multus-U600] — YAX_LOOP_GAIN_11 | Default: 18.000 | Linked Fault: ERR-OKU600-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Okuma-Multus-U600.

PRM_44072 [Okuma-Multus-U600] — CAX_LOOP_GAIN_12 | Default: 156.000 | Linked Fault: ERR-OKU600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U600.

PRM_44073 [Okuma-Multus-U600] — LUB_LOOP_GAIN_13 | Default: 44.000 | Linked Fault: ERR-OKU600-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Okuma-Multus-U600.

PRM_44074 [Okuma-Multus-U600] — SPN_LOOP_GAIN_14 | Default: 182.000 | Linked Fault: ERR-OKU600-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Okuma-Multus-U600.

PRM_44075 [Okuma-Multus-U600] — CLN_LOOP_GAIN_15 | Default: 70.000 | Linked Fault: ERR-OKU600-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Okuma-Multus-U600.

PRM_44076 [Okuma-Multus-U600] — ZAX_LOOP_GAIN_16 | Default: 208.000 | Linked Fault: ERR-OKU600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U600.

CNC/PLC SERVO PARAMETER DICTIONARY — Okuma-Multus-U600 (BANK 4)

Machine Model: Okuma-Multus-U600 | Controller: OSP-P500A Aerospace. Use these parameter registers when diagnosing servo following errors or thermal drift on Okuma-Multus-U600.

PRM_44081 [Okuma-Multus-U600] — XAX_LOOP_GAIN_1 | Default: 184.000 | Linked Fault: ERR-OKU600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U600.

PRM_44082 [Okuma-Multus-U600] — CAX_LOOP_GAIN_2 | Default: 118.000 | Linked Fault: ERR-OKU600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U600.

PRM_44083 [Okuma-Multus-U600] — VAC_LOOP_GAIN_3 | Default: 52.000 | Linked Fault: ERR-OKU600-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U600.

PRM_44084 [Okuma-Multus-U600] — HYD_LOOP_GAIN_4 | Default: 236.000 | Linked Fault: ERR-OKU600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U600.

PRM_44085 [Okuma-Multus-U600] — ZAX_LOOP_GAIN_5 | Default: 170.000 | Linked Fault: ERR-OKU600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U600.

PRM_44086 [Okuma-Multus-U600] — PLC_LOOP_GAIN_6 | Default: 104.000 | Linked Fault: ERR-OKU600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U600.

PRM_44087 [Okuma-Multus-U600] — SPN_LOOP_GAIN_7 | Default: 38.000 | Linked Fault: ERR-OKU600-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Okuma-Multus-U600.

PRM_44088 [Okuma-Multus-U600] — XAX_LOOP_GAIN_8 | Default: 222.000 | Linked Fault: ERR-OKU600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U600.

PRM_44089 [Okuma-Multus-U600] — CAX_LOOP_GAIN_9 | Default: 156.000 | Linked Fault: ERR-OKU600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U600.

PRM_44090 [Okuma-Multus-U600] — VAC_LOOP_GAIN_10 | Default: 90.000 | Linked Fault: ERR-OKU600-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U600.

PRM_44091 [Okuma-Multus-U600] — HYD_LOOP_GAIN_11 | Default: 24.000 | Linked Fault: ERR-OKU600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U600.

PRM_44092 [Okuma-Multus-U600] — ZAX_LOOP_GAIN_12 | Default: 208.000 | Linked Fault: ERR-OKU600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U600.

PRM_44093 [Okuma-Multus-U600] — PLC_LOOP_GAIN_13 | Default: 142.000 | Linked Fault: ERR-OKU600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U600.

PRM_44094 [Okuma-Multus-U600] — SPN_LOOP_GAIN_14 | Default: 76.000 | Linked Fault: ERR-OKU600-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Okuma-Multus-U600.

PRM_44095 [Okuma-Multus-U600] — XAX_LOOP_GAIN_15 | Default: 10.000 | Linked Fault: ERR-OKU600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U600.

PRM_44096 [Okuma-Multus-U600] — CAX_LOOP_GAIN_16 | Default: 194.000 | Linked Fault: ERR-OKU600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U600.

CNC/PLC SERVO PARAMETER DICTIONARY — Okuma-Multus-U600 (BANK 5)

Machine Model: Okuma-Multus-U600 | Controller: OSP-P500A Aerospace. Use these parameter registers when diagnosing servo following errors or thermal drift on Okuma-Multus-U600.

PRM_44101 [Okuma-Multus-U600] — YAX_LOOP_GAIN_1 | Default: 230.000 | Linked Fault: ERR-OKU600-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Okuma-Multus-U600.

PRM_44102 [Okuma-Multus-U600] — PLC_LOOP_GAIN_2 | Default: 210.000 | Linked Fault: ERR-OKU600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U600.

PRM_44103 [Okuma-Multus-U600] — ATC_LOOP_GAIN_3 | Default: 190.000 | Linked Fault: ERR-OKU600-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Okuma-Multus-U600.

PRM_44104 [Okuma-Multus-U600] — ZAX_LOOP_GAIN_4 | Default: 170.000 | Linked Fault: ERR-OKU600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Okuma-Multus-U600.

PRM_44105 [Okuma-Multus-U600] — LUB_LOOP_GAIN_5 | Default: 150.000 | Linked Fault: ERR-OKU600-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Okuma-Multus-U600.

PRM_44106 [Okuma-Multus-U600] — HYD_LOOP_GAIN_6 | Default: 130.000 | Linked Fault: ERR-OKU600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Okuma-Multus-U600.

PRM_44107 [Okuma-Multus-U600] — BAX_LOOP_GAIN_7 | Default: 110.000 | Linked Fault: ERR-OKU600-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Okuma-Multus-U600.

PRM_44108 [Okuma-Multus-U600] — VAC_LOOP_GAIN_8 | Default: 90.000 | Linked Fault: ERR-OKU600-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Okuma-Multus-U600.

PRM_44109 [Okuma-Multus-U600] — CLN_LOOP_GAIN_9 | Default: 70.000 | Linked Fault: ERR-OKU600-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Okuma-Multus-U600.

PRM_44110 [Okuma-Multus-U600] — CAX_LOOP_GAIN_10 | Default: 50.000 | Linked Fault: ERR-OKU600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Okuma-Multus-U600.

PRM_44111 [Okuma-Multus-U600] — THM_LOOP_GAIN_11 | Default: 30.000 | Linked Fault: ERR-OKU600-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Okuma-Multus-U600.

PRM_44112 [Okuma-Multus-U600] — XAX_LOOP_GAIN_12 | Default: 10.000 | Linked Fault: ERR-OKU600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Okuma-Multus-U600.

PRM_44113 [Okuma-Multus-U600] — OPT_LOOP_GAIN_13 | Default: 240.000 | Linked Fault: ERR-OKU600-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Okuma-Multus-U600.

PRM_44114 [Okuma-Multus-U600] — SPN_LOOP_GAIN_14 | Default: 220.000 | Linked Fault: ERR-OKU600-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Okuma-Multus-U600.

PRM_44115 [Okuma-Multus-U600] — YAX_LOOP_GAIN_15 | Default: 200.000 | Linked Fault: ERR-OKU600-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Okuma-Multus-U600.

PRM_44116 [Okuma-Multus-U600] — PLC_LOOP_GAIN_16 | Default: 180.000 | Linked Fault: ERR-OKU600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Okuma-Multus-U600.

CNC/PLC SERVO PARAMETER DICTIONARY — EOS-Additive-M150 (BANK 1)

Machine Model: EOS-Additive-M150 | Controller: EOSYSTEM Dual-PLC. Use these parameter registers when diagnosing servo following errors or thermal drift on EOS-Additive-M150.

PRM_44121 [EOS-Additive-M150] — ATC_LOOP_GAIN_1 | Default: 47.150 | Linked Fault: ERR-EOS150-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on EOS-Additive-M150.

PRM_44122 [EOS-Additive-M150] — HYD_LOOP_GAIN_2 | Default: 94.300 | Linked Fault: ERR-EOS150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M150.

PRM_44123 [EOS-Additive-M150] — CLN_LOOP_GAIN_3 | Default: 141.450 | Linked Fault: ERR-EOS150-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on EOS-Additive-M150.

PRM_44124 [EOS-Additive-M150] — XAX_LOOP_GAIN_4 | Default: 188.600 | Linked Fault: ERR-EOS150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M150.

PRM_44125 [EOS-Additive-M150] — YAX_LOOP_GAIN_5 | Default: 235.750 | Linked Fault: ERR-EOS150-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on EOS-Additive-M150.

PRM_44126 [EOS-Additive-M150] — ZAX_LOOP_GAIN_6 | Default: 32.900 | Linked Fault: ERR-EOS150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M150.

PRM_44127 [EOS-Additive-M150] — BAX_LOOP_GAIN_7 | Default: 80.050 | Linked Fault: ERR-EOS150-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on EOS-Additive-M150.

PRM_44128 [EOS-Additive-M150] — CAX_LOOP_GAIN_8 | Default: 127.200 | Linked Fault: ERR-EOS150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M150.

PRM_44129 [EOS-Additive-M150] — OPT_LOOP_GAIN_9 | Default: 174.350 | Linked Fault: ERR-EOS150-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on EOS-Additive-M150.

PRM_44130 [EOS-Additive-M150] — PLC_LOOP_GAIN_10 | Default: 221.500 | Linked Fault: ERR-EOS150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M150.

PRM_44131 [EOS-Additive-M150] — LUB_LOOP_GAIN_11 | Default: 18.650 | Linked Fault: ERR-EOS150-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on EOS-Additive-M150.

PRM_44132 [EOS-Additive-M150] — VAC_LOOP_GAIN_12 | Default: 65.800 | Linked Fault: ERR-EOS150-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M150.

PRM_44133 [EOS-Additive-M150] — THM_LOOP_GAIN_13 | Default: 112.950 | Linked Fault: ERR-EOS150-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on EOS-Additive-M150.

PRM_44134 [EOS-Additive-M150] — SPN_LOOP_GAIN_14 | Default: 160.100 | Linked Fault: ERR-EOS150-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on EOS-Additive-M150.

PRM_44135 [EOS-Additive-M150] — ATC_LOOP_GAIN_15 | Default: 207.250 | Linked Fault: ERR-EOS150-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on EOS-Additive-M150.

PRM_44136 [EOS-Additive-M150] — HYD_LOOP_GAIN_16 | Default: 4.400 | Linked Fault: ERR-EOS150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M150.

CNC/PLC SERVO PARAMETER DICTIONARY — EOS-Additive-M150 (BANK 2)

Machine Model: EOS-Additive-M150 | Controller: EOSYSTEM Dual-PLC. Use these parameter registers when diagnosing servo following errors or thermal drift on EOS-Additive-M150.

PRM_44141 [EOS-Additive-M150] — HYD_LOOP_GAIN_1 | Default: 94.300 | Linked Fault: ERR-EOS150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M150.

PRM_44142 [EOS-Additive-M150] — XAX_LOOP_GAIN_2 | Default: 188.600 | Linked Fault: ERR-EOS150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M150.

PRM_44143 [EOS-Additive-M150] — ZAX_LOOP_GAIN_3 | Default: 32.900 | Linked Fault: ERR-EOS150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M150.

PRM_44144 [EOS-Additive-M150] — CAX_LOOP_GAIN_4 | Default: 127.200 | Linked Fault: ERR-EOS150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M150.

PRM_44145 [EOS-Additive-M150] — PLC_LOOP_GAIN_5 | Default: 221.500 | Linked Fault: ERR-EOS150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M150.

PRM_44146 [EOS-Additive-M150] — VAC_LOOP_GAIN_6 | Default: 65.800 | Linked Fault: ERR-EOS150-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M150.

PRM_44147 [EOS-Additive-M150] — SPN_LOOP_GAIN_7 | Default: 160.100 | Linked Fault: ERR-EOS150-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on EOS-Additive-M150.

PRM_44148 [EOS-Additive-M150] — HYD_LOOP_GAIN_8 | Default: 4.400 | Linked Fault: ERR-EOS150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M150.

PRM_44149 [EOS-Additive-M150] — XAX_LOOP_GAIN_9 | Default: 98.700 | Linked Fault: ERR-EOS150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M150.

PRM_44150 [EOS-Additive-M150] — ZAX_LOOP_GAIN_10 | Default: 193.000 | Linked Fault: ERR-EOS150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M150.

PRM_44151 [EOS-Additive-M150] — CAX_LOOP_GAIN_11 | Default: 37.300 | Linked Fault: ERR-EOS150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M150.

PRM_44152 [EOS-Additive-M150] — PLC_LOOP_GAIN_12 | Default: 131.600 | Linked Fault: ERR-EOS150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M150.

PRM_44153 [EOS-Additive-M150] — VAC_LOOP_GAIN_13 | Default: 225.900 | Linked Fault: ERR-EOS150-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M150.

PRM_44154 [EOS-Additive-M150] — SPN_LOOP_GAIN_14 | Default: 70.200 | Linked Fault: ERR-EOS150-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on EOS-Additive-M150.

PRM_44155 [EOS-Additive-M150] — HYD_LOOP_GAIN_15 | Default: 164.500 | Linked Fault: ERR-EOS150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M150.

PRM_44156 [EOS-Additive-M150] — XAX_LOOP_GAIN_16 | Default: 8.800 | Linked Fault: ERR-EOS150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M150.

CNC/PLC SERVO PARAMETER DICTIONARY — EOS-Additive-M150 (BANK 3)

Machine Model: EOS-Additive-M150 | Controller: EOSYSTEM Dual-PLC. Use these parameter registers when diagnosing servo following errors or thermal drift on EOS-Additive-M150.

PRM_44161 [EOS-Additive-M150] — CLN_LOOP_GAIN_1 | Default: 141.450 | Linked Fault: ERR-EOS150-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on EOS-Additive-M150.

PRM_44162 [EOS-Additive-M150] — ZAX_LOOP_GAIN_2 | Default: 32.900 | Linked Fault: ERR-EOS150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M150.

PRM_44163 [EOS-Additive-M150] — OPT_LOOP_GAIN_3 | Default: 174.350 | Linked Fault: ERR-EOS150-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on EOS-Additive-M150.

PRM_44164 [EOS-Additive-M150] — VAC_LOOP_GAIN_4 | Default: 65.800 | Linked Fault: ERR-EOS150-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M150.

PRM_44165 [EOS-Additive-M150] — ATC_LOOP_GAIN_5 | Default: 207.250 | Linked Fault: ERR-EOS150-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on EOS-Additive-M150.

PRM_44166 [EOS-Additive-M150] — XAX_LOOP_GAIN_6 | Default: 98.700 | Linked Fault: ERR-EOS150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M150.

PRM_44167 [EOS-Additive-M150] — BAX_LOOP_GAIN_7 | Default: 240.150 | Linked Fault: ERR-EOS150-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on EOS-Additive-M150.

PRM_44168 [EOS-Additive-M150] — PLC_LOOP_GAIN_8 | Default: 131.600 | Linked Fault: ERR-EOS150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M150.

PRM_44169 [EOS-Additive-M150] — THM_LOOP_GAIN_9 | Default: 23.050 | Linked Fault: ERR-EOS150-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on EOS-Additive-M150.

PRM_44170 [EOS-Additive-M150] — HYD_LOOP_GAIN_10 | Default: 164.500 | Linked Fault: ERR-EOS150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M150.

PRM_44171 [EOS-Additive-M150] — YAX_LOOP_GAIN_11 | Default: 55.950 | Linked Fault: ERR-EOS150-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on EOS-Additive-M150.

PRM_44172 [EOS-Additive-M150] — CAX_LOOP_GAIN_12 | Default: 197.400 | Linked Fault: ERR-EOS150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M150.

PRM_44173 [EOS-Additive-M150] — LUB_LOOP_GAIN_13 | Default: 88.850 | Linked Fault: ERR-EOS150-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on EOS-Additive-M150.

PRM_44174 [EOS-Additive-M150] — SPN_LOOP_GAIN_14 | Default: 230.300 | Linked Fault: ERR-EOS150-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on EOS-Additive-M150.

PRM_44175 [EOS-Additive-M150] — CLN_LOOP_GAIN_15 | Default: 121.750 | Linked Fault: ERR-EOS150-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on EOS-Additive-M150.

PRM_44176 [EOS-Additive-M150] — ZAX_LOOP_GAIN_16 | Default: 13.200 | Linked Fault: ERR-EOS150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M150.

CNC/PLC SERVO PARAMETER DICTIONARY — EOS-Additive-M150 (BANK 4)

Machine Model: EOS-Additive-M150 | Controller: EOSYSTEM Dual-PLC. Use these parameter registers when diagnosing servo following errors or thermal drift on EOS-Additive-M150.

PRM_44181 [EOS-Additive-M150] — XAX_LOOP_GAIN_1 | Default: 188.600 | Linked Fault: ERR-EOS150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M150.

PRM_44182 [EOS-Additive-M150] — CAX_LOOP_GAIN_2 | Default: 127.200 | Linked Fault: ERR-EOS150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M150.

PRM_44183 [EOS-Additive-M150] — VAC_LOOP_GAIN_3 | Default: 65.800 | Linked Fault: ERR-EOS150-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M150.

PRM_44184 [EOS-Additive-M150] — HYD_LOOP_GAIN_4 | Default: 4.400 | Linked Fault: ERR-EOS150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M150.

PRM_44185 [EOS-Additive-M150] — ZAX_LOOP_GAIN_5 | Default: 193.000 | Linked Fault: ERR-EOS150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M150.

PRM_44186 [EOS-Additive-M150] — PLC_LOOP_GAIN_6 | Default: 131.600 | Linked Fault: ERR-EOS150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M150.

PRM_44187 [EOS-Additive-M150] — SPN_LOOP_GAIN_7 | Default: 70.200 | Linked Fault: ERR-EOS150-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on EOS-Additive-M150.

PRM_44188 [EOS-Additive-M150] — XAX_LOOP_GAIN_8 | Default: 8.800 | Linked Fault: ERR-EOS150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M150.

PRM_44189 [EOS-Additive-M150] — CAX_LOOP_GAIN_9 | Default: 197.400 | Linked Fault: ERR-EOS150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M150.

PRM_44190 [EOS-Additive-M150] — VAC_LOOP_GAIN_10 | Default: 136.000 | Linked Fault: ERR-EOS150-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M150.

PRM_44191 [EOS-Additive-M150] — HYD_LOOP_GAIN_11 | Default: 74.600 | Linked Fault: ERR-EOS150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M150.

PRM_44192 [EOS-Additive-M150] — ZAX_LOOP_GAIN_12 | Default: 13.200 | Linked Fault: ERR-EOS150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M150.

PRM_44193 [EOS-Additive-M150] — PLC_LOOP_GAIN_13 | Default: 201.800 | Linked Fault: ERR-EOS150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M150.

PRM_44194 [EOS-Additive-M150] — SPN_LOOP_GAIN_14 | Default: 140.400 | Linked Fault: ERR-EOS150-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on EOS-Additive-M150.

PRM_44195 [EOS-Additive-M150] — XAX_LOOP_GAIN_15 | Default: 79.000 | Linked Fault: ERR-EOS150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M150.

PRM_44196 [EOS-Additive-M150] — CAX_LOOP_GAIN_16 | Default: 17.600 | Linked Fault: ERR-EOS150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M150.

CNC/PLC SERVO PARAMETER DICTIONARY — EOS-Additive-M150 (BANK 5)

Machine Model: EOS-Additive-M150 | Controller: EOSYSTEM Dual-PLC. Use these parameter registers when diagnosing servo following errors or thermal drift on EOS-Additive-M150.

PRM_44201 [EOS-Additive-M150] — YAX_LOOP_GAIN_1 | Default: 235.750 | Linked Fault: ERR-EOS150-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on EOS-Additive-M150.

PRM_44202 [EOS-Additive-M150] — PLC_LOOP_GAIN_2 | Default: 221.500 | Linked Fault: ERR-EOS150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M150.

PRM_44203 [EOS-Additive-M150] — ATC_LOOP_GAIN_3 | Default: 207.250 | Linked Fault: ERR-EOS150-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on EOS-Additive-M150.

PRM_44204 [EOS-Additive-M150] — ZAX_LOOP_GAIN_4 | Default: 193.000 | Linked Fault: ERR-EOS150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M150.

PRM_44205 [EOS-Additive-M150] — LUB_LOOP_GAIN_5 | Default: 178.750 | Linked Fault: ERR-EOS150-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on EOS-Additive-M150.

PRM_44206 [EOS-Additive-M150] — HYD_LOOP_GAIN_6 | Default: 164.500 | Linked Fault: ERR-EOS150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M150.

PRM_44207 [EOS-Additive-M150] — BAX_LOOP_GAIN_7 | Default: 150.250 | Linked Fault: ERR-EOS150-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on EOS-Additive-M150.

PRM_44208 [EOS-Additive-M150] — VAC_LOOP_GAIN_8 | Default: 136.000 | Linked Fault: ERR-EOS150-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M150.

PRM_44209 [EOS-Additive-M150] — CLN_LOOP_GAIN_9 | Default: 121.750 | Linked Fault: ERR-EOS150-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on EOS-Additive-M150.

PRM_44210 [EOS-Additive-M150] — CAX_LOOP_GAIN_10 | Default: 107.500 | Linked Fault: ERR-EOS150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M150.

PRM_44211 [EOS-Additive-M150] — THM_LOOP_GAIN_11 | Default: 93.250 | Linked Fault: ERR-EOS150-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on EOS-Additive-M150.

PRM_44212 [EOS-Additive-M150] — XAX_LOOP_GAIN_12 | Default: 79.000 | Linked Fault: ERR-EOS150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M150.

PRM_44213 [EOS-Additive-M150] — OPT_LOOP_GAIN_13 | Default: 64.750 | Linked Fault: ERR-EOS150-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on EOS-Additive-M150.

PRM_44214 [EOS-Additive-M150] — SPN_LOOP_GAIN_14 | Default: 50.500 | Linked Fault: ERR-EOS150-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on EOS-Additive-M150.

PRM_44215 [EOS-Additive-M150] — YAX_LOOP_GAIN_15 | Default: 36.250 | Linked Fault: ERR-EOS150-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on EOS-Additive-M150.

PRM_44216 [EOS-Additive-M150] — PLC_LOOP_GAIN_16 | Default: 22.000 | Linked Fault: ERR-EOS150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M150.

CNC/PLC SERVO PARAMETER DICTIONARY — EOS-Additive-M200 (BANK 1)

Machine Model: EOS-Additive-M200 | Controller: EOSYSTEM Dual-PLC. Use these parameter registers when diagnosing servo following errors or thermal drift on EOS-Additive-M200.

PRM_44221 [EOS-Additive-M200] — ATC_LOOP_GAIN_1 | Default: 48.300 | Linked Fault: ERR-EOS200-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on EOS-Additive-M200.

PRM_44222 [EOS-Additive-M200] — HYD_LOOP_GAIN_2 | Default: 96.600 | Linked Fault: ERR-EOS200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M200.

PRM_44223 [EOS-Additive-M200] — CLN_LOOP_GAIN_3 | Default: 144.900 | Linked Fault: ERR-EOS200-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on EOS-Additive-M200.

PRM_44224 [EOS-Additive-M200] — XAX_LOOP_GAIN_4 | Default: 193.200 | Linked Fault: ERR-EOS200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M200.

PRM_44225 [EOS-Additive-M200] — YAX_LOOP_GAIN_5 | Default: 241.500 | Linked Fault: ERR-EOS200-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on EOS-Additive-M200.

PRM_44226 [EOS-Additive-M200] — ZAX_LOOP_GAIN_6 | Default: 39.800 | Linked Fault: ERR-EOS200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M200.

PRM_44227 [EOS-Additive-M200] — BAX_LOOP_GAIN_7 | Default: 88.100 | Linked Fault: ERR-EOS200-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on EOS-Additive-M200.

PRM_44228 [EOS-Additive-M200] — CAX_LOOP_GAIN_8 | Default: 136.400 | Linked Fault: ERR-EOS200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M200.

PRM_44229 [EOS-Additive-M200] — OPT_LOOP_GAIN_9 | Default: 184.700 | Linked Fault: ERR-EOS200-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on EOS-Additive-M200.

PRM_44230 [EOS-Additive-M200] — PLC_LOOP_GAIN_10 | Default: 233.000 | Linked Fault: ERR-EOS200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M200.

PRM_44231 [EOS-Additive-M200] — LUB_LOOP_GAIN_11 | Default: 31.300 | Linked Fault: ERR-EOS200-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on EOS-Additive-M200.

PRM_44232 [EOS-Additive-M200] — VAC_LOOP_GAIN_12 | Default: 79.600 | Linked Fault: ERR-EOS200-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M200.

PRM_44233 [EOS-Additive-M200] — THM_LOOP_GAIN_13 | Default: 127.900 | Linked Fault: ERR-EOS200-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on EOS-Additive-M200.

PRM_44234 [EOS-Additive-M200] — SPN_LOOP_GAIN_14 | Default: 176.200 | Linked Fault: ERR-EOS200-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on EOS-Additive-M200.

PRM_44235 [EOS-Additive-M200] — ATC_LOOP_GAIN_15 | Default: 224.500 | Linked Fault: ERR-EOS200-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on EOS-Additive-M200.

PRM_44236 [EOS-Additive-M200] — HYD_LOOP_GAIN_16 | Default: 22.800 | Linked Fault: ERR-EOS200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M200.

CNC/PLC SERVO PARAMETER DICTIONARY — EOS-Additive-M200 (BANK 2)

Machine Model: EOS-Additive-M200 | Controller: EOSYSTEM Dual-PLC. Use these parameter registers when diagnosing servo following errors or thermal drift on EOS-Additive-M200.

PRM_44241 [EOS-Additive-M200] — HYD_LOOP_GAIN_1 | Default: 96.600 | Linked Fault: ERR-EOS200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M200.

PRM_44242 [EOS-Additive-M200] — XAX_LOOP_GAIN_2 | Default: 193.200 | Linked Fault: ERR-EOS200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M200.

PRM_44243 [EOS-Additive-M200] — ZAX_LOOP_GAIN_3 | Default: 39.800 | Linked Fault: ERR-EOS200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M200.

PRM_44244 [EOS-Additive-M200] — CAX_LOOP_GAIN_4 | Default: 136.400 | Linked Fault: ERR-EOS200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M200.

PRM_44245 [EOS-Additive-M200] — PLC_LOOP_GAIN_5 | Default: 233.000 | Linked Fault: ERR-EOS200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M200.

PRM_44246 [EOS-Additive-M200] — VAC_LOOP_GAIN_6 | Default: 79.600 | Linked Fault: ERR-EOS200-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M200.

PRM_44247 [EOS-Additive-M200] — SPN_LOOP_GAIN_7 | Default: 176.200 | Linked Fault: ERR-EOS200-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on EOS-Additive-M200.

PRM_44248 [EOS-Additive-M200] — HYD_LOOP_GAIN_8 | Default: 22.800 | Linked Fault: ERR-EOS200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M200.

PRM_44249 [EOS-Additive-M200] — XAX_LOOP_GAIN_9 | Default: 119.400 | Linked Fault: ERR-EOS200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M200.

PRM_44250 [EOS-Additive-M200] — ZAX_LOOP_GAIN_10 | Default: 216.000 | Linked Fault: ERR-EOS200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M200.

PRM_44251 [EOS-Additive-M200] — CAX_LOOP_GAIN_11 | Default: 62.600 | Linked Fault: ERR-EOS200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M200.

PRM_44252 [EOS-Additive-M200] — PLC_LOOP_GAIN_12 | Default: 159.200 | Linked Fault: ERR-EOS200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M200.

PRM_44253 [EOS-Additive-M200] — VAC_LOOP_GAIN_13 | Default: 5.800 | Linked Fault: ERR-EOS200-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M200.

PRM_44254 [EOS-Additive-M200] — SPN_LOOP_GAIN_14 | Default: 102.400 | Linked Fault: ERR-EOS200-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on EOS-Additive-M200.

PRM_44255 [EOS-Additive-M200] — HYD_LOOP_GAIN_15 | Default: 199.000 | Linked Fault: ERR-EOS200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M200.

PRM_44256 [EOS-Additive-M200] — XAX_LOOP_GAIN_16 | Default: 45.600 | Linked Fault: ERR-EOS200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M200.

CNC/PLC SERVO PARAMETER DICTIONARY — EOS-Additive-M200 (BANK 3)

Machine Model: EOS-Additive-M200 | Controller: EOSYSTEM Dual-PLC. Use these parameter registers when diagnosing servo following errors or thermal drift on EOS-Additive-M200.

PRM_44261 [EOS-Additive-M200] — CLN_LOOP_GAIN_1 | Default: 144.900 | Linked Fault: ERR-EOS200-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on EOS-Additive-M200.

PRM_44262 [EOS-Additive-M200] — ZAX_LOOP_GAIN_2 | Default: 39.800 | Linked Fault: ERR-EOS200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M200.

PRM_44263 [EOS-Additive-M200] — OPT_LOOP_GAIN_3 | Default: 184.700 | Linked Fault: ERR-EOS200-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on EOS-Additive-M200.

PRM_44264 [EOS-Additive-M200] — VAC_LOOP_GAIN_4 | Default: 79.600 | Linked Fault: ERR-EOS200-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M200.

PRM_44265 [EOS-Additive-M200] — ATC_LOOP_GAIN_5 | Default: 224.500 | Linked Fault: ERR-EOS200-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on EOS-Additive-M200.

PRM_44266 [EOS-Additive-M200] — XAX_LOOP_GAIN_6 | Default: 119.400 | Linked Fault: ERR-EOS200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M200.

PRM_44267 [EOS-Additive-M200] — BAX_LOOP_GAIN_7 | Default: 14.300 | Linked Fault: ERR-EOS200-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on EOS-Additive-M200.

PRM_44268 [EOS-Additive-M200] — PLC_LOOP_GAIN_8 | Default: 159.200 | Linked Fault: ERR-EOS200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M200.

PRM_44269 [EOS-Additive-M200] — THM_LOOP_GAIN_9 | Default: 54.100 | Linked Fault: ERR-EOS200-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on EOS-Additive-M200.

PRM_44270 [EOS-Additive-M200] — HYD_LOOP_GAIN_10 | Default: 199.000 | Linked Fault: ERR-EOS200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M200.

PRM_44271 [EOS-Additive-M200] — YAX_LOOP_GAIN_11 | Default: 93.900 | Linked Fault: ERR-EOS200-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on EOS-Additive-M200.

PRM_44272 [EOS-Additive-M200] — CAX_LOOP_GAIN_12 | Default: 238.800 | Linked Fault: ERR-EOS200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M200.

PRM_44273 [EOS-Additive-M200] — LUB_LOOP_GAIN_13 | Default: 133.700 | Linked Fault: ERR-EOS200-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on EOS-Additive-M200.

PRM_44274 [EOS-Additive-M200] — SPN_LOOP_GAIN_14 | Default: 28.600 | Linked Fault: ERR-EOS200-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on EOS-Additive-M200.

PRM_44275 [EOS-Additive-M200] — CLN_LOOP_GAIN_15 | Default: 173.500 | Linked Fault: ERR-EOS200-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on EOS-Additive-M200.

PRM_44276 [EOS-Additive-M200] — ZAX_LOOP_GAIN_16 | Default: 68.400 | Linked Fault: ERR-EOS200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M200.

CNC/PLC SERVO PARAMETER DICTIONARY — EOS-Additive-M200 (BANK 4)

Machine Model: EOS-Additive-M200 | Controller: EOSYSTEM Dual-PLC. Use these parameter registers when diagnosing servo following errors or thermal drift on EOS-Additive-M200.

PRM_44281 [EOS-Additive-M200] — XAX_LOOP_GAIN_1 | Default: 193.200 | Linked Fault: ERR-EOS200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M200.

PRM_44282 [EOS-Additive-M200] — CAX_LOOP_GAIN_2 | Default: 136.400 | Linked Fault: ERR-EOS200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M200.

PRM_44283 [EOS-Additive-M200] — VAC_LOOP_GAIN_3 | Default: 79.600 | Linked Fault: ERR-EOS200-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M200.

PRM_44284 [EOS-Additive-M200] — HYD_LOOP_GAIN_4 | Default: 22.800 | Linked Fault: ERR-EOS200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M200.

PRM_44285 [EOS-Additive-M200] — ZAX_LOOP_GAIN_5 | Default: 216.000 | Linked Fault: ERR-EOS200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M200.

PRM_44286 [EOS-Additive-M200] — PLC_LOOP_GAIN_6 | Default: 159.200 | Linked Fault: ERR-EOS200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M200.

PRM_44287 [EOS-Additive-M200] — SPN_LOOP_GAIN_7 | Default: 102.400 | Linked Fault: ERR-EOS200-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on EOS-Additive-M200.

PRM_44288 [EOS-Additive-M200] — XAX_LOOP_GAIN_8 | Default: 45.600 | Linked Fault: ERR-EOS200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M200.

PRM_44289 [EOS-Additive-M200] — CAX_LOOP_GAIN_9 | Default: 238.800 | Linked Fault: ERR-EOS200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M200.

PRM_44290 [EOS-Additive-M200] — VAC_LOOP_GAIN_10 | Default: 182.000 | Linked Fault: ERR-EOS200-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M200.

PRM_44291 [EOS-Additive-M200] — HYD_LOOP_GAIN_11 | Default: 125.200 | Linked Fault: ERR-EOS200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M200.

PRM_44292 [EOS-Additive-M200] — ZAX_LOOP_GAIN_12 | Default: 68.400 | Linked Fault: ERR-EOS200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M200.

PRM_44293 [EOS-Additive-M200] — PLC_LOOP_GAIN_13 | Default: 11.600 | Linked Fault: ERR-EOS200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M200.

PRM_44294 [EOS-Additive-M200] — SPN_LOOP_GAIN_14 | Default: 204.800 | Linked Fault: ERR-EOS200-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on EOS-Additive-M200.

PRM_44295 [EOS-Additive-M200] — XAX_LOOP_GAIN_15 | Default: 148.000 | Linked Fault: ERR-EOS200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M200.

PRM_44296 [EOS-Additive-M200] — CAX_LOOP_GAIN_16 | Default: 91.200 | Linked Fault: ERR-EOS200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M200.

CNC/PLC SERVO PARAMETER DICTIONARY — EOS-Additive-M200 (BANK 5)

Machine Model: EOS-Additive-M200 | Controller: EOSYSTEM Dual-PLC. Use these parameter registers when diagnosing servo following errors or thermal drift on EOS-Additive-M200.

PRM_44301 [EOS-Additive-M200] — YAX_LOOP_GAIN_1 | Default: 241.500 | Linked Fault: ERR-EOS200-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on EOS-Additive-M200.

PRM_44302 [EOS-Additive-M200] — PLC_LOOP_GAIN_2 | Default: 233.000 | Linked Fault: ERR-EOS200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M200.

PRM_44303 [EOS-Additive-M200] — ATC_LOOP_GAIN_3 | Default: 224.500 | Linked Fault: ERR-EOS200-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on EOS-Additive-M200.

PRM_44304 [EOS-Additive-M200] — ZAX_LOOP_GAIN_4 | Default: 216.000 | Linked Fault: ERR-EOS200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M200.

PRM_44305 [EOS-Additive-M200] — LUB_LOOP_GAIN_5 | Default: 207.500 | Linked Fault: ERR-EOS200-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on EOS-Additive-M200.

PRM_44306 [EOS-Additive-M200] — HYD_LOOP_GAIN_6 | Default: 199.000 | Linked Fault: ERR-EOS200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M200.

PRM_44307 [EOS-Additive-M200] — BAX_LOOP_GAIN_7 | Default: 190.500 | Linked Fault: ERR-EOS200-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on EOS-Additive-M200.

PRM_44308 [EOS-Additive-M200] — VAC_LOOP_GAIN_8 | Default: 182.000 | Linked Fault: ERR-EOS200-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M200.

PRM_44309 [EOS-Additive-M200] — CLN_LOOP_GAIN_9 | Default: 173.500 | Linked Fault: ERR-EOS200-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on EOS-Additive-M200.

PRM_44310 [EOS-Additive-M200] — CAX_LOOP_GAIN_10 | Default: 165.000 | Linked Fault: ERR-EOS200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M200.

PRM_44311 [EOS-Additive-M200] — THM_LOOP_GAIN_11 | Default: 156.500 | Linked Fault: ERR-EOS200-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on EOS-Additive-M200.

PRM_44312 [EOS-Additive-M200] — XAX_LOOP_GAIN_12 | Default: 148.000 | Linked Fault: ERR-EOS200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M200.

PRM_44313 [EOS-Additive-M200] — OPT_LOOP_GAIN_13 | Default: 139.500 | Linked Fault: ERR-EOS200-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on EOS-Additive-M200.

PRM_44314 [EOS-Additive-M200] — SPN_LOOP_GAIN_14 | Default: 131.000 | Linked Fault: ERR-EOS200-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on EOS-Additive-M200.

PRM_44315 [EOS-Additive-M200] — YAX_LOOP_GAIN_15 | Default: 122.500 | Linked Fault: ERR-EOS200-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on EOS-Additive-M200.

PRM_44316 [EOS-Additive-M200] — PLC_LOOP_GAIN_16 | Default: 114.000 | Linked Fault: ERR-EOS200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M200.

CNC/PLC SERVO PARAMETER DICTIONARY — EOS-Additive-M250 (BANK 1)

Machine Model: EOS-Additive-M250 | Controller: EOSYSTEM Dual-PLC. Use these parameter registers when diagnosing servo following errors or thermal drift on EOS-Additive-M250.

PRM_44321 [EOS-Additive-M250] — ATC_LOOP_GAIN_1 | Default: 49.450 | Linked Fault: ERR-EOS250-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on EOS-Additive-M250.

PRM_44322 [EOS-Additive-M250] — HYD_LOOP_GAIN_2 | Default: 98.900 | Linked Fault: ERR-EOS250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M250.

PRM_44323 [EOS-Additive-M250] — CLN_LOOP_GAIN_3 | Default: 148.350 | Linked Fault: ERR-EOS250-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on EOS-Additive-M250.

PRM_44324 [EOS-Additive-M250] — XAX_LOOP_GAIN_4 | Default: 197.800 | Linked Fault: ERR-EOS250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M250.

PRM_44325 [EOS-Additive-M250] — YAX_LOOP_GAIN_5 | Default: 247.250 | Linked Fault: ERR-EOS250-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on EOS-Additive-M250.

PRM_44326 [EOS-Additive-M250] — ZAX_LOOP_GAIN_6 | Default: 46.700 | Linked Fault: ERR-EOS250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M250.

PRM_44327 [EOS-Additive-M250] — BAX_LOOP_GAIN_7 | Default: 96.150 | Linked Fault: ERR-EOS250-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on EOS-Additive-M250.

PRM_44328 [EOS-Additive-M250] — CAX_LOOP_GAIN_8 | Default: 145.600 | Linked Fault: ERR-EOS250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M250.

PRM_44329 [EOS-Additive-M250] — OPT_LOOP_GAIN_9 | Default: 195.050 | Linked Fault: ERR-EOS250-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on EOS-Additive-M250.

PRM_44330 [EOS-Additive-M250] — PLC_LOOP_GAIN_10 | Default: 244.500 | Linked Fault: ERR-EOS250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M250.

PRM_44331 [EOS-Additive-M250] — LUB_LOOP_GAIN_11 | Default: 43.950 | Linked Fault: ERR-EOS250-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on EOS-Additive-M250.

PRM_44332 [EOS-Additive-M250] — VAC_LOOP_GAIN_12 | Default: 93.400 | Linked Fault: ERR-EOS250-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M250.

PRM_44333 [EOS-Additive-M250] — THM_LOOP_GAIN_13 | Default: 142.850 | Linked Fault: ERR-EOS250-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on EOS-Additive-M250.

PRM_44334 [EOS-Additive-M250] — SPN_LOOP_GAIN_14 | Default: 192.300 | Linked Fault: ERR-EOS250-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on EOS-Additive-M250.

PRM_44335 [EOS-Additive-M250] — ATC_LOOP_GAIN_15 | Default: 241.750 | Linked Fault: ERR-EOS250-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on EOS-Additive-M250.

PRM_44336 [EOS-Additive-M250] — HYD_LOOP_GAIN_16 | Default: 41.200 | Linked Fault: ERR-EOS250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M250.

CNC/PLC SERVO PARAMETER DICTIONARY — EOS-Additive-M250 (BANK 2)

Machine Model: EOS-Additive-M250 | Controller: EOSYSTEM Dual-PLC. Use these parameter registers when diagnosing servo following errors or thermal drift on EOS-Additive-M250.

PRM_44341 [EOS-Additive-M250] — HYD_LOOP_GAIN_1 | Default: 98.900 | Linked Fault: ERR-EOS250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M250.

PRM_44342 [EOS-Additive-M250] — XAX_LOOP_GAIN_2 | Default: 197.800 | Linked Fault: ERR-EOS250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M250.

PRM_44343 [EOS-Additive-M250] — ZAX_LOOP_GAIN_3 | Default: 46.700 | Linked Fault: ERR-EOS250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M250.

PRM_44344 [EOS-Additive-M250] — CAX_LOOP_GAIN_4 | Default: 145.600 | Linked Fault: ERR-EOS250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M250.

PRM_44345 [EOS-Additive-M250] — PLC_LOOP_GAIN_5 | Default: 244.500 | Linked Fault: ERR-EOS250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M250.

PRM_44346 [EOS-Additive-M250] — VAC_LOOP_GAIN_6 | Default: 93.400 | Linked Fault: ERR-EOS250-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M250.

PRM_44347 [EOS-Additive-M250] — SPN_LOOP_GAIN_7 | Default: 192.300 | Linked Fault: ERR-EOS250-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on EOS-Additive-M250.

PRM_44348 [EOS-Additive-M250] — HYD_LOOP_GAIN_8 | Default: 41.200 | Linked Fault: ERR-EOS250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M250.

PRM_44349 [EOS-Additive-M250] — XAX_LOOP_GAIN_9 | Default: 140.100 | Linked Fault: ERR-EOS250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M250.

PRM_44350 [EOS-Additive-M250] — ZAX_LOOP_GAIN_10 | Default: 239.000 | Linked Fault: ERR-EOS250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M250.

PRM_44351 [EOS-Additive-M250] — CAX_LOOP_GAIN_11 | Default: 87.900 | Linked Fault: ERR-EOS250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M250.

PRM_44352 [EOS-Additive-M250] — PLC_LOOP_GAIN_12 | Default: 186.800 | Linked Fault: ERR-EOS250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M250.

PRM_44353 [EOS-Additive-M250] — VAC_LOOP_GAIN_13 | Default: 35.700 | Linked Fault: ERR-EOS250-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M250.

PRM_44354 [EOS-Additive-M250] — SPN_LOOP_GAIN_14 | Default: 134.600 | Linked Fault: ERR-EOS250-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on EOS-Additive-M250.

PRM_44355 [EOS-Additive-M250] — HYD_LOOP_GAIN_15 | Default: 233.500 | Linked Fault: ERR-EOS250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M250.

PRM_44356 [EOS-Additive-M250] — XAX_LOOP_GAIN_16 | Default: 82.400 | Linked Fault: ERR-EOS250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M250.

CNC/PLC SERVO PARAMETER DICTIONARY — EOS-Additive-M250 (BANK 3)

Machine Model: EOS-Additive-M250 | Controller: EOSYSTEM Dual-PLC. Use these parameter registers when diagnosing servo following errors or thermal drift on EOS-Additive-M250.

PRM_44361 [EOS-Additive-M250] — CLN_LOOP_GAIN_1 | Default: 148.350 | Linked Fault: ERR-EOS250-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on EOS-Additive-M250.

PRM_44362 [EOS-Additive-M250] — ZAX_LOOP_GAIN_2 | Default: 46.700 | Linked Fault: ERR-EOS250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M250.

PRM_44363 [EOS-Additive-M250] — OPT_LOOP_GAIN_3 | Default: 195.050 | Linked Fault: ERR-EOS250-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on EOS-Additive-M250.

PRM_44364 [EOS-Additive-M250] — VAC_LOOP_GAIN_4 | Default: 93.400 | Linked Fault: ERR-EOS250-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M250.

PRM_44365 [EOS-Additive-M250] — ATC_LOOP_GAIN_5 | Default: 241.750 | Linked Fault: ERR-EOS250-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on EOS-Additive-M250.

PRM_44366 [EOS-Additive-M250] — XAX_LOOP_GAIN_6 | Default: 140.100 | Linked Fault: ERR-EOS250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M250.

PRM_44367 [EOS-Additive-M250] — BAX_LOOP_GAIN_7 | Default: 38.450 | Linked Fault: ERR-EOS250-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on EOS-Additive-M250.

PRM_44368 [EOS-Additive-M250] — PLC_LOOP_GAIN_8 | Default: 186.800 | Linked Fault: ERR-EOS250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M250.

PRM_44369 [EOS-Additive-M250] — THM_LOOP_GAIN_9 | Default: 85.150 | Linked Fault: ERR-EOS250-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on EOS-Additive-M250.

PRM_44370 [EOS-Additive-M250] — HYD_LOOP_GAIN_10 | Default: 233.500 | Linked Fault: ERR-EOS250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M250.

PRM_44371 [EOS-Additive-M250] — YAX_LOOP_GAIN_11 | Default: 131.850 | Linked Fault: ERR-EOS250-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on EOS-Additive-M250.

PRM_44372 [EOS-Additive-M250] — CAX_LOOP_GAIN_12 | Default: 30.200 | Linked Fault: ERR-EOS250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M250.

PRM_44373 [EOS-Additive-M250] — LUB_LOOP_GAIN_13 | Default: 178.550 | Linked Fault: ERR-EOS250-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on EOS-Additive-M250.

PRM_44374 [EOS-Additive-M250] — SPN_LOOP_GAIN_14 | Default: 76.900 | Linked Fault: ERR-EOS250-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on EOS-Additive-M250.

PRM_44375 [EOS-Additive-M250] — CLN_LOOP_GAIN_15 | Default: 225.250 | Linked Fault: ERR-EOS250-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on EOS-Additive-M250.

PRM_44376 [EOS-Additive-M250] — ZAX_LOOP_GAIN_16 | Default: 123.600 | Linked Fault: ERR-EOS250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M250.

CNC/PLC SERVO PARAMETER DICTIONARY — EOS-Additive-M250 (BANK 4)

Machine Model: EOS-Additive-M250 | Controller: EOSYSTEM Dual-PLC. Use these parameter registers when diagnosing servo following errors or thermal drift on EOS-Additive-M250.

PRM_44381 [EOS-Additive-M250] — XAX_LOOP_GAIN_1 | Default: 197.800 | Linked Fault: ERR-EOS250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M250.

PRM_44382 [EOS-Additive-M250] — CAX_LOOP_GAIN_2 | Default: 145.600 | Linked Fault: ERR-EOS250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M250.

PRM_44383 [EOS-Additive-M250] — VAC_LOOP_GAIN_3 | Default: 93.400 | Linked Fault: ERR-EOS250-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M250.

PRM_44384 [EOS-Additive-M250] — HYD_LOOP_GAIN_4 | Default: 41.200 | Linked Fault: ERR-EOS250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M250.

PRM_44385 [EOS-Additive-M250] — ZAX_LOOP_GAIN_5 | Default: 239.000 | Linked Fault: ERR-EOS250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M250.

PRM_44386 [EOS-Additive-M250] — PLC_LOOP_GAIN_6 | Default: 186.800 | Linked Fault: ERR-EOS250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M250.

PRM_44387 [EOS-Additive-M250] — SPN_LOOP_GAIN_7 | Default: 134.600 | Linked Fault: ERR-EOS250-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on EOS-Additive-M250.

PRM_44388 [EOS-Additive-M250] — XAX_LOOP_GAIN_8 | Default: 82.400 | Linked Fault: ERR-EOS250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M250.

PRM_44389 [EOS-Additive-M250] — CAX_LOOP_GAIN_9 | Default: 30.200 | Linked Fault: ERR-EOS250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M250.

PRM_44390 [EOS-Additive-M250] — VAC_LOOP_GAIN_10 | Default: 228.000 | Linked Fault: ERR-EOS250-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M250.

PRM_44391 [EOS-Additive-M250] — HYD_LOOP_GAIN_11 | Default: 175.800 | Linked Fault: ERR-EOS250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M250.

PRM_44392 [EOS-Additive-M250] — ZAX_LOOP_GAIN_12 | Default: 123.600 | Linked Fault: ERR-EOS250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M250.

PRM_44393 [EOS-Additive-M250] — PLC_LOOP_GAIN_13 | Default: 71.400 | Linked Fault: ERR-EOS250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M250.

PRM_44394 [EOS-Additive-M250] — SPN_LOOP_GAIN_14 | Default: 19.200 | Linked Fault: ERR-EOS250-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on EOS-Additive-M250.

PRM_44395 [EOS-Additive-M250] — XAX_LOOP_GAIN_15 | Default: 217.000 | Linked Fault: ERR-EOS250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M250.

PRM_44396 [EOS-Additive-M250] — CAX_LOOP_GAIN_16 | Default: 164.800 | Linked Fault: ERR-EOS250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M250.

CNC/PLC SERVO PARAMETER DICTIONARY — EOS-Additive-M250 (BANK 5)

Machine Model: EOS-Additive-M250 | Controller: EOSYSTEM Dual-PLC. Use these parameter registers when diagnosing servo following errors or thermal drift on EOS-Additive-M250.

PRM_44401 [EOS-Additive-M250] — YAX_LOOP_GAIN_1 | Default: 247.250 | Linked Fault: ERR-EOS250-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on EOS-Additive-M250.

PRM_44402 [EOS-Additive-M250] — PLC_LOOP_GAIN_2 | Default: 244.500 | Linked Fault: ERR-EOS250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M250.

PRM_44403 [EOS-Additive-M250] — ATC_LOOP_GAIN_3 | Default: 241.750 | Linked Fault: ERR-EOS250-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on EOS-Additive-M250.

PRM_44404 [EOS-Additive-M250] — ZAX_LOOP_GAIN_4 | Default: 239.000 | Linked Fault: ERR-EOS250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M250.

PRM_44405 [EOS-Additive-M250] — LUB_LOOP_GAIN_5 | Default: 236.250 | Linked Fault: ERR-EOS250-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on EOS-Additive-M250.

PRM_44406 [EOS-Additive-M250] — HYD_LOOP_GAIN_6 | Default: 233.500 | Linked Fault: ERR-EOS250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M250.

PRM_44407 [EOS-Additive-M250] — BAX_LOOP_GAIN_7 | Default: 230.750 | Linked Fault: ERR-EOS250-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on EOS-Additive-M250.

PRM_44408 [EOS-Additive-M250] — VAC_LOOP_GAIN_8 | Default: 228.000 | Linked Fault: ERR-EOS250-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M250.

PRM_44409 [EOS-Additive-M250] — CLN_LOOP_GAIN_9 | Default: 225.250 | Linked Fault: ERR-EOS250-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on EOS-Additive-M250.

PRM_44410 [EOS-Additive-M250] — CAX_LOOP_GAIN_10 | Default: 222.500 | Linked Fault: ERR-EOS250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M250.

PRM_44411 [EOS-Additive-M250] — THM_LOOP_GAIN_11 | Default: 219.750 | Linked Fault: ERR-EOS250-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on EOS-Additive-M250.

PRM_44412 [EOS-Additive-M250] — XAX_LOOP_GAIN_12 | Default: 217.000 | Linked Fault: ERR-EOS250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M250.

PRM_44413 [EOS-Additive-M250] — OPT_LOOP_GAIN_13 | Default: 214.250 | Linked Fault: ERR-EOS250-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on EOS-Additive-M250.

PRM_44414 [EOS-Additive-M250] — SPN_LOOP_GAIN_14 | Default: 211.500 | Linked Fault: ERR-EOS250-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on EOS-Additive-M250.

PRM_44415 [EOS-Additive-M250] — YAX_LOOP_GAIN_15 | Default: 208.750 | Linked Fault: ERR-EOS250-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on EOS-Additive-M250.

PRM_44416 [EOS-Additive-M250] — PLC_LOOP_GAIN_16 | Default: 206.000 | Linked Fault: ERR-EOS250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M250.

CNC/PLC SERVO PARAMETER DICTIONARY — EOS-Additive-M300 (BANK 1)

Machine Model: EOS-Additive-M300 | Controller: EOSYSTEM Dual-PLC. Use these parameter registers when diagnosing servo following errors or thermal drift on EOS-Additive-M300.

PRM_44421 [EOS-Additive-M300] — ATC_LOOP_GAIN_1 | Default: 50.600 | Linked Fault: ERR-EOS300-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on EOS-Additive-M300.

PRM_44422 [EOS-Additive-M300] — HYD_LOOP_GAIN_2 | Default: 101.200 | Linked Fault: ERR-EOS300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M300.

PRM_44423 [EOS-Additive-M300] — CLN_LOOP_GAIN_3 | Default: 151.800 | Linked Fault: ERR-EOS300-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on EOS-Additive-M300.

PRM_44424 [EOS-Additive-M300] — XAX_LOOP_GAIN_4 | Default: 202.400 | Linked Fault: ERR-EOS300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M300.

PRM_44425 [EOS-Additive-M300] — YAX_LOOP_GAIN_5 | Default: 3.000 | Linked Fault: ERR-EOS300-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on EOS-Additive-M300.

PRM_44426 [EOS-Additive-M300] — ZAX_LOOP_GAIN_6 | Default: 53.600 | Linked Fault: ERR-EOS300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M300.

PRM_44427 [EOS-Additive-M300] — BAX_LOOP_GAIN_7 | Default: 104.200 | Linked Fault: ERR-EOS300-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on EOS-Additive-M300.

PRM_44428 [EOS-Additive-M300] — CAX_LOOP_GAIN_8 | Default: 154.800 | Linked Fault: ERR-EOS300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M300.

PRM_44429 [EOS-Additive-M300] — OPT_LOOP_GAIN_9 | Default: 205.400 | Linked Fault: ERR-EOS300-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on EOS-Additive-M300.

PRM_44430 [EOS-Additive-M300] — PLC_LOOP_GAIN_10 | Default: 6.000 | Linked Fault: ERR-EOS300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M300.

PRM_44431 [EOS-Additive-M300] — LUB_LOOP_GAIN_11 | Default: 56.600 | Linked Fault: ERR-EOS300-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on EOS-Additive-M300.

PRM_44432 [EOS-Additive-M300] — VAC_LOOP_GAIN_12 | Default: 107.200 | Linked Fault: ERR-EOS300-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M300.

PRM_44433 [EOS-Additive-M300] — THM_LOOP_GAIN_13 | Default: 157.800 | Linked Fault: ERR-EOS300-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on EOS-Additive-M300.

PRM_44434 [EOS-Additive-M300] — SPN_LOOP_GAIN_14 | Default: 208.400 | Linked Fault: ERR-EOS300-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on EOS-Additive-M300.

PRM_44435 [EOS-Additive-M300] — ATC_LOOP_GAIN_15 | Default: 9.000 | Linked Fault: ERR-EOS300-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on EOS-Additive-M300.

PRM_44436 [EOS-Additive-M300] — HYD_LOOP_GAIN_16 | Default: 59.600 | Linked Fault: ERR-EOS300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M300.

CNC/PLC SERVO PARAMETER DICTIONARY — EOS-Additive-M300 (BANK 2)

Machine Model: EOS-Additive-M300 | Controller: EOSYSTEM Dual-PLC. Use these parameter registers when diagnosing servo following errors or thermal drift on EOS-Additive-M300.

PRM_44441 [EOS-Additive-M300] — HYD_LOOP_GAIN_1 | Default: 101.200 | Linked Fault: ERR-EOS300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M300.

PRM_44442 [EOS-Additive-M300] — XAX_LOOP_GAIN_2 | Default: 202.400 | Linked Fault: ERR-EOS300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M300.

PRM_44443 [EOS-Additive-M300] — ZAX_LOOP_GAIN_3 | Default: 53.600 | Linked Fault: ERR-EOS300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M300.

PRM_44444 [EOS-Additive-M300] — CAX_LOOP_GAIN_4 | Default: 154.800 | Linked Fault: ERR-EOS300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M300.

PRM_44445 [EOS-Additive-M300] — PLC_LOOP_GAIN_5 | Default: 6.000 | Linked Fault: ERR-EOS300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M300.

PRM_44446 [EOS-Additive-M300] — VAC_LOOP_GAIN_6 | Default: 107.200 | Linked Fault: ERR-EOS300-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M300.

PRM_44447 [EOS-Additive-M300] — SPN_LOOP_GAIN_7 | Default: 208.400 | Linked Fault: ERR-EOS300-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on EOS-Additive-M300.

PRM_44448 [EOS-Additive-M300] — HYD_LOOP_GAIN_8 | Default: 59.600 | Linked Fault: ERR-EOS300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M300.

PRM_44449 [EOS-Additive-M300] — XAX_LOOP_GAIN_9 | Default: 160.800 | Linked Fault: ERR-EOS300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M300.

PRM_44450 [EOS-Additive-M300] — ZAX_LOOP_GAIN_10 | Default: 12.000 | Linked Fault: ERR-EOS300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M300.

PRM_44451 [EOS-Additive-M300] — CAX_LOOP_GAIN_11 | Default: 113.200 | Linked Fault: ERR-EOS300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M300.

PRM_44452 [EOS-Additive-M300] — PLC_LOOP_GAIN_12 | Default: 214.400 | Linked Fault: ERR-EOS300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M300.

PRM_44453 [EOS-Additive-M300] — VAC_LOOP_GAIN_13 | Default: 65.600 | Linked Fault: ERR-EOS300-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M300.

PRM_44454 [EOS-Additive-M300] — SPN_LOOP_GAIN_14 | Default: 166.800 | Linked Fault: ERR-EOS300-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on EOS-Additive-M300.

PRM_44455 [EOS-Additive-M300] — HYD_LOOP_GAIN_15 | Default: 18.000 | Linked Fault: ERR-EOS300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M300.

PRM_44456 [EOS-Additive-M300] — XAX_LOOP_GAIN_16 | Default: 119.200 | Linked Fault: ERR-EOS300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M300.

CNC/PLC SERVO PARAMETER DICTIONARY — EOS-Additive-M300 (BANK 3)

Machine Model: EOS-Additive-M300 | Controller: EOSYSTEM Dual-PLC. Use these parameter registers when diagnosing servo following errors or thermal drift on EOS-Additive-M300.

PRM_44461 [EOS-Additive-M300] — CLN_LOOP_GAIN_1 | Default: 151.800 | Linked Fault: ERR-EOS300-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on EOS-Additive-M300.

PRM_44462 [EOS-Additive-M300] — ZAX_LOOP_GAIN_2 | Default: 53.600 | Linked Fault: ERR-EOS300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M300.

PRM_44463 [EOS-Additive-M300] — OPT_LOOP_GAIN_3 | Default: 205.400 | Linked Fault: ERR-EOS300-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on EOS-Additive-M300.

PRM_44464 [EOS-Additive-M300] — VAC_LOOP_GAIN_4 | Default: 107.200 | Linked Fault: ERR-EOS300-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M300.

PRM_44465 [EOS-Additive-M300] — ATC_LOOP_GAIN_5 | Default: 9.000 | Linked Fault: ERR-EOS300-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on EOS-Additive-M300.

PRM_44466 [EOS-Additive-M300] — XAX_LOOP_GAIN_6 | Default: 160.800 | Linked Fault: ERR-EOS300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M300.

PRM_44467 [EOS-Additive-M300] — BAX_LOOP_GAIN_7 | Default: 62.600 | Linked Fault: ERR-EOS300-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on EOS-Additive-M300.

PRM_44468 [EOS-Additive-M300] — PLC_LOOP_GAIN_8 | Default: 214.400 | Linked Fault: ERR-EOS300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M300.

PRM_44469 [EOS-Additive-M300] — THM_LOOP_GAIN_9 | Default: 116.200 | Linked Fault: ERR-EOS300-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on EOS-Additive-M300.

PRM_44470 [EOS-Additive-M300] — HYD_LOOP_GAIN_10 | Default: 18.000 | Linked Fault: ERR-EOS300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M300.

PRM_44471 [EOS-Additive-M300] — YAX_LOOP_GAIN_11 | Default: 169.800 | Linked Fault: ERR-EOS300-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on EOS-Additive-M300.

PRM_44472 [EOS-Additive-M300] — CAX_LOOP_GAIN_12 | Default: 71.600 | Linked Fault: ERR-EOS300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M300.

PRM_44473 [EOS-Additive-M300] — LUB_LOOP_GAIN_13 | Default: 223.400 | Linked Fault: ERR-EOS300-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on EOS-Additive-M300.

PRM_44474 [EOS-Additive-M300] — SPN_LOOP_GAIN_14 | Default: 125.200 | Linked Fault: ERR-EOS300-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on EOS-Additive-M300.

PRM_44475 [EOS-Additive-M300] — CLN_LOOP_GAIN_15 | Default: 27.000 | Linked Fault: ERR-EOS300-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on EOS-Additive-M300.

PRM_44476 [EOS-Additive-M300] — ZAX_LOOP_GAIN_16 | Default: 178.800 | Linked Fault: ERR-EOS300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M300.

CNC/PLC SERVO PARAMETER DICTIONARY — EOS-Additive-M300 (BANK 4)

Machine Model: EOS-Additive-M300 | Controller: EOSYSTEM Dual-PLC. Use these parameter registers when diagnosing servo following errors or thermal drift on EOS-Additive-M300.

PRM_44481 [EOS-Additive-M300] — XAX_LOOP_GAIN_1 | Default: 202.400 | Linked Fault: ERR-EOS300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M300.

PRM_44482 [EOS-Additive-M300] — CAX_LOOP_GAIN_2 | Default: 154.800 | Linked Fault: ERR-EOS300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M300.

PRM_44483 [EOS-Additive-M300] — VAC_LOOP_GAIN_3 | Default: 107.200 | Linked Fault: ERR-EOS300-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M300.

PRM_44484 [EOS-Additive-M300] — HYD_LOOP_GAIN_4 | Default: 59.600 | Linked Fault: ERR-EOS300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M300.

PRM_44485 [EOS-Additive-M300] — ZAX_LOOP_GAIN_5 | Default: 12.000 | Linked Fault: ERR-EOS300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M300.

PRM_44486 [EOS-Additive-M300] — PLC_LOOP_GAIN_6 | Default: 214.400 | Linked Fault: ERR-EOS300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M300.

PRM_44487 [EOS-Additive-M300] — SPN_LOOP_GAIN_7 | Default: 166.800 | Linked Fault: ERR-EOS300-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on EOS-Additive-M300.

PRM_44488 [EOS-Additive-M300] — XAX_LOOP_GAIN_8 | Default: 119.200 | Linked Fault: ERR-EOS300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M300.

PRM_44489 [EOS-Additive-M300] — CAX_LOOP_GAIN_9 | Default: 71.600 | Linked Fault: ERR-EOS300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M300.

PRM_44490 [EOS-Additive-M300] — VAC_LOOP_GAIN_10 | Default: 24.000 | Linked Fault: ERR-EOS300-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M300.

PRM_44491 [EOS-Additive-M300] — HYD_LOOP_GAIN_11 | Default: 226.400 | Linked Fault: ERR-EOS300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M300.

PRM_44492 [EOS-Additive-M300] — ZAX_LOOP_GAIN_12 | Default: 178.800 | Linked Fault: ERR-EOS300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M300.

PRM_44493 [EOS-Additive-M300] — PLC_LOOP_GAIN_13 | Default: 131.200 | Linked Fault: ERR-EOS300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M300.

PRM_44494 [EOS-Additive-M300] — SPN_LOOP_GAIN_14 | Default: 83.600 | Linked Fault: ERR-EOS300-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on EOS-Additive-M300.

PRM_44495 [EOS-Additive-M300] — XAX_LOOP_GAIN_15 | Default: 36.000 | Linked Fault: ERR-EOS300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M300.

PRM_44496 [EOS-Additive-M300] — CAX_LOOP_GAIN_16 | Default: 238.400 | Linked Fault: ERR-EOS300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M300.

CNC/PLC SERVO PARAMETER DICTIONARY — EOS-Additive-M300 (BANK 5)

Machine Model: EOS-Additive-M300 | Controller: EOSYSTEM Dual-PLC. Use these parameter registers when diagnosing servo following errors or thermal drift on EOS-Additive-M300.

PRM_44501 [EOS-Additive-M300] — YAX_LOOP_GAIN_1 | Default: 3.000 | Linked Fault: ERR-EOS300-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on EOS-Additive-M300.

PRM_44502 [EOS-Additive-M300] — PLC_LOOP_GAIN_2 | Default: 6.000 | Linked Fault: ERR-EOS300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M300.

PRM_44503 [EOS-Additive-M300] — ATC_LOOP_GAIN_3 | Default: 9.000 | Linked Fault: ERR-EOS300-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on EOS-Additive-M300.

PRM_44504 [EOS-Additive-M300] — ZAX_LOOP_GAIN_4 | Default: 12.000 | Linked Fault: ERR-EOS300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M300.

PRM_44505 [EOS-Additive-M300] — LUB_LOOP_GAIN_5 | Default: 15.000 | Linked Fault: ERR-EOS300-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on EOS-Additive-M300.

PRM_44506 [EOS-Additive-M300] — HYD_LOOP_GAIN_6 | Default: 18.000 | Linked Fault: ERR-EOS300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M300.

PRM_44507 [EOS-Additive-M300] — BAX_LOOP_GAIN_7 | Default: 21.000 | Linked Fault: ERR-EOS300-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on EOS-Additive-M300.

PRM_44508 [EOS-Additive-M300] — VAC_LOOP_GAIN_8 | Default: 24.000 | Linked Fault: ERR-EOS300-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M300.

PRM_44509 [EOS-Additive-M300] — CLN_LOOP_GAIN_9 | Default: 27.000 | Linked Fault: ERR-EOS300-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on EOS-Additive-M300.

PRM_44510 [EOS-Additive-M300] — CAX_LOOP_GAIN_10 | Default: 30.000 | Linked Fault: ERR-EOS300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M300.

PRM_44511 [EOS-Additive-M300] — THM_LOOP_GAIN_11 | Default: 33.000 | Linked Fault: ERR-EOS300-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on EOS-Additive-M300.

PRM_44512 [EOS-Additive-M300] — XAX_LOOP_GAIN_12 | Default: 36.000 | Linked Fault: ERR-EOS300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M300.

PRM_44513 [EOS-Additive-M300] — OPT_LOOP_GAIN_13 | Default: 39.000 | Linked Fault: ERR-EOS300-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on EOS-Additive-M300.

PRM_44514 [EOS-Additive-M300] — SPN_LOOP_GAIN_14 | Default: 42.000 | Linked Fault: ERR-EOS300-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on EOS-Additive-M300.

PRM_44515 [EOS-Additive-M300] — YAX_LOOP_GAIN_15 | Default: 45.000 | Linked Fault: ERR-EOS300-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on EOS-Additive-M300.

PRM_44516 [EOS-Additive-M300] — PLC_LOOP_GAIN_16 | Default: 48.000 | Linked Fault: ERR-EOS300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M300.

CNC/PLC SERVO PARAMETER DICTIONARY — EOS-Additive-M350 (BANK 1)

Machine Model: EOS-Additive-M350 | Controller: EOSYSTEM Dual-PLC. Use these parameter registers when diagnosing servo following errors or thermal drift on EOS-Additive-M350.

PRM_44521 [EOS-Additive-M350] — ATC_LOOP_GAIN_1 | Default: 51.750 | Linked Fault: ERR-EOS350-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on EOS-Additive-M350.

PRM_44522 [EOS-Additive-M350] — HYD_LOOP_GAIN_2 | Default: 103.500 | Linked Fault: ERR-EOS350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M350.

PRM_44523 [EOS-Additive-M350] — CLN_LOOP_GAIN_3 | Default: 155.250 | Linked Fault: ERR-EOS350-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on EOS-Additive-M350.

PRM_44524 [EOS-Additive-M350] — XAX_LOOP_GAIN_4 | Default: 207.000 | Linked Fault: ERR-EOS350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M350.

PRM_44525 [EOS-Additive-M350] — YAX_LOOP_GAIN_5 | Default: 8.750 | Linked Fault: ERR-EOS350-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on EOS-Additive-M350.

PRM_44526 [EOS-Additive-M350] — ZAX_LOOP_GAIN_6 | Default: 60.500 | Linked Fault: ERR-EOS350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M350.

PRM_44527 [EOS-Additive-M350] — BAX_LOOP_GAIN_7 | Default: 112.250 | Linked Fault: ERR-EOS350-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on EOS-Additive-M350.

PRM_44528 [EOS-Additive-M350] — CAX_LOOP_GAIN_8 | Default: 164.000 | Linked Fault: ERR-EOS350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M350.

PRM_44529 [EOS-Additive-M350] — OPT_LOOP_GAIN_9 | Default: 215.750 | Linked Fault: ERR-EOS350-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on EOS-Additive-M350.

PRM_44530 [EOS-Additive-M350] — PLC_LOOP_GAIN_10 | Default: 17.500 | Linked Fault: ERR-EOS350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M350.

PRM_44531 [EOS-Additive-M350] — LUB_LOOP_GAIN_11 | Default: 69.250 | Linked Fault: ERR-EOS350-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on EOS-Additive-M350.

PRM_44532 [EOS-Additive-M350] — VAC_LOOP_GAIN_12 | Default: 121.000 | Linked Fault: ERR-EOS350-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M350.

PRM_44533 [EOS-Additive-M350] — THM_LOOP_GAIN_13 | Default: 172.750 | Linked Fault: ERR-EOS350-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on EOS-Additive-M350.

PRM_44534 [EOS-Additive-M350] — SPN_LOOP_GAIN_14 | Default: 224.500 | Linked Fault: ERR-EOS350-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on EOS-Additive-M350.

PRM_44535 [EOS-Additive-M350] — ATC_LOOP_GAIN_15 | Default: 26.250 | Linked Fault: ERR-EOS350-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on EOS-Additive-M350.

PRM_44536 [EOS-Additive-M350] — HYD_LOOP_GAIN_16 | Default: 78.000 | Linked Fault: ERR-EOS350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M350.

CNC/PLC SERVO PARAMETER DICTIONARY — EOS-Additive-M350 (BANK 2)

Machine Model: EOS-Additive-M350 | Controller: EOSYSTEM Dual-PLC. Use these parameter registers when diagnosing servo following errors or thermal drift on EOS-Additive-M350.

PRM_44541 [EOS-Additive-M350] — HYD_LOOP_GAIN_1 | Default: 103.500 | Linked Fault: ERR-EOS350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M350.

PRM_44542 [EOS-Additive-M350] — XAX_LOOP_GAIN_2 | Default: 207.000 | Linked Fault: ERR-EOS350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M350.

PRM_44543 [EOS-Additive-M350] — ZAX_LOOP_GAIN_3 | Default: 60.500 | Linked Fault: ERR-EOS350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M350.

PRM_44544 [EOS-Additive-M350] — CAX_LOOP_GAIN_4 | Default: 164.000 | Linked Fault: ERR-EOS350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M350.

PRM_44545 [EOS-Additive-M350] — PLC_LOOP_GAIN_5 | Default: 17.500 | Linked Fault: ERR-EOS350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M350.

PRM_44546 [EOS-Additive-M350] — VAC_LOOP_GAIN_6 | Default: 121.000 | Linked Fault: ERR-EOS350-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M350.

PRM_44547 [EOS-Additive-M350] — SPN_LOOP_GAIN_7 | Default: 224.500 | Linked Fault: ERR-EOS350-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on EOS-Additive-M350.

PRM_44548 [EOS-Additive-M350] — HYD_LOOP_GAIN_8 | Default: 78.000 | Linked Fault: ERR-EOS350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M350.

PRM_44549 [EOS-Additive-M350] — XAX_LOOP_GAIN_9 | Default: 181.500 | Linked Fault: ERR-EOS350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M350.

PRM_44550 [EOS-Additive-M350] — ZAX_LOOP_GAIN_10 | Default: 35.000 | Linked Fault: ERR-EOS350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M350.

PRM_44551 [EOS-Additive-M350] — CAX_LOOP_GAIN_11 | Default: 138.500 | Linked Fault: ERR-EOS350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M350.

PRM_44552 [EOS-Additive-M350] — PLC_LOOP_GAIN_12 | Default: 242.000 | Linked Fault: ERR-EOS350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M350.

PRM_44553 [EOS-Additive-M350] — VAC_LOOP_GAIN_13 | Default: 95.500 | Linked Fault: ERR-EOS350-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M350.

PRM_44554 [EOS-Additive-M350] — SPN_LOOP_GAIN_14 | Default: 199.000 | Linked Fault: ERR-EOS350-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on EOS-Additive-M350.

PRM_44555 [EOS-Additive-M350] — HYD_LOOP_GAIN_15 | Default: 52.500 | Linked Fault: ERR-EOS350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M350.

PRM_44556 [EOS-Additive-M350] — XAX_LOOP_GAIN_16 | Default: 156.000 | Linked Fault: ERR-EOS350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M350.

CNC/PLC SERVO PARAMETER DICTIONARY — EOS-Additive-M350 (BANK 3)

Machine Model: EOS-Additive-M350 | Controller: EOSYSTEM Dual-PLC. Use these parameter registers when diagnosing servo following errors or thermal drift on EOS-Additive-M350.

PRM_44561 [EOS-Additive-M350] — CLN_LOOP_GAIN_1 | Default: 155.250 | Linked Fault: ERR-EOS350-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on EOS-Additive-M350.

PRM_44562 [EOS-Additive-M350] — ZAX_LOOP_GAIN_2 | Default: 60.500 | Linked Fault: ERR-EOS350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M350.

PRM_44563 [EOS-Additive-M350] — OPT_LOOP_GAIN_3 | Default: 215.750 | Linked Fault: ERR-EOS350-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on EOS-Additive-M350.

PRM_44564 [EOS-Additive-M350] — VAC_LOOP_GAIN_4 | Default: 121.000 | Linked Fault: ERR-EOS350-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M350.

PRM_44565 [EOS-Additive-M350] — ATC_LOOP_GAIN_5 | Default: 26.250 | Linked Fault: ERR-EOS350-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on EOS-Additive-M350.

PRM_44566 [EOS-Additive-M350] — XAX_LOOP_GAIN_6 | Default: 181.500 | Linked Fault: ERR-EOS350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M350.

PRM_44567 [EOS-Additive-M350] — BAX_LOOP_GAIN_7 | Default: 86.750 | Linked Fault: ERR-EOS350-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on EOS-Additive-M350.

PRM_44568 [EOS-Additive-M350] — PLC_LOOP_GAIN_8 | Default: 242.000 | Linked Fault: ERR-EOS350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M350.

PRM_44569 [EOS-Additive-M350] — THM_LOOP_GAIN_9 | Default: 147.250 | Linked Fault: ERR-EOS350-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on EOS-Additive-M350.

PRM_44570 [EOS-Additive-M350] — HYD_LOOP_GAIN_10 | Default: 52.500 | Linked Fault: ERR-EOS350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M350.

PRM_44571 [EOS-Additive-M350] — YAX_LOOP_GAIN_11 | Default: 207.750 | Linked Fault: ERR-EOS350-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on EOS-Additive-M350.

PRM_44572 [EOS-Additive-M350] — CAX_LOOP_GAIN_12 | Default: 113.000 | Linked Fault: ERR-EOS350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M350.

PRM_44573 [EOS-Additive-M350] — LUB_LOOP_GAIN_13 | Default: 18.250 | Linked Fault: ERR-EOS350-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on EOS-Additive-M350.

PRM_44574 [EOS-Additive-M350] — SPN_LOOP_GAIN_14 | Default: 173.500 | Linked Fault: ERR-EOS350-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on EOS-Additive-M350.

PRM_44575 [EOS-Additive-M350] — CLN_LOOP_GAIN_15 | Default: 78.750 | Linked Fault: ERR-EOS350-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on EOS-Additive-M350.

PRM_44576 [EOS-Additive-M350] — ZAX_LOOP_GAIN_16 | Default: 234.000 | Linked Fault: ERR-EOS350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M350.

CNC/PLC SERVO PARAMETER DICTIONARY — EOS-Additive-M350 (BANK 4)

Machine Model: EOS-Additive-M350 | Controller: EOSYSTEM Dual-PLC. Use these parameter registers when diagnosing servo following errors or thermal drift on EOS-Additive-M350.

PRM_44581 [EOS-Additive-M350] — XAX_LOOP_GAIN_1 | Default: 207.000 | Linked Fault: ERR-EOS350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M350.

PRM_44582 [EOS-Additive-M350] — CAX_LOOP_GAIN_2 | Default: 164.000 | Linked Fault: ERR-EOS350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M350.

PRM_44583 [EOS-Additive-M350] — VAC_LOOP_GAIN_3 | Default: 121.000 | Linked Fault: ERR-EOS350-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M350.

PRM_44584 [EOS-Additive-M350] — HYD_LOOP_GAIN_4 | Default: 78.000 | Linked Fault: ERR-EOS350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M350.

PRM_44585 [EOS-Additive-M350] — ZAX_LOOP_GAIN_5 | Default: 35.000 | Linked Fault: ERR-EOS350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M350.

PRM_44586 [EOS-Additive-M350] — PLC_LOOP_GAIN_6 | Default: 242.000 | Linked Fault: ERR-EOS350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M350.

PRM_44587 [EOS-Additive-M350] — SPN_LOOP_GAIN_7 | Default: 199.000 | Linked Fault: ERR-EOS350-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on EOS-Additive-M350.

PRM_44588 [EOS-Additive-M350] — XAX_LOOP_GAIN_8 | Default: 156.000 | Linked Fault: ERR-EOS350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M350.

PRM_44589 [EOS-Additive-M350] — CAX_LOOP_GAIN_9 | Default: 113.000 | Linked Fault: ERR-EOS350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M350.

PRM_44590 [EOS-Additive-M350] — VAC_LOOP_GAIN_10 | Default: 70.000 | Linked Fault: ERR-EOS350-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M350.

PRM_44591 [EOS-Additive-M350] — HYD_LOOP_GAIN_11 | Default: 27.000 | Linked Fault: ERR-EOS350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M350.

PRM_44592 [EOS-Additive-M350] — ZAX_LOOP_GAIN_12 | Default: 234.000 | Linked Fault: ERR-EOS350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M350.

PRM_44593 [EOS-Additive-M350] — PLC_LOOP_GAIN_13 | Default: 191.000 | Linked Fault: ERR-EOS350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M350.

PRM_44594 [EOS-Additive-M350] — SPN_LOOP_GAIN_14 | Default: 148.000 | Linked Fault: ERR-EOS350-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on EOS-Additive-M350.

PRM_44595 [EOS-Additive-M350] — XAX_LOOP_GAIN_15 | Default: 105.000 | Linked Fault: ERR-EOS350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M350.

PRM_44596 [EOS-Additive-M350] — CAX_LOOP_GAIN_16 | Default: 62.000 | Linked Fault: ERR-EOS350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M350.

CNC/PLC SERVO PARAMETER DICTIONARY — EOS-Additive-M350 (BANK 5)

Machine Model: EOS-Additive-M350 | Controller: EOSYSTEM Dual-PLC. Use these parameter registers when diagnosing servo following errors or thermal drift on EOS-Additive-M350.

PRM_44601 [EOS-Additive-M350] — YAX_LOOP_GAIN_1 | Default: 8.750 | Linked Fault: ERR-EOS350-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on EOS-Additive-M350.

PRM_44602 [EOS-Additive-M350] — PLC_LOOP_GAIN_2 | Default: 17.500 | Linked Fault: ERR-EOS350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M350.

PRM_44603 [EOS-Additive-M350] — ATC_LOOP_GAIN_3 | Default: 26.250 | Linked Fault: ERR-EOS350-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on EOS-Additive-M350.

PRM_44604 [EOS-Additive-M350] — ZAX_LOOP_GAIN_4 | Default: 35.000 | Linked Fault: ERR-EOS350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M350.

PRM_44605 [EOS-Additive-M350] — LUB_LOOP_GAIN_5 | Default: 43.750 | Linked Fault: ERR-EOS350-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on EOS-Additive-M350.

PRM_44606 [EOS-Additive-M350] — HYD_LOOP_GAIN_6 | Default: 52.500 | Linked Fault: ERR-EOS350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M350.

PRM_44607 [EOS-Additive-M350] — BAX_LOOP_GAIN_7 | Default: 61.250 | Linked Fault: ERR-EOS350-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on EOS-Additive-M350.

PRM_44608 [EOS-Additive-M350] — VAC_LOOP_GAIN_8 | Default: 70.000 | Linked Fault: ERR-EOS350-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M350.

PRM_44609 [EOS-Additive-M350] — CLN_LOOP_GAIN_9 | Default: 78.750 | Linked Fault: ERR-EOS350-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on EOS-Additive-M350.

PRM_44610 [EOS-Additive-M350] — CAX_LOOP_GAIN_10 | Default: 87.500 | Linked Fault: ERR-EOS350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M350.

PRM_44611 [EOS-Additive-M350] — THM_LOOP_GAIN_11 | Default: 96.250 | Linked Fault: ERR-EOS350-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on EOS-Additive-M350.

PRM_44612 [EOS-Additive-M350] — XAX_LOOP_GAIN_12 | Default: 105.000 | Linked Fault: ERR-EOS350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M350.

PRM_44613 [EOS-Additive-M350] — OPT_LOOP_GAIN_13 | Default: 113.750 | Linked Fault: ERR-EOS350-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on EOS-Additive-M350.

PRM_44614 [EOS-Additive-M350] — SPN_LOOP_GAIN_14 | Default: 122.500 | Linked Fault: ERR-EOS350-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on EOS-Additive-M350.

PRM_44615 [EOS-Additive-M350] — YAX_LOOP_GAIN_15 | Default: 131.250 | Linked Fault: ERR-EOS350-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on EOS-Additive-M350.

PRM_44616 [EOS-Additive-M350] — PLC_LOOP_GAIN_16 | Default: 140.000 | Linked Fault: ERR-EOS350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M350.

CNC/PLC SERVO PARAMETER DICTIONARY — EOS-Additive-M400 (BANK 1)

Machine Model: EOS-Additive-M400 | Controller: EOSYSTEM Dual-PLC. Use these parameter registers when diagnosing servo following errors or thermal drift on EOS-Additive-M400.

PRM_44621 [EOS-Additive-M400] — ATC_LOOP_GAIN_1 | Default: 52.900 | Linked Fault: ERR-EOS400-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on EOS-Additive-M400.

PRM_44622 [EOS-Additive-M400] — HYD_LOOP_GAIN_2 | Default: 105.800 | Linked Fault: ERR-EOS400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M400.

PRM_44623 [EOS-Additive-M400] — CLN_LOOP_GAIN_3 | Default: 158.700 | Linked Fault: ERR-EOS400-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on EOS-Additive-M400.

PRM_44624 [EOS-Additive-M400] — XAX_LOOP_GAIN_4 | Default: 211.600 | Linked Fault: ERR-EOS400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M400.

PRM_44625 [EOS-Additive-M400] — YAX_LOOP_GAIN_5 | Default: 14.500 | Linked Fault: ERR-EOS400-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on EOS-Additive-M400.

PRM_44626 [EOS-Additive-M400] — ZAX_LOOP_GAIN_6 | Default: 67.400 | Linked Fault: ERR-EOS400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M400.

PRM_44627 [EOS-Additive-M400] — BAX_LOOP_GAIN_7 | Default: 120.300 | Linked Fault: ERR-EOS400-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on EOS-Additive-M400.

PRM_44628 [EOS-Additive-M400] — CAX_LOOP_GAIN_8 | Default: 173.200 | Linked Fault: ERR-EOS400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M400.

PRM_44629 [EOS-Additive-M400] — OPT_LOOP_GAIN_9 | Default: 226.100 | Linked Fault: ERR-EOS400-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on EOS-Additive-M400.

PRM_44630 [EOS-Additive-M400] — PLC_LOOP_GAIN_10 | Default: 29.000 | Linked Fault: ERR-EOS400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M400.

PRM_44631 [EOS-Additive-M400] — LUB_LOOP_GAIN_11 | Default: 81.900 | Linked Fault: ERR-EOS400-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on EOS-Additive-M400.

PRM_44632 [EOS-Additive-M400] — VAC_LOOP_GAIN_12 | Default: 134.800 | Linked Fault: ERR-EOS400-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M400.

PRM_44633 [EOS-Additive-M400] — THM_LOOP_GAIN_13 | Default: 187.700 | Linked Fault: ERR-EOS400-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on EOS-Additive-M400.

PRM_44634 [EOS-Additive-M400] — SPN_LOOP_GAIN_14 | Default: 240.600 | Linked Fault: ERR-EOS400-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on EOS-Additive-M400.

PRM_44635 [EOS-Additive-M400] — ATC_LOOP_GAIN_15 | Default: 43.500 | Linked Fault: ERR-EOS400-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on EOS-Additive-M400.

PRM_44636 [EOS-Additive-M400] — HYD_LOOP_GAIN_16 | Default: 96.400 | Linked Fault: ERR-EOS400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M400.

CNC/PLC SERVO PARAMETER DICTIONARY — EOS-Additive-M400 (BANK 2)

Machine Model: EOS-Additive-M400 | Controller: EOSYSTEM Dual-PLC. Use these parameter registers when diagnosing servo following errors or thermal drift on EOS-Additive-M400.

PRM_44641 [EOS-Additive-M400] — HYD_LOOP_GAIN_1 | Default: 105.800 | Linked Fault: ERR-EOS400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M400.

PRM_44642 [EOS-Additive-M400] — XAX_LOOP_GAIN_2 | Default: 211.600 | Linked Fault: ERR-EOS400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M400.

PRM_44643 [EOS-Additive-M400] — ZAX_LOOP_GAIN_3 | Default: 67.400 | Linked Fault: ERR-EOS400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M400.

PRM_44644 [EOS-Additive-M400] — CAX_LOOP_GAIN_4 | Default: 173.200 | Linked Fault: ERR-EOS400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M400.

PRM_44645 [EOS-Additive-M400] — PLC_LOOP_GAIN_5 | Default: 29.000 | Linked Fault: ERR-EOS400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M400.

PRM_44646 [EOS-Additive-M400] — VAC_LOOP_GAIN_6 | Default: 134.800 | Linked Fault: ERR-EOS400-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M400.

PRM_44647 [EOS-Additive-M400] — SPN_LOOP_GAIN_7 | Default: 240.600 | Linked Fault: ERR-EOS400-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on EOS-Additive-M400.

PRM_44648 [EOS-Additive-M400] — HYD_LOOP_GAIN_8 | Default: 96.400 | Linked Fault: ERR-EOS400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M400.

PRM_44649 [EOS-Additive-M400] — XAX_LOOP_GAIN_9 | Default: 202.200 | Linked Fault: ERR-EOS400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M400.

PRM_44650 [EOS-Additive-M400] — ZAX_LOOP_GAIN_10 | Default: 58.000 | Linked Fault: ERR-EOS400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M400.

PRM_44651 [EOS-Additive-M400] — CAX_LOOP_GAIN_11 | Default: 163.800 | Linked Fault: ERR-EOS400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M400.

PRM_44652 [EOS-Additive-M400] — PLC_LOOP_GAIN_12 | Default: 19.600 | Linked Fault: ERR-EOS400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M400.

PRM_44653 [EOS-Additive-M400] — VAC_LOOP_GAIN_13 | Default: 125.400 | Linked Fault: ERR-EOS400-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M400.

PRM_44654 [EOS-Additive-M400] — SPN_LOOP_GAIN_14 | Default: 231.200 | Linked Fault: ERR-EOS400-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on EOS-Additive-M400.

PRM_44655 [EOS-Additive-M400] — HYD_LOOP_GAIN_15 | Default: 87.000 | Linked Fault: ERR-EOS400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M400.

PRM_44656 [EOS-Additive-M400] — XAX_LOOP_GAIN_16 | Default: 192.800 | Linked Fault: ERR-EOS400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M400.

CNC/PLC SERVO PARAMETER DICTIONARY — EOS-Additive-M400 (BANK 3)

Machine Model: EOS-Additive-M400 | Controller: EOSYSTEM Dual-PLC. Use these parameter registers when diagnosing servo following errors or thermal drift on EOS-Additive-M400.

PRM_44661 [EOS-Additive-M400] — CLN_LOOP_GAIN_1 | Default: 158.700 | Linked Fault: ERR-EOS400-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on EOS-Additive-M400.

PRM_44662 [EOS-Additive-M400] — ZAX_LOOP_GAIN_2 | Default: 67.400 | Linked Fault: ERR-EOS400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M400.

PRM_44663 [EOS-Additive-M400] — OPT_LOOP_GAIN_3 | Default: 226.100 | Linked Fault: ERR-EOS400-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on EOS-Additive-M400.

PRM_44664 [EOS-Additive-M400] — VAC_LOOP_GAIN_4 | Default: 134.800 | Linked Fault: ERR-EOS400-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M400.

PRM_44665 [EOS-Additive-M400] — ATC_LOOP_GAIN_5 | Default: 43.500 | Linked Fault: ERR-EOS400-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on EOS-Additive-M400.

PRM_44666 [EOS-Additive-M400] — XAX_LOOP_GAIN_6 | Default: 202.200 | Linked Fault: ERR-EOS400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M400.

PRM_44667 [EOS-Additive-M400] — BAX_LOOP_GAIN_7 | Default: 110.900 | Linked Fault: ERR-EOS400-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on EOS-Additive-M400.

PRM_44668 [EOS-Additive-M400] — PLC_LOOP_GAIN_8 | Default: 19.600 | Linked Fault: ERR-EOS400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M400.

PRM_44669 [EOS-Additive-M400] — THM_LOOP_GAIN_9 | Default: 178.300 | Linked Fault: ERR-EOS400-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on EOS-Additive-M400.

PRM_44670 [EOS-Additive-M400] — HYD_LOOP_GAIN_10 | Default: 87.000 | Linked Fault: ERR-EOS400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M400.

PRM_44671 [EOS-Additive-M400] — YAX_LOOP_GAIN_11 | Default: 245.700 | Linked Fault: ERR-EOS400-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on EOS-Additive-M400.

PRM_44672 [EOS-Additive-M400] — CAX_LOOP_GAIN_12 | Default: 154.400 | Linked Fault: ERR-EOS400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M400.

PRM_44673 [EOS-Additive-M400] — LUB_LOOP_GAIN_13 | Default: 63.100 | Linked Fault: ERR-EOS400-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on EOS-Additive-M400.

PRM_44674 [EOS-Additive-M400] — SPN_LOOP_GAIN_14 | Default: 221.800 | Linked Fault: ERR-EOS400-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on EOS-Additive-M400.

PRM_44675 [EOS-Additive-M400] — CLN_LOOP_GAIN_15 | Default: 130.500 | Linked Fault: ERR-EOS400-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on EOS-Additive-M400.

PRM_44676 [EOS-Additive-M400] — ZAX_LOOP_GAIN_16 | Default: 39.200 | Linked Fault: ERR-EOS400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M400.

CNC/PLC SERVO PARAMETER DICTIONARY — EOS-Additive-M400 (BANK 4)

Machine Model: EOS-Additive-M400 | Controller: EOSYSTEM Dual-PLC. Use these parameter registers when diagnosing servo following errors or thermal drift on EOS-Additive-M400.

PRM_44681 [EOS-Additive-M400] — XAX_LOOP_GAIN_1 | Default: 211.600 | Linked Fault: ERR-EOS400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M400.

PRM_44682 [EOS-Additive-M400] — CAX_LOOP_GAIN_2 | Default: 173.200 | Linked Fault: ERR-EOS400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M400.

PRM_44683 [EOS-Additive-M400] — VAC_LOOP_GAIN_3 | Default: 134.800 | Linked Fault: ERR-EOS400-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M400.

PRM_44684 [EOS-Additive-M400] — HYD_LOOP_GAIN_4 | Default: 96.400 | Linked Fault: ERR-EOS400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M400.

PRM_44685 [EOS-Additive-M400] — ZAX_LOOP_GAIN_5 | Default: 58.000 | Linked Fault: ERR-EOS400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M400.

PRM_44686 [EOS-Additive-M400] — PLC_LOOP_GAIN_6 | Default: 19.600 | Linked Fault: ERR-EOS400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M400.

PRM_44687 [EOS-Additive-M400] — SPN_LOOP_GAIN_7 | Default: 231.200 | Linked Fault: ERR-EOS400-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on EOS-Additive-M400.

PRM_44688 [EOS-Additive-M400] — XAX_LOOP_GAIN_8 | Default: 192.800 | Linked Fault: ERR-EOS400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M400.

PRM_44689 [EOS-Additive-M400] — CAX_LOOP_GAIN_9 | Default: 154.400 | Linked Fault: ERR-EOS400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M400.

PRM_44690 [EOS-Additive-M400] — VAC_LOOP_GAIN_10 | Default: 116.000 | Linked Fault: ERR-EOS400-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M400.

PRM_44691 [EOS-Additive-M400] — HYD_LOOP_GAIN_11 | Default: 77.600 | Linked Fault: ERR-EOS400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M400.

PRM_44692 [EOS-Additive-M400] — ZAX_LOOP_GAIN_12 | Default: 39.200 | Linked Fault: ERR-EOS400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M400.

PRM_44693 [EOS-Additive-M400] — PLC_LOOP_GAIN_13 | Default: 0.800 | Linked Fault: ERR-EOS400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M400.

PRM_44694 [EOS-Additive-M400] — SPN_LOOP_GAIN_14 | Default: 212.400 | Linked Fault: ERR-EOS400-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on EOS-Additive-M400.

PRM_44695 [EOS-Additive-M400] — XAX_LOOP_GAIN_15 | Default: 174.000 | Linked Fault: ERR-EOS400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M400.

PRM_44696 [EOS-Additive-M400] — CAX_LOOP_GAIN_16 | Default: 135.600 | Linked Fault: ERR-EOS400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M400.

CNC/PLC SERVO PARAMETER DICTIONARY — EOS-Additive-M400 (BANK 5)

Machine Model: EOS-Additive-M400 | Controller: EOSYSTEM Dual-PLC. Use these parameter registers when diagnosing servo following errors or thermal drift on EOS-Additive-M400.

PRM_44701 [EOS-Additive-M400] — YAX_LOOP_GAIN_1 | Default: 14.500 | Linked Fault: ERR-EOS400-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on EOS-Additive-M400.

PRM_44702 [EOS-Additive-M400] — PLC_LOOP_GAIN_2 | Default: 29.000 | Linked Fault: ERR-EOS400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M400.

PRM_44703 [EOS-Additive-M400] — ATC_LOOP_GAIN_3 | Default: 43.500 | Linked Fault: ERR-EOS400-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on EOS-Additive-M400.

PRM_44704 [EOS-Additive-M400] — ZAX_LOOP_GAIN_4 | Default: 58.000 | Linked Fault: ERR-EOS400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M400.

PRM_44705 [EOS-Additive-M400] — LUB_LOOP_GAIN_5 | Default: 72.500 | Linked Fault: ERR-EOS400-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on EOS-Additive-M400.

PRM_44706 [EOS-Additive-M400] — HYD_LOOP_GAIN_6 | Default: 87.000 | Linked Fault: ERR-EOS400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M400.

PRM_44707 [EOS-Additive-M400] — BAX_LOOP_GAIN_7 | Default: 101.500 | Linked Fault: ERR-EOS400-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on EOS-Additive-M400.

PRM_44708 [EOS-Additive-M400] — VAC_LOOP_GAIN_8 | Default: 116.000 | Linked Fault: ERR-EOS400-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M400.

PRM_44709 [EOS-Additive-M400] — CLN_LOOP_GAIN_9 | Default: 130.500 | Linked Fault: ERR-EOS400-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on EOS-Additive-M400.

PRM_44710 [EOS-Additive-M400] — CAX_LOOP_GAIN_10 | Default: 145.000 | Linked Fault: ERR-EOS400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M400.

PRM_44711 [EOS-Additive-M400] — THM_LOOP_GAIN_11 | Default: 159.500 | Linked Fault: ERR-EOS400-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on EOS-Additive-M400.

PRM_44712 [EOS-Additive-M400] — XAX_LOOP_GAIN_12 | Default: 174.000 | Linked Fault: ERR-EOS400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M400.

PRM_44713 [EOS-Additive-M400] — OPT_LOOP_GAIN_13 | Default: 188.500 | Linked Fault: ERR-EOS400-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on EOS-Additive-M400.

PRM_44714 [EOS-Additive-M400] — SPN_LOOP_GAIN_14 | Default: 203.000 | Linked Fault: ERR-EOS400-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on EOS-Additive-M400.

PRM_44715 [EOS-Additive-M400] — YAX_LOOP_GAIN_15 | Default: 217.500 | Linked Fault: ERR-EOS400-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on EOS-Additive-M400.

PRM_44716 [EOS-Additive-M400] — PLC_LOOP_GAIN_16 | Default: 232.000 | Linked Fault: ERR-EOS400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M400.

CNC/PLC SERVO PARAMETER DICTIONARY — EOS-Additive-M450 (BANK 1)

Machine Model: EOS-Additive-M450 | Controller: EOSYSTEM Dual-PLC. Use these parameter registers when diagnosing servo following errors or thermal drift on EOS-Additive-M450.

PRM_44721 [EOS-Additive-M450] — ATC_LOOP_GAIN_1 | Default: 54.050 | Linked Fault: ERR-EOS450-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on EOS-Additive-M450.

PRM_44722 [EOS-Additive-M450] — HYD_LOOP_GAIN_2 | Default: 108.100 | Linked Fault: ERR-EOS450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M450.

PRM_44723 [EOS-Additive-M450] — CLN_LOOP_GAIN_3 | Default: 162.150 | Linked Fault: ERR-EOS450-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on EOS-Additive-M450.

PRM_44724 [EOS-Additive-M450] — XAX_LOOP_GAIN_4 | Default: 216.200 | Linked Fault: ERR-EOS450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M450.

PRM_44725 [EOS-Additive-M450] — YAX_LOOP_GAIN_5 | Default: 20.250 | Linked Fault: ERR-EOS450-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on EOS-Additive-M450.

PRM_44726 [EOS-Additive-M450] — ZAX_LOOP_GAIN_6 | Default: 74.300 | Linked Fault: ERR-EOS450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M450.

PRM_44727 [EOS-Additive-M450] — BAX_LOOP_GAIN_7 | Default: 128.350 | Linked Fault: ERR-EOS450-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on EOS-Additive-M450.

PRM_44728 [EOS-Additive-M450] — CAX_LOOP_GAIN_8 | Default: 182.400 | Linked Fault: ERR-EOS450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M450.

PRM_44729 [EOS-Additive-M450] — OPT_LOOP_GAIN_9 | Default: 236.450 | Linked Fault: ERR-EOS450-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on EOS-Additive-M450.

PRM_44730 [EOS-Additive-M450] — PLC_LOOP_GAIN_10 | Default: 40.500 | Linked Fault: ERR-EOS450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M450.

PRM_44731 [EOS-Additive-M450] — LUB_LOOP_GAIN_11 | Default: 94.550 | Linked Fault: ERR-EOS450-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on EOS-Additive-M450.

PRM_44732 [EOS-Additive-M450] — VAC_LOOP_GAIN_12 | Default: 148.600 | Linked Fault: ERR-EOS450-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M450.

PRM_44733 [EOS-Additive-M450] — THM_LOOP_GAIN_13 | Default: 202.650 | Linked Fault: ERR-EOS450-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on EOS-Additive-M450.

PRM_44734 [EOS-Additive-M450] — SPN_LOOP_GAIN_14 | Default: 6.700 | Linked Fault: ERR-EOS450-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on EOS-Additive-M450.

PRM_44735 [EOS-Additive-M450] — ATC_LOOP_GAIN_15 | Default: 60.750 | Linked Fault: ERR-EOS450-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on EOS-Additive-M450.

PRM_44736 [EOS-Additive-M450] — HYD_LOOP_GAIN_16 | Default: 114.800 | Linked Fault: ERR-EOS450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M450.

CNC/PLC SERVO PARAMETER DICTIONARY — EOS-Additive-M450 (BANK 2)

Machine Model: EOS-Additive-M450 | Controller: EOSYSTEM Dual-PLC. Use these parameter registers when diagnosing servo following errors or thermal drift on EOS-Additive-M450.

PRM_44741 [EOS-Additive-M450] — HYD_LOOP_GAIN_1 | Default: 108.100 | Linked Fault: ERR-EOS450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M450.

PRM_44742 [EOS-Additive-M450] — XAX_LOOP_GAIN_2 | Default: 216.200 | Linked Fault: ERR-EOS450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M450.

PRM_44743 [EOS-Additive-M450] — ZAX_LOOP_GAIN_3 | Default: 74.300 | Linked Fault: ERR-EOS450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M450.

PRM_44744 [EOS-Additive-M450] — CAX_LOOP_GAIN_4 | Default: 182.400 | Linked Fault: ERR-EOS450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M450.

PRM_44745 [EOS-Additive-M450] — PLC_LOOP_GAIN_5 | Default: 40.500 | Linked Fault: ERR-EOS450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M450.

PRM_44746 [EOS-Additive-M450] — VAC_LOOP_GAIN_6 | Default: 148.600 | Linked Fault: ERR-EOS450-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M450.

PRM_44747 [EOS-Additive-M450] — SPN_LOOP_GAIN_7 | Default: 6.700 | Linked Fault: ERR-EOS450-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on EOS-Additive-M450.

PRM_44748 [EOS-Additive-M450] — HYD_LOOP_GAIN_8 | Default: 114.800 | Linked Fault: ERR-EOS450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M450.

PRM_44749 [EOS-Additive-M450] — XAX_LOOP_GAIN_9 | Default: 222.900 | Linked Fault: ERR-EOS450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M450.

PRM_44750 [EOS-Additive-M450] — ZAX_LOOP_GAIN_10 | Default: 81.000 | Linked Fault: ERR-EOS450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M450.

PRM_44751 [EOS-Additive-M450] — CAX_LOOP_GAIN_11 | Default: 189.100 | Linked Fault: ERR-EOS450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M450.

PRM_44752 [EOS-Additive-M450] — PLC_LOOP_GAIN_12 | Default: 47.200 | Linked Fault: ERR-EOS450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M450.

PRM_44753 [EOS-Additive-M450] — VAC_LOOP_GAIN_13 | Default: 155.300 | Linked Fault: ERR-EOS450-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M450.

PRM_44754 [EOS-Additive-M450] — SPN_LOOP_GAIN_14 | Default: 13.400 | Linked Fault: ERR-EOS450-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on EOS-Additive-M450.

PRM_44755 [EOS-Additive-M450] — HYD_LOOP_GAIN_15 | Default: 121.500 | Linked Fault: ERR-EOS450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M450.

PRM_44756 [EOS-Additive-M450] — XAX_LOOP_GAIN_16 | Default: 229.600 | Linked Fault: ERR-EOS450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M450.

CNC/PLC SERVO PARAMETER DICTIONARY — EOS-Additive-M450 (BANK 3)

Machine Model: EOS-Additive-M450 | Controller: EOSYSTEM Dual-PLC. Use these parameter registers when diagnosing servo following errors or thermal drift on EOS-Additive-M450.

PRM_44761 [EOS-Additive-M450] — CLN_LOOP_GAIN_1 | Default: 162.150 | Linked Fault: ERR-EOS450-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on EOS-Additive-M450.

PRM_44762 [EOS-Additive-M450] — ZAX_LOOP_GAIN_2 | Default: 74.300 | Linked Fault: ERR-EOS450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M450.

PRM_44763 [EOS-Additive-M450] — OPT_LOOP_GAIN_3 | Default: 236.450 | Linked Fault: ERR-EOS450-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on EOS-Additive-M450.

PRM_44764 [EOS-Additive-M450] — VAC_LOOP_GAIN_4 | Default: 148.600 | Linked Fault: ERR-EOS450-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M450.

PRM_44765 [EOS-Additive-M450] — ATC_LOOP_GAIN_5 | Default: 60.750 | Linked Fault: ERR-EOS450-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on EOS-Additive-M450.

PRM_44766 [EOS-Additive-M450] — XAX_LOOP_GAIN_6 | Default: 222.900 | Linked Fault: ERR-EOS450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M450.

PRM_44767 [EOS-Additive-M450] — BAX_LOOP_GAIN_7 | Default: 135.050 | Linked Fault: ERR-EOS450-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on EOS-Additive-M450.

PRM_44768 [EOS-Additive-M450] — PLC_LOOP_GAIN_8 | Default: 47.200 | Linked Fault: ERR-EOS450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M450.

PRM_44769 [EOS-Additive-M450] — THM_LOOP_GAIN_9 | Default: 209.350 | Linked Fault: ERR-EOS450-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on EOS-Additive-M450.

PRM_44770 [EOS-Additive-M450] — HYD_LOOP_GAIN_10 | Default: 121.500 | Linked Fault: ERR-EOS450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M450.

PRM_44771 [EOS-Additive-M450] — YAX_LOOP_GAIN_11 | Default: 33.650 | Linked Fault: ERR-EOS450-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on EOS-Additive-M450.

PRM_44772 [EOS-Additive-M450] — CAX_LOOP_GAIN_12 | Default: 195.800 | Linked Fault: ERR-EOS450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M450.

PRM_44773 [EOS-Additive-M450] — LUB_LOOP_GAIN_13 | Default: 107.950 | Linked Fault: ERR-EOS450-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on EOS-Additive-M450.

PRM_44774 [EOS-Additive-M450] — SPN_LOOP_GAIN_14 | Default: 20.100 | Linked Fault: ERR-EOS450-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on EOS-Additive-M450.

PRM_44775 [EOS-Additive-M450] — CLN_LOOP_GAIN_15 | Default: 182.250 | Linked Fault: ERR-EOS450-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on EOS-Additive-M450.

PRM_44776 [EOS-Additive-M450] — ZAX_LOOP_GAIN_16 | Default: 94.400 | Linked Fault: ERR-EOS450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M450.

CNC/PLC SERVO PARAMETER DICTIONARY — EOS-Additive-M450 (BANK 4)

Machine Model: EOS-Additive-M450 | Controller: EOSYSTEM Dual-PLC. Use these parameter registers when diagnosing servo following errors or thermal drift on EOS-Additive-M450.

PRM_44781 [EOS-Additive-M450] — XAX_LOOP_GAIN_1 | Default: 216.200 | Linked Fault: ERR-EOS450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M450.

PRM_44782 [EOS-Additive-M450] — CAX_LOOP_GAIN_2 | Default: 182.400 | Linked Fault: ERR-EOS450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M450.

PRM_44783 [EOS-Additive-M450] — VAC_LOOP_GAIN_3 | Default: 148.600 | Linked Fault: ERR-EOS450-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M450.

PRM_44784 [EOS-Additive-M450] — HYD_LOOP_GAIN_4 | Default: 114.800 | Linked Fault: ERR-EOS450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M450.

PRM_44785 [EOS-Additive-M450] — ZAX_LOOP_GAIN_5 | Default: 81.000 | Linked Fault: ERR-EOS450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M450.

PRM_44786 [EOS-Additive-M450] — PLC_LOOP_GAIN_6 | Default: 47.200 | Linked Fault: ERR-EOS450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M450.

PRM_44787 [EOS-Additive-M450] — SPN_LOOP_GAIN_7 | Default: 13.400 | Linked Fault: ERR-EOS450-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on EOS-Additive-M450.

PRM_44788 [EOS-Additive-M450] — XAX_LOOP_GAIN_8 | Default: 229.600 | Linked Fault: ERR-EOS450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M450.

PRM_44789 [EOS-Additive-M450] — CAX_LOOP_GAIN_9 | Default: 195.800 | Linked Fault: ERR-EOS450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M450.

PRM_44790 [EOS-Additive-M450] — VAC_LOOP_GAIN_10 | Default: 162.000 | Linked Fault: ERR-EOS450-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M450.

PRM_44791 [EOS-Additive-M450] — HYD_LOOP_GAIN_11 | Default: 128.200 | Linked Fault: ERR-EOS450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M450.

PRM_44792 [EOS-Additive-M450] — ZAX_LOOP_GAIN_12 | Default: 94.400 | Linked Fault: ERR-EOS450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M450.

PRM_44793 [EOS-Additive-M450] — PLC_LOOP_GAIN_13 | Default: 60.600 | Linked Fault: ERR-EOS450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M450.

PRM_44794 [EOS-Additive-M450] — SPN_LOOP_GAIN_14 | Default: 26.800 | Linked Fault: ERR-EOS450-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on EOS-Additive-M450.

PRM_44795 [EOS-Additive-M450] — XAX_LOOP_GAIN_15 | Default: 243.000 | Linked Fault: ERR-EOS450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M450.

PRM_44796 [EOS-Additive-M450] — CAX_LOOP_GAIN_16 | Default: 209.200 | Linked Fault: ERR-EOS450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M450.

CNC/PLC SERVO PARAMETER DICTIONARY — EOS-Additive-M450 (BANK 5)

Machine Model: EOS-Additive-M450 | Controller: EOSYSTEM Dual-PLC. Use these parameter registers when diagnosing servo following errors or thermal drift on EOS-Additive-M450.

PRM_44801 [EOS-Additive-M450] — YAX_LOOP_GAIN_1 | Default: 20.250 | Linked Fault: ERR-EOS450-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on EOS-Additive-M450.

PRM_44802 [EOS-Additive-M450] — PLC_LOOP_GAIN_2 | Default: 40.500 | Linked Fault: ERR-EOS450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M450.

PRM_44803 [EOS-Additive-M450] — ATC_LOOP_GAIN_3 | Default: 60.750 | Linked Fault: ERR-EOS450-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on EOS-Additive-M450.

PRM_44804 [EOS-Additive-M450] — ZAX_LOOP_GAIN_4 | Default: 81.000 | Linked Fault: ERR-EOS450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M450.

PRM_44805 [EOS-Additive-M450] — LUB_LOOP_GAIN_5 | Default: 101.250 | Linked Fault: ERR-EOS450-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on EOS-Additive-M450.

PRM_44806 [EOS-Additive-M450] — HYD_LOOP_GAIN_6 | Default: 121.500 | Linked Fault: ERR-EOS450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M450.

PRM_44807 [EOS-Additive-M450] — BAX_LOOP_GAIN_7 | Default: 141.750 | Linked Fault: ERR-EOS450-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on EOS-Additive-M450.

PRM_44808 [EOS-Additive-M450] — VAC_LOOP_GAIN_8 | Default: 162.000 | Linked Fault: ERR-EOS450-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M450.

PRM_44809 [EOS-Additive-M450] — CLN_LOOP_GAIN_9 | Default: 182.250 | Linked Fault: ERR-EOS450-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on EOS-Additive-M450.

PRM_44810 [EOS-Additive-M450] — CAX_LOOP_GAIN_10 | Default: 202.500 | Linked Fault: ERR-EOS450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M450.

PRM_44811 [EOS-Additive-M450] — THM_LOOP_GAIN_11 | Default: 222.750 | Linked Fault: ERR-EOS450-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on EOS-Additive-M450.

PRM_44812 [EOS-Additive-M450] — XAX_LOOP_GAIN_12 | Default: 243.000 | Linked Fault: ERR-EOS450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M450.

PRM_44813 [EOS-Additive-M450] — OPT_LOOP_GAIN_13 | Default: 13.250 | Linked Fault: ERR-EOS450-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on EOS-Additive-M450.

PRM_44814 [EOS-Additive-M450] — SPN_LOOP_GAIN_14 | Default: 33.500 | Linked Fault: ERR-EOS450-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on EOS-Additive-M450.

PRM_44815 [EOS-Additive-M450] — YAX_LOOP_GAIN_15 | Default: 53.750 | Linked Fault: ERR-EOS450-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on EOS-Additive-M450.

PRM_44816 [EOS-Additive-M450] — PLC_LOOP_GAIN_16 | Default: 74.000 | Linked Fault: ERR-EOS450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M450.

CNC/PLC SERVO PARAMETER DICTIONARY — EOS-Additive-M500 (BANK 1)

Machine Model: EOS-Additive-M500 | Controller: EOSYSTEM Dual-PLC. Use these parameter registers when diagnosing servo following errors or thermal drift on EOS-Additive-M500.

PRM_44821 [EOS-Additive-M500] — ATC_LOOP_GAIN_1 | Default: 55.200 | Linked Fault: ERR-EOS500-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on EOS-Additive-M500.

PRM_44822 [EOS-Additive-M500] — HYD_LOOP_GAIN_2 | Default: 110.400 | Linked Fault: ERR-EOS500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M500.

PRM_44823 [EOS-Additive-M500] — CLN_LOOP_GAIN_3 | Default: 165.600 | Linked Fault: ERR-EOS500-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on EOS-Additive-M500.

PRM_44824 [EOS-Additive-M500] — XAX_LOOP_GAIN_4 | Default: 220.800 | Linked Fault: ERR-EOS500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M500.

PRM_44825 [EOS-Additive-M500] — YAX_LOOP_GAIN_5 | Default: 26.000 | Linked Fault: ERR-EOS500-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on EOS-Additive-M500.

PRM_44826 [EOS-Additive-M500] — ZAX_LOOP_GAIN_6 | Default: 81.200 | Linked Fault: ERR-EOS500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M500.

PRM_44827 [EOS-Additive-M500] — BAX_LOOP_GAIN_7 | Default: 136.400 | Linked Fault: ERR-EOS500-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on EOS-Additive-M500.

PRM_44828 [EOS-Additive-M500] — CAX_LOOP_GAIN_8 | Default: 191.600 | Linked Fault: ERR-EOS500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M500.

PRM_44829 [EOS-Additive-M500] — OPT_LOOP_GAIN_9 | Default: 246.800 | Linked Fault: ERR-EOS500-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on EOS-Additive-M500.

PRM_44830 [EOS-Additive-M500] — PLC_LOOP_GAIN_10 | Default: 52.000 | Linked Fault: ERR-EOS500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M500.

PRM_44831 [EOS-Additive-M500] — LUB_LOOP_GAIN_11 | Default: 107.200 | Linked Fault: ERR-EOS500-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on EOS-Additive-M500.

PRM_44832 [EOS-Additive-M500] — VAC_LOOP_GAIN_12 | Default: 162.400 | Linked Fault: ERR-EOS500-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M500.

PRM_44833 [EOS-Additive-M500] — THM_LOOP_GAIN_13 | Default: 217.600 | Linked Fault: ERR-EOS500-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on EOS-Additive-M500.

PRM_44834 [EOS-Additive-M500] — SPN_LOOP_GAIN_14 | Default: 22.800 | Linked Fault: ERR-EOS500-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on EOS-Additive-M500.

PRM_44835 [EOS-Additive-M500] — ATC_LOOP_GAIN_15 | Default: 78.000 | Linked Fault: ERR-EOS500-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on EOS-Additive-M500.

PRM_44836 [EOS-Additive-M500] — HYD_LOOP_GAIN_16 | Default: 133.200 | Linked Fault: ERR-EOS500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M500.

CNC/PLC SERVO PARAMETER DICTIONARY — EOS-Additive-M500 (BANK 2)

Machine Model: EOS-Additive-M500 | Controller: EOSYSTEM Dual-PLC. Use these parameter registers when diagnosing servo following errors or thermal drift on EOS-Additive-M500.

PRM_44841 [EOS-Additive-M500] — HYD_LOOP_GAIN_1 | Default: 110.400 | Linked Fault: ERR-EOS500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M500.

PRM_44842 [EOS-Additive-M500] — XAX_LOOP_GAIN_2 | Default: 220.800 | Linked Fault: ERR-EOS500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M500.

PRM_44843 [EOS-Additive-M500] — ZAX_LOOP_GAIN_3 | Default: 81.200 | Linked Fault: ERR-EOS500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M500.

PRM_44844 [EOS-Additive-M500] — CAX_LOOP_GAIN_4 | Default: 191.600 | Linked Fault: ERR-EOS500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M500.

PRM_44845 [EOS-Additive-M500] — PLC_LOOP_GAIN_5 | Default: 52.000 | Linked Fault: ERR-EOS500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M500.

PRM_44846 [EOS-Additive-M500] — VAC_LOOP_GAIN_6 | Default: 162.400 | Linked Fault: ERR-EOS500-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M500.

PRM_44847 [EOS-Additive-M500] — SPN_LOOP_GAIN_7 | Default: 22.800 | Linked Fault: ERR-EOS500-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on EOS-Additive-M500.

PRM_44848 [EOS-Additive-M500] — HYD_LOOP_GAIN_8 | Default: 133.200 | Linked Fault: ERR-EOS500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M500.

PRM_44849 [EOS-Additive-M500] — XAX_LOOP_GAIN_9 | Default: 243.600 | Linked Fault: ERR-EOS500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M500.

PRM_44850 [EOS-Additive-M500] — ZAX_LOOP_GAIN_10 | Default: 104.000 | Linked Fault: ERR-EOS500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M500.

PRM_44851 [EOS-Additive-M500] — CAX_LOOP_GAIN_11 | Default: 214.400 | Linked Fault: ERR-EOS500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M500.

PRM_44852 [EOS-Additive-M500] — PLC_LOOP_GAIN_12 | Default: 74.800 | Linked Fault: ERR-EOS500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M500.

PRM_44853 [EOS-Additive-M500] — VAC_LOOP_GAIN_13 | Default: 185.200 | Linked Fault: ERR-EOS500-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M500.

PRM_44854 [EOS-Additive-M500] — SPN_LOOP_GAIN_14 | Default: 45.600 | Linked Fault: ERR-EOS500-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on EOS-Additive-M500.

PRM_44855 [EOS-Additive-M500] — HYD_LOOP_GAIN_15 | Default: 156.000 | Linked Fault: ERR-EOS500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M500.

PRM_44856 [EOS-Additive-M500] — XAX_LOOP_GAIN_16 | Default: 16.400 | Linked Fault: ERR-EOS500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M500.

CNC/PLC SERVO PARAMETER DICTIONARY — EOS-Additive-M500 (BANK 3)

Machine Model: EOS-Additive-M500 | Controller: EOSYSTEM Dual-PLC. Use these parameter registers when diagnosing servo following errors or thermal drift on EOS-Additive-M500.

PRM_44861 [EOS-Additive-M500] — CLN_LOOP_GAIN_1 | Default: 165.600 | Linked Fault: ERR-EOS500-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on EOS-Additive-M500.

PRM_44862 [EOS-Additive-M500] — ZAX_LOOP_GAIN_2 | Default: 81.200 | Linked Fault: ERR-EOS500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M500.

PRM_44863 [EOS-Additive-M500] — OPT_LOOP_GAIN_3 | Default: 246.800 | Linked Fault: ERR-EOS500-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on EOS-Additive-M500.

PRM_44864 [EOS-Additive-M500] — VAC_LOOP_GAIN_4 | Default: 162.400 | Linked Fault: ERR-EOS500-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M500.

PRM_44865 [EOS-Additive-M500] — ATC_LOOP_GAIN_5 | Default: 78.000 | Linked Fault: ERR-EOS500-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on EOS-Additive-M500.

PRM_44866 [EOS-Additive-M500] — XAX_LOOP_GAIN_6 | Default: 243.600 | Linked Fault: ERR-EOS500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M500.

PRM_44867 [EOS-Additive-M500] — BAX_LOOP_GAIN_7 | Default: 159.200 | Linked Fault: ERR-EOS500-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on EOS-Additive-M500.

PRM_44868 [EOS-Additive-M500] — PLC_LOOP_GAIN_8 | Default: 74.800 | Linked Fault: ERR-EOS500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M500.

PRM_44869 [EOS-Additive-M500] — THM_LOOP_GAIN_9 | Default: 240.400 | Linked Fault: ERR-EOS500-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on EOS-Additive-M500.

PRM_44870 [EOS-Additive-M500] — HYD_LOOP_GAIN_10 | Default: 156.000 | Linked Fault: ERR-EOS500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M500.

PRM_44871 [EOS-Additive-M500] — YAX_LOOP_GAIN_11 | Default: 71.600 | Linked Fault: ERR-EOS500-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on EOS-Additive-M500.

PRM_44872 [EOS-Additive-M500] — CAX_LOOP_GAIN_12 | Default: 237.200 | Linked Fault: ERR-EOS500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M500.

PRM_44873 [EOS-Additive-M500] — LUB_LOOP_GAIN_13 | Default: 152.800 | Linked Fault: ERR-EOS500-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on EOS-Additive-M500.

PRM_44874 [EOS-Additive-M500] — SPN_LOOP_GAIN_14 | Default: 68.400 | Linked Fault: ERR-EOS500-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on EOS-Additive-M500.

PRM_44875 [EOS-Additive-M500] — CLN_LOOP_GAIN_15 | Default: 234.000 | Linked Fault: ERR-EOS500-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on EOS-Additive-M500.

PRM_44876 [EOS-Additive-M500] — ZAX_LOOP_GAIN_16 | Default: 149.600 | Linked Fault: ERR-EOS500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M500.

CNC/PLC SERVO PARAMETER DICTIONARY — EOS-Additive-M500 (BANK 4)

Machine Model: EOS-Additive-M500 | Controller: EOSYSTEM Dual-PLC. Use these parameter registers when diagnosing servo following errors or thermal drift on EOS-Additive-M500.

PRM_44881 [EOS-Additive-M500] — XAX_LOOP_GAIN_1 | Default: 220.800 | Linked Fault: ERR-EOS500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M500.

PRM_44882 [EOS-Additive-M500] — CAX_LOOP_GAIN_2 | Default: 191.600 | Linked Fault: ERR-EOS500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M500.

PRM_44883 [EOS-Additive-M500] — VAC_LOOP_GAIN_3 | Default: 162.400 | Linked Fault: ERR-EOS500-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M500.

PRM_44884 [EOS-Additive-M500] — HYD_LOOP_GAIN_4 | Default: 133.200 | Linked Fault: ERR-EOS500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M500.

PRM_44885 [EOS-Additive-M500] — ZAX_LOOP_GAIN_5 | Default: 104.000 | Linked Fault: ERR-EOS500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M500.

PRM_44886 [EOS-Additive-M500] — PLC_LOOP_GAIN_6 | Default: 74.800 | Linked Fault: ERR-EOS500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M500.

PRM_44887 [EOS-Additive-M500] — SPN_LOOP_GAIN_7 | Default: 45.600 | Linked Fault: ERR-EOS500-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on EOS-Additive-M500.

PRM_44888 [EOS-Additive-M500] — XAX_LOOP_GAIN_8 | Default: 16.400 | Linked Fault: ERR-EOS500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M500.

PRM_44889 [EOS-Additive-M500] — CAX_LOOP_GAIN_9 | Default: 237.200 | Linked Fault: ERR-EOS500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M500.

PRM_44890 [EOS-Additive-M500] — VAC_LOOP_GAIN_10 | Default: 208.000 | Linked Fault: ERR-EOS500-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M500.

PRM_44891 [EOS-Additive-M500] — HYD_LOOP_GAIN_11 | Default: 178.800 | Linked Fault: ERR-EOS500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M500.

PRM_44892 [EOS-Additive-M500] — ZAX_LOOP_GAIN_12 | Default: 149.600 | Linked Fault: ERR-EOS500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M500.

PRM_44893 [EOS-Additive-M500] — PLC_LOOP_GAIN_13 | Default: 120.400 | Linked Fault: ERR-EOS500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M500.

PRM_44894 [EOS-Additive-M500] — SPN_LOOP_GAIN_14 | Default: 91.200 | Linked Fault: ERR-EOS500-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on EOS-Additive-M500.

PRM_44895 [EOS-Additive-M500] — XAX_LOOP_GAIN_15 | Default: 62.000 | Linked Fault: ERR-EOS500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M500.

PRM_44896 [EOS-Additive-M500] — CAX_LOOP_GAIN_16 | Default: 32.800 | Linked Fault: ERR-EOS500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M500.

CNC/PLC SERVO PARAMETER DICTIONARY — EOS-Additive-M500 (BANK 5)

Machine Model: EOS-Additive-M500 | Controller: EOSYSTEM Dual-PLC. Use these parameter registers when diagnosing servo following errors or thermal drift on EOS-Additive-M500.

PRM_44901 [EOS-Additive-M500] — YAX_LOOP_GAIN_1 | Default: 26.000 | Linked Fault: ERR-EOS500-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on EOS-Additive-M500.

PRM_44902 [EOS-Additive-M500] — PLC_LOOP_GAIN_2 | Default: 52.000 | Linked Fault: ERR-EOS500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M500.

PRM_44903 [EOS-Additive-M500] — ATC_LOOP_GAIN_3 | Default: 78.000 | Linked Fault: ERR-EOS500-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on EOS-Additive-M500.

PRM_44904 [EOS-Additive-M500] — ZAX_LOOP_GAIN_4 | Default: 104.000 | Linked Fault: ERR-EOS500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M500.

PRM_44905 [EOS-Additive-M500] — LUB_LOOP_GAIN_5 | Default: 130.000 | Linked Fault: ERR-EOS500-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on EOS-Additive-M500.

PRM_44906 [EOS-Additive-M500] — HYD_LOOP_GAIN_6 | Default: 156.000 | Linked Fault: ERR-EOS500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M500.

PRM_44907 [EOS-Additive-M500] — BAX_LOOP_GAIN_7 | Default: 182.000 | Linked Fault: ERR-EOS500-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on EOS-Additive-M500.

PRM_44908 [EOS-Additive-M500] — VAC_LOOP_GAIN_8 | Default: 208.000 | Linked Fault: ERR-EOS500-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M500.

PRM_44909 [EOS-Additive-M500] — CLN_LOOP_GAIN_9 | Default: 234.000 | Linked Fault: ERR-EOS500-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on EOS-Additive-M500.

PRM_44910 [EOS-Additive-M500] — CAX_LOOP_GAIN_10 | Default: 10.000 | Linked Fault: ERR-EOS500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M500.

PRM_44911 [EOS-Additive-M500] — THM_LOOP_GAIN_11 | Default: 36.000 | Linked Fault: ERR-EOS500-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on EOS-Additive-M500.

PRM_44912 [EOS-Additive-M500] — XAX_LOOP_GAIN_12 | Default: 62.000 | Linked Fault: ERR-EOS500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M500.

PRM_44913 [EOS-Additive-M500] — OPT_LOOP_GAIN_13 | Default: 88.000 | Linked Fault: ERR-EOS500-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on EOS-Additive-M500.

PRM_44914 [EOS-Additive-M500] — SPN_LOOP_GAIN_14 | Default: 114.000 | Linked Fault: ERR-EOS500-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on EOS-Additive-M500.

PRM_44915 [EOS-Additive-M500] — YAX_LOOP_GAIN_15 | Default: 140.000 | Linked Fault: ERR-EOS500-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on EOS-Additive-M500.

PRM_44916 [EOS-Additive-M500] — PLC_LOOP_GAIN_16 | Default: 166.000 | Linked Fault: ERR-EOS500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M500.

CNC/PLC SERVO PARAMETER DICTIONARY — EOS-Additive-M550 (BANK 1)

Machine Model: EOS-Additive-M550 | Controller: EOSYSTEM Dual-PLC. Use these parameter registers when diagnosing servo following errors or thermal drift on EOS-Additive-M550.

PRM_44921 [EOS-Additive-M550] — ATC_LOOP_GAIN_1 | Default: 56.350 | Linked Fault: ERR-EOS550-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on EOS-Additive-M550.

PRM_44922 [EOS-Additive-M550] — HYD_LOOP_GAIN_2 | Default: 112.700 | Linked Fault: ERR-EOS550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M550.

PRM_44923 [EOS-Additive-M550] — CLN_LOOP_GAIN_3 | Default: 169.050 | Linked Fault: ERR-EOS550-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on EOS-Additive-M550.

PRM_44924 [EOS-Additive-M550] — XAX_LOOP_GAIN_4 | Default: 225.400 | Linked Fault: ERR-EOS550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M550.

PRM_44925 [EOS-Additive-M550] — YAX_LOOP_GAIN_5 | Default: 31.750 | Linked Fault: ERR-EOS550-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on EOS-Additive-M550.

PRM_44926 [EOS-Additive-M550] — ZAX_LOOP_GAIN_6 | Default: 88.100 | Linked Fault: ERR-EOS550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M550.

PRM_44927 [EOS-Additive-M550] — BAX_LOOP_GAIN_7 | Default: 144.450 | Linked Fault: ERR-EOS550-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on EOS-Additive-M550.

PRM_44928 [EOS-Additive-M550] — CAX_LOOP_GAIN_8 | Default: 200.800 | Linked Fault: ERR-EOS550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M550.

PRM_44929 [EOS-Additive-M550] — OPT_LOOP_GAIN_9 | Default: 7.150 | Linked Fault: ERR-EOS550-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on EOS-Additive-M550.

PRM_44930 [EOS-Additive-M550] — PLC_LOOP_GAIN_10 | Default: 63.500 | Linked Fault: ERR-EOS550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M550.

PRM_44931 [EOS-Additive-M550] — LUB_LOOP_GAIN_11 | Default: 119.850 | Linked Fault: ERR-EOS550-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on EOS-Additive-M550.

PRM_44932 [EOS-Additive-M550] — VAC_LOOP_GAIN_12 | Default: 176.200 | Linked Fault: ERR-EOS550-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M550.

PRM_44933 [EOS-Additive-M550] — THM_LOOP_GAIN_13 | Default: 232.550 | Linked Fault: ERR-EOS550-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on EOS-Additive-M550.

PRM_44934 [EOS-Additive-M550] — SPN_LOOP_GAIN_14 | Default: 38.900 | Linked Fault: ERR-EOS550-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on EOS-Additive-M550.

PRM_44935 [EOS-Additive-M550] — ATC_LOOP_GAIN_15 | Default: 95.250 | Linked Fault: ERR-EOS550-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on EOS-Additive-M550.

PRM_44936 [EOS-Additive-M550] — HYD_LOOP_GAIN_16 | Default: 151.600 | Linked Fault: ERR-EOS550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M550.

CNC/PLC SERVO PARAMETER DICTIONARY — EOS-Additive-M550 (BANK 2)

Machine Model: EOS-Additive-M550 | Controller: EOSYSTEM Dual-PLC. Use these parameter registers when diagnosing servo following errors or thermal drift on EOS-Additive-M550.

PRM_44941 [EOS-Additive-M550] — HYD_LOOP_GAIN_1 | Default: 112.700 | Linked Fault: ERR-EOS550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M550.

PRM_44942 [EOS-Additive-M550] — XAX_LOOP_GAIN_2 | Default: 225.400 | Linked Fault: ERR-EOS550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M550.

PRM_44943 [EOS-Additive-M550] — ZAX_LOOP_GAIN_3 | Default: 88.100 | Linked Fault: ERR-EOS550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M550.

PRM_44944 [EOS-Additive-M550] — CAX_LOOP_GAIN_4 | Default: 200.800 | Linked Fault: ERR-EOS550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M550.

PRM_44945 [EOS-Additive-M550] — PLC_LOOP_GAIN_5 | Default: 63.500 | Linked Fault: ERR-EOS550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M550.

PRM_44946 [EOS-Additive-M550] — VAC_LOOP_GAIN_6 | Default: 176.200 | Linked Fault: ERR-EOS550-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M550.

PRM_44947 [EOS-Additive-M550] — SPN_LOOP_GAIN_7 | Default: 38.900 | Linked Fault: ERR-EOS550-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on EOS-Additive-M550.

PRM_44948 [EOS-Additive-M550] — HYD_LOOP_GAIN_8 | Default: 151.600 | Linked Fault: ERR-EOS550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M550.

PRM_44949 [EOS-Additive-M550] — XAX_LOOP_GAIN_9 | Default: 14.300 | Linked Fault: ERR-EOS550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M550.

PRM_44950 [EOS-Additive-M550] — ZAX_LOOP_GAIN_10 | Default: 127.000 | Linked Fault: ERR-EOS550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M550.

PRM_44951 [EOS-Additive-M550] — CAX_LOOP_GAIN_11 | Default: 239.700 | Linked Fault: ERR-EOS550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M550.

PRM_44952 [EOS-Additive-M550] — PLC_LOOP_GAIN_12 | Default: 102.400 | Linked Fault: ERR-EOS550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M550.

PRM_44953 [EOS-Additive-M550] — VAC_LOOP_GAIN_13 | Default: 215.100 | Linked Fault: ERR-EOS550-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M550.

PRM_44954 [EOS-Additive-M550] — SPN_LOOP_GAIN_14 | Default: 77.800 | Linked Fault: ERR-EOS550-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on EOS-Additive-M550.

PRM_44955 [EOS-Additive-M550] — HYD_LOOP_GAIN_15 | Default: 190.500 | Linked Fault: ERR-EOS550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M550.

PRM_44956 [EOS-Additive-M550] — XAX_LOOP_GAIN_16 | Default: 53.200 | Linked Fault: ERR-EOS550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M550.

CNC/PLC SERVO PARAMETER DICTIONARY — EOS-Additive-M550 (BANK 3)

Machine Model: EOS-Additive-M550 | Controller: EOSYSTEM Dual-PLC. Use these parameter registers when diagnosing servo following errors or thermal drift on EOS-Additive-M550.

PRM_44961 [EOS-Additive-M550] — CLN_LOOP_GAIN_1 | Default: 169.050 | Linked Fault: ERR-EOS550-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on EOS-Additive-M550.

PRM_44962 [EOS-Additive-M550] — ZAX_LOOP_GAIN_2 | Default: 88.100 | Linked Fault: ERR-EOS550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M550.

PRM_44963 [EOS-Additive-M550] — OPT_LOOP_GAIN_3 | Default: 7.150 | Linked Fault: ERR-EOS550-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on EOS-Additive-M550.

PRM_44964 [EOS-Additive-M550] — VAC_LOOP_GAIN_4 | Default: 176.200 | Linked Fault: ERR-EOS550-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M550.

PRM_44965 [EOS-Additive-M550] — ATC_LOOP_GAIN_5 | Default: 95.250 | Linked Fault: ERR-EOS550-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on EOS-Additive-M550.

PRM_44966 [EOS-Additive-M550] — XAX_LOOP_GAIN_6 | Default: 14.300 | Linked Fault: ERR-EOS550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M550.

PRM_44967 [EOS-Additive-M550] — BAX_LOOP_GAIN_7 | Default: 183.350 | Linked Fault: ERR-EOS550-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on EOS-Additive-M550.

PRM_44968 [EOS-Additive-M550] — PLC_LOOP_GAIN_8 | Default: 102.400 | Linked Fault: ERR-EOS550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M550.

PRM_44969 [EOS-Additive-M550] — THM_LOOP_GAIN_9 | Default: 21.450 | Linked Fault: ERR-EOS550-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on EOS-Additive-M550.

PRM_44970 [EOS-Additive-M550] — HYD_LOOP_GAIN_10 | Default: 190.500 | Linked Fault: ERR-EOS550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M550.

PRM_44971 [EOS-Additive-M550] — YAX_LOOP_GAIN_11 | Default: 109.550 | Linked Fault: ERR-EOS550-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on EOS-Additive-M550.

PRM_44972 [EOS-Additive-M550] — CAX_LOOP_GAIN_12 | Default: 28.600 | Linked Fault: ERR-EOS550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M550.

PRM_44973 [EOS-Additive-M550] — LUB_LOOP_GAIN_13 | Default: 197.650 | Linked Fault: ERR-EOS550-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on EOS-Additive-M550.

PRM_44974 [EOS-Additive-M550] — SPN_LOOP_GAIN_14 | Default: 116.700 | Linked Fault: ERR-EOS550-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on EOS-Additive-M550.

PRM_44975 [EOS-Additive-M550] — CLN_LOOP_GAIN_15 | Default: 35.750 | Linked Fault: ERR-EOS550-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on EOS-Additive-M550.

PRM_44976 [EOS-Additive-M550] — ZAX_LOOP_GAIN_16 | Default: 204.800 | Linked Fault: ERR-EOS550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M550.

CNC/PLC SERVO PARAMETER DICTIONARY — EOS-Additive-M550 (BANK 4)

Machine Model: EOS-Additive-M550 | Controller: EOSYSTEM Dual-PLC. Use these parameter registers when diagnosing servo following errors or thermal drift on EOS-Additive-M550.

PRM_44981 [EOS-Additive-M550] — XAX_LOOP_GAIN_1 | Default: 225.400 | Linked Fault: ERR-EOS550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M550.

PRM_44982 [EOS-Additive-M550] — CAX_LOOP_GAIN_2 | Default: 200.800 | Linked Fault: ERR-EOS550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M550.

PRM_44983 [EOS-Additive-M550] — VAC_LOOP_GAIN_3 | Default: 176.200 | Linked Fault: ERR-EOS550-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M550.

PRM_44984 [EOS-Additive-M550] — HYD_LOOP_GAIN_4 | Default: 151.600 | Linked Fault: ERR-EOS550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M550.

PRM_44985 [EOS-Additive-M550] — ZAX_LOOP_GAIN_5 | Default: 127.000 | Linked Fault: ERR-EOS550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M550.

PRM_44986 [EOS-Additive-M550] — PLC_LOOP_GAIN_6 | Default: 102.400 | Linked Fault: ERR-EOS550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M550.

PRM_44987 [EOS-Additive-M550] — SPN_LOOP_GAIN_7 | Default: 77.800 | Linked Fault: ERR-EOS550-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on EOS-Additive-M550.

PRM_44988 [EOS-Additive-M550] — XAX_LOOP_GAIN_8 | Default: 53.200 | Linked Fault: ERR-EOS550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M550.

PRM_44989 [EOS-Additive-M550] — CAX_LOOP_GAIN_9 | Default: 28.600 | Linked Fault: ERR-EOS550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M550.

PRM_44990 [EOS-Additive-M550] — VAC_LOOP_GAIN_10 | Default: 4.000 | Linked Fault: ERR-EOS550-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M550.

PRM_44991 [EOS-Additive-M550] — HYD_LOOP_GAIN_11 | Default: 229.400 | Linked Fault: ERR-EOS550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M550.

PRM_44992 [EOS-Additive-M550] — ZAX_LOOP_GAIN_12 | Default: 204.800 | Linked Fault: ERR-EOS550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M550.

PRM_44993 [EOS-Additive-M550] — PLC_LOOP_GAIN_13 | Default: 180.200 | Linked Fault: ERR-EOS550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M550.

PRM_44994 [EOS-Additive-M550] — SPN_LOOP_GAIN_14 | Default: 155.600 | Linked Fault: ERR-EOS550-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on EOS-Additive-M550.

PRM_44995 [EOS-Additive-M550] — XAX_LOOP_GAIN_15 | Default: 131.000 | Linked Fault: ERR-EOS550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M550.

PRM_44996 [EOS-Additive-M550] — CAX_LOOP_GAIN_16 | Default: 106.400 | Linked Fault: ERR-EOS550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M550.

CNC/PLC SERVO PARAMETER DICTIONARY — EOS-Additive-M550 (BANK 5)

Machine Model: EOS-Additive-M550 | Controller: EOSYSTEM Dual-PLC. Use these parameter registers when diagnosing servo following errors or thermal drift on EOS-Additive-M550.

PRM_45001 [EOS-Additive-M550] — YAX_LOOP_GAIN_1 | Default: 31.750 | Linked Fault: ERR-EOS550-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on EOS-Additive-M550.

PRM_45002 [EOS-Additive-M550] — PLC_LOOP_GAIN_2 | Default: 63.500 | Linked Fault: ERR-EOS550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M550.

PRM_45003 [EOS-Additive-M550] — ATC_LOOP_GAIN_3 | Default: 95.250 | Linked Fault: ERR-EOS550-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on EOS-Additive-M550.

PRM_45004 [EOS-Additive-M550] — ZAX_LOOP_GAIN_4 | Default: 127.000 | Linked Fault: ERR-EOS550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M550.

PRM_45005 [EOS-Additive-M550] — LUB_LOOP_GAIN_5 | Default: 158.750 | Linked Fault: ERR-EOS550-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on EOS-Additive-M550.

PRM_45006 [EOS-Additive-M550] — HYD_LOOP_GAIN_6 | Default: 190.500 | Linked Fault: ERR-EOS550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M550.

PRM_45007 [EOS-Additive-M550] — BAX_LOOP_GAIN_7 | Default: 222.250 | Linked Fault: ERR-EOS550-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on EOS-Additive-M550.

PRM_45008 [EOS-Additive-M550] — VAC_LOOP_GAIN_8 | Default: 4.000 | Linked Fault: ERR-EOS550-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M550.

PRM_45009 [EOS-Additive-M550] — CLN_LOOP_GAIN_9 | Default: 35.750 | Linked Fault: ERR-EOS550-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on EOS-Additive-M550.

PRM_45010 [EOS-Additive-M550] — CAX_LOOP_GAIN_10 | Default: 67.500 | Linked Fault: ERR-EOS550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M550.

PRM_45011 [EOS-Additive-M550] — THM_LOOP_GAIN_11 | Default: 99.250 | Linked Fault: ERR-EOS550-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on EOS-Additive-M550.

PRM_45012 [EOS-Additive-M550] — XAX_LOOP_GAIN_12 | Default: 131.000 | Linked Fault: ERR-EOS550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M550.

PRM_45013 [EOS-Additive-M550] — OPT_LOOP_GAIN_13 | Default: 162.750 | Linked Fault: ERR-EOS550-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on EOS-Additive-M550.

PRM_45014 [EOS-Additive-M550] — SPN_LOOP_GAIN_14 | Default: 194.500 | Linked Fault: ERR-EOS550-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on EOS-Additive-M550.

PRM_45015 [EOS-Additive-M550] — YAX_LOOP_GAIN_15 | Default: 226.250 | Linked Fault: ERR-EOS550-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on EOS-Additive-M550.

PRM_45016 [EOS-Additive-M550] — PLC_LOOP_GAIN_16 | Default: 8.000 | Linked Fault: ERR-EOS550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M550.

CNC/PLC SERVO PARAMETER DICTIONARY — EOS-Additive-M600 (BANK 1)

Machine Model: EOS-Additive-M600 | Controller: EOSYSTEM Dual-PLC. Use these parameter registers when diagnosing servo following errors or thermal drift on EOS-Additive-M600.

PRM_45021 [EOS-Additive-M600] — ATC_LOOP_GAIN_1 | Default: 57.500 | Linked Fault: ERR-EOS600-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on EOS-Additive-M600.

PRM_45022 [EOS-Additive-M600] — HYD_LOOP_GAIN_2 | Default: 115.000 | Linked Fault: ERR-EOS600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M600.

PRM_45023 [EOS-Additive-M600] — CLN_LOOP_GAIN_3 | Default: 172.500 | Linked Fault: ERR-EOS600-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on EOS-Additive-M600.

PRM_45024 [EOS-Additive-M600] — XAX_LOOP_GAIN_4 | Default: 230.000 | Linked Fault: ERR-EOS600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M600.

PRM_45025 [EOS-Additive-M600] — YAX_LOOP_GAIN_5 | Default: 37.500 | Linked Fault: ERR-EOS600-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on EOS-Additive-M600.

PRM_45026 [EOS-Additive-M600] — ZAX_LOOP_GAIN_6 | Default: 95.000 | Linked Fault: ERR-EOS600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M600.

PRM_45027 [EOS-Additive-M600] — BAX_LOOP_GAIN_7 | Default: 152.500 | Linked Fault: ERR-EOS600-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on EOS-Additive-M600.

PRM_45028 [EOS-Additive-M600] — CAX_LOOP_GAIN_8 | Default: 210.000 | Linked Fault: ERR-EOS600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M600.

PRM_45029 [EOS-Additive-M600] — OPT_LOOP_GAIN_9 | Default: 17.500 | Linked Fault: ERR-EOS600-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on EOS-Additive-M600.

PRM_45030 [EOS-Additive-M600] — PLC_LOOP_GAIN_10 | Default: 75.000 | Linked Fault: ERR-EOS600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M600.

PRM_45031 [EOS-Additive-M600] — LUB_LOOP_GAIN_11 | Default: 132.500 | Linked Fault: ERR-EOS600-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on EOS-Additive-M600.

PRM_45032 [EOS-Additive-M600] — VAC_LOOP_GAIN_12 | Default: 190.000 | Linked Fault: ERR-EOS600-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M600.

PRM_45033 [EOS-Additive-M600] — THM_LOOP_GAIN_13 | Default: 247.500 | Linked Fault: ERR-EOS600-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on EOS-Additive-M600.

PRM_45034 [EOS-Additive-M600] — SPN_LOOP_GAIN_14 | Default: 55.000 | Linked Fault: ERR-EOS600-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on EOS-Additive-M600.

PRM_45035 [EOS-Additive-M600] — ATC_LOOP_GAIN_15 | Default: 112.500 | Linked Fault: ERR-EOS600-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on EOS-Additive-M600.

PRM_45036 [EOS-Additive-M600] — HYD_LOOP_GAIN_16 | Default: 170.000 | Linked Fault: ERR-EOS600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M600.

CNC/PLC SERVO PARAMETER DICTIONARY — EOS-Additive-M600 (BANK 2)

Machine Model: EOS-Additive-M600 | Controller: EOSYSTEM Dual-PLC. Use these parameter registers when diagnosing servo following errors or thermal drift on EOS-Additive-M600.

PRM_45041 [EOS-Additive-M600] — HYD_LOOP_GAIN_1 | Default: 115.000 | Linked Fault: ERR-EOS600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M600.

PRM_45042 [EOS-Additive-M600] — XAX_LOOP_GAIN_2 | Default: 230.000 | Linked Fault: ERR-EOS600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M600.

PRM_45043 [EOS-Additive-M600] — ZAX_LOOP_GAIN_3 | Default: 95.000 | Linked Fault: ERR-EOS600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M600.

PRM_45044 [EOS-Additive-M600] — CAX_LOOP_GAIN_4 | Default: 210.000 | Linked Fault: ERR-EOS600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M600.

PRM_45045 [EOS-Additive-M600] — PLC_LOOP_GAIN_5 | Default: 75.000 | Linked Fault: ERR-EOS600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M600.

PRM_45046 [EOS-Additive-M600] — VAC_LOOP_GAIN_6 | Default: 190.000 | Linked Fault: ERR-EOS600-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M600.

PRM_45047 [EOS-Additive-M600] — SPN_LOOP_GAIN_7 | Default: 55.000 | Linked Fault: ERR-EOS600-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on EOS-Additive-M600.

PRM_45048 [EOS-Additive-M600] — HYD_LOOP_GAIN_8 | Default: 170.000 | Linked Fault: ERR-EOS600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M600.

PRM_45049 [EOS-Additive-M600] — XAX_LOOP_GAIN_9 | Default: 35.000 | Linked Fault: ERR-EOS600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M600.

PRM_45050 [EOS-Additive-M600] — ZAX_LOOP_GAIN_10 | Default: 150.000 | Linked Fault: ERR-EOS600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M600.

PRM_45051 [EOS-Additive-M600] — CAX_LOOP_GAIN_11 | Default: 15.000 | Linked Fault: ERR-EOS600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M600.

PRM_45052 [EOS-Additive-M600] — PLC_LOOP_GAIN_12 | Default: 130.000 | Linked Fault: ERR-EOS600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M600.

PRM_45053 [EOS-Additive-M600] — VAC_LOOP_GAIN_13 | Default: 245.000 | Linked Fault: ERR-EOS600-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M600.

PRM_45054 [EOS-Additive-M600] — SPN_LOOP_GAIN_14 | Default: 110.000 | Linked Fault: ERR-EOS600-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on EOS-Additive-M600.

PRM_45055 [EOS-Additive-M600] — HYD_LOOP_GAIN_15 | Default: 225.000 | Linked Fault: ERR-EOS600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M600.

PRM_45056 [EOS-Additive-M600] — XAX_LOOP_GAIN_16 | Default: 90.000 | Linked Fault: ERR-EOS600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M600.

CNC/PLC SERVO PARAMETER DICTIONARY — EOS-Additive-M600 (BANK 3)

Machine Model: EOS-Additive-M600 | Controller: EOSYSTEM Dual-PLC. Use these parameter registers when diagnosing servo following errors or thermal drift on EOS-Additive-M600.

PRM_45061 [EOS-Additive-M600] — CLN_LOOP_GAIN_1 | Default: 172.500 | Linked Fault: ERR-EOS600-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on EOS-Additive-M600.

PRM_45062 [EOS-Additive-M600] — ZAX_LOOP_GAIN_2 | Default: 95.000 | Linked Fault: ERR-EOS600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M600.

PRM_45063 [EOS-Additive-M600] — OPT_LOOP_GAIN_3 | Default: 17.500 | Linked Fault: ERR-EOS600-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on EOS-Additive-M600.

PRM_45064 [EOS-Additive-M600] — VAC_LOOP_GAIN_4 | Default: 190.000 | Linked Fault: ERR-EOS600-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M600.

PRM_45065 [EOS-Additive-M600] — ATC_LOOP_GAIN_5 | Default: 112.500 | Linked Fault: ERR-EOS600-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on EOS-Additive-M600.

PRM_45066 [EOS-Additive-M600] — XAX_LOOP_GAIN_6 | Default: 35.000 | Linked Fault: ERR-EOS600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M600.

PRM_45067 [EOS-Additive-M600] — BAX_LOOP_GAIN_7 | Default: 207.500 | Linked Fault: ERR-EOS600-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on EOS-Additive-M600.

PRM_45068 [EOS-Additive-M600] — PLC_LOOP_GAIN_8 | Default: 130.000 | Linked Fault: ERR-EOS600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M600.

PRM_45069 [EOS-Additive-M600] — THM_LOOP_GAIN_9 | Default: 52.500 | Linked Fault: ERR-EOS600-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on EOS-Additive-M600.

PRM_45070 [EOS-Additive-M600] — HYD_LOOP_GAIN_10 | Default: 225.000 | Linked Fault: ERR-EOS600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M600.

PRM_45071 [EOS-Additive-M600] — YAX_LOOP_GAIN_11 | Default: 147.500 | Linked Fault: ERR-EOS600-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on EOS-Additive-M600.

PRM_45072 [EOS-Additive-M600] — CAX_LOOP_GAIN_12 | Default: 70.000 | Linked Fault: ERR-EOS600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M600.

PRM_45073 [EOS-Additive-M600] — LUB_LOOP_GAIN_13 | Default: 242.500 | Linked Fault: ERR-EOS600-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on EOS-Additive-M600.

PRM_45074 [EOS-Additive-M600] — SPN_LOOP_GAIN_14 | Default: 165.000 | Linked Fault: ERR-EOS600-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on EOS-Additive-M600.

PRM_45075 [EOS-Additive-M600] — CLN_LOOP_GAIN_15 | Default: 87.500 | Linked Fault: ERR-EOS600-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on EOS-Additive-M600.

PRM_45076 [EOS-Additive-M600] — ZAX_LOOP_GAIN_16 | Default: 10.000 | Linked Fault: ERR-EOS600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M600.

CNC/PLC SERVO PARAMETER DICTIONARY — EOS-Additive-M600 (BANK 4)

Machine Model: EOS-Additive-M600 | Controller: EOSYSTEM Dual-PLC. Use these parameter registers when diagnosing servo following errors or thermal drift on EOS-Additive-M600.

PRM_45081 [EOS-Additive-M600] — XAX_LOOP_GAIN_1 | Default: 230.000 | Linked Fault: ERR-EOS600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M600.

PRM_45082 [EOS-Additive-M600] — CAX_LOOP_GAIN_2 | Default: 210.000 | Linked Fault: ERR-EOS600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M600.

PRM_45083 [EOS-Additive-M600] — VAC_LOOP_GAIN_3 | Default: 190.000 | Linked Fault: ERR-EOS600-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M600.

PRM_45084 [EOS-Additive-M600] — HYD_LOOP_GAIN_4 | Default: 170.000 | Linked Fault: ERR-EOS600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M600.

PRM_45085 [EOS-Additive-M600] — ZAX_LOOP_GAIN_5 | Default: 150.000 | Linked Fault: ERR-EOS600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M600.

PRM_45086 [EOS-Additive-M600] — PLC_LOOP_GAIN_6 | Default: 130.000 | Linked Fault: ERR-EOS600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M600.

PRM_45087 [EOS-Additive-M600] — SPN_LOOP_GAIN_7 | Default: 110.000 | Linked Fault: ERR-EOS600-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on EOS-Additive-M600.

PRM_45088 [EOS-Additive-M600] — XAX_LOOP_GAIN_8 | Default: 90.000 | Linked Fault: ERR-EOS600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M600.

PRM_45089 [EOS-Additive-M600] — CAX_LOOP_GAIN_9 | Default: 70.000 | Linked Fault: ERR-EOS600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M600.

PRM_45090 [EOS-Additive-M600] — VAC_LOOP_GAIN_10 | Default: 50.000 | Linked Fault: ERR-EOS600-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M600.

PRM_45091 [EOS-Additive-M600] — HYD_LOOP_GAIN_11 | Default: 30.000 | Linked Fault: ERR-EOS600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M600.

PRM_45092 [EOS-Additive-M600] — ZAX_LOOP_GAIN_12 | Default: 10.000 | Linked Fault: ERR-EOS600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M600.

PRM_45093 [EOS-Additive-M600] — PLC_LOOP_GAIN_13 | Default: 240.000 | Linked Fault: ERR-EOS600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M600.

PRM_45094 [EOS-Additive-M600] — SPN_LOOP_GAIN_14 | Default: 220.000 | Linked Fault: ERR-EOS600-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on EOS-Additive-M600.

PRM_45095 [EOS-Additive-M600] — XAX_LOOP_GAIN_15 | Default: 200.000 | Linked Fault: ERR-EOS600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M600.

PRM_45096 [EOS-Additive-M600] — CAX_LOOP_GAIN_16 | Default: 180.000 | Linked Fault: ERR-EOS600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M600.

CNC/PLC SERVO PARAMETER DICTIONARY — EOS-Additive-M600 (BANK 5)

Machine Model: EOS-Additive-M600 | Controller: EOSYSTEM Dual-PLC. Use these parameter registers when diagnosing servo following errors or thermal drift on EOS-Additive-M600.

PRM_45101 [EOS-Additive-M600] — YAX_LOOP_GAIN_1 | Default: 37.500 | Linked Fault: ERR-EOS600-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on EOS-Additive-M600.

PRM_45102 [EOS-Additive-M600] — PLC_LOOP_GAIN_2 | Default: 75.000 | Linked Fault: ERR-EOS600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M600.

PRM_45103 [EOS-Additive-M600] — ATC_LOOP_GAIN_3 | Default: 112.500 | Linked Fault: ERR-EOS600-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on EOS-Additive-M600.

PRM_45104 [EOS-Additive-M600] — ZAX_LOOP_GAIN_4 | Default: 150.000 | Linked Fault: ERR-EOS600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on EOS-Additive-M600.

PRM_45105 [EOS-Additive-M600] — LUB_LOOP_GAIN_5 | Default: 187.500 | Linked Fault: ERR-EOS600-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on EOS-Additive-M600.

PRM_45106 [EOS-Additive-M600] — HYD_LOOP_GAIN_6 | Default: 225.000 | Linked Fault: ERR-EOS600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on EOS-Additive-M600.

PRM_45107 [EOS-Additive-M600] — BAX_LOOP_GAIN_7 | Default: 12.500 | Linked Fault: ERR-EOS600-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on EOS-Additive-M600.

PRM_45108 [EOS-Additive-M600] — VAC_LOOP_GAIN_8 | Default: 50.000 | Linked Fault: ERR-EOS600-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on EOS-Additive-M600.

PRM_45109 [EOS-Additive-M600] — CLN_LOOP_GAIN_9 | Default: 87.500 | Linked Fault: ERR-EOS600-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on EOS-Additive-M600.

PRM_45110 [EOS-Additive-M600] — CAX_LOOP_GAIN_10 | Default: 125.000 | Linked Fault: ERR-EOS600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on EOS-Additive-M600.

PRM_45111 [EOS-Additive-M600] — THM_LOOP_GAIN_11 | Default: 162.500 | Linked Fault: ERR-EOS600-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on EOS-Additive-M600.

PRM_45112 [EOS-Additive-M600] — XAX_LOOP_GAIN_12 | Default: 200.000 | Linked Fault: ERR-EOS600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on EOS-Additive-M600.

PRM_45113 [EOS-Additive-M600] — OPT_LOOP_GAIN_13 | Default: 237.500 | Linked Fault: ERR-EOS600-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on EOS-Additive-M600.

PRM_45114 [EOS-Additive-M600] — SPN_LOOP_GAIN_14 | Default: 25.000 | Linked Fault: ERR-EOS600-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on EOS-Additive-M600.

PRM_45115 [EOS-Additive-M600] — YAX_LOOP_GAIN_15 | Default: 62.500 | Linked Fault: ERR-EOS600-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on EOS-Additive-M600.

PRM_45116 [EOS-Additive-M600] — PLC_LOOP_GAIN_16 | Default: 100.000 | Linked Fault: ERR-EOS600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on EOS-Additive-M600.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Mazak-Integrex-i400 (GROUP 1)

Certified Aerospace Bill of Materials for Mazak-Integrex-i400. Verify part numbers before closing Work Orders.

BOM-26-1.1: PN-MAZ400-CAM-GRIP-2602 / Kit KIT-MAZ400-ATC-2 (Mazak-Integrex-i400) | PM Interval: 500 Hrs | Torque: 68 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-MAZ400-ATC-2 within tolerance ±0.0060 mm TIR.

BOM-26-1.2: PN-MAZ400-VLV-PROP-2603 / Kit KIT-MAZ400-HYD-3 (Mazak-Integrex-i400) | PM Interval: 750 Hrs | Torque: 77 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAZ400-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-26-1.3: PN-MAZ400-PMP-PLG-2604 / Kit KIT-MAZ400-CLN-4 (Mazak-Integrex-i400) | PM Interval: 1000 Hrs | Torque: 86 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-MAZ400-CLN-4 within tolerance ±0.0010 mm TIR.

BOM-26-1.4: PN-MAZ400-LIN-COIL-2605 / Kit KIT-MAZ400-XAX-5 (Mazak-Integrex-i400) | PM Interval: 1250 Hrs | Torque: 95 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAZ400-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-26-1.5: PN-MAZ400-BSW-NUT-2606 / Kit KIT-MAZ400-YAX-6 (Mazak-Integrex-i400) | PM Interval: 1500 Hrs | Torque: 104 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-MAZ400-YAX-6 within tolerance ±0.0020 mm TIR.

BOM-26-1.6: PN-MAZ400-CYL-SEAL-2607 / Kit KIT-MAZ400-ZAX-7 (Mazak-Integrex-i400) | PM Interval: 250 Hrs | Torque: 113 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAZ400-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-26-1.7: PN-MAZ400-TRQ-STAT-2608 / Kit KIT-MAZ400-BAX-8 (Mazak-Integrex-i400) | PM Interval: 500 Hrs | Torque: 122 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-MAZ400-BAX-8 within tolerance ±0.0030 mm TIR.

BOM-26-1.8: PN-MAZ400-BRK-DISC-2609 / Kit KIT-MAZ400-CAX-9 (Mazak-Integrex-i400) | PM Interval: 750 Hrs | Torque: 131 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAZ400-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-26-1.9: PN-MAZ400-GLASS-RD-2610 / Kit KIT-MAZ400-OPT-10 (Mazak-Integrex-i400) | PM Interval: 1000 Hrs | Torque: 140 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-MAZ400-OPT-10 within tolerance ±0.0040 mm TIR.

BOM-26-1.10: PN-MAZ400-IGBT-MOD-2611 / Kit KIT-MAZ400-PLC-11 (Mazak-Integrex-i400) | PM Interval: 1250 Hrs | Torque: 149 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAZ400-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-26-1.11: PN-MAZ400-INJ-MET-2612 / Kit KIT-MAZ400-LUB-12 (Mazak-Integrex-i400) | PM Interval: 1500 Hrs | Torque: 158 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-MAZ400-LUB-12 within tolerance ±0.0050 mm TIR.

BOM-26-1.12: PN-MAZ400-GASK-VIT-2613 / Kit KIT-MAZ400-VAC-13 (Mazak-Integrex-i400) | PM Interval: 250 Hrs | Torque: 167 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAZ400-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-26-1.13: PN-MAZ400-HEX-CORE-2614 / Kit KIT-MAZ400-THM-14 (Mazak-Integrex-i400) | PM Interval: 500 Hrs | Torque: 176 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-MAZ400-THM-14 within tolerance ±0.0060 mm TIR.

BOM-26-1.14: PN-MAZ400-BRG-CER-2601 / Kit KIT-MAZ400-SPN-1 (Mazak-Integrex-i400) | PM Interval: 750 Hrs | Torque: 59 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAZ400-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-26-1.15: PN-MAZ400-CAM-GRIP-2602 / Kit KIT-MAZ400-ATC-2 (Mazak-Integrex-i400) | PM Interval: 1000 Hrs | Torque: 68 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-MAZ400-ATC-2 within tolerance ±0.0060 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Mazak-Integrex-i400 (GROUP 2)

Certified Aerospace Bill of Materials for Mazak-Integrex-i400. Verify part numbers before closing Work Orders.

BOM-26-2.1: PN-MAZ400-VLV-PROP-2603 / Kit KIT-MAZ400-HYD-3 (Mazak-Integrex-i400) | PM Interval: 500 Hrs | Torque: 77 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAZ400-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-26-2.2: PN-MAZ400-LIN-COIL-2605 / Kit KIT-MAZ400-XAX-5 (Mazak-Integrex-i400) | PM Interval: 750 Hrs | Torque: 95 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAZ400-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-26-2.3: PN-MAZ400-CYL-SEAL-2607 / Kit KIT-MAZ400-ZAX-7 (Mazak-Integrex-i400) | PM Interval: 1000 Hrs | Torque: 113 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAZ400-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-26-2.4: PN-MAZ400-BRK-DISC-2609 / Kit KIT-MAZ400-CAX-9 (Mazak-Integrex-i400) | PM Interval: 1250 Hrs | Torque: 131 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAZ400-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-26-2.5: PN-MAZ400-IGBT-MOD-2611 / Kit KIT-MAZ400-PLC-11 (Mazak-Integrex-i400) | PM Interval: 1500 Hrs | Torque: 149 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAZ400-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-26-2.6: PN-MAZ400-GASK-VIT-2613 / Kit KIT-MAZ400-VAC-13 (Mazak-Integrex-i400) | PM Interval: 250 Hrs | Torque: 167 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAZ400-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-26-2.7: PN-MAZ400-BRG-CER-2601 / Kit KIT-MAZ400-SPN-1 (Mazak-Integrex-i400) | PM Interval: 500 Hrs | Torque: 59 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAZ400-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-26-2.8: PN-MAZ400-VLV-PROP-2603 / Kit KIT-MAZ400-HYD-3 (Mazak-Integrex-i400) | PM Interval: 750 Hrs | Torque: 77 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAZ400-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-26-2.9: PN-MAZ400-LIN-COIL-2605 / Kit KIT-MAZ400-XAX-5 (Mazak-Integrex-i400) | PM Interval: 1000 Hrs | Torque: 95 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAZ400-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-26-2.10: PN-MAZ400-CYL-SEAL-2607 / Kit KIT-MAZ400-ZAX-7 (Mazak-Integrex-i400) | PM Interval: 1250 Hrs | Torque: 113 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAZ400-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-26-2.11: PN-MAZ400-BRK-DISC-2609 / Kit KIT-MAZ400-CAX-9 (Mazak-Integrex-i400) | PM Interval: 1500 Hrs | Torque: 131 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAZ400-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-26-2.12: PN-MAZ400-IGBT-MOD-2611 / Kit KIT-MAZ400-PLC-11 (Mazak-Integrex-i400) | PM Interval: 250 Hrs | Torque: 149 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAZ400-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-26-2.13: PN-MAZ400-GASK-VIT-2613 / Kit KIT-MAZ400-VAC-13 (Mazak-Integrex-i400) | PM Interval: 500 Hrs | Torque: 167 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAZ400-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-26-2.14: PN-MAZ400-BRG-CER-2601 / Kit KIT-MAZ400-SPN-1 (Mazak-Integrex-i400) | PM Interval: 750 Hrs | Torque: 59 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAZ400-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-26-2.15: PN-MAZ400-VLV-PROP-2603 / Kit KIT-MAZ400-HYD-3 (Mazak-Integrex-i400) | PM Interval: 1000 Hrs | Torque: 77 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAZ400-HYD-3 within tolerance ±0.0065 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Mazak-Integrex-i400 (GROUP 3)

Certified Aerospace Bill of Materials for Mazak-Integrex-i400. Verify part numbers before closing Work Orders.

BOM-26-3.1: PN-MAZ400-PMP-PLG-2604 / Kit KIT-MAZ400-CLN-4 (Mazak-Integrex-i400) | PM Interval: 500 Hrs | Torque: 86 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-MAZ400-CLN-4 within tolerance ±0.0010 mm TIR.

BOM-26-3.2: PN-MAZ400-CYL-SEAL-2607 / Kit KIT-MAZ400-ZAX-7 (Mazak-Integrex-i400) | PM Interval: 750 Hrs | Torque: 113 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAZ400-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-26-3.3: PN-MAZ400-GLASS-RD-2610 / Kit KIT-MAZ400-OPT-10 (Mazak-Integrex-i400) | PM Interval: 1000 Hrs | Torque: 140 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-MAZ400-OPT-10 within tolerance ±0.0040 mm TIR.

BOM-26-3.4: PN-MAZ400-GASK-VIT-2613 / Kit KIT-MAZ400-VAC-13 (Mazak-Integrex-i400) | PM Interval: 1250 Hrs | Torque: 167 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAZ400-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-26-3.5: PN-MAZ400-CAM-GRIP-2602 / Kit KIT-MAZ400-ATC-2 (Mazak-Integrex-i400) | PM Interval: 1500 Hrs | Torque: 68 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-MAZ400-ATC-2 within tolerance ±0.0060 mm TIR.

BOM-26-3.6: PN-MAZ400-LIN-COIL-2605 / Kit KIT-MAZ400-XAX-5 (Mazak-Integrex-i400) | PM Interval: 250 Hrs | Torque: 95 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAZ400-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-26-3.7: PN-MAZ400-TRQ-STAT-2608 / Kit KIT-MAZ400-BAX-8 (Mazak-Integrex-i400) | PM Interval: 500 Hrs | Torque: 122 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-MAZ400-BAX-8 within tolerance ±0.0030 mm TIR.

BOM-26-3.8: PN-MAZ400-IGBT-MOD-2611 / Kit KIT-MAZ400-PLC-11 (Mazak-Integrex-i400) | PM Interval: 750 Hrs | Torque: 149 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAZ400-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-26-3.9: PN-MAZ400-HEX-CORE-2614 / Kit KIT-MAZ400-THM-14 (Mazak-Integrex-i400) | PM Interval: 1000 Hrs | Torque: 176 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-MAZ400-THM-14 within tolerance ±0.0060 mm TIR.

BOM-26-3.10: PN-MAZ400-VLV-PROP-2603 / Kit KIT-MAZ400-HYD-3 (Mazak-Integrex-i400) | PM Interval: 1250 Hrs | Torque: 77 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAZ400-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-26-3.11: PN-MAZ400-BSW-NUT-2606 / Kit KIT-MAZ400-YAX-6 (Mazak-Integrex-i400) | PM Interval: 1500 Hrs | Torque: 104 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-MAZ400-YAX-6 within tolerance ±0.0020 mm TIR.

BOM-26-3.12: PN-MAZ400-BRK-DISC-2609 / Kit KIT-MAZ400-CAX-9 (Mazak-Integrex-i400) | PM Interval: 250 Hrs | Torque: 131 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAZ400-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-26-3.13: PN-MAZ400-INJ-MET-2612 / Kit KIT-MAZ400-LUB-12 (Mazak-Integrex-i400) | PM Interval: 500 Hrs | Torque: 158 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-MAZ400-LUB-12 within tolerance ±0.0050 mm TIR.

BOM-26-3.14: PN-MAZ400-BRG-CER-2601 / Kit KIT-MAZ400-SPN-1 (Mazak-Integrex-i400) | PM Interval: 750 Hrs | Torque: 59 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAZ400-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-26-3.15: PN-MAZ400-PMP-PLG-2604 / Kit KIT-MAZ400-CLN-4 (Mazak-Integrex-i400) | PM Interval: 1000 Hrs | Torque: 86 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-MAZ400-CLN-4 within tolerance ±0.0010 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Mazak-Integrex-i450 (GROUP 1)

Certified Aerospace Bill of Materials for Mazak-Integrex-i450. Verify part numbers before closing Work Orders.

BOM-27-1.1: PN-MAZ450-CAM-GRIP-2702 / Kit KIT-MAZ450-ATC-2 (Mazak-Integrex-i450) | PM Interval: 500 Hrs | Torque: 208 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-MAZ450-ATC-2 within tolerance ±0.0020 mm TIR.

BOM-27-1.2: PN-MAZ450-VLV-PROP-2703 / Kit KIT-MAZ450-HYD-3 (Mazak-Integrex-i450) | PM Interval: 750 Hrs | Torque: 27 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAZ450-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-27-1.3: PN-MAZ450-PMP-PLG-2704 / Kit KIT-MAZ450-CLN-4 (Mazak-Integrex-i450) | PM Interval: 1000 Hrs | Torque: 36 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-MAZ450-CLN-4 within tolerance ±0.0030 mm TIR.

BOM-27-1.4: PN-MAZ450-LIN-COIL-2705 / Kit KIT-MAZ450-XAX-5 (Mazak-Integrex-i450) | PM Interval: 1250 Hrs | Torque: 45 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAZ450-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-27-1.5: PN-MAZ450-BSW-NUT-2706 / Kit KIT-MAZ450-YAX-6 (Mazak-Integrex-i450) | PM Interval: 1500 Hrs | Torque: 54 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-MAZ450-YAX-6 within tolerance ±0.0040 mm TIR.

BOM-27-1.6: PN-MAZ450-CYL-SEAL-2707 / Kit KIT-MAZ450-ZAX-7 (Mazak-Integrex-i450) | PM Interval: 250 Hrs | Torque: 63 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAZ450-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-27-1.7: PN-MAZ450-TRQ-STAT-2708 / Kit KIT-MAZ450-BAX-8 (Mazak-Integrex-i450) | PM Interval: 500 Hrs | Torque: 72 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-MAZ450-BAX-8 within tolerance ±0.0050 mm TIR.

BOM-27-1.8: PN-MAZ450-BRK-DISC-2709 / Kit KIT-MAZ450-CAX-9 (Mazak-Integrex-i450) | PM Interval: 750 Hrs | Torque: 81 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAZ450-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-27-1.9: PN-MAZ450-GLASS-RD-2710 / Kit KIT-MAZ450-OPT-10 (Mazak-Integrex-i450) | PM Interval: 1000 Hrs | Torque: 90 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-MAZ450-OPT-10 within tolerance ±0.0060 mm TIR.

BOM-27-1.10: PN-MAZ450-IGBT-MOD-2711 / Kit KIT-MAZ450-PLC-11 (Mazak-Integrex-i450) | PM Interval: 1250 Hrs | Torque: 99 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAZ450-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-27-1.11: PN-MAZ450-INJ-MET-2712 / Kit KIT-MAZ450-LUB-12 (Mazak-Integrex-i450) | PM Interval: 1500 Hrs | Torque: 108 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-MAZ450-LUB-12 within tolerance ±0.0010 mm TIR.

BOM-27-1.12: PN-MAZ450-GASK-VIT-2713 / Kit KIT-MAZ450-VAC-13 (Mazak-Integrex-i450) | PM Interval: 250 Hrs | Torque: 117 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAZ450-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-27-1.13: PN-MAZ450-HEX-CORE-2714 / Kit KIT-MAZ450-THM-14 (Mazak-Integrex-i450) | PM Interval: 500 Hrs | Torque: 126 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-MAZ450-THM-14 within tolerance ±0.0020 mm TIR.

BOM-27-1.14: PN-MAZ450-BRG-CER-2701 / Kit KIT-MAZ450-SPN-1 (Mazak-Integrex-i450) | PM Interval: 750 Hrs | Torque: 199 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAZ450-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-27-1.15: PN-MAZ450-CAM-GRIP-2702 / Kit KIT-MAZ450-ATC-2 (Mazak-Integrex-i450) | PM Interval: 1000 Hrs | Torque: 208 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-MAZ450-ATC-2 within tolerance ±0.0020 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Mazak-Integrex-i450 (GROUP 2)

Certified Aerospace Bill of Materials for Mazak-Integrex-i450. Verify part numbers before closing Work Orders.

BOM-27-2.1: PN-MAZ450-VLV-PROP-2703 / Kit KIT-MAZ450-HYD-3 (Mazak-Integrex-i450) | PM Interval: 500 Hrs | Torque: 27 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAZ450-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-27-2.2: PN-MAZ450-LIN-COIL-2705 / Kit KIT-MAZ450-XAX-5 (Mazak-Integrex-i450) | PM Interval: 750 Hrs | Torque: 45 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAZ450-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-27-2.3: PN-MAZ450-CYL-SEAL-2707 / Kit KIT-MAZ450-ZAX-7 (Mazak-Integrex-i450) | PM Interval: 1000 Hrs | Torque: 63 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAZ450-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-27-2.4: PN-MAZ450-BRK-DISC-2709 / Kit KIT-MAZ450-CAX-9 (Mazak-Integrex-i450) | PM Interval: 1250 Hrs | Torque: 81 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAZ450-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-27-2.5: PN-MAZ450-IGBT-MOD-2711 / Kit KIT-MAZ450-PLC-11 (Mazak-Integrex-i450) | PM Interval: 1500 Hrs | Torque: 99 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAZ450-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-27-2.6: PN-MAZ450-GASK-VIT-2713 / Kit KIT-MAZ450-VAC-13 (Mazak-Integrex-i450) | PM Interval: 250 Hrs | Torque: 117 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAZ450-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-27-2.7: PN-MAZ450-BRG-CER-2701 / Kit KIT-MAZ450-SPN-1 (Mazak-Integrex-i450) | PM Interval: 500 Hrs | Torque: 199 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAZ450-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-27-2.8: PN-MAZ450-VLV-PROP-2703 / Kit KIT-MAZ450-HYD-3 (Mazak-Integrex-i450) | PM Interval: 750 Hrs | Torque: 27 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAZ450-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-27-2.9: PN-MAZ450-LIN-COIL-2705 / Kit KIT-MAZ450-XAX-5 (Mazak-Integrex-i450) | PM Interval: 1000 Hrs | Torque: 45 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAZ450-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-27-2.10: PN-MAZ450-CYL-SEAL-2707 / Kit KIT-MAZ450-ZAX-7 (Mazak-Integrex-i450) | PM Interval: 1250 Hrs | Torque: 63 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAZ450-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-27-2.11: PN-MAZ450-BRK-DISC-2709 / Kit KIT-MAZ450-CAX-9 (Mazak-Integrex-i450) | PM Interval: 1500 Hrs | Torque: 81 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAZ450-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-27-2.12: PN-MAZ450-IGBT-MOD-2711 / Kit KIT-MAZ450-PLC-11 (Mazak-Integrex-i450) | PM Interval: 250 Hrs | Torque: 99 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAZ450-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-27-2.13: PN-MAZ450-GASK-VIT-2713 / Kit KIT-MAZ450-VAC-13 (Mazak-Integrex-i450) | PM Interval: 500 Hrs | Torque: 117 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAZ450-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-27-2.14: PN-MAZ450-BRG-CER-2701 / Kit KIT-MAZ450-SPN-1 (Mazak-Integrex-i450) | PM Interval: 750 Hrs | Torque: 199 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAZ450-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-27-2.15: PN-MAZ450-VLV-PROP-2703 / Kit KIT-MAZ450-HYD-3 (Mazak-Integrex-i450) | PM Interval: 1000 Hrs | Torque: 27 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAZ450-HYD-3 within tolerance ±0.0025 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Mazak-Integrex-i450 (GROUP 3)

Certified Aerospace Bill of Materials for Mazak-Integrex-i450. Verify part numbers before closing Work Orders.

BOM-27-3.1: PN-MAZ450-PMP-PLG-2704 / Kit KIT-MAZ450-CLN-4 (Mazak-Integrex-i450) | PM Interval: 500 Hrs | Torque: 36 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-MAZ450-CLN-4 within tolerance ±0.0030 mm TIR.

BOM-27-3.2: PN-MAZ450-CYL-SEAL-2707 / Kit KIT-MAZ450-ZAX-7 (Mazak-Integrex-i450) | PM Interval: 750 Hrs | Torque: 63 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAZ450-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-27-3.3: PN-MAZ450-GLASS-RD-2710 / Kit KIT-MAZ450-OPT-10 (Mazak-Integrex-i450) | PM Interval: 1000 Hrs | Torque: 90 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-MAZ450-OPT-10 within tolerance ±0.0060 mm TIR.

BOM-27-3.4: PN-MAZ450-GASK-VIT-2713 / Kit KIT-MAZ450-VAC-13 (Mazak-Integrex-i450) | PM Interval: 1250 Hrs | Torque: 117 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAZ450-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-27-3.5: PN-MAZ450-CAM-GRIP-2702 / Kit KIT-MAZ450-ATC-2 (Mazak-Integrex-i450) | PM Interval: 1500 Hrs | Torque: 208 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-MAZ450-ATC-2 within tolerance ±0.0020 mm TIR.

BOM-27-3.6: PN-MAZ450-LIN-COIL-2705 / Kit KIT-MAZ450-XAX-5 (Mazak-Integrex-i450) | PM Interval: 250 Hrs | Torque: 45 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAZ450-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-27-3.7: PN-MAZ450-TRQ-STAT-2708 / Kit KIT-MAZ450-BAX-8 (Mazak-Integrex-i450) | PM Interval: 500 Hrs | Torque: 72 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-MAZ450-BAX-8 within tolerance ±0.0050 mm TIR.

BOM-27-3.8: PN-MAZ450-IGBT-MOD-2711 / Kit KIT-MAZ450-PLC-11 (Mazak-Integrex-i450) | PM Interval: 750 Hrs | Torque: 99 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAZ450-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-27-3.9: PN-MAZ450-HEX-CORE-2714 / Kit KIT-MAZ450-THM-14 (Mazak-Integrex-i450) | PM Interval: 1000 Hrs | Torque: 126 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-MAZ450-THM-14 within tolerance ±0.0020 mm TIR.

BOM-27-3.10: PN-MAZ450-VLV-PROP-2703 / Kit KIT-MAZ450-HYD-3 (Mazak-Integrex-i450) | PM Interval: 1250 Hrs | Torque: 27 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAZ450-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-27-3.11: PN-MAZ450-BSW-NUT-2706 / Kit KIT-MAZ450-YAX-6 (Mazak-Integrex-i450) | PM Interval: 1500 Hrs | Torque: 54 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-MAZ450-YAX-6 within tolerance ±0.0040 mm TIR.

BOM-27-3.12: PN-MAZ450-BRK-DISC-2709 / Kit KIT-MAZ450-CAX-9 (Mazak-Integrex-i450) | PM Interval: 250 Hrs | Torque: 81 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAZ450-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-27-3.13: PN-MAZ450-INJ-MET-2712 / Kit KIT-MAZ450-LUB-12 (Mazak-Integrex-i450) | PM Interval: 500 Hrs | Torque: 108 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-MAZ450-LUB-12 within tolerance ±0.0010 mm TIR.

BOM-27-3.14: PN-MAZ450-BRG-CER-2701 / Kit KIT-MAZ450-SPN-1 (Mazak-Integrex-i450) | PM Interval: 750 Hrs | Torque: 199 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAZ450-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-27-3.15: PN-MAZ450-PMP-PLG-2704 / Kit KIT-MAZ450-CLN-4 (Mazak-Integrex-i450) | PM Interval: 1000 Hrs | Torque: 36 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-MAZ450-CLN-4 within tolerance ±0.0030 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Mazak-Integrex-i500 (GROUP 1)

Certified Aerospace Bill of Materials for Mazak-Integrex-i500. Verify part numbers before closing Work Orders.

BOM-28-1.1: PN-MAZ500-CAM-GRIP-2802 / Kit KIT-MAZ500-ATC-2 (Mazak-Integrex-i500) | PM Interval: 500 Hrs | Torque: 158 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-MAZ500-ATC-2 within tolerance ±0.0040 mm TIR.

BOM-28-1.2: PN-MAZ500-VLV-PROP-2803 / Kit KIT-MAZ500-HYD-3 (Mazak-Integrex-i500) | PM Interval: 750 Hrs | Torque: 167 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAZ500-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-28-1.3: PN-MAZ500-PMP-PLG-2804 / Kit KIT-MAZ500-CLN-4 (Mazak-Integrex-i500) | PM Interval: 1000 Hrs | Torque: 176 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-MAZ500-CLN-4 within tolerance ±0.0050 mm TIR.

BOM-28-1.4: PN-MAZ500-LIN-COIL-2805 / Kit KIT-MAZ500-XAX-5 (Mazak-Integrex-i500) | PM Interval: 1250 Hrs | Torque: 185 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAZ500-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-28-1.5: PN-MAZ500-BSW-NUT-2806 / Kit KIT-MAZ500-YAX-6 (Mazak-Integrex-i500) | PM Interval: 1500 Hrs | Torque: 194 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-MAZ500-YAX-6 within tolerance ±0.0060 mm TIR.

BOM-28-1.6: PN-MAZ500-CYL-SEAL-2807 / Kit KIT-MAZ500-ZAX-7 (Mazak-Integrex-i500) | PM Interval: 250 Hrs | Torque: 203 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAZ500-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-28-1.7: PN-MAZ500-TRQ-STAT-2808 / Kit KIT-MAZ500-BAX-8 (Mazak-Integrex-i500) | PM Interval: 500 Hrs | Torque: 22 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-MAZ500-BAX-8 within tolerance ±0.0010 mm TIR.

BOM-28-1.8: PN-MAZ500-BRK-DISC-2809 / Kit KIT-MAZ500-CAX-9 (Mazak-Integrex-i500) | PM Interval: 750 Hrs | Torque: 31 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAZ500-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-28-1.9: PN-MAZ500-GLASS-RD-2810 / Kit KIT-MAZ500-OPT-10 (Mazak-Integrex-i500) | PM Interval: 1000 Hrs | Torque: 40 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-MAZ500-OPT-10 within tolerance ±0.0020 mm TIR.

BOM-28-1.10: PN-MAZ500-IGBT-MOD-2811 / Kit KIT-MAZ500-PLC-11 (Mazak-Integrex-i500) | PM Interval: 1250 Hrs | Torque: 49 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAZ500-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-28-1.11: PN-MAZ500-INJ-MET-2812 / Kit KIT-MAZ500-LUB-12 (Mazak-Integrex-i500) | PM Interval: 1500 Hrs | Torque: 58 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-MAZ500-LUB-12 within tolerance ±0.0030 mm TIR.

BOM-28-1.12: PN-MAZ500-GASK-VIT-2813 / Kit KIT-MAZ500-VAC-13 (Mazak-Integrex-i500) | PM Interval: 250 Hrs | Torque: 67 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAZ500-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-28-1.13: PN-MAZ500-HEX-CORE-2814 / Kit KIT-MAZ500-THM-14 (Mazak-Integrex-i500) | PM Interval: 500 Hrs | Torque: 76 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-MAZ500-THM-14 within tolerance ±0.0040 mm TIR.

BOM-28-1.14: PN-MAZ500-BRG-CER-2801 / Kit KIT-MAZ500-SPN-1 (Mazak-Integrex-i500) | PM Interval: 750 Hrs | Torque: 149 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAZ500-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-28-1.15: PN-MAZ500-CAM-GRIP-2802 / Kit KIT-MAZ500-ATC-2 (Mazak-Integrex-i500) | PM Interval: 1000 Hrs | Torque: 158 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-MAZ500-ATC-2 within tolerance ±0.0040 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Mazak-Integrex-i500 (GROUP 2)

Certified Aerospace Bill of Materials for Mazak-Integrex-i500. Verify part numbers before closing Work Orders.

BOM-28-2.1: PN-MAZ500-VLV-PROP-2803 / Kit KIT-MAZ500-HYD-3 (Mazak-Integrex-i500) | PM Interval: 500 Hrs | Torque: 167 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAZ500-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-28-2.2: PN-MAZ500-LIN-COIL-2805 / Kit KIT-MAZ500-XAX-5 (Mazak-Integrex-i500) | PM Interval: 750 Hrs | Torque: 185 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAZ500-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-28-2.3: PN-MAZ500-CYL-SEAL-2807 / Kit KIT-MAZ500-ZAX-7 (Mazak-Integrex-i500) | PM Interval: 1000 Hrs | Torque: 203 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAZ500-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-28-2.4: PN-MAZ500-BRK-DISC-2809 / Kit KIT-MAZ500-CAX-9 (Mazak-Integrex-i500) | PM Interval: 1250 Hrs | Torque: 31 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAZ500-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-28-2.5: PN-MAZ500-IGBT-MOD-2811 / Kit KIT-MAZ500-PLC-11 (Mazak-Integrex-i500) | PM Interval: 1500 Hrs | Torque: 49 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAZ500-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-28-2.6: PN-MAZ500-GASK-VIT-2813 / Kit KIT-MAZ500-VAC-13 (Mazak-Integrex-i500) | PM Interval: 250 Hrs | Torque: 67 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAZ500-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-28-2.7: PN-MAZ500-BRG-CER-2801 / Kit KIT-MAZ500-SPN-1 (Mazak-Integrex-i500) | PM Interval: 500 Hrs | Torque: 149 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAZ500-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-28-2.8: PN-MAZ500-VLV-PROP-2803 / Kit KIT-MAZ500-HYD-3 (Mazak-Integrex-i500) | PM Interval: 750 Hrs | Torque: 167 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAZ500-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-28-2.9: PN-MAZ500-LIN-COIL-2805 / Kit KIT-MAZ500-XAX-5 (Mazak-Integrex-i500) | PM Interval: 1000 Hrs | Torque: 185 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAZ500-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-28-2.10: PN-MAZ500-CYL-SEAL-2807 / Kit KIT-MAZ500-ZAX-7 (Mazak-Integrex-i500) | PM Interval: 1250 Hrs | Torque: 203 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAZ500-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-28-2.11: PN-MAZ500-BRK-DISC-2809 / Kit KIT-MAZ500-CAX-9 (Mazak-Integrex-i500) | PM Interval: 1500 Hrs | Torque: 31 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAZ500-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-28-2.12: PN-MAZ500-IGBT-MOD-2811 / Kit KIT-MAZ500-PLC-11 (Mazak-Integrex-i500) | PM Interval: 250 Hrs | Torque: 49 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAZ500-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-28-2.13: PN-MAZ500-GASK-VIT-2813 / Kit KIT-MAZ500-VAC-13 (Mazak-Integrex-i500) | PM Interval: 500 Hrs | Torque: 67 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAZ500-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-28-2.14: PN-MAZ500-BRG-CER-2801 / Kit KIT-MAZ500-SPN-1 (Mazak-Integrex-i500) | PM Interval: 750 Hrs | Torque: 149 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAZ500-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-28-2.15: PN-MAZ500-VLV-PROP-2803 / Kit KIT-MAZ500-HYD-3 (Mazak-Integrex-i500) | PM Interval: 1000 Hrs | Torque: 167 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAZ500-HYD-3 within tolerance ±0.0045 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Mazak-Integrex-i500 (GROUP 3)

Certified Aerospace Bill of Materials for Mazak-Integrex-i500. Verify part numbers before closing Work Orders.

BOM-28-3.1: PN-MAZ500-PMP-PLG-2804 / Kit KIT-MAZ500-CLN-4 (Mazak-Integrex-i500) | PM Interval: 500 Hrs | Torque: 176 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-MAZ500-CLN-4 within tolerance ±0.0050 mm TIR.

BOM-28-3.2: PN-MAZ500-CYL-SEAL-2807 / Kit KIT-MAZ500-ZAX-7 (Mazak-Integrex-i500) | PM Interval: 750 Hrs | Torque: 203 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAZ500-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-28-3.3: PN-MAZ500-GLASS-RD-2810 / Kit KIT-MAZ500-OPT-10 (Mazak-Integrex-i500) | PM Interval: 1000 Hrs | Torque: 40 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-MAZ500-OPT-10 within tolerance ±0.0020 mm TIR.

BOM-28-3.4: PN-MAZ500-GASK-VIT-2813 / Kit KIT-MAZ500-VAC-13 (Mazak-Integrex-i500) | PM Interval: 1250 Hrs | Torque: 67 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAZ500-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-28-3.5: PN-MAZ500-CAM-GRIP-2802 / Kit KIT-MAZ500-ATC-2 (Mazak-Integrex-i500) | PM Interval: 1500 Hrs | Torque: 158 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-MAZ500-ATC-2 within tolerance ±0.0040 mm TIR.

BOM-28-3.6: PN-MAZ500-LIN-COIL-2805 / Kit KIT-MAZ500-XAX-5 (Mazak-Integrex-i500) | PM Interval: 250 Hrs | Torque: 185 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAZ500-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-28-3.7: PN-MAZ500-TRQ-STAT-2808 / Kit KIT-MAZ500-BAX-8 (Mazak-Integrex-i500) | PM Interval: 500 Hrs | Torque: 22 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-MAZ500-BAX-8 within tolerance ±0.0010 mm TIR.

BOM-28-3.8: PN-MAZ500-IGBT-MOD-2811 / Kit KIT-MAZ500-PLC-11 (Mazak-Integrex-i500) | PM Interval: 750 Hrs | Torque: 49 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAZ500-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-28-3.9: PN-MAZ500-HEX-CORE-2814 / Kit KIT-MAZ500-THM-14 (Mazak-Integrex-i500) | PM Interval: 1000 Hrs | Torque: 76 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-MAZ500-THM-14 within tolerance ±0.0040 mm TIR.

BOM-28-3.10: PN-MAZ500-VLV-PROP-2803 / Kit KIT-MAZ500-HYD-3 (Mazak-Integrex-i500) | PM Interval: 1250 Hrs | Torque: 167 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAZ500-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-28-3.11: PN-MAZ500-BSW-NUT-2806 / Kit KIT-MAZ500-YAX-6 (Mazak-Integrex-i500) | PM Interval: 1500 Hrs | Torque: 194 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-MAZ500-YAX-6 within tolerance ±0.0060 mm TIR.

BOM-28-3.12: PN-MAZ500-BRK-DISC-2809 / Kit KIT-MAZ500-CAX-9 (Mazak-Integrex-i500) | PM Interval: 250 Hrs | Torque: 31 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAZ500-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-28-3.13: PN-MAZ500-INJ-MET-2812 / Kit KIT-MAZ500-LUB-12 (Mazak-Integrex-i500) | PM Interval: 500 Hrs | Torque: 58 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-MAZ500-LUB-12 within tolerance ±0.0030 mm TIR.

BOM-28-3.14: PN-MAZ500-BRG-CER-2801 / Kit KIT-MAZ500-SPN-1 (Mazak-Integrex-i500) | PM Interval: 750 Hrs | Torque: 149 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAZ500-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-28-3.15: PN-MAZ500-PMP-PLG-2804 / Kit KIT-MAZ500-CLN-4 (Mazak-Integrex-i500) | PM Interval: 1000 Hrs | Torque: 176 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-MAZ500-CLN-4 within tolerance ±0.0050 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Mazak-Integrex-i550 (GROUP 1)

Certified Aerospace Bill of Materials for Mazak-Integrex-i550. Verify part numbers before closing Work Orders.

BOM-29-1.1: PN-MAZ550-CAM-GRIP-2902 / Kit KIT-MAZ550-ATC-2 (Mazak-Integrex-i550) | PM Interval: 500 Hrs | Torque: 108 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-MAZ550-ATC-2 within tolerance ±0.0060 mm TIR.

BOM-29-1.2: PN-MAZ550-VLV-PROP-2903 / Kit KIT-MAZ550-HYD-3 (Mazak-Integrex-i550) | PM Interval: 750 Hrs | Torque: 117 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAZ550-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-29-1.3: PN-MAZ550-PMP-PLG-2904 / Kit KIT-MAZ550-CLN-4 (Mazak-Integrex-i550) | PM Interval: 1000 Hrs | Torque: 126 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-MAZ550-CLN-4 within tolerance ±0.0010 mm TIR.

BOM-29-1.4: PN-MAZ550-LIN-COIL-2905 / Kit KIT-MAZ550-XAX-5 (Mazak-Integrex-i550) | PM Interval: 1250 Hrs | Torque: 135 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAZ550-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-29-1.5: PN-MAZ550-BSW-NUT-2906 / Kit KIT-MAZ550-YAX-6 (Mazak-Integrex-i550) | PM Interval: 1500 Hrs | Torque: 144 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-MAZ550-YAX-6 within tolerance ±0.0020 mm TIR.

BOM-29-1.6: PN-MAZ550-CYL-SEAL-2907 / Kit KIT-MAZ550-ZAX-7 (Mazak-Integrex-i550) | PM Interval: 250 Hrs | Torque: 153 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAZ550-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-29-1.7: PN-MAZ550-TRQ-STAT-2908 / Kit KIT-MAZ550-BAX-8 (Mazak-Integrex-i550) | PM Interval: 500 Hrs | Torque: 162 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-MAZ550-BAX-8 within tolerance ±0.0030 mm TIR.

BOM-29-1.8: PN-MAZ550-BRK-DISC-2909 / Kit KIT-MAZ550-CAX-9 (Mazak-Integrex-i550) | PM Interval: 750 Hrs | Torque: 171 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAZ550-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-29-1.9: PN-MAZ550-GLASS-RD-2910 / Kit KIT-MAZ550-OPT-10 (Mazak-Integrex-i550) | PM Interval: 1000 Hrs | Torque: 180 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-MAZ550-OPT-10 within tolerance ±0.0040 mm TIR.

BOM-29-1.10: PN-MAZ550-IGBT-MOD-2911 / Kit KIT-MAZ550-PLC-11 (Mazak-Integrex-i550) | PM Interval: 1250 Hrs | Torque: 189 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAZ550-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-29-1.11: PN-MAZ550-INJ-MET-2912 / Kit KIT-MAZ550-LUB-12 (Mazak-Integrex-i550) | PM Interval: 1500 Hrs | Torque: 198 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-MAZ550-LUB-12 within tolerance ±0.0050 mm TIR.

BOM-29-1.12: PN-MAZ550-GASK-VIT-2913 / Kit KIT-MAZ550-VAC-13 (Mazak-Integrex-i550) | PM Interval: 250 Hrs | Torque: 207 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAZ550-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-29-1.13: PN-MAZ550-HEX-CORE-2914 / Kit KIT-MAZ550-THM-14 (Mazak-Integrex-i550) | PM Interval: 500 Hrs | Torque: 26 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-MAZ550-THM-14 within tolerance ±0.0060 mm TIR.

BOM-29-1.14: PN-MAZ550-BRG-CER-2901 / Kit KIT-MAZ550-SPN-1 (Mazak-Integrex-i550) | PM Interval: 750 Hrs | Torque: 99 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAZ550-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-29-1.15: PN-MAZ550-CAM-GRIP-2902 / Kit KIT-MAZ550-ATC-2 (Mazak-Integrex-i550) | PM Interval: 1000 Hrs | Torque: 108 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-MAZ550-ATC-2 within tolerance ±0.0060 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Mazak-Integrex-i550 (GROUP 2)

Certified Aerospace Bill of Materials for Mazak-Integrex-i550. Verify part numbers before closing Work Orders.

BOM-29-2.1: PN-MAZ550-VLV-PROP-2903 / Kit KIT-MAZ550-HYD-3 (Mazak-Integrex-i550) | PM Interval: 500 Hrs | Torque: 117 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAZ550-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-29-2.2: PN-MAZ550-LIN-COIL-2905 / Kit KIT-MAZ550-XAX-5 (Mazak-Integrex-i550) | PM Interval: 750 Hrs | Torque: 135 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAZ550-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-29-2.3: PN-MAZ550-CYL-SEAL-2907 / Kit KIT-MAZ550-ZAX-7 (Mazak-Integrex-i550) | PM Interval: 1000 Hrs | Torque: 153 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAZ550-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-29-2.4: PN-MAZ550-BRK-DISC-2909 / Kit KIT-MAZ550-CAX-9 (Mazak-Integrex-i550) | PM Interval: 1250 Hrs | Torque: 171 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAZ550-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-29-2.5: PN-MAZ550-IGBT-MOD-2911 / Kit KIT-MAZ550-PLC-11 (Mazak-Integrex-i550) | PM Interval: 1500 Hrs | Torque: 189 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAZ550-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-29-2.6: PN-MAZ550-GASK-VIT-2913 / Kit KIT-MAZ550-VAC-13 (Mazak-Integrex-i550) | PM Interval: 250 Hrs | Torque: 207 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAZ550-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-29-2.7: PN-MAZ550-BRG-CER-2901 / Kit KIT-MAZ550-SPN-1 (Mazak-Integrex-i550) | PM Interval: 500 Hrs | Torque: 99 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAZ550-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-29-2.8: PN-MAZ550-VLV-PROP-2903 / Kit KIT-MAZ550-HYD-3 (Mazak-Integrex-i550) | PM Interval: 750 Hrs | Torque: 117 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAZ550-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-29-2.9: PN-MAZ550-LIN-COIL-2905 / Kit KIT-MAZ550-XAX-5 (Mazak-Integrex-i550) | PM Interval: 1000 Hrs | Torque: 135 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAZ550-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-29-2.10: PN-MAZ550-CYL-SEAL-2907 / Kit KIT-MAZ550-ZAX-7 (Mazak-Integrex-i550) | PM Interval: 1250 Hrs | Torque: 153 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAZ550-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-29-2.11: PN-MAZ550-BRK-DISC-2909 / Kit KIT-MAZ550-CAX-9 (Mazak-Integrex-i550) | PM Interval: 1500 Hrs | Torque: 171 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAZ550-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-29-2.12: PN-MAZ550-IGBT-MOD-2911 / Kit KIT-MAZ550-PLC-11 (Mazak-Integrex-i550) | PM Interval: 250 Hrs | Torque: 189 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAZ550-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-29-2.13: PN-MAZ550-GASK-VIT-2913 / Kit KIT-MAZ550-VAC-13 (Mazak-Integrex-i550) | PM Interval: 500 Hrs | Torque: 207 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAZ550-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-29-2.14: PN-MAZ550-BRG-CER-2901 / Kit KIT-MAZ550-SPN-1 (Mazak-Integrex-i550) | PM Interval: 750 Hrs | Torque: 99 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAZ550-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-29-2.15: PN-MAZ550-VLV-PROP-2903 / Kit KIT-MAZ550-HYD-3 (Mazak-Integrex-i550) | PM Interval: 1000 Hrs | Torque: 117 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAZ550-HYD-3 within tolerance ±0.0065 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Mazak-Integrex-i550 (GROUP 3)

Certified Aerospace Bill of Materials for Mazak-Integrex-i550. Verify part numbers before closing Work Orders.

BOM-29-3.1: PN-MAZ550-PMP-PLG-2904 / Kit KIT-MAZ550-CLN-4 (Mazak-Integrex-i550) | PM Interval: 500 Hrs | Torque: 126 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-MAZ550-CLN-4 within tolerance ±0.0010 mm TIR.

BOM-29-3.2: PN-MAZ550-CYL-SEAL-2907 / Kit KIT-MAZ550-ZAX-7 (Mazak-Integrex-i550) | PM Interval: 750 Hrs | Torque: 153 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAZ550-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-29-3.3: PN-MAZ550-GLASS-RD-2910 / Kit KIT-MAZ550-OPT-10 (Mazak-Integrex-i550) | PM Interval: 1000 Hrs | Torque: 180 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-MAZ550-OPT-10 within tolerance ±0.0040 mm TIR.

BOM-29-3.4: PN-MAZ550-GASK-VIT-2913 / Kit KIT-MAZ550-VAC-13 (Mazak-Integrex-i550) | PM Interval: 1250 Hrs | Torque: 207 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAZ550-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-29-3.5: PN-MAZ550-CAM-GRIP-2902 / Kit KIT-MAZ550-ATC-2 (Mazak-Integrex-i550) | PM Interval: 1500 Hrs | Torque: 108 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-MAZ550-ATC-2 within tolerance ±0.0060 mm TIR.

BOM-29-3.6: PN-MAZ550-LIN-COIL-2905 / Kit KIT-MAZ550-XAX-5 (Mazak-Integrex-i550) | PM Interval: 250 Hrs | Torque: 135 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAZ550-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-29-3.7: PN-MAZ550-TRQ-STAT-2908 / Kit KIT-MAZ550-BAX-8 (Mazak-Integrex-i550) | PM Interval: 500 Hrs | Torque: 162 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-MAZ550-BAX-8 within tolerance ±0.0030 mm TIR.

BOM-29-3.8: PN-MAZ550-IGBT-MOD-2911 / Kit KIT-MAZ550-PLC-11 (Mazak-Integrex-i550) | PM Interval: 750 Hrs | Torque: 189 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAZ550-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-29-3.9: PN-MAZ550-HEX-CORE-2914 / Kit KIT-MAZ550-THM-14 (Mazak-Integrex-i550) | PM Interval: 1000 Hrs | Torque: 26 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-MAZ550-THM-14 within tolerance ±0.0060 mm TIR.

BOM-29-3.10: PN-MAZ550-VLV-PROP-2903 / Kit KIT-MAZ550-HYD-3 (Mazak-Integrex-i550) | PM Interval: 1250 Hrs | Torque: 117 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAZ550-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-29-3.11: PN-MAZ550-BSW-NUT-2906 / Kit KIT-MAZ550-YAX-6 (Mazak-Integrex-i550) | PM Interval: 1500 Hrs | Torque: 144 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-MAZ550-YAX-6 within tolerance ±0.0020 mm TIR.

BOM-29-3.12: PN-MAZ550-BRK-DISC-2909 / Kit KIT-MAZ550-CAX-9 (Mazak-Integrex-i550) | PM Interval: 250 Hrs | Torque: 171 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAZ550-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-29-3.13: PN-MAZ550-INJ-MET-2912 / Kit KIT-MAZ550-LUB-12 (Mazak-Integrex-i550) | PM Interval: 500 Hrs | Torque: 198 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-MAZ550-LUB-12 within tolerance ±0.0050 mm TIR.

BOM-29-3.14: PN-MAZ550-BRG-CER-2901 / Kit KIT-MAZ550-SPN-1 (Mazak-Integrex-i550) | PM Interval: 750 Hrs | Torque: 99 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAZ550-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-29-3.15: PN-MAZ550-PMP-PLG-2904 / Kit KIT-MAZ550-CLN-4 (Mazak-Integrex-i550) | PM Interval: 1000 Hrs | Torque: 126 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-MAZ550-CLN-4 within tolerance ±0.0010 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Mazak-Integrex-i600 (GROUP 1)

Certified Aerospace Bill of Materials for Mazak-Integrex-i600. Verify part numbers before closing Work Orders.

BOM-30-1.1: PN-MAZ600-CAM-GRIP-3002 / Kit KIT-MAZ600-ATC-2 (Mazak-Integrex-i600) | PM Interval: 500 Hrs | Torque: 58 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-MAZ600-ATC-2 within tolerance ±0.0020 mm TIR.

BOM-30-1.2: PN-MAZ600-VLV-PROP-3003 / Kit KIT-MAZ600-HYD-3 (Mazak-Integrex-i600) | PM Interval: 750 Hrs | Torque: 67 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAZ600-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-30-1.3: PN-MAZ600-PMP-PLG-3004 / Kit KIT-MAZ600-CLN-4 (Mazak-Integrex-i600) | PM Interval: 1000 Hrs | Torque: 76 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-MAZ600-CLN-4 within tolerance ±0.0030 mm TIR.

BOM-30-1.4: PN-MAZ600-LIN-COIL-3005 / Kit KIT-MAZ600-XAX-5 (Mazak-Integrex-i600) | PM Interval: 1250 Hrs | Torque: 85 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAZ600-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-30-1.5: PN-MAZ600-BSW-NUT-3006 / Kit KIT-MAZ600-YAX-6 (Mazak-Integrex-i600) | PM Interval: 1500 Hrs | Torque: 94 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-MAZ600-YAX-6 within tolerance ±0.0040 mm TIR.

BOM-30-1.6: PN-MAZ600-CYL-SEAL-3007 / Kit KIT-MAZ600-ZAX-7 (Mazak-Integrex-i600) | PM Interval: 250 Hrs | Torque: 103 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAZ600-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-30-1.7: PN-MAZ600-TRQ-STAT-3008 / Kit KIT-MAZ600-BAX-8 (Mazak-Integrex-i600) | PM Interval: 500 Hrs | Torque: 112 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-MAZ600-BAX-8 within tolerance ±0.0050 mm TIR.

BOM-30-1.8: PN-MAZ600-BRK-DISC-3009 / Kit KIT-MAZ600-CAX-9 (Mazak-Integrex-i600) | PM Interval: 750 Hrs | Torque: 121 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAZ600-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-30-1.9: PN-MAZ600-GLASS-RD-3010 / Kit KIT-MAZ600-OPT-10 (Mazak-Integrex-i600) | PM Interval: 1000 Hrs | Torque: 130 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-MAZ600-OPT-10 within tolerance ±0.0060 mm TIR.

BOM-30-1.10: PN-MAZ600-IGBT-MOD-3011 / Kit KIT-MAZ600-PLC-11 (Mazak-Integrex-i600) | PM Interval: 1250 Hrs | Torque: 139 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAZ600-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-30-1.11: PN-MAZ600-INJ-MET-3012 / Kit KIT-MAZ600-LUB-12 (Mazak-Integrex-i600) | PM Interval: 1500 Hrs | Torque: 148 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-MAZ600-LUB-12 within tolerance ±0.0010 mm TIR.

BOM-30-1.12: PN-MAZ600-GASK-VIT-3013 / Kit KIT-MAZ600-VAC-13 (Mazak-Integrex-i600) | PM Interval: 250 Hrs | Torque: 157 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAZ600-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-30-1.13: PN-MAZ600-HEX-CORE-3014 / Kit KIT-MAZ600-THM-14 (Mazak-Integrex-i600) | PM Interval: 500 Hrs | Torque: 166 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-MAZ600-THM-14 within tolerance ±0.0020 mm TIR.

BOM-30-1.14: PN-MAZ600-BRG-CER-3001 / Kit KIT-MAZ600-SPN-1 (Mazak-Integrex-i600) | PM Interval: 750 Hrs | Torque: 49 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAZ600-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-30-1.15: PN-MAZ600-CAM-GRIP-3002 / Kit KIT-MAZ600-ATC-2 (Mazak-Integrex-i600) | PM Interval: 1000 Hrs | Torque: 58 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-MAZ600-ATC-2 within tolerance ±0.0020 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Mazak-Integrex-i600 (GROUP 2)

Certified Aerospace Bill of Materials for Mazak-Integrex-i600. Verify part numbers before closing Work Orders.

BOM-30-2.1: PN-MAZ600-VLV-PROP-3003 / Kit KIT-MAZ600-HYD-3 (Mazak-Integrex-i600) | PM Interval: 500 Hrs | Torque: 67 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAZ600-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-30-2.2: PN-MAZ600-LIN-COIL-3005 / Kit KIT-MAZ600-XAX-5 (Mazak-Integrex-i600) | PM Interval: 750 Hrs | Torque: 85 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAZ600-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-30-2.3: PN-MAZ600-CYL-SEAL-3007 / Kit KIT-MAZ600-ZAX-7 (Mazak-Integrex-i600) | PM Interval: 1000 Hrs | Torque: 103 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAZ600-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-30-2.4: PN-MAZ600-BRK-DISC-3009 / Kit KIT-MAZ600-CAX-9 (Mazak-Integrex-i600) | PM Interval: 1250 Hrs | Torque: 121 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAZ600-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-30-2.5: PN-MAZ600-IGBT-MOD-3011 / Kit KIT-MAZ600-PLC-11 (Mazak-Integrex-i600) | PM Interval: 1500 Hrs | Torque: 139 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAZ600-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-30-2.6: PN-MAZ600-GASK-VIT-3013 / Kit KIT-MAZ600-VAC-13 (Mazak-Integrex-i600) | PM Interval: 250 Hrs | Torque: 157 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAZ600-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-30-2.7: PN-MAZ600-BRG-CER-3001 / Kit KIT-MAZ600-SPN-1 (Mazak-Integrex-i600) | PM Interval: 500 Hrs | Torque: 49 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAZ600-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-30-2.8: PN-MAZ600-VLV-PROP-3003 / Kit KIT-MAZ600-HYD-3 (Mazak-Integrex-i600) | PM Interval: 750 Hrs | Torque: 67 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAZ600-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-30-2.9: PN-MAZ600-LIN-COIL-3005 / Kit KIT-MAZ600-XAX-5 (Mazak-Integrex-i600) | PM Interval: 1000 Hrs | Torque: 85 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAZ600-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-30-2.10: PN-MAZ600-CYL-SEAL-3007 / Kit KIT-MAZ600-ZAX-7 (Mazak-Integrex-i600) | PM Interval: 1250 Hrs | Torque: 103 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAZ600-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-30-2.11: PN-MAZ600-BRK-DISC-3009 / Kit KIT-MAZ600-CAX-9 (Mazak-Integrex-i600) | PM Interval: 1500 Hrs | Torque: 121 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAZ600-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-30-2.12: PN-MAZ600-IGBT-MOD-3011 / Kit KIT-MAZ600-PLC-11 (Mazak-Integrex-i600) | PM Interval: 250 Hrs | Torque: 139 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAZ600-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-30-2.13: PN-MAZ600-GASK-VIT-3013 / Kit KIT-MAZ600-VAC-13 (Mazak-Integrex-i600) | PM Interval: 500 Hrs | Torque: 157 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAZ600-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-30-2.14: PN-MAZ600-BRG-CER-3001 / Kit KIT-MAZ600-SPN-1 (Mazak-Integrex-i600) | PM Interval: 750 Hrs | Torque: 49 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAZ600-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-30-2.15: PN-MAZ600-VLV-PROP-3003 / Kit KIT-MAZ600-HYD-3 (Mazak-Integrex-i600) | PM Interval: 1000 Hrs | Torque: 67 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAZ600-HYD-3 within tolerance ±0.0025 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Mazak-Integrex-i600 (GROUP 3)

Certified Aerospace Bill of Materials for Mazak-Integrex-i600. Verify part numbers before closing Work Orders.

BOM-30-3.1: PN-MAZ600-PMP-PLG-3004 / Kit KIT-MAZ600-CLN-4 (Mazak-Integrex-i600) | PM Interval: 500 Hrs | Torque: 76 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-MAZ600-CLN-4 within tolerance ±0.0030 mm TIR.

BOM-30-3.2: PN-MAZ600-CYL-SEAL-3007 / Kit KIT-MAZ600-ZAX-7 (Mazak-Integrex-i600) | PM Interval: 750 Hrs | Torque: 103 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAZ600-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-30-3.3: PN-MAZ600-GLASS-RD-3010 / Kit KIT-MAZ600-OPT-10 (Mazak-Integrex-i600) | PM Interval: 1000 Hrs | Torque: 130 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-MAZ600-OPT-10 within tolerance ±0.0060 mm TIR.

BOM-30-3.4: PN-MAZ600-GASK-VIT-3013 / Kit KIT-MAZ600-VAC-13 (Mazak-Integrex-i600) | PM Interval: 1250 Hrs | Torque: 157 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAZ600-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-30-3.5: PN-MAZ600-CAM-GRIP-3002 / Kit KIT-MAZ600-ATC-2 (Mazak-Integrex-i600) | PM Interval: 1500 Hrs | Torque: 58 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-MAZ600-ATC-2 within tolerance ±0.0020 mm TIR.

BOM-30-3.6: PN-MAZ600-LIN-COIL-3005 / Kit KIT-MAZ600-XAX-5 (Mazak-Integrex-i600) | PM Interval: 250 Hrs | Torque: 85 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAZ600-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-30-3.7: PN-MAZ600-TRQ-STAT-3008 / Kit KIT-MAZ600-BAX-8 (Mazak-Integrex-i600) | PM Interval: 500 Hrs | Torque: 112 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-MAZ600-BAX-8 within tolerance ±0.0050 mm TIR.

BOM-30-3.8: PN-MAZ600-IGBT-MOD-3011 / Kit KIT-MAZ600-PLC-11 (Mazak-Integrex-i600) | PM Interval: 750 Hrs | Torque: 139 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAZ600-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-30-3.9: PN-MAZ600-HEX-CORE-3014 / Kit KIT-MAZ600-THM-14 (Mazak-Integrex-i600) | PM Interval: 1000 Hrs | Torque: 166 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-MAZ600-THM-14 within tolerance ±0.0020 mm TIR.

BOM-30-3.10: PN-MAZ600-VLV-PROP-3003 / Kit KIT-MAZ600-HYD-3 (Mazak-Integrex-i600) | PM Interval: 1250 Hrs | Torque: 67 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAZ600-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-30-3.11: PN-MAZ600-BSW-NUT-3006 / Kit KIT-MAZ600-YAX-6 (Mazak-Integrex-i600) | PM Interval: 1500 Hrs | Torque: 94 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-MAZ600-YAX-6 within tolerance ±0.0040 mm TIR.

BOM-30-3.12: PN-MAZ600-BRK-DISC-3009 / Kit KIT-MAZ600-CAX-9 (Mazak-Integrex-i600) | PM Interval: 250 Hrs | Torque: 121 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAZ600-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-30-3.13: PN-MAZ600-INJ-MET-3012 / Kit KIT-MAZ600-LUB-12 (Mazak-Integrex-i600) | PM Interval: 500 Hrs | Torque: 148 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-MAZ600-LUB-12 within tolerance ±0.0010 mm TIR.

BOM-30-3.14: PN-MAZ600-BRG-CER-3001 / Kit KIT-MAZ600-SPN-1 (Mazak-Integrex-i600) | PM Interval: 750 Hrs | Torque: 49 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAZ600-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-30-3.15: PN-MAZ600-PMP-PLG-3004 / Kit KIT-MAZ600-CLN-4 (Mazak-Integrex-i600) | PM Interval: 1000 Hrs | Torque: 76 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-MAZ600-CLN-4 within tolerance ±0.0030 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Okuma-Multus-U150 (GROUP 1)

Certified Aerospace Bill of Materials for Okuma-Multus-U150. Verify part numbers before closing Work Orders.

BOM-31-1.1: PN-OKU150-CAM-GRIP-3102 / Kit KIT-OKU150-ATC-2 (Okuma-Multus-U150) | PM Interval: 500 Hrs | Torque: 198 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-OKU150-ATC-2 within tolerance ±0.0040 mm TIR.

BOM-31-1.2: PN-OKU150-VLV-PROP-3103 / Kit KIT-OKU150-HYD-3 (Okuma-Multus-U150) | PM Interval: 750 Hrs | Torque: 207 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-OKU150-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-31-1.3: PN-OKU150-PMP-PLG-3104 / Kit KIT-OKU150-CLN-4 (Okuma-Multus-U150) | PM Interval: 1000 Hrs | Torque: 26 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-OKU150-CLN-4 within tolerance ±0.0050 mm TIR.

BOM-31-1.4: PN-OKU150-LIN-COIL-3105 / Kit KIT-OKU150-XAX-5 (Okuma-Multus-U150) | PM Interval: 1250 Hrs | Torque: 35 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-OKU150-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-31-1.5: PN-OKU150-BSW-NUT-3106 / Kit KIT-OKU150-YAX-6 (Okuma-Multus-U150) | PM Interval: 1500 Hrs | Torque: 44 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-OKU150-YAX-6 within tolerance ±0.0060 mm TIR.

BOM-31-1.6: PN-OKU150-CYL-SEAL-3107 / Kit KIT-OKU150-ZAX-7 (Okuma-Multus-U150) | PM Interval: 250 Hrs | Torque: 53 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-OKU150-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-31-1.7: PN-OKU150-TRQ-STAT-3108 / Kit KIT-OKU150-BAX-8 (Okuma-Multus-U150) | PM Interval: 500 Hrs | Torque: 62 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-OKU150-BAX-8 within tolerance ±0.0010 mm TIR.

BOM-31-1.8: PN-OKU150-BRK-DISC-3109 / Kit KIT-OKU150-CAX-9 (Okuma-Multus-U150) | PM Interval: 750 Hrs | Torque: 71 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-OKU150-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-31-1.9: PN-OKU150-GLASS-RD-3110 / Kit KIT-OKU150-OPT-10 (Okuma-Multus-U150) | PM Interval: 1000 Hrs | Torque: 80 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-OKU150-OPT-10 within tolerance ±0.0020 mm TIR.

BOM-31-1.10: PN-OKU150-IGBT-MOD-3111 / Kit KIT-OKU150-PLC-11 (Okuma-Multus-U150) | PM Interval: 1250 Hrs | Torque: 89 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-OKU150-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-31-1.11: PN-OKU150-INJ-MET-3112 / Kit KIT-OKU150-LUB-12 (Okuma-Multus-U150) | PM Interval: 1500 Hrs | Torque: 98 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-OKU150-LUB-12 within tolerance ±0.0030 mm TIR.

BOM-31-1.12: PN-OKU150-GASK-VIT-3113 / Kit KIT-OKU150-VAC-13 (Okuma-Multus-U150) | PM Interval: 250 Hrs | Torque: 107 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-OKU150-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-31-1.13: PN-OKU150-HEX-CORE-3114 / Kit KIT-OKU150-THM-14 (Okuma-Multus-U150) | PM Interval: 500 Hrs | Torque: 116 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-OKU150-THM-14 within tolerance ±0.0040 mm TIR.

BOM-31-1.14: PN-OKU150-BRG-CER-3101 / Kit KIT-OKU150-SPN-1 (Okuma-Multus-U150) | PM Interval: 750 Hrs | Torque: 189 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-OKU150-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-31-1.15: PN-OKU150-CAM-GRIP-3102 / Kit KIT-OKU150-ATC-2 (Okuma-Multus-U150) | PM Interval: 1000 Hrs | Torque: 198 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-OKU150-ATC-2 within tolerance ±0.0040 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Okuma-Multus-U150 (GROUP 2)

Certified Aerospace Bill of Materials for Okuma-Multus-U150. Verify part numbers before closing Work Orders.

BOM-31-2.1: PN-OKU150-VLV-PROP-3103 / Kit KIT-OKU150-HYD-3 (Okuma-Multus-U150) | PM Interval: 500 Hrs | Torque: 207 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-OKU150-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-31-2.2: PN-OKU150-LIN-COIL-3105 / Kit KIT-OKU150-XAX-5 (Okuma-Multus-U150) | PM Interval: 750 Hrs | Torque: 35 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-OKU150-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-31-2.3: PN-OKU150-CYL-SEAL-3107 / Kit KIT-OKU150-ZAX-7 (Okuma-Multus-U150) | PM Interval: 1000 Hrs | Torque: 53 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-OKU150-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-31-2.4: PN-OKU150-BRK-DISC-3109 / Kit KIT-OKU150-CAX-9 (Okuma-Multus-U150) | PM Interval: 1250 Hrs | Torque: 71 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-OKU150-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-31-2.5: PN-OKU150-IGBT-MOD-3111 / Kit KIT-OKU150-PLC-11 (Okuma-Multus-U150) | PM Interval: 1500 Hrs | Torque: 89 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-OKU150-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-31-2.6: PN-OKU150-GASK-VIT-3113 / Kit KIT-OKU150-VAC-13 (Okuma-Multus-U150) | PM Interval: 250 Hrs | Torque: 107 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-OKU150-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-31-2.7: PN-OKU150-BRG-CER-3101 / Kit KIT-OKU150-SPN-1 (Okuma-Multus-U150) | PM Interval: 500 Hrs | Torque: 189 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-OKU150-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-31-2.8: PN-OKU150-VLV-PROP-3103 / Kit KIT-OKU150-HYD-3 (Okuma-Multus-U150) | PM Interval: 750 Hrs | Torque: 207 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-OKU150-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-31-2.9: PN-OKU150-LIN-COIL-3105 / Kit KIT-OKU150-XAX-5 (Okuma-Multus-U150) | PM Interval: 1000 Hrs | Torque: 35 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-OKU150-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-31-2.10: PN-OKU150-CYL-SEAL-3107 / Kit KIT-OKU150-ZAX-7 (Okuma-Multus-U150) | PM Interval: 1250 Hrs | Torque: 53 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-OKU150-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-31-2.11: PN-OKU150-BRK-DISC-3109 / Kit KIT-OKU150-CAX-9 (Okuma-Multus-U150) | PM Interval: 1500 Hrs | Torque: 71 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-OKU150-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-31-2.12: PN-OKU150-IGBT-MOD-3111 / Kit KIT-OKU150-PLC-11 (Okuma-Multus-U150) | PM Interval: 250 Hrs | Torque: 89 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-OKU150-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-31-2.13: PN-OKU150-GASK-VIT-3113 / Kit KIT-OKU150-VAC-13 (Okuma-Multus-U150) | PM Interval: 500 Hrs | Torque: 107 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-OKU150-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-31-2.14: PN-OKU150-BRG-CER-3101 / Kit KIT-OKU150-SPN-1 (Okuma-Multus-U150) | PM Interval: 750 Hrs | Torque: 189 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-OKU150-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-31-2.15: PN-OKU150-VLV-PROP-3103 / Kit KIT-OKU150-HYD-3 (Okuma-Multus-U150) | PM Interval: 1000 Hrs | Torque: 207 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-OKU150-HYD-3 within tolerance ±0.0045 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Okuma-Multus-U150 (GROUP 3)

Certified Aerospace Bill of Materials for Okuma-Multus-U150. Verify part numbers before closing Work Orders.

BOM-31-3.1: PN-OKU150-PMP-PLG-3104 / Kit KIT-OKU150-CLN-4 (Okuma-Multus-U150) | PM Interval: 500 Hrs | Torque: 26 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-OKU150-CLN-4 within tolerance ±0.0050 mm TIR.

BOM-31-3.2: PN-OKU150-CYL-SEAL-3107 / Kit KIT-OKU150-ZAX-7 (Okuma-Multus-U150) | PM Interval: 750 Hrs | Torque: 53 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-OKU150-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-31-3.3: PN-OKU150-GLASS-RD-3110 / Kit KIT-OKU150-OPT-10 (Okuma-Multus-U150) | PM Interval: 1000 Hrs | Torque: 80 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-OKU150-OPT-10 within tolerance ±0.0020 mm TIR.

BOM-31-3.4: PN-OKU150-GASK-VIT-3113 / Kit KIT-OKU150-VAC-13 (Okuma-Multus-U150) | PM Interval: 1250 Hrs | Torque: 107 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-OKU150-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-31-3.5: PN-OKU150-CAM-GRIP-3102 / Kit KIT-OKU150-ATC-2 (Okuma-Multus-U150) | PM Interval: 1500 Hrs | Torque: 198 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-OKU150-ATC-2 within tolerance ±0.0040 mm TIR.

BOM-31-3.6: PN-OKU150-LIN-COIL-3105 / Kit KIT-OKU150-XAX-5 (Okuma-Multus-U150) | PM Interval: 250 Hrs | Torque: 35 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-OKU150-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-31-3.7: PN-OKU150-TRQ-STAT-3108 / Kit KIT-OKU150-BAX-8 (Okuma-Multus-U150) | PM Interval: 500 Hrs | Torque: 62 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-OKU150-BAX-8 within tolerance ±0.0010 mm TIR.

BOM-31-3.8: PN-OKU150-IGBT-MOD-3111 / Kit KIT-OKU150-PLC-11 (Okuma-Multus-U150) | PM Interval: 750 Hrs | Torque: 89 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-OKU150-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-31-3.9: PN-OKU150-HEX-CORE-3114 / Kit KIT-OKU150-THM-14 (Okuma-Multus-U150) | PM Interval: 1000 Hrs | Torque: 116 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-OKU150-THM-14 within tolerance ±0.0040 mm TIR.

BOM-31-3.10: PN-OKU150-VLV-PROP-3103 / Kit KIT-OKU150-HYD-3 (Okuma-Multus-U150) | PM Interval: 1250 Hrs | Torque: 207 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-OKU150-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-31-3.11: PN-OKU150-BSW-NUT-3106 / Kit KIT-OKU150-YAX-6 (Okuma-Multus-U150) | PM Interval: 1500 Hrs | Torque: 44 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-OKU150-YAX-6 within tolerance ±0.0060 mm TIR.

BOM-31-3.12: PN-OKU150-BRK-DISC-3109 / Kit KIT-OKU150-CAX-9 (Okuma-Multus-U150) | PM Interval: 250 Hrs | Torque: 71 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-OKU150-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-31-3.13: PN-OKU150-INJ-MET-3112 / Kit KIT-OKU150-LUB-12 (Okuma-Multus-U150) | PM Interval: 500 Hrs | Torque: 98 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-OKU150-LUB-12 within tolerance ±0.0030 mm TIR.

BOM-31-3.14: PN-OKU150-BRG-CER-3101 / Kit KIT-OKU150-SPN-1 (Okuma-Multus-U150) | PM Interval: 750 Hrs | Torque: 189 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-OKU150-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-31-3.15: PN-OKU150-PMP-PLG-3104 / Kit KIT-OKU150-CLN-4 (Okuma-Multus-U150) | PM Interval: 1000 Hrs | Torque: 26 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-OKU150-CLN-4 within tolerance ±0.0050 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Okuma-Multus-U200 (GROUP 1)

Certified Aerospace Bill of Materials for Okuma-Multus-U200. Verify part numbers before closing Work Orders.

BOM-32-1.1: PN-OKU200-CAM-GRIP-3202 / Kit KIT-OKU200-ATC-2 (Okuma-Multus-U200) | PM Interval: 500 Hrs | Torque: 148 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-OKU200-ATC-2 within tolerance ±0.0060 mm TIR.

BOM-32-1.2: PN-OKU200-VLV-PROP-3203 / Kit KIT-OKU200-HYD-3 (Okuma-Multus-U200) | PM Interval: 750 Hrs | Torque: 157 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-OKU200-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-32-1.3: PN-OKU200-PMP-PLG-3204 / Kit KIT-OKU200-CLN-4 (Okuma-Multus-U200) | PM Interval: 1000 Hrs | Torque: 166 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-OKU200-CLN-4 within tolerance ±0.0010 mm TIR.

BOM-32-1.4: PN-OKU200-LIN-COIL-3205 / Kit KIT-OKU200-XAX-5 (Okuma-Multus-U200) | PM Interval: 1250 Hrs | Torque: 175 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-OKU200-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-32-1.5: PN-OKU200-BSW-NUT-3206 / Kit KIT-OKU200-YAX-6 (Okuma-Multus-U200) | PM Interval: 1500 Hrs | Torque: 184 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-OKU200-YAX-6 within tolerance ±0.0020 mm TIR.

BOM-32-1.6: PN-OKU200-CYL-SEAL-3207 / Kit KIT-OKU200-ZAX-7 (Okuma-Multus-U200) | PM Interval: 250 Hrs | Torque: 193 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-OKU200-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-32-1.7: PN-OKU200-TRQ-STAT-3208 / Kit KIT-OKU200-BAX-8 (Okuma-Multus-U200) | PM Interval: 500 Hrs | Torque: 202 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-OKU200-BAX-8 within tolerance ±0.0030 mm TIR.

BOM-32-1.8: PN-OKU200-BRK-DISC-3209 / Kit KIT-OKU200-CAX-9 (Okuma-Multus-U200) | PM Interval: 750 Hrs | Torque: 21 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-OKU200-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-32-1.9: PN-OKU200-GLASS-RD-3210 / Kit KIT-OKU200-OPT-10 (Okuma-Multus-U200) | PM Interval: 1000 Hrs | Torque: 30 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-OKU200-OPT-10 within tolerance ±0.0040 mm TIR.

BOM-32-1.10: PN-OKU200-IGBT-MOD-3211 / Kit KIT-OKU200-PLC-11 (Okuma-Multus-U200) | PM Interval: 1250 Hrs | Torque: 39 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-OKU200-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-32-1.11: PN-OKU200-INJ-MET-3212 / Kit KIT-OKU200-LUB-12 (Okuma-Multus-U200) | PM Interval: 1500 Hrs | Torque: 48 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-OKU200-LUB-12 within tolerance ±0.0050 mm TIR.

BOM-32-1.12: PN-OKU200-GASK-VIT-3213 / Kit KIT-OKU200-VAC-13 (Okuma-Multus-U200) | PM Interval: 250 Hrs | Torque: 57 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-OKU200-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-32-1.13: PN-OKU200-HEX-CORE-3214 / Kit KIT-OKU200-THM-14 (Okuma-Multus-U200) | PM Interval: 500 Hrs | Torque: 66 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-OKU200-THM-14 within tolerance ±0.0060 mm TIR.

BOM-32-1.14: PN-OKU200-BRG-CER-3201 / Kit KIT-OKU200-SPN-1 (Okuma-Multus-U200) | PM Interval: 750 Hrs | Torque: 139 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-OKU200-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-32-1.15: PN-OKU200-CAM-GRIP-3202 / Kit KIT-OKU200-ATC-2 (Okuma-Multus-U200) | PM Interval: 1000 Hrs | Torque: 148 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-OKU200-ATC-2 within tolerance ±0.0060 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Okuma-Multus-U200 (GROUP 2)

Certified Aerospace Bill of Materials for Okuma-Multus-U200. Verify part numbers before closing Work Orders.

BOM-32-2.1: PN-OKU200-VLV-PROP-3203 / Kit KIT-OKU200-HYD-3 (Okuma-Multus-U200) | PM Interval: 500 Hrs | Torque: 157 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-OKU200-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-32-2.2: PN-OKU200-LIN-COIL-3205 / Kit KIT-OKU200-XAX-5 (Okuma-Multus-U200) | PM Interval: 750 Hrs | Torque: 175 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-OKU200-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-32-2.3: PN-OKU200-CYL-SEAL-3207 / Kit KIT-OKU200-ZAX-7 (Okuma-Multus-U200) | PM Interval: 1000 Hrs | Torque: 193 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-OKU200-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-32-2.4: PN-OKU200-BRK-DISC-3209 / Kit KIT-OKU200-CAX-9 (Okuma-Multus-U200) | PM Interval: 1250 Hrs | Torque: 21 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-OKU200-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-32-2.5: PN-OKU200-IGBT-MOD-3211 / Kit KIT-OKU200-PLC-11 (Okuma-Multus-U200) | PM Interval: 1500 Hrs | Torque: 39 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-OKU200-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-32-2.6: PN-OKU200-GASK-VIT-3213 / Kit KIT-OKU200-VAC-13 (Okuma-Multus-U200) | PM Interval: 250 Hrs | Torque: 57 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-OKU200-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-32-2.7: PN-OKU200-BRG-CER-3201 / Kit KIT-OKU200-SPN-1 (Okuma-Multus-U200) | PM Interval: 500 Hrs | Torque: 139 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-OKU200-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-32-2.8: PN-OKU200-VLV-PROP-3203 / Kit KIT-OKU200-HYD-3 (Okuma-Multus-U200) | PM Interval: 750 Hrs | Torque: 157 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-OKU200-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-32-2.9: PN-OKU200-LIN-COIL-3205 / Kit KIT-OKU200-XAX-5 (Okuma-Multus-U200) | PM Interval: 1000 Hrs | Torque: 175 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-OKU200-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-32-2.10: PN-OKU200-CYL-SEAL-3207 / Kit KIT-OKU200-ZAX-7 (Okuma-Multus-U200) | PM Interval: 1250 Hrs | Torque: 193 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-OKU200-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-32-2.11: PN-OKU200-BRK-DISC-3209 / Kit KIT-OKU200-CAX-9 (Okuma-Multus-U200) | PM Interval: 1500 Hrs | Torque: 21 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-OKU200-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-32-2.12: PN-OKU200-IGBT-MOD-3211 / Kit KIT-OKU200-PLC-11 (Okuma-Multus-U200) | PM Interval: 250 Hrs | Torque: 39 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-OKU200-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-32-2.13: PN-OKU200-GASK-VIT-3213 / Kit KIT-OKU200-VAC-13 (Okuma-Multus-U200) | PM Interval: 500 Hrs | Torque: 57 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-OKU200-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-32-2.14: PN-OKU200-BRG-CER-3201 / Kit KIT-OKU200-SPN-1 (Okuma-Multus-U200) | PM Interval: 750 Hrs | Torque: 139 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-OKU200-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-32-2.15: PN-OKU200-VLV-PROP-3203 / Kit KIT-OKU200-HYD-3 (Okuma-Multus-U200) | PM Interval: 1000 Hrs | Torque: 157 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-OKU200-HYD-3 within tolerance ±0.0065 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Okuma-Multus-U200 (GROUP 3)

Certified Aerospace Bill of Materials for Okuma-Multus-U200. Verify part numbers before closing Work Orders.

BOM-32-3.1: PN-OKU200-PMP-PLG-3204 / Kit KIT-OKU200-CLN-4 (Okuma-Multus-U200) | PM Interval: 500 Hrs | Torque: 166 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-OKU200-CLN-4 within tolerance ±0.0010 mm TIR.

BOM-32-3.2: PN-OKU200-CYL-SEAL-3207 / Kit KIT-OKU200-ZAX-7 (Okuma-Multus-U200) | PM Interval: 750 Hrs | Torque: 193 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-OKU200-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-32-3.3: PN-OKU200-GLASS-RD-3210 / Kit KIT-OKU200-OPT-10 (Okuma-Multus-U200) | PM Interval: 1000 Hrs | Torque: 30 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-OKU200-OPT-10 within tolerance ±0.0040 mm TIR.

BOM-32-3.4: PN-OKU200-GASK-VIT-3213 / Kit KIT-OKU200-VAC-13 (Okuma-Multus-U200) | PM Interval: 1250 Hrs | Torque: 57 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-OKU200-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-32-3.5: PN-OKU200-CAM-GRIP-3202 / Kit KIT-OKU200-ATC-2 (Okuma-Multus-U200) | PM Interval: 1500 Hrs | Torque: 148 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-OKU200-ATC-2 within tolerance ±0.0060 mm TIR.

BOM-32-3.6: PN-OKU200-LIN-COIL-3205 / Kit KIT-OKU200-XAX-5 (Okuma-Multus-U200) | PM Interval: 250 Hrs | Torque: 175 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-OKU200-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-32-3.7: PN-OKU200-TRQ-STAT-3208 / Kit KIT-OKU200-BAX-8 (Okuma-Multus-U200) | PM Interval: 500 Hrs | Torque: 202 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-OKU200-BAX-8 within tolerance ±0.0030 mm TIR.

BOM-32-3.8: PN-OKU200-IGBT-MOD-3211 / Kit KIT-OKU200-PLC-11 (Okuma-Multus-U200) | PM Interval: 750 Hrs | Torque: 39 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-OKU200-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-32-3.9: PN-OKU200-HEX-CORE-3214 / Kit KIT-OKU200-THM-14 (Okuma-Multus-U200) | PM Interval: 1000 Hrs | Torque: 66 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-OKU200-THM-14 within tolerance ±0.0060 mm TIR.

BOM-32-3.10: PN-OKU200-VLV-PROP-3203 / Kit KIT-OKU200-HYD-3 (Okuma-Multus-U200) | PM Interval: 1250 Hrs | Torque: 157 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-OKU200-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-32-3.11: PN-OKU200-BSW-NUT-3206 / Kit KIT-OKU200-YAX-6 (Okuma-Multus-U200) | PM Interval: 1500 Hrs | Torque: 184 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-OKU200-YAX-6 within tolerance ±0.0020 mm TIR.

BOM-32-3.12: PN-OKU200-BRK-DISC-3209 / Kit KIT-OKU200-CAX-9 (Okuma-Multus-U200) | PM Interval: 250 Hrs | Torque: 21 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-OKU200-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-32-3.13: PN-OKU200-INJ-MET-3212 / Kit KIT-OKU200-LUB-12 (Okuma-Multus-U200) | PM Interval: 500 Hrs | Torque: 48 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-OKU200-LUB-12 within tolerance ±0.0050 mm TIR.

BOM-32-3.14: PN-OKU200-BRG-CER-3201 / Kit KIT-OKU200-SPN-1 (Okuma-Multus-U200) | PM Interval: 750 Hrs | Torque: 139 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-OKU200-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-32-3.15: PN-OKU200-PMP-PLG-3204 / Kit KIT-OKU200-CLN-4 (Okuma-Multus-U200) | PM Interval: 1000 Hrs | Torque: 166 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-OKU200-CLN-4 within tolerance ±0.0010 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Okuma-Multus-U250 (GROUP 1)

Certified Aerospace Bill of Materials for Okuma-Multus-U250. Verify part numbers before closing Work Orders.

BOM-33-1.1: PN-OKU250-CAM-GRIP-3302 / Kit KIT-OKU250-ATC-2 (Okuma-Multus-U250) | PM Interval: 500 Hrs | Torque: 98 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-OKU250-ATC-2 within tolerance ±0.0020 mm TIR.

BOM-33-1.2: PN-OKU250-VLV-PROP-3303 / Kit KIT-OKU250-HYD-3 (Okuma-Multus-U250) | PM Interval: 750 Hrs | Torque: 107 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-OKU250-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-33-1.3: PN-OKU250-PMP-PLG-3304 / Kit KIT-OKU250-CLN-4 (Okuma-Multus-U250) | PM Interval: 1000 Hrs | Torque: 116 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-OKU250-CLN-4 within tolerance ±0.0030 mm TIR.

BOM-33-1.4: PN-OKU250-LIN-COIL-3305 / Kit KIT-OKU250-XAX-5 (Okuma-Multus-U250) | PM Interval: 1250 Hrs | Torque: 125 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-OKU250-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-33-1.5: PN-OKU250-BSW-NUT-3306 / Kit KIT-OKU250-YAX-6 (Okuma-Multus-U250) | PM Interval: 1500 Hrs | Torque: 134 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-OKU250-YAX-6 within tolerance ±0.0040 mm TIR.

BOM-33-1.6: PN-OKU250-CYL-SEAL-3307 / Kit KIT-OKU250-ZAX-7 (Okuma-Multus-U250) | PM Interval: 250 Hrs | Torque: 143 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-OKU250-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-33-1.7: PN-OKU250-TRQ-STAT-3308 / Kit KIT-OKU250-BAX-8 (Okuma-Multus-U250) | PM Interval: 500 Hrs | Torque: 152 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-OKU250-BAX-8 within tolerance ±0.0050 mm TIR.

BOM-33-1.8: PN-OKU250-BRK-DISC-3309 / Kit KIT-OKU250-CAX-9 (Okuma-Multus-U250) | PM Interval: 750 Hrs | Torque: 161 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-OKU250-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-33-1.9: PN-OKU250-GLASS-RD-3310 / Kit KIT-OKU250-OPT-10 (Okuma-Multus-U250) | PM Interval: 1000 Hrs | Torque: 170 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-OKU250-OPT-10 within tolerance ±0.0060 mm TIR.

BOM-33-1.10: PN-OKU250-IGBT-MOD-3311 / Kit KIT-OKU250-PLC-11 (Okuma-Multus-U250) | PM Interval: 1250 Hrs | Torque: 179 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-OKU250-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-33-1.11: PN-OKU250-INJ-MET-3312 / Kit KIT-OKU250-LUB-12 (Okuma-Multus-U250) | PM Interval: 1500 Hrs | Torque: 188 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-OKU250-LUB-12 within tolerance ±0.0010 mm TIR.

BOM-33-1.12: PN-OKU250-GASK-VIT-3313 / Kit KIT-OKU250-VAC-13 (Okuma-Multus-U250) | PM Interval: 250 Hrs | Torque: 197 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-OKU250-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-33-1.13: PN-OKU250-HEX-CORE-3314 / Kit KIT-OKU250-THM-14 (Okuma-Multus-U250) | PM Interval: 500 Hrs | Torque: 206 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-OKU250-THM-14 within tolerance ±0.0020 mm TIR.

BOM-33-1.14: PN-OKU250-BRG-CER-3301 / Kit KIT-OKU250-SPN-1 (Okuma-Multus-U250) | PM Interval: 750 Hrs | Torque: 89 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-OKU250-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-33-1.15: PN-OKU250-CAM-GRIP-3302 / Kit KIT-OKU250-ATC-2 (Okuma-Multus-U250) | PM Interval: 1000 Hrs | Torque: 98 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-OKU250-ATC-2 within tolerance ±0.0020 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Okuma-Multus-U250 (GROUP 2)

Certified Aerospace Bill of Materials for Okuma-Multus-U250. Verify part numbers before closing Work Orders.

BOM-33-2.1: PN-OKU250-VLV-PROP-3303 / Kit KIT-OKU250-HYD-3 (Okuma-Multus-U250) | PM Interval: 500 Hrs | Torque: 107 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-OKU250-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-33-2.2: PN-OKU250-LIN-COIL-3305 / Kit KIT-OKU250-XAX-5 (Okuma-Multus-U250) | PM Interval: 750 Hrs | Torque: 125 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-OKU250-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-33-2.3: PN-OKU250-CYL-SEAL-3307 / Kit KIT-OKU250-ZAX-7 (Okuma-Multus-U250) | PM Interval: 1000 Hrs | Torque: 143 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-OKU250-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-33-2.4: PN-OKU250-BRK-DISC-3309 / Kit KIT-OKU250-CAX-9 (Okuma-Multus-U250) | PM Interval: 1250 Hrs | Torque: 161 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-OKU250-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-33-2.5: PN-OKU250-IGBT-MOD-3311 / Kit KIT-OKU250-PLC-11 (Okuma-Multus-U250) | PM Interval: 1500 Hrs | Torque: 179 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-OKU250-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-33-2.6: PN-OKU250-GASK-VIT-3313 / Kit KIT-OKU250-VAC-13 (Okuma-Multus-U250) | PM Interval: 250 Hrs | Torque: 197 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-OKU250-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-33-2.7: PN-OKU250-BRG-CER-3301 / Kit KIT-OKU250-SPN-1 (Okuma-Multus-U250) | PM Interval: 500 Hrs | Torque: 89 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-OKU250-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-33-2.8: PN-OKU250-VLV-PROP-3303 / Kit KIT-OKU250-HYD-3 (Okuma-Multus-U250) | PM Interval: 750 Hrs | Torque: 107 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-OKU250-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-33-2.9: PN-OKU250-LIN-COIL-3305 / Kit KIT-OKU250-XAX-5 (Okuma-Multus-U250) | PM Interval: 1000 Hrs | Torque: 125 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-OKU250-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-33-2.10: PN-OKU250-CYL-SEAL-3307 / Kit KIT-OKU250-ZAX-7 (Okuma-Multus-U250) | PM Interval: 1250 Hrs | Torque: 143 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-OKU250-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-33-2.11: PN-OKU250-BRK-DISC-3309 / Kit KIT-OKU250-CAX-9 (Okuma-Multus-U250) | PM Interval: 1500 Hrs | Torque: 161 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-OKU250-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-33-2.12: PN-OKU250-IGBT-MOD-3311 / Kit KIT-OKU250-PLC-11 (Okuma-Multus-U250) | PM Interval: 250 Hrs | Torque: 179 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-OKU250-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-33-2.13: PN-OKU250-GASK-VIT-3313 / Kit KIT-OKU250-VAC-13 (Okuma-Multus-U250) | PM Interval: 500 Hrs | Torque: 197 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-OKU250-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-33-2.14: PN-OKU250-BRG-CER-3301 / Kit KIT-OKU250-SPN-1 (Okuma-Multus-U250) | PM Interval: 750 Hrs | Torque: 89 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-OKU250-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-33-2.15: PN-OKU250-VLV-PROP-3303 / Kit KIT-OKU250-HYD-3 (Okuma-Multus-U250) | PM Interval: 1000 Hrs | Torque: 107 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-OKU250-HYD-3 within tolerance ±0.0025 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Okuma-Multus-U250 (GROUP 3)

Certified Aerospace Bill of Materials for Okuma-Multus-U250. Verify part numbers before closing Work Orders.

BOM-33-3.1: PN-OKU250-PMP-PLG-3304 / Kit KIT-OKU250-CLN-4 (Okuma-Multus-U250) | PM Interval: 500 Hrs | Torque: 116 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-OKU250-CLN-4 within tolerance ±0.0030 mm TIR.

BOM-33-3.2: PN-OKU250-CYL-SEAL-3307 / Kit KIT-OKU250-ZAX-7 (Okuma-Multus-U250) | PM Interval: 750 Hrs | Torque: 143 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-OKU250-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-33-3.3: PN-OKU250-GLASS-RD-3310 / Kit KIT-OKU250-OPT-10 (Okuma-Multus-U250) | PM Interval: 1000 Hrs | Torque: 170 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-OKU250-OPT-10 within tolerance ±0.0060 mm TIR.

BOM-33-3.4: PN-OKU250-GASK-VIT-3313 / Kit KIT-OKU250-VAC-13 (Okuma-Multus-U250) | PM Interval: 1250 Hrs | Torque: 197 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-OKU250-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-33-3.5: PN-OKU250-CAM-GRIP-3302 / Kit KIT-OKU250-ATC-2 (Okuma-Multus-U250) | PM Interval: 1500 Hrs | Torque: 98 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-OKU250-ATC-2 within tolerance ±0.0020 mm TIR.

BOM-33-3.6: PN-OKU250-LIN-COIL-3305 / Kit KIT-OKU250-XAX-5 (Okuma-Multus-U250) | PM Interval: 250 Hrs | Torque: 125 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-OKU250-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-33-3.7: PN-OKU250-TRQ-STAT-3308 / Kit KIT-OKU250-BAX-8 (Okuma-Multus-U250) | PM Interval: 500 Hrs | Torque: 152 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-OKU250-BAX-8 within tolerance ±0.0050 mm TIR.

BOM-33-3.8: PN-OKU250-IGBT-MOD-3311 / Kit KIT-OKU250-PLC-11 (Okuma-Multus-U250) | PM Interval: 750 Hrs | Torque: 179 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-OKU250-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-33-3.9: PN-OKU250-HEX-CORE-3314 / Kit KIT-OKU250-THM-14 (Okuma-Multus-U250) | PM Interval: 1000 Hrs | Torque: 206 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-OKU250-THM-14 within tolerance ±0.0020 mm TIR.

BOM-33-3.10: PN-OKU250-VLV-PROP-3303 / Kit KIT-OKU250-HYD-3 (Okuma-Multus-U250) | PM Interval: 1250 Hrs | Torque: 107 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-OKU250-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-33-3.11: PN-OKU250-BSW-NUT-3306 / Kit KIT-OKU250-YAX-6 (Okuma-Multus-U250) | PM Interval: 1500 Hrs | Torque: 134 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-OKU250-YAX-6 within tolerance ±0.0040 mm TIR.

BOM-33-3.12: PN-OKU250-BRK-DISC-3309 / Kit KIT-OKU250-CAX-9 (Okuma-Multus-U250) | PM Interval: 250 Hrs | Torque: 161 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-OKU250-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-33-3.13: PN-OKU250-INJ-MET-3312 / Kit KIT-OKU250-LUB-12 (Okuma-Multus-U250) | PM Interval: 500 Hrs | Torque: 188 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-OKU250-LUB-12 within tolerance ±0.0010 mm TIR.

BOM-33-3.14: PN-OKU250-BRG-CER-3301 / Kit KIT-OKU250-SPN-1 (Okuma-Multus-U250) | PM Interval: 750 Hrs | Torque: 89 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-OKU250-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-33-3.15: PN-OKU250-PMP-PLG-3304 / Kit KIT-OKU250-CLN-4 (Okuma-Multus-U250) | PM Interval: 1000 Hrs | Torque: 116 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-OKU250-CLN-4 within tolerance ±0.0030 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Okuma-Multus-U300 (GROUP 1)

Certified Aerospace Bill of Materials for Okuma-Multus-U300. Verify part numbers before closing Work Orders.

BOM-34-1.1: PN-OKU300-CAM-GRIP-3402 / Kit KIT-OKU300-ATC-2 (Okuma-Multus-U300) | PM Interval: 500 Hrs | Torque: 48 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-OKU300-ATC-2 within tolerance ±0.0040 mm TIR.

BOM-34-1.2: PN-OKU300-VLV-PROP-3403 / Kit KIT-OKU300-HYD-3 (Okuma-Multus-U300) | PM Interval: 750 Hrs | Torque: 57 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-OKU300-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-34-1.3: PN-OKU300-PMP-PLG-3404 / Kit KIT-OKU300-CLN-4 (Okuma-Multus-U300) | PM Interval: 1000 Hrs | Torque: 66 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-OKU300-CLN-4 within tolerance ±0.0050 mm TIR.

BOM-34-1.4: PN-OKU300-LIN-COIL-3405 / Kit KIT-OKU300-XAX-5 (Okuma-Multus-U300) | PM Interval: 1250 Hrs | Torque: 75 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-OKU300-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-34-1.5: PN-OKU300-BSW-NUT-3406 / Kit KIT-OKU300-YAX-6 (Okuma-Multus-U300) | PM Interval: 1500 Hrs | Torque: 84 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-OKU300-YAX-6 within tolerance ±0.0060 mm TIR.

BOM-34-1.6: PN-OKU300-CYL-SEAL-3407 / Kit KIT-OKU300-ZAX-7 (Okuma-Multus-U300) | PM Interval: 250 Hrs | Torque: 93 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-OKU300-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-34-1.7: PN-OKU300-TRQ-STAT-3408 / Kit KIT-OKU300-BAX-8 (Okuma-Multus-U300) | PM Interval: 500 Hrs | Torque: 102 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-OKU300-BAX-8 within tolerance ±0.0010 mm TIR.

BOM-34-1.8: PN-OKU300-BRK-DISC-3409 / Kit KIT-OKU300-CAX-9 (Okuma-Multus-U300) | PM Interval: 750 Hrs | Torque: 111 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-OKU300-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-34-1.9: PN-OKU300-GLASS-RD-3410 / Kit KIT-OKU300-OPT-10 (Okuma-Multus-U300) | PM Interval: 1000 Hrs | Torque: 120 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-OKU300-OPT-10 within tolerance ±0.0020 mm TIR.

BOM-34-1.10: PN-OKU300-IGBT-MOD-3411 / Kit KIT-OKU300-PLC-11 (Okuma-Multus-U300) | PM Interval: 1250 Hrs | Torque: 129 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-OKU300-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-34-1.11: PN-OKU300-INJ-MET-3412 / Kit KIT-OKU300-LUB-12 (Okuma-Multus-U300) | PM Interval: 1500 Hrs | Torque: 138 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-OKU300-LUB-12 within tolerance ±0.0030 mm TIR.

BOM-34-1.12: PN-OKU300-GASK-VIT-3413 / Kit KIT-OKU300-VAC-13 (Okuma-Multus-U300) | PM Interval: 250 Hrs | Torque: 147 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-OKU300-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-34-1.13: PN-OKU300-HEX-CORE-3414 / Kit KIT-OKU300-THM-14 (Okuma-Multus-U300) | PM Interval: 500 Hrs | Torque: 156 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-OKU300-THM-14 within tolerance ±0.0040 mm TIR.

BOM-34-1.14: PN-OKU300-BRG-CER-3401 / Kit KIT-OKU300-SPN-1 (Okuma-Multus-U300) | PM Interval: 750 Hrs | Torque: 39 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-OKU300-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-34-1.15: PN-OKU300-CAM-GRIP-3402 / Kit KIT-OKU300-ATC-2 (Okuma-Multus-U300) | PM Interval: 1000 Hrs | Torque: 48 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-OKU300-ATC-2 within tolerance ±0.0040 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Okuma-Multus-U300 (GROUP 2)

Certified Aerospace Bill of Materials for Okuma-Multus-U300. Verify part numbers before closing Work Orders.

BOM-34-2.1: PN-OKU300-VLV-PROP-3403 / Kit KIT-OKU300-HYD-3 (Okuma-Multus-U300) | PM Interval: 500 Hrs | Torque: 57 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-OKU300-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-34-2.2: PN-OKU300-LIN-COIL-3405 / Kit KIT-OKU300-XAX-5 (Okuma-Multus-U300) | PM Interval: 750 Hrs | Torque: 75 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-OKU300-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-34-2.3: PN-OKU300-CYL-SEAL-3407 / Kit KIT-OKU300-ZAX-7 (Okuma-Multus-U300) | PM Interval: 1000 Hrs | Torque: 93 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-OKU300-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-34-2.4: PN-OKU300-BRK-DISC-3409 / Kit KIT-OKU300-CAX-9 (Okuma-Multus-U300) | PM Interval: 1250 Hrs | Torque: 111 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-OKU300-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-34-2.5: PN-OKU300-IGBT-MOD-3411 / Kit KIT-OKU300-PLC-11 (Okuma-Multus-U300) | PM Interval: 1500 Hrs | Torque: 129 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-OKU300-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-34-2.6: PN-OKU300-GASK-VIT-3413 / Kit KIT-OKU300-VAC-13 (Okuma-Multus-U300) | PM Interval: 250 Hrs | Torque: 147 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-OKU300-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-34-2.7: PN-OKU300-BRG-CER-3401 / Kit KIT-OKU300-SPN-1 (Okuma-Multus-U300) | PM Interval: 500 Hrs | Torque: 39 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-OKU300-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-34-2.8: PN-OKU300-VLV-PROP-3403 / Kit KIT-OKU300-HYD-3 (Okuma-Multus-U300) | PM Interval: 750 Hrs | Torque: 57 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-OKU300-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-34-2.9: PN-OKU300-LIN-COIL-3405 / Kit KIT-OKU300-XAX-5 (Okuma-Multus-U300) | PM Interval: 1000 Hrs | Torque: 75 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-OKU300-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-34-2.10: PN-OKU300-CYL-SEAL-3407 / Kit KIT-OKU300-ZAX-7 (Okuma-Multus-U300) | PM Interval: 1250 Hrs | Torque: 93 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-OKU300-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-34-2.11: PN-OKU300-BRK-DISC-3409 / Kit KIT-OKU300-CAX-9 (Okuma-Multus-U300) | PM Interval: 1500 Hrs | Torque: 111 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-OKU300-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-34-2.12: PN-OKU300-IGBT-MOD-3411 / Kit KIT-OKU300-PLC-11 (Okuma-Multus-U300) | PM Interval: 250 Hrs | Torque: 129 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-OKU300-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-34-2.13: PN-OKU300-GASK-VIT-3413 / Kit KIT-OKU300-VAC-13 (Okuma-Multus-U300) | PM Interval: 500 Hrs | Torque: 147 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-OKU300-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-34-2.14: PN-OKU300-BRG-CER-3401 / Kit KIT-OKU300-SPN-1 (Okuma-Multus-U300) | PM Interval: 750 Hrs | Torque: 39 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-OKU300-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-34-2.15: PN-OKU300-VLV-PROP-3403 / Kit KIT-OKU300-HYD-3 (Okuma-Multus-U300) | PM Interval: 1000 Hrs | Torque: 57 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-OKU300-HYD-3 within tolerance ±0.0045 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Okuma-Multus-U300 (GROUP 3)

Certified Aerospace Bill of Materials for Okuma-Multus-U300. Verify part numbers before closing Work Orders.

BOM-34-3.1: PN-OKU300-PMP-PLG-3404 / Kit KIT-OKU300-CLN-4 (Okuma-Multus-U300) | PM Interval: 500 Hrs | Torque: 66 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-OKU300-CLN-4 within tolerance ±0.0050 mm TIR.

BOM-34-3.2: PN-OKU300-CYL-SEAL-3407 / Kit KIT-OKU300-ZAX-7 (Okuma-Multus-U300) | PM Interval: 750 Hrs | Torque: 93 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-OKU300-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-34-3.3: PN-OKU300-GLASS-RD-3410 / Kit KIT-OKU300-OPT-10 (Okuma-Multus-U300) | PM Interval: 1000 Hrs | Torque: 120 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-OKU300-OPT-10 within tolerance ±0.0020 mm TIR.

BOM-34-3.4: PN-OKU300-GASK-VIT-3413 / Kit KIT-OKU300-VAC-13 (Okuma-Multus-U300) | PM Interval: 1250 Hrs | Torque: 147 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-OKU300-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-34-3.5: PN-OKU300-CAM-GRIP-3402 / Kit KIT-OKU300-ATC-2 (Okuma-Multus-U300) | PM Interval: 1500 Hrs | Torque: 48 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-OKU300-ATC-2 within tolerance ±0.0040 mm TIR.

BOM-34-3.6: PN-OKU300-LIN-COIL-3405 / Kit KIT-OKU300-XAX-5 (Okuma-Multus-U300) | PM Interval: 250 Hrs | Torque: 75 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-OKU300-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-34-3.7: PN-OKU300-TRQ-STAT-3408 / Kit KIT-OKU300-BAX-8 (Okuma-Multus-U300) | PM Interval: 500 Hrs | Torque: 102 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-OKU300-BAX-8 within tolerance ±0.0010 mm TIR.

BOM-34-3.8: PN-OKU300-IGBT-MOD-3411 / Kit KIT-OKU300-PLC-11 (Okuma-Multus-U300) | PM Interval: 750 Hrs | Torque: 129 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-OKU300-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-34-3.9: PN-OKU300-HEX-CORE-3414 / Kit KIT-OKU300-THM-14 (Okuma-Multus-U300) | PM Interval: 1000 Hrs | Torque: 156 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-OKU300-THM-14 within tolerance ±0.0040 mm TIR.

BOM-34-3.10: PN-OKU300-VLV-PROP-3403 / Kit KIT-OKU300-HYD-3 (Okuma-Multus-U300) | PM Interval: 1250 Hrs | Torque: 57 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-OKU300-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-34-3.11: PN-OKU300-BSW-NUT-3406 / Kit KIT-OKU300-YAX-6 (Okuma-Multus-U300) | PM Interval: 1500 Hrs | Torque: 84 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-OKU300-YAX-6 within tolerance ±0.0060 mm TIR.

BOM-34-3.12: PN-OKU300-BRK-DISC-3409 / Kit KIT-OKU300-CAX-9 (Okuma-Multus-U300) | PM Interval: 250 Hrs | Torque: 111 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-OKU300-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-34-3.13: PN-OKU300-INJ-MET-3412 / Kit KIT-OKU300-LUB-12 (Okuma-Multus-U300) | PM Interval: 500 Hrs | Torque: 138 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-OKU300-LUB-12 within tolerance ±0.0030 mm TIR.

BOM-34-3.14: PN-OKU300-BRG-CER-3401 / Kit KIT-OKU300-SPN-1 (Okuma-Multus-U300) | PM Interval: 750 Hrs | Torque: 39 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-OKU300-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-34-3.15: PN-OKU300-PMP-PLG-3404 / Kit KIT-OKU300-CLN-4 (Okuma-Multus-U300) | PM Interval: 1000 Hrs | Torque: 66 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-OKU300-CLN-4 within tolerance ±0.0050 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Okuma-Multus-U350 (GROUP 1)

Certified Aerospace Bill of Materials for Okuma-Multus-U350. Verify part numbers before closing Work Orders.

BOM-35-1.1: PN-OKU350-CAM-GRIP-3502 / Kit KIT-OKU350-ATC-2 (Okuma-Multus-U350) | PM Interval: 500 Hrs | Torque: 188 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-OKU350-ATC-2 within tolerance ±0.0060 mm TIR.

BOM-35-1.2: PN-OKU350-VLV-PROP-3503 / Kit KIT-OKU350-HYD-3 (Okuma-Multus-U350) | PM Interval: 750 Hrs | Torque: 197 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-OKU350-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-35-1.3: PN-OKU350-PMP-PLG-3504 / Kit KIT-OKU350-CLN-4 (Okuma-Multus-U350) | PM Interval: 1000 Hrs | Torque: 206 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-OKU350-CLN-4 within tolerance ±0.0010 mm TIR.

BOM-35-1.4: PN-OKU350-LIN-COIL-3505 / Kit KIT-OKU350-XAX-5 (Okuma-Multus-U350) | PM Interval: 1250 Hrs | Torque: 25 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-OKU350-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-35-1.5: PN-OKU350-BSW-NUT-3506 / Kit KIT-OKU350-YAX-6 (Okuma-Multus-U350) | PM Interval: 1500 Hrs | Torque: 34 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-OKU350-YAX-6 within tolerance ±0.0020 mm TIR.

BOM-35-1.6: PN-OKU350-CYL-SEAL-3507 / Kit KIT-OKU350-ZAX-7 (Okuma-Multus-U350) | PM Interval: 250 Hrs | Torque: 43 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-OKU350-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-35-1.7: PN-OKU350-TRQ-STAT-3508 / Kit KIT-OKU350-BAX-8 (Okuma-Multus-U350) | PM Interval: 500 Hrs | Torque: 52 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-OKU350-BAX-8 within tolerance ±0.0030 mm TIR.

BOM-35-1.8: PN-OKU350-BRK-DISC-3509 / Kit KIT-OKU350-CAX-9 (Okuma-Multus-U350) | PM Interval: 750 Hrs | Torque: 61 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-OKU350-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-35-1.9: PN-OKU350-GLASS-RD-3510 / Kit KIT-OKU350-OPT-10 (Okuma-Multus-U350) | PM Interval: 1000 Hrs | Torque: 70 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-OKU350-OPT-10 within tolerance ±0.0040 mm TIR.

BOM-35-1.10: PN-OKU350-IGBT-MOD-3511 / Kit KIT-OKU350-PLC-11 (Okuma-Multus-U350) | PM Interval: 1250 Hrs | Torque: 79 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-OKU350-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-35-1.11: PN-OKU350-INJ-MET-3512 / Kit KIT-OKU350-LUB-12 (Okuma-Multus-U350) | PM Interval: 1500 Hrs | Torque: 88 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-OKU350-LUB-12 within tolerance ±0.0050 mm TIR.

BOM-35-1.12: PN-OKU350-GASK-VIT-3513 / Kit KIT-OKU350-VAC-13 (Okuma-Multus-U350) | PM Interval: 250 Hrs | Torque: 97 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-OKU350-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-35-1.13: PN-OKU350-HEX-CORE-3514 / Kit KIT-OKU350-THM-14 (Okuma-Multus-U350) | PM Interval: 500 Hrs | Torque: 106 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-OKU350-THM-14 within tolerance ±0.0060 mm TIR.

BOM-35-1.14: PN-OKU350-BRG-CER-3501 / Kit KIT-OKU350-SPN-1 (Okuma-Multus-U350) | PM Interval: 750 Hrs | Torque: 179 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-OKU350-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-35-1.15: PN-OKU350-CAM-GRIP-3502 / Kit KIT-OKU350-ATC-2 (Okuma-Multus-U350) | PM Interval: 1000 Hrs | Torque: 188 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-OKU350-ATC-2 within tolerance ±0.0060 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Okuma-Multus-U350 (GROUP 2)

Certified Aerospace Bill of Materials for Okuma-Multus-U350. Verify part numbers before closing Work Orders.

BOM-35-2.1: PN-OKU350-VLV-PROP-3503 / Kit KIT-OKU350-HYD-3 (Okuma-Multus-U350) | PM Interval: 500 Hrs | Torque: 197 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-OKU350-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-35-2.2: PN-OKU350-LIN-COIL-3505 / Kit KIT-OKU350-XAX-5 (Okuma-Multus-U350) | PM Interval: 750 Hrs | Torque: 25 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-OKU350-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-35-2.3: PN-OKU350-CYL-SEAL-3507 / Kit KIT-OKU350-ZAX-7 (Okuma-Multus-U350) | PM Interval: 1000 Hrs | Torque: 43 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-OKU350-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-35-2.4: PN-OKU350-BRK-DISC-3509 / Kit KIT-OKU350-CAX-9 (Okuma-Multus-U350) | PM Interval: 1250 Hrs | Torque: 61 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-OKU350-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-35-2.5: PN-OKU350-IGBT-MOD-3511 / Kit KIT-OKU350-PLC-11 (Okuma-Multus-U350) | PM Interval: 1500 Hrs | Torque: 79 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-OKU350-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-35-2.6: PN-OKU350-GASK-VIT-3513 / Kit KIT-OKU350-VAC-13 (Okuma-Multus-U350) | PM Interval: 250 Hrs | Torque: 97 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-OKU350-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-35-2.7: PN-OKU350-BRG-CER-3501 / Kit KIT-OKU350-SPN-1 (Okuma-Multus-U350) | PM Interval: 500 Hrs | Torque: 179 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-OKU350-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-35-2.8: PN-OKU350-VLV-PROP-3503 / Kit KIT-OKU350-HYD-3 (Okuma-Multus-U350) | PM Interval: 750 Hrs | Torque: 197 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-OKU350-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-35-2.9: PN-OKU350-LIN-COIL-3505 / Kit KIT-OKU350-XAX-5 (Okuma-Multus-U350) | PM Interval: 1000 Hrs | Torque: 25 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-OKU350-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-35-2.10: PN-OKU350-CYL-SEAL-3507 / Kit KIT-OKU350-ZAX-7 (Okuma-Multus-U350) | PM Interval: 1250 Hrs | Torque: 43 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-OKU350-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-35-2.11: PN-OKU350-BRK-DISC-3509 / Kit KIT-OKU350-CAX-9 (Okuma-Multus-U350) | PM Interval: 1500 Hrs | Torque: 61 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-OKU350-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-35-2.12: PN-OKU350-IGBT-MOD-3511 / Kit KIT-OKU350-PLC-11 (Okuma-Multus-U350) | PM Interval: 250 Hrs | Torque: 79 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-OKU350-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-35-2.13: PN-OKU350-GASK-VIT-3513 / Kit KIT-OKU350-VAC-13 (Okuma-Multus-U350) | PM Interval: 500 Hrs | Torque: 97 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-OKU350-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-35-2.14: PN-OKU350-BRG-CER-3501 / Kit KIT-OKU350-SPN-1 (Okuma-Multus-U350) | PM Interval: 750 Hrs | Torque: 179 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-OKU350-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-35-2.15: PN-OKU350-VLV-PROP-3503 / Kit KIT-OKU350-HYD-3 (Okuma-Multus-U350) | PM Interval: 1000 Hrs | Torque: 197 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-OKU350-HYD-3 within tolerance ±0.0065 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Okuma-Multus-U350 (GROUP 3)

Certified Aerospace Bill of Materials for Okuma-Multus-U350. Verify part numbers before closing Work Orders.

BOM-35-3.1: PN-OKU350-PMP-PLG-3504 / Kit KIT-OKU350-CLN-4 (Okuma-Multus-U350) | PM Interval: 500 Hrs | Torque: 206 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-OKU350-CLN-4 within tolerance ±0.0010 mm TIR.

BOM-35-3.2: PN-OKU350-CYL-SEAL-3507 / Kit KIT-OKU350-ZAX-7 (Okuma-Multus-U350) | PM Interval: 750 Hrs | Torque: 43 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-OKU350-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-35-3.3: PN-OKU350-GLASS-RD-3510 / Kit KIT-OKU350-OPT-10 (Okuma-Multus-U350) | PM Interval: 1000 Hrs | Torque: 70 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-OKU350-OPT-10 within tolerance ±0.0040 mm TIR.

BOM-35-3.4: PN-OKU350-GASK-VIT-3513 / Kit KIT-OKU350-VAC-13 (Okuma-Multus-U350) | PM Interval: 1250 Hrs | Torque: 97 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-OKU350-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-35-3.5: PN-OKU350-CAM-GRIP-3502 / Kit KIT-OKU350-ATC-2 (Okuma-Multus-U350) | PM Interval: 1500 Hrs | Torque: 188 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-OKU350-ATC-2 within tolerance ±0.0060 mm TIR.

BOM-35-3.6: PN-OKU350-LIN-COIL-3505 / Kit KIT-OKU350-XAX-5 (Okuma-Multus-U350) | PM Interval: 250 Hrs | Torque: 25 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-OKU350-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-35-3.7: PN-OKU350-TRQ-STAT-3508 / Kit KIT-OKU350-BAX-8 (Okuma-Multus-U350) | PM Interval: 500 Hrs | Torque: 52 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-OKU350-BAX-8 within tolerance ±0.0030 mm TIR.

BOM-35-3.8: PN-OKU350-IGBT-MOD-3511 / Kit KIT-OKU350-PLC-11 (Okuma-Multus-U350) | PM Interval: 750 Hrs | Torque: 79 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-OKU350-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-35-3.9: PN-OKU350-HEX-CORE-3514 / Kit KIT-OKU350-THM-14 (Okuma-Multus-U350) | PM Interval: 1000 Hrs | Torque: 106 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-OKU350-THM-14 within tolerance ±0.0060 mm TIR.

BOM-35-3.10: PN-OKU350-VLV-PROP-3503 / Kit KIT-OKU350-HYD-3 (Okuma-Multus-U350) | PM Interval: 1250 Hrs | Torque: 197 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-OKU350-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-35-3.11: PN-OKU350-BSW-NUT-3506 / Kit KIT-OKU350-YAX-6 (Okuma-Multus-U350) | PM Interval: 1500 Hrs | Torque: 34 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-OKU350-YAX-6 within tolerance ±0.0020 mm TIR.

BOM-35-3.12: PN-OKU350-BRK-DISC-3509 / Kit KIT-OKU350-CAX-9 (Okuma-Multus-U350) | PM Interval: 250 Hrs | Torque: 61 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-OKU350-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-35-3.13: PN-OKU350-INJ-MET-3512 / Kit KIT-OKU350-LUB-12 (Okuma-Multus-U350) | PM Interval: 500 Hrs | Torque: 88 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-OKU350-LUB-12 within tolerance ±0.0050 mm TIR.

BOM-35-3.14: PN-OKU350-BRG-CER-3501 / Kit KIT-OKU350-SPN-1 (Okuma-Multus-U350) | PM Interval: 750 Hrs | Torque: 179 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-OKU350-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-35-3.15: PN-OKU350-PMP-PLG-3504 / Kit KIT-OKU350-CLN-4 (Okuma-Multus-U350) | PM Interval: 1000 Hrs | Torque: 206 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-OKU350-CLN-4 within tolerance ±0.0010 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Okuma-Multus-U400 (GROUP 1)

Certified Aerospace Bill of Materials for Okuma-Multus-U400. Verify part numbers before closing Work Orders.

BOM-36-1.1: PN-OKU400-CAM-GRIP-3602 / Kit KIT-OKU400-ATC-2 (Okuma-Multus-U400) | PM Interval: 500 Hrs | Torque: 138 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-OKU400-ATC-2 within tolerance ±0.0020 mm TIR.

BOM-36-1.2: PN-OKU400-VLV-PROP-3603 / Kit KIT-OKU400-HYD-3 (Okuma-Multus-U400) | PM Interval: 750 Hrs | Torque: 147 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-OKU400-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-36-1.3: PN-OKU400-PMP-PLG-3604 / Kit KIT-OKU400-CLN-4 (Okuma-Multus-U400) | PM Interval: 1000 Hrs | Torque: 156 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-OKU400-CLN-4 within tolerance ±0.0030 mm TIR.

BOM-36-1.4: PN-OKU400-LIN-COIL-3605 / Kit KIT-OKU400-XAX-5 (Okuma-Multus-U400) | PM Interval: 1250 Hrs | Torque: 165 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-OKU400-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-36-1.5: PN-OKU400-BSW-NUT-3606 / Kit KIT-OKU400-YAX-6 (Okuma-Multus-U400) | PM Interval: 1500 Hrs | Torque: 174 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-OKU400-YAX-6 within tolerance ±0.0040 mm TIR.

BOM-36-1.6: PN-OKU400-CYL-SEAL-3607 / Kit KIT-OKU400-ZAX-7 (Okuma-Multus-U400) | PM Interval: 250 Hrs | Torque: 183 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-OKU400-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-36-1.7: PN-OKU400-TRQ-STAT-3608 / Kit KIT-OKU400-BAX-8 (Okuma-Multus-U400) | PM Interval: 500 Hrs | Torque: 192 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-OKU400-BAX-8 within tolerance ±0.0050 mm TIR.

BOM-36-1.8: PN-OKU400-BRK-DISC-3609 / Kit KIT-OKU400-CAX-9 (Okuma-Multus-U400) | PM Interval: 750 Hrs | Torque: 201 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-OKU400-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-36-1.9: PN-OKU400-GLASS-RD-3610 / Kit KIT-OKU400-OPT-10 (Okuma-Multus-U400) | PM Interval: 1000 Hrs | Torque: 20 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-OKU400-OPT-10 within tolerance ±0.0060 mm TIR.

BOM-36-1.10: PN-OKU400-IGBT-MOD-3611 / Kit KIT-OKU400-PLC-11 (Okuma-Multus-U400) | PM Interval: 1250 Hrs | Torque: 29 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-OKU400-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-36-1.11: PN-OKU400-INJ-MET-3612 / Kit KIT-OKU400-LUB-12 (Okuma-Multus-U400) | PM Interval: 1500 Hrs | Torque: 38 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-OKU400-LUB-12 within tolerance ±0.0010 mm TIR.

BOM-36-1.12: PN-OKU400-GASK-VIT-3613 / Kit KIT-OKU400-VAC-13 (Okuma-Multus-U400) | PM Interval: 250 Hrs | Torque: 47 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-OKU400-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-36-1.13: PN-OKU400-HEX-CORE-3614 / Kit KIT-OKU400-THM-14 (Okuma-Multus-U400) | PM Interval: 500 Hrs | Torque: 56 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-OKU400-THM-14 within tolerance ±0.0020 mm TIR.

BOM-36-1.14: PN-OKU400-BRG-CER-3601 / Kit KIT-OKU400-SPN-1 (Okuma-Multus-U400) | PM Interval: 750 Hrs | Torque: 129 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-OKU400-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-36-1.15: PN-OKU400-CAM-GRIP-3602 / Kit KIT-OKU400-ATC-2 (Okuma-Multus-U400) | PM Interval: 1000 Hrs | Torque: 138 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-OKU400-ATC-2 within tolerance ±0.0020 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Okuma-Multus-U400 (GROUP 2)

Certified Aerospace Bill of Materials for Okuma-Multus-U400. Verify part numbers before closing Work Orders.

BOM-36-2.1: PN-OKU400-VLV-PROP-3603 / Kit KIT-OKU400-HYD-3 (Okuma-Multus-U400) | PM Interval: 500 Hrs | Torque: 147 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-OKU400-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-36-2.2: PN-OKU400-LIN-COIL-3605 / Kit KIT-OKU400-XAX-5 (Okuma-Multus-U400) | PM Interval: 750 Hrs | Torque: 165 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-OKU400-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-36-2.3: PN-OKU400-CYL-SEAL-3607 / Kit KIT-OKU400-ZAX-7 (Okuma-Multus-U400) | PM Interval: 1000 Hrs | Torque: 183 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-OKU400-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-36-2.4: PN-OKU400-BRK-DISC-3609 / Kit KIT-OKU400-CAX-9 (Okuma-Multus-U400) | PM Interval: 1250 Hrs | Torque: 201 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-OKU400-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-36-2.5: PN-OKU400-IGBT-MOD-3611 / Kit KIT-OKU400-PLC-11 (Okuma-Multus-U400) | PM Interval: 1500 Hrs | Torque: 29 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-OKU400-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-36-2.6: PN-OKU400-GASK-VIT-3613 / Kit KIT-OKU400-VAC-13 (Okuma-Multus-U400) | PM Interval: 250 Hrs | Torque: 47 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-OKU400-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-36-2.7: PN-OKU400-BRG-CER-3601 / Kit KIT-OKU400-SPN-1 (Okuma-Multus-U400) | PM Interval: 500 Hrs | Torque: 129 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-OKU400-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-36-2.8: PN-OKU400-VLV-PROP-3603 / Kit KIT-OKU400-HYD-3 (Okuma-Multus-U400) | PM Interval: 750 Hrs | Torque: 147 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-OKU400-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-36-2.9: PN-OKU400-LIN-COIL-3605 / Kit KIT-OKU400-XAX-5 (Okuma-Multus-U400) | PM Interval: 1000 Hrs | Torque: 165 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-OKU400-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-36-2.10: PN-OKU400-CYL-SEAL-3607 / Kit KIT-OKU400-ZAX-7 (Okuma-Multus-U400) | PM Interval: 1250 Hrs | Torque: 183 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-OKU400-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-36-2.11: PN-OKU400-BRK-DISC-3609 / Kit KIT-OKU400-CAX-9 (Okuma-Multus-U400) | PM Interval: 1500 Hrs | Torque: 201 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-OKU400-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-36-2.12: PN-OKU400-IGBT-MOD-3611 / Kit KIT-OKU400-PLC-11 (Okuma-Multus-U400) | PM Interval: 250 Hrs | Torque: 29 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-OKU400-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-36-2.13: PN-OKU400-GASK-VIT-3613 / Kit KIT-OKU400-VAC-13 (Okuma-Multus-U400) | PM Interval: 500 Hrs | Torque: 47 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-OKU400-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-36-2.14: PN-OKU400-BRG-CER-3601 / Kit KIT-OKU400-SPN-1 (Okuma-Multus-U400) | PM Interval: 750 Hrs | Torque: 129 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-OKU400-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-36-2.15: PN-OKU400-VLV-PROP-3603 / Kit KIT-OKU400-HYD-3 (Okuma-Multus-U400) | PM Interval: 1000 Hrs | Torque: 147 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-OKU400-HYD-3 within tolerance ±0.0025 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Okuma-Multus-U400 (GROUP 3)

Certified Aerospace Bill of Materials for Okuma-Multus-U400. Verify part numbers before closing Work Orders.

BOM-36-3.1: PN-OKU400-PMP-PLG-3604 / Kit KIT-OKU400-CLN-4 (Okuma-Multus-U400) | PM Interval: 500 Hrs | Torque: 156 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-OKU400-CLN-4 within tolerance ±0.0030 mm TIR.

BOM-36-3.2: PN-OKU400-CYL-SEAL-3607 / Kit KIT-OKU400-ZAX-7 (Okuma-Multus-U400) | PM Interval: 750 Hrs | Torque: 183 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-OKU400-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-36-3.3: PN-OKU400-GLASS-RD-3610 / Kit KIT-OKU400-OPT-10 (Okuma-Multus-U400) | PM Interval: 1000 Hrs | Torque: 20 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-OKU400-OPT-10 within tolerance ±0.0060 mm TIR.

BOM-36-3.4: PN-OKU400-GASK-VIT-3613 / Kit KIT-OKU400-VAC-13 (Okuma-Multus-U400) | PM Interval: 1250 Hrs | Torque: 47 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-OKU400-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-36-3.5: PN-OKU400-CAM-GRIP-3602 / Kit KIT-OKU400-ATC-2 (Okuma-Multus-U400) | PM Interval: 1500 Hrs | Torque: 138 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-OKU400-ATC-2 within tolerance ±0.0020 mm TIR.

BOM-36-3.6: PN-OKU400-LIN-COIL-3605 / Kit KIT-OKU400-XAX-5 (Okuma-Multus-U400) | PM Interval: 250 Hrs | Torque: 165 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-OKU400-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-36-3.7: PN-OKU400-TRQ-STAT-3608 / Kit KIT-OKU400-BAX-8 (Okuma-Multus-U400) | PM Interval: 500 Hrs | Torque: 192 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-OKU400-BAX-8 within tolerance ±0.0050 mm TIR.

BOM-36-3.8: PN-OKU400-IGBT-MOD-3611 / Kit KIT-OKU400-PLC-11 (Okuma-Multus-U400) | PM Interval: 750 Hrs | Torque: 29 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-OKU400-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-36-3.9: PN-OKU400-HEX-CORE-3614 / Kit KIT-OKU400-THM-14 (Okuma-Multus-U400) | PM Interval: 1000 Hrs | Torque: 56 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-OKU400-THM-14 within tolerance ±0.0020 mm TIR.

BOM-36-3.10: PN-OKU400-VLV-PROP-3603 / Kit KIT-OKU400-HYD-3 (Okuma-Multus-U400) | PM Interval: 1250 Hrs | Torque: 147 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-OKU400-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-36-3.11: PN-OKU400-BSW-NUT-3606 / Kit KIT-OKU400-YAX-6 (Okuma-Multus-U400) | PM Interval: 1500 Hrs | Torque: 174 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-OKU400-YAX-6 within tolerance ±0.0040 mm TIR.

BOM-36-3.12: PN-OKU400-BRK-DISC-3609 / Kit KIT-OKU400-CAX-9 (Okuma-Multus-U400) | PM Interval: 250 Hrs | Torque: 201 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-OKU400-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-36-3.13: PN-OKU400-INJ-MET-3612 / Kit KIT-OKU400-LUB-12 (Okuma-Multus-U400) | PM Interval: 500 Hrs | Torque: 38 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-OKU400-LUB-12 within tolerance ±0.0010 mm TIR.

BOM-36-3.14: PN-OKU400-BRG-CER-3601 / Kit KIT-OKU400-SPN-1 (Okuma-Multus-U400) | PM Interval: 750 Hrs | Torque: 129 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-OKU400-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-36-3.15: PN-OKU400-PMP-PLG-3604 / Kit KIT-OKU400-CLN-4 (Okuma-Multus-U400) | PM Interval: 1000 Hrs | Torque: 156 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-OKU400-CLN-4 within tolerance ±0.0030 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Okuma-Multus-U450 (GROUP 1)

Certified Aerospace Bill of Materials for Okuma-Multus-U450. Verify part numbers before closing Work Orders.

BOM-37-1.1: PN-OKU450-CAM-GRIP-3702 / Kit KIT-OKU450-ATC-2 (Okuma-Multus-U450) | PM Interval: 500 Hrs | Torque: 88 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-OKU450-ATC-2 within tolerance ±0.0040 mm TIR.

BOM-37-1.2: PN-OKU450-VLV-PROP-3703 / Kit KIT-OKU450-HYD-3 (Okuma-Multus-U450) | PM Interval: 750 Hrs | Torque: 97 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-OKU450-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-37-1.3: PN-OKU450-PMP-PLG-3704 / Kit KIT-OKU450-CLN-4 (Okuma-Multus-U450) | PM Interval: 1000 Hrs | Torque: 106 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-OKU450-CLN-4 within tolerance ±0.0050 mm TIR.

BOM-37-1.4: PN-OKU450-LIN-COIL-3705 / Kit KIT-OKU450-XAX-5 (Okuma-Multus-U450) | PM Interval: 1250 Hrs | Torque: 115 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-OKU450-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-37-1.5: PN-OKU450-BSW-NUT-3706 / Kit KIT-OKU450-YAX-6 (Okuma-Multus-U450) | PM Interval: 1500 Hrs | Torque: 124 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-OKU450-YAX-6 within tolerance ±0.0060 mm TIR.

BOM-37-1.6: PN-OKU450-CYL-SEAL-3707 / Kit KIT-OKU450-ZAX-7 (Okuma-Multus-U450) | PM Interval: 250 Hrs | Torque: 133 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-OKU450-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-37-1.7: PN-OKU450-TRQ-STAT-3708 / Kit KIT-OKU450-BAX-8 (Okuma-Multus-U450) | PM Interval: 500 Hrs | Torque: 142 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-OKU450-BAX-8 within tolerance ±0.0010 mm TIR.

BOM-37-1.8: PN-OKU450-BRK-DISC-3709 / Kit KIT-OKU450-CAX-9 (Okuma-Multus-U450) | PM Interval: 750 Hrs | Torque: 151 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-OKU450-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-37-1.9: PN-OKU450-GLASS-RD-3710 / Kit KIT-OKU450-OPT-10 (Okuma-Multus-U450) | PM Interval: 1000 Hrs | Torque: 160 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-OKU450-OPT-10 within tolerance ±0.0020 mm TIR.

BOM-37-1.10: PN-OKU450-IGBT-MOD-3711 / Kit KIT-OKU450-PLC-11 (Okuma-Multus-U450) | PM Interval: 1250 Hrs | Torque: 169 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-OKU450-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-37-1.11: PN-OKU450-INJ-MET-3712 / Kit KIT-OKU450-LUB-12 (Okuma-Multus-U450) | PM Interval: 1500 Hrs | Torque: 178 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-OKU450-LUB-12 within tolerance ±0.0030 mm TIR.

BOM-37-1.12: PN-OKU450-GASK-VIT-3713 / Kit KIT-OKU450-VAC-13 (Okuma-Multus-U450) | PM Interval: 250 Hrs | Torque: 187 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-OKU450-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-37-1.13: PN-OKU450-HEX-CORE-3714 / Kit KIT-OKU450-THM-14 (Okuma-Multus-U450) | PM Interval: 500 Hrs | Torque: 196 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-OKU450-THM-14 within tolerance ±0.0040 mm TIR.

BOM-37-1.14: PN-OKU450-BRG-CER-3701 / Kit KIT-OKU450-SPN-1 (Okuma-Multus-U450) | PM Interval: 750 Hrs | Torque: 79 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-OKU450-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-37-1.15: PN-OKU450-CAM-GRIP-3702 / Kit KIT-OKU450-ATC-2 (Okuma-Multus-U450) | PM Interval: 1000 Hrs | Torque: 88 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-OKU450-ATC-2 within tolerance ±0.0040 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Okuma-Multus-U450 (GROUP 2)

Certified Aerospace Bill of Materials for Okuma-Multus-U450. Verify part numbers before closing Work Orders.

BOM-37-2.1: PN-OKU450-VLV-PROP-3703 / Kit KIT-OKU450-HYD-3 (Okuma-Multus-U450) | PM Interval: 500 Hrs | Torque: 97 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-OKU450-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-37-2.2: PN-OKU450-LIN-COIL-3705 / Kit KIT-OKU450-XAX-5 (Okuma-Multus-U450) | PM Interval: 750 Hrs | Torque: 115 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-OKU450-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-37-2.3: PN-OKU450-CYL-SEAL-3707 / Kit KIT-OKU450-ZAX-7 (Okuma-Multus-U450) | PM Interval: 1000 Hrs | Torque: 133 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-OKU450-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-37-2.4: PN-OKU450-BRK-DISC-3709 / Kit KIT-OKU450-CAX-9 (Okuma-Multus-U450) | PM Interval: 1250 Hrs | Torque: 151 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-OKU450-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-37-2.5: PN-OKU450-IGBT-MOD-3711 / Kit KIT-OKU450-PLC-11 (Okuma-Multus-U450) | PM Interval: 1500 Hrs | Torque: 169 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-OKU450-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-37-2.6: PN-OKU450-GASK-VIT-3713 / Kit KIT-OKU450-VAC-13 (Okuma-Multus-U450) | PM Interval: 250 Hrs | Torque: 187 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-OKU450-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-37-2.7: PN-OKU450-BRG-CER-3701 / Kit KIT-OKU450-SPN-1 (Okuma-Multus-U450) | PM Interval: 500 Hrs | Torque: 79 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-OKU450-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-37-2.8: PN-OKU450-VLV-PROP-3703 / Kit KIT-OKU450-HYD-3 (Okuma-Multus-U450) | PM Interval: 750 Hrs | Torque: 97 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-OKU450-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-37-2.9: PN-OKU450-LIN-COIL-3705 / Kit KIT-OKU450-XAX-5 (Okuma-Multus-U450) | PM Interval: 1000 Hrs | Torque: 115 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-OKU450-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-37-2.10: PN-OKU450-CYL-SEAL-3707 / Kit KIT-OKU450-ZAX-7 (Okuma-Multus-U450) | PM Interval: 1250 Hrs | Torque: 133 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-OKU450-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-37-2.11: PN-OKU450-BRK-DISC-3709 / Kit KIT-OKU450-CAX-9 (Okuma-Multus-U450) | PM Interval: 1500 Hrs | Torque: 151 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-OKU450-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-37-2.12: PN-OKU450-IGBT-MOD-3711 / Kit KIT-OKU450-PLC-11 (Okuma-Multus-U450) | PM Interval: 250 Hrs | Torque: 169 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-OKU450-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-37-2.13: PN-OKU450-GASK-VIT-3713 / Kit KIT-OKU450-VAC-13 (Okuma-Multus-U450) | PM Interval: 500 Hrs | Torque: 187 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-OKU450-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-37-2.14: PN-OKU450-BRG-CER-3701 / Kit KIT-OKU450-SPN-1 (Okuma-Multus-U450) | PM Interval: 750 Hrs | Torque: 79 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-OKU450-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-37-2.15: PN-OKU450-VLV-PROP-3703 / Kit KIT-OKU450-HYD-3 (Okuma-Multus-U450) | PM Interval: 1000 Hrs | Torque: 97 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-OKU450-HYD-3 within tolerance ±0.0045 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Okuma-Multus-U450 (GROUP 3)

Certified Aerospace Bill of Materials for Okuma-Multus-U450. Verify part numbers before closing Work Orders.

BOM-37-3.1: PN-OKU450-PMP-PLG-3704 / Kit KIT-OKU450-CLN-4 (Okuma-Multus-U450) | PM Interval: 500 Hrs | Torque: 106 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-OKU450-CLN-4 within tolerance ±0.0050 mm TIR.

BOM-37-3.2: PN-OKU450-CYL-SEAL-3707 / Kit KIT-OKU450-ZAX-7 (Okuma-Multus-U450) | PM Interval: 750 Hrs | Torque: 133 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-OKU450-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-37-3.3: PN-OKU450-GLASS-RD-3710 / Kit KIT-OKU450-OPT-10 (Okuma-Multus-U450) | PM Interval: 1000 Hrs | Torque: 160 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-OKU450-OPT-10 within tolerance ±0.0020 mm TIR.

BOM-37-3.4: PN-OKU450-GASK-VIT-3713 / Kit KIT-OKU450-VAC-13 (Okuma-Multus-U450) | PM Interval: 1250 Hrs | Torque: 187 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-OKU450-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-37-3.5: PN-OKU450-CAM-GRIP-3702 / Kit KIT-OKU450-ATC-2 (Okuma-Multus-U450) | PM Interval: 1500 Hrs | Torque: 88 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-OKU450-ATC-2 within tolerance ±0.0040 mm TIR.

BOM-37-3.6: PN-OKU450-LIN-COIL-3705 / Kit KIT-OKU450-XAX-5 (Okuma-Multus-U450) | PM Interval: 250 Hrs | Torque: 115 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-OKU450-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-37-3.7: PN-OKU450-TRQ-STAT-3708 / Kit KIT-OKU450-BAX-8 (Okuma-Multus-U450) | PM Interval: 500 Hrs | Torque: 142 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-OKU450-BAX-8 within tolerance ±0.0010 mm TIR.

BOM-37-3.8: PN-OKU450-IGBT-MOD-3711 / Kit KIT-OKU450-PLC-11 (Okuma-Multus-U450) | PM Interval: 750 Hrs | Torque: 169 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-OKU450-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-37-3.9: PN-OKU450-HEX-CORE-3714 / Kit KIT-OKU450-THM-14 (Okuma-Multus-U450) | PM Interval: 1000 Hrs | Torque: 196 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-OKU450-THM-14 within tolerance ±0.0040 mm TIR.

BOM-37-3.10: PN-OKU450-VLV-PROP-3703 / Kit KIT-OKU450-HYD-3 (Okuma-Multus-U450) | PM Interval: 1250 Hrs | Torque: 97 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-OKU450-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-37-3.11: PN-OKU450-BSW-NUT-3706 / Kit KIT-OKU450-YAX-6 (Okuma-Multus-U450) | PM Interval: 1500 Hrs | Torque: 124 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-OKU450-YAX-6 within tolerance ±0.0060 mm TIR.

BOM-37-3.12: PN-OKU450-BRK-DISC-3709 / Kit KIT-OKU450-CAX-9 (Okuma-Multus-U450) | PM Interval: 250 Hrs | Torque: 151 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-OKU450-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-37-3.13: PN-OKU450-INJ-MET-3712 / Kit KIT-OKU450-LUB-12 (Okuma-Multus-U450) | PM Interval: 500 Hrs | Torque: 178 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-OKU450-LUB-12 within tolerance ±0.0030 mm TIR.

BOM-37-3.14: PN-OKU450-BRG-CER-3701 / Kit KIT-OKU450-SPN-1 (Okuma-Multus-U450) | PM Interval: 750 Hrs | Torque: 79 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-OKU450-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-37-3.15: PN-OKU450-PMP-PLG-3704 / Kit KIT-OKU450-CLN-4 (Okuma-Multus-U450) | PM Interval: 1000 Hrs | Torque: 106 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-OKU450-CLN-4 within tolerance ±0.0050 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Okuma-Multus-U500 (GROUP 1)

Certified Aerospace Bill of Materials for Okuma-Multus-U500. Verify part numbers before closing Work Orders.

BOM-38-1.1: PN-OKU500-CAM-GRIP-3802 / Kit KIT-OKU500-ATC-2 (Okuma-Multus-U500) | PM Interval: 500 Hrs | Torque: 38 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-OKU500-ATC-2 within tolerance ±0.0060 mm TIR.

BOM-38-1.2: PN-OKU500-VLV-PROP-3803 / Kit KIT-OKU500-HYD-3 (Okuma-Multus-U500) | PM Interval: 750 Hrs | Torque: 47 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-OKU500-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-38-1.3: PN-OKU500-PMP-PLG-3804 / Kit KIT-OKU500-CLN-4 (Okuma-Multus-U500) | PM Interval: 1000 Hrs | Torque: 56 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-OKU500-CLN-4 within tolerance ±0.0010 mm TIR.

BOM-38-1.4: PN-OKU500-LIN-COIL-3805 / Kit KIT-OKU500-XAX-5 (Okuma-Multus-U500) | PM Interval: 1250 Hrs | Torque: 65 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-OKU500-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-38-1.5: PN-OKU500-BSW-NUT-3806 / Kit KIT-OKU500-YAX-6 (Okuma-Multus-U500) | PM Interval: 1500 Hrs | Torque: 74 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-OKU500-YAX-6 within tolerance ±0.0020 mm TIR.

BOM-38-1.6: PN-OKU500-CYL-SEAL-3807 / Kit KIT-OKU500-ZAX-7 (Okuma-Multus-U500) | PM Interval: 250 Hrs | Torque: 83 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-OKU500-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-38-1.7: PN-OKU500-TRQ-STAT-3808 / Kit KIT-OKU500-BAX-8 (Okuma-Multus-U500) | PM Interval: 500 Hrs | Torque: 92 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-OKU500-BAX-8 within tolerance ±0.0030 mm TIR.

BOM-38-1.8: PN-OKU500-BRK-DISC-3809 / Kit KIT-OKU500-CAX-9 (Okuma-Multus-U500) | PM Interval: 750 Hrs | Torque: 101 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-OKU500-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-38-1.9: PN-OKU500-GLASS-RD-3810 / Kit KIT-OKU500-OPT-10 (Okuma-Multus-U500) | PM Interval: 1000 Hrs | Torque: 110 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-OKU500-OPT-10 within tolerance ±0.0040 mm TIR.

BOM-38-1.10: PN-OKU500-IGBT-MOD-3811 / Kit KIT-OKU500-PLC-11 (Okuma-Multus-U500) | PM Interval: 1250 Hrs | Torque: 119 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-OKU500-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-38-1.11: PN-OKU500-INJ-MET-3812 / Kit KIT-OKU500-LUB-12 (Okuma-Multus-U500) | PM Interval: 1500 Hrs | Torque: 128 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-OKU500-LUB-12 within tolerance ±0.0050 mm TIR.

BOM-38-1.12: PN-OKU500-GASK-VIT-3813 / Kit KIT-OKU500-VAC-13 (Okuma-Multus-U500) | PM Interval: 250 Hrs | Torque: 137 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-OKU500-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-38-1.13: PN-OKU500-HEX-CORE-3814 / Kit KIT-OKU500-THM-14 (Okuma-Multus-U500) | PM Interval: 500 Hrs | Torque: 146 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-OKU500-THM-14 within tolerance ±0.0060 mm TIR.

BOM-38-1.14: PN-OKU500-BRG-CER-3801 / Kit KIT-OKU500-SPN-1 (Okuma-Multus-U500) | PM Interval: 750 Hrs | Torque: 29 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-OKU500-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-38-1.15: PN-OKU500-CAM-GRIP-3802 / Kit KIT-OKU500-ATC-2 (Okuma-Multus-U500) | PM Interval: 1000 Hrs | Torque: 38 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-OKU500-ATC-2 within tolerance ±0.0060 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Okuma-Multus-U500 (GROUP 2)

Certified Aerospace Bill of Materials for Okuma-Multus-U500. Verify part numbers before closing Work Orders.

BOM-38-2.1: PN-OKU500-VLV-PROP-3803 / Kit KIT-OKU500-HYD-3 (Okuma-Multus-U500) | PM Interval: 500 Hrs | Torque: 47 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-OKU500-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-38-2.2: PN-OKU500-LIN-COIL-3805 / Kit KIT-OKU500-XAX-5 (Okuma-Multus-U500) | PM Interval: 750 Hrs | Torque: 65 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-OKU500-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-38-2.3: PN-OKU500-CYL-SEAL-3807 / Kit KIT-OKU500-ZAX-7 (Okuma-Multus-U500) | PM Interval: 1000 Hrs | Torque: 83 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-OKU500-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-38-2.4: PN-OKU500-BRK-DISC-3809 / Kit KIT-OKU500-CAX-9 (Okuma-Multus-U500) | PM Interval: 1250 Hrs | Torque: 101 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-OKU500-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-38-2.5: PN-OKU500-IGBT-MOD-3811 / Kit KIT-OKU500-PLC-11 (Okuma-Multus-U500) | PM Interval: 1500 Hrs | Torque: 119 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-OKU500-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-38-2.6: PN-OKU500-GASK-VIT-3813 / Kit KIT-OKU500-VAC-13 (Okuma-Multus-U500) | PM Interval: 250 Hrs | Torque: 137 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-OKU500-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-38-2.7: PN-OKU500-BRG-CER-3801 / Kit KIT-OKU500-SPN-1 (Okuma-Multus-U500) | PM Interval: 500 Hrs | Torque: 29 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-OKU500-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-38-2.8: PN-OKU500-VLV-PROP-3803 / Kit KIT-OKU500-HYD-3 (Okuma-Multus-U500) | PM Interval: 750 Hrs | Torque: 47 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-OKU500-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-38-2.9: PN-OKU500-LIN-COIL-3805 / Kit KIT-OKU500-XAX-5 (Okuma-Multus-U500) | PM Interval: 1000 Hrs | Torque: 65 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-OKU500-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-38-2.10: PN-OKU500-CYL-SEAL-3807 / Kit KIT-OKU500-ZAX-7 (Okuma-Multus-U500) | PM Interval: 1250 Hrs | Torque: 83 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-OKU500-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-38-2.11: PN-OKU500-BRK-DISC-3809 / Kit KIT-OKU500-CAX-9 (Okuma-Multus-U500) | PM Interval: 1500 Hrs | Torque: 101 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-OKU500-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-38-2.12: PN-OKU500-IGBT-MOD-3811 / Kit KIT-OKU500-PLC-11 (Okuma-Multus-U500) | PM Interval: 250 Hrs | Torque: 119 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-OKU500-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-38-2.13: PN-OKU500-GASK-VIT-3813 / Kit KIT-OKU500-VAC-13 (Okuma-Multus-U500) | PM Interval: 500 Hrs | Torque: 137 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-OKU500-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-38-2.14: PN-OKU500-BRG-CER-3801 / Kit KIT-OKU500-SPN-1 (Okuma-Multus-U500) | PM Interval: 750 Hrs | Torque: 29 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-OKU500-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-38-2.15: PN-OKU500-VLV-PROP-3803 / Kit KIT-OKU500-HYD-3 (Okuma-Multus-U500) | PM Interval: 1000 Hrs | Torque: 47 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-OKU500-HYD-3 within tolerance ±0.0065 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Okuma-Multus-U500 (GROUP 3)

Certified Aerospace Bill of Materials for Okuma-Multus-U500. Verify part numbers before closing Work Orders.

BOM-38-3.1: PN-OKU500-PMP-PLG-3804 / Kit KIT-OKU500-CLN-4 (Okuma-Multus-U500) | PM Interval: 500 Hrs | Torque: 56 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-OKU500-CLN-4 within tolerance ±0.0010 mm TIR.

BOM-38-3.2: PN-OKU500-CYL-SEAL-3807 / Kit KIT-OKU500-ZAX-7 (Okuma-Multus-U500) | PM Interval: 750 Hrs | Torque: 83 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-OKU500-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-38-3.3: PN-OKU500-GLASS-RD-3810 / Kit KIT-OKU500-OPT-10 (Okuma-Multus-U500) | PM Interval: 1000 Hrs | Torque: 110 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-OKU500-OPT-10 within tolerance ±0.0040 mm TIR.

BOM-38-3.4: PN-OKU500-GASK-VIT-3813 / Kit KIT-OKU500-VAC-13 (Okuma-Multus-U500) | PM Interval: 1250 Hrs | Torque: 137 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-OKU500-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-38-3.5: PN-OKU500-CAM-GRIP-3802 / Kit KIT-OKU500-ATC-2 (Okuma-Multus-U500) | PM Interval: 1500 Hrs | Torque: 38 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-OKU500-ATC-2 within tolerance ±0.0060 mm TIR.

BOM-38-3.6: PN-OKU500-LIN-COIL-3805 / Kit KIT-OKU500-XAX-5 (Okuma-Multus-U500) | PM Interval: 250 Hrs | Torque: 65 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-OKU500-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-38-3.7: PN-OKU500-TRQ-STAT-3808 / Kit KIT-OKU500-BAX-8 (Okuma-Multus-U500) | PM Interval: 500 Hrs | Torque: 92 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-OKU500-BAX-8 within tolerance ±0.0030 mm TIR.

BOM-38-3.8: PN-OKU500-IGBT-MOD-3811 / Kit KIT-OKU500-PLC-11 (Okuma-Multus-U500) | PM Interval: 750 Hrs | Torque: 119 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-OKU500-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-38-3.9: PN-OKU500-HEX-CORE-3814 / Kit KIT-OKU500-THM-14 (Okuma-Multus-U500) | PM Interval: 1000 Hrs | Torque: 146 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-OKU500-THM-14 within tolerance ±0.0060 mm TIR.

BOM-38-3.10: PN-OKU500-VLV-PROP-3803 / Kit KIT-OKU500-HYD-3 (Okuma-Multus-U500) | PM Interval: 1250 Hrs | Torque: 47 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-OKU500-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-38-3.11: PN-OKU500-BSW-NUT-3806 / Kit KIT-OKU500-YAX-6 (Okuma-Multus-U500) | PM Interval: 1500 Hrs | Torque: 74 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-OKU500-YAX-6 within tolerance ±0.0020 mm TIR.

BOM-38-3.12: PN-OKU500-BRK-DISC-3809 / Kit KIT-OKU500-CAX-9 (Okuma-Multus-U500) | PM Interval: 250 Hrs | Torque: 101 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-OKU500-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-38-3.13: PN-OKU500-INJ-MET-3812 / Kit KIT-OKU500-LUB-12 (Okuma-Multus-U500) | PM Interval: 500 Hrs | Torque: 128 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-OKU500-LUB-12 within tolerance ±0.0050 mm TIR.

BOM-38-3.14: PN-OKU500-BRG-CER-3801 / Kit KIT-OKU500-SPN-1 (Okuma-Multus-U500) | PM Interval: 750 Hrs | Torque: 29 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-OKU500-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-38-3.15: PN-OKU500-PMP-PLG-3804 / Kit KIT-OKU500-CLN-4 (Okuma-Multus-U500) | PM Interval: 1000 Hrs | Torque: 56 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-OKU500-CLN-4 within tolerance ±0.0010 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Okuma-Multus-U550 (GROUP 1)

Certified Aerospace Bill of Materials for Okuma-Multus-U550. Verify part numbers before closing Work Orders.

BOM-39-1.1: PN-OKU550-CAM-GRIP-3902 / Kit KIT-OKU550-ATC-2 (Okuma-Multus-U550) | PM Interval: 500 Hrs | Torque: 178 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-OKU550-ATC-2 within tolerance ±0.0020 mm TIR.

BOM-39-1.2: PN-OKU550-VLV-PROP-3903 / Kit KIT-OKU550-HYD-3 (Okuma-Multus-U550) | PM Interval: 750 Hrs | Torque: 187 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-OKU550-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-39-1.3: PN-OKU550-PMP-PLG-3904 / Kit KIT-OKU550-CLN-4 (Okuma-Multus-U550) | PM Interval: 1000 Hrs | Torque: 196 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-OKU550-CLN-4 within tolerance ±0.0030 mm TIR.

BOM-39-1.4: PN-OKU550-LIN-COIL-3905 / Kit KIT-OKU550-XAX-5 (Okuma-Multus-U550) | PM Interval: 1250 Hrs | Torque: 205 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-OKU550-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-39-1.5: PN-OKU550-BSW-NUT-3906 / Kit KIT-OKU550-YAX-6 (Okuma-Multus-U550) | PM Interval: 1500 Hrs | Torque: 24 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-OKU550-YAX-6 within tolerance ±0.0040 mm TIR.

BOM-39-1.6: PN-OKU550-CYL-SEAL-3907 / Kit KIT-OKU550-ZAX-7 (Okuma-Multus-U550) | PM Interval: 250 Hrs | Torque: 33 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-OKU550-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-39-1.7: PN-OKU550-TRQ-STAT-3908 / Kit KIT-OKU550-BAX-8 (Okuma-Multus-U550) | PM Interval: 500 Hrs | Torque: 42 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-OKU550-BAX-8 within tolerance ±0.0050 mm TIR.

BOM-39-1.8: PN-OKU550-BRK-DISC-3909 / Kit KIT-OKU550-CAX-9 (Okuma-Multus-U550) | PM Interval: 750 Hrs | Torque: 51 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-OKU550-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-39-1.9: PN-OKU550-GLASS-RD-3910 / Kit KIT-OKU550-OPT-10 (Okuma-Multus-U550) | PM Interval: 1000 Hrs | Torque: 60 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-OKU550-OPT-10 within tolerance ±0.0060 mm TIR.

BOM-39-1.10: PN-OKU550-IGBT-MOD-3911 / Kit KIT-OKU550-PLC-11 (Okuma-Multus-U550) | PM Interval: 1250 Hrs | Torque: 69 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-OKU550-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-39-1.11: PN-OKU550-INJ-MET-3912 / Kit KIT-OKU550-LUB-12 (Okuma-Multus-U550) | PM Interval: 1500 Hrs | Torque: 78 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-OKU550-LUB-12 within tolerance ±0.0010 mm TIR.

BOM-39-1.12: PN-OKU550-GASK-VIT-3913 / Kit KIT-OKU550-VAC-13 (Okuma-Multus-U550) | PM Interval: 250 Hrs | Torque: 87 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-OKU550-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-39-1.13: PN-OKU550-HEX-CORE-3914 / Kit KIT-OKU550-THM-14 (Okuma-Multus-U550) | PM Interval: 500 Hrs | Torque: 96 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-OKU550-THM-14 within tolerance ±0.0020 mm TIR.

BOM-39-1.14: PN-OKU550-BRG-CER-3901 / Kit KIT-OKU550-SPN-1 (Okuma-Multus-U550) | PM Interval: 750 Hrs | Torque: 169 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-OKU550-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-39-1.15: PN-OKU550-CAM-GRIP-3902 / Kit KIT-OKU550-ATC-2 (Okuma-Multus-U550) | PM Interval: 1000 Hrs | Torque: 178 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-OKU550-ATC-2 within tolerance ±0.0020 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Okuma-Multus-U550 (GROUP 2)

Certified Aerospace Bill of Materials for Okuma-Multus-U550. Verify part numbers before closing Work Orders.

BOM-39-2.1: PN-OKU550-VLV-PROP-3903 / Kit KIT-OKU550-HYD-3 (Okuma-Multus-U550) | PM Interval: 500 Hrs | Torque: 187 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-OKU550-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-39-2.2: PN-OKU550-LIN-COIL-3905 / Kit KIT-OKU550-XAX-5 (Okuma-Multus-U550) | PM Interval: 750 Hrs | Torque: 205 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-OKU550-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-39-2.3: PN-OKU550-CYL-SEAL-3907 / Kit KIT-OKU550-ZAX-7 (Okuma-Multus-U550) | PM Interval: 1000 Hrs | Torque: 33 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-OKU550-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-39-2.4: PN-OKU550-BRK-DISC-3909 / Kit KIT-OKU550-CAX-9 (Okuma-Multus-U550) | PM Interval: 1250 Hrs | Torque: 51 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-OKU550-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-39-2.5: PN-OKU550-IGBT-MOD-3911 / Kit KIT-OKU550-PLC-11 (Okuma-Multus-U550) | PM Interval: 1500 Hrs | Torque: 69 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-OKU550-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-39-2.6: PN-OKU550-GASK-VIT-3913 / Kit KIT-OKU550-VAC-13 (Okuma-Multus-U550) | PM Interval: 250 Hrs | Torque: 87 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-OKU550-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-39-2.7: PN-OKU550-BRG-CER-3901 / Kit KIT-OKU550-SPN-1 (Okuma-Multus-U550) | PM Interval: 500 Hrs | Torque: 169 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-OKU550-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-39-2.8: PN-OKU550-VLV-PROP-3903 / Kit KIT-OKU550-HYD-3 (Okuma-Multus-U550) | PM Interval: 750 Hrs | Torque: 187 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-OKU550-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-39-2.9: PN-OKU550-LIN-COIL-3905 / Kit KIT-OKU550-XAX-5 (Okuma-Multus-U550) | PM Interval: 1000 Hrs | Torque: 205 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-OKU550-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-39-2.10: PN-OKU550-CYL-SEAL-3907 / Kit KIT-OKU550-ZAX-7 (Okuma-Multus-U550) | PM Interval: 1250 Hrs | Torque: 33 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-OKU550-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-39-2.11: PN-OKU550-BRK-DISC-3909 / Kit KIT-OKU550-CAX-9 (Okuma-Multus-U550) | PM Interval: 1500 Hrs | Torque: 51 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-OKU550-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-39-2.12: PN-OKU550-IGBT-MOD-3911 / Kit KIT-OKU550-PLC-11 (Okuma-Multus-U550) | PM Interval: 250 Hrs | Torque: 69 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-OKU550-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-39-2.13: PN-OKU550-GASK-VIT-3913 / Kit KIT-OKU550-VAC-13 (Okuma-Multus-U550) | PM Interval: 500 Hrs | Torque: 87 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-OKU550-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-39-2.14: PN-OKU550-BRG-CER-3901 / Kit KIT-OKU550-SPN-1 (Okuma-Multus-U550) | PM Interval: 750 Hrs | Torque: 169 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-OKU550-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-39-2.15: PN-OKU550-VLV-PROP-3903 / Kit KIT-OKU550-HYD-3 (Okuma-Multus-U550) | PM Interval: 1000 Hrs | Torque: 187 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-OKU550-HYD-3 within tolerance ±0.0025 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Okuma-Multus-U550 (GROUP 3)

Certified Aerospace Bill of Materials for Okuma-Multus-U550. Verify part numbers before closing Work Orders.

BOM-39-3.1: PN-OKU550-PMP-PLG-3904 / Kit KIT-OKU550-CLN-4 (Okuma-Multus-U550) | PM Interval: 500 Hrs | Torque: 196 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-OKU550-CLN-4 within tolerance ±0.0030 mm TIR.

BOM-39-3.2: PN-OKU550-CYL-SEAL-3907 / Kit KIT-OKU550-ZAX-7 (Okuma-Multus-U550) | PM Interval: 750 Hrs | Torque: 33 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-OKU550-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-39-3.3: PN-OKU550-GLASS-RD-3910 / Kit KIT-OKU550-OPT-10 (Okuma-Multus-U550) | PM Interval: 1000 Hrs | Torque: 60 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-OKU550-OPT-10 within tolerance ±0.0060 mm TIR.

BOM-39-3.4: PN-OKU550-GASK-VIT-3913 / Kit KIT-OKU550-VAC-13 (Okuma-Multus-U550) | PM Interval: 1250 Hrs | Torque: 87 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-OKU550-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-39-3.5: PN-OKU550-CAM-GRIP-3902 / Kit KIT-OKU550-ATC-2 (Okuma-Multus-U550) | PM Interval: 1500 Hrs | Torque: 178 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-OKU550-ATC-2 within tolerance ±0.0020 mm TIR.

BOM-39-3.6: PN-OKU550-LIN-COIL-3905 / Kit KIT-OKU550-XAX-5 (Okuma-Multus-U550) | PM Interval: 250 Hrs | Torque: 205 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-OKU550-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-39-3.7: PN-OKU550-TRQ-STAT-3908 / Kit KIT-OKU550-BAX-8 (Okuma-Multus-U550) | PM Interval: 500 Hrs | Torque: 42 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-OKU550-BAX-8 within tolerance ±0.0050 mm TIR.

BOM-39-3.8: PN-OKU550-IGBT-MOD-3911 / Kit KIT-OKU550-PLC-11 (Okuma-Multus-U550) | PM Interval: 750 Hrs | Torque: 69 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-OKU550-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-39-3.9: PN-OKU550-HEX-CORE-3914 / Kit KIT-OKU550-THM-14 (Okuma-Multus-U550) | PM Interval: 1000 Hrs | Torque: 96 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-OKU550-THM-14 within tolerance ±0.0020 mm TIR.

BOM-39-3.10: PN-OKU550-VLV-PROP-3903 / Kit KIT-OKU550-HYD-3 (Okuma-Multus-U550) | PM Interval: 1250 Hrs | Torque: 187 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-OKU550-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-39-3.11: PN-OKU550-BSW-NUT-3906 / Kit KIT-OKU550-YAX-6 (Okuma-Multus-U550) | PM Interval: 1500 Hrs | Torque: 24 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-OKU550-YAX-6 within tolerance ±0.0040 mm TIR.

BOM-39-3.12: PN-OKU550-BRK-DISC-3909 / Kit KIT-OKU550-CAX-9 (Okuma-Multus-U550) | PM Interval: 250 Hrs | Torque: 51 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-OKU550-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-39-3.13: PN-OKU550-INJ-MET-3912 / Kit KIT-OKU550-LUB-12 (Okuma-Multus-U550) | PM Interval: 500 Hrs | Torque: 78 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-OKU550-LUB-12 within tolerance ±0.0010 mm TIR.

BOM-39-3.14: PN-OKU550-BRG-CER-3901 / Kit KIT-OKU550-SPN-1 (Okuma-Multus-U550) | PM Interval: 750 Hrs | Torque: 169 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-OKU550-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-39-3.15: PN-OKU550-PMP-PLG-3904 / Kit KIT-OKU550-CLN-4 (Okuma-Multus-U550) | PM Interval: 1000 Hrs | Torque: 196 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-OKU550-CLN-4 within tolerance ±0.0030 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Okuma-Multus-U600 (GROUP 1)

Certified Aerospace Bill of Materials for Okuma-Multus-U600. Verify part numbers before closing Work Orders.

BOM-40-1.1: PN-OKU600-CAM-GRIP-4002 / Kit KIT-OKU600-ATC-2 (Okuma-Multus-U600) | PM Interval: 500 Hrs | Torque: 128 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-OKU600-ATC-2 within tolerance ±0.0040 mm TIR.

BOM-40-1.2: PN-OKU600-VLV-PROP-4003 / Kit KIT-OKU600-HYD-3 (Okuma-Multus-U600) | PM Interval: 750 Hrs | Torque: 137 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-OKU600-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-40-1.3: PN-OKU600-PMP-PLG-4004 / Kit KIT-OKU600-CLN-4 (Okuma-Multus-U600) | PM Interval: 1000 Hrs | Torque: 146 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-OKU600-CLN-4 within tolerance ±0.0050 mm TIR.

BOM-40-1.4: PN-OKU600-LIN-COIL-4005 / Kit KIT-OKU600-XAX-5 (Okuma-Multus-U600) | PM Interval: 1250 Hrs | Torque: 155 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-OKU600-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-40-1.5: PN-OKU600-BSW-NUT-4006 / Kit KIT-OKU600-YAX-6 (Okuma-Multus-U600) | PM Interval: 1500 Hrs | Torque: 164 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-OKU600-YAX-6 within tolerance ±0.0060 mm TIR.

BOM-40-1.6: PN-OKU600-CYL-SEAL-4007 / Kit KIT-OKU600-ZAX-7 (Okuma-Multus-U600) | PM Interval: 250 Hrs | Torque: 173 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-OKU600-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-40-1.7: PN-OKU600-TRQ-STAT-4008 / Kit KIT-OKU600-BAX-8 (Okuma-Multus-U600) | PM Interval: 500 Hrs | Torque: 182 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-OKU600-BAX-8 within tolerance ±0.0010 mm TIR.

BOM-40-1.8: PN-OKU600-BRK-DISC-4009 / Kit KIT-OKU600-CAX-9 (Okuma-Multus-U600) | PM Interval: 750 Hrs | Torque: 191 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-OKU600-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-40-1.9: PN-OKU600-GLASS-RD-4010 / Kit KIT-OKU600-OPT-10 (Okuma-Multus-U600) | PM Interval: 1000 Hrs | Torque: 200 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-OKU600-OPT-10 within tolerance ±0.0020 mm TIR.

BOM-40-1.10: PN-OKU600-IGBT-MOD-4011 / Kit KIT-OKU600-PLC-11 (Okuma-Multus-U600) | PM Interval: 1250 Hrs | Torque: 209 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-OKU600-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-40-1.11: PN-OKU600-INJ-MET-4012 / Kit KIT-OKU600-LUB-12 (Okuma-Multus-U600) | PM Interval: 1500 Hrs | Torque: 28 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-OKU600-LUB-12 within tolerance ±0.0030 mm TIR.

BOM-40-1.12: PN-OKU600-GASK-VIT-4013 / Kit KIT-OKU600-VAC-13 (Okuma-Multus-U600) | PM Interval: 250 Hrs | Torque: 37 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-OKU600-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-40-1.13: PN-OKU600-HEX-CORE-4014 / Kit KIT-OKU600-THM-14 (Okuma-Multus-U600) | PM Interval: 500 Hrs | Torque: 46 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-OKU600-THM-14 within tolerance ±0.0040 mm TIR.

BOM-40-1.14: PN-OKU600-BRG-CER-4001 / Kit KIT-OKU600-SPN-1 (Okuma-Multus-U600) | PM Interval: 750 Hrs | Torque: 119 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-OKU600-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-40-1.15: PN-OKU600-CAM-GRIP-4002 / Kit KIT-OKU600-ATC-2 (Okuma-Multus-U600) | PM Interval: 1000 Hrs | Torque: 128 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-OKU600-ATC-2 within tolerance ±0.0040 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Okuma-Multus-U600 (GROUP 2)

Certified Aerospace Bill of Materials for Okuma-Multus-U600. Verify part numbers before closing Work Orders.

BOM-40-2.1: PN-OKU600-VLV-PROP-4003 / Kit KIT-OKU600-HYD-3 (Okuma-Multus-U600) | PM Interval: 500 Hrs | Torque: 137 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-OKU600-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-40-2.2: PN-OKU600-LIN-COIL-4005 / Kit KIT-OKU600-XAX-5 (Okuma-Multus-U600) | PM Interval: 750 Hrs | Torque: 155 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-OKU600-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-40-2.3: PN-OKU600-CYL-SEAL-4007 / Kit KIT-OKU600-ZAX-7 (Okuma-Multus-U600) | PM Interval: 1000 Hrs | Torque: 173 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-OKU600-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-40-2.4: PN-OKU600-BRK-DISC-4009 / Kit KIT-OKU600-CAX-9 (Okuma-Multus-U600) | PM Interval: 1250 Hrs | Torque: 191 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-OKU600-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-40-2.5: PN-OKU600-IGBT-MOD-4011 / Kit KIT-OKU600-PLC-11 (Okuma-Multus-U600) | PM Interval: 1500 Hrs | Torque: 209 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-OKU600-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-40-2.6: PN-OKU600-GASK-VIT-4013 / Kit KIT-OKU600-VAC-13 (Okuma-Multus-U600) | PM Interval: 250 Hrs | Torque: 37 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-OKU600-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-40-2.7: PN-OKU600-BRG-CER-4001 / Kit KIT-OKU600-SPN-1 (Okuma-Multus-U600) | PM Interval: 500 Hrs | Torque: 119 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-OKU600-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-40-2.8: PN-OKU600-VLV-PROP-4003 / Kit KIT-OKU600-HYD-3 (Okuma-Multus-U600) | PM Interval: 750 Hrs | Torque: 137 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-OKU600-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-40-2.9: PN-OKU600-LIN-COIL-4005 / Kit KIT-OKU600-XAX-5 (Okuma-Multus-U600) | PM Interval: 1000 Hrs | Torque: 155 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-OKU600-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-40-2.10: PN-OKU600-CYL-SEAL-4007 / Kit KIT-OKU600-ZAX-7 (Okuma-Multus-U600) | PM Interval: 1250 Hrs | Torque: 173 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-OKU600-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-40-2.11: PN-OKU600-BRK-DISC-4009 / Kit KIT-OKU600-CAX-9 (Okuma-Multus-U600) | PM Interval: 1500 Hrs | Torque: 191 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-OKU600-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-40-2.12: PN-OKU600-IGBT-MOD-4011 / Kit KIT-OKU600-PLC-11 (Okuma-Multus-U600) | PM Interval: 250 Hrs | Torque: 209 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-OKU600-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-40-2.13: PN-OKU600-GASK-VIT-4013 / Kit KIT-OKU600-VAC-13 (Okuma-Multus-U600) | PM Interval: 500 Hrs | Torque: 37 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-OKU600-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-40-2.14: PN-OKU600-BRG-CER-4001 / Kit KIT-OKU600-SPN-1 (Okuma-Multus-U600) | PM Interval: 750 Hrs | Torque: 119 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-OKU600-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-40-2.15: PN-OKU600-VLV-PROP-4003 / Kit KIT-OKU600-HYD-3 (Okuma-Multus-U600) | PM Interval: 1000 Hrs | Torque: 137 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-OKU600-HYD-3 within tolerance ±0.0045 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Okuma-Multus-U600 (GROUP 3)

Certified Aerospace Bill of Materials for Okuma-Multus-U600. Verify part numbers before closing Work Orders.

BOM-40-3.1: PN-OKU600-PMP-PLG-4004 / Kit KIT-OKU600-CLN-4 (Okuma-Multus-U600) | PM Interval: 500 Hrs | Torque: 146 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-OKU600-CLN-4 within tolerance ±0.0050 mm TIR.

BOM-40-3.2: PN-OKU600-CYL-SEAL-4007 / Kit KIT-OKU600-ZAX-7 (Okuma-Multus-U600) | PM Interval: 750 Hrs | Torque: 173 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-OKU600-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-40-3.3: PN-OKU600-GLASS-RD-4010 / Kit KIT-OKU600-OPT-10 (Okuma-Multus-U600) | PM Interval: 1000 Hrs | Torque: 200 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-OKU600-OPT-10 within tolerance ±0.0020 mm TIR.

BOM-40-3.4: PN-OKU600-GASK-VIT-4013 / Kit KIT-OKU600-VAC-13 (Okuma-Multus-U600) | PM Interval: 1250 Hrs | Torque: 37 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-OKU600-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-40-3.5: PN-OKU600-CAM-GRIP-4002 / Kit KIT-OKU600-ATC-2 (Okuma-Multus-U600) | PM Interval: 1500 Hrs | Torque: 128 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-OKU600-ATC-2 within tolerance ±0.0040 mm TIR.

BOM-40-3.6: PN-OKU600-LIN-COIL-4005 / Kit KIT-OKU600-XAX-5 (Okuma-Multus-U600) | PM Interval: 250 Hrs | Torque: 155 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-OKU600-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-40-3.7: PN-OKU600-TRQ-STAT-4008 / Kit KIT-OKU600-BAX-8 (Okuma-Multus-U600) | PM Interval: 500 Hrs | Torque: 182 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-OKU600-BAX-8 within tolerance ±0.0010 mm TIR.

BOM-40-3.8: PN-OKU600-IGBT-MOD-4011 / Kit KIT-OKU600-PLC-11 (Okuma-Multus-U600) | PM Interval: 750 Hrs | Torque: 209 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-OKU600-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-40-3.9: PN-OKU600-HEX-CORE-4014 / Kit KIT-OKU600-THM-14 (Okuma-Multus-U600) | PM Interval: 1000 Hrs | Torque: 46 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-OKU600-THM-14 within tolerance ±0.0040 mm TIR.

BOM-40-3.10: PN-OKU600-VLV-PROP-4003 / Kit KIT-OKU600-HYD-3 (Okuma-Multus-U600) | PM Interval: 1250 Hrs | Torque: 137 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-OKU600-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-40-3.11: PN-OKU600-BSW-NUT-4006 / Kit KIT-OKU600-YAX-6 (Okuma-Multus-U600) | PM Interval: 1500 Hrs | Torque: 164 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-OKU600-YAX-6 within tolerance ±0.0060 mm TIR.

BOM-40-3.12: PN-OKU600-BRK-DISC-4009 / Kit KIT-OKU600-CAX-9 (Okuma-Multus-U600) | PM Interval: 250 Hrs | Torque: 191 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-OKU600-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-40-3.13: PN-OKU600-INJ-MET-4012 / Kit KIT-OKU600-LUB-12 (Okuma-Multus-U600) | PM Interval: 500 Hrs | Torque: 28 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-OKU600-LUB-12 within tolerance ±0.0030 mm TIR.

BOM-40-3.14: PN-OKU600-BRG-CER-4001 / Kit KIT-OKU600-SPN-1 (Okuma-Multus-U600) | PM Interval: 750 Hrs | Torque: 119 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-OKU600-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-40-3.15: PN-OKU600-PMP-PLG-4004 / Kit KIT-OKU600-CLN-4 (Okuma-Multus-U600) | PM Interval: 1000 Hrs | Torque: 146 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-OKU600-CLN-4 within tolerance ±0.0050 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — EOS-Additive-M150 (GROUP 1)

Certified Aerospace Bill of Materials for EOS-Additive-M150. Verify part numbers before closing Work Orders.

BOM-41-1.1: PN-EOS150-CAM-GRIP-4102 / Kit KIT-EOS150-ATC-2 (EOS-Additive-M150) | PM Interval: 500 Hrs | Torque: 78 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-EOS150-ATC-2 within tolerance ±0.0060 mm TIR.

BOM-41-1.2: PN-EOS150-VLV-PROP-4103 / Kit KIT-EOS150-HYD-3 (EOS-Additive-M150) | PM Interval: 750 Hrs | Torque: 87 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-EOS150-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-41-1.3: PN-EOS150-PMP-PLG-4104 / Kit KIT-EOS150-CLN-4 (EOS-Additive-M150) | PM Interval: 1000 Hrs | Torque: 96 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-EOS150-CLN-4 within tolerance ±0.0010 mm TIR.

BOM-41-1.4: PN-EOS150-LIN-COIL-4105 / Kit KIT-EOS150-XAX-5 (EOS-Additive-M150) | PM Interval: 1250 Hrs | Torque: 105 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-EOS150-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-41-1.5: PN-EOS150-BSW-NUT-4106 / Kit KIT-EOS150-YAX-6 (EOS-Additive-M150) | PM Interval: 1500 Hrs | Torque: 114 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-EOS150-YAX-6 within tolerance ±0.0020 mm TIR.

BOM-41-1.6: PN-EOS150-CYL-SEAL-4107 / Kit KIT-EOS150-ZAX-7 (EOS-Additive-M150) | PM Interval: 250 Hrs | Torque: 123 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-EOS150-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-41-1.7: PN-EOS150-TRQ-STAT-4108 / Kit KIT-EOS150-BAX-8 (EOS-Additive-M150) | PM Interval: 500 Hrs | Torque: 132 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-EOS150-BAX-8 within tolerance ±0.0030 mm TIR.

BOM-41-1.8: PN-EOS150-BRK-DISC-4109 / Kit KIT-EOS150-CAX-9 (EOS-Additive-M150) | PM Interval: 750 Hrs | Torque: 141 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-EOS150-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-41-1.9: PN-EOS150-GLASS-RD-4110 / Kit KIT-EOS150-OPT-10 (EOS-Additive-M150) | PM Interval: 1000 Hrs | Torque: 150 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-EOS150-OPT-10 within tolerance ±0.0040 mm TIR.

BOM-41-1.10: PN-EOS150-IGBT-MOD-4111 / Kit KIT-EOS150-PLC-11 (EOS-Additive-M150) | PM Interval: 1250 Hrs | Torque: 159 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-EOS150-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-41-1.11: PN-EOS150-INJ-MET-4112 / Kit KIT-EOS150-LUB-12 (EOS-Additive-M150) | PM Interval: 1500 Hrs | Torque: 168 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-EOS150-LUB-12 within tolerance ±0.0050 mm TIR.

BOM-41-1.12: PN-EOS150-GASK-VIT-4113 / Kit KIT-EOS150-VAC-13 (EOS-Additive-M150) | PM Interval: 250 Hrs | Torque: 177 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-EOS150-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-41-1.13: PN-EOS150-HEX-CORE-4114 / Kit KIT-EOS150-THM-14 (EOS-Additive-M150) | PM Interval: 500 Hrs | Torque: 186 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-EOS150-THM-14 within tolerance ±0.0060 mm TIR.

BOM-41-1.14: PN-EOS150-BRG-CER-4101 / Kit KIT-EOS150-SPN-1 (EOS-Additive-M150) | PM Interval: 750 Hrs | Torque: 69 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-EOS150-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-41-1.15: PN-EOS150-CAM-GRIP-4102 / Kit KIT-EOS150-ATC-2 (EOS-Additive-M150) | PM Interval: 1000 Hrs | Torque: 78 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-EOS150-ATC-2 within tolerance ±0.0060 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — EOS-Additive-M150 (GROUP 2)

Certified Aerospace Bill of Materials for EOS-Additive-M150. Verify part numbers before closing Work Orders.

BOM-41-2.1: PN-EOS150-VLV-PROP-4103 / Kit KIT-EOS150-HYD-3 (EOS-Additive-M150) | PM Interval: 500 Hrs | Torque: 87 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-EOS150-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-41-2.2: PN-EOS150-LIN-COIL-4105 / Kit KIT-EOS150-XAX-5 (EOS-Additive-M150) | PM Interval: 750 Hrs | Torque: 105 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-EOS150-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-41-2.3: PN-EOS150-CYL-SEAL-4107 / Kit KIT-EOS150-ZAX-7 (EOS-Additive-M150) | PM Interval: 1000 Hrs | Torque: 123 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-EOS150-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-41-2.4: PN-EOS150-BRK-DISC-4109 / Kit KIT-EOS150-CAX-9 (EOS-Additive-M150) | PM Interval: 1250 Hrs | Torque: 141 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-EOS150-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-41-2.5: PN-EOS150-IGBT-MOD-4111 / Kit KIT-EOS150-PLC-11 (EOS-Additive-M150) | PM Interval: 1500 Hrs | Torque: 159 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-EOS150-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-41-2.6: PN-EOS150-GASK-VIT-4113 / Kit KIT-EOS150-VAC-13 (EOS-Additive-M150) | PM Interval: 250 Hrs | Torque: 177 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-EOS150-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-41-2.7: PN-EOS150-BRG-CER-4101 / Kit KIT-EOS150-SPN-1 (EOS-Additive-M150) | PM Interval: 500 Hrs | Torque: 69 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-EOS150-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-41-2.8: PN-EOS150-VLV-PROP-4103 / Kit KIT-EOS150-HYD-3 (EOS-Additive-M150) | PM Interval: 750 Hrs | Torque: 87 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-EOS150-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-41-2.9: PN-EOS150-LIN-COIL-4105 / Kit KIT-EOS150-XAX-5 (EOS-Additive-M150) | PM Interval: 1000 Hrs | Torque: 105 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-EOS150-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-41-2.10: PN-EOS150-CYL-SEAL-4107 / Kit KIT-EOS150-ZAX-7 (EOS-Additive-M150) | PM Interval: 1250 Hrs | Torque: 123 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-EOS150-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-41-2.11: PN-EOS150-BRK-DISC-4109 / Kit KIT-EOS150-CAX-9 (EOS-Additive-M150) | PM Interval: 1500 Hrs | Torque: 141 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-EOS150-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-41-2.12: PN-EOS150-IGBT-MOD-4111 / Kit KIT-EOS150-PLC-11 (EOS-Additive-M150) | PM Interval: 250 Hrs | Torque: 159 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-EOS150-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-41-2.13: PN-EOS150-GASK-VIT-4113 / Kit KIT-EOS150-VAC-13 (EOS-Additive-M150) | PM Interval: 500 Hrs | Torque: 177 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-EOS150-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-41-2.14: PN-EOS150-BRG-CER-4101 / Kit KIT-EOS150-SPN-1 (EOS-Additive-M150) | PM Interval: 750 Hrs | Torque: 69 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-EOS150-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-41-2.15: PN-EOS150-VLV-PROP-4103 / Kit KIT-EOS150-HYD-3 (EOS-Additive-M150) | PM Interval: 1000 Hrs | Torque: 87 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-EOS150-HYD-3 within tolerance ±0.0065 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — EOS-Additive-M150 (GROUP 3)

Certified Aerospace Bill of Materials for EOS-Additive-M150. Verify part numbers before closing Work Orders.

BOM-41-3.1: PN-EOS150-PMP-PLG-4104 / Kit KIT-EOS150-CLN-4 (EOS-Additive-M150) | PM Interval: 500 Hrs | Torque: 96 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-EOS150-CLN-4 within tolerance ±0.0010 mm TIR.

BOM-41-3.2: PN-EOS150-CYL-SEAL-4107 / Kit KIT-EOS150-ZAX-7 (EOS-Additive-M150) | PM Interval: 750 Hrs | Torque: 123 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-EOS150-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-41-3.3: PN-EOS150-GLASS-RD-4110 / Kit KIT-EOS150-OPT-10 (EOS-Additive-M150) | PM Interval: 1000 Hrs | Torque: 150 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-EOS150-OPT-10 within tolerance ±0.0040 mm TIR.

BOM-41-3.4: PN-EOS150-GASK-VIT-4113 / Kit KIT-EOS150-VAC-13 (EOS-Additive-M150) | PM Interval: 1250 Hrs | Torque: 177 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-EOS150-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-41-3.5: PN-EOS150-CAM-GRIP-4102 / Kit KIT-EOS150-ATC-2 (EOS-Additive-M150) | PM Interval: 1500 Hrs | Torque: 78 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-EOS150-ATC-2 within tolerance ±0.0060 mm TIR.

BOM-41-3.6: PN-EOS150-LIN-COIL-4105 / Kit KIT-EOS150-XAX-5 (EOS-Additive-M150) | PM Interval: 250 Hrs | Torque: 105 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-EOS150-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-41-3.7: PN-EOS150-TRQ-STAT-4108 / Kit KIT-EOS150-BAX-8 (EOS-Additive-M150) | PM Interval: 500 Hrs | Torque: 132 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-EOS150-BAX-8 within tolerance ±0.0030 mm TIR.

BOM-41-3.8: PN-EOS150-IGBT-MOD-4111 / Kit KIT-EOS150-PLC-11 (EOS-Additive-M150) | PM Interval: 750 Hrs | Torque: 159 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-EOS150-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-41-3.9: PN-EOS150-HEX-CORE-4114 / Kit KIT-EOS150-THM-14 (EOS-Additive-M150) | PM Interval: 1000 Hrs | Torque: 186 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-EOS150-THM-14 within tolerance ±0.0060 mm TIR.

BOM-41-3.10: PN-EOS150-VLV-PROP-4103 / Kit KIT-EOS150-HYD-3 (EOS-Additive-M150) | PM Interval: 1250 Hrs | Torque: 87 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-EOS150-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-41-3.11: PN-EOS150-BSW-NUT-4106 / Kit KIT-EOS150-YAX-6 (EOS-Additive-M150) | PM Interval: 1500 Hrs | Torque: 114 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-EOS150-YAX-6 within tolerance ±0.0020 mm TIR.

BOM-41-3.12: PN-EOS150-BRK-DISC-4109 / Kit KIT-EOS150-CAX-9 (EOS-Additive-M150) | PM Interval: 250 Hrs | Torque: 141 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-EOS150-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-41-3.13: PN-EOS150-INJ-MET-4112 / Kit KIT-EOS150-LUB-12 (EOS-Additive-M150) | PM Interval: 500 Hrs | Torque: 168 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-EOS150-LUB-12 within tolerance ±0.0050 mm TIR.

BOM-41-3.14: PN-EOS150-BRG-CER-4101 / Kit KIT-EOS150-SPN-1 (EOS-Additive-M150) | PM Interval: 750 Hrs | Torque: 69 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-EOS150-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-41-3.15: PN-EOS150-PMP-PLG-4104 / Kit KIT-EOS150-CLN-4 (EOS-Additive-M150) | PM Interval: 1000 Hrs | Torque: 96 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-EOS150-CLN-4 within tolerance ±0.0010 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — EOS-Additive-M200 (GROUP 1)

Certified Aerospace Bill of Materials for EOS-Additive-M200. Verify part numbers before closing Work Orders.

BOM-42-1.1: PN-EOS200-CAM-GRIP-4202 / Kit KIT-EOS200-ATC-2 (EOS-Additive-M200) | PM Interval: 500 Hrs | Torque: 28 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-EOS200-ATC-2 within tolerance ±0.0020 mm TIR.

BOM-42-1.2: PN-EOS200-VLV-PROP-4203 / Kit KIT-EOS200-HYD-3 (EOS-Additive-M200) | PM Interval: 750 Hrs | Torque: 37 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-EOS200-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-42-1.3: PN-EOS200-PMP-PLG-4204 / Kit KIT-EOS200-CLN-4 (EOS-Additive-M200) | PM Interval: 1000 Hrs | Torque: 46 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-EOS200-CLN-4 within tolerance ±0.0030 mm TIR.

BOM-42-1.4: PN-EOS200-LIN-COIL-4205 / Kit KIT-EOS200-XAX-5 (EOS-Additive-M200) | PM Interval: 1250 Hrs | Torque: 55 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-EOS200-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-42-1.5: PN-EOS200-BSW-NUT-4206 / Kit KIT-EOS200-YAX-6 (EOS-Additive-M200) | PM Interval: 1500 Hrs | Torque: 64 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-EOS200-YAX-6 within tolerance ±0.0040 mm TIR.

BOM-42-1.6: PN-EOS200-CYL-SEAL-4207 / Kit KIT-EOS200-ZAX-7 (EOS-Additive-M200) | PM Interval: 250 Hrs | Torque: 73 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-EOS200-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-42-1.7: PN-EOS200-TRQ-STAT-4208 / Kit KIT-EOS200-BAX-8 (EOS-Additive-M200) | PM Interval: 500 Hrs | Torque: 82 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-EOS200-BAX-8 within tolerance ±0.0050 mm TIR.

BOM-42-1.8: PN-EOS200-BRK-DISC-4209 / Kit KIT-EOS200-CAX-9 (EOS-Additive-M200) | PM Interval: 750 Hrs | Torque: 91 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-EOS200-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-42-1.9: PN-EOS200-GLASS-RD-4210 / Kit KIT-EOS200-OPT-10 (EOS-Additive-M200) | PM Interval: 1000 Hrs | Torque: 100 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-EOS200-OPT-10 within tolerance ±0.0060 mm TIR.

BOM-42-1.10: PN-EOS200-IGBT-MOD-4211 / Kit KIT-EOS200-PLC-11 (EOS-Additive-M200) | PM Interval: 1250 Hrs | Torque: 109 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-EOS200-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-42-1.11: PN-EOS200-INJ-MET-4212 / Kit KIT-EOS200-LUB-12 (EOS-Additive-M200) | PM Interval: 1500 Hrs | Torque: 118 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-EOS200-LUB-12 within tolerance ±0.0010 mm TIR.

BOM-42-1.12: PN-EOS200-GASK-VIT-4213 / Kit KIT-EOS200-VAC-13 (EOS-Additive-M200) | PM Interval: 250 Hrs | Torque: 127 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-EOS200-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-42-1.13: PN-EOS200-HEX-CORE-4214 / Kit KIT-EOS200-THM-14 (EOS-Additive-M200) | PM Interval: 500 Hrs | Torque: 136 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-EOS200-THM-14 within tolerance ±0.0020 mm TIR.

BOM-42-1.14: PN-EOS200-BRG-CER-4201 / Kit KIT-EOS200-SPN-1 (EOS-Additive-M200) | PM Interval: 750 Hrs | Torque: 209 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-EOS200-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-42-1.15: PN-EOS200-CAM-GRIP-4202 / Kit KIT-EOS200-ATC-2 (EOS-Additive-M200) | PM Interval: 1000 Hrs | Torque: 28 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-EOS200-ATC-2 within tolerance ±0.0020 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — EOS-Additive-M200 (GROUP 2)

Certified Aerospace Bill of Materials for EOS-Additive-M200. Verify part numbers before closing Work Orders.

BOM-42-2.1: PN-EOS200-VLV-PROP-4203 / Kit KIT-EOS200-HYD-3 (EOS-Additive-M200) | PM Interval: 500 Hrs | Torque: 37 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-EOS200-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-42-2.2: PN-EOS200-LIN-COIL-4205 / Kit KIT-EOS200-XAX-5 (EOS-Additive-M200) | PM Interval: 750 Hrs | Torque: 55 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-EOS200-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-42-2.3: PN-EOS200-CYL-SEAL-4207 / Kit KIT-EOS200-ZAX-7 (EOS-Additive-M200) | PM Interval: 1000 Hrs | Torque: 73 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-EOS200-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-42-2.4: PN-EOS200-BRK-DISC-4209 / Kit KIT-EOS200-CAX-9 (EOS-Additive-M200) | PM Interval: 1250 Hrs | Torque: 91 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-EOS200-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-42-2.5: PN-EOS200-IGBT-MOD-4211 / Kit KIT-EOS200-PLC-11 (EOS-Additive-M200) | PM Interval: 1500 Hrs | Torque: 109 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-EOS200-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-42-2.6: PN-EOS200-GASK-VIT-4213 / Kit KIT-EOS200-VAC-13 (EOS-Additive-M200) | PM Interval: 250 Hrs | Torque: 127 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-EOS200-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-42-2.7: PN-EOS200-BRG-CER-4201 / Kit KIT-EOS200-SPN-1 (EOS-Additive-M200) | PM Interval: 500 Hrs | Torque: 209 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-EOS200-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-42-2.8: PN-EOS200-VLV-PROP-4203 / Kit KIT-EOS200-HYD-3 (EOS-Additive-M200) | PM Interval: 750 Hrs | Torque: 37 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-EOS200-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-42-2.9: PN-EOS200-LIN-COIL-4205 / Kit KIT-EOS200-XAX-5 (EOS-Additive-M200) | PM Interval: 1000 Hrs | Torque: 55 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-EOS200-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-42-2.10: PN-EOS200-CYL-SEAL-4207 / Kit KIT-EOS200-ZAX-7 (EOS-Additive-M200) | PM Interval: 1250 Hrs | Torque: 73 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-EOS200-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-42-2.11: PN-EOS200-BRK-DISC-4209 / Kit KIT-EOS200-CAX-9 (EOS-Additive-M200) | PM Interval: 1500 Hrs | Torque: 91 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-EOS200-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-42-2.12: PN-EOS200-IGBT-MOD-4211 / Kit KIT-EOS200-PLC-11 (EOS-Additive-M200) | PM Interval: 250 Hrs | Torque: 109 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-EOS200-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-42-2.13: PN-EOS200-GASK-VIT-4213 / Kit KIT-EOS200-VAC-13 (EOS-Additive-M200) | PM Interval: 500 Hrs | Torque: 127 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-EOS200-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-42-2.14: PN-EOS200-BRG-CER-4201 / Kit KIT-EOS200-SPN-1 (EOS-Additive-M200) | PM Interval: 750 Hrs | Torque: 209 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-EOS200-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-42-2.15: PN-EOS200-VLV-PROP-4203 / Kit KIT-EOS200-HYD-3 (EOS-Additive-M200) | PM Interval: 1000 Hrs | Torque: 37 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-EOS200-HYD-3 within tolerance ±0.0025 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — EOS-Additive-M200 (GROUP 3)

Certified Aerospace Bill of Materials for EOS-Additive-M200. Verify part numbers before closing Work Orders.

BOM-42-3.1: PN-EOS200-PMP-PLG-4204 / Kit KIT-EOS200-CLN-4 (EOS-Additive-M200) | PM Interval: 500 Hrs | Torque: 46 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-EOS200-CLN-4 within tolerance ±0.0030 mm TIR.

BOM-42-3.2: PN-EOS200-CYL-SEAL-4207 / Kit KIT-EOS200-ZAX-7 (EOS-Additive-M200) | PM Interval: 750 Hrs | Torque: 73 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-EOS200-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-42-3.3: PN-EOS200-GLASS-RD-4210 / Kit KIT-EOS200-OPT-10 (EOS-Additive-M200) | PM Interval: 1000 Hrs | Torque: 100 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-EOS200-OPT-10 within tolerance ±0.0060 mm TIR.

BOM-42-3.4: PN-EOS200-GASK-VIT-4213 / Kit KIT-EOS200-VAC-13 (EOS-Additive-M200) | PM Interval: 1250 Hrs | Torque: 127 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-EOS200-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-42-3.5: PN-EOS200-CAM-GRIP-4202 / Kit KIT-EOS200-ATC-2 (EOS-Additive-M200) | PM Interval: 1500 Hrs | Torque: 28 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-EOS200-ATC-2 within tolerance ±0.0020 mm TIR.

BOM-42-3.6: PN-EOS200-LIN-COIL-4205 / Kit KIT-EOS200-XAX-5 (EOS-Additive-M200) | PM Interval: 250 Hrs | Torque: 55 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-EOS200-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-42-3.7: PN-EOS200-TRQ-STAT-4208 / Kit KIT-EOS200-BAX-8 (EOS-Additive-M200) | PM Interval: 500 Hrs | Torque: 82 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-EOS200-BAX-8 within tolerance ±0.0050 mm TIR.

BOM-42-3.8: PN-EOS200-IGBT-MOD-4211 / Kit KIT-EOS200-PLC-11 (EOS-Additive-M200) | PM Interval: 750 Hrs | Torque: 109 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-EOS200-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-42-3.9: PN-EOS200-HEX-CORE-4214 / Kit KIT-EOS200-THM-14 (EOS-Additive-M200) | PM Interval: 1000 Hrs | Torque: 136 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-EOS200-THM-14 within tolerance ±0.0020 mm TIR.

BOM-42-3.10: PN-EOS200-VLV-PROP-4203 / Kit KIT-EOS200-HYD-3 (EOS-Additive-M200) | PM Interval: 1250 Hrs | Torque: 37 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-EOS200-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-42-3.11: PN-EOS200-BSW-NUT-4206 / Kit KIT-EOS200-YAX-6 (EOS-Additive-M200) | PM Interval: 1500 Hrs | Torque: 64 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-EOS200-YAX-6 within tolerance ±0.0040 mm TIR.

BOM-42-3.12: PN-EOS200-BRK-DISC-4209 / Kit KIT-EOS200-CAX-9 (EOS-Additive-M200) | PM Interval: 250 Hrs | Torque: 91 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-EOS200-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-42-3.13: PN-EOS200-INJ-MET-4212 / Kit KIT-EOS200-LUB-12 (EOS-Additive-M200) | PM Interval: 500 Hrs | Torque: 118 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-EOS200-LUB-12 within tolerance ±0.0010 mm TIR.

BOM-42-3.14: PN-EOS200-BRG-CER-4201 / Kit KIT-EOS200-SPN-1 (EOS-Additive-M200) | PM Interval: 750 Hrs | Torque: 209 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-EOS200-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-42-3.15: PN-EOS200-PMP-PLG-4204 / Kit KIT-EOS200-CLN-4 (EOS-Additive-M200) | PM Interval: 1000 Hrs | Torque: 46 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-EOS200-CLN-4 within tolerance ±0.0030 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — EOS-Additive-M250 (GROUP 1)

Certified Aerospace Bill of Materials for EOS-Additive-M250. Verify part numbers before closing Work Orders.

BOM-43-1.1: PN-EOS250-CAM-GRIP-4302 / Kit KIT-EOS250-ATC-2 (EOS-Additive-M250) | PM Interval: 500 Hrs | Torque: 168 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-EOS250-ATC-2 within tolerance ±0.0040 mm TIR.

BOM-43-1.2: PN-EOS250-VLV-PROP-4303 / Kit KIT-EOS250-HYD-3 (EOS-Additive-M250) | PM Interval: 750 Hrs | Torque: 177 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-EOS250-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-43-1.3: PN-EOS250-PMP-PLG-4304 / Kit KIT-EOS250-CLN-4 (EOS-Additive-M250) | PM Interval: 1000 Hrs | Torque: 186 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-EOS250-CLN-4 within tolerance ±0.0050 mm TIR.

BOM-43-1.4: PN-EOS250-LIN-COIL-4305 / Kit KIT-EOS250-XAX-5 (EOS-Additive-M250) | PM Interval: 1250 Hrs | Torque: 195 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-EOS250-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-43-1.5: PN-EOS250-BSW-NUT-4306 / Kit KIT-EOS250-YAX-6 (EOS-Additive-M250) | PM Interval: 1500 Hrs | Torque: 204 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-EOS250-YAX-6 within tolerance ±0.0060 mm TIR.

BOM-43-1.6: PN-EOS250-CYL-SEAL-4307 / Kit KIT-EOS250-ZAX-7 (EOS-Additive-M250) | PM Interval: 250 Hrs | Torque: 23 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-EOS250-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-43-1.7: PN-EOS250-TRQ-STAT-4308 / Kit KIT-EOS250-BAX-8 (EOS-Additive-M250) | PM Interval: 500 Hrs | Torque: 32 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-EOS250-BAX-8 within tolerance ±0.0010 mm TIR.

BOM-43-1.8: PN-EOS250-BRK-DISC-4309 / Kit KIT-EOS250-CAX-9 (EOS-Additive-M250) | PM Interval: 750 Hrs | Torque: 41 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-EOS250-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-43-1.9: PN-EOS250-GLASS-RD-4310 / Kit KIT-EOS250-OPT-10 (EOS-Additive-M250) | PM Interval: 1000 Hrs | Torque: 50 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-EOS250-OPT-10 within tolerance ±0.0020 mm TIR.

BOM-43-1.10: PN-EOS250-IGBT-MOD-4311 / Kit KIT-EOS250-PLC-11 (EOS-Additive-M250) | PM Interval: 1250 Hrs | Torque: 59 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-EOS250-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-43-1.11: PN-EOS250-INJ-MET-4312 / Kit KIT-EOS250-LUB-12 (EOS-Additive-M250) | PM Interval: 1500 Hrs | Torque: 68 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-EOS250-LUB-12 within tolerance ±0.0030 mm TIR.

BOM-43-1.12: PN-EOS250-GASK-VIT-4313 / Kit KIT-EOS250-VAC-13 (EOS-Additive-M250) | PM Interval: 250 Hrs | Torque: 77 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-EOS250-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-43-1.13: PN-EOS250-HEX-CORE-4314 / Kit KIT-EOS250-THM-14 (EOS-Additive-M250) | PM Interval: 500 Hrs | Torque: 86 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-EOS250-THM-14 within tolerance ±0.0040 mm TIR.

BOM-43-1.14: PN-EOS250-BRG-CER-4301 / Kit KIT-EOS250-SPN-1 (EOS-Additive-M250) | PM Interval: 750 Hrs | Torque: 159 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-EOS250-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-43-1.15: PN-EOS250-CAM-GRIP-4302 / Kit KIT-EOS250-ATC-2 (EOS-Additive-M250) | PM Interval: 1000 Hrs | Torque: 168 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-EOS250-ATC-2 within tolerance ±0.0040 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — EOS-Additive-M250 (GROUP 2)

Certified Aerospace Bill of Materials for EOS-Additive-M250. Verify part numbers before closing Work Orders.

BOM-43-2.1: PN-EOS250-VLV-PROP-4303 / Kit KIT-EOS250-HYD-3 (EOS-Additive-M250) | PM Interval: 500 Hrs | Torque: 177 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-EOS250-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-43-2.2: PN-EOS250-LIN-COIL-4305 / Kit KIT-EOS250-XAX-5 (EOS-Additive-M250) | PM Interval: 750 Hrs | Torque: 195 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-EOS250-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-43-2.3: PN-EOS250-CYL-SEAL-4307 / Kit KIT-EOS250-ZAX-7 (EOS-Additive-M250) | PM Interval: 1000 Hrs | Torque: 23 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-EOS250-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-43-2.4: PN-EOS250-BRK-DISC-4309 / Kit KIT-EOS250-CAX-9 (EOS-Additive-M250) | PM Interval: 1250 Hrs | Torque: 41 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-EOS250-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-43-2.5: PN-EOS250-IGBT-MOD-4311 / Kit KIT-EOS250-PLC-11 (EOS-Additive-M250) | PM Interval: 1500 Hrs | Torque: 59 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-EOS250-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-43-2.6: PN-EOS250-GASK-VIT-4313 / Kit KIT-EOS250-VAC-13 (EOS-Additive-M250) | PM Interval: 250 Hrs | Torque: 77 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-EOS250-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-43-2.7: PN-EOS250-BRG-CER-4301 / Kit KIT-EOS250-SPN-1 (EOS-Additive-M250) | PM Interval: 500 Hrs | Torque: 159 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-EOS250-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-43-2.8: PN-EOS250-VLV-PROP-4303 / Kit KIT-EOS250-HYD-3 (EOS-Additive-M250) | PM Interval: 750 Hrs | Torque: 177 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-EOS250-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-43-2.9: PN-EOS250-LIN-COIL-4305 / Kit KIT-EOS250-XAX-5 (EOS-Additive-M250) | PM Interval: 1000 Hrs | Torque: 195 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-EOS250-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-43-2.10: PN-EOS250-CYL-SEAL-4307 / Kit KIT-EOS250-ZAX-7 (EOS-Additive-M250) | PM Interval: 1250 Hrs | Torque: 23 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-EOS250-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-43-2.11: PN-EOS250-BRK-DISC-4309 / Kit KIT-EOS250-CAX-9 (EOS-Additive-M250) | PM Interval: 1500 Hrs | Torque: 41 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-EOS250-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-43-2.12: PN-EOS250-IGBT-MOD-4311 / Kit KIT-EOS250-PLC-11 (EOS-Additive-M250) | PM Interval: 250 Hrs | Torque: 59 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-EOS250-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-43-2.13: PN-EOS250-GASK-VIT-4313 / Kit KIT-EOS250-VAC-13 (EOS-Additive-M250) | PM Interval: 500 Hrs | Torque: 77 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-EOS250-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-43-2.14: PN-EOS250-BRG-CER-4301 / Kit KIT-EOS250-SPN-1 (EOS-Additive-M250) | PM Interval: 750 Hrs | Torque: 159 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-EOS250-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-43-2.15: PN-EOS250-VLV-PROP-4303 / Kit KIT-EOS250-HYD-3 (EOS-Additive-M250) | PM Interval: 1000 Hrs | Torque: 177 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-EOS250-HYD-3 within tolerance ±0.0045 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — EOS-Additive-M250 (GROUP 3)

Certified Aerospace Bill of Materials for EOS-Additive-M250. Verify part numbers before closing Work Orders.

BOM-43-3.1: PN-EOS250-PMP-PLG-4304 / Kit KIT-EOS250-CLN-4 (EOS-Additive-M250) | PM Interval: 500 Hrs | Torque: 186 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-EOS250-CLN-4 within tolerance ±0.0050 mm TIR.

BOM-43-3.2: PN-EOS250-CYL-SEAL-4307 / Kit KIT-EOS250-ZAX-7 (EOS-Additive-M250) | PM Interval: 750 Hrs | Torque: 23 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-EOS250-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-43-3.3: PN-EOS250-GLASS-RD-4310 / Kit KIT-EOS250-OPT-10 (EOS-Additive-M250) | PM Interval: 1000 Hrs | Torque: 50 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-EOS250-OPT-10 within tolerance ±0.0020 mm TIR.

BOM-43-3.4: PN-EOS250-GASK-VIT-4313 / Kit KIT-EOS250-VAC-13 (EOS-Additive-M250) | PM Interval: 1250 Hrs | Torque: 77 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-EOS250-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-43-3.5: PN-EOS250-CAM-GRIP-4302 / Kit KIT-EOS250-ATC-2 (EOS-Additive-M250) | PM Interval: 1500 Hrs | Torque: 168 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-EOS250-ATC-2 within tolerance ±0.0040 mm TIR.

BOM-43-3.6: PN-EOS250-LIN-COIL-4305 / Kit KIT-EOS250-XAX-5 (EOS-Additive-M250) | PM Interval: 250 Hrs | Torque: 195 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-EOS250-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-43-3.7: PN-EOS250-TRQ-STAT-4308 / Kit KIT-EOS250-BAX-8 (EOS-Additive-M250) | PM Interval: 500 Hrs | Torque: 32 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-EOS250-BAX-8 within tolerance ±0.0010 mm TIR.

BOM-43-3.8: PN-EOS250-IGBT-MOD-4311 / Kit KIT-EOS250-PLC-11 (EOS-Additive-M250) | PM Interval: 750 Hrs | Torque: 59 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-EOS250-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-43-3.9: PN-EOS250-HEX-CORE-4314 / Kit KIT-EOS250-THM-14 (EOS-Additive-M250) | PM Interval: 1000 Hrs | Torque: 86 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-EOS250-THM-14 within tolerance ±0.0040 mm TIR.

BOM-43-3.10: PN-EOS250-VLV-PROP-4303 / Kit KIT-EOS250-HYD-3 (EOS-Additive-M250) | PM Interval: 1250 Hrs | Torque: 177 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-EOS250-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-43-3.11: PN-EOS250-BSW-NUT-4306 / Kit KIT-EOS250-YAX-6 (EOS-Additive-M250) | PM Interval: 1500 Hrs | Torque: 204 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-EOS250-YAX-6 within tolerance ±0.0060 mm TIR.

BOM-43-3.12: PN-EOS250-BRK-DISC-4309 / Kit KIT-EOS250-CAX-9 (EOS-Additive-M250) | PM Interval: 250 Hrs | Torque: 41 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-EOS250-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-43-3.13: PN-EOS250-INJ-MET-4312 / Kit KIT-EOS250-LUB-12 (EOS-Additive-M250) | PM Interval: 500 Hrs | Torque: 68 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-EOS250-LUB-12 within tolerance ±0.0030 mm TIR.

BOM-43-3.14: PN-EOS250-BRG-CER-4301 / Kit KIT-EOS250-SPN-1 (EOS-Additive-M250) | PM Interval: 750 Hrs | Torque: 159 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-EOS250-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-43-3.15: PN-EOS250-PMP-PLG-4304 / Kit KIT-EOS250-CLN-4 (EOS-Additive-M250) | PM Interval: 1000 Hrs | Torque: 186 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-EOS250-CLN-4 within tolerance ±0.0050 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — EOS-Additive-M300 (GROUP 1)

Certified Aerospace Bill of Materials for EOS-Additive-M300. Verify part numbers before closing Work Orders.

BOM-44-1.1: PN-EOS300-CAM-GRIP-4402 / Kit KIT-EOS300-ATC-2 (EOS-Additive-M300) | PM Interval: 500 Hrs | Torque: 118 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-EOS300-ATC-2 within tolerance ±0.0060 mm TIR.

BOM-44-1.2: PN-EOS300-VLV-PROP-4403 / Kit KIT-EOS300-HYD-3 (EOS-Additive-M300) | PM Interval: 750 Hrs | Torque: 127 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-EOS300-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-44-1.3: PN-EOS300-PMP-PLG-4404 / Kit KIT-EOS300-CLN-4 (EOS-Additive-M300) | PM Interval: 1000 Hrs | Torque: 136 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-EOS300-CLN-4 within tolerance ±0.0010 mm TIR.

BOM-44-1.4: PN-EOS300-LIN-COIL-4405 / Kit KIT-EOS300-XAX-5 (EOS-Additive-M300) | PM Interval: 1250 Hrs | Torque: 145 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-EOS300-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-44-1.5: PN-EOS300-BSW-NUT-4406 / Kit KIT-EOS300-YAX-6 (EOS-Additive-M300) | PM Interval: 1500 Hrs | Torque: 154 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-EOS300-YAX-6 within tolerance ±0.0020 mm TIR.

BOM-44-1.6: PN-EOS300-CYL-SEAL-4407 / Kit KIT-EOS300-ZAX-7 (EOS-Additive-M300) | PM Interval: 250 Hrs | Torque: 163 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-EOS300-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-44-1.7: PN-EOS300-TRQ-STAT-4408 / Kit KIT-EOS300-BAX-8 (EOS-Additive-M300) | PM Interval: 500 Hrs | Torque: 172 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-EOS300-BAX-8 within tolerance ±0.0030 mm TIR.

BOM-44-1.8: PN-EOS300-BRK-DISC-4409 / Kit KIT-EOS300-CAX-9 (EOS-Additive-M300) | PM Interval: 750 Hrs | Torque: 181 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-EOS300-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-44-1.9: PN-EOS300-GLASS-RD-4410 / Kit KIT-EOS300-OPT-10 (EOS-Additive-M300) | PM Interval: 1000 Hrs | Torque: 190 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-EOS300-OPT-10 within tolerance ±0.0040 mm TIR.

BOM-44-1.10: PN-EOS300-IGBT-MOD-4411 / Kit KIT-EOS300-PLC-11 (EOS-Additive-M300) | PM Interval: 1250 Hrs | Torque: 199 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-EOS300-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-44-1.11: PN-EOS300-INJ-MET-4412 / Kit KIT-EOS300-LUB-12 (EOS-Additive-M300) | PM Interval: 1500 Hrs | Torque: 208 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-EOS300-LUB-12 within tolerance ±0.0050 mm TIR.

BOM-44-1.12: PN-EOS300-GASK-VIT-4413 / Kit KIT-EOS300-VAC-13 (EOS-Additive-M300) | PM Interval: 250 Hrs | Torque: 27 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-EOS300-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-44-1.13: PN-EOS300-HEX-CORE-4414 / Kit KIT-EOS300-THM-14 (EOS-Additive-M300) | PM Interval: 500 Hrs | Torque: 36 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-EOS300-THM-14 within tolerance ±0.0060 mm TIR.

BOM-44-1.14: PN-EOS300-BRG-CER-4401 / Kit KIT-EOS300-SPN-1 (EOS-Additive-M300) | PM Interval: 750 Hrs | Torque: 109 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-EOS300-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-44-1.15: PN-EOS300-CAM-GRIP-4402 / Kit KIT-EOS300-ATC-2 (EOS-Additive-M300) | PM Interval: 1000 Hrs | Torque: 118 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-EOS300-ATC-2 within tolerance ±0.0060 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — EOS-Additive-M300 (GROUP 2)

Certified Aerospace Bill of Materials for EOS-Additive-M300. Verify part numbers before closing Work Orders.

BOM-44-2.1: PN-EOS300-VLV-PROP-4403 / Kit KIT-EOS300-HYD-3 (EOS-Additive-M300) | PM Interval: 500 Hrs | Torque: 127 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-EOS300-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-44-2.2: PN-EOS300-LIN-COIL-4405 / Kit KIT-EOS300-XAX-5 (EOS-Additive-M300) | PM Interval: 750 Hrs | Torque: 145 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-EOS300-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-44-2.3: PN-EOS300-CYL-SEAL-4407 / Kit KIT-EOS300-ZAX-7 (EOS-Additive-M300) | PM Interval: 1000 Hrs | Torque: 163 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-EOS300-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-44-2.4: PN-EOS300-BRK-DISC-4409 / Kit KIT-EOS300-CAX-9 (EOS-Additive-M300) | PM Interval: 1250 Hrs | Torque: 181 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-EOS300-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-44-2.5: PN-EOS300-IGBT-MOD-4411 / Kit KIT-EOS300-PLC-11 (EOS-Additive-M300) | PM Interval: 1500 Hrs | Torque: 199 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-EOS300-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-44-2.6: PN-EOS300-GASK-VIT-4413 / Kit KIT-EOS300-VAC-13 (EOS-Additive-M300) | PM Interval: 250 Hrs | Torque: 27 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-EOS300-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-44-2.7: PN-EOS300-BRG-CER-4401 / Kit KIT-EOS300-SPN-1 (EOS-Additive-M300) | PM Interval: 500 Hrs | Torque: 109 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-EOS300-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-44-2.8: PN-EOS300-VLV-PROP-4403 / Kit KIT-EOS300-HYD-3 (EOS-Additive-M300) | PM Interval: 750 Hrs | Torque: 127 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-EOS300-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-44-2.9: PN-EOS300-LIN-COIL-4405 / Kit KIT-EOS300-XAX-5 (EOS-Additive-M300) | PM Interval: 1000 Hrs | Torque: 145 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-EOS300-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-44-2.10: PN-EOS300-CYL-SEAL-4407 / Kit KIT-EOS300-ZAX-7 (EOS-Additive-M300) | PM Interval: 1250 Hrs | Torque: 163 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-EOS300-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-44-2.11: PN-EOS300-BRK-DISC-4409 / Kit KIT-EOS300-CAX-9 (EOS-Additive-M300) | PM Interval: 1500 Hrs | Torque: 181 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-EOS300-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-44-2.12: PN-EOS300-IGBT-MOD-4411 / Kit KIT-EOS300-PLC-11 (EOS-Additive-M300) | PM Interval: 250 Hrs | Torque: 199 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-EOS300-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-44-2.13: PN-EOS300-GASK-VIT-4413 / Kit KIT-EOS300-VAC-13 (EOS-Additive-M300) | PM Interval: 500 Hrs | Torque: 27 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-EOS300-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-44-2.14: PN-EOS300-BRG-CER-4401 / Kit KIT-EOS300-SPN-1 (EOS-Additive-M300) | PM Interval: 750 Hrs | Torque: 109 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-EOS300-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-44-2.15: PN-EOS300-VLV-PROP-4403 / Kit KIT-EOS300-HYD-3 (EOS-Additive-M300) | PM Interval: 1000 Hrs | Torque: 127 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-EOS300-HYD-3 within tolerance ±0.0065 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — EOS-Additive-M300 (GROUP 3)

Certified Aerospace Bill of Materials for EOS-Additive-M300. Verify part numbers before closing Work Orders.

BOM-44-3.1: PN-EOS300-PMP-PLG-4404 / Kit KIT-EOS300-CLN-4 (EOS-Additive-M300) | PM Interval: 500 Hrs | Torque: 136 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-EOS300-CLN-4 within tolerance ±0.0010 mm TIR.

BOM-44-3.2: PN-EOS300-CYL-SEAL-4407 / Kit KIT-EOS300-ZAX-7 (EOS-Additive-M300) | PM Interval: 750 Hrs | Torque: 163 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-EOS300-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-44-3.3: PN-EOS300-GLASS-RD-4410 / Kit KIT-EOS300-OPT-10 (EOS-Additive-M300) | PM Interval: 1000 Hrs | Torque: 190 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-EOS300-OPT-10 within tolerance ±0.0040 mm TIR.

BOM-44-3.4: PN-EOS300-GASK-VIT-4413 / Kit KIT-EOS300-VAC-13 (EOS-Additive-M300) | PM Interval: 1250 Hrs | Torque: 27 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-EOS300-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-44-3.5: PN-EOS300-CAM-GRIP-4402 / Kit KIT-EOS300-ATC-2 (EOS-Additive-M300) | PM Interval: 1500 Hrs | Torque: 118 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-EOS300-ATC-2 within tolerance ±0.0060 mm TIR.

BOM-44-3.6: PN-EOS300-LIN-COIL-4405 / Kit KIT-EOS300-XAX-5 (EOS-Additive-M300) | PM Interval: 250 Hrs | Torque: 145 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-EOS300-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-44-3.7: PN-EOS300-TRQ-STAT-4408 / Kit KIT-EOS300-BAX-8 (EOS-Additive-M300) | PM Interval: 500 Hrs | Torque: 172 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-EOS300-BAX-8 within tolerance ±0.0030 mm TIR.

BOM-44-3.8: PN-EOS300-IGBT-MOD-4411 / Kit KIT-EOS300-PLC-11 (EOS-Additive-M300) | PM Interval: 750 Hrs | Torque: 199 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-EOS300-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-44-3.9: PN-EOS300-HEX-CORE-4414 / Kit KIT-EOS300-THM-14 (EOS-Additive-M300) | PM Interval: 1000 Hrs | Torque: 36 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-EOS300-THM-14 within tolerance ±0.0060 mm TIR.

BOM-44-3.10: PN-EOS300-VLV-PROP-4403 / Kit KIT-EOS300-HYD-3 (EOS-Additive-M300) | PM Interval: 1250 Hrs | Torque: 127 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-EOS300-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-44-3.11: PN-EOS300-BSW-NUT-4406 / Kit KIT-EOS300-YAX-6 (EOS-Additive-M300) | PM Interval: 1500 Hrs | Torque: 154 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-EOS300-YAX-6 within tolerance ±0.0020 mm TIR.

BOM-44-3.12: PN-EOS300-BRK-DISC-4409 / Kit KIT-EOS300-CAX-9 (EOS-Additive-M300) | PM Interval: 250 Hrs | Torque: 181 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-EOS300-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-44-3.13: PN-EOS300-INJ-MET-4412 / Kit KIT-EOS300-LUB-12 (EOS-Additive-M300) | PM Interval: 500 Hrs | Torque: 208 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-EOS300-LUB-12 within tolerance ±0.0050 mm TIR.

BOM-44-3.14: PN-EOS300-BRG-CER-4401 / Kit KIT-EOS300-SPN-1 (EOS-Additive-M300) | PM Interval: 750 Hrs | Torque: 109 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-EOS300-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-44-3.15: PN-EOS300-PMP-PLG-4404 / Kit KIT-EOS300-CLN-4 (EOS-Additive-M300) | PM Interval: 1000 Hrs | Torque: 136 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-EOS300-CLN-4 within tolerance ±0.0010 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — EOS-Additive-M350 (GROUP 1)

Certified Aerospace Bill of Materials for EOS-Additive-M350. Verify part numbers before closing Work Orders.

BOM-45-1.1: PN-EOS350-CAM-GRIP-4502 / Kit KIT-EOS350-ATC-2 (EOS-Additive-M350) | PM Interval: 500 Hrs | Torque: 68 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-EOS350-ATC-2 within tolerance ±0.0020 mm TIR.

BOM-45-1.2: PN-EOS350-VLV-PROP-4503 / Kit KIT-EOS350-HYD-3 (EOS-Additive-M350) | PM Interval: 750 Hrs | Torque: 77 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-EOS350-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-45-1.3: PN-EOS350-PMP-PLG-4504 / Kit KIT-EOS350-CLN-4 (EOS-Additive-M350) | PM Interval: 1000 Hrs | Torque: 86 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-EOS350-CLN-4 within tolerance ±0.0030 mm TIR.

BOM-45-1.4: PN-EOS350-LIN-COIL-4505 / Kit KIT-EOS350-XAX-5 (EOS-Additive-M350) | PM Interval: 1250 Hrs | Torque: 95 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-EOS350-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-45-1.5: PN-EOS350-BSW-NUT-4506 / Kit KIT-EOS350-YAX-6 (EOS-Additive-M350) | PM Interval: 1500 Hrs | Torque: 104 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-EOS350-YAX-6 within tolerance ±0.0040 mm TIR.

BOM-45-1.6: PN-EOS350-CYL-SEAL-4507 / Kit KIT-EOS350-ZAX-7 (EOS-Additive-M350) | PM Interval: 250 Hrs | Torque: 113 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-EOS350-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-45-1.7: PN-EOS350-TRQ-STAT-4508 / Kit KIT-EOS350-BAX-8 (EOS-Additive-M350) | PM Interval: 500 Hrs | Torque: 122 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-EOS350-BAX-8 within tolerance ±0.0050 mm TIR.

BOM-45-1.8: PN-EOS350-BRK-DISC-4509 / Kit KIT-EOS350-CAX-9 (EOS-Additive-M350) | PM Interval: 750 Hrs | Torque: 131 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-EOS350-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-45-1.9: PN-EOS350-GLASS-RD-4510 / Kit KIT-EOS350-OPT-10 (EOS-Additive-M350) | PM Interval: 1000 Hrs | Torque: 140 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-EOS350-OPT-10 within tolerance ±0.0060 mm TIR.

BOM-45-1.10: PN-EOS350-IGBT-MOD-4511 / Kit KIT-EOS350-PLC-11 (EOS-Additive-M350) | PM Interval: 1250 Hrs | Torque: 149 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-EOS350-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-45-1.11: PN-EOS350-INJ-MET-4512 / Kit KIT-EOS350-LUB-12 (EOS-Additive-M350) | PM Interval: 1500 Hrs | Torque: 158 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-EOS350-LUB-12 within tolerance ±0.0010 mm TIR.

BOM-45-1.12: PN-EOS350-GASK-VIT-4513 / Kit KIT-EOS350-VAC-13 (EOS-Additive-M350) | PM Interval: 250 Hrs | Torque: 167 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-EOS350-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-45-1.13: PN-EOS350-HEX-CORE-4514 / Kit KIT-EOS350-THM-14 (EOS-Additive-M350) | PM Interval: 500 Hrs | Torque: 176 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-EOS350-THM-14 within tolerance ±0.0020 mm TIR.

BOM-45-1.14: PN-EOS350-BRG-CER-4501 / Kit KIT-EOS350-SPN-1 (EOS-Additive-M350) | PM Interval: 750 Hrs | Torque: 59 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-EOS350-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-45-1.15: PN-EOS350-CAM-GRIP-4502 / Kit KIT-EOS350-ATC-2 (EOS-Additive-M350) | PM Interval: 1000 Hrs | Torque: 68 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-EOS350-ATC-2 within tolerance ±0.0020 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — EOS-Additive-M350 (GROUP 2)

Certified Aerospace Bill of Materials for EOS-Additive-M350. Verify part numbers before closing Work Orders.

BOM-45-2.1: PN-EOS350-VLV-PROP-4503 / Kit KIT-EOS350-HYD-3 (EOS-Additive-M350) | PM Interval: 500 Hrs | Torque: 77 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-EOS350-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-45-2.2: PN-EOS350-LIN-COIL-4505 / Kit KIT-EOS350-XAX-5 (EOS-Additive-M350) | PM Interval: 750 Hrs | Torque: 95 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-EOS350-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-45-2.3: PN-EOS350-CYL-SEAL-4507 / Kit KIT-EOS350-ZAX-7 (EOS-Additive-M350) | PM Interval: 1000 Hrs | Torque: 113 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-EOS350-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-45-2.4: PN-EOS350-BRK-DISC-4509 / Kit KIT-EOS350-CAX-9 (EOS-Additive-M350) | PM Interval: 1250 Hrs | Torque: 131 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-EOS350-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-45-2.5: PN-EOS350-IGBT-MOD-4511 / Kit KIT-EOS350-PLC-11 (EOS-Additive-M350) | PM Interval: 1500 Hrs | Torque: 149 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-EOS350-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-45-2.6: PN-EOS350-GASK-VIT-4513 / Kit KIT-EOS350-VAC-13 (EOS-Additive-M350) | PM Interval: 250 Hrs | Torque: 167 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-EOS350-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-45-2.7: PN-EOS350-BRG-CER-4501 / Kit KIT-EOS350-SPN-1 (EOS-Additive-M350) | PM Interval: 500 Hrs | Torque: 59 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-EOS350-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-45-2.8: PN-EOS350-VLV-PROP-4503 / Kit KIT-EOS350-HYD-3 (EOS-Additive-M350) | PM Interval: 750 Hrs | Torque: 77 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-EOS350-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-45-2.9: PN-EOS350-LIN-COIL-4505 / Kit KIT-EOS350-XAX-5 (EOS-Additive-M350) | PM Interval: 1000 Hrs | Torque: 95 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-EOS350-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-45-2.10: PN-EOS350-CYL-SEAL-4507 / Kit KIT-EOS350-ZAX-7 (EOS-Additive-M350) | PM Interval: 1250 Hrs | Torque: 113 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-EOS350-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-45-2.11: PN-EOS350-BRK-DISC-4509 / Kit KIT-EOS350-CAX-9 (EOS-Additive-M350) | PM Interval: 1500 Hrs | Torque: 131 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-EOS350-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-45-2.12: PN-EOS350-IGBT-MOD-4511 / Kit KIT-EOS350-PLC-11 (EOS-Additive-M350) | PM Interval: 250 Hrs | Torque: 149 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-EOS350-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-45-2.13: PN-EOS350-GASK-VIT-4513 / Kit KIT-EOS350-VAC-13 (EOS-Additive-M350) | PM Interval: 500 Hrs | Torque: 167 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-EOS350-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-45-2.14: PN-EOS350-BRG-CER-4501 / Kit KIT-EOS350-SPN-1 (EOS-Additive-M350) | PM Interval: 750 Hrs | Torque: 59 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-EOS350-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-45-2.15: PN-EOS350-VLV-PROP-4503 / Kit KIT-EOS350-HYD-3 (EOS-Additive-M350) | PM Interval: 1000 Hrs | Torque: 77 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-EOS350-HYD-3 within tolerance ±0.0025 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — EOS-Additive-M350 (GROUP 3)

Certified Aerospace Bill of Materials for EOS-Additive-M350. Verify part numbers before closing Work Orders.

BOM-45-3.1: PN-EOS350-PMP-PLG-4504 / Kit KIT-EOS350-CLN-4 (EOS-Additive-M350) | PM Interval: 500 Hrs | Torque: 86 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-EOS350-CLN-4 within tolerance ±0.0030 mm TIR.

BOM-45-3.2: PN-EOS350-CYL-SEAL-4507 / Kit KIT-EOS350-ZAX-7 (EOS-Additive-M350) | PM Interval: 750 Hrs | Torque: 113 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-EOS350-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-45-3.3: PN-EOS350-GLASS-RD-4510 / Kit KIT-EOS350-OPT-10 (EOS-Additive-M350) | PM Interval: 1000 Hrs | Torque: 140 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-EOS350-OPT-10 within tolerance ±0.0060 mm TIR.

BOM-45-3.4: PN-EOS350-GASK-VIT-4513 / Kit KIT-EOS350-VAC-13 (EOS-Additive-M350) | PM Interval: 1250 Hrs | Torque: 167 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-EOS350-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-45-3.5: PN-EOS350-CAM-GRIP-4502 / Kit KIT-EOS350-ATC-2 (EOS-Additive-M350) | PM Interval: 1500 Hrs | Torque: 68 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-EOS350-ATC-2 within tolerance ±0.0020 mm TIR.

BOM-45-3.6: PN-EOS350-LIN-COIL-4505 / Kit KIT-EOS350-XAX-5 (EOS-Additive-M350) | PM Interval: 250 Hrs | Torque: 95 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-EOS350-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-45-3.7: PN-EOS350-TRQ-STAT-4508 / Kit KIT-EOS350-BAX-8 (EOS-Additive-M350) | PM Interval: 500 Hrs | Torque: 122 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-EOS350-BAX-8 within tolerance ±0.0050 mm TIR.

BOM-45-3.8: PN-EOS350-IGBT-MOD-4511 / Kit KIT-EOS350-PLC-11 (EOS-Additive-M350) | PM Interval: 750 Hrs | Torque: 149 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-EOS350-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-45-3.9: PN-EOS350-HEX-CORE-4514 / Kit KIT-EOS350-THM-14 (EOS-Additive-M350) | PM Interval: 1000 Hrs | Torque: 176 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-EOS350-THM-14 within tolerance ±0.0020 mm TIR.

BOM-45-3.10: PN-EOS350-VLV-PROP-4503 / Kit KIT-EOS350-HYD-3 (EOS-Additive-M350) | PM Interval: 1250 Hrs | Torque: 77 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-EOS350-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-45-3.11: PN-EOS350-BSW-NUT-4506 / Kit KIT-EOS350-YAX-6 (EOS-Additive-M350) | PM Interval: 1500 Hrs | Torque: 104 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-EOS350-YAX-6 within tolerance ±0.0040 mm TIR.

BOM-45-3.12: PN-EOS350-BRK-DISC-4509 / Kit KIT-EOS350-CAX-9 (EOS-Additive-M350) | PM Interval: 250 Hrs | Torque: 131 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-EOS350-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-45-3.13: PN-EOS350-INJ-MET-4512 / Kit KIT-EOS350-LUB-12 (EOS-Additive-M350) | PM Interval: 500 Hrs | Torque: 158 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-EOS350-LUB-12 within tolerance ±0.0010 mm TIR.

BOM-45-3.14: PN-EOS350-BRG-CER-4501 / Kit KIT-EOS350-SPN-1 (EOS-Additive-M350) | PM Interval: 750 Hrs | Torque: 59 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-EOS350-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-45-3.15: PN-EOS350-PMP-PLG-4504 / Kit KIT-EOS350-CLN-4 (EOS-Additive-M350) | PM Interval: 1000 Hrs | Torque: 86 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-EOS350-CLN-4 within tolerance ±0.0030 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — EOS-Additive-M400 (GROUP 1)

Certified Aerospace Bill of Materials for EOS-Additive-M400. Verify part numbers before closing Work Orders.

BOM-46-1.1: PN-EOS400-CAM-GRIP-4602 / Kit KIT-EOS400-ATC-2 (EOS-Additive-M400) | PM Interval: 500 Hrs | Torque: 208 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-EOS400-ATC-2 within tolerance ±0.0040 mm TIR.

BOM-46-1.2: PN-EOS400-VLV-PROP-4603 / Kit KIT-EOS400-HYD-3 (EOS-Additive-M400) | PM Interval: 750 Hrs | Torque: 27 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-EOS400-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-46-1.3: PN-EOS400-PMP-PLG-4604 / Kit KIT-EOS400-CLN-4 (EOS-Additive-M400) | PM Interval: 1000 Hrs | Torque: 36 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-EOS400-CLN-4 within tolerance ±0.0050 mm TIR.

BOM-46-1.4: PN-EOS400-LIN-COIL-4605 / Kit KIT-EOS400-XAX-5 (EOS-Additive-M400) | PM Interval: 1250 Hrs | Torque: 45 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-EOS400-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-46-1.5: PN-EOS400-BSW-NUT-4606 / Kit KIT-EOS400-YAX-6 (EOS-Additive-M400) | PM Interval: 1500 Hrs | Torque: 54 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-EOS400-YAX-6 within tolerance ±0.0060 mm TIR.

BOM-46-1.6: PN-EOS400-CYL-SEAL-4607 / Kit KIT-EOS400-ZAX-7 (EOS-Additive-M400) | PM Interval: 250 Hrs | Torque: 63 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-EOS400-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-46-1.7: PN-EOS400-TRQ-STAT-4608 / Kit KIT-EOS400-BAX-8 (EOS-Additive-M400) | PM Interval: 500 Hrs | Torque: 72 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-EOS400-BAX-8 within tolerance ±0.0010 mm TIR.

BOM-46-1.8: PN-EOS400-BRK-DISC-4609 / Kit KIT-EOS400-CAX-9 (EOS-Additive-M400) | PM Interval: 750 Hrs | Torque: 81 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-EOS400-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-46-1.9: PN-EOS400-GLASS-RD-4610 / Kit KIT-EOS400-OPT-10 (EOS-Additive-M400) | PM Interval: 1000 Hrs | Torque: 90 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-EOS400-OPT-10 within tolerance ±0.0020 mm TIR.

BOM-46-1.10: PN-EOS400-IGBT-MOD-4611 / Kit KIT-EOS400-PLC-11 (EOS-Additive-M400) | PM Interval: 1250 Hrs | Torque: 99 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-EOS400-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-46-1.11: PN-EOS400-INJ-MET-4612 / Kit KIT-EOS400-LUB-12 (EOS-Additive-M400) | PM Interval: 1500 Hrs | Torque: 108 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-EOS400-LUB-12 within tolerance ±0.0030 mm TIR.

BOM-46-1.12: PN-EOS400-GASK-VIT-4613 / Kit KIT-EOS400-VAC-13 (EOS-Additive-M400) | PM Interval: 250 Hrs | Torque: 117 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-EOS400-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-46-1.13: PN-EOS400-HEX-CORE-4614 / Kit KIT-EOS400-THM-14 (EOS-Additive-M400) | PM Interval: 500 Hrs | Torque: 126 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-EOS400-THM-14 within tolerance ±0.0040 mm TIR.

BOM-46-1.14: PN-EOS400-BRG-CER-4601 / Kit KIT-EOS400-SPN-1 (EOS-Additive-M400) | PM Interval: 750 Hrs | Torque: 199 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-EOS400-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-46-1.15: PN-EOS400-CAM-GRIP-4602 / Kit KIT-EOS400-ATC-2 (EOS-Additive-M400) | PM Interval: 1000 Hrs | Torque: 208 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-EOS400-ATC-2 within tolerance ±0.0040 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — EOS-Additive-M400 (GROUP 2)

Certified Aerospace Bill of Materials for EOS-Additive-M400. Verify part numbers before closing Work Orders.

BOM-46-2.1: PN-EOS400-VLV-PROP-4603 / Kit KIT-EOS400-HYD-3 (EOS-Additive-M400) | PM Interval: 500 Hrs | Torque: 27 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-EOS400-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-46-2.2: PN-EOS400-LIN-COIL-4605 / Kit KIT-EOS400-XAX-5 (EOS-Additive-M400) | PM Interval: 750 Hrs | Torque: 45 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-EOS400-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-46-2.3: PN-EOS400-CYL-SEAL-4607 / Kit KIT-EOS400-ZAX-7 (EOS-Additive-M400) | PM Interval: 1000 Hrs | Torque: 63 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-EOS400-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-46-2.4: PN-EOS400-BRK-DISC-4609 / Kit KIT-EOS400-CAX-9 (EOS-Additive-M400) | PM Interval: 1250 Hrs | Torque: 81 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-EOS400-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-46-2.5: PN-EOS400-IGBT-MOD-4611 / Kit KIT-EOS400-PLC-11 (EOS-Additive-M400) | PM Interval: 1500 Hrs | Torque: 99 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-EOS400-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-46-2.6: PN-EOS400-GASK-VIT-4613 / Kit KIT-EOS400-VAC-13 (EOS-Additive-M400) | PM Interval: 250 Hrs | Torque: 117 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-EOS400-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-46-2.7: PN-EOS400-BRG-CER-4601 / Kit KIT-EOS400-SPN-1 (EOS-Additive-M400) | PM Interval: 500 Hrs | Torque: 199 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-EOS400-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-46-2.8: PN-EOS400-VLV-PROP-4603 / Kit KIT-EOS400-HYD-3 (EOS-Additive-M400) | PM Interval: 750 Hrs | Torque: 27 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-EOS400-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-46-2.9: PN-EOS400-LIN-COIL-4605 / Kit KIT-EOS400-XAX-5 (EOS-Additive-M400) | PM Interval: 1000 Hrs | Torque: 45 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-EOS400-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-46-2.10: PN-EOS400-CYL-SEAL-4607 / Kit KIT-EOS400-ZAX-7 (EOS-Additive-M400) | PM Interval: 1250 Hrs | Torque: 63 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-EOS400-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-46-2.11: PN-EOS400-BRK-DISC-4609 / Kit KIT-EOS400-CAX-9 (EOS-Additive-M400) | PM Interval: 1500 Hrs | Torque: 81 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-EOS400-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-46-2.12: PN-EOS400-IGBT-MOD-4611 / Kit KIT-EOS400-PLC-11 (EOS-Additive-M400) | PM Interval: 250 Hrs | Torque: 99 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-EOS400-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-46-2.13: PN-EOS400-GASK-VIT-4613 / Kit KIT-EOS400-VAC-13 (EOS-Additive-M400) | PM Interval: 500 Hrs | Torque: 117 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-EOS400-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-46-2.14: PN-EOS400-BRG-CER-4601 / Kit KIT-EOS400-SPN-1 (EOS-Additive-M400) | PM Interval: 750 Hrs | Torque: 199 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-EOS400-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-46-2.15: PN-EOS400-VLV-PROP-4603 / Kit KIT-EOS400-HYD-3 (EOS-Additive-M400) | PM Interval: 1000 Hrs | Torque: 27 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-EOS400-HYD-3 within tolerance ±0.0045 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — EOS-Additive-M400 (GROUP 3)

Certified Aerospace Bill of Materials for EOS-Additive-M400. Verify part numbers before closing Work Orders.

BOM-46-3.1: PN-EOS400-PMP-PLG-4604 / Kit KIT-EOS400-CLN-4 (EOS-Additive-M400) | PM Interval: 500 Hrs | Torque: 36 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-EOS400-CLN-4 within tolerance ±0.0050 mm TIR.

BOM-46-3.2: PN-EOS400-CYL-SEAL-4607 / Kit KIT-EOS400-ZAX-7 (EOS-Additive-M400) | PM Interval: 750 Hrs | Torque: 63 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-EOS400-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-46-3.3: PN-EOS400-GLASS-RD-4610 / Kit KIT-EOS400-OPT-10 (EOS-Additive-M400) | PM Interval: 1000 Hrs | Torque: 90 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-EOS400-OPT-10 within tolerance ±0.0020 mm TIR.

BOM-46-3.4: PN-EOS400-GASK-VIT-4613 / Kit KIT-EOS400-VAC-13 (EOS-Additive-M400) | PM Interval: 1250 Hrs | Torque: 117 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-EOS400-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-46-3.5: PN-EOS400-CAM-GRIP-4602 / Kit KIT-EOS400-ATC-2 (EOS-Additive-M400) | PM Interval: 1500 Hrs | Torque: 208 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-EOS400-ATC-2 within tolerance ±0.0040 mm TIR.

BOM-46-3.6: PN-EOS400-LIN-COIL-4605 / Kit KIT-EOS400-XAX-5 (EOS-Additive-M400) | PM Interval: 250 Hrs | Torque: 45 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-EOS400-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-46-3.7: PN-EOS400-TRQ-STAT-4608 / Kit KIT-EOS400-BAX-8 (EOS-Additive-M400) | PM Interval: 500 Hrs | Torque: 72 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-EOS400-BAX-8 within tolerance ±0.0010 mm TIR.

BOM-46-3.8: PN-EOS400-IGBT-MOD-4611 / Kit KIT-EOS400-PLC-11 (EOS-Additive-M400) | PM Interval: 750 Hrs | Torque: 99 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-EOS400-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-46-3.9: PN-EOS400-HEX-CORE-4614 / Kit KIT-EOS400-THM-14 (EOS-Additive-M400) | PM Interval: 1000 Hrs | Torque: 126 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-EOS400-THM-14 within tolerance ±0.0040 mm TIR.

BOM-46-3.10: PN-EOS400-VLV-PROP-4603 / Kit KIT-EOS400-HYD-3 (EOS-Additive-M400) | PM Interval: 1250 Hrs | Torque: 27 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-EOS400-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-46-3.11: PN-EOS400-BSW-NUT-4606 / Kit KIT-EOS400-YAX-6 (EOS-Additive-M400) | PM Interval: 1500 Hrs | Torque: 54 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-EOS400-YAX-6 within tolerance ±0.0060 mm TIR.

BOM-46-3.12: PN-EOS400-BRK-DISC-4609 / Kit KIT-EOS400-CAX-9 (EOS-Additive-M400) | PM Interval: 250 Hrs | Torque: 81 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-EOS400-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-46-3.13: PN-EOS400-INJ-MET-4612 / Kit KIT-EOS400-LUB-12 (EOS-Additive-M400) | PM Interval: 500 Hrs | Torque: 108 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-EOS400-LUB-12 within tolerance ±0.0030 mm TIR.

BOM-46-3.14: PN-EOS400-BRG-CER-4601 / Kit KIT-EOS400-SPN-1 (EOS-Additive-M400) | PM Interval: 750 Hrs | Torque: 199 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-EOS400-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-46-3.15: PN-EOS400-PMP-PLG-4604 / Kit KIT-EOS400-CLN-4 (EOS-Additive-M400) | PM Interval: 1000 Hrs | Torque: 36 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-EOS400-CLN-4 within tolerance ±0.0050 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — EOS-Additive-M450 (GROUP 1)

Certified Aerospace Bill of Materials for EOS-Additive-M450. Verify part numbers before closing Work Orders.

BOM-47-1.1: PN-EOS450-CAM-GRIP-4702 / Kit KIT-EOS450-ATC-2 (EOS-Additive-M450) | PM Interval: 500 Hrs | Torque: 158 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-EOS450-ATC-2 within tolerance ±0.0060 mm TIR.

BOM-47-1.2: PN-EOS450-VLV-PROP-4703 / Kit KIT-EOS450-HYD-3 (EOS-Additive-M450) | PM Interval: 750 Hrs | Torque: 167 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-EOS450-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-47-1.3: PN-EOS450-PMP-PLG-4704 / Kit KIT-EOS450-CLN-4 (EOS-Additive-M450) | PM Interval: 1000 Hrs | Torque: 176 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-EOS450-CLN-4 within tolerance ±0.0010 mm TIR.

BOM-47-1.4: PN-EOS450-LIN-COIL-4705 / Kit KIT-EOS450-XAX-5 (EOS-Additive-M450) | PM Interval: 1250 Hrs | Torque: 185 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-EOS450-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-47-1.5: PN-EOS450-BSW-NUT-4706 / Kit KIT-EOS450-YAX-6 (EOS-Additive-M450) | PM Interval: 1500 Hrs | Torque: 194 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-EOS450-YAX-6 within tolerance ±0.0020 mm TIR.

BOM-47-1.6: PN-EOS450-CYL-SEAL-4707 / Kit KIT-EOS450-ZAX-7 (EOS-Additive-M450) | PM Interval: 250 Hrs | Torque: 203 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-EOS450-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-47-1.7: PN-EOS450-TRQ-STAT-4708 / Kit KIT-EOS450-BAX-8 (EOS-Additive-M450) | PM Interval: 500 Hrs | Torque: 22 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-EOS450-BAX-8 within tolerance ±0.0030 mm TIR.

BOM-47-1.8: PN-EOS450-BRK-DISC-4709 / Kit KIT-EOS450-CAX-9 (EOS-Additive-M450) | PM Interval: 750 Hrs | Torque: 31 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-EOS450-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-47-1.9: PN-EOS450-GLASS-RD-4710 / Kit KIT-EOS450-OPT-10 (EOS-Additive-M450) | PM Interval: 1000 Hrs | Torque: 40 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-EOS450-OPT-10 within tolerance ±0.0040 mm TIR.

BOM-47-1.10: PN-EOS450-IGBT-MOD-4711 / Kit KIT-EOS450-PLC-11 (EOS-Additive-M450) | PM Interval: 1250 Hrs | Torque: 49 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-EOS450-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-47-1.11: PN-EOS450-INJ-MET-4712 / Kit KIT-EOS450-LUB-12 (EOS-Additive-M450) | PM Interval: 1500 Hrs | Torque: 58 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-EOS450-LUB-12 within tolerance ±0.0050 mm TIR.

BOM-47-1.12: PN-EOS450-GASK-VIT-4713 / Kit KIT-EOS450-VAC-13 (EOS-Additive-M450) | PM Interval: 250 Hrs | Torque: 67 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-EOS450-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-47-1.13: PN-EOS450-HEX-CORE-4714 / Kit KIT-EOS450-THM-14 (EOS-Additive-M450) | PM Interval: 500 Hrs | Torque: 76 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-EOS450-THM-14 within tolerance ±0.0060 mm TIR.

BOM-47-1.14: PN-EOS450-BRG-CER-4701 / Kit KIT-EOS450-SPN-1 (EOS-Additive-M450) | PM Interval: 750 Hrs | Torque: 149 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-EOS450-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-47-1.15: PN-EOS450-CAM-GRIP-4702 / Kit KIT-EOS450-ATC-2 (EOS-Additive-M450) | PM Interval: 1000 Hrs | Torque: 158 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-EOS450-ATC-2 within tolerance ±0.0060 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — EOS-Additive-M450 (GROUP 2)

Certified Aerospace Bill of Materials for EOS-Additive-M450. Verify part numbers before closing Work Orders.

BOM-47-2.1: PN-EOS450-VLV-PROP-4703 / Kit KIT-EOS450-HYD-3 (EOS-Additive-M450) | PM Interval: 500 Hrs | Torque: 167 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-EOS450-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-47-2.2: PN-EOS450-LIN-COIL-4705 / Kit KIT-EOS450-XAX-5 (EOS-Additive-M450) | PM Interval: 750 Hrs | Torque: 185 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-EOS450-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-47-2.3: PN-EOS450-CYL-SEAL-4707 / Kit KIT-EOS450-ZAX-7 (EOS-Additive-M450) | PM Interval: 1000 Hrs | Torque: 203 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-EOS450-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-47-2.4: PN-EOS450-BRK-DISC-4709 / Kit KIT-EOS450-CAX-9 (EOS-Additive-M450) | PM Interval: 1250 Hrs | Torque: 31 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-EOS450-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-47-2.5: PN-EOS450-IGBT-MOD-4711 / Kit KIT-EOS450-PLC-11 (EOS-Additive-M450) | PM Interval: 1500 Hrs | Torque: 49 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-EOS450-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-47-2.6: PN-EOS450-GASK-VIT-4713 / Kit KIT-EOS450-VAC-13 (EOS-Additive-M450) | PM Interval: 250 Hrs | Torque: 67 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-EOS450-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-47-2.7: PN-EOS450-BRG-CER-4701 / Kit KIT-EOS450-SPN-1 (EOS-Additive-M450) | PM Interval: 500 Hrs | Torque: 149 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-EOS450-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-47-2.8: PN-EOS450-VLV-PROP-4703 / Kit KIT-EOS450-HYD-3 (EOS-Additive-M450) | PM Interval: 750 Hrs | Torque: 167 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-EOS450-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-47-2.9: PN-EOS450-LIN-COIL-4705 / Kit KIT-EOS450-XAX-5 (EOS-Additive-M450) | PM Interval: 1000 Hrs | Torque: 185 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-EOS450-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-47-2.10: PN-EOS450-CYL-SEAL-4707 / Kit KIT-EOS450-ZAX-7 (EOS-Additive-M450) | PM Interval: 1250 Hrs | Torque: 203 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-EOS450-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-47-2.11: PN-EOS450-BRK-DISC-4709 / Kit KIT-EOS450-CAX-9 (EOS-Additive-M450) | PM Interval: 1500 Hrs | Torque: 31 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-EOS450-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-47-2.12: PN-EOS450-IGBT-MOD-4711 / Kit KIT-EOS450-PLC-11 (EOS-Additive-M450) | PM Interval: 250 Hrs | Torque: 49 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-EOS450-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-47-2.13: PN-EOS450-GASK-VIT-4713 / Kit KIT-EOS450-VAC-13 (EOS-Additive-M450) | PM Interval: 500 Hrs | Torque: 67 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-EOS450-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-47-2.14: PN-EOS450-BRG-CER-4701 / Kit KIT-EOS450-SPN-1 (EOS-Additive-M450) | PM Interval: 750 Hrs | Torque: 149 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-EOS450-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-47-2.15: PN-EOS450-VLV-PROP-4703 / Kit KIT-EOS450-HYD-3 (EOS-Additive-M450) | PM Interval: 1000 Hrs | Torque: 167 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-EOS450-HYD-3 within tolerance ±0.0065 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — EOS-Additive-M450 (GROUP 3)

Certified Aerospace Bill of Materials for EOS-Additive-M450. Verify part numbers before closing Work Orders.

BOM-47-3.1: PN-EOS450-PMP-PLG-4704 / Kit KIT-EOS450-CLN-4 (EOS-Additive-M450) | PM Interval: 500 Hrs | Torque: 176 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-EOS450-CLN-4 within tolerance ±0.0010 mm TIR.

BOM-47-3.2: PN-EOS450-CYL-SEAL-4707 / Kit KIT-EOS450-ZAX-7 (EOS-Additive-M450) | PM Interval: 750 Hrs | Torque: 203 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-EOS450-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-47-3.3: PN-EOS450-GLASS-RD-4710 / Kit KIT-EOS450-OPT-10 (EOS-Additive-M450) | PM Interval: 1000 Hrs | Torque: 40 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-EOS450-OPT-10 within tolerance ±0.0040 mm TIR.

BOM-47-3.4: PN-EOS450-GASK-VIT-4713 / Kit KIT-EOS450-VAC-13 (EOS-Additive-M450) | PM Interval: 1250 Hrs | Torque: 67 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-EOS450-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-47-3.5: PN-EOS450-CAM-GRIP-4702 / Kit KIT-EOS450-ATC-2 (EOS-Additive-M450) | PM Interval: 1500 Hrs | Torque: 158 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-EOS450-ATC-2 within tolerance ±0.0060 mm TIR.

BOM-47-3.6: PN-EOS450-LIN-COIL-4705 / Kit KIT-EOS450-XAX-5 (EOS-Additive-M450) | PM Interval: 250 Hrs | Torque: 185 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-EOS450-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-47-3.7: PN-EOS450-TRQ-STAT-4708 / Kit KIT-EOS450-BAX-8 (EOS-Additive-M450) | PM Interval: 500 Hrs | Torque: 22 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-EOS450-BAX-8 within tolerance ±0.0030 mm TIR.

BOM-47-3.8: PN-EOS450-IGBT-MOD-4711 / Kit KIT-EOS450-PLC-11 (EOS-Additive-M450) | PM Interval: 750 Hrs | Torque: 49 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-EOS450-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-47-3.9: PN-EOS450-HEX-CORE-4714 / Kit KIT-EOS450-THM-14 (EOS-Additive-M450) | PM Interval: 1000 Hrs | Torque: 76 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-EOS450-THM-14 within tolerance ±0.0060 mm TIR.

BOM-47-3.10: PN-EOS450-VLV-PROP-4703 / Kit KIT-EOS450-HYD-3 (EOS-Additive-M450) | PM Interval: 1250 Hrs | Torque: 167 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-EOS450-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-47-3.11: PN-EOS450-BSW-NUT-4706 / Kit KIT-EOS450-YAX-6 (EOS-Additive-M450) | PM Interval: 1500 Hrs | Torque: 194 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-EOS450-YAX-6 within tolerance ±0.0020 mm TIR.

BOM-47-3.12: PN-EOS450-BRK-DISC-4709 / Kit KIT-EOS450-CAX-9 (EOS-Additive-M450) | PM Interval: 250 Hrs | Torque: 31 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-EOS450-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-47-3.13: PN-EOS450-INJ-MET-4712 / Kit KIT-EOS450-LUB-12 (EOS-Additive-M450) | PM Interval: 500 Hrs | Torque: 58 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-EOS450-LUB-12 within tolerance ±0.0050 mm TIR.

BOM-47-3.14: PN-EOS450-BRG-CER-4701 / Kit KIT-EOS450-SPN-1 (EOS-Additive-M450) | PM Interval: 750 Hrs | Torque: 149 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-EOS450-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-47-3.15: PN-EOS450-PMP-PLG-4704 / Kit KIT-EOS450-CLN-4 (EOS-Additive-M450) | PM Interval: 1000 Hrs | Torque: 176 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-EOS450-CLN-4 within tolerance ±0.0010 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — EOS-Additive-M500 (GROUP 1)

Certified Aerospace Bill of Materials for EOS-Additive-M500. Verify part numbers before closing Work Orders.

BOM-48-1.1: PN-EOS500-CAM-GRIP-4802 / Kit KIT-EOS500-ATC-2 (EOS-Additive-M500) | PM Interval: 500 Hrs | Torque: 108 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-EOS500-ATC-2 within tolerance ±0.0020 mm TIR.

BOM-48-1.2: PN-EOS500-VLV-PROP-4803 / Kit KIT-EOS500-HYD-3 (EOS-Additive-M500) | PM Interval: 750 Hrs | Torque: 117 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-EOS500-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-48-1.3: PN-EOS500-PMP-PLG-4804 / Kit KIT-EOS500-CLN-4 (EOS-Additive-M500) | PM Interval: 1000 Hrs | Torque: 126 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-EOS500-CLN-4 within tolerance ±0.0030 mm TIR.

BOM-48-1.4: PN-EOS500-LIN-COIL-4805 / Kit KIT-EOS500-XAX-5 (EOS-Additive-M500) | PM Interval: 1250 Hrs | Torque: 135 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-EOS500-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-48-1.5: PN-EOS500-BSW-NUT-4806 / Kit KIT-EOS500-YAX-6 (EOS-Additive-M500) | PM Interval: 1500 Hrs | Torque: 144 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-EOS500-YAX-6 within tolerance ±0.0040 mm TIR.

BOM-48-1.6: PN-EOS500-CYL-SEAL-4807 / Kit KIT-EOS500-ZAX-7 (EOS-Additive-M500) | PM Interval: 250 Hrs | Torque: 153 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-EOS500-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-48-1.7: PN-EOS500-TRQ-STAT-4808 / Kit KIT-EOS500-BAX-8 (EOS-Additive-M500) | PM Interval: 500 Hrs | Torque: 162 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-EOS500-BAX-8 within tolerance ±0.0050 mm TIR.

BOM-48-1.8: PN-EOS500-BRK-DISC-4809 / Kit KIT-EOS500-CAX-9 (EOS-Additive-M500) | PM Interval: 750 Hrs | Torque: 171 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-EOS500-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-48-1.9: PN-EOS500-GLASS-RD-4810 / Kit KIT-EOS500-OPT-10 (EOS-Additive-M500) | PM Interval: 1000 Hrs | Torque: 180 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-EOS500-OPT-10 within tolerance ±0.0060 mm TIR.

BOM-48-1.10: PN-EOS500-IGBT-MOD-4811 / Kit KIT-EOS500-PLC-11 (EOS-Additive-M500) | PM Interval: 1250 Hrs | Torque: 189 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-EOS500-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-48-1.11: PN-EOS500-INJ-MET-4812 / Kit KIT-EOS500-LUB-12 (EOS-Additive-M500) | PM Interval: 1500 Hrs | Torque: 198 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-EOS500-LUB-12 within tolerance ±0.0010 mm TIR.

BOM-48-1.12: PN-EOS500-GASK-VIT-4813 / Kit KIT-EOS500-VAC-13 (EOS-Additive-M500) | PM Interval: 250 Hrs | Torque: 207 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-EOS500-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-48-1.13: PN-EOS500-HEX-CORE-4814 / Kit KIT-EOS500-THM-14 (EOS-Additive-M500) | PM Interval: 500 Hrs | Torque: 26 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-EOS500-THM-14 within tolerance ±0.0020 mm TIR.

BOM-48-1.14: PN-EOS500-BRG-CER-4801 / Kit KIT-EOS500-SPN-1 (EOS-Additive-M500) | PM Interval: 750 Hrs | Torque: 99 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-EOS500-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-48-1.15: PN-EOS500-CAM-GRIP-4802 / Kit KIT-EOS500-ATC-2 (EOS-Additive-M500) | PM Interval: 1000 Hrs | Torque: 108 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-EOS500-ATC-2 within tolerance ±0.0020 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — EOS-Additive-M500 (GROUP 2)

Certified Aerospace Bill of Materials for EOS-Additive-M500. Verify part numbers before closing Work Orders.

BOM-48-2.1: PN-EOS500-VLV-PROP-4803 / Kit KIT-EOS500-HYD-3 (EOS-Additive-M500) | PM Interval: 500 Hrs | Torque: 117 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-EOS500-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-48-2.2: PN-EOS500-LIN-COIL-4805 / Kit KIT-EOS500-XAX-5 (EOS-Additive-M500) | PM Interval: 750 Hrs | Torque: 135 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-EOS500-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-48-2.3: PN-EOS500-CYL-SEAL-4807 / Kit KIT-EOS500-ZAX-7 (EOS-Additive-M500) | PM Interval: 1000 Hrs | Torque: 153 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-EOS500-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-48-2.4: PN-EOS500-BRK-DISC-4809 / Kit KIT-EOS500-CAX-9 (EOS-Additive-M500) | PM Interval: 1250 Hrs | Torque: 171 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-EOS500-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-48-2.5: PN-EOS500-IGBT-MOD-4811 / Kit KIT-EOS500-PLC-11 (EOS-Additive-M500) | PM Interval: 1500 Hrs | Torque: 189 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-EOS500-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-48-2.6: PN-EOS500-GASK-VIT-4813 / Kit KIT-EOS500-VAC-13 (EOS-Additive-M500) | PM Interval: 250 Hrs | Torque: 207 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-EOS500-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-48-2.7: PN-EOS500-BRG-CER-4801 / Kit KIT-EOS500-SPN-1 (EOS-Additive-M500) | PM Interval: 500 Hrs | Torque: 99 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-EOS500-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-48-2.8: PN-EOS500-VLV-PROP-4803 / Kit KIT-EOS500-HYD-3 (EOS-Additive-M500) | PM Interval: 750 Hrs | Torque: 117 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-EOS500-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-48-2.9: PN-EOS500-LIN-COIL-4805 / Kit KIT-EOS500-XAX-5 (EOS-Additive-M500) | PM Interval: 1000 Hrs | Torque: 135 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-EOS500-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-48-2.10: PN-EOS500-CYL-SEAL-4807 / Kit KIT-EOS500-ZAX-7 (EOS-Additive-M500) | PM Interval: 1250 Hrs | Torque: 153 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-EOS500-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-48-2.11: PN-EOS500-BRK-DISC-4809 / Kit KIT-EOS500-CAX-9 (EOS-Additive-M500) | PM Interval: 1500 Hrs | Torque: 171 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-EOS500-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-48-2.12: PN-EOS500-IGBT-MOD-4811 / Kit KIT-EOS500-PLC-11 (EOS-Additive-M500) | PM Interval: 250 Hrs | Torque: 189 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-EOS500-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-48-2.13: PN-EOS500-GASK-VIT-4813 / Kit KIT-EOS500-VAC-13 (EOS-Additive-M500) | PM Interval: 500 Hrs | Torque: 207 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-EOS500-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-48-2.14: PN-EOS500-BRG-CER-4801 / Kit KIT-EOS500-SPN-1 (EOS-Additive-M500) | PM Interval: 750 Hrs | Torque: 99 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-EOS500-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-48-2.15: PN-EOS500-VLV-PROP-4803 / Kit KIT-EOS500-HYD-3 (EOS-Additive-M500) | PM Interval: 1000 Hrs | Torque: 117 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-EOS500-HYD-3 within tolerance ±0.0025 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — EOS-Additive-M500 (GROUP 3)

Certified Aerospace Bill of Materials for EOS-Additive-M500. Verify part numbers before closing Work Orders.

BOM-48-3.1: PN-EOS500-PMP-PLG-4804 / Kit KIT-EOS500-CLN-4 (EOS-Additive-M500) | PM Interval: 500 Hrs | Torque: 126 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-EOS500-CLN-4 within tolerance ±0.0030 mm TIR.

BOM-48-3.2: PN-EOS500-CYL-SEAL-4807 / Kit KIT-EOS500-ZAX-7 (EOS-Additive-M500) | PM Interval: 750 Hrs | Torque: 153 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-EOS500-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-48-3.3: PN-EOS500-GLASS-RD-4810 / Kit KIT-EOS500-OPT-10 (EOS-Additive-M500) | PM Interval: 1000 Hrs | Torque: 180 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-EOS500-OPT-10 within tolerance ±0.0060 mm TIR.

BOM-48-3.4: PN-EOS500-GASK-VIT-4813 / Kit KIT-EOS500-VAC-13 (EOS-Additive-M500) | PM Interval: 1250 Hrs | Torque: 207 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-EOS500-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-48-3.5: PN-EOS500-CAM-GRIP-4802 / Kit KIT-EOS500-ATC-2 (EOS-Additive-M500) | PM Interval: 1500 Hrs | Torque: 108 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-EOS500-ATC-2 within tolerance ±0.0020 mm TIR.

BOM-48-3.6: PN-EOS500-LIN-COIL-4805 / Kit KIT-EOS500-XAX-5 (EOS-Additive-M500) | PM Interval: 250 Hrs | Torque: 135 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-EOS500-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-48-3.7: PN-EOS500-TRQ-STAT-4808 / Kit KIT-EOS500-BAX-8 (EOS-Additive-M500) | PM Interval: 500 Hrs | Torque: 162 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-EOS500-BAX-8 within tolerance ±0.0050 mm TIR.

BOM-48-3.8: PN-EOS500-IGBT-MOD-4811 / Kit KIT-EOS500-PLC-11 (EOS-Additive-M500) | PM Interval: 750 Hrs | Torque: 189 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-EOS500-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-48-3.9: PN-EOS500-HEX-CORE-4814 / Kit KIT-EOS500-THM-14 (EOS-Additive-M500) | PM Interval: 1000 Hrs | Torque: 26 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-EOS500-THM-14 within tolerance ±0.0020 mm TIR.

BOM-48-3.10: PN-EOS500-VLV-PROP-4803 / Kit KIT-EOS500-HYD-3 (EOS-Additive-M500) | PM Interval: 1250 Hrs | Torque: 117 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-EOS500-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-48-3.11: PN-EOS500-BSW-NUT-4806 / Kit KIT-EOS500-YAX-6 (EOS-Additive-M500) | PM Interval: 1500 Hrs | Torque: 144 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-EOS500-YAX-6 within tolerance ±0.0040 mm TIR.

BOM-48-3.12: PN-EOS500-BRK-DISC-4809 / Kit KIT-EOS500-CAX-9 (EOS-Additive-M500) | PM Interval: 250 Hrs | Torque: 171 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-EOS500-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-48-3.13: PN-EOS500-INJ-MET-4812 / Kit KIT-EOS500-LUB-12 (EOS-Additive-M500) | PM Interval: 500 Hrs | Torque: 198 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-EOS500-LUB-12 within tolerance ±0.0010 mm TIR.

BOM-48-3.14: PN-EOS500-BRG-CER-4801 / Kit KIT-EOS500-SPN-1 (EOS-Additive-M500) | PM Interval: 750 Hrs | Torque: 99 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-EOS500-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-48-3.15: PN-EOS500-PMP-PLG-4804 / Kit KIT-EOS500-CLN-4 (EOS-Additive-M500) | PM Interval: 1000 Hrs | Torque: 126 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-EOS500-CLN-4 within tolerance ±0.0030 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — EOS-Additive-M550 (GROUP 1)

Certified Aerospace Bill of Materials for EOS-Additive-M550. Verify part numbers before closing Work Orders.

BOM-49-1.1: PN-EOS550-CAM-GRIP-4902 / Kit KIT-EOS550-ATC-2 (EOS-Additive-M550) | PM Interval: 500 Hrs | Torque: 58 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-EOS550-ATC-2 within tolerance ±0.0040 mm TIR.

BOM-49-1.2: PN-EOS550-VLV-PROP-4903 / Kit KIT-EOS550-HYD-3 (EOS-Additive-M550) | PM Interval: 750 Hrs | Torque: 67 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-EOS550-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-49-1.3: PN-EOS550-PMP-PLG-4904 / Kit KIT-EOS550-CLN-4 (EOS-Additive-M550) | PM Interval: 1000 Hrs | Torque: 76 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-EOS550-CLN-4 within tolerance ±0.0050 mm TIR.

BOM-49-1.4: PN-EOS550-LIN-COIL-4905 / Kit KIT-EOS550-XAX-5 (EOS-Additive-M550) | PM Interval: 1250 Hrs | Torque: 85 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-EOS550-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-49-1.5: PN-EOS550-BSW-NUT-4906 / Kit KIT-EOS550-YAX-6 (EOS-Additive-M550) | PM Interval: 1500 Hrs | Torque: 94 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-EOS550-YAX-6 within tolerance ±0.0060 mm TIR.

BOM-49-1.6: PN-EOS550-CYL-SEAL-4907 / Kit KIT-EOS550-ZAX-7 (EOS-Additive-M550) | PM Interval: 250 Hrs | Torque: 103 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-EOS550-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-49-1.7: PN-EOS550-TRQ-STAT-4908 / Kit KIT-EOS550-BAX-8 (EOS-Additive-M550) | PM Interval: 500 Hrs | Torque: 112 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-EOS550-BAX-8 within tolerance ±0.0010 mm TIR.

BOM-49-1.8: PN-EOS550-BRK-DISC-4909 / Kit KIT-EOS550-CAX-9 (EOS-Additive-M550) | PM Interval: 750 Hrs | Torque: 121 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-EOS550-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-49-1.9: PN-EOS550-GLASS-RD-4910 / Kit KIT-EOS550-OPT-10 (EOS-Additive-M550) | PM Interval: 1000 Hrs | Torque: 130 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-EOS550-OPT-10 within tolerance ±0.0020 mm TIR.

BOM-49-1.10: PN-EOS550-IGBT-MOD-4911 / Kit KIT-EOS550-PLC-11 (EOS-Additive-M550) | PM Interval: 1250 Hrs | Torque: 139 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-EOS550-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-49-1.11: PN-EOS550-INJ-MET-4912 / Kit KIT-EOS550-LUB-12 (EOS-Additive-M550) | PM Interval: 1500 Hrs | Torque: 148 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-EOS550-LUB-12 within tolerance ±0.0030 mm TIR.

BOM-49-1.12: PN-EOS550-GASK-VIT-4913 / Kit KIT-EOS550-VAC-13 (EOS-Additive-M550) | PM Interval: 250 Hrs | Torque: 157 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-EOS550-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-49-1.13: PN-EOS550-HEX-CORE-4914 / Kit KIT-EOS550-THM-14 (EOS-Additive-M550) | PM Interval: 500 Hrs | Torque: 166 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-EOS550-THM-14 within tolerance ±0.0040 mm TIR.

BOM-49-1.14: PN-EOS550-BRG-CER-4901 / Kit KIT-EOS550-SPN-1 (EOS-Additive-M550) | PM Interval: 750 Hrs | Torque: 49 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-EOS550-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-49-1.15: PN-EOS550-CAM-GRIP-4902 / Kit KIT-EOS550-ATC-2 (EOS-Additive-M550) | PM Interval: 1000 Hrs | Torque: 58 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-EOS550-ATC-2 within tolerance ±0.0040 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — EOS-Additive-M550 (GROUP 2)

Certified Aerospace Bill of Materials for EOS-Additive-M550. Verify part numbers before closing Work Orders.

BOM-49-2.1: PN-EOS550-VLV-PROP-4903 / Kit KIT-EOS550-HYD-3 (EOS-Additive-M550) | PM Interval: 500 Hrs | Torque: 67 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-EOS550-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-49-2.2: PN-EOS550-LIN-COIL-4905 / Kit KIT-EOS550-XAX-5 (EOS-Additive-M550) | PM Interval: 750 Hrs | Torque: 85 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-EOS550-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-49-2.3: PN-EOS550-CYL-SEAL-4907 / Kit KIT-EOS550-ZAX-7 (EOS-Additive-M550) | PM Interval: 1000 Hrs | Torque: 103 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-EOS550-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-49-2.4: PN-EOS550-BRK-DISC-4909 / Kit KIT-EOS550-CAX-9 (EOS-Additive-M550) | PM Interval: 1250 Hrs | Torque: 121 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-EOS550-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-49-2.5: PN-EOS550-IGBT-MOD-4911 / Kit KIT-EOS550-PLC-11 (EOS-Additive-M550) | PM Interval: 1500 Hrs | Torque: 139 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-EOS550-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-49-2.6: PN-EOS550-GASK-VIT-4913 / Kit KIT-EOS550-VAC-13 (EOS-Additive-M550) | PM Interval: 250 Hrs | Torque: 157 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-EOS550-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-49-2.7: PN-EOS550-BRG-CER-4901 / Kit KIT-EOS550-SPN-1 (EOS-Additive-M550) | PM Interval: 500 Hrs | Torque: 49 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-EOS550-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-49-2.8: PN-EOS550-VLV-PROP-4903 / Kit KIT-EOS550-HYD-3 (EOS-Additive-M550) | PM Interval: 750 Hrs | Torque: 67 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-EOS550-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-49-2.9: PN-EOS550-LIN-COIL-4905 / Kit KIT-EOS550-XAX-5 (EOS-Additive-M550) | PM Interval: 1000 Hrs | Torque: 85 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-EOS550-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-49-2.10: PN-EOS550-CYL-SEAL-4907 / Kit KIT-EOS550-ZAX-7 (EOS-Additive-M550) | PM Interval: 1250 Hrs | Torque: 103 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-EOS550-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-49-2.11: PN-EOS550-BRK-DISC-4909 / Kit KIT-EOS550-CAX-9 (EOS-Additive-M550) | PM Interval: 1500 Hrs | Torque: 121 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-EOS550-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-49-2.12: PN-EOS550-IGBT-MOD-4911 / Kit KIT-EOS550-PLC-11 (EOS-Additive-M550) | PM Interval: 250 Hrs | Torque: 139 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-EOS550-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-49-2.13: PN-EOS550-GASK-VIT-4913 / Kit KIT-EOS550-VAC-13 (EOS-Additive-M550) | PM Interval: 500 Hrs | Torque: 157 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-EOS550-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-49-2.14: PN-EOS550-BRG-CER-4901 / Kit KIT-EOS550-SPN-1 (EOS-Additive-M550) | PM Interval: 750 Hrs | Torque: 49 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-EOS550-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-49-2.15: PN-EOS550-VLV-PROP-4903 / Kit KIT-EOS550-HYD-3 (EOS-Additive-M550) | PM Interval: 1000 Hrs | Torque: 67 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-EOS550-HYD-3 within tolerance ±0.0045 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — EOS-Additive-M550 (GROUP 3)

Certified Aerospace Bill of Materials for EOS-Additive-M550. Verify part numbers before closing Work Orders.

BOM-49-3.1: PN-EOS550-PMP-PLG-4904 / Kit KIT-EOS550-CLN-4 (EOS-Additive-M550) | PM Interval: 500 Hrs | Torque: 76 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-EOS550-CLN-4 within tolerance ±0.0050 mm TIR.

BOM-49-3.2: PN-EOS550-CYL-SEAL-4907 / Kit KIT-EOS550-ZAX-7 (EOS-Additive-M550) | PM Interval: 750 Hrs | Torque: 103 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-EOS550-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-49-3.3: PN-EOS550-GLASS-RD-4910 / Kit KIT-EOS550-OPT-10 (EOS-Additive-M550) | PM Interval: 1000 Hrs | Torque: 130 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-EOS550-OPT-10 within tolerance ±0.0020 mm TIR.

BOM-49-3.4: PN-EOS550-GASK-VIT-4913 / Kit KIT-EOS550-VAC-13 (EOS-Additive-M550) | PM Interval: 1250 Hrs | Torque: 157 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-EOS550-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-49-3.5: PN-EOS550-CAM-GRIP-4902 / Kit KIT-EOS550-ATC-2 (EOS-Additive-M550) | PM Interval: 1500 Hrs | Torque: 58 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-EOS550-ATC-2 within tolerance ±0.0040 mm TIR.

BOM-49-3.6: PN-EOS550-LIN-COIL-4905 / Kit KIT-EOS550-XAX-5 (EOS-Additive-M550) | PM Interval: 250 Hrs | Torque: 85 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-EOS550-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-49-3.7: PN-EOS550-TRQ-STAT-4908 / Kit KIT-EOS550-BAX-8 (EOS-Additive-M550) | PM Interval: 500 Hrs | Torque: 112 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-EOS550-BAX-8 within tolerance ±0.0010 mm TIR.

BOM-49-3.8: PN-EOS550-IGBT-MOD-4911 / Kit KIT-EOS550-PLC-11 (EOS-Additive-M550) | PM Interval: 750 Hrs | Torque: 139 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-EOS550-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-49-3.9: PN-EOS550-HEX-CORE-4914 / Kit KIT-EOS550-THM-14 (EOS-Additive-M550) | PM Interval: 1000 Hrs | Torque: 166 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-EOS550-THM-14 within tolerance ±0.0040 mm TIR.

BOM-49-3.10: PN-EOS550-VLV-PROP-4903 / Kit KIT-EOS550-HYD-3 (EOS-Additive-M550) | PM Interval: 1250 Hrs | Torque: 67 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-EOS550-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-49-3.11: PN-EOS550-BSW-NUT-4906 / Kit KIT-EOS550-YAX-6 (EOS-Additive-M550) | PM Interval: 1500 Hrs | Torque: 94 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-EOS550-YAX-6 within tolerance ±0.0060 mm TIR.

BOM-49-3.12: PN-EOS550-BRK-DISC-4909 / Kit KIT-EOS550-CAX-9 (EOS-Additive-M550) | PM Interval: 250 Hrs | Torque: 121 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-EOS550-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-49-3.13: PN-EOS550-INJ-MET-4912 / Kit KIT-EOS550-LUB-12 (EOS-Additive-M550) | PM Interval: 500 Hrs | Torque: 148 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-EOS550-LUB-12 within tolerance ±0.0030 mm TIR.

BOM-49-3.14: PN-EOS550-BRG-CER-4901 / Kit KIT-EOS550-SPN-1 (EOS-Additive-M550) | PM Interval: 750 Hrs | Torque: 49 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-EOS550-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-49-3.15: PN-EOS550-PMP-PLG-4904 / Kit KIT-EOS550-CLN-4 (EOS-Additive-M550) | PM Interval: 1000 Hrs | Torque: 76 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-EOS550-CLN-4 within tolerance ±0.0050 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — EOS-Additive-M600 (GROUP 1)

Certified Aerospace Bill of Materials for EOS-Additive-M600. Verify part numbers before closing Work Orders.

BOM-50-1.1: PN-EOS600-CAM-GRIP-5002 / Kit KIT-EOS600-ATC-2 (EOS-Additive-M600) | PM Interval: 500 Hrs | Torque: 198 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-EOS600-ATC-2 within tolerance ±0.0060 mm TIR.

BOM-50-1.2: PN-EOS600-VLV-PROP-5003 / Kit KIT-EOS600-HYD-3 (EOS-Additive-M600) | PM Interval: 750 Hrs | Torque: 207 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-EOS600-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-50-1.3: PN-EOS600-PMP-PLG-5004 / Kit KIT-EOS600-CLN-4 (EOS-Additive-M600) | PM Interval: 1000 Hrs | Torque: 26 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-EOS600-CLN-4 within tolerance ±0.0010 mm TIR.

BOM-50-1.4: PN-EOS600-LIN-COIL-5005 / Kit KIT-EOS600-XAX-5 (EOS-Additive-M600) | PM Interval: 1250 Hrs | Torque: 35 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-EOS600-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-50-1.5: PN-EOS600-BSW-NUT-5006 / Kit KIT-EOS600-YAX-6 (EOS-Additive-M600) | PM Interval: 1500 Hrs | Torque: 44 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-EOS600-YAX-6 within tolerance ±0.0020 mm TIR.

BOM-50-1.6: PN-EOS600-CYL-SEAL-5007 / Kit KIT-EOS600-ZAX-7 (EOS-Additive-M600) | PM Interval: 250 Hrs | Torque: 53 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-EOS600-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-50-1.7: PN-EOS600-TRQ-STAT-5008 / Kit KIT-EOS600-BAX-8 (EOS-Additive-M600) | PM Interval: 500 Hrs | Torque: 62 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-EOS600-BAX-8 within tolerance ±0.0030 mm TIR.

BOM-50-1.8: PN-EOS600-BRK-DISC-5009 / Kit KIT-EOS600-CAX-9 (EOS-Additive-M600) | PM Interval: 750 Hrs | Torque: 71 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-EOS600-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-50-1.9: PN-EOS600-GLASS-RD-5010 / Kit KIT-EOS600-OPT-10 (EOS-Additive-M600) | PM Interval: 1000 Hrs | Torque: 80 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-EOS600-OPT-10 within tolerance ±0.0040 mm TIR.

BOM-50-1.10: PN-EOS600-IGBT-MOD-5011 / Kit KIT-EOS600-PLC-11 (EOS-Additive-M600) | PM Interval: 1250 Hrs | Torque: 89 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-EOS600-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-50-1.11: PN-EOS600-INJ-MET-5012 / Kit KIT-EOS600-LUB-12 (EOS-Additive-M600) | PM Interval: 1500 Hrs | Torque: 98 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-EOS600-LUB-12 within tolerance ±0.0050 mm TIR.

BOM-50-1.12: PN-EOS600-GASK-VIT-5013 / Kit KIT-EOS600-VAC-13 (EOS-Additive-M600) | PM Interval: 250 Hrs | Torque: 107 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-EOS600-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-50-1.13: PN-EOS600-HEX-CORE-5014 / Kit KIT-EOS600-THM-14 (EOS-Additive-M600) | PM Interval: 500 Hrs | Torque: 116 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-EOS600-THM-14 within tolerance ±0.0060 mm TIR.

BOM-50-1.14: PN-EOS600-BRG-CER-5001 / Kit KIT-EOS600-SPN-1 (EOS-Additive-M600) | PM Interval: 750 Hrs | Torque: 189 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-EOS600-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-50-1.15: PN-EOS600-CAM-GRIP-5002 / Kit KIT-EOS600-ATC-2 (EOS-Additive-M600) | PM Interval: 1000 Hrs | Torque: 198 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-EOS600-ATC-2 within tolerance ±0.0060 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — EOS-Additive-M600 (GROUP 2)

Certified Aerospace Bill of Materials for EOS-Additive-M600. Verify part numbers before closing Work Orders.

BOM-50-2.1: PN-EOS600-VLV-PROP-5003 / Kit KIT-EOS600-HYD-3 (EOS-Additive-M600) | PM Interval: 500 Hrs | Torque: 207 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-EOS600-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-50-2.2: PN-EOS600-LIN-COIL-5005 / Kit KIT-EOS600-XAX-5 (EOS-Additive-M600) | PM Interval: 750 Hrs | Torque: 35 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-EOS600-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-50-2.3: PN-EOS600-CYL-SEAL-5007 / Kit KIT-EOS600-ZAX-7 (EOS-Additive-M600) | PM Interval: 1000 Hrs | Torque: 53 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-EOS600-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-50-2.4: PN-EOS600-BRK-DISC-5009 / Kit KIT-EOS600-CAX-9 (EOS-Additive-M600) | PM Interval: 1250 Hrs | Torque: 71 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-EOS600-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-50-2.5: PN-EOS600-IGBT-MOD-5011 / Kit KIT-EOS600-PLC-11 (EOS-Additive-M600) | PM Interval: 1500 Hrs | Torque: 89 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-EOS600-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-50-2.6: PN-EOS600-GASK-VIT-5013 / Kit KIT-EOS600-VAC-13 (EOS-Additive-M600) | PM Interval: 250 Hrs | Torque: 107 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-EOS600-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-50-2.7: PN-EOS600-BRG-CER-5001 / Kit KIT-EOS600-SPN-1 (EOS-Additive-M600) | PM Interval: 500 Hrs | Torque: 189 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-EOS600-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-50-2.8: PN-EOS600-VLV-PROP-5003 / Kit KIT-EOS600-HYD-3 (EOS-Additive-M600) | PM Interval: 750 Hrs | Torque: 207 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-EOS600-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-50-2.9: PN-EOS600-LIN-COIL-5005 / Kit KIT-EOS600-XAX-5 (EOS-Additive-M600) | PM Interval: 1000 Hrs | Torque: 35 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-EOS600-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-50-2.10: PN-EOS600-CYL-SEAL-5007 / Kit KIT-EOS600-ZAX-7 (EOS-Additive-M600) | PM Interval: 1250 Hrs | Torque: 53 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-EOS600-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-50-2.11: PN-EOS600-BRK-DISC-5009 / Kit KIT-EOS600-CAX-9 (EOS-Additive-M600) | PM Interval: 1500 Hrs | Torque: 71 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-EOS600-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-50-2.12: PN-EOS600-IGBT-MOD-5011 / Kit KIT-EOS600-PLC-11 (EOS-Additive-M600) | PM Interval: 250 Hrs | Torque: 89 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-EOS600-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-50-2.13: PN-EOS600-GASK-VIT-5013 / Kit KIT-EOS600-VAC-13 (EOS-Additive-M600) | PM Interval: 500 Hrs | Torque: 107 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-EOS600-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-50-2.14: PN-EOS600-BRG-CER-5001 / Kit KIT-EOS600-SPN-1 (EOS-Additive-M600) | PM Interval: 750 Hrs | Torque: 189 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-EOS600-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-50-2.15: PN-EOS600-VLV-PROP-5003 / Kit KIT-EOS600-HYD-3 (EOS-Additive-M600) | PM Interval: 1000 Hrs | Torque: 207 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-EOS600-HYD-3 within tolerance ±0.0065 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — EOS-Additive-M600 (GROUP 3)

Certified Aerospace Bill of Materials for EOS-Additive-M600. Verify part numbers before closing Work Orders.

BOM-50-3.1: PN-EOS600-PMP-PLG-5004 / Kit KIT-EOS600-CLN-4 (EOS-Additive-M600) | PM Interval: 500 Hrs | Torque: 26 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-EOS600-CLN-4 within tolerance ±0.0010 mm TIR.

BOM-50-3.2: PN-EOS600-CYL-SEAL-5007 / Kit KIT-EOS600-ZAX-7 (EOS-Additive-M600) | PM Interval: 750 Hrs | Torque: 53 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-EOS600-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-50-3.3: PN-EOS600-GLASS-RD-5010 / Kit KIT-EOS600-OPT-10 (EOS-Additive-M600) | PM Interval: 1000 Hrs | Torque: 80 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-EOS600-OPT-10 within tolerance ±0.0040 mm TIR.

BOM-50-3.4: PN-EOS600-GASK-VIT-5013 / Kit KIT-EOS600-VAC-13 (EOS-Additive-M600) | PM Interval: 1250 Hrs | Torque: 107 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-EOS600-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-50-3.5: PN-EOS600-CAM-GRIP-5002 / Kit KIT-EOS600-ATC-2 (EOS-Additive-M600) | PM Interval: 1500 Hrs | Torque: 198 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-EOS600-ATC-2 within tolerance ±0.0060 mm TIR.

BOM-50-3.6: PN-EOS600-LIN-COIL-5005 / Kit KIT-EOS600-XAX-5 (EOS-Additive-M600) | PM Interval: 250 Hrs | Torque: 35 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-EOS600-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-50-3.7: PN-EOS600-TRQ-STAT-5008 / Kit KIT-EOS600-BAX-8 (EOS-Additive-M600) | PM Interval: 500 Hrs | Torque: 62 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-EOS600-BAX-8 within tolerance ±0.0030 mm TIR.

BOM-50-3.8: PN-EOS600-IGBT-MOD-5011 / Kit KIT-EOS600-PLC-11 (EOS-Additive-M600) | PM Interval: 750 Hrs | Torque: 89 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-EOS600-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-50-3.9: PN-EOS600-HEX-CORE-5014 / Kit KIT-EOS600-THM-14 (EOS-Additive-M600) | PM Interval: 1000 Hrs | Torque: 116 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-EOS600-THM-14 within tolerance ±0.0060 mm TIR.

BOM-50-3.10: PN-EOS600-VLV-PROP-5003 / Kit KIT-EOS600-HYD-3 (EOS-Additive-M600) | PM Interval: 1250 Hrs | Torque: 207 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-EOS600-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-50-3.11: PN-EOS600-BSW-NUT-5006 / Kit KIT-EOS600-YAX-6 (EOS-Additive-M600) | PM Interval: 1500 Hrs | Torque: 44 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-EOS600-YAX-6 within tolerance ±0.0020 mm TIR.

BOM-50-3.12: PN-EOS600-BRK-DISC-5009 / Kit KIT-EOS600-CAX-9 (EOS-Additive-M600) | PM Interval: 250 Hrs | Torque: 71 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-EOS600-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-50-3.13: PN-EOS600-INJ-MET-5012 / Kit KIT-EOS600-LUB-12 (EOS-Additive-M600) | PM Interval: 500 Hrs | Torque: 98 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-EOS600-LUB-12 within tolerance ±0.0050 mm TIR.

BOM-50-3.14: PN-EOS600-BRG-CER-5001 / Kit KIT-EOS600-SPN-1 (EOS-Additive-M600) | PM Interval: 750 Hrs | Torque: 189 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-EOS600-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-50-3.15: PN-EOS600-PMP-PLG-5004 / Kit KIT-EOS600-CLN-4 (EOS-Additive-M600) | PM Interval: 1000 Hrs | Torque: 26 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-EOS600-CLN-4 within tolerance ±0.0010 mm TIR.